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Constantin P. Niculescu
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Convex Functions and Their Applications

A Contemporary Approach

Third Edition



Springer

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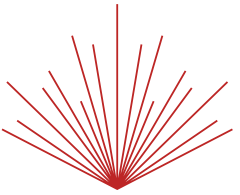
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Constantin P. Niculescu · Lars-Erik Persson

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A Contemporary Approach

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To Liliana and Lena

Preface

Convexity is a simple and natural notion which can be traced back to Archimedes (circa 250 B.C.), in connection with his famous estimate of the value of π (by using inscribed and circumscribed regular polygons). He noticed the important fact that the perimeter of a convex figure is smaller than the perimeter of any other convex figure surrounding it.

As a matter of fact, we experience convexity all the time and in many ways. The most prosaic example is our upright position, which is secured as long as the vertical projection of our center of gravity lies inside the convex envelope of our feet. Also, convexity has a great impact on our everyday life through numerous applications in industry, business, medicine, and art. So do the problems of optimum allocation of resources, estimation and signal processing, statistics, and finance to name just a few.

The recognition of convex functions as a subject that deserves to be studied in its own right is generally ascribed to J. L. W. V. Jensen [273], [274]. However, he was not the first to deal with such functions. Among his predecessors, we should recall here Ch. Hermite [251], O. Hölder [266], and O. Stolz [555]. During the whole twentieth century, there was intense research activity and many significant results were obtained in geometric functional analysis, mathematical economics, convex analysis, and nonlinear optimization. The classic book by G. H. Hardy, J. E. Littlewood, and G. Pólya [247] played a prominent role in the popularization of the subject of convex functions.

What motivates the constant interest in this subject?

First of all, its elegance and the possibility to prove deep results even with simple mathematical tools.

Secondly, many problems raised by science, engineering, economics, informatics, etc., fall in the area of convex analysis. More and more mathematicians are seeking the hidden convexity, as a way to unveil the true nature of certain intricate problems. There are two basic properties of convex functions that make them so widely used in theoretical and applied mathematics:

The maximum is attained at a boundary point.

Any local minimum is a global one. Moreover, a strictly convex function admits at most one minimum.

The modern viewpoint on convex functions entails a powerful and elegant interaction between analysis and geometry. In a memorable paper dedicated

to the Brunn–Minkowski inequality, R. J. Gardner [214, p. 358], described this reality in beautiful phrases: [convexity] "appears like an octopus, tentacles reaching far and wide, its shape and color changing as it roams from one area to the next. It is quite clear that research opportunities abound."

Over the years, a number of notable books dedicated to the theory and applications of convex functions appeared. We mention here: H. H. Bauschke and P. L. Combettes [43], J. M. Borwein and J. Vanderwerff [90], S. Boyd and L. Vandenberghe [95], J.-B. Hiriart-Urruty and C. Lemaréchal [257], L. Hörmander [268], M. A. Krasnosel'skii and Ya. B. Rutickii [307], J. E. Pečarić, F. Proschan and Y. C. Tong [463], R. R. Phelps [473], [474], A. W. Roberts and D. E. Varberg [504], R. T. Rockafellar [505], R. T. Rockafellar and R. J.-B. Wets [506], and B. Simon [542], [543]. The references at the end of this book include many other fine books pertaining to various aspects of the theory.

The title of L. Hörmander's book, *Notions of Convexity,* is very suggestive for the present state of the art. In fact, nowadays, the study of convex functions has evolved into a larger theory about functions which are adapted to other geometries of the domain and/or obey other laws of comparison of means. Examples are log-convex functions, multiplicatively convex functions, subharmonic functions, and functions which are convex with respect to a subgroup of the linear group.

Our book aims to be a thorough introduction to contemporary convex function theory. It covers a large variety of subjects, from the one real variable case to the infinite-dimensional case, including Jensen's inequality and its ramifications, the Hardy–Littlewood–Pólya theory of majorization, the Borell–Brascamp–Lieb form of the Prékopa–Leindler inequality (as well as its connection with isoperimetric inequalities), the Legendre–Fenchel duality, Alexandrov's result on the second differentiability of convex functions, the highlights of Choquet's theory, and many more. It is certainly a book where inequalities play a central role but in no case a book on inequalities. Many results are new, and the whole book reflects our own experience, both in teaching and research.

The necessary background is advanced calculus, linear algebra, and some elements of real analysis. When necessary, the reader will be guided to the most pertinent sources, where the quoted results are presented as transparently as possible.

This book aims to serve many purposes, ranging from honors options for undergraduate students to one-semester graduate course on Convex Functions and Applications.

For example, Chapters 1 and 2 together with Appendix A offer a quick introduction to generalized convexity, a subject which has became very popular during the last decades. The same combination works nicely as a supplementary material for a seminar debating heuristics of mathematical research. Chapters 3–8 together with Appendices B and E could be used as a reference text for a graduate course. And the options can continue.

In order to avoid any confusion relative to our notation, a comprehensive symbol index was added for the convenience of the reader.

A word of caution is necessary. In this book,

$$\mathbb{N} = \{0, 1, 2, ...\} \quad \text{and} \quad \mathbb{N}^* = \{1, 2, 3, ...\}.$$

According to a recent tendency in mathematical terminology, we call a number x *positive* if $x \geq 0$ and *strictly positive* if $x > 0$. A function f is called *increasing* if $x \leq y$ implies $f(x) \leq f(y)$ and *strictly increasing* if $x < y$ implies $f(x) < f(y)$.

Notice also that our book deals only with *real* linear spaces and *all* Borel measures under attention are assumed to be *regular*.

This third edition of our book is motivated by our desire to update the content of the book with many new results that have appeared in recent years, and also to shift the focus to facts that illustrate the utility of convex analysis methods in real-world problems. Furthermore, we have taken great care to make the text accessible to a wider audience, including students and researchers outside mathematics departments. This imposed a rearrangement of the whole material into nine chapters and five appendices.

The former Chapter 1 was split into two new chapters, *Convex Functions at a First Glance* and *More on Convex Functions on Intervals*, offering together a panorama of the one variable convex functions theory. Both chapters include many new examples, comments, and exercises. Moreover, Sections 2.5 and 2.6 are new. Section 2.5 is an introduction to the topic of stochastic orderings and includes a proof of the important cut-criterion of S. Karlin and A. Novikoff [288]. Section 2.6 is devoted to the higher order convex functions (in the sense of E. Hopf [259] and T. Popoviciu [483]). One good reason for this addition is the ubiquity of functions combining different orders of convexity (see, for example, the case of completely monotone functions). A number of very recent functional inequalities that appear here for the first time supports this remark.

In Chapter 3, Section 3.4 (on *Ordered Banach spaces*), and Section 3.5 (on the *Spectral properties of symmetric matrices*) were entirely rewritten to ensure the necessary background for the next chapters.

The former Chapter 3 was split into two new chapters. The present Chapter 4, focuses on the continuity and the subdifferentiability of convex functions and also includes the famous inequalities of Jensen and of Prékopa and Leindler. Section 4.8 is new and concerns the extension of the subdifferential calculus to the framework of vector-valued convex functions acting on ordered Banach spaces with an application to the generalized Hahn–Banach extension theorem.

The new Chapter 5 is dedicated entirely to the applications of differentiable convex functions. Two new sections have been added. Section 5.5, discusses the connections between the classes of strongly convex, strongly smooth, and semiconvex functions and includes a thorough presentation of the gradient descent algorithm. Section 5.6 describes various aspects of the theory of supermodular convex functions, based on recent results due to C. P. Niculescu and his collaborators.

Chapter 6 is devoted to the highlights of the theory of majorization and starts with the classical majorization theorems of Hardy–Littlewood–Pólya and Tomić–Weyl, both put here in a much broader perspective. The new Section 6.5, *Convexity versus positive differences*, outlines an important difference between one real variable case and the case of several variables. While in the first case, the continuous functions with positive differences are precisely the convex functions of higher order, in the case of several variables not every convex function on $\mathbb{R}^N$ has positive differences of second order. However, the property of having positive differences of higher order is still fruitful, being related to the property of complete monotonicity. See Section 9.5, a new addition in this edition.

Chapter 8, on *Duality and Optimization*, was enlarged by adding new results in Section 8.2 concerning the duality between strong convexity and strong smoothness.

Chapter 9, dedicated to some *Special Topics in Majorization Theory*, includes the new Section 9.5 which offers a brief presentation of an important application of Choquet's theorem, the integral representation of completely monotone functions defined on cones.

Last but not least dozens of new exercises and new remarks were added. Besides, several typographical errors and reference omissions that were present in the previous editions have been corrected. These errors were discovered through feedback from students and the authors' re-reading of the manuscript.

The present text was drafted by the first-named author and put in final form by both authors.

While we did our best in composing this new edition, we are quite sure that much room for corrections and additions has remained. To inform the reader about them, we provided a page at the address

http : //math.ucv.ro/niculescu/

Any corrections (including typos) and other comments and suggestions from the readers are most welcome at

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Târgu Mureş and Luleå
January 2025

Constantin P. Niculescu
Lars-Erik Persson

List of Symbols

$\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$: The classical numerical sets (naturals, integers, etc.)

$\mathbb{N}^\star$: The set of strictly positive integers

$\mathbb{R}_+$: The set of positive real numbers

$\mathbb{R}_{++}$: The set of strictly positive real numbers

$\overline{\mathbb{R}}$: The set of extended real numbers

$\cup, \cap, \setminus, \complement$: Union, intersection, difference, complement

$\emptyset$: Empty set

∂A: Boundary of A

$\overline{A}$: Closure of A

$\mathrm{int} A$: Interior of A

A°: Polar of A

$\mathcal{C}^*$: Dual cone

E^*: Dual space

$\mathrm{rbd}(A)$: Relative boundary of A

$\mathrm{ri}(A)$: Relative interior of A

$B_r(a)$: Open ball with center a and radius r

$\overline{B}_r(a)$: Closed ball with center a and radius r

$[x, y]$: Line segment

aff(A): Affine hull of A

conv(A): Convex hull of A

dim$E$: Dimension of E

$\overline{\text{conv}}(A)$: Closed convex hull of A

ext$K$: Extreme points of K

span(A): Linear span of A

$\overline{\text{span}}(A)$: Closed linear span of A

$\lambda A + \mu B = \{\lambda x + \mu y : x \in A,\ y \in B\}$

$|A|$: Cardinality of A

diam(A): Diameter of A

$\text{Vol}_N(K)$: N-dimensional volume of K

χ_A: Characteristic function of A

cl$f$: Closure of f

dom$f$: Domain of f

I, I_E: Identity of the space

ι_C: Indicator function of C

h_C: Support function of C

$f|_K$: Restriction of f to K

f^*: Legendre–Fenchel transform of f (convex conjugate of f)

$d_U(x) = d(x, U)$: Distance from x to U

$d_H(A, B)$: Pompeiu–Hausdorff distance

$P_K(x)$: Orthogonal projection

dom(f): Effective domain of f

epi(f): Epigraph of f

graph(f): Graph of f

$\partial f(a)$: Subdifferential of f at a

supp(f): Support of f

$f * g$: Convolution

$f \square g$: Infimal convolution

$\mathbb{R}^N$: Euclidean N-space

$\mathbb{R}^N_+ = \{(x_1, \ldots, x_N) \in \mathbb{R}^N : x_1, \ldots, x_N \geq 0\}$, the positive orthant

$\mathbb{R}^N_{++} = \{(x_1, \ldots, x_N) \in \mathbb{R}^N : x_1, \ldots, x_N > 0\}$

$\mathbb{R}^N_{\geq} = \{(x_1, \ldots, x_N) \in \mathbb{R}^N : x_1 \geq \cdots \geq x_N\}$

Δ^N: The unit N-simplex

$\langle x, y \rangle$, $x \cdot y$: Inner product

$\mathrm{M}_N(\mathbb{R})$, $\mathrm{M}_N(\mathbb{C})$: Spaces of $N \times N$-dimensional matrices

$\mathrm{GL}(n, \mathbb{R})$: The group of $n \times n$ invertible matrices of real numbers

$\mathrm{O}(N)$: Orthogonal group

$\Pi(N)$: Permutation group

$\mathrm{Sym}(N, \mathbb{R})$: The space of all symmetric matrices of $M_N(\mathbb{R})$

$\mathrm{Sym}^+(N, \mathbb{R})$: The cone of all positive (i.e. positive semidefinite) matrices

$\mathrm{Sym}^{++}(N, \mathbb{R})$: The open cone of all strictly positive (i.e. positive definite) matrices

$\mathcal{A}(H)$: Space of all symmetric operators from the real Hilbert space H into itself

diag($\mathbf{x}$): Diagonal matrix displaying the components of x

A^T: Transpose of A

det$A$: Determinant of A

ker$A$: Kernel (null space) of A

rng$A$: Range of A

$\sigma(A)$: Spectrum of A

trace$A$: Trace of A

$\delta(A, B)$: The trace metric between the strictly positive matrices A and B

$\mathcal{O}(A)$: Orbit of A

$A\#_{1/2}B$: Geometric mean of the matrices A and B

$\liminf_{x\to a} f(x) = \lim_{r\to 0} \inf\{f(x) : x \in dom(f) \cap B_r(a)\}$: Lower limit

$\limsup_{x\to a} f(x) = \lim_{r\to 0} \sup\{f(x) : x \in dom(f) \cap B_r(a)\}$: Upper limit

$f'_+(a; v)$ and $f'_-(a; v)$: Sided directional derivatives

$f'(a; v)$: First Gâteaux differential

$f''(a; v, w)$: Second Gâteaux differential

df: First-order Fréchet differential

d$^2 f$: Second-order Fréchet differential

∇: Gradient

∇^2, $Hess$: Hessian matrix

$A(K)$: Space of real-valued continuous and affine functions

$Cvx(K)$: Cone of real-valued continuous and convex functions

$Cvx_{\{n_1,n_2,\ldots,n_m\}}(I)$: Cone of functions (defined on I) which mix various types of convexity

$\Gamma(E)$: Cone of all lower semicontinuous proper convex functions defined on the Banach space E

$C(K)$: Space of real-valued continuous functions

$C^m(\Omega) = \{f : D^\alpha f \in C(\Omega) \text{ for all } |\alpha| \le m\}$

$C^m(\overline{\Omega}) = \{f : D^\alpha f$ uniformly continuous on Ω for all $|\alpha| \leq m\}$

$C_c^\infty(\Omega)$: space of functions of class C^∞ with compact support

$L^p(\Omega)$: Space of p-th-power Lebesgue integrable functions on Ω

$L^p(\mu)$: Space of p-th-power μ-integrable functions

M_p: Power mean of order p

$\|f\|_{L^p}$: L^p-norm

ℓ^p: Space of p-th-power absolutely summing real sequences

$\mathrm{Lip}(f)$: Lipschitz constant

$W^{m,p}(\Omega)$: Sobolev space on Ω

$\|f\|_{W^{m,p}}$: Sobolev norm

$W_0^{m,p}(\Omega)$: Norm closure of $C_c^\infty(\Omega)$ in $W^{m,p}(\Omega)$

$\mathcal{B}(X)$: Borel σ-algebra associated to X

$\mathrm{Prob}(X)$: Set of Borel probability measures on X

δ_a: Dirac measure concentrated at a

$\mathcal{L}$: Lebesgue measure

$\mathrm{bar}(\mu)$: Barycenter of μ

$\mathbb{E}f$: Conditional expectation (mean value) of f

$\mathrm{var}(f)$: Variance of f

$\prec_{Ch}, \prec_{HLP}, \prec_{Sch}, \prec_{SPM}, \prec_{\mathrm{ds}}, \prec_{\mathrm{uds}}$: Various types of majorization

■: End of the proof

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Chapter 1

Convex Functions at a First Glance

The study of convex functions of one real variable offers an excellent glimpse of the beauty and fascination of advanced mathematics. The reader will find here a large variety of results based on simple and intuitive arguments that have remarkable applications. At the same time, they provide the starting point of deep generalizations that will be discussed in the next chapters.

1.1 Definitions and Elementary Properties

Throughout this book the letter I will denote a nondegenerate interval (that is, an interval containing an infinity of points).

1.1.1 Definition *A function $f\colon I \to \mathbb{R}$ is called* convex *if*

$$f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y) \tag{1.1}$$

for all points x and y in I and all $\lambda \in [0,1]$. It is called strictly convex *if the inequality (1.1) holds strictly whenever x and y are distinct points and $\lambda \in (0,1)$. If $-f$ is convex (respectively, strictly convex), then we say that f is* concave (*respectively,* strictly concave). *If f is both convex and concave, then f is said to be* affine.

The affine functions are precisely the functions of the form $mx+n$, for suitable real constants m and n. One can easily prove that the following three functions are convex (though not strictly convex): the positive part $x^+ = \max\{x, 0\}$, the negative part $x^- = \max\{-x, 0\}$, and the absolute value $|x| = \max\{-x, x\}$. Together with the affine functions they provide the building blocks for the entire class of convex functions on intervals. See Lemma 2.4.1 and Lemma 2.4.2.

Simple computations show that the square function x^2 is strictly convex on $\mathbb{R}$ and the square root function $\sqrt{x}$ is strictly concave on $\mathbb{R}_+$. In many cases of

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interest, the convexity is established via the second derivative test. See Corollary 1.4.9. Some other criteria of convexity related to basic theory of convex functions will be presented in what follows.

The convexity of a function $f\colon I \to \mathbb{R}$ means geometrically that the points of the graph of $f|_{[u,v]}$ are under (or on) the chord joining the endpoints $(u, f(u))$ and $(v, f(v))$, for all $u, v \in I$, $u < v$; see Figure 1.1. Thus the inequality (1.1) is equivalent to

$$f(x) \le f(u) + \frac{f(v) - f(u)}{v - u}(x - u) \tag{1.2}$$

for all $x \in [u, v]$, and all $u, v \in I$, $u < v$.

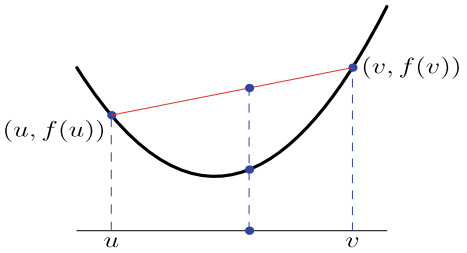


Figure 1.1: Convex function: the graph is under the chord.

This remark shows that the convex functions are majorized by affine functions on any compact subinterval. The existence of affine minorants will be discussed in Section 2.1.

Every convex function f is bounded on each compact subinterval $[u, v]$ of its interval of definition. In fact, $f(x) \le M = \max\{f(u), f(v)\}$ on $[u, v]$ and writing an arbitrary point $x \in [u, v]$ as $x = (u+v)/2 + t$ for some t with $|t| \le (v-u)/2$, we easily infer that

$$\begin{aligned} f(x) &= f\Big(\frac{u+v}{2} + t\Big) \ge 2f\Big(\frac{u+v}{2}\Big) - f\Big(\frac{u+v}{2} - t\Big) \\ &\ge 2f\Big(\frac{u+v}{2}\Big) - M. \end{aligned}$$

1.1.2 Theorem *A convex function $f\colon I \to \mathbb{R}$ is continuous at each interior point of I.*

Proof. Suppose that $a \in \operatorname{int} I$ and choose $\varepsilon > 0$ such that $[a - \varepsilon, a + \varepsilon] \subset I$. Then

$$f(a) \le \frac{1}{2} f(a - \varepsilon) + \frac{1}{2} f(a + \varepsilon),$$

and

$$f(a \pm t\varepsilon) = f\left((1-t)a + t(a \pm \varepsilon)\right) \le (1-t) f(a) + t f(a \pm \varepsilon)$$

for every $t \in [0,1]$. Therefore

$$t\,(f(a \pm \varepsilon) - f(a)) \geq f(a \pm t\varepsilon) - f(a) \geq -t\,(f(a \mp \varepsilon) - f(a))\,,$$

which yields

$$|f(a \pm t\varepsilon) - f(a)| \leq t \max\{|f(a-\varepsilon) - f(a)|\,, |f(a+\varepsilon) - f(a)|\}$$

for every $t \in [0,1]$. The continuity of f at a is now clear. ■

Simple examples such as $f(x) = 0$ if $x \in (0,1)$, and $f(0) = f(1) = 1$, show that upward jumps can appear at the endpoints of the interval of definition of a convex function. Fortunately, these possible discontinuities are removable:

1.1.3 Proposition *If $f\colon [a,b] \to \mathbb{R}$ is a convex function, then the limits $f(a+) = \lim_{x \searrow a} f(x)$ and $f(b-) = \lim_{x \nearrow b} f(x)$ exist in $\mathbb{R}$ and*

$$\widetilde{f}(x) = \begin{cases} f(a+) & \text{if } x = a \\ f(x) & \text{if } x \in (a,b) \\ f(b-) & \text{if } x = b \end{cases}$$

is a continuous convex function.

This result is a consequence of the following:

1.1.4 Lemma *If $f\colon I \to \mathbb{R}$ is convex, then either f is monotonic on $\operatorname{int} I$, or there exists a point $\xi \in \operatorname{int} I$ such that f is decreasing on the interval $(-\infty, \xi] \cap I$ and increasing on the interval $[\xi, \infty) \cap I$.*

Proof. Choose $a < b$ arbitrarily among the interior points of I and put $m = \inf\{f(x) : x \in [a,b]\}$. Since f is continuous on $[a,b]$, this infimum is attained at a point $\xi \in [a,b]$, that is,

$$m = f(\xi).$$

If $a \leq x < y < \xi$, then y is a convex combination of x and ξ, precisely, $y = \frac{\xi - y}{\xi - x} x + \frac{y - x}{\xi - x} \xi$. Since f is convex,

$$f(y) \leq \frac{\xi - y}{\xi - x} f(x) + \frac{y - x}{\xi - x} f(\xi) \leq f(x)$$

and thus f is decreasing on the interval $[a, \xi]$. If $\xi < b$, a similar argument shows that f is increasing on $[\xi, b]$.

The proof ends with a gluing process (to the left of a and the right of b), observing that the property of convexity makes impossible the existence of three numbers $u < v < w$ in I such that $f(u) < f(v) > f(w)$. ■

1.1.5 Corollary (a) *Every increasing convex function defined on an interval $[a,b)$ is continuous and the same occurs for every decreasing convex function defined on an interval $(a,b]$.*

(b) *Every convex function $f\colon I \to \mathbb{R}$ which is not monotonic on $\operatorname{int} I$ has an interior global minimum.*

(c) *If a convex function $f\colon \mathbb{R} \to \mathbb{R}$ is bounded from above, then it is constant.*

Attaining supremum at endpoints is not a property characteristic of convex functions, but a variant of it does the job.

1.1.6 Theorem (S. Saks [517]) *Let f be a real-valued function defined on an interval I. Then f is (strictly) convex if and only if for every compact subinterval J of I, and every affine function L, the supremum of $f+L$ on J is attained at an endpoint (and only there).*

This statement remains valid if the perturbations L are supposed to be linear (that is, of the form $L(x) = mx$ for suitable $m \in \mathbb{R}$). See Exercise 1 for examples of convex functions derived from Theorem 1.1.6.

Proof. We will restrict ourselves to the case of convex functions. The case of strictly convex functions can be treated in the same manner.

Necessity: If f is convex, so is the sum $F = f + L$. Since every point of a subinterval $J = [x, y]$ is a convex combination $z = (1-\lambda)x + \lambda y$ of x and y, we have

$$\begin{aligned}\sup_{z\in J} F(z) &= \sup_{\lambda\in[0,1]} F((1-\lambda)x+\lambda y)\\ &\le \sup_{\lambda\in[0,1]} [(1-\lambda)F(x)+\lambda F(y)] = \max\{F(x), F(y)\}.\end{aligned}$$

Sufficiency: Given a compact subinterval $J = [x, y]$ of I, there exists an affine function $L(x) = mx + n$ which agrees with f at the two endpoints x and y. Then

$$\sup_{\lambda\in[0,1]} [(f-L)((1-\lambda)x+\lambda y)] = 0,$$

which yields

$$\begin{aligned}0 &\ge f((1-\lambda)x+\lambda y) - L((1-\lambda)x+\lambda y)\\ &= f((1-\lambda)x+\lambda y) - (1-\lambda)L(x) - \lambda L(y)\\ &= f((1-\lambda)x+\lambda y) - (1-\lambda)f(x) - \lambda f(y)\end{aligned}$$

for every $\lambda \in [0, 1]$. ■

1.1.7 Remark *Call a function $f : I \to \mathbb{R}$* quasiconvex *if*

$$f((1-\lambda)x+\lambda y) \le \max\{f(x), f(y)\}$$

for all $x, y \in I$ and $\lambda \in [0, 1]$. For example, so are the monotonic functions and the functions that admit interior points $c \in I$ such that they are decreasing on $(-\infty, c] \cap I$ and increasing on $[c, \infty) \cap I$. Unlike the case of convex functions, the sum of a quasiconvex function and a linear function may not be quasiconvex. See the case of the sum $x^3 - 3x$. The floor function, $\lfloor x \rfloor = \max\{m \in \mathbb{Z} : m \le x\}$, *is an example of quasiconvex function that is neither convex nor continuous. Related to quasiconvex functions are the* quasiconcave *functions*

$$f((1-\lambda)x+\lambda y) \ge \min\{f(x), f(y)\}$$

for all $x, y \in I$ and $\lambda \in [0, 1]$. See Exercises 13 and 14 for more information.

The following characterization of convexity within the class of continuous functions also proves useful in checking convexity.

1.1.8 Theorem (J. L. W. V. Jensen [274]) *A function $f\colon I \to \mathbb{R}$ is convex if and only if it verifies the following two conditions:*

(a) *f is continuous at each interior point of I and*

(b) *f is midpoint convex, that is,*

$$f\Big(\frac{x+y}{2}\Big) \le \frac{f(x)+f(y)}{2} \quad \text{for all } \ x, y \in I.$$

Proof. The necessity follows from Theorem 1.1.2. The sufficiency is proved by *reductio ad absurdum*. If f were not convex, then there would exist a subinterval $[a,b]$ such that the graph of $f|_{[a,b]}$ is not under the chord joining $(a, f(a))$ and $(b, f(b))$; that is, the function

$$\varphi(x) = -f(x) + f(a) + \frac{f(b)-f(a)}{b-a}(x-a), \quad x \in [a,b]$$

verifies $\gamma = \inf\{\varphi(x) : x \in [a,b]\} < 0$. Notice that $-\varphi$ is midpoint convex, continuous, and $\varphi(a) = \varphi(b) = 0$. Put $c = \inf\{x \in [a,b] : \varphi(x) = \gamma\}$; then necessarily $\varphi(c) = \gamma$ and $c \in (a,b)$. By the definition of c, for every $h > 0$ for which $c \pm h \in (a,b)$ we have

$$\varphi(c-h) > \varphi(c) \quad \text{and} \quad \varphi(c+h) \ge \varphi(c)$$

so that

$$-\varphi(c) > \frac{-\varphi(c-h) - \varphi(c+h)}{2}$$

in contradiction with the fact that $-\varphi$ is midpoint convex. ∎

1.1.9 Corollary *Suppose that $f\colon I \to \mathbb{R}$ is a continuous function. Then f is strictly convex if (and only if) it is strictly midpoint convex, that is,*

$$f\Big(\frac{x+y}{2}\Big) < \frac{f(x)+f(y)}{2} \quad \text{for all } \ x, y \in I, \ x \ne y.$$

Equivalently, f is strictly convex if (and only if)

$$f(x+h) + f(x-h) - 2f(x) > 0,$$

for all $x \in I$ and all $h > 0$ such that both $x+h$ and $x-h$ are in I.

Proof. Suppose that $x \ne y$ are points in I and $\lambda \in (0,1)$. If $(1-\lambda)x + \lambda y = (x+y)/2$, then necessarily $\lambda = 1/2$ and thus

$$f((1-\lambda)x + \lambda y) = f\left((x+y)/2\right) < \frac{f(x)+f(y)}{2}.$$

Suppose now that $x < (1-\lambda)x + \lambda y < (x+y)/2$. Then there exists $t \in (0,1)$ such that $(1-\lambda)x + \lambda y = (1-t)x + t((x+y)/2)$. Necessarily $t = 2\lambda$ and by Theorem 1.1.8,

$$f((1-\lambda)x + \lambda y) = f\left((1-t)x + t\frac{x+y}{2})\right) \leq (1-t)f(x) + tf\left(\frac{x+y}{2}\right)$$
$$< (1-2\lambda)f(x) + 2\lambda\frac{f(x)+f(y)}{2} = (1-\lambda)f(x) + \lambda f(y).$$

The case when $(x+y)/2 < (1-\lambda)x+\lambda y < y$ can be treated in the same manner. ■

Corollary 1.1.9 allows us to check immediately the strict convexity/concavity of some very common functions, such as the exponential function, the logarithmic function, and the restriction of the sine function to [0,π]. Indeed, in the first case, the fact that

$$a, b > 0,\ a \neq b, \quad \text{implies} \quad \frac{a+b}{2} > \sqrt{ab}$$

is equivalent to

$$\mathrm{e}^{x+h} + \mathrm{e}^{x-h} - 2\mathrm{e}^{x} > 0$$

for all $x \in \mathbb{R}$ and all $h > 0$.

Many other examples can be deduced using the closure under functional operations with convex/concave functions.

1.1.10 Proposition (The operations with convex functions)

(a) *Adding two convex functions (defined on the same interval) we obtain a convex function; if one of them is strictly convex, then the sum is also strictly convex.*

(b) *Multiplying a (strictly) convex function by a (strictly) positive scalar we obtain also a (strictly) convex function.*

(c) *Suppose that f and g are two positive convex functions defined on an interval I. Then their product is convex provided that they are synchronous in the sense that*

$$(f(x) - f(y))\,(g(x) - g(y)) \geq 0$$

for all $x, y \in I$; for example, this condition occurs if f and g are both decreasing or both increasing.

(d) *The restriction of every (strictly) convex function to a subinterval of its domain is also a (strictly) convex function.*

(e) *If f is a convex and increasing (decreasing) function and g is a convex (concave) function, then their composition $f \circ g$ is a convex function. The variant where convexity is strict and monotonicity is strict works as well. Moreover, the roles of convexity and concavity can be interchanged.*

(f) *Suppose that f is a bijection between two intervals I and J. If f is strictly increasing, then f is (strictly) convex if and only if f^{-1} is (strictly)* concave. *If f is a decreasing bijection, then f and f^{-1} are of the same type of convexity.*

(g) *If f is a strictly positive concave function, then $1/f$ is a convex function. Here the role of concavity and convexity cannot be changed to each other.*

(h) *The maximum of two (strictly) convex functions $f, g : I \to \mathbb{R}$,*

$$\max\{f, g\}(x) = \max\{f(x), g(x)\}$$

is also a (strictly) convex function.

(i) *The superposition $f(ax + b)$, of a convex function f and an affine function $ax + b$, is a convex function.*

The proof of Proposition 1.1.10 is straightforward, so we omit the details.

We next discuss the extension of the inequality of convexity (1.1) to arbitrarily long finite families of points. The basic remark in this respect is the fact that the intervals are closed under arbitrary convex combinations, that is,

$$\sum_{k=1}^{n} \lambda_k x_k \in I$$

for all $x_1, \ldots, x_n \in I$ and all $\lambda_1, \ldots, \lambda_n \in [0, 1]$ with $\sum_{k=1}^{n} \lambda_k = 1$. This can be proved by induction on the number n of points involved in the convex combinations. The case $n = 1$ is trivial, while for $n = 2$ it follows from the definition of a convex set. Assuming the result is true for all convex combinations with at most $n \geq 2$ points, let us pass to the case of combinations with $n + 1$ points, $x = \sum_{k=1}^{n+1} \lambda_k x_k$. The nontrivial case is when all coefficients λ_k lie in $(0, 1)$. But in this case, due to our induction hypothesis, x can be represented as a convex combination of two elements of I,

$$x = (1 - \lambda_{n+1})\Big(\sum_{k=1}^{n} \frac{\lambda_k}{1 - \lambda_{n+1}} x_k\Big) + \lambda_{n+1} x_{n+1},$$

hence x belongs to I.

The above remark on intervals has a notable counterpart for convex functions:

1.1.11 Lemma (The discrete case of Jensen's inequality) *A real-valued function f defined on an interval I is convex if and only if for all points $x_1, \ldots, x_n$ in I and all scalars $\lambda_1, \ldots, \lambda_n$ in $[0, 1]$ with $\sum_{k=1}^{n} \lambda_k = 1$ we have*

$$f\Big(\sum_{k=1}^{n} \lambda_k x_k\Big) \leq \sum_{k=1}^{n} \lambda_k f(x_k).$$

If f is strictly convex, then the above inequality is strict provided that the points x_k are not all equal and the scalars λ_k are strictly positive.

Proof. The first assertion follows by mathematical induction. As concerns the second assertion, suppose that the function f is strictly convex and

$$f\left(\sum_{k=1}^{n} \lambda_k x_k\right) = \sum_{k=1}^{n} \lambda_k f(x_k) \tag{1.3}$$

for some points $x_1, \dots, x_n \in I$ and some scalars $\lambda_1, \dots, \lambda_n \in (0,1)$ that sum up to 1. If $x_1, \dots, x_n$ are not all equal, the set $S = \{k : x_k < \max\{x_1, \dots, x_n\}\}$ will be a proper subset of $\{1, ..., n\}$ and $\lambda_S = \sum_{k \in S} \lambda_k \in (0,1)$. Since f is strictly convex, we get

$$\begin{aligned} f\left(\sum_{k=1}^{n} \lambda_k x_k\right) &= f\left(\lambda_S \left(\sum_{k\in S} \frac{\lambda_k}{\lambda_S} x_k\right) + (1-\lambda_S)\left(\sum_{k\notin S} \frac{\lambda_k}{1-\lambda_S} x_k\right)\right) \\ &< \lambda_S f\left(\sum_{k\in S} \frac{\lambda_k}{\lambda_S} x_k\right) + (1-\lambda_S) f\left(\sum_{k\notin S} \frac{\lambda_k}{1-\lambda_S} x_k\right) \\ &< \lambda_S \sum_{k\in S} \frac{\lambda_k}{\lambda_S} f(x_k) + (1-\lambda_S) \sum_{k\notin S} \frac{\lambda_k}{1-\lambda_S} f(x_k) = \sum_{k=1}^{n} \lambda_k f(x_k), \end{aligned}$$

which contradicts our hypothesis (1.3). Therefore all points x_k should coincide. ∎

The extension of Lemma 1.1.11 to the case of arbitrary finite measure spaces (as well as an estimate of its precision) will be discussed in Section 1.7.

An immediate consequence of Lemma 1.1.11 (when applied to the exponential function) is the following result which extends the well-known *AM–GM* inequality (that is, the inequality between the arithmetic mean and the geometric mean):

1.1.12 Theorem (The weighted form of the AM–GM inequality; L. J. Rogers [509]) *If $x_1, \dots, x_n \in (0, \infty)$ and $\lambda_1, \dots, \lambda_n \in (0,1)$, $\sum_{k=1}^{n} \lambda_k = 1$, then*

$$\sum_{k=1}^{n} \lambda_k x_k > x_1^{\lambda_1} \cdots x_n^{\lambda_n}$$

unless $x_1 = \cdots = x_n$.

Replacing x_k by $1/x_k$ in the last inequality, we get

$$x_1^{\lambda_1} \cdots x_n^{\lambda_n} > 1 \Big/ \sum_{k=1}^{n} \frac{\lambda_k}{x_k}$$

unless $x_1 = \cdots = x_n$. This represents the weighted form of the *geometric mean–harmonic mean inequality* (that is, of *GM–HM* inequality).

For $\lambda_1 = \cdots = \lambda_n = 1/n$ we recover the usual *AM–GM–HM inequality, which asserts that for every family $x_1, \dots, x_n$ of strictly positive numbers, not all equal to each other, we have*

$$\frac{x_1 + \cdots + x_n}{n} > \sqrt[n]{x_1 \cdots x_n} > \frac{n}{\frac{1}{x_1} + \cdots + \frac{1}{x_n}}.$$

A stronger concept of convexity, called logarithmic convexity (or log-convexity), will be discussed in Section 1.3.

Exercises

1. Infer from Theorem 1.1.6 that the following differentiable functions are strictly convex:

   (a) $-\sin x$ on $[0, \pi]$ and $-\cos x$ on $[-\pi/2, \pi/2]$;

   (b) x^p on $[0, \infty)$ if $p > 1$; x^p on $(0, \infty)$ if $p < 0$; $-x^p$ on $[0, \infty)$ if $p \in (0, 1)$;

   (c) $(1 + x^p)^{1/p}$ on $[0, \infty)$ if $p > 1$.

2. (New from old) Assume that $f(x)$ is a (strictly) convex function on an interval $I \subset (0, \infty)$. Prove that $xf(1/x)$ is (strictly) convex on any interval J for which $x \in J$ implies $1/x \in I$. Then infer that $x \log x$ is a strictly convex function on $[0, \infty)$ and $x \sin(1/x)$ is strictly concave on $[1/\pi, \infty)$.

3. (Reshaping convex functions) Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a convex function and P and Q are distinct points of its graph. Prove that:

   (a) if we modify the graph of f by replacing the portion joining P and Q with the corresponding linear segment, the result is still the graph of a convex function g;

   (b) all points of the graph of g are on or above the secant line joining P and Q.

4. Using the fact that all points of an interval $[a, b]$ are convex combinations of its endpoints prove the following assertions:

   (a) *If* $f : [0, \infty) \to \mathbb{R}$ is a convex function, then for all $x, y > 0$,

   $$f(x) + f(y) \leq f(x + y) + f(0).$$

   The inequality is strict if f is strictly convex. Infer that every convex function $f : [0, \infty) \to \mathbb{R}$ for which $f(0) \leq 0$ is superadditive (that is, $f(x) + f(y) \leq f(x + y)$ for all $x, y \geq 0$) and every concave function $f : [0, \infty) \to \mathbb{R}$ for which $f(0) \geq 0$ is subadditive (that is, $f(x) + f(y) \geq f(x + y)$ for all $x, y \geq 0$).

   (b) If $f : [0, \infty) \to \mathbb{R}$ is a positive concave function, then f is increasing (and subadditive).

   (c) If $f : [0, \infty) \to \mathbb{R}$ is a concave function and $f(0) \geq 0$, then the function $f(x)/x$ is decreasing on $(0, \infty)$. Equivalently, for all $a, x \in [0, \infty)$,

   $$f(ax) \geq af(x) \text{ if } a \leq 1 \text{ and } f(ax) \leq af(x) \text{ if } a \geq 1.$$

   (d) Every function $f : [0, \infty) \to \mathbb{R}$ with $f(0) \geq 0$ for which the conclusion of the assertion (c) holds is subadditive (though not necessarily concave).

   (e) If $f : [0, \infty) \to \mathbb{R}$ is a convex and increasing function such that $f(0) = 0$, then f is superadditive. Note that every positive and superadditive function is increasing but not necessarily convex.

5. Assume $f : [0,\infty) \to \mathbb{R}$ is a concave function and $a, b \in (0,\infty)$, $b > a$. Then $f(b+x) + f(a-x)$ is decreasing on $[0,a]$, while the function $f(b-x) + f(a+x)$ is increasing on $[0,(b-a)/2]$.

6. Assume $f_1, \ldots, f_n$ are positive concave functions with the same domain of definition. Prove that $(f_1 \cdots f_n)^{1/n}$ is also a concave function.

7. (From discrete to continuous) Prove that the discrete form of Jensen's inequality implies (and is implied by) the following integral form of this inequality: If $f : [c,d] \to \mathbb{R}$ is a convex function and $g : [a,b] \to [c,d]$ is a Riemann integrable function, then

$$f\left(\frac{1}{b-a}\int_a^b g(x)\mathrm{d}x\right) \leq \frac{1}{b-a}\int_a^b f\left(g(x)\right)\mathrm{d}x.$$

8. (Optimization without calculus) The *AM–GM* inequality offers a very convenient way to minimize a sum when the product of its terms is constant (or to maximize a product whose factors sum to a constant). Infer from Theorem 1.1.12 that

$$\min_{x,y>0}\left(x+y+\frac{1}{x^2y}\right) = \frac{4}{\sqrt{2}} \quad \text{and} \quad \max_{6|x|+5|y|+|z|=4} x^2yz = \frac{1}{45}.$$

9. (a) Prove that Theorem 1.1.8 remains true if the condition of midpoint convexity is replaced by the following one: $f((1-\alpha)x+\alpha y) \leq (1-\alpha)f(x)+\alpha f(y)$ for some fixed parameter $\alpha \in (0,1)$, and for all $x, y \in I$.

   (b) Prove that Theorem 1.1.8 remains true if the condition of continuity is replaced by boundedness from above on every compact subinterval.

10. Let $f\colon I \to \mathbb{R}$ be a convex function and let $x_1, \ldots, x_n \in I$ $(n \geq 2)$. Prove that

$$(n-1)\Big[\frac{f(x_1)+\cdots+f(x_{n-1})}{n-1} - f\Big(\frac{x_1+\cdots+x_{n-1}}{n-1}\Big)\Big]$$

cannot exceed

$$n\Big[\frac{f(x_1)+\cdots+f(x_n)}{n} - f\Big(\frac{x_1+\cdots+x_n}{n}\Big)\Big].$$

11. (Two more proofs of the *AM–GM* inequality) Let $x_1, \ldots, x_n > 0$ $(n \geq 2)$ and for each $1 \leq k \leq n$ put

$$A_k = \frac{x_1+\cdots+x_k}{k} \quad \text{and} \quad G_k = (x_1 \cdots x_k)^{1/k}.$$

   (a) (T. Popoviciu) Prove that

$$\Big(\frac{A_n}{G_n}\Big)^n \geq \Big(\frac{A_{n-1}}{G_{n-1}}\Big)^{n-1} \geq \cdots \geq \Big(\frac{A_1}{G_1}\Big)^1 = 1.$$

(b) (R. Rado) Prove that

$$n(A_n - G_n) \geq (n-1)(A_{n-1} - G_{n-1}) \geq \cdots \geq 1 \cdot (A_1 - G_1) = 0.$$

[*Hint*: Apply the result of Exercise 10, respectively, to $f = -\log$ and $f = \exp$.]

12. (The power means in the discrete case) Let $x = (x_1, \ldots, x_n)$ and $\lambda = (\lambda_1, \ldots, \lambda_n)$ be two n-tuples of strictly positive numbers such that $\sum_{k=1}^{n} \lambda_k = 1$. The (weighted) *power mean* of order t is defined as

$$M_t(x;\lambda) = \Big(\sum_{k=1}^{n} \lambda_k x_k^t\Big)^{1/t} \quad \text{for } t \in \mathbb{R} \setminus \{0\}$$

and

$$M_0(x;\lambda) = \prod_{k=1}^{n} x_k^{\lambda_k}. \tag{1.4}$$

We also define $M_{-\infty}(x;\lambda) = \min\{x_k : k = 1, ..., n\}$ and $M_{\infty}(x,\lambda) = \max\{x_k : k = 1, ..., n\}$. Notice that M_1 is the arithmetic mean, M_0 is the geometric mean, and M_{-1} is the harmonic mean. Moreover, $M_{-t}(x;\lambda) = M_t(x^{-1};\lambda)^{-1}$. Prove that:

(*a*) $M_s(x;\lambda) \leq M_t(x;\lambda)$ whenever $s \leq t$ in $\mathbb{R}$;

(*b*) $\lim_{t\to 0} M_t(x,\lambda) = \prod_{k=1}^{n} x_k^{\lambda_k}$;

(*c*) $\lim_{t\to-\infty} M_t(x;\lambda) = M_{-\infty}(x;\lambda)$ and $\lim_{t\to\infty} M_t(x;\lambda) = M_{\infty}(x;\lambda)$.

[*Hint:* (a) According to the weighted form of the *AM–GM* inequality, $M_0(x;\lambda) \leq M_t(x;\lambda)$ for all $t \geq 0$. Then apply Jensen's inequality to the function $x^{t/s}$ to infer that $M_s(x;\lambda) \leq M_t(x;\lambda)$ whenever $0 < s \leq t$. To end the proof of (a) use the formula $M_{-t}(x;\lambda) = M_t(x^{-1};\lambda)^{-1}$.]

Remark. The power means discussed above are special cases of integral power means that will be presented in Exercise 1, Section 2.2. Indeed, an n-tuple $\lambda = (\lambda_1, \ldots, \lambda_n)$ of strictly positive numbers that sum to unity can be identified with the discrete probability measure on the set $\Omega = \{1, 2, ..., n\}$ defined by $\lambda(A) = \sum_{k\in A} \lambda_k$ for $A \in \mathcal{P}(\Omega)$. The λ-integrable functions $x : \Omega \to \mathbb{R}$, $x(k) = x_k$, are nothing but the strings $x = (x_1, \ldots, x_n)$ of real numbers, so that

$$M_t(x;\lambda) = \left(\int_{\Omega} x^t(k)\mathrm{d}\lambda(k)\right)^{1/t} \quad \text{for } t \in \mathbb{R}\setminus\{0\}$$

and $M_0(x;\lambda) = \exp\left(\int_{\Omega} \log x(k)\mathrm{d}\lambda(k)\right)$.

13. Prove that:

(a) a function $f : I \to \mathbb{R}$ is quasiconvex if and only if all sublevel sets $L_\lambda = \{x : f(x) \leq \lambda\}$ are intervals;

(b) the product of two positive concave functions is a quasiconcave function;

(c) if f is positive and convex, then $1/f$ is quasiconcave;

(d) if f is negative and quasiconvex then $1/f$ is quasiconcave;

(e) if f is a positive convex function and g is a positive concave function, then f/g is quasiconvex.

14. (S. Johansen [277]) Suppose that f is a continuous real-valued function defined on an interval I. Prove that f is quasiconvex if and only if it is either monotonic or there exists a point $c \in I$ such that f verifies one of the following two alternatives:

(a) f is decreasing on $(-\infty, c] \cap I$ and increasing on $(c, \infty) \cap I$;

(b) f is decreasing on $(-\infty, c) \cap I$ and increasing on $[c, \infty) \cap I$.

15. A sequence of real numbers $a_0, a_1, \ldots, a_n$ (with $n \geq 2$) is said to be *convex* provided that

$$\Delta^2 a_k = a_k - 2a_{k+1} + a_{k+2} \geq 0$$

for all $k = 0, \ldots, n-2$; it is said to be *concave* provided $\Delta^2 a_k \leq 0$ for all k for all $k = 0, \ldots, n-2$.

(a) Solve the system

$$\Delta^2 a_k = b_k \quad \text{for } k = 0, \ldots, n-2$$

(in the unknowns a_k) to prove that the general form of a convex sequence $\mathbf{a} = (a_0, a_1, \ldots, a_n)$ with $a_0 = a_n = 0$ is given by the formula:

$$\mathbf{a} = \sum_{j=1}^{n-1} c_j \mathbf{w}^j,$$

where $c_j = 2a_j - a_{j-1} - a_{j+1}$ and $\mathbf{w}^j$ has the components

$$w_k^j = \begin{cases} k(n-j)/n, & \text{for } k = 0, \ldots, j \\ j(n-k)/n, & \text{for } k = j, \ldots, n. \end{cases}$$

(b) Prove that the general form of a convex sequence $\mathbf{a} = (a_0, a_1, \ldots, a_n)$ is $\mathbf{a} = \sum_{j=0}^{n} c_j \mathbf{w}^j$, where c_j and $\mathbf{w}^j$ are as in the case (a) for $j = 1, \ldots, n-1$. The other coefficients and components are

$$c_0 = a_0, c_n = a_n, w_k^0 = (n-k)/n \text{ and } w_k^n = k/n \text{ for } k = 0, \ldots, n).$$

Remark. The theory of convex sequences can be subordinated to that of convex functions. If $f\colon [0, n] \to \mathbb{R}$ is a convex function, then $f(0), ..., f(n)$ is a convex sequence; conversely, if $a_0, ..., a_n$ is a convex sequence, then the piecewise linear function $f\colon [0, n] \to \mathbb{R}$ obtained by joining the points (k, a_k) is convex.

1.2 Young's Inequality and Its Consequences

The following special case of the weighted form of the *AM–GM* inequality is known as *Young's inequality*:

$$ab \le \frac{a^p}{p} + \frac{b^q}{q} \quad \text{for all } a, b \ge 0, \tag{1.5}$$

whenever $p, q \in (1, \infty)$ and $1/p + 1/q = 1$; the equality holds if and only if $a^p = b^q$. Young's inequality can be also obtained as a consequence of strict convexity of the exponential function. In fact,

$$\begin{aligned} ab &= \mathrm{e}^{\log ab} = \mathrm{e}^{(1/p)\log a^p + (1/q)\log b^q} \\ &< \frac{1}{p}\,\mathrm{e}^{\log a^p} + \frac{1}{q}\,\mathrm{e}^{\log b^q} = \frac{a^p}{p} + \frac{b^q}{q} \end{aligned}$$

for all $a, b > 0$ with $a^p \ne b^q$. Yet another argument is provided by the study of variation of the differentiable function:

$$F(a) = \frac{a^p}{p} + \frac{b^q}{q} - ab, \quad a \ge 0,$$

where $b \ge 0$ is a parameter. This function attains at $a = b^{q/p}$ its strict global minimum, which yields $F(a) > F(b^{q/p}) = 0$ for all $a \ge 0$, $a \ne b^{q/p}$. A refinement of the inequality (1.5) is presented in Exercise 2, at the end of this section.

W. H. Young [592] actually proved a much more general inequality which covers (1.5) for $f(x) = x^{p-1}$:

1.2.1 Theorem (Young's inequality) *Suppose that* $f\colon [0, \infty) \to [0, \infty)$ *is a strictly increasing continuous function such that* $f(0) = 0$ *and* $\lim_{x\to\infty} f(x) = \infty$. *Then*

$$uv \le \int_0^u f(x)\,\mathrm{d}x + \int_0^v f^{-1}(y)\,\mathrm{d}y$$

for all $u, v \ge 0$, *and equality occurs if and only if* $v = f(u)$.

Proof. Using the definition of the derivative one can easily prove that the function

$$F(u) = \int_0^u f(x)\,\mathrm{d}x + \int_0^{f(u)} f^{-1}(y)\,\mathrm{d}y - uf(u), \quad u \in [0, \infty),$$

is differentiable, with F' identically 0. Thus $F(u) = F(0) = 0$ for all $u \ge 0$.

If $u, v \ge 0$ and $v \ge f(u)$, then

$$\begin{aligned} uv &= uf(u) + u(v - f(u)) = \int_0^u f(x)\,\mathrm{d}x + \int_0^{f(u)} f^{-1}(y)\,\mathrm{d}y + u(v - f(u)) \\ &= \int_0^u f(x)\,\mathrm{d}x + \int_0^v f^{-1}(y)\,\mathrm{d}y + \left[u(v - f(u)) - \int_{f(u)}^v f^{-1}(y)\,\mathrm{d}y\right] \\ &\le \int_0^u f(x)\,\mathrm{d}x + \int_0^v f^{-1}(y)\,\mathrm{d}y. \end{aligned}$$

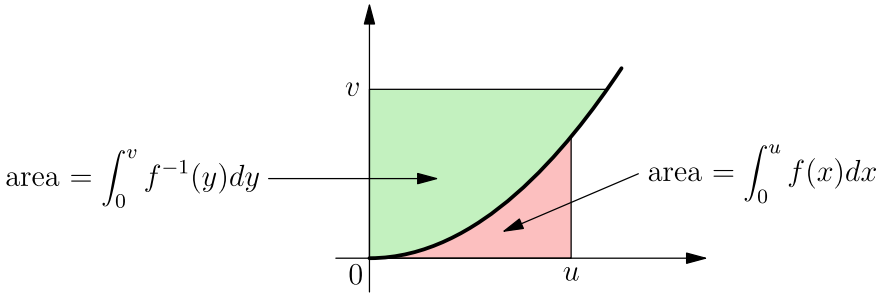


Figure 1.2: The area of the union of the two curvilinear triangles exceeds the area of the rectangle with sides u and v.

The other case, where $v \leq f(u)$ can be treated in a similar way. ■

A pictorial illustration of Theorem 1.2.1 is suggested by Figure 1.2.

An alternative proof of Theorem 1.2.1 based on the Legendre–Fenchel duality makes the objective of Exercise 7, Section 2.1.

Young's inequality (1.5) is the source of many important integral inequalities, most of them valid in the context of arbitrary measure spaces. The interested reader may find a thorough presentation of the basic elements of measure theory in the monograph of E. Hewitt and K. Stromberg [252], Sections 12 and 13.

1.2.2 Theorem (The Rogers–Hölder inequality for $p > 1$) *Let $p, q \in (1, \infty)$ with $1/p + 1/q = 1$, and let $f \in L^p(\mu)$ and $g \in L^q(\mu)$. Then fg is in $L^1(\mu)$ and we have*

$$\left| \int_\Omega fg \, \mathrm{d}\mu \right| \leq \int_\Omega |fg| \, \mathrm{d}\mu \tag{1.6}$$

and

$$\int_\Omega |fg| \, \mathrm{d}\mu \leq \|f\|_{L^p} \, \|g\|_{L^q}. \tag{1.7}$$

As a consequence,

$$\left| \int_\Omega fg \, \mathrm{d}\mu \right| \leq \|f\|_{L^p} \, \|g\|_{L^q}. \tag{1.8}$$

The above result extends in a straightforward manner to the pairs $p = 1$, $q = \infty$ and $p = \infty$, $q = 1$. In the complementary domain, $p \in (-\infty, 1) \backslash \{0\}$ and $1/p + 1/q = 1$, the inequality sign in (1.6)–(1.8) should be reversed. See Exercise 3.

From the Rogers–Hölder inequality it follows that for all $p, q, r \in (1, \infty)$ with $1/p + 1/q = 1/r$ and all $f \in L^p(\mu)$ and $g \in L^q(\mu)$ we have $fg \in L^r(\mu)$ and

$$\|fg\|_{L^r} \leq \|f\|_{L^p} \, \|g\|_{L^q}. \tag{1.9}$$

The special case of inequality (1.8) where $p = q = 2$ is known as the *Cauchy–Bunyakovsky–Schwarz inequality*. See Remark 1.2.6.

Proof. The first inequality is trivial. If f or g is zero μ-almost everywhere, then the second inequality is trivial. Otherwise, using Young's inequality, we have

$$\frac{|f(x)|}{\|f\|_{L^p}} \cdot \frac{|g(x)|}{\|g\|_{L^q}} \le \frac{1}{p} \cdot \frac{|f(x)|^p}{\|f\|_{L^p}^p} + \frac{1}{q} \cdot \frac{|g(x)|^q}{\|g\|_{L^q}^q}$$

for all x in Ω, such that $fg \in L^1(\mu)$. Thus

$$\frac{1}{\|f\|_{L^p}\|g\|_{L^q}} \int_\Omega |fg| \, \mathrm{d}\mu \le 1$$

and the proof of (1.7) is done. The inequality (1.8) is immediate. ∎

1.2.3 Remark (Conditions for equality in Theorem 1.2.2) *The basic observation is the fact that*

$$f \ge 0 \quad \textit{and} \quad \int_\Omega f \, \mathrm{d}\mu = 0 \quad \textit{imply} \quad f = 0 \mu\textit{-almost everywhere.}$$

Consequently we have equality in (1.6) *if and only if*

$$f(x)g(x) = \mathrm{e}^{\mathrm{i}\theta}|f(x)g(x)|$$

for some real constant θ *and for* μ*-almost every* x*.*

Suppose that $p, q \in (1, \infty)$ *and* f *and* g *are not zero* μ*-almost everywhere. In order to get equality in* (1.7) *it is necessary and sufficient to have*

$$\frac{|f(x)|}{\|f\|_{L^p}} \cdot \frac{|g(x)|}{\|g\|_{L^q}} = \frac{1}{p} \cdot \frac{|f(x)|^p}{\|f\|_{L^p}^p} + \frac{1}{q} \cdot \frac{|g(x)|^q}{\|g\|_{L^q}^q}$$

μ*-almost everywhere. The equality case in Young's inequality shows that this is equivalent to* $|f(x)|^p/\|f\|_{L^p}^p = |g(x)|^q/\|g\|_{L^q}^q$ μ*-almost everywhere, that is,*

$$A|f(x)|^p = B|g(x)|^q \quad \mu\textit{-almost everywhere}$$

for some positive numbers A *and* B*.*

If $p = 1$ *and* $q = \infty$*, we have equality in* (1.7) *if and only if there is a constant* $\lambda \ge 0$ *such that* $|g(x)| \le \lambda$ μ*-almost everywhere, and* $|g(x)| = \lambda$ μ*-almost everywhere in the set* $\{x : f(x) \ne 0\}$*.*

1.2.4 Theorem (Minkowski's inequality) *For* $1 \le p < \infty$ *and* $f, g \in L^p(\mu)$ *we have*

$$\|f + g\|_{L^p} \le \|f\|_{L^p} + \|g\|_{L^p}. \tag{1.10}$$

In the discrete case, using the notation of Exercise 12 in Section 1.1, this inequality reads

$$M_p(x + y, \lambda) \le M_p(x, \lambda) + M_p(y, \lambda). \tag{1.11}$$

The integral analogue of Minkowski's inequality for $0 < p < 1$ holds in reversed direction. See Exercise 7. The case $p = 0$ is presented in Section 3.1, Example 4.1.5.

Proof. For $p = 1$, the inequality (1.10) follows immediately by integrating the inequality $|f + g| \le |f| + |g|$. For $p \in (1, \infty)$ we have

$$\begin{aligned} |f+g|^p &\le (|f| + |g|)^p \le (2 \sup\{|f|, |g|\})^p \\ &\le 2^p (|f|^p + |g|^p), \end{aligned}$$

which shows that $f + g \in L^p(\mu)$. Moreover, according to Theorem 1.2.2,

$$\begin{aligned} \|f+g\|_{L^p}^p &= \int_\Omega |f+g|^p \, \mathrm{d}\mu \le \int_\Omega |f+g|^{p-1} \, |f| \, \mathrm{d}\mu + \int_\Omega |f+g|^{p-1} \, |g| \, \mathrm{d}\mu \\ &\le \left(\int_\Omega |f|^p \, \mathrm{d}\mu \right)^{1/p} \left(\int_\Omega |f+g|^{(p-1)q} \, \mathrm{d}\mu \right)^{1/q} \\ &\quad + \left(\int_\Omega |g|^p \, \mathrm{d}\mu \right)^{1/p} \left(\int_\Omega |f+g|^{(p-1)q} \, \mathrm{d}\mu \right)^{1/q} \\ &= (\|f\|_{L^p} + \|g\|_{L^p}) \|f+g\|_{L^p}^{p/q}, \end{aligned}$$

where $1/p + 1/q = 1$, and it remains to observe that $p - p/q = 1$. ∎

1.2.5 Remark *If $p = 1$, we obtain equality in (1.10) if and only if there is a positive measurable function φ such that*

$$f(x)\varphi(x) = g(x)$$

μ-almost everywhere on the set $\{x : f(x)g(x) \neq 0\}$.

If $p \in (1, \infty)$ and f is not 0 almost everywhere, then we have equality in (1.10) if and only if $g = \lambda f$ almost everywhere, for some $\lambda \ge 0$.

In the particular case where (Ω, Σ, μ) is the measure space associated with the counting measure on a finite set,

$$\mu \colon \mathcal{P}(\{1, \dots, n\}) \to \mathbb{N}, \quad \mu(A) = |A|,$$

we retrieve the classical discrete forms of the above inequalities. For example, the discrete version of the Rogers–Hölder inequality can be read

$$\left| \sum_{k=1}^n \xi_k \eta_k \right| \le \left(\sum_{k=1}^n |\xi_k|^p \right)^{1/p} \left(\sum_{k=1}^n |\eta_k|^q \right)^{1/q}$$

for all ξ_k, $\eta_k \in \mathbb{C}$, $k \in \{1, \dots, n\}$. On the other hand, a moment's reflection shows that we can pass immediately from these discrete inequalities to their integral analogues, corresponding to finite measure spaces.

1.2.6 Remark (More on Cauchy–Bunyakovsky–Schwarz inequality) *A.-L. Cauchy, in his famous* Cours d'Analyse [124]*, derived the discrete case of this inequality from Lagrange's algebraic identity,*

$$\left(\sum_{k=1}^n a_k^2 \right) \left(\sum_{k=1}^n b_k^2 \right) = \sum_{1 \le j < k \le n} (a_j b_k - a_k b_j)^2 + \left(\sum_{k=1}^n a_k b_k \right)^2.$$

Thus

$$\left|\sum_{k=1}^{n} a_k b_k\right| \leq \left(\sum_{k=1}^{n} a_k^2\right)^{1/2} \left(\sum_{k=1}^{n} b_k^2\right)^{1/2}$$

for all families of real numbers $a_1, \ldots, a_n$, $b_1, \ldots, b_n$. *The equality case is straightforward. The corresponding inequality for integrals was proved independently by V. Y. Bunyakovsky* [94] *and H. A. Schwarz* [527].

In 1890, H. Poincaré [476] *noticed the integral version of Lagrange's algebraic identity (that yields the Cauchy–Bunyakovsky–Schwarz inequality in full generality): If* μ *is a probability measure on a space* Ω *and* f *and* g *are two functions belonging to the space* $L^2(\mu)$, *then*

$$\left(\int_\Omega f^2 \mathrm{d}\mu\right)\left(\int_\Omega g^2 \mathrm{d}\mu\right) - \left(\int_\Omega fg \mathrm{d}\mu\right)^2 = \frac{1}{2}\int_\Omega \int_\Omega \left(f(x)g(y) - f(y)g(x)\right)^2 \mathrm{d}\mu(x)\mathrm{d}\mu(y).$$

He used this integral identity to derive the one-dimensional case of an inequality bearing his name. See Exercise 8.

Another simple proof of the Cauchy–Bunyakovsky–Schwarz inequality is provided by an identity equivalent to the law of cosines: for every pair of nonzero vectors x *and* y *in a real inner vector space,*

$$\left\| \frac{x}{\|x\|} - \frac{y}{\|y\|} \right\|^2 = 2 - 2\frac{\langle x, y\rangle}{\|x\|\,\|y\|}.$$

The book of J. M. Steele [548] *offers many other proofs and a bulk of interesting applications (including Heisenberg's Uncertainty Principle and van der Corput's inequality). Still another application (motivated by information theory) makes the object of the next remark.*

1.2.7 Remark (The Kullback–Leibler divergence) *Denote by*

$$Prob\,(n) = \left\{ (\lambda_1, ..., \lambda_n) \in [0,1]^n : \sum_{k=1}^{n} \lambda_k = 1 \right\}$$

the set of all probability distributions of order n *and suppose there are given two such distributions* $Q = (q_1, ..., q_n)$ *and* $P = (p_1, ..., p_n)$. *With the convention* $\log 0 = -\infty$, $\log \frac{0}{0} = \infty$ *and* $0 \cdot (\pm\infty) = 0$, *the* Kullback–Leibler divergence *from* Q *to* P *is defined by the formula:*

$$D_{KL}(P||Q) = \begin{cases} \sum_{k=1}^{n} p_k \log \frac{p_k}{q_k} & \text{if } q_k = 0 \text{ implies } p_k = 0 \\ \infty & \text{otherwise} \end{cases}$$

and represents a measure of how one probability distribution diverges from a second expected probability distribution. The Kullback–Leibler divergence is also known as the relative entropy.

From Jensen's inequality, it follows that $D_{KL}(P||Q) \geq 0$ *(with equality if and only if* $P = Q$*). This is known as* Gibbs' inequality. *A better lower bound is provided by* Pinsker's inequality,

$$D_{KL}(P||Q) \geq \frac{1}{2}\left(\sum_{k=1}^{n} |p_k - q_k|\right)^2, \tag{1.12}$$

the constant $1/2$ *being optimal (see* [145]*). Following the argument of J. M. Borwein and A. S. Lewis* [89]*, p.* 63*, this can be derived from the following double inequality:*

$$2\sum_{k=1}^{n} p_k \log \frac{p_k}{q_k} \geq 3\sum_{k=1}^{n} \frac{(p_k - q_k)^2}{p_k + 2q_k} \geq \left(\sum_{k=1}^{n} |p_k - q_k|\right)^2. \tag{1.13}$$

The right-hand side inequality follows from the Cauchy–Bunyakovsky–Schwarz inequality. Next, notice that $3(x-1)^2 \leq 2\,(x+2)\,(x\log x - x + 1)$ *for all* $x > 0$ *and infer from it the inequality*

$$3(u-v)^2 \leq 2\,(u+2v)\left(u\log\frac{u}{v} - u + v\right),$$

for all $u, v > 0$*. Then put* $u = p_k$ *and* $v = q_k$ *and sum up for* k *from 1 to* n *to get the left-hand side inequality in* 1.13*.*

Exercises

1. (The Bernoulli inequality) (a) Prove that for all $x > -1$ we have

$$(1+x)^p \geq 1 + px \quad \text{if } p \in (-\infty, 0] \cup [1, \infty)$$

and

$$(1+x)^p \leq 1 + px \quad \text{if } p \in [0, 1];$$

if $p \notin \{0, 1\}$, the equality occurs only for $x = 0$.

(b) Use the substitution $1 + x = a/b^{q-1}$ to infer Young's inequality for full range of parameters, that is, for all $x, y > 0$,

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q} \quad \text{if } p \in (1, \infty) \text{ and } 1/p + 1/q = 1$$

and

$$xy \geq \frac{x^p}{p} + \frac{y^q}{q} \quad \text{if } p \in (0, 1) \text{ and } 1/p + 1/q = 1.$$

Remark. The last inequality easily yields the following version of Rogers–Hölder inequality for $p \in (0, 1)$ and $1/p + 1/q = 1$: If $f \in L^p(\mu)$, $f \geq 0$, and $g \in L^q(\mu)$, $g > 0$, then

$$\int_\Omega f g \, \mathrm{d}\mu \geq \|f\|_{L^p} \, \|g\|_{L^q}. \tag{1.14}$$

Alternatively, this case follows from the usual one (corresponding to $p > 1$) via a change of function. See the monograph of E. Hewitt and K. Stromberg [252], Theorem 13.6, p. 191. For more general results, see L.-E. Persson [470].

2. (J. M. Aldaz [10]) Use calculus to prove the following refinement of the inequality (1.5): If $1 < p \le 2$ and $1/p + 1/q = 1$, then for all $a, b \ge 0$

$$\frac{1}{q}\left(a^{p/2} - b^{q/2}\right)^2 \le \frac{a^p}{p} + \frac{b^q}{q} - ab \le \frac{1}{p}\left(a^{p/2} - b^{q/2}\right)^2.$$

This fact yields a refinement of the Rogers–Hölder inequality and also a new proof of the uniform convexity of real L^p-spaces for $1 < p < \infty$.

Remark. Another refinement of Young's inequality was proposed by Y. Al-Manasrah and F. Kittaneh [13]: $(1-\mu)a + \mu b \ge a^{1-\mu} b^{\mu} + \min\{\mu, 1-\mu\} \cdot \left(\sqrt{a} - \sqrt{b}\right)^2$ for all $a, b \in [0, \infty)$ and $\mu \in [0, 1]$.

3. (A symmetric form of the Rogers–Hölder inequality)Let p, q, r be nonzero real numbers such that $1/p + 1/q = 1/r$.

(a) Prove that the inequality

$$\|fg\|_{L^r} \le \|f\|_{L^p} \, \|g\|_{L^q}$$

holds in each of the following three cases:

$$p > 0,\ q > 0,\ r > 0 \quad \text{or} \quad p < 0,\ q > 0,\ r < 0 \quad \text{or} \quad p > 0,\ q < 0,\ r < 0.$$

(b) Prove that the preceding inequality holds in the reversed direction in each of the following three cases:

$$p > 0,\ q < 0,\ r > 0 \quad \text{or} \quad p < 0,\ q > 0,\ r > 0 \quad \text{or} \quad p < 0,\ q < 0,\ r < 0.$$

Remark. For more general results, see L.-E. Persson [470] and the references therein.

4. (a) Young's inequality extends to more than two variables. Prove that if $p_1, \ldots, p_n \in (1, \infty)$ and $1/p_1 + \cdots + 1/p_n = 1$, then

$$\prod_{k=1}^{n} x_k \le \sum_{k=1}^{n} \frac{x_k^{p_k}}{p_k}$$

for all $x_1, \ldots, x_n \ge 0$.

(b) Infer the following version of the Rogers–Hölder inequality: If (Ω, Σ, μ) is a measure space and $f_1, \ldots, f_n$ are functions such that $f_k \in L^{p_k}(\mu)$ for some $p_k \in [1, \infty]$, and $\sum_{k=1}^{n} 1/p_k = 1$, then

$$\left| \int_{\Omega} \left(\prod_{k=1}^{n} f_k \right) \mathrm{d}\mu \right| \le \prod_{k=1}^{n} \|f_k\|_{L^{p_k}}.$$

5. (Hilbert's inequality) Suppose that $p, q \in (1, \infty)$, $1/p + 1/q = 1$, and f, g are two positive functions such $f \in L^p(0, \infty)$ and $g \in L^q(0, \infty)$. Infer from the Rogers–Hölder inequality that

$$\int_0^\infty \int_0^\infty \frac{f(x)\, g(y)}{x+y} \mathrm{d}x\mathrm{d}y \leq \frac{\pi}{\sin \frac{\pi}{p}} \left(\int_0^\infty f^p(x)\, \mathrm{d}x \right)^{1/p} \left(\int_0^\infty g^q(y)\, \mathrm{d}y \right)^{1/q},$$

the constant $\pi / \sin(\pi/p)$ being sharp.

[*Hint*: Notice that the left-hand side equals

$$\int_0^\infty \int_0^\infty \frac{f(x)}{(x+y)^{1/p}} \left(\frac{x}{y} \right)^{1/pq} \frac{g(y)}{(x+y)^{1/q}} \left(\frac{y}{x} \right)^{1/pq} \mathrm{d}x\mathrm{d}y. \ ]$$

6. (A general form of Minkowski's inequality) Suppose that $(X, \mathcal{M}, \mu)$ and $(Y, \mathcal{N}, \nu)$ are two σ-finite measure spaces, f is a positive function on $X \times Y$ which is $\mu \times \nu$-measurable, and let $p \in [1, \infty)$. Prove that

$$\left(\int_X \left(\int_Y f(x,y)\, \mathrm{d}\nu(y) \right)^p \mathrm{d}\mu(x) \right)^{1/p} \leq \int_Y \left(\int_X f(x,y)^p\, \mathrm{d}\mu(x) \right)^{1/p} \mathrm{d}\nu(y).$$

7. (The Minkowski inequality for $0 < p < 1$) Suppose that $0 < p < 1$ and $f, g \in L^P(\mu)$ are two positive functions. Prove that

$$\|f\|_{L^p} + \|g\|_{L^p} \leq \|f + g\|_{L^p} \leq 2^{(1-p)/p} \left(\|f\|_{L^p} + \|g\|_{L^p} \right).$$

[*Hint:* The function x^p is concave on $[0,\infty)$ if $0 < p < 1$ and every concave function $f : [0, \infty) \to \mathbb{R}$ with $f(0) \geq 0$ verifies the inequality $f(x + y) \leq f(x) + f(y)$ for all $x, y \geq 0$.]

8. (The one-dimensional case of Poincaré inequality [476]) Infer from the integral version of Lagrange's algebraic identity the following inequality: If $f : [0, 1] \to \mathbb{R}$ is a function of class C^1 that verifies the condition $\int_0^1 f \mathrm{d}x = 0$, then

$$\int_0^1 f^2 \mathrm{d}x \leq \frac{1}{2} \int_0^1 f'^2 \mathrm{d}x.$$

The n-dimensional variant of the Poincaré inequality plays a major role in partial differential equations and their applications. See [183].

9. Let (Ω, Σ, μ) be a measure space and let $f \colon \Omega \to \mathbb{C}$ be a measurable function, which belongs to $L^t(\mu)$ for t in a subinterval I of $(0, \infty)$. Infer from the Cauchy–Bunyakovsky–Schwarz inequality that the function $t \to \log \int_\Omega |f|^t \mathrm{d}\mu$ is convex on I.

Remark. The result of this exercise is equivalent to *Lyapunov's inequality* [347]: *If $a \geq b \geq c$, then*

$$\left(\int_\Omega |f|^b \mathrm{d}\mu \right)^{a-c} \leq \left(\int_\Omega |f|^c \mathrm{d}\mu \right)^{a-b} \left(\int_\Omega |f|^a \mathrm{d}\mu \right)^{b-c}$$

(provided the integrability aspects are fixed). Equality holds if and only if one of the following four conditions holds:

(1) f is constant on some subset of Ω and 0 elsewhere;

(2) $a = b$;

(3) $b = c$;

(4) $c(2a - b) = ab$.

10. (a) Prove the *discrete Berwald inequality*:

$$\frac{1}{n+1}\sum_{k=0}^{n} a_k \geq \Big(\frac{3(n-1)}{4(n+1)}\Big)^{1/2}\Big(\frac{1}{n+1}\sum_{k=0}^{n} a_k^2\Big)^{1/2}$$

for every concave sequence $a_0, a_1, \ldots, a_n$ of positive numbers.

(b) Derive from (a) the integral form of this inequality: if f is a concave positive function on $[a, b]$, then

$$\left(\frac{1}{b-a}\int_a^b f^2(x)\mathrm{d}x\right)^{1/2} \leq \frac{2\sqrt{3}}{3}\left(\frac{1}{b-a}\int_a^b f(x)\mathrm{d}x\right).$$

[*Hint*: By Minkowski's inequality (Theorem 1.2.4 above), if the Berwald inequality works for two concave sequences, then it also works for all linear combinations of them, with positive coefficients. Then apply the assertion (b) of Exercise 15, Section 1.1.]

1.3 Log-convex Functions

This section is aimed to give a brief discussion on a stronger concept of convexity.

1.3.1 Definition *A strictly positive function $f : I \to (0, \infty)$ is called log-convex (respectively log-concave) if $\log f$ (respectively $-\log f$) is a convex function.*

Equivalently, the condition of log-convexity of f means

$$x, y \in I \text{ and } \lambda \in (0, 1) \implies f((1-\lambda)x + \lambda y) \leq f(x)^{1-\lambda} f(y)^{\lambda}.$$

The following result collects some immediate consequences of the generalized *AM–GM* inequality:

1.3.2 Lemma (a) *The multiplicative inverse of any strictly positive concave function is a log-convex function.*

(b) *Every log-convex function is also convex.*

(c) *Every strictly positive concave function is also log-concave.*

According to this lemma, $1/x$ and $1/\sin x$ (as well as their exponentials) are log-convex, respectively, on $(0,\infty)$ and $(0,\pi)$.

Notice that the function e^{-x^2} is log-concave although it is neither convex nor concave.

An important example of a log-convex function is the gamma function,

$$\Gamma\colon (0,\infty)\to\mathbb{R},\quad \Gamma(x)=\int_0^\infty t^{x-1}\mathrm{e}^{-t}\,\mathrm{d}t\quad \text{for } x>0.$$

See A. D. R. Choudary and C. P. Niculescu [134], pp. 352–360, for its basic theory.

The fact that Γ is log-convex is a consequence of Rogers–Hölder inequality. Indeed, for every $x,y>0$ and $\lambda\in[0,1]$ we have

$$\begin{aligned}\Gamma((1-\lambda)x+\lambda y)&=\int_0^\infty t^{(1-\lambda)x+\lambda y-1}\mathrm{e}^{-t}\mathrm{d}t=\int_0^\infty (t^{x-1}\mathrm{e}^{-t})^{1-\lambda}(t^{y-1}\mathrm{e}^{-t})^{\lambda}\,\mathrm{d}t\\&\le\left(\int_0^\infty t^{x-1}\mathrm{e}^{-t}\,\mathrm{d}t\right)^{1-\lambda}\left(\int_0^\infty t^{y-1}\mathrm{e}^{-t}\,\mathrm{d}t\right)^{\lambda}=\Gamma^{1-\lambda}(x)\Gamma^{\lambda}(y).\end{aligned}$$

Remarkably, Γ is the unique log-convex extension of the factorial function.

1.3.3 Theorem (H. Bohr and J. Mollerup [75], [23]) *The gamma function is the only function $f\colon(0,\infty)\to\mathbb{R}$ that satisfies the following three conditions:*

(a) $f(x+1)=xf(x)$ *for all* $x>0$;

(*b*) $f(1)=1$;

(*c*) f *is* log-*convex.*

An important class of log-convex functions is that of completely monotonic functions.

A function $f\colon(0,\infty)\to\mathbb{R}$ is called *completely monotonic* if f has derivatives of all orders and satisfies $(-1)^n f^{(n)}(x)\ge 0$ for all $x>0$ and $n\in\mathbb{N}$.

Necessarily, every completely monotonic function is positive, decreasing, and convex.

One can prove easily that the functions

$$\mathrm{e}^{-x},\mathrm{e}^{-1/x},\frac{1}{(1+x)^2}\ \text{and}\ \frac{\ln(1+x)}{x}$$

(as well as their sums, products, and derivatives of even order) are completely monotonic on $(0,\infty)$. More interesting examples can be found in the papers of M. Merkle [377] and K. S. Miller and S. G. Samko [383].

S. N. Bernstein has proved in 1928 that a necessary and sufficient condition that a function $f(x)$ to be completely monotonic is the existence of a representation of the form:

$$f(x)=\int_0^\infty \mathrm{e}^{-xt}\mathrm{d}\mu(t),\tag{1.15}$$

where μ is a positive Borel measure on $(0,\infty)$ and the integral converges for $0<x<\infty$. A simple proof based on the Krein–Milman Theorem (see Appendix B,

Theorem B.3.2) can be found in B. Simon [542], pp. 143-152. His argument also covers the fact (known to Bernstein) that every completely monotonic function is the restriction to $(0,\infty)$ of an analytic function in the right semi-plan $\operatorname{Re} z > 0$.

F. Qi and C.-P. Chen [490] have noticed that if the logarithm of a function is completely monotonic, then the function itself is completely monotonic. So is the case of the function $(\Gamma(1+x))^{1/x}/x$.

1.3.4 Theorem (D. V. Widder [586], p. 167) *Every completely monotonic function is log-convex.*

Proof. It suffices to prove that every completely monotonic function f verifies the inequality $f(x)f''(x) \geq (f'(x))^2$. See Exercise 1. Taking into account the aforementioned integral representation due to Bernstein, this inequality is equivalent to

$$\int_0^\infty \mathrm{e}^{-xt_1}\mathrm{d}\mu(t_1)\int_0^\infty t_2^2\mathrm{e}^{-xt_2}\mathrm{d}\mu(t_2) \geq \int_0^\infty t_1\mathrm{e}^{-xt_1}\mathrm{d}\mu(t_1)\int_0^\infty t_2\mathrm{e}^{-xt_2}\mathrm{d}\mu(t_2)$$

and also to

$$\int_0^\infty\int_0^\infty t_2^2\mathrm{e}^{-x(t_1+t_2)}\mathrm{d}\mu(t_1)\mathrm{d}\mu(t_2) \geq \int_0^\infty\int_0^\infty t_1t_2\mathrm{e}^{-x(t_1+t_2)}\mathrm{d}\mu(t_1)\mathrm{d}\mu(t_2),$$

which is clear because by symmetry

$$\int_0^\infty\int_0^\infty t_2^2\mathrm{e}^{-x(t_1+t_2)}\mathrm{d}\mu(t_1)\mathrm{d}\mu(t_2) = \frac{1}{2}\int_0^\infty\int_0^\infty \left(t_1^2+t_2^2\right)\mathrm{e}^{-x(t_1+t_2)}\mathrm{d}\mu(t_1)\mathrm{d}\mu(t_2).$$

■

The convex functions, log-convex functions, and quasiconvex functions are all just special cases of a considerably more general concept, that of convex function with respect to a pair of means. This makes the subject of Appendix A.1.

Exercises

1. (The second derivative test of log-convexity) Prove that a twice differentiable function $f : I \to (0,\infty)$ is log-convex if and only if

$$f(x)f''(x) \geq (f'(x))^2 \quad \text{for all } x \in I.$$

   Infer from this result the log-convexity of the gamma function.

2. (The log-convex analogue of Theorem 1.1.8) Prove that a strictly positive continuous function f defined on an interval I is log-convex if and only if $f\left(\frac{x+y}{2}\right) \leq \sqrt{f(x)f(y)}$ for all $x, y \in I$.

3. (Operations with log-convex functions) (a) Clearly, a product of log-convex functions is also a log-convex function. Prove that the same happens for sums.

(b) Suppose that $x = (x_1, \dots, x_n)$ and $\lambda = (\lambda_1, \dots, \lambda_n)$ are n-tuples of strictly positive numbers such that $\sum_{k=1}^n \lambda_k = 1$). Infer from the assertion (a) the convexity of the function $t \to t \log M_t(x; \lambda)$ on $\mathbb{R}$ and of the function $t \to \log M_t(x; \lambda)$ on $(0, \infty)$. The last assertion yields

$$M_{\alpha s+\beta t}(x; \lambda) \leq M_s^{\alpha}(x; \lambda) M_t^{\beta}(x; \lambda)$$

whenever $\alpha, \beta \in [0, 1]$, $\alpha + \beta = 1$ and $s, t > 0$.

(c) Suppose that f is a convex function and g is an increasing log-convex function. Prove that $g \circ f$ is a log-convex function.

(d) Prove that the harmonic mean of two log-concave functions is also log-concave.

[*Hint*: (a) Note that this assertion is equivalent to the following inequality for positive numbers: $a^{\alpha} b^{\beta} + c^{\alpha} d^{\beta} \leq (a + c)^{\alpha} (b + d)^{\beta}$.]

4. (Operations with log-concave functions) (a) Prove that the product of log-concave functions is also log-concave.

   (b) Prove that the convolution preserves log-concavity, that is, if f and g are two integrable log-concave functions defined on $\mathbb{R}$, then the function

   $$(f * g)(x) = \int_{\mathbb{R}} f(x - y) g(y) \mathrm{d}y$$

   is log-concave too.

   Remark. The statement (b) of Exercise 4 is a particular case of the Prékopa–Leindler inequality. See Section 4.7.

5. (Ch. Borell [78]) If $H : [0, \infty] \to [0, \infty]$ is log-concave and decreasing, then the function $h : \mathbb{R} \to [0, \infty]$ defined by $h(x) = H(\exp(x)) \exp(x)$ is log-concave.

6. (P. Montel [394]) Let I be an interval. Prove that the following assertions are equivalent for every function $f \colon I \to (0, \infty)$:

   (a) f is log-convex;

   (b) the function $x \to \mathrm{e}^{\alpha x} f(x)$ is convex on I for all $\alpha \in \mathbb{R}$;

   (c) the function $x \to [f(x)]^{\alpha}$ is convex on I for all $\alpha > 0$.

   [*Hint*: For $(c) \Rightarrow (a)$, note that $([f(x)]^{\alpha} - 1)/\alpha$ is convex for all $\alpha > 0$ and $\log f(x) = \lim_{\alpha \to 0+} ([f(x)]^{\alpha} - 1)/\alpha$. The limit of a sequence of convex functions is itself convex.]

7. Prove that the function $\frac{e^x \Gamma(x)}{x^x}$ is log-convex.

8. (Some geometric consequences of log-concavity)

   (a) A convex quadrilateral $ABCD$ is inscribed in the unit circle. Prove that $ABCD$ is a square if its sides satisfy the inequality

   $$AB \cdot BC \cdot CD \cdot DA \geq 4.$$

(b) Suppose that A, B, C are the angles of a triangle, expressed in radians. Prove that

$$\sin A \sin B \sin C < \left(\frac{3\sqrt{3}}{2\pi}\right)^3 ABC < \left(\frac{\sqrt{3}}{2}\right)^3,$$

unless $A = B = C$.

[*Hint*: Note that the sine function is log-concave, while $x/\sin x$ is log-convex on $(0, \pi)$.]

9. (E. Artin [23]) Let U be an open convex subset of $\mathbb{R}^n$ and let μ be a Borel measure on an interval I. Consider the integral transform:

$$F(x) = \int_I K(x,t)\,\mathrm{d}\mu(t),$$

where the kernel $K(x,t)\colon U \times I \to [0,\infty)$ satisfies the following two conditions:

(a) $K(x,t)$ is μ-integrable in t for each fixed x;

(b) $K(x,t)$ is log-convex in x for each fixed t.

Prove that F is log-convex on U.

[*Hint*: Apply the Rogers–Hölder inequality, noticing that

$$K((1-\lambda)x + \lambda y, t) \le (K(x,t))^{1-\lambda}(K(y,t))^{\lambda}. \quad]$$

Remark. The *Laplace transform* of a function $f \in L^1(0,\infty)$ is given by the formula $(\mathcal{L}f)(x) = \int_0^\infty f(t)\mathrm{e}^{-tx}\,\mathrm{d}t$. By Exercise 9, the Laplace transform of any positive function is log-convex. In the same way, one can show that the moment $\mu_\alpha = \int_0^\infty t^\alpha f(t)\,\mathrm{d}t$, of any random variable with probability density f, is a log-convex function in α on each subinterval of $[0,\infty)$ where it is finite.

10. (Combinatorial properties of sequences) Call a sequence $(a_n)_n$ of strictly positive numbers log-*concave* (respectively log-*convex*) if $a_{n-1}a_{n+1} \le a_n^2$ for all $n \ge 1$ (respectively $a_{n-1}a_{n+1} \ge a_n^2$ for all $n \ge 1$).

(a) Let $(a_n)_n$ be a strictly positive and strictly decreasing sequence of numbers and let $A_n = \sum_{k=0}^n a_k$. Prove that $(A_n)_n$ is log-concave. Illustrate this result using the Maclaurin expansion of $\log(1-x)$ on $(0,1)$.

(b) Prove that the sequence of binomial coefficients $\binom{n}{k}$ is log-concave in k for fixed n.

(c) (I. Newton) Suppose that $P(x) = \sum_{k=0}^n \binom{n}{k} a_k x^k$ is a real polynomial with real zeros. Prove that $a_0, a_1, \ldots, a_n$ is a log-convex sequence.

(d) (H. Davenport - G. Pólya) If both $(x_n)_n$ and $(y_n)_n$ are log-convex, then so is the sequence of their binomial convolution,

$$z_n = \sum_{k=0}^n \binom{n}{k} x_k y_{n-k}.$$

Remark. More information on log-convex/log-concave sequences are available in the papers of L. L. Liu and Y. Wang [345] and R. P. Stanley [546]. See also Appendix C.

1.4 Smoothness Properties of Convex Functions

The starting point is the following restatement of Definition 1.1.1. A function $f\colon I \to \mathbb{R}$ is convex if and only if

$$f(x) \le \frac{b-x}{b-a} \cdot f(a) + \frac{x-a}{b-a} \cdot f(b), \tag{1.16}$$

equivalently,

$$\begin{vmatrix} 1 & a & f(a) \\ 1 & x & f(x) \\ 1 & b & f(b) \end{vmatrix} \ge 0, \tag{1.17}$$

whenever $a < x < b$ in I. Indeed, every point x belonging to an interval $[a, b]$ can be written uniquely as a convex combination of a and b, more precisely,

$$x = \frac{b-x}{b-a} \cdot a + \frac{x-a}{b-a} \cdot b.$$

Subtracting $f(a)$ from both sides of the inequality (1.16) and repeating the operation with $f(b)$ instead of $f(a)$, we obtain that f verifies the *three chords inequality*,

$$\frac{f(x)-f(a)}{x-a} \le \frac{f(b)-f(a)}{b-a} \le \frac{f(b)-f(x)}{b-x} \tag{1.18}$$

whenever $a < x < b$ in I. See Figure 1.3. Clearly, this inequality actually characterizes the convexity of f. Moreover, the three chords inequality with strict inequalities provides a characterization of strict convexity. These facts were first noticed by L. Galvani [211] and we will refer to them as the *Galvani characterization of convexity.*

1.4.1 Remark *The three chords inequality can be strengthened as follows: If $f : I \to \mathbb{R}$ is a convex function and x, y, a, b are points in the interval I such that $x \le a$, $y \le b$, $x \ne y$ and $a \ne b$, then*

$$\frac{f(x)-f(y)}{x-y} \le \frac{f(a)-f(b)}{a-b}.$$

1.4.2 Theorem (O. Stolz [555]) *Let $f\colon I \to \mathbb{R}$ be a convex function. Then f has finite left and right derivatives at each interior point of I and $x < y$ in* $\operatorname{int} I$ *implies*

$$f'_-(x) \le f'_+(x) \le \frac{f(y)-f(x)}{y-x} \le f'_-(y) \le f'_+(y).$$

Moreover, on $\operatorname{int} I$, *f'_- is left continuous and f'_+ is right continuous.*

Therefore, if a convex function is differentiable on $\operatorname{int} I$, *then it is also continuously differentiable.*

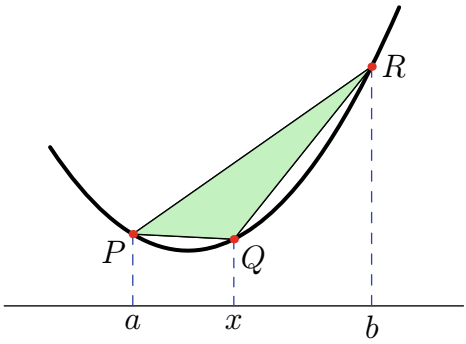


Figure 1.3: *slope* $PQ \le$ *slope* $PR \le$ *slope* QR.

Proof. Indeed, according to the three chords inequality, we have

$$\frac{f(x)-f(a)}{x-a} \le \frac{f(y)-f(a)}{y-a} \le \frac{f(z)-f(a)}{z-a}$$

for all $x \le y < a < z$ in I. This fact assures us that the left derivative at a exists and

$$f'_-(a) \le \frac{f(z)-f(a)}{z-a}.$$

A symmetric argument yields the existence of $f'_+(a)$ and the relation $f'_-(a) \le f'_+(a)$. On the other hand, starting with $x < u \le v < y$ in $\operatorname{int} I$, we infer from the three chords inequality the fact that

$$\frac{f(u)-f(x)}{u-x} \le \frac{f(v)-f(x)}{v-x} \le \frac{f(v)-f(y)}{v-y},$$

so letting $u \to x+$ and $v \to y-$, it follows that $f'_+(x) \le f'_-(y)$.

For the continuity of the one-sided derivatives, let us notice that from the continuity of f on $\operatorname{int} I$ we infer that

$$\frac{f(y)-f(x)}{y-x} = \lim_{z \searrow x} \frac{f(y)-f(z)}{y-z} \ge \lim_{z \searrow x} f'_+(z)$$

whenever $x < z < y$. Passing to the limit as $y \searrow x$ we obtain

$$f'_+(x) \ge \lim_{z \searrow x} f'_+(z).$$

Since f'_+ is increasing, the reverse inequality also holds. Thus f'_+ is right continuous on $\operatorname{int} I$. The left continuity of f'_- can be proved in a similar way. ■

The statement of Theorem 1.4.2 can be strengthened as follows:

1.4.3 Theorem *A differentiable function f defined on an open interval I is (strictly) convex if and only if its derivative is a (strictly) increasing function.*

Proof. The fact that the derivative of f is increasing if f is convex follows from Theorem 1.4.2. Consider now the case when f is strictly convex and choose arbitrarily the points $a < b$ in I. Then insert the points x_1, x_2, x_3 between a and b such that $a < x_1 < x_2 < x_3 < b$ and infer from the strict version of the three chords inequality 1.3 that

$$\frac{f(x_1) - f(a)}{x_1 - a} < \frac{f(x_2) - f(x_1)}{x_2 - x_1} < \frac{f(x_3) - f(x_2)}{x_3 - x_2} < \frac{f(b) - f(x_3)}{b - x_3}.$$

Passing to the limit as $x_1 \to a+$ and $x_3 \to b-$ one obtains $f'(a) < f'(b)$, that is, f' is strictly increasing.

The converse implications follow from the mean value theorem. Indeed, if f' is strictly increasing and $a < x < b$ are points in I, then the mean value theorem implies the existence of points $c_1 \in (a, x)$ and $c_2 \in (x, b)$ such that

$$\frac{f(x) - f(a)}{x - a} = f'(c_1) \quad \text{and} \quad \frac{f(b) - f(x)}{b - x} = f'(c_2).$$

Since $f'(c_1) < f'(c_2)$, it follows that

$$\frac{f(x) - f(a)}{x - a} < \frac{f(b) - f(x)}{b - x},$$

hence f is strictly convex. When f' is just increasing, the last inequality is nonstrict, which means that f is only convex. ■

By Theorem 1.4.2, every continuous convex function $f : [a, b] \to \mathbb{R}$ admits one-sided derivatives at the endpoints, but they can be infinite:

$$-\infty \le f'_+(a) < \infty \quad \text{and} \quad -\infty < f'_-(b) \le \infty.$$

This theorem also yields the fact that a continuous convex function $f\colon [a, b] \to \mathbb{R}$ can be extended to a convex function on $\mathbb{R}$ if and only if $f'_+(a)$ and $f'_-(b)$ exist and are finite.

How "nondifferentiable" can a convex function be?

Theorem 1.4.2 implies that every convex function $f\colon I \to \mathbb{R}$ is differentiable except for an enumerable subset. In fact, by considering the set of nondifferentiability,

$$I_{\mathrm{nd}} = \{x : f'_-(x) < f'_+(x)\},$$

and choosing for each $x \in I_{\mathrm{nd}}$ a rational point $r_x \in (f'_-(x), f'_+(x))$ we get a one-to-one function $\varphi\colon x \to r_x$ from I_{nd} into $\mathbb{Q}$. Consequently, I_{nd} is at most countable. Notice that this reasoning depends on the axiom of choice. An example of a convex function on $\mathbb{R}$ for which I_{nd} is infinitely countable is $f(x) = \sum_{n=0}^{\infty} |x - n|/2^n$.

Theorem 1.4.2 provides an alternative argument for the property of convex functions of being locally Lipschitz on open intervals. Indeed, if $f\colon I \to \mathbb{R}$ is a convex function and $[a, b]$ is a compact interval contained in the interior of I, then

$$f'_+(a) \le f'_+(x) \le \frac{f(y) - f(x)}{y - x} \le f'_-(y) \le f'_-(b)$$

for all $x, y \in [a,b]$ with $x < y$. Therefore $f|_{[a,b]}$ verifies the Lipschitz condition $|f(x) - f(y)| \leq L\,|x - y|$ with $L = \max\{|f'_+(a)|, |f'_-(b)|\}$. An immediate consequence is the following result:

1.4.4 Theorem *If $(f_n)_n$ is a pointwise converging sequence of convex functions defined on an open interval I, then its limit f is also convex. Moreover, the convergence is uniform on any compact subinterval and*

$$f'_-(a) \leq \liminf_{n\to\infty} (f_n)'_-(a) \leq \limsup_{n\to\infty} (f_n)'_+(a) \leq f'_+(a) \text{ for all } a \in I.$$

Proof. The convexity of f is trivial. By the three chords inequality (1.18), for every $h > 0$ with $a + h \in I$,

$$(f_n)'_+(a) \leq \frac{f_n(a+h) - f_n(a)}{h}$$

so that

$$\limsup_{n\to\infty} (f_n)'_+(a) \leq \limsup_{n\to\infty} \frac{f_n(a+h) - f_n(a)}{h} = \frac{f(a+h) - f(a)}{h}.$$

Letting $h \to 0$, we obtain the right-hand side inequality in the statement of Theorem 1.4.4. The left-hand side inequality can be proved in a similar way. Taking into account an above remark about local Lipschitzianity of convex functions, one can easily infer the uniform convergence on compact subintervals of I. ∎

Equality may not hold in the above theorem. To see this, consider the sequence of convex functions $f_n(x) = |x|^{1+1/n}$ that converges on $\mathbb{R}$ to the function $f(x) = |x|$. Then $(f_n)'_+(0) = 0$ for all n while $f'_+(0) = 1$.

The next result (which exhibits the power of one-variable techniques in a several-variable context) yields an upper estimate of Jensen's inequality:

1.4.5 Theorem *Let $f\colon [a,b] \to \mathbb{R}$ be a convex function and let*

$$[m_1, M_1], \ldots, [m_n, M_n]$$

be compact subintervals of $[a,b]$. Given $\lambda_1, \ldots, \lambda_n$ in $[0,1]$, with $\sum_{k=1}^n \lambda_k = 1$, the function

$$E(x_1, \ldots, x_n) = \sum_{k=1}^n \lambda_k f(x_k) - f\Big(\sum_{k=1}^n \lambda_k x_k\Big)$$

attains its maximum on $\Omega = [m_1, M_1] \times \cdots \times [m_n, M_n]$ at a vertex, that is, at a point of $\{m_1, M_1\} \times \cdots \times \{m_n, M_n\}$.

The proof depends upon the following refinement of Lagrange's mean value theorem:

1.4.6 Lemma *Let $h\colon [a,b] \to \mathbb{R}$ be a continuous function. Then there exists a point $c \in (a,b)$ such that*

$$\underline{D}h(c) \leq \frac{h(b) - h(a)}{b - a} \leq \overline{D}h(c).$$

Here

$$\underline{D}h(c) = \liminf_{x\to c} \frac{h(x)-h(c)}{x-c} \quad \text{and} \quad \overline{D}h(c) = \limsup_{x\to c} \frac{h(x)-h(c)}{x-c}$$

are, respectively, the *lower derivative* and the *upper derivative* of h at c. According to Theorem 1.4.2, in the case of convex functions, $\underline{D}h(c) = h'_-(c)$ and $\overline{D}h(c) = h'_+(c)$.

Proof. As in the smooth case, we consider the function

$$H(x) = h(x) - \frac{h(b)-h(a)}{b-a}\,(x-a), \quad x \in [a,b].$$

Clearly, H is continuous and $H(a) = H(b)$. If H attains its supremum at $c \in (a,b)$, then $\underline{D}H(c) \le 0 \le \overline{D}H(c)$ and the conclusion of Lemma 1.4.6 is immediate. The same is true when H attains its infimum at an interior point of $[a,b]$. If both extremes are attained at the endpoints, then H is constant and the conclusion of Lemma 1.4.6 works for all c in (a,b). ■

Proof of Theorem 1.4.4. Clearly, we may assume that f is also continuous. We shall show (by *reductio ad absurdum*) that

$$E(x_1,\dots,x_k,\dots,x_n) \le \sup\{E(x_1,\dots,m_k,\dots,x_n), E(x_1,\dots,M_k,\dots,x_n)\}$$

for all $(x_1,x_2,\dots,x_n) \in \Omega$ and all $k \in \{1,\dots,n\}$. In fact, if

$$E(x_1,x_2,\dots,x_n) > \sup\{E(m_1,x_2,\dots,x_n), E(M_1,x_2,\dots,x_n)\}$$

for some $(x_1,x_2,\dots,x_n) \in \Omega$, we consider the function

$$h\colon [m_1,M_1] \to \mathbb{R}, \quad h(x) = E(x,x_2,\dots,x_n).$$

According to Lemma 1.4.6, there exists $\xi \in (m_1,x_1)$ such that

$$h(x_1) - h(m_1) \le (x_1-m_1)\overline{D}h(\xi).$$

Since $h(x_1) > h(m_1)$, it follows that $\overline{D}h(\xi) > 0$, equivalently,

$$\overline{D}f(\xi) > \overline{D}f(\lambda_1\xi + \lambda_2x_2 + \cdots + \lambda_nx_n).$$

Or, $\overline{D}f = f'_+$ is an increasing function on (a,b), which yields

$$\xi > \lambda_1\xi + \lambda_2x_2 + \cdots + \lambda_nx_n,$$

and thus $\xi > (\lambda_2x_2 + \cdots + \lambda_nx_n)/(\lambda_2 + \cdots + \lambda_n)$.

A new appeal to Lemma 1.4.6 (applied this time to $h|_{[x_1,M_1]}$) yields the existence of an $\eta \in (x_1,M_1)$ such that $\eta < (\lambda_2x_2 + \cdots + \lambda_nx_n)/(\lambda_2 + \cdots + \lambda_n)$. But this contradicts the fact that $\xi < \eta$. ■

1.4.7 Corollary *Let $f\colon [a,b] \to \mathbb{R}$ be a convex function. Then*

$$(1-\lambda)f(a)+\lambda f(b)-f\Big((1-\lambda)a+\lambda b\Big) \geq (1-\lambda)f(c)+\lambda f(d)-f\Big((1-\lambda)c+\lambda d\Big)$$

for all $a \leq c \leq d \leq b$ and $\lambda \in [0,1]$. In particular, if $a+b=c+d$, then

$$f(a)+f(b) \geq f(c)+f(d).$$

Corollary 1.4.7 is a first step toward the inequality of majorization. See Section 6.1. It can be easily turned into a characterization of convexity within the class of continuous functions. For this, use Theorem 1.1.8.

Since the first derivative of a convex function may not exist at a dense subset, a characterization of convexity in terms of second-order derivatives is not possible unless we relax the concept of twice differentiability.

The *upper* and the *lower second symmetric derivatives* of f at x are, respectively, defined by the formulas:

$$\overline{\mathcal{D}}^2 f(x) = \limsup_{h \downarrow 0} \frac{f(x+h)+f(x-h)-2f(x)}{h^2}$$

$$\underline{\mathcal{D}}^2 f(x) = \liminf_{h \downarrow 0} \frac{f(x+h)+f(x-h)-2f(x)}{h^2}.$$

It is not difficult to check that if f is twice differentiable at a point x, then

$$\overline{\mathcal{D}}^2 f(x) = \underline{\mathcal{D}}^2 f(x) = f''(x); \tag{1.19}$$

however, $\overline{\mathcal{D}}^2 f(x)$ and $\underline{\mathcal{D}}^2 f(x)$ can exist even at points of discontinuity; for example, consider the case of the signum function and the point $x=0$. The basic tool to prove the relations (1.19) is the formula

$$f(x+h)-f(x)-hf'(x)-\frac{h^2}{2}f''(x) = o(h^2)$$

that works at any point x at which f is twice differentiable. Indeed, by denoting the left-hand side as $g(h)$, we infer from the mean value theorem that $g(h) = g(h)-g(0) = hg'(k)$ and $g'(k) = g'(k)-g'(0) = kg''(0)+o(k) = o(k)$. Therefore $g(h) = ho(k) = o(h^2)$.

1.4.8 Theorem *Suppose that I is an open interval. A real-valued function f is convex on I if and only if f is continuous and $\overline{\mathcal{D}}^2 f \geq 0$.*

As a consequence, if a function $f\colon I \to \mathbb{R}$ is convex in the neighborhood of each point of I, then it is convex on the whole interval I.

Proof. If f is convex, then clearly $\overline{\mathcal{D}}^2 f \geq \underline{\mathcal{D}}^2 f \geq 0$. The continuity of f follows from Theorem 1.1.2.

Now, suppose that $\overline{\mathcal{D}}^2 f > 0$ on I. If f is not convex, then we can find a point x_0 such that $\overline{\mathcal{D}}^2 f(x_0) \leq 0$, which will be a contradiction. In fact, in this

case, there exists a subinterval $I_0 = [a_0, b_0]$ such that $f((a_0+b_0)/2) > (f(a_0) + f(b_0))/2$. A moment's reflection shows that one of the intervals $[a_0, (a_0+b_0)/2]$, $[(3a_0+b_0)/4, (a_0+3b_0)/4]$, $[(a_0+b_0)/2, b_0]$ can be chosen to replace I_0 by a smaller interval $I_1 = [a_1, b_1]$, with $b_1 - a_1 = (b_0 - a_0)/2$ and $f((a_1+b_1)/2) > (f(a_1) + f(b_1))/2$. Proceeding by induction, we arrive at a situation where the principle of included intervals gives us the point x_0.

In the general case, consider the sequence of functions

$$f_n(x) = f(x) + \frac{1}{n}\,x^2.$$

Then $\overline{\mathcal{D}}^2 f_n > 0$, and the above reasoning shows us that f_n is convex. Clearly $f_n(x) \to f(x)$ for each $x \in I$, so that the convexity of f is a consequence of Theorem 1.4.4. ∎

1.4.9 Corollary (The second derivative test) *Suppose that $f\colon I \to \mathbb{R}$ is a twice differentiable function defined on an open interval. Then:*

(a) *f is convex if and only if $f'' \geq 0$;*

(b) *f is strictly convex if and only if $f'' \geq 0$ and the set of points where f'' vanishes does not include intervals of positive length.*

The following variant of the second derivative test plays an important role in optimization:

1.4.10 Theorem (A second-order condition of optimization) *If $f\colon I \to \mathbb{R}$ is a twice differentiable function and $f'' \geq C$ for a suitable constant $C > 0$, then every critical point is a global minimizer.*

Theorem 1.4.10 follows easily from Taylor's formula. See [134] p. 235.

The condition $f'' \geq C$ (for $C \in \mathbb{R}$) is equivalent to the convexity of the function $f - \frac{C}{2}x^2$, which in turn is equivalent to strong convexity of f if $C > 0$. See Exercise 5. The case $C < 0$ is that of *semiconvexity* and a first encounter with it will be provided by Proposition 2.2.7 (that estimates the gap in Jensen's inequality).

Exercises

1. (An application of the second derivative test of convexity). Prove that:

   (a) $\frac{x}{e^x-1}$ and $\frac{e^x-1}{x}$ are convex and positive on $\mathbb{R}$;

   (b) the functions $\log((e^{ax}-1)/(e^x-1))$ and $\log(\sinh ax/\sinh x)$ are convex on $\mathbb{R}$ if $a \geq 1$;

   (c) the function $b\log\cos(x/\sqrt{b}) - a\log\cos(x/\sqrt{a})$ is convex on $(0, \pi/2)$ if $b \geq a \geq 1$;

   (d) $\binom{x}{a} = \frac{x(x-1)\cdots(x-a+1)}{a!}$ is a convex function on $[a, \infty)$, whenever a is a positive integer;

   (e) the *running average* $F(x) = \frac{1}{x}\int_0^x f(t)\mathrm{d}t$, of a differentiable convex function $f : (0, \infty) \to \mathbb{R}$ is convex too.

2. Suppose that $0 < a < b < c$ (or $0 < b < c < a$, or $0 < c < b < a$). Prove the following inequalities:

   (a) $ab^p + bc^p + ca^p > ac^p + ba^p + cb^p$ for $p \geq 1$;

   (b) $a^b b^c c^a > a^c c^b b^a$;

   (c) $\dfrac{a(c-b)}{(c+b)(2a+b+c)} + \dfrac{b(a-c)}{(a+c)(a+2b+c)} + \dfrac{c(b-a)}{(b+a)(a+b+2c)} > 0.$

   [*Hint*: Notice that the condition of convexity (1.16) can be reformulated as $(z-y)f(x) - (z-x)f(y) + (y-x)f(z) \geq 0$ for all $x < y < z$ in I. For concave functions, this inequality holds in reversed direction.]

3. (More on Galvani's characterization of convexity) Given a function f defined on an interval I and a point $a \in I$, one can associate to them the *slope* function,

$$s_a \colon I \backslash \{a\} \to \mathbb{R}, \quad s_a(x) = \frac{f(x) - f(a)}{x - a},$$

   whose value at x is the slope of the chord joining the points $(a, f(a))$ and $(x, f(x))$ of the graph of f. Prove that f is convex (respectively, strictly convex) if and only if its associated slope functions s_a are increasing (respectively, strictly increasing).

4. Infer from Exercise 3 that every positive concave function f defined on $[0, \infty)$ is increasing and subadditive.

5. According to B. T. Polyak [480], a function $f : I \to \mathbb{R}$ is called *strongly convex* if there exists a constant $\alpha > 0$ (called *parameter of strong convexity*) such that

$$f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y) - \frac{\alpha}{2}\lambda\,(1-\lambda)\,(x-y)^2 \qquad (1.20)$$

   for all points x and y in I and all $\lambda \in [0, 1]$. Every strongly convex function is strictly convex. The converse fails, as shows the case of the function $f(x) = x^4$, $x \geq 0$. Prove that:

   (a) f is strongly convex (with parameter $\alpha > 0$) if and only if $g = f - (\alpha/2)x^2$ is convex.

   (b) For a strongly convex function (with parameter $\alpha > 0$), Jensen's inequality takes the form

$$f\Big(\sum_{k=1}^{n} \lambda_k x_k\Big) \leq \sum_{k=1}^{n} \lambda_k f(x_k) - \frac{\alpha}{2} \sum_{k=1}^{n} \lambda_k \,(x_k - m)^2\,,$$

   where $m = \sum_{k=1}^{n} \lambda_k x_k$.

   (c) A twice differentiable function f is strongly convex (with parameter $\alpha > 0$) if and only if $f'' \geq \alpha$.

6. Derive Corollary 1.4.7 directly from Remark 1.4.1.

7. (An integral analogue of Corollary 1.4.7) Suppose that $f : [a, b] \to \mathbb{R}$ is a convex function and $[c, d] \subset [a, b]$. Prove that

$$\frac{1}{b-a}\int_a^b f(x)\mathrm{d}x - f\left(\frac{a+b}{2}\right) \geq \frac{1}{d-c}\int_c^d f(x)\mathrm{d}x - f\left(\frac{c+d}{2}\right)$$

and illustrate this result in the case of functions exp and log.

8. (G. Chiti [129]) Suppose that $f : [0, \infty) \to \mathbb{R}$ is an increasing convex function and $x_1 \geq x_2$, $y_1 \geq y_2$ are real points. Prove that

$$f(|x_1 - y_1|) + f(|x_2 - y_2|) \leq f(|x_1 - y_2|) + f(|x_2 - y_1|).$$

[*Hint:* Notice that if $f : \mathbb{R} \to \mathbb{R}$ is a convex function, then $f(x)-f(x-b) \leq f(x+a) - f(x+a-b)$ whenever $x \in \mathbb{R}$ and $a, b \geq 0$.]

9. (H. L. Montgomery) Prove that $1/\sin^2 x < 1/x^2 + 1$ for all $x \in (0, \pi/2]$. This inequality is not sharp. Try to find a better upper bound.

10. Suppose that $(f_n)_n$ is a sequence of differentiable convex functions defined on an open interval I that converges pointwise to a differentiable function f. Prove that $(f_n')_n$ converges uniformly to f' on each compact subinterval of I.

11. (Kantorovich's inequality) Let $m, M, a_1, \ldots, a_n$ be positive numbers, with $m < M$. Prove that the maximum of

$$f(x_1, \ldots, x_n) = \Big(\sum_{k=1}^n a_k x_k\Big)\Big(\sum_{k=1}^n a_k/x_k\Big)$$

for $x_1, \ldots, x_n \in [m, M]$ is equal to

$$\frac{(M+m)^2}{4Mm}\Big(\sum_{k=1}^n a_k\Big)^2 - \frac{(M-m)^2}{4Mm}\min_{X\subset\{1,\ldots,n\}}\Big(\sum_{k\in X} a_k - \sum_{k\in \complement X} a_k\Big)^2.$$

Remark. The following particular case of Kantorovich's inequality,

$$\Big(\frac{1}{n}\sum_{k=1}^n x_k\Big)\Big(\frac{1}{n}\sum_{k=1}^n \frac{1}{x_k}\Big) \leq \frac{(M+m)^2}{4Mm} - \frac{(1+(-1)^{n+1})(M-m)^2}{8Mmn^2},$$

represents an improvement on Schweitzer's inequality for odd n.

12. Let $a_k, b_k, c_k, m_k, M_k, m_k', M_k'$ be positive numbers with $m_k < M_k$ and $m_k' < M_k'$ for $k \in \{1, \ldots, n\}$ and let $p > 1$. Prove that the maximum of

$$\Big(\sum_{k=1}^n a_k x_k^p\Big)\Big(\sum_{k=1}^n b_k y_k^p\Big)\Big/\Big(\sum_{k=1}^n c_k x_k y_k\Big)^p$$

for $x_k \in [m_k, M_k]$ and $y_k \in [m_k', M_k']$ $(k \in \{1, \ldots, n\})$ is attained at a $2n$-tuple whose components are endpoints.

13. Assume that $f : I \to \mathbb{R}$ is strictly convex and continuous and $g\colon I \to \mathbb{R}$ is continuous. For $a_1, \dots, a_n > 0$ and $m_k, M_k \in I$, with $m_k < M_k$ for $k \in \{1, \dots, n\}$, consider the function

$$h(x_1, \dots, x_n) = \sum_{k=1}^{n} a_k f(x_k) + g\Big(\sum_{k=1}^{n} a_k x_k \Big/ \sum_{k=1}^{n} a_k\Big)$$

defined on $\prod_{k=1}^{n} [m_k, M_k]$. Prove that a necessary condition for a point $(y_1, \dots, y_n)$ to be a point of maximum is that at most one component y_k is inside the corresponding interval $[m_k, M_k]$.

1.5 Absolute Continuity of Convex Functions

The fact that differentiation and integration are operations inverse to each other induces a duality between the class of continuous convex functions defined on an interval I and the class of increasing functions on that interval.

Given an increasing function $\varphi\colon I \to \mathbb{R}$ and a point $c \in I$ one can attach to them a new function f, given by

$$f(x) = \int_c^x \varphi(t)\,\mathrm{d}t.$$

As φ is bounded on bounded intervals, it follows that f is locally Lipschitz (and thus continuous). It is also a convex function. In fact, according to Theorem 1.1.4, it suffices to show that f is midpoint convex. In fact, for $x \le y$ in I, we have

$$\frac{f(x)+f(y)}{2} - f\Big(\frac{x+y}{2}\Big) = \frac{1}{2}\left(\int_{(x+y)/2}^{y} \varphi(t)\,\mathrm{d}t - \int_x^{(x+y)/2} \varphi(t)\,\mathrm{d}t\right) \ge 0$$

since φ is increasing.

Notice that f is differentiable at all points of continuity of φ and $f' = \varphi$ at those points.

1.5.1 Remark *There exist convex functions whose first derivative fails to exist on a dense set. For this, let $r_1, r_2, r_3, \dots$ be an enumeration of the rational numbers in $[0, 1]$ and put*

$$\varphi(t) = \sum_{\{k : r_k \le t\}} \frac{1}{2^k}.$$

Then

$$f(x) = \int_0^x \varphi(t)\,\mathrm{d}t$$

is a continuous convex function whose first derivative does not exist at the points r_k. F. Riesz exhibited an example of strictly increasing function φ with $\varphi' = 0$ almost everywhere. See [252], *Example* 18.8, *pp.* 278-282. *The corresponding*

function f in his example is strictly convex though $f'' = 0$ almost everywhere. As we shall see in the Comments at the end of this chapter, Riesz's example is typical from the generic point of view.

Remarkably, every continuous convex function admits an integral representation as above.

1.5.2 Theorem *If $f : I \to \mathbb{R}$ is a continuous convex function, then for all $a < b$ in I we have*

$$f(b) - f(a) = \int_a^b f'_+(t)\,\mathrm{d}t.$$

This formula also works by replacing $f'_+(t)$ with $f'_-(t)$ (as well as with any function φ such that $\varphi(t) \in [f'_-(t), f'_+(t)]$ for $t \in (a, b)$).

Proof. Choose arbitrarily u and v such that $a < u < v < b$. If $u = t_0 < t_1 < \cdots < t_n = v$ is a division of $[u, v]$, then

$$f'_+(t_{k-1}) \leq \frac{f(t_k) - f(t_{k-1})}{t_k - t_{k-1}} \leq f'_+(t_k)$$

for all $k = 1, ..., n$. This yields

$$\sum_{k=1}^{n} f'_+(t_{k-1})\,(t_k - t_{k-1}) \leq f(v) - f(u) = \sum_{k=1}^{n} [f(t_k) - f(t_{k-1})]$$
$$\leq \sum_{k=1}^{n} f'_+(t_k)\,(t_k - t_{k-1})\,,$$

and taking into account that f'_+ is an increasing function we infer that

$$f(v) - f(u) = \int_u^v f'_+(t)\,\mathrm{d}t.$$

Let u decrease to a. Since f is continuous at a and $f'_+(t) \leq f'_+(v)$ on $(a, v]$, an appeal to Beppo Levi's theorem (that is, the monotone convergence theorem for integrals, [134], Theorem 11.3.10, p. 378) gives us the Lebesgue integrability of f'_+ on $[a, v]$ and the formula:

$$f(v) - f(a) = \int_a^v f'_+(t)\,\mathrm{d}t.$$

Then let v increase to b to deduce, via a similar argument, the asserted integral expressions for $f(b) - f(a)$, and the integrability of f'_+ on $[a, b]$. ■

In real analysis, a function $f : [a, b] \to \mathbb{R}$ is called *absolutely continuous* if it is differentiable almost everywhere and f' is Lebesgue integrable on $[a, b]$. Lebesgue has proved that this fact is equivalent to the existence of an integral representation of the form

$$f(x) = f(a) + \int_a^x g(t)\mathrm{d}t$$

for a suitable Lebesgue integrable function g. Thus, Theorem 1.5.2 shows that every continuous convex function is absolutely continuous. One can also prove that every Lipschitz continuous function is absolutely continuous.

The basic theory of absolutely continuous functions can be found in the monographs [134], pp. 495–500 and [252], Chapter V, Section 18.

Exercises

1. Suppose that $f : [a, b] \to \mathbb{R}$ is an absolutely continuous function. Prove that for every $\varepsilon > 0$ there is a $\delta > 0$ such that for every finite family $([a_k, b_k])_{k=1}^n$ of pairwise disjoint subintervals of $[a, b]$ such that $\sum_{k=1}^n (b_k - a_k) < \delta$ we have
$$\sum_{k=1}^n (f(b_k) - f(a_k)) < \varepsilon.$$
Remark. The property mentioned in Exercise 1 actually characterizes the absolute continuity of f. See [134].

2. Prove that every absolutely continuous function is uniformly continuous. The converse is not true. See the case of the function
$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ x \sin(1/x) & \text{if } x \in (0, 1]. \end{cases}$$

3. (Integration by parts for absolutely continuous functions) Suppose that f and g are two absolutely continuous functions on $[a, b]$. Prove that
$$\int_a^b f(t)g'(t)\mathrm{d}t = f(b)g(b) - f(a)g(a) - \int_a^b f'(t)g(t)\mathrm{d}t.$$

1.6 The Hermite–Hadamard Inequality

As stated in Proposition 1.1.3, every convex function f defined on a nondegenerate interval $[a, b]$ can be modified at the endpoints to become convex and continuous. An immediate consequence of this fact is the (Riemann) integrability of f. The arithmetic mean of f can be estimated (both from below and from above) by the *Hermite–Hadamard double inequality*,
$$f\Big(\frac{a+b}{2}\Big) \le \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x \le \frac{f(a)+f(b)}{2}. \tag{1.21}$$

The left-hand side inequality (denoted (LHH)) is nothing but a special case of Jensen's inequality. See Corollary 2.2.4. A very simple argument (which does not use that inequality) is as follows:

$$\frac{1}{b-a}\int_a^b f(x)\mathrm{d}x = \frac{1}{b-a}\left(\int_a^{(a+b)/2} f(x)\mathrm{d}x + \int_{(a+b)/2}^b f(x)\mathrm{d}x\right)$$
$$= \frac{1}{2}\int_0^1 \left[f\left(\frac{a+b-t(b-a)}{2}\right) + f\left(\frac{a+b+t(b-a)}{2}\right)\right]\mathrm{d}t$$
$$\geq f\left(\frac{a+b}{2}\right).$$

Equality occurs if and only if f is an affine function. There are many other ways to establish (LHH), for example, by noticing that

$$y(x) = f\left(\frac{a+b}{2}\right) + f'_+\left(\frac{a+b}{2}\right)\left(x - \frac{a+b}{2}\right)$$

is a supporting line of f at (a+b)/2 and to integrate the inequality $f(x) \geq y(x)$ over $[a,b]$. This is related to the property of subdifferentiability of convex functions and will be detailed in Section 2.1. A third proof of the inequality (LHH), based on the Legendre–Fenchel duality, will make the objective of Remark 2.1.9.

The right-hand side inequality (denoted (RHH)) follows by integrating the inequality (1.2) in Section 1.1 (which says that the graph is under the chord joining the endpoints). As above, the equality case occurs only when f coincides with an affine function on (a,b).

It is worth noticing that each of the two sides of (1.21) actually characterizes convex functions. More precisely, *if I is an interval and $f\colon I \to \mathbb{R}$ is a function continuous on* $\operatorname{int} I$, *whose restriction to each compact subinterval $[a,b]$ verifies (LHH), then f is convex.* The same works when (LHH) is replaced by (RHH). See Exercises 1 and 2 at the end of this section.

Some examples illustrating the usefulness of Hermite–Hadamard inequality are presented below.

1.6.1 Example (Ch. Hermite [251]) *For $f(x) = 1/(1+x)$, $x \geq 0$, the inequality (1.21) becomes*

$$x - \frac{x^2}{2+x} < \log(1+x) < x - \frac{x^2}{2+2x}.$$

Particularly,

$$\frac{1}{n+1/2} < \log(n+1) - \log n < \frac{1}{2}\left(\frac{1}{n} + \frac{1}{n+1}\right) \tag{1.22}$$

for all $n \in \mathbb{N}^*$, and this fact is instrumental in deriving Stirling's formula,

$$n! \sim \sqrt{2\pi}\cdot n^{n+1/2}\mathrm{e}^{-n}.$$

For details, see [134], *p.* 299.

1.6.2 Example *For $f = \exp$, the inequality (1.21) yields*

$$\mathrm{e}^{(a+b)/2} < \frac{\mathrm{e}^b - \mathrm{e}^a}{b-a} < \frac{\mathrm{e}^a + \mathrm{e}^b}{2} \quad \text{for } a \neq b \text{ in } \mathbb{R},$$

that is,

$$\sqrt{xy} < L(x,y) = \frac{x-y}{\log x - \log y} < \frac{x+y}{2} \quad \text{for } x \neq y \text{ in } (0,\infty), \tag{1.23}$$

which represents the geometric–logarithmic–arithmetic mean inequality. For $f = \log$, we obtain a similar inequality, where the role of the logarithmic mean is taken by the identric mean. See Appendix A, Section A.1, for more on these means.

1.6.3 Example *For $f(x) = \sin x$, $x \in [0,\pi]$, we obtain*

$$\frac{\sin a + \sin b}{2} < \frac{\cos a - \cos b}{b-a} < \sin\Big(\frac{a+b}{2}\Big) \quad \text{for } a \neq b$$

and this implies the well-known inequalities $\tan x > x > \sin x$ *(for x in $(0,\pi/2)$).*

The following result yields an estimate of the precision in the Hermite–Hadamard inequality:

1.6.4 Lemma *Let $f\colon [a,b] \to \mathbb{R}$ be a twice differentiable function for which there exist real constants m and M such that*

$$m \le f'' \le M.$$

Then

$$m \cdot \frac{(b-a)^2}{24} \le \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x - f\Big(\frac{a+b}{2}\Big) \le M \cdot \frac{(b-a)^2}{24},$$

and

$$m \cdot \frac{(b-a)^2}{12} \le \frac{f(a)+f(b)}{2} - \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x \le M \cdot \frac{(b-a)^2}{12}.$$

Proof. It suffices to apply the Hermite–Hadamard inequality to the convex functions $f - mx^2/2$ and $Mx^2/2 - f$. ∎

For other estimates of the Hermite–Hadamard inequality see the Comments at the end of this chapter.

1.6.5 Remark (An improvement of the Hermite–Hadamard inequality) *Suppose that $f\colon [a,b] \to \mathbb{R}$ is a convex function. By applying the Hermite–Hadamard inequality on each of the intervals $[a,(a+b)/2]$ and $[(a+b)/2,b]$, we get*

$$f\Big(\frac{3a+b}{4}\Big) \le \frac{2}{b-a}\int_a^{(a+b)/2} f(x)\,\mathrm{d}x \le \frac{1}{2}\Big(f(a) + f\Big(\frac{a+b}{2}\Big)\Big)$$

and

$$f\Big(\frac{a+3b}{4}\Big) \le \frac{2}{b-a}\int_{(a+b)/2}^{b} f(x)\,\mathrm{d}x \le \frac{1}{2}\Big(f\Big(\frac{a+b}{2}\Big)+f(b)\Big).$$

Summing up (side by side), we obtain the following refinement of (1.21) *due to P. C. Hammer* [235]*:*

$$\begin{aligned} f\Big(\frac{a+b}{2}\Big) &\le \frac{1}{2}\Big(f\Big(\frac{3a+b}{4}\Big)+f\Big(\frac{a+3b}{4}\Big)\Big) \\ &\le \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x \le \frac{1}{2}\Big[f\Big(\frac{a+b}{2}\Big)+\frac{f(a)+f(b)}{2}\Big] \\ &\le \frac{1}{2}(f(a)+f(b)). \end{aligned}$$

By continuing the division process, the arithmetic mean of f can be approximated as close as desired by convex combinations of values of f at suitable dyadic points of $[a,b]$. For more inequalities improving the Hermite–Hadamard inequality, see T. Szostok [562].

1.6.6 Remark (The case of log-convex functions) *If $f\colon [a,b]\to(0,\infty)$ is a log-convex function, then the Hermite–Hadamard inequality yields*

$$f\Big(\frac{a+b}{2}\Big) \le \exp\left(\frac{1}{b-a}\int_a^b \log f(x)\,\mathrm{d}x\right) \le \sqrt{f(a)f(b)}.$$

The middle term is the geometric mean of f, so that it can be bounded above by the arithmetic mean of f. As was noticed by P. M. Gill, C. E. M. Pearce, and J. E. Pečarić [220]*, the arithmetic mean of f is bounded from above by the logarithmic mean of $f(a)$ and $f(b)$:*

$$\frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x = \int_0^1 f((1-t)\,a+tb)\,\mathrm{d}t \le \int_0^1 f(a)^{1-t}f\,(b)^t\ \mathrm{d}t = L(f(a),f(b)).$$

The Hermite–Hadamard inequality is an archetype for many interesting results in higher dimensions. Indeed, a careful look at the formula (1.21) shows that by considering the compact interval $K=[a,b]$ endowed with the probability measure $\mu=\frac{1}{b-a}dx$, the integral of any convex function f over K lies between the value of f at the barycenter of μ and the integral of f with respect to the probability measure $\nu=\frac{1}{2}\delta_a+\frac{1}{2}\delta_b$, supported by the boundary $\partial K=\{a,b\}$ of K. This fact extends to all Borel probability measures defined on compact convex subsets of locally convex Hausdorff spaces. The details constitute the core of Choquet's theory and will be presented in Chapter 9. At the same time, the Hermite–Hadamard inequality draws attention to an order relation $\prec$ on the Borel probability measures on K. This order relation, known as the *relation of majorization,* is associated to the cone $\mathrm{Conv}(K)$, of continuous convex functions defined on K, via the formula

$$\lambda_1 \prec \lambda_2 \text{ if and only if } \lambda_1(f) \le \lambda_2(f) \text{ for all } f\in \mathrm{Conv}(K).$$

According to Remark 1.6.5,

$$\delta_{(a+b)/2} \prec \frac{1}{2}\left(\delta_{(3a+b)/4} + \delta_{(a+3b)/4}\right) \prec \frac{1}{b-a}\mathrm{d}x$$
$$\prec \frac{1}{2\,(b-a)}\mathrm{d}x + \frac{1}{4}\delta_a + \frac{1}{4}\delta_b \prec \frac{1}{2}\delta_a + \frac{1}{2}\delta_b.$$

The theory of majorization was created by the contribution of I. Schur [526], G. H. Hardy, J. E. Littlewood and G. Pólya [246], [247] and J. Karamata [287]). Its main features will be presented in Section 6.1.

Exercises

1. Infer from Theorem 1.1.6 that a (necessary and) sufficient condition for a continuous function f to be convex on an open interval I is that

$$f(x) \leq \frac{1}{2h}\int_{x-h}^{x+h} f(t)\,\mathrm{d}t$$

for all x and h with $[x-h, x+h] \subset I$.

2. Let f be a real-valued continuous function defined on an open interval I. Prove that f is convex if all restrictions $f|_{[a,b]}$ (for $[a,b] \subset I$) verify the right-hand side inequality in (1.21).

3. (An improvement of the left-hand side of formula (1.21)) Let f be a convex function defined on the interval $[a,b]$. Use the existence of supporting lines to show that

$$\frac{1}{2}\left(f\left(\frac{a+b}{2} - c\right) + f\left(\frac{a+b}{2} + c\right)\right) \leq \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x$$

for all $c \in [0, (b-a)/4]$, and that $c = (b-a)/4$ is maximal within the class of convex functions on $[a,b]$.

4. (Identities that yield the Hermite–Hadamard inequality). Suppose that $f \in C^2([a,b], \mathbb{R})$. Prove the following two identities related to trapezoidal and midpoint rules of quadrature:

$$\frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x = \frac{1}{2}\left[f(a) + f(b)\right] - \frac{1}{b-a}\int_a^b \frac{(b-x)(x-a)}{2} f''(x)\,\mathrm{d}x$$

and

$$\frac{1}{b-a}\int_a^b f(x)\mathrm{d}x = f\left(\frac{a+b}{2}\right) + \frac{1}{b-a}\int_a^b \varphi(x) f''(x)\mathrm{d}x,$$

where

$$\varphi(x) = \begin{cases} \frac{(x-a)^2}{2} & \text{if } x \in [a, (a+b)/2] \\ \frac{(b-x)^2}{2} & \text{if } x \in [(a+b)/2, b]. \end{cases}$$

Then, use an approximation argument to cover the general case of the Hermite–Hadamard inequality.

5. (Bullen–Simpson inequality [105]). Prove that every function $f \in C^4([a,b])$ with $f^{(4)}(x) \geq 0$ for all x verifies the double inequality

$$0 \leq \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x - \frac{1}{3}\left[2f\left(\frac{3a+b}{4}\right) - f\left(\frac{a+b}{2}\right) + 2f\left(\frac{a+3b}{4}\right)\right]$$
$$\leq \frac{1}{6}\left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b)\right] - \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x.$$

[*Hint:* Use Simpson's quadrature formulas

$$\int_0^1 f(x)\mathrm{d}x = \frac{1}{6}\left[f(0) + 4f\left(\frac{1}{2}\right) + f(1)\right] - \frac{1}{2880}f^{(4)}(\xi)$$

and

$$\int_0^1 f(x)\mathrm{d}x = \frac{1}{3}\left[2f\left(\frac{1}{4}\right) - f\left(\frac{1}{2}\right) + 2f\left(\frac{3}{4}\right)\right] + \frac{7}{23040}f^{(4)}(\eta).\ ]$$

6. (L. Fejér [192]) Suppose that $f\colon [a,b] \to \mathbb{R}$ is a convex function and $w : [a,b] \to [0,\infty)$ is an integrable function such that $W = \int_a^b w(x)\mathrm{d}x > 0$ and $w(\frac{a+b}{2} - x) = w(\frac{a+b}{2} + x)$ for all $x \in [0,(a+b)/2]$. Prove that

$$f\Big(\frac{a+b}{2}\Big) \leq \frac{1}{W}\int_a^b f(x)\,w(x)\mathrm{d}x \leq \frac{f(a)+f(b)}{2},$$

For more general results see Section 9.4 and the papers by A. Florea and C. P. Niculescu [200] and D. S. Marinescu and C. P. Niculescu [361].

7. (S. S. Dragomir, J. Pecarić and L.-E. Persson [166]) Let $P(I)$ denote the set of all real-valued functions f defined on an interval I such that $f((1-\lambda)x + \lambda y)) \leq f(x) + f(y)$ for all $x, y \in I$ and $\lambda \in [0,1]$. Notice that $P(I)$ contains all positive convex functions and also all quasiconvex functions defined on I. Prove that each function $f \in P(I)$ verifies the following variant of the Hermite–Hadamard inequality:

$$f\left(\frac{a+b}{2}\right) \leq \frac{2}{b-a}\int_a^b f(t)\mathrm{d}t \leq 2(f(a)+f(b)),$$

whenever $a, b \in I$, $a < b$, and that the constants $2/(b-a)$ and 2 are best possible.

1.7 Comments

The recognition of convex functions as a class of functions to be studied in its own right generally can be traced back to J. L. W. V. Jensen [274]. However, he was not the first one to deal with convex functions. The discrete form of Jensen's inequality was first proved by O. Hölder [266] in 1889, under the stronger hypothesis that the second derivative is positive. Moreover, O. Stolz [555] proved in 1893 that every midpoint convex and continuous function $f\colon [a,b] \to \mathbb{R}$ has left and right derivatives at each point of (a,b). While the usual convex functions are continuous at all interior points (a fact due to J. L. W. V. Jensen [274]), the midpoint convex functions may be discontinuous everywhere. In fact, regard $\mathbb{R}$ as a vector space over $\mathbb{Q}$ and choose (via the axiom of choice) a basis $(b_i)_{i\in I}$ of $\mathbb{R}$ over $\mathbb{Q}$, that is, a maximal linearly independent set. Then every element x of $\mathbb{R}$ has a unique representation $x = \sum_{i\in I} c_i(x)b_i$ with coefficients $c_i(x)$ in $\mathbb{Q}$ and $c_i(x) = 0$ except for finitely many indices i. The uniqueness of this representation gives rise, for each $i \in I$, to a surjective coordinate projection $\mathrm{pr}_i\colon x \to c_i(x)$, from $\mathbb{R}$ to $\mathbb{Q}$. As G. Hamel [234] observed in 1905, the functions pr_i are discontinuous everywhere and

$$\mathrm{pr}_i(\alpha x + \beta y) = \alpha\,\mathrm{pr}_i(x) + \beta\,\mathrm{pr}_i(y),$$

for all $x, y \in \mathbb{R}$ and all $\alpha, \beta \in \mathbb{Q}$. H. Blumberg [71] and W. Sierpiński [537] have noted independently that if $f\colon (a,b) \to \mathbb{R}$ is measurable and midpoint convex, then f is also continuous (and thus convex). See [504, pp. 220–221] for related results. The complete understanding of midpoint convexity is due to G. Rodé [508], who proved that a real-valued function is midpoint convex if and only if it is the pointwise supremum of a family of functions of the form $a(x)+c$, where $a(x)$ is additive and c is a real constant.

A nice graphical proof of Jensen's inequality can be found at the blog of M. Reid (post of November 17, 2008).

The classical *AM-GM* inequality can be traced back to C. Maclaurin [348] who obtained it as part of a chain of inequalities concerning the elementary symmetric functions. See Appendix C. The famous *Cours d'Analyse* of A.-L. Cauchy includes a proof of this inequality by forward–backward induction. See [124], Théorème 17, page 457. Later, in 1891, A. Hurwitz made the striking remark that the *AM-GM* inequality follows from an identity. Indeed, he proved that

$$\frac{x_1^{2n} + x_2^{2n} + \cdots + x_n^{2n}}{n} - x_1^2 x_2^2 \cdots x_n^2$$

can be expressed as a sum of squares of real polynomials in the real variables $x_1, x_2, ..., x_n$. For example,

$$\frac{x_1^4 + x_2^4 + x_3^4 + x_4^4}{4} - x_1x_2x_3x_4 = \frac{\left(x_1^2 - x_2^2\right)^2 + (x_3^2 - x_4^2)^2}{4} + \frac{(x_1x_2 - x_3x_4)^2}{2}.$$

Can every real multivariate polynomial that takes only positive values over the reals be represented as a sum of squares of real polynomials? The answer

is negative but the first concrete example was produced only in 1966 by T. S. Motzkin [399], who noticed that the polynomial $1 + x^4y^2 + x^2y^4 - 3x^2y^2$ (positive, according to the *AM-GM* inequality) can be only represented as a sum of squares of rational functions:

$$1 + x^4y^2 + x^2y^4 - 3x^2y^2 = \left(\frac{x^2y(x^2+y^2-2)}{x^2+y^2}\right)^2 + \left(\frac{xy^2(x^2+y^2-2)}{x^2+y^2}\right)^2 + \left(\frac{xy(x^2+y^2-2)}{x^2+y^2}\right)^2 + \left(\frac{x^2-y^2}{x^2+y^2}\right)^2.$$

See, for details, J. M. Steele [548], p. 45.

In 1900, D. Hilbert gave an address in Paris to the second International Congress of Mathematicians, describing a list of problems which he believed to be worth the attention of the world's mathematicians. Hilbert's 17th Problem asked whether every real multivariate polynomial that takes only positive values over the reals can be represented as a sum of squares of real rational functions. This was answered affirmatively in 1927 by E. Artin.

The relationship between inequalities and identities is much more subtle than saying that $a \leq b$ is always the consequence of $a + c = b$ for some $c \geq 0$. Indeed, the problem of finding c in a reasonable class of objects related to a and b and to ascribe to c a meaningful interpretation needs more attention. On the other hand, we use inequalities (even when identities are available) because they extract, in a certain context, the useful part of the identities lying behind them. See, for example, the case of Cauchy–Bunyakovsky–Schwarz inequality.

The restriction to strict monotonicity and continuity in Young's inequality is not necessary if we use the concept of generalized inverse. Suppose that $f : [0,\infty) \longrightarrow [0,\infty)$ is an increasing function such that $f(0) = 0$ and $\lim_{x\to\infty} f(x) = \infty$. The *generalized inverse* of f is the function

$$f^{-1} : [0,\infty) \longrightarrow [0,\infty), \quad f^{-1}(y) = \inf\{x \geq 0 : f(x) \geq y\}.$$

Clearly, f^{-1} is increasing, left continuous, and admits a limit from the right at $y > 0$. Moreover $f^{-1}(f(x)) \leq x$ for all x (with equality if f is strictly increasing). A thorough presentation of the theory of generalized inverses may be found in P. Embrechts and M. Hofert [179]. The paper by F.-C. Mitroi and C. P. Niculescu [391] includes several generalizations of Young's inequality in the context of generalized inverses, using some ideas from mass transport theory.

The Rogers–Hölder inequality (known to most mathematicians as the Hölder inequality) was proved in 1888 by L. J. Rogers [509] in a slightly different, but equivalent form. The basic ingredient was his weighted form of the *AM-GM* inequality (as stated in Theorem 1.1.6). One year later, O. Hölder [266] clearly wrote that he, after Rogers, proved the inequality

$$\Big(\sum_{k=1}^{n} a_k b_k\Big)^t \leq \Big(\sum_{k=1}^{n} a_k\Big)^{t-1}\Big(\sum_{k=1}^{n} a_k b_k^t\Big),$$

valid for all $t > 1$, and all $a_k > 0$, $b_k > 0$, $k = 1, \dots, n$, $n \in \mathbb{N}^*$. His idea was to apply Jensen's inequality to the function $f(x) = x^t$, $x > 0$. However, F. Riesz was the first who stated and used the Rogers–Hölder inequality as we did in Section 1.2. See the paper of L. Maligranda [355] for the complete history.

The inequality of J. Bernoulli, $(1+x)^n \geq 1+nx$, for all $x \geq -1$ and n in $\mathbb{N}$, appeared in [55]. The generalized form (see Exercise 1, Section 1.2) is due to O. Stolz and J. A. Gmeiner; see L. Maligranda [355]. L. Maligranda also noticed that the classical Bernoulli inequality, the classical $AM-GM$ inequality, and the generalized $AM-GM$ inequality of Rogers are all equivalent (that is, each one can be used to prove the other ones).

The upper estimate for the discrete form of Jensen's inequality given in Theorem 1.4.5 follows the paper of C. P. Niculescu [418].

The dramatic story of the Hermite–Hadamard inequality is told in a short note by D. S. Mitrinović and I. B. Lacković [387]: In a letter sent on November 22, 1881, to *Mathesis* (and published there in 1883), Ch. Hermite [251] noted that every convex function $f\colon [a,b] \to \mathbb{R}$ satisfies the inequalities

$$f\Big(\frac{a+b}{2}\Big) \leq \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x \leq \frac{f(a)+f(b)}{2}$$

and illustrated this with Example 1.9.1 in our text. Ten years later, the left-hand side inequality was rediscovered by J. Hadamard [233]. However, the priority of Ch. Hermite was not recorded and his note was not even mentioned in Hermite's *Collected Papers* (published by E. Picard). The precision in the Hermite–Hadamard inequality can be estimated via two classical inequalities that work in the Lipschitz function framework. Suppose that $f\colon [a,b] \to \mathbb{R}$ is a Lipschitz function, with *Lipschitz constant*

$$\mathrm{Lip}(f) = \sup\Big\{\Big|\frac{f(x)-f(y)}{x-y}\Big| \;\Big|\; x \neq y\Big\}.$$

Then the left Hermite–Hadamard inequality can be estimated by the *inequality of Ostrowski*,

$$\left|f(x) - \frac{1}{b-a}\int_a^b f(t)\,\mathrm{d}t\right| \leq M\Big[\frac{1}{4} + \Big(\frac{x-\frac{a+b}{2}}{b-a}\Big)^2\Big](b-a),$$

while the right Hermite–Hadamard inequality can be estimated by the *inequality of Iyengar*,

$$\left|\frac{f(a)+f(b)}{2} - \frac{1}{b-a}\int_a^b f(t)\,\mathrm{d}t\right| \leq \frac{M(b-a)}{4} - \frac{1}{4M(b-a)}\,(f(b)-f(a))^2,$$

where $M = \mathrm{Lip}(f)$. The first inequality is a direct consequence of the triangle inequality. The second one makes the objective of Exercise 5, Section 7.1.

Many classical results related to the Hermite–Hadamard inequality can be found in the monograph of J. E. Pečarić, F. Proschan, and Y. C. Tong [463]. The

contributions of Niculescu and his collaborators to this subject are available in [417], [419], [425], [428], [431], and [432].

We end this chapter with a brief discussion on the differentiability properties of convex functions from a generic point of view. Let $\mathcal{P}$ be a property which refers to the elements of a complete metric space X. We say that $\mathcal{P}$ is *generic* (or that most elements of X enjoy $\mathcal{P}$) if those elements not enjoying the property $\mathcal{P}$ form a set of first Baire category, that is, a countable union of nowhere dense sets. The space $C[0,1]$, of all continuous real functions on $[0,1]$, endowed with the usual sup norm, is complete. The same is true for $\mathrm{Conv}([0,1])$, the subset of all continuous convex functions on $[0,1]$. A well-known elegant proof of S. Banach shows that most functions in $C[0,1]$ are nowhere differentiable. The situation in $\mathrm{Conv}([0,1])$ is different. In fact, as noted by V. Klee [299], *most convex functions in* $C[0,1]$ *are differentiable*. The generic aspects of the second differentiability of convex functions are described by T. Zamfirescu [594]: *For most convex functions* $f\colon [0,1] \to \mathbb{R}$,

$$\underline{D}f' = 0 \quad \text{or} \quad \overline{D}f' = \infty \quad \text{everywhere.}$$

Moreover, for most convex functions f*, the second derivative* f'' *vanishes wherever it exists, that is, almost everywhere.* Thus, the situation mentioned in Remark 1.5.1 is rather the rule, not the exception.



Chapter 2

More on Convex Functions on Intervals

In the previous chapter, the convex functions were presented by analyzing their epigraphs. Here we analyze them via a dual characterization based on the existing of supporting lines.

2.1 The Subdifferential

Our definition of a convex function took into consideration the position of the graph relative to the line segment between any two points on the graph. The aim of this section is to present a dual characterization, based on suitable substitutes of tangent lines.

2.1.1 Definition *A function f admits a supporting line at a point $a \in I$ if there exists a real number λ (called the* subgradient *of f at a) such that*

$$f(x) \geq f(a) + \lambda(x - a), \quad \text{for all } x \in I. \tag{2.1}$$

The set of subgradients of f at the point a is called the subdifferential of f at a, and is denoted $\partial f(a)$.

Geometrically, the subdifferential gives us the slope of the supporting lines for the graph of f. See Figure 2.1. The subdifferential at a point is always a convex set, possibly empty.

The convex functions have the remarkable property that $\partial f(x) \neq \emptyset$ at all interior points. However, even in their case, the subdifferential could be empty at the endpoints. An example is given by the continuous convex function $f(x) = 1 - \sqrt{1 - x^2}$, $x \in [-1, 1]$, which fails to have a support line at $x = \pm 1$.

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CMS/CAIMS Books in Mathematics 14, https://doi.org/10.1007/978-3-031-71967-7_2

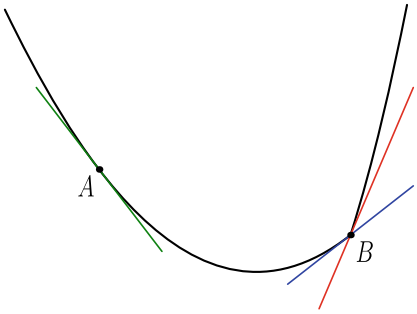


Figure 2.1: Convexity: the existence of supporting lines at interior points.

2.1.2 Theorem *If f is a convex function on an interval I, then $\partial f(a) = \left[f'_-(a), f'_+(a)\right]$ at all interior points a of I. As a consequence,*

$$f(x) \geq f(a) + (x-a)f'_+(a) \quad \text{for all } x \in I \text{ and } a \in \operatorname{int} I.$$

The conclusion above includes the endpoints of I provided that f is differentiable there. As a consequence, the differentiability of a convex function f at a point means that f admits a unique supporting line at that point (Figure 2.2).

Proof. A point λ belongs to $\partial f(a)$ if and only if

$$\frac{f(x)-f(a)}{x-a} \leq \lambda \leq \frac{f(y)-f(a)}{y-a}$$

whenever $x < a < y$ in I; we use the fact that $x - a < 0$. According to the inequality of three chords, the first ratio increases to $f'_-(a)$ as x increases to a, while the second ratio decreases to $f'_+(a)$ as y decreases to a. Therefore λ belongs to $\partial f(a)$ if and only if $\lambda \in \left[f'_-(a), f'_+(a)\right]$. ∎

Many elementary inequalities like $\mathrm{e}^x \geq 1 + x$ (on $\mathbb{R}$) and $\sin x \leq x$ (on $\mathbb{R}_+$) are just illustrations of Theorem 2.1.2.

Combining Theorems 1.4.2 and 2.1.2 we obtain the following result:

2.1.3 Corollary *If $f : I \to \mathbb{R}$ is a convex function, then every function φ from I to $\mathbb{R}$ that verifies the condition $\varphi(x) \in \partial f(x)$ whenever $x \in \operatorname{int} I$ is increasing on* int *I.*

2.1.4 Remark *The convex functions are the only functions f that satisfy the condition $\partial f(x) \neq \emptyset$ at all interior points of the interval of definition.*

To prove this, let $u, v \in I$ with $u \neq v$ and $t \in (0,1)$. Then $(1-t)u+tv \in \operatorname{int} I$, so that for all $\lambda \in \partial f((1-t)u+tv)$ we get

$$f(u) \geq f((1-t)u+tv) + t(u-v)\cdot\lambda,$$
$$f(v) \geq f((1-t)u+tv) - (1-t)(u-v)\cdot\lambda.$$

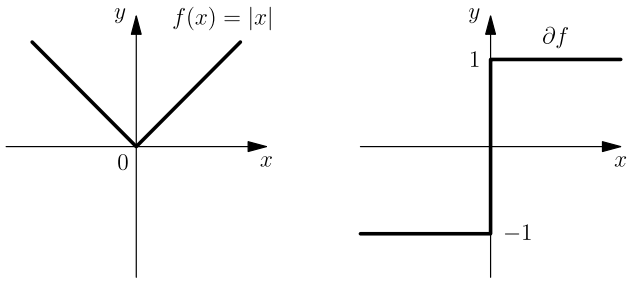


Figure 2.2: The absolute value function and its subdifferential.

By multiplying the first inequality by $1-t$*, the second one by* t*, and then adding them side by side, we get* $(1-t)f(u)+tf(v) \geq f((1-t)u+tv)$*, hence* f *is a convex function.*

Every continuous convex function is the upper envelope of its supporting lines:

2.1.5 Theorem *Suppose that* f *is a continuous convex function on a nondegenerate interval* I *and* $\varphi\colon I \to \mathbb{R}$ *is a function such that* $\varphi(x)$ *belongs to* $\partial f(x)$ *for all* $x \in \operatorname{int} I$*. Then*

$$f(z) = \sup\{f(x) + (z-x)\varphi(x) : x \in \operatorname{int} I\} \quad \textit{for all } z \in I.$$

Proof. The case of interior points follows from Theorem 2.1.2. Suppose now that z is an endpoint, say the left one. We already noticed that

$$f(z+t) - f(z) \leq t\varphi(z+t) \leq f(z+2t) - f(z+t)$$

for $t > 0$ small enough, which yields $\lim_{t\to 0+} t\varphi(z+t) = 0$. Given $\varepsilon > 0$, there is $\delta > 0$ such that $|f(z) - f(z+t)| < \varepsilon/2$ and $|t\varphi(z+t)| < \varepsilon/2$ for $0 < t < \delta$. This shows that $f(z+t) - t\varphi(z+t) > f(z) + \varepsilon$ for $0 < t < \delta$ and the result follows. ∎

The fact that convex functions admit supporting lines at the interior points has many important consequences. In what follows, we will discuss the Legendre–Fenchel duality of convex functions from $\mathbb{R}$ to $\mathbb{R} \cup \{\infty\}$.

2.1.6 Definition *A function* $f : \mathbb{R} \to \mathbb{R} \cup \{\infty\}$ *is called convex if*

$$f((1-\lambda)x + \lambda y) \leq (1-\lambda)f(x) + \lambda f(y)$$

for all $x, y \in \mathbb{R}$ *and all* $\lambda \in (0,1)$*. The function* f *is called strictly convex if this inequality is strict for* $x \neq y$ *and* $\lambda \in (0,1)$*.*

The function f *is a proper function if its domain* $\operatorname{dom} f = \{x : f(x) \in \mathbb{R}\}$ *is nonempty.*

Every real-valued convex function defined on an interval $I \neq \mathbb{R}$ can be turned into a proper convex function in the sense of Definition 2.1.6 by extending it with ∞ outside I. The domain of every proper convex function $g : \mathbb{R} \to \mathbb{R} \cup \{\infty\}$ is a nonempty interval and the restriction g to this set is a usual convex function.

Given a proper convex function $f : \mathbb{R} \to \mathbb{R} \cup \{\infty\}$, one defines its *convex conjugate* (or *Legendre–Fenchel conjugate*) as

$$f^* : \mathbb{R} \to \mathbb{R} \cup \{\infty\}, \quad f^*(y) = \sup\{xy - f(x) : x \in \mathbb{R}\}.$$

Notice that f^* is a *proper convex function*. Indeed, if $\operatorname{dom} f$ is a singleton, say $\{x_0\}$, then $f^*(y) = x_0 y - f(x_0)$ for every $y \in \mathbb{R}$. If $\operatorname{dom} f$ is a nondegenerate interval, then for each $a \in \operatorname{int}(\operatorname{dom} f)$ there is an $y \in \mathbb{R}$ such that $f(x) \geq f(a) + y(x - a)$, which yields

$$xy - f(x) \leq ay - f(a),$$

so, $y \in \operatorname{dom} f^*$. Thus, $\operatorname{dom} f^*$ includes $\bigcup_{x \in \mathbb{R}} \partial f(x)$.

If $\lambda \in (0, 1)$ and $y, z \in \operatorname{dom} f^*$, then

$$\begin{aligned} f^*((1-\lambda)y + \lambda z) &= \sup\{x\,[(1-\lambda)y + \lambda z] - f(x) : x \in I\} \\ &\leq (1-\lambda)\sup\{xy - f(x) : x \in I\} + \lambda \sup\{xz - f(x) : x \in I\} \\ &= (1-\lambda)f^*(y) + \lambda f^*(z) \end{aligned}$$

and the convexity of f^* follows.

The function f^* is also *lower semicontinuous* in the sense that all its *level sets* $L(\lambda) = \{y : f^*(y) \leq \lambda\}$ are closed sets. Since f^* is the upper envelope of a family of continuous functions,

$$L(\lambda) = \{y : f^*(y) \leq \lambda\} = \bigcap\nolimits_{x \in \mathbb{R}} \{y : xy - f(x) \leq \lambda\},$$

and an intersection of closed sets is also a closed set.

An immediate consequence of the definition of convex conjugation is the *Fenchel–Young inequality*,

$$xy \leq f(x) + f^*(y) \quad \text{for all } x, y \in \mathbb{R}.$$

This inequality extends the previous form of Young's inequality stated as equation (1.5). Indeed, the convex conjugate of the function $f(x) = |x|^p / p$, $x \in \mathbb{R}$ $(p > 1)$ is the function $f^*(y) = |y|^q / q$, $y \in \mathbb{R}$ $(1/p + 1/q = 1)$.

2.1.7 Lemma *Suppose that $f : \mathbb{R} \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper convex function. Then, for all $x, y \in \mathbb{R}$,*

$$xy = f(x) + f^*(y) \text{ if, and only if, } y \in \partial f(x).$$

Proof. Indeed, $xy = f(x) + f^*(y)$ if and only if $xy \geq f(x) + f^*(y)$, which in turn is equivalent to $xy \geq f(x) + yz - f(z)$ for all $z \in \mathbb{R}$. The latter inequality

can be rewritten as $f(z) \geq f(x) + y(z-x)$ for all $z \in \mathbb{R}$, which means that $y \in \partial f(x)$. ■

In general

$$f \leq g \text{ implies } g^* \leq f^*,$$

but for lower semicontinuous proper convex functions this becomes an equivalence.

2.1.8 Theorem (Legendre–Fenchel duality for lower semicontinuous proper convex functions) *Let $f : \mathbb{R} \to \mathbb{R} \cup \{\infty\}$ be a proper function. The following assertions are equivalent:*

(a) f is convex and lower semicontinuous;

(b) $f = g^*$ for some proper function g;

(c) $f^{**} = f$.

We will come back to this result in Section 8.1, where its several-variable variants are presented with full details.

2.1.9 Remark *It is worth mentioning that the left-hand side of the Hermite–Hadamard inequality can be easily obtained by using the Legendre–Fenchel duality. An outline of the proof is as follows:*

$$\begin{aligned} f\left(\frac{a+b}{2}\right) &= f\left(\frac{1}{b-a}\int_a^b t\mathrm{d}t\right) \\ &= \sup_x \left[\frac{x}{b-a}\int_a^b t\mathrm{d}t - f^*(x)\right] \\ &= \sup_x \left[\frac{1}{b-a}\int_a^b (xt - f^*(x))\,\mathrm{d}t\right] \\ &\leq \frac{1}{b-a}\int_a^b \sup_x (xt - f^*(x))\,\mathrm{d}t \\ &= \frac{1}{b-a}\int_a^b f(t)\mathrm{d}t. \end{aligned}$$

2.1.10 Remark (Orlicz spaces) *An* Orlicz function *is any convex function $\Phi : [0,\infty) \to \mathbb{R}$ such that:*

(Φ1) $\Phi(0) = 0$, $\Phi(x) > 0$ for $x > 0$;

(Φ2) $\Phi(x)/x \to 0$ as $x \to 0$ and $\Phi(x)/x \to \infty$ as $x \to \infty$;

(Φ3) there exists a positive constant K such that $\Phi(2x) \leq K\Phi(x)$ for every $x \geq 0$.

Let (X, Σ, μ) be a complete σ-finite measure space and let $S(\mu)$ be the vector space of all equivalence classes of μ-measurable real-valued functions defined on X. The Orlicz space $L^\Phi(X)$ *is the subspace of all $f \in S(\mu)$ such that*

$$I_\Phi(f/\lambda) = \int_X \Phi(|f(x)|/\lambda)\,\mathrm{d}\mu < \infty$$

for some $\lambda > 0$. One can prove that $L^\Phi(X)$ is a linear space such that

$$|f| \le |g| \text{ and } g \in L^\Phi(X) \text{ implies } f \in L^\Phi(X).$$

Moreover, it constitutes a Banach space when endowed with the norm

$$\|f\|_\Phi = \inf\{\lambda > 0 \,|\, I_\Phi(f/\lambda) \le 1\}$$

and the dual space of $L^\Phi(X)$ is $L^\Psi(X)$, where Ψ is the conjugate of Φ.

The Orlicz spaces extend the $L^p(\mu)$ spaces. Their theory is exposed in books like that of M. A. Krasnosel'skii and Ya. B. Rutickii [307]. *The Orlicz space $\mathcal{L}\log^+\mathcal{L}$ (corresponding to the Lebesgue measure on $[0,\infty)$ and the function $\Phi(t) = t\,(\log t)^+$) plays an important role in Fourier Analysis. See, for example, the book of A. Zygmund* [604]. *Applications to the interpolation theory are given by C. Bennett and R. Sharpley* [51].

Exercises

1. Let $f\colon I \to \mathbb{R}$ be a convex function. Show that:
   (a) any local minimum of f is a global one;
   (b) f attains a global minimum at a if and only if $0 \in \partial f(a)$;
   (c) if f has a global maximum at an interior point of I, then f is constant.

2. (Convex mean value theorem) Consider a continuous convex function $f\colon [a,b] \to \mathbb{R}$. Prove that $(f(b) - f(a))/(b-a) \in \partial f(c)$ for some point $c \in (a,b)$.

3. Prove that a function $f : \mathbb{R} \to \mathbb{R} \cup \{\infty\}$ is lower semicontinuous if and only if $x_n \to x$ in $\mathbb{R}$ implies $f(x) \le \liminf_{n\to\infty} f(x_n)$. Infer that the extension $\widetilde{f}$ of a convex function $f : [a,b) \to \mathbb{R}$ with ∞ outside $[a,b)$ is lower semicontinuous if and only if f is continuous at a and $\lim_{x\to b} f(x) = \infty$.

4. Compute ∂f, ∂f^* and f^* for $f(x) = |x|$, $x \in \mathbb{R}$.

5. Prove that:
   (*a*) the conjugate of $f(x) = x^p/p$, $x \ge 0$, is $f^*(y) = (y^+)^q/q$, $y \in \mathbb{R}$ ($p > 1$, $1/p + 1/q = 1$);
   (*b*) the conjugate of $f(x) = \mathrm{e}^x$, $x \in \mathbb{R}$, is the function $f^*(y) = y\log y - y$ for $y > 0$ and $f^*(0) = 0$;
   (*c*) the conjugate of $f(x) = -\log x$, $x > 0$, is the function $f^*(y) = -1 - \log(-y)$, $y < 0$.

6. Prove that the function $f(x) = \frac{1}{2}x^2$ is the unique function $g : \mathbb{R} \to \mathbb{R}$ such that $g = g^*$.
   [*Hint:* Use Young's inequality.]

7. (An alternative proof of Theorem 1.2.1) Suppose that $f\colon [0,\infty) \to [0,\infty)$ is a strictly increasing continuous function such that $f(0) = 0$ and $\lim_{x\to\infty} f(x) = \infty$. Prove that the function $\Phi(u) = \int_0^u f(x)\,\mathrm{d}x$, $u \geq 0$, is strictly convex and differentiable and the convex conjugate of Φ is the function $\Psi(v) = \int_0^v f^{-1}(x)\,\mathrm{d}x$, $v \geq 0$. Then, infer the conclusion of Theorem 1.2.1,
$$uv \leq \int_0^u f(x)\,\mathrm{d}x + \int_0^v f^{-1}(x)\,\mathrm{d}x$$
for all $u, v \geq 0$, with equality if and only if $v = f(u)$.

8. (Reshaping convex functions) Suppose that $f : I \to \mathbb{R}$ is a convex function and $a \in \operatorname{int} I$. Prove that the function obtained from f by replacing its restriction to $I \cap (-\infty, a]$ with the semi-tangent line to the left, that is, with $f(a) + f'_-(a)(x-a)$, is also convex.

2.2 The Integral Form of Jensen's Inequality

The aim of this section is to extend Jensen's discrete inequality to the general framework of finite measure spaces. Recall that a *finite measure space* is any triplet (Ω, Σ, μ) consisting of an abstract nonempty set Ω, a σ-algebra Σ of subsets of Ω, and a σ-additive measure $\mu : \Sigma \to \mathbb{R}_+$ such that $\mu(\Omega) > 0$. By replacing μ by $\mu/\mu(\Omega)$, one can reduce the study of finite measure spaces to that of *probability spaces*, characterized by the fact that $\mu(\Omega) = 1$.

2.2.1 Remark *The language of measure-theoretic probability theory is slightly different from that of traditional measure theory. Most books on probability theory use the notation* (Ω, Σ, P) *for a probability space consisting of a* sample space Ω, *a* σ*-algebra* Σ *of* events (*viewed as subsets of* Ω) *and a probability measure* $P : \Sigma \to \mathbb{R}_+$. *In the probabilistic context, the real measurable functions* $X : \Omega \to \mathbb{R}$ *are called* random variables. *Their definitory property is that* $X^{-1}(A) \in \Sigma$ *for every Borel subset* A *of* $\mathbb{R}$. *An important point in probability theory is the use of the notion of independence* (*both for events and random variables*).

Jensen's inequality relates two important concepts that can be attached to a finite measure space (Ω, Σ, μ): the integral arithmetic mean and the barycenter. As will be shown below, another probabilistic concept, that of variance, provides an efficient way for evaluating the gap in Jensen's inequality.

The *integral arithmetic mean* (or the *mean value*) of a μ-integrable function $f\colon \Omega \to \mathbb{R}$ is defined by the formula:
$$M_1(f) = \frac{1}{\mu(\Omega)} \int_\Omega f(x)\,\mathrm{d}\mu(x).$$

In the context of probability theory, this number is also called the *expectation* (or *expected value*) of f and is denoted $E(f)$. The expectation generates a

functional $E : L^1(\mu) \to \mathbb{R}$ having the following three properties:

- linearity, $E(\alpha f + \beta g) = \alpha E(f) + \beta E(g)$;
- positivity, $f \geq 0$ implies $E(f) \geq 0$;
- calibration, $E(1) = 1$.

A very convenient way to introduce the expectation of a real-valued function $f \in L^1(\mu)$ is provided by the Riemann–Stieltjes integral. The key ingredient is the *cumulative distribution function* of f, which is defined by the formula:

$$F : \mathbb{R} \to [0,1], \quad F(x) = \mu\left(\{\omega : f(\omega) \leq x\}\right);$$

on several occasions we will use the notation F_f instead of F, to specify the integrable function under attention.

This function is increasing and its limits at infinity are $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$. Also, the cumulative distribution function is right continuous, that is,

$$\lim_{x\to x_0+} F(x) = F(x_0) \text{ at every } x_0 \in \mathbb{R}.$$

The distribution function allows us to introduce the expectation of a random variable by a formula that avoids the use of measure theory:

$$E(f) = \int_{-\infty}^{\infty} x \mathrm{d}F_f(x).$$

For convenience, the concept of barycenter will be introduced in the context of probability measures μ defined on the σ-algebra $\mathcal{B}(I)$ of Borel subsets of an interval I (the so-called Borel probability measures on the interval I). Precisely, we will consider the class

$$\mathcal{P}^1(I) = \left\{\mu : \mu \text{ Borel probability measure on } I \text{ and } \int_I |x| \, \mathrm{d}\mu(x) < \infty\right\}.$$

This class includes all Borel probability measures null outside a bounded subinterval.

2.2.2 Definition *The barycenter (or resultant) of a Borel probability measure $\mu \in \mathcal{P}^1(I)$ is the real point*

$$\mathrm{bar}(\mu) = \int_I x \mathrm{d}\mu(x).$$

Necessarily, when the barycenter $\mathrm{bar}(\mu)$ exists, it must be in I. Indeed, if $\mathrm{bar}(\mu) \notin I$, then either $\mathrm{bar}(\mu)$ is an upper bound for I or it is a lower bound. Since

$$\int_I [x - \mathrm{bar}(\mu)] \, \mathrm{d}\mu(x) = 0,$$

this situation will impose $x - \mathrm{bar}(\mu) = 0$ μ-almost everywhere, which is not possible because $\mu(I) = 1$.

A first example illustrating Definition 2.2.2 concerns the case of a discrete probability measure $\lambda = \sum_{k=1}^{n} \lambda_k \delta_{x_k}$ concentrated at the points $x_1, ..., x_n \in \mathbb{R}$. Here δ_z represents the Dirac measure concentrated at z, that is, the measure given by

$$\delta_z(A) = 1 \text{ if } z \in A \text{ and } \delta_z(A) = 0 \text{ otherwise.}$$

In this case,

$$E(f) = \int_{\mathbb{R}} f(x) \mathrm{d}\lambda(x) = \sum_{k=1}^{n} \lambda_k f(x_k)$$

for every function $f : \mathbb{R} \to \mathbb{R}$. Therefore

$$\mathrm{bar}(\lambda) = \int_{\mathbb{R}} x \mathrm{d}\lambda(x) = \sum_{k=1}^{n} \lambda_k x_k.$$

The barycenter of the restriction of Lebesgue measure to an interval $[a, b]$ is the middle point because

$$\frac{1}{b-a} \int_a^b x \mathrm{d}x = \frac{a+b}{2}.$$

The barycenter of the Gaussian probability measure $\frac{1}{\sqrt{2\pi}} e^{-x^2/2} \mathrm{d}x$ is the origin. Indeed,

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x e^{-x^2/2} \mathrm{d}x = 0.$$

2.2.3 Theorem (The integral form of Jensen's inequality) *Let (Ω, Σ, μ) be a probability space and let $g \colon \Omega \to \mathbb{R}$ be a μ-integrable function. If f is a convex function defined on an interval I that includes the image of g, then $E(g) \in I$ and*

$$f(E(g)) \leq \int_{\Omega} f(g(x)) \mathrm{d}\mu(x).$$

Notice that the right-hand side integral always exists but it might be ∞ if the μ-integrability of $f \circ g$ is not expressly asked. When both functions g and $f \circ g$ are μ-integrable, then the above inequality becomes

$$f(E(g)) \leq E(f \circ g).$$

Moreover, if in addition f is strictly convex, then this inequality becomes an equality if and only if g is constant μ-almost everywhere.

In order to avoid unnecessary repetitions, the details are postponed till Section 4.6, where the more general case of several variables will be presented.

2.2.4 Corollary *If $\mu \in \mathcal{P}^1(I)$ and $f : I \to \mathbb{R}$ is a μ-integrable convex function, then*

$$f(\mathrm{bar}(\mu)) \leq \int_I f(x) \mathrm{d}\mu(x).$$

Proof. Apply Theorem 2.2.3 for g the identity of the interval I. An alternative proof can be obtained by adapting the argument of Remark 2.1.9. ∎

2.2.5 Corollary (The Hermite–Hadamard inequality for arbitrary Borel probability measures; A. M. Fink [197]) *Suppose that $f\colon [a,b] \to \mathbb{R}$ is a convex function and μ is Borel probability measure on $[a,b]$. Then*

$$f(\operatorname{bar}(\mu)) \le \int_a^b f(x)\,\mathrm{d}\mu(x) \le \frac{b-\operatorname{bar}(\mu)}{b-a}\cdot f(a) + \frac{\operatorname{bar}(\mu)-a}{b-a}\cdot f(b).$$

Proof. The left-hand side follows from Corollary 2.2.4. For the right-hand side, proceed as in the classical case of Lebesgue measure. ∎

2.2.6 Remark (*Differential entropy*) *As above, we assume that (Ω,Σ,P) is a probability space. By applying Jensen's inequality to the convex function $f(x) = x\log x$ in $[0,\infty)$ we obtain the following upper estimate for the differential entropy $h(g) = -\int_\Omega g\log g\,d\mu$ of a positive μ-integrable function g:*

$$h(g) \le -\left(\int_\Omega g\,\mathrm{d}\mu\right)\log\left(\int_\Omega g\,\mathrm{d}\mu\right).$$

If $\int_\Omega g\,d\mu = 1$, then $h(g) \le 0$ even though the function f has a variable sign (it attains the minimum value $-1/\mathrm{e}$ at $z = 1/\mathrm{e}$).

A natural question is how large is the *Jensen gap*

$$E(f\circ g) - f(E(g)).$$

The following result noticed by O. Hölder [266] provides the answer in an important special case.

2.2.7 Proposition *Suppose that $f\colon [a,b] \to \mathbb{R}$ is a twice differentiable function for which there exist real constants m and M such that $m \le f'' \le M$. Then*

$$\begin{aligned}\frac{m}{2}\sum_{1\le i<j\le n}\lambda_i\lambda_j\,(x_i-x_j)^2 &\le \sum_{k=1}^n \lambda_k f(x_k) - f\Big(\sum_{k=1}^n \lambda_k x_k\Big)\\ &\le \frac{M}{2}\sum_{1\le i<j\le n}\lambda_i\lambda_j\,(x_i-x_j)^2,\end{aligned} \tag{2.2}$$

whenever $x_1,\dots,x_n \in [a,b]$, $\lambda_1,\dots,\lambda_n \in [0,1]$ and $\sum_{k=1}^n \lambda_k = 1$.

Proof. This is a consequence of the discrete Jensen's inequality when applied to the convex functions $f - mx^2/2$ and $Mx^2/2 - f$. ∎

2.2.8 Corollary (The gap in the *AM-GM* inequality) *If* $0 < m \leq x_1, ..., x_n \leq M$, $\lambda_1, \ldots, \lambda_n \in [0,1]$ *and* $\sum_{k=1}^n \lambda_k = 1$, *then*

$$\frac{m}{2n^2} \sum_{1\leq j<k\leq n} (\log x_j - \log x_k)^2 \leq \frac{1}{n}\sum_{k=1}^n x_k - \left(\prod_{k=1}^n x_k\right)^{1/n} \leq \frac{M}{2n^2} \sum_{1\leq j<k\leq n} (\log x_j - \log x_k)^2.$$

Proposition 2.2.7 exhibits the role of the variance in estimating the precision in Jensen's inequality.

If (Ω, Σ, μ) is a probability space, the *variance* of a function $g \in L^2(\mu)$ is defined by the formula:

$$\begin{aligned} \operatorname{var}(g) &= E\left(\left(g - E(g)\right)^2\right) \\ &= E\left(g^2\right) - \left(E(g)\right)^2. \end{aligned} \tag{2.3}$$

The variance is an indicator of how much the values of g are spread out. A variance of zero indicates that all the values of g are identical (except possibly for a subset of Ω of probability zero).

The square root of the variance is called the *standard deviation*.

Since a probability measure is a finite measure, the space $L^2(\mu)$ is included in $L^1(\mu)$. Thus expectation and variance applies to every function that belongs to $L^2(\mu)$.

Let us consider the discrete probability space $\Omega = \{1, ..., n\}$, $\Sigma = \mathcal{P}(\Omega)$ and $\mu = \sum_{k=1}^n \lambda_k \delta_k$. The variance of the function $g : \{1, ..., n\} \to \mathbb{R}$ defined by $g(k) = x_k$ for $k = 1, ..., n$, is

$$\begin{aligned} \operatorname{var}(g) &= E\big((g - E(g))^2\big) \\ &= E\left(g^2\right) - \left(E(g)\right)^2 \\ &= \sum_{n=1}^n \lambda_n x_n^2 - \left(\sum_{n=1}^n \lambda_n x_n\right)^2 \\ &= \frac{1}{2} \sum_{1\leq i,j\leq n} \lambda_i \lambda_j \left(x_i - x_j\right)^2 = \sum_{1\leq i<j\leq n} \lambda_i \lambda_j \left(x_i - x_j\right)^2, \end{aligned}$$

and thus the result of Proposition 2.2.7 can be reformulated as

$$\frac{m}{2} \operatorname{var}(g) \leq E(f(g)) - f\left(E\left(g\right)\right) \leq \frac{M}{2} \operatorname{var}(g).$$

It is worth noticing that this double estimate works in general. See Exercise 5.

In probability theory and statistics, an important role is played by the so-called continuous random variables. A random variable X (attached to a probability measure space (Ω, Σ, P)) is called *continuous* if its cumulative distribution function is of the form

$$F_X(x) = P\left(\{\omega : X(\omega) < x\}\right) = \int_{-\infty}^x w(t)\, \mathrm{d}t,$$

for a suitable Lebesgue integrable function $w \in L^1(\mathbb{R})$, called the *density* of F_X. In this case, the probability that X takes a value α is 0 and

$$P\left(\{\omega : a \leq X(\omega) \leq b\}\right) = \int_a^b w(t)\,\mathrm{d}t \quad \text{for all } -\infty \leq a \leq b \leq \infty.$$

Moreover, the computation of the expectation and of the variance of X reduces to the computation of certain Lebesgue integrals:

$$E(X) = \int_{-\infty}^{\infty} xw(x)\mathrm{d}x \text{ and } \operatorname{var}(X) = \int_{-\infty}^{\infty} (x - E(X))^2\, w(x)\mathrm{d}x.$$

A continuous random variable X is called *normal* if its distribution function is associated to a density of the form:

$$w(t, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}.$$

In this case, the values of the parameters $\mu \in \mathbb{R}$ and $\sigma > 0$ are precisely the expectation and the standard deviation of X.

2.2.9 Remark (Upper bounds on the variance) *As was noticed by D. S. Mitrinović, J. E. Pečarić and A. M. Fink in* [389]*, p.* 296*, if X is a random variable such that $\alpha \leq X \leq \beta$ for two suitable constants α and β, then*

$$\operatorname{var}(X) \leq (\beta - E(X))\,(E(X) - \alpha)\,.$$

This remark improves the upper bound previously indicated by T. Popoviciu [484]*,*

$$\operatorname{var}(X) \leq \frac{(\beta - \alpha)^2}{4}.$$

The bound found by Mitrinović, Pečarić and Fink follows easily from the general properties of expectation:

$$\begin{aligned} 0 &\leq E\left((\beta - X)\,(X - \alpha)\right) = -\alpha\beta + (\alpha + \beta)\,E(X) - E\left(X^2\right) \\ &= (\beta - E(X))\,(E(X) - \alpha) - \operatorname{var}(X). \end{aligned}$$

2.2.10 Remark (Chebyshev's probabilistic inequality) *If X is a random variable (associated to a probability measure space (Ω, Σ, μ)), then*

$$\mu\left(\{|X - E(X)| \geq \varepsilon\}\right) \leq \frac{\operatorname{var}(X)}{\varepsilon^2}$$

for all $\varepsilon > 0$.

The *covariance* of two real random variables $X, Y \in L^2(P)$ is defined by

$$\begin{aligned} \operatorname{cov}(X, Y) &= E\left((X - E(X))\,(Y - E(Y))\right) \\ &= E\left(XY\right) - E(X)E(Y). \end{aligned}$$

An instance when the covariance is positive is mentioned in Exercise 8.

Two random variables X and Y whose covariance is zero are called *uncorrelated*. If X and Y are independent, then their covariance is zero. This follows because under independence, $E(XY) = E(X)E(Y)$. However, uncorrelation does not imply in general independence.

The concept of covariance allows us to indicate a new upper estimate of the Jensen gap.

2.2.11 Theorem (The covariance form of Jensen's inequality) *Let (Ω, Σ, μ) be a probability space and let $g\colon \Omega \to \mathbb{R}$ be a μ-integrable function. Suppose also that f is a convex function defined on an interval I that includes the image of g and $\varphi\colon I \to \mathbb{R}$ is a function such that*

(a) $\varphi(x) \in \partial f(x)$ for every $x \in I$ and

(b) $\varphi \circ g$ and $g \cdot (\varphi \circ g)$ are μ-integrable functions.

Then

$$0 \leq E(f \circ g) - f(E(g)) \leq \operatorname{cov}(g, \varphi \circ g).$$

If f is concave, then the last two inequalities work in the reversed direction.

Proof. The first inequality is motivated by Theorem 2.2.3. The second inequality follows by integrating the inequality

$$f(E(g)) \geq f(g(x)) + (E(g) - g(x)) \cdot \varphi(g(x)) \quad \text{for all } x \in \Omega.$$

■

2.2.12 Corollary (S. S. Dragomir and N. M. Ionescu [164]) *If f is a differentiable convex function defined on an open interval I, then*

$$\begin{aligned} 0 &\leq \sum_{k=1}^{n} \lambda_k f(x_k) - f\Big(\sum_{k=1}^{n} \lambda_k x_k\Big) \\ &\leq \sum_{k=1}^{n} \lambda_k x_k f'(x_k) - \Big(\sum_{k=1}^{n} \lambda_k x_k\Big)\Big(\sum_{k=1}^{n} \lambda_k f'(x_k)\Big) \end{aligned}$$

for all $x_1, \ldots, x_n \in I$ and all $\lambda_1, \ldots, \lambda_n \in [0,1]$, with $\sum_{k=1}^{n} \lambda_k = 1$.

The covariance defines a Hermitian product; it differs from a scalar product by the fact that $\operatorname{cov}(X, X) = 0$ implies only that X is constant almost everywhere. Since the Cauchy–Bunyakovsky–Schwarz inequality still works for such products, we have the inequality

$$|\operatorname{cov}(X, Y)| \leq (\operatorname{var}(X))^{1/2} (\operatorname{var}(Y))^{1/2} \tag{2.4}$$

known as the *covariance form of Cauchy–Bunyakovsky–Schwarz inequality*. This inequality shows that *Pearson's correlation coefficient* of X and Y, that is,

$$\rho_{X,Y} = \frac{\operatorname{cov}(X, Y)}{(\operatorname{var}(X))^{1/2} (\operatorname{var}(Y))^{1/2}}$$

takes values in the interval $[-1, 1]$. A value of 1 (respectively -1) for $\rho_{X,Y}$ implies that the relationship between X and Y is described by a linear equation, for which Y increases (respectively decreases) as X increases. A value of 0 implies that there is no linear correlation between the two random variables.

From the covariance form of Cauchy–Bunyakovsky–Schwarz inequality and Remark 2.2.9, we infer the following classical result:

2.2.13 Theorem (Grüss Inequality [231]) *Suppose that the random variables X and Y are bounded, precisely, $\alpha \le X \le \beta$ and $\delta \le Y \le \gamma$. Then*

$$|\operatorname{cov}(X, Y)| \le \frac{1}{4}(\beta - \alpha)(\gamma - \delta)$$

and the constant $1/4$ being sharp.

Exercises

1. (The power means; see Exercise 12 in Section 1.1, for the discrete case) Consider a finite measure space (Ω, Σ, μ) and a function $f \in L^1(\mu)$ such that $f \ge 0$. Adopting the convention $\log 0 = -\infty$ and $\mathrm{e}^{-\infty} = 0$, we define the geometric mean of f by the formula:

$$M_0(f; \mu) = \exp\left(\frac{1}{\mu(\Omega)} \int_\Omega \log f(x)\, \mathrm{d}\mu\right).$$

According to Jensen's inequality,

$$\frac{1}{\mu(\Omega)} \int_\Omega \log f(x)\, \mathrm{d}\mu \le \log\left(\frac{1}{\mu(\Omega)} \int_\Omega f(x)\, \mathrm{d}\mu\right),$$

the inequality being strict except for the case when f is a constant function μ-almost everywhere. This fact can be restated as $M_0(f; \mu) \le M_1(f; \mu)$, and represents the integral form of the $AM-GM$ inequality.
The *power mean* of order $t \in \mathbb{R} \setminus \{0\}$ is defined for all positive measurable functions f with $f^t \in L^1(\mu)$ via the formula

$$M_t(f; \mu) = \left(\frac{1}{\mu(\Omega)} \int_\Omega f^t\, \mathrm{d}\mu\right)^{1/t}.$$

We also define

$$M_{-\infty}(f; \mu) = \sup\{\alpha \ge 0 \mid \mu(\{x \in \Omega : f(x) < \alpha\}) = 0\}$$
$$M_{\infty}(f; \mu) = \inf\{\alpha \ge 0 \mid \mu(\{x \in \Omega : f(x) > \alpha\}) = 0\}$$

for $f \in L^\infty(\mu)$, $f \ge 0$.

(a) (Jensen's inequality for power means) Suppose that $-\infty \le s \le t \le \infty$ and $M_t(f;\mu) < \infty$. Prove that

$$M_s(f;\mu) \le M_t(f;\mu).$$

(b) Suppose that $f \in L^\infty(\mu)$, $f \ge 0$. Prove that

$$\lim_{t\to-\infty} M_t(f;\mu) = M_{-\infty}(f;\mu) \quad \text{and} \quad \lim_{t\to\infty} M_t(f;\mu) = M_\infty(f;\mu).$$

(c) Suppose that $f \in L^\infty(\mu)$, $f \ge 0$. Prove the convexity of the function $t \mapsto t \log M_t(f;\mu)$ on $\mathbb{R}$.
(d) Notice that $(t^r - 1)/r$ decreases to $\log t$ as $r \downarrow 0$ and apply the dominated convergence theorem of Lebesgue to conclude that

$$\lim_{r\to 0+} M_r(f;\mu) = M_0(f;\mu)$$

for all $f \in L^1(\mu)$, $f \ge 0$.
A special case of Exercise 1 (a) is the integral form of the *arithmetic–geometric–harmonic mean inequality*,

$$M_{-1}(f;\mu) \le M_0(f;\mu) \le M_1(f;\mu).$$

2. Prove that $L(a,b) < I(a,b) < A(a,b)$ for all $a, b > 0$, $a \ne b$. Here

$$L(a,b) = \frac{a-b}{\log a - \log b},\ I(a,b) = \frac{1}{\mathrm{e}}\Big(\frac{b^b}{a^a}\Big)^{1/(b-a)},\ \text{and } A(a,b) = \frac{a+b}{2}$$

are, respectively, the logarithmic, the identric, and the arithmetic mean of a and b.

3. (Jensen's inequality for series) Suppose that $f : I \to \mathbb{R}$ is a convex function, $(x_n)_n$ is a sequence in I, and $(\lambda_n)_n$ is a sequence in $[0,1]$ such that $\sum_{n=1}^\infty \lambda_n = 1$. If the series $\sum_{n=1}^\infty \lambda_n x_n$ is absolutely convergent, then the series $\sum_{n=1}^\infty \lambda_n f(x_n)$ is either convergent or has infinite sum. In both cases,

$$f\left(\sum_{n=1}^\infty \lambda_n x_n\right) \le \sum_{n=1}^\infty \lambda_n f(x_n).$$

4. Let f be a convex function defined on $\mathbb{R}$ and X a random variable such that $g(x) = E\left(f\left(x - X\right)\right)$ exists and is finite for all $x \in \mathbb{R}$. Prove that $g(\cdot)$ is a convex function on $\mathbb{R}$.

5. (An estimate of the Jensen gap) Let (X, Σ, μ) be a finite measure space and let $g \in L^2(\mu)$. If f is a twice differentiable function given on an interval I that includes the image of g and $\alpha \le f''/2 \le \beta$, then

$$\alpha \operatorname{var}(g) \le M_1(f \circ g) - f(M_1(g)) \le \beta \operatorname{var}(g).$$

Here $\mathrm{var}(g)$ denotes the *variance* of g.
Remark. More information about the Jensen gap may be found in the paper [4] by S. Abramovich and L.-E. Persson.
[*Hint*: Apply Taylor's formula to infer that

$$\begin{aligned}\alpha(g(x)-M_1(g))^2 &\le f(g(x))-f(M_1(g))-f'(M_1(g))(g(x)-M_1(g))\\ &\le \beta(g(x)-M_1(g))^2.]\end{aligned}$$

6. Consider the normal random variable X whose density of its cumulative distribution function is

$$w(t,\mu,\sigma)=\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{(t-\mu)^2}{2\sigma^2}}.$$

Prove that $E(X)=\mu$ and $\mathrm{var}(X)=\sigma^2$.

7. (The Korkine–Andréief identity) Suppose that (Ω,Σ,μ) is a probability measure space and $f,g\in L^2(\mu)$. Prove the identity

$$\begin{aligned}&\int_\Omega f(x)g(x)\mathrm{d}\mu(x)-\left(\int_\Omega f(x)\mathrm{d}\mu(x)\right)\left(\int_\Omega g(x)\mathrm{d}\mu(x)\right)\\ &\qquad=\frac{1}{2}\int_\Omega\int_\Omega\left(f(x)-f(y)\right)\left(g(x)-g(y)\right)\mathrm{d}\mu(x)\mathrm{d}\mu(y).\end{aligned}$$

8. (Chebyshev's algebraic inequality) Suppose that (Ω,Σ,μ) is a probability measure space and $f,g\in L^2(\mu)$ are two functions synchronous in the sense that

$$(g(x)-g(y))(h(x)-h(y))\ge 0$$

for all $x,y\in\Omega$. Infer from the preceding exercise that $E(g)E(h)\le E(gh)$ (that is, $\mathrm{cov}\,(f,g)\ge 0$).

9. (Jensen's inequality at a point; see C. P. Niculescu and H. Stephan [449]) Let I be an interval of $\mathbb{R}$ endowed with a discrete measure $\mu=\sum_{k=1}^n p_k\delta_{x_k}$, whose weights p_k are all nonzero and sum to 1. We assume that the barycenter of μ, $b_\mu=\sum_{k=1}^n p_k x_k$, belongs to $I\setminus\{x_1,x_2,...,x_n\}$.
(a) Prove that every function $f:I\to\mathbb{R}$ verifies the identity

$$\sum_{k=1}^n p_k f(x_k)-f\left(b_\mu\right)=\sum_{1\le k<j\le n}p_k p_j\left(s(x_k)-s(x_j)\right)\left(x_k-x_j\right),$$

where

$$s(x)=\frac{f(x)-f(b_\mu)}{x-b_\mu}\quad\text{for } x\in I\setminus\{b_\mu\}$$

is the slope function of the segment joining the points of abscissas x and b_μ.

(b) Assume that all weights p_k are positive. Prove that if $s(x)$ is increasing, then

$$\sum_{k=1}^{n} p_k f(x_k) \geq f\left(b_\mu\right),$$

while if $s(x)$ is decreasing, this inequality works in the reversed direction.
(c) Infer that

$$\max\left\{\sum_{k=1}^{n} \frac{x_k}{x_k^3+8} : x_1, ..., x_n > -2 \text{ and } \frac{1}{n}\sum_{k=1}^{n} x_k = 1\right\} = \frac{n}{9}.$$

10. (The connection of Jensen's inequality with subdifferentiability) Suppose that μ is a Borel probability measure μ supported by an interval I and having the barycenter $\text{bar}(\mu)$. Prove that every μ-integrable function $f : I \to \mathbb{R}$ that admits a supporting line at $\text{bar}(\mu)$ verifies the inequality

$$f(\text{bar}(\mu)) \leq \int_I f(x)\mathrm{d}\mu(x).$$

This result allows the use of Jensen's inequality in the context of mixed convex functions like xe^x, x^2e^{-x}, $\log^2 x$, $\frac{\log x}{x}$, etc. See C. P. Niculescu and I. Roventa [441].

2.3 Two More Applications of Jensen's Inequality

Jensen's inequality is instrumental in deriving many useful inequalities, including the Rogers–Hölder inequality and Hardy's inequality. In the first case, we have to notice that Jensen's inequality (when applied to the function x^p on $(0,\infty)$ and $p \geq 1$) implies that every positive measurable function h (attached to a finite measure space (Ω, Σ, μ) with $\mu(\Omega) = 1$) verifies the inequality

$$\left(\int_\Omega h(x)\mathrm{d}\mu(x)\right)^p \leq \int_\Omega h^p(x)\mathrm{d}\mu(x). \tag{2.5}$$

Suppose now that $p, q \in (1,\infty)$ with $1/p + 1/q = 1$, and f, g are two measurable functions such that $f \in L^p(\mu)$ and $g \in L^q(\mu)$. To prove the Rogers–Hölder inequality

$$\int_\Omega |fg|\,\mathrm{d}\mu \leq \|f\|_{L^p}\,\|g\|_{L^q}$$

it suffices to consider the case where f and g are positive and $\int_\Omega g^q(x)\mathrm{d}\mu(x) = 1$. Then the measure

$$\nu(A) = \int_A g^q(x)\mathrm{d}\mu(x)\,, \quad A \in \Sigma,$$

is a probability measure that acts according to the formula:

$$\int_\Omega h\mathrm{d}\nu = \int_\Omega hg^q\mathrm{d}\mu.$$

Taking into account the formula (2.5) we infer that

$$\begin{aligned}\int_\Omega fg\mathrm{d}\mu &= \int_\Omega fg^{1-q}g^q\mathrm{d}\mu = \int_\Omega fg^{1-q}\mathrm{d}\nu \\ &\le \left(\int_\Omega f^p g^{p(1-q)}\mathrm{d}\nu\right)^{1/p} = \left(\int_\Omega f^p\mathrm{d}\mu\right)^{1/p},\end{aligned}$$

and the Rogers–Hölder inequality follows.

We next deal with Hardy's inequality.

2.3.1 Theorem (Hardy's inequality [243], cf. also [244] and [245]) *Suppose that $f \in L^p(0,\infty)$, $f \ge 0$, where $p \in (1,\infty)$. Then the function*

$$F(x) = \frac{1}{x}\int_0^x f(t)\,\mathrm{d}t, \quad x > 0,$$

also belongs to $L^p(0,\infty)$ and

$$\|F\|_{L^p} \le \frac{p}{p-1}\,\|f\|_{L^p}; \tag{2.6}$$

equality occurs if and only if $f = 0$ almost everywhere. The constant $p/(p-1)$ is sharp.

Hardy's inequality provides the norm of the averaging operator

$$H : L^p(0,\infty) \to L^p(0,\infty), \quad H(f)(x) = \frac{1}{x}\int_0^x f(t)\,\mathrm{d}t.$$

We have $\|H\| = p/(p-1)$, but the constant $p/(p-1)$, though the best possible, is unattained in inequality (2.6). The optimality can easily be checked by considering the sequence of functions $f_\varepsilon(t) = (t^{-1/p+\varepsilon})\chi_{(0,1]}(t)$, and letting $\varepsilon \to 0+$.

2.3.2 Lemma (L.-E. Persson and N. Samko [472]) *Suppose that $0 < b \le \infty$ and $-\infty \le a < c \le \infty$. If u is a positive convex function on (a,c), then*

$$\int_0^b u\left(\frac{1}{x}\int_0^x h(t)\,\mathrm{d}t\right)\frac{\mathrm{d}x}{x} \le \int_0^b u(h(x))\left(1-\frac{x}{b}\right)\frac{\mathrm{d}x}{x}$$

for all integrable functions $h\colon (0,b) \to (a,c)$. If u is instead concave, then the inequality holds in the reversed direction.

Proof. It suffices to consider the case when u is a convex function. In this case, by Jensen's inequality,

$$\begin{aligned}\int_0^b u\left(\frac{1}{x}\int_0^x h(t)\,\mathrm{d}t\right)\frac{\mathrm{d}x}{x} &\le \int_0^b \left(\frac{1}{x}\int_0^x u(h(t))\,\mathrm{d}t\right)\frac{\mathrm{d}x}{x}\\ &= \int_0^b \frac{1}{x^2}\left(\int_0^b u(h(t))\chi_{[0,x]}(t)\,\mathrm{d}t\right)\mathrm{d}x\\ &= \int_0^b u(h(t))\left(\int_t^b \frac{1}{x^2}\,\mathrm{d}x\right)\mathrm{d}t\\ &= \int_0^b u(h(t))\left(1-\frac{t}{b}\right)\frac{\mathrm{d}t}{t}\end{aligned}$$

and the proof is complete. ∎

For $u(x) = |x|^p$ and $0 < a \le \infty$ the result of Lemma 2.3.2 can be put (via an obvious substitution) in the form:

$$\int_0^a \left|\frac{1}{x}\int_0^x f(t)\,\mathrm{d}t\right|^p \mathrm{d}x \le \left(\frac{p}{p-1}\right)^p \int_0^a |f(x)|^p\left(1-\left(\frac{x}{a}\right)^{(p-1)/p}\right)\mathrm{d}x, \qquad (2.7)$$

where $a = b^{p/(p-1)}$ and $f(x) = h(x^{1-1/p})x^{-1/p}$. This yields a generalization of Hardy's inequality for functions $f \in L^p(0,a)$, where $0 < a \le \infty$. Hardy's inequality is just the case $\alpha \to \infty$.

It is worth noticing that the inequality (2.7) also holds for $p < 0$ if we assume that $f(x) > 0$ for $x \in (0,a)$.

The equality case in Theorem 2.3.1 implies that F and f are proportional, which makes f of the form Cx^r. Since $f \in L^p(0,\infty)$, this is possible only for $C = 0$.

There are known today more than ten different proofs of Theorem 2.3.1. Let us just mention the following one which is built on the well-known fact that $C_c(0,\infty)$ (the space of all continuous functions $f\colon (0,\infty) \to \mathbb{C}$ with compact support) is dense in $L^p(0,\infty)$. This allows us to restrict ourselves to the case where $f \in C_c(0,\infty)$. Then $(xF(x))' = f(x)$, which yields

$$\begin{aligned}\int_0^\infty F^p(t)\,\mathrm{d}t &= -p\int_0^\infty F^{p-1}(t)tF'(t)\,\mathrm{d}t\\ &= -p\int_0^\infty F^{p-1}(t)(f(t)-F(t))\,\mathrm{d}t,\end{aligned}$$

that is, $\int_0^\infty F^p(t)\,\mathrm{d}t = \frac{p}{p-1}\int_0^\infty F^{p-1}(t)f(t)\,\mathrm{d}t$. The proof ends by taking into account the integral form of the Rogers–Hölder inequality.

For a concrete application of Hardy's inequality, let us consider the *Schrödinger operator*

$$H_\lambda = -\frac{\mathrm{d}^2}{\mathrm{d}x^2} - \frac{\lambda}{x^2}$$

on $(0,\infty)$. The natural domain of this operator is the Sobolev space $H_0^1(0,\infty)$, the completion of $C_c(0,\infty)$ with respect to the norm

$$\|f\|_{H^1} = \left(\int_0^\infty \left(|f|^2 + |f'|^2\right) \mathrm{d}x\right)^{1/2}.$$

See Appendix E for a quick introduction to these spaces. The interested reader may found a detailed presentation of the theory of Sobolev spaces in the book of R. A. Adams and J. J. F. Fournier [8].

A natural question is to find out the range of parameter λ for which the operator H_λ is positive as an operator acting on $L^2(0,\infty)$, that is,

$$\langle H_\lambda \varphi, \varphi\rangle \geq 0 \quad \text{for all } \varphi \in C_c(0,\infty).$$

Since the last inequality is equivalent to

$$\int_0^\infty |\varphi'(x)|^2 \,\mathrm{d}x \geq \lambda \int_0^\infty \left|\frac{\varphi(x)}{x}\right|^2 \mathrm{d}x,$$

the answer is $\lambda \geq 1/4$ and follows from Hardy's inequality for $p = 2$.

Exercises

1. Infer from Theorem 2.3.1 the discrete form of Hardy's inequality:

$$\left(\sum_{n=1}^\infty \left(\frac{1}{n}\sum_{k=1}^n a_k\right)^p\right)^{1/p} < \frac{p}{p-1}\left(\sum_{k=1}^\infty a_k^p\right)^{1/p},$$

for every sequence $(a_n)_n$ of positive numbers (not all zero) and every $p \in (1,\infty)$.

2. (The Pólya–Knopp inequality; see [247], [302]) Prove the following limiting case of Hardy's inequality: for every $f \in L^1(0,\infty)$, $f \geq 0$ and f not identically zero,

$$\int_0^\infty \exp\left(\frac{1}{x}\int_0^x \log f(t)\,\mathrm{d}t\right)\mathrm{d}x < e\int_0^\infty f(x)\,\mathrm{d}x.$$

The discrete form of this inequality was previously noticed by T. Carleman [116]:

$$\sum_{n=1}^\infty (a_1 a_2 \cdots a_n)^{1/n} < e\sum_{n=1}^\infty a_n$$

for $a_1, a_2, a_3, \ldots \geq 0$, not all zero.
[*Hint*: Apply Lemma 2.3.2 for $h(x) = \log f(x)$.]

3. Formulate and prove the analogue of the Pólya–Knopp inequality for functions defined on bounded intervals of the form $(0,\alpha)$.

4. (L.-E. Persson and N. Samko [472]) Let $0 < \ell \le \infty$ and $p \in \mathbb{R}\backslash\{0\}$.
(a) Prove that the following power weighted form of Hardy's inequality (2.7),

$$\int_0^\ell \Big(\frac{1}{x}\int_0^x f(t)\,\mathrm{d}t\Big)^p x^\alpha \mathrm{d}x \le \Big(\frac{p}{p-1-\alpha}\Big)^p \int_0^\ell f^p(x)\Big(1-\Big(\frac{x}{\ell}\Big)^{(p-\alpha-1)/p}\Big)\,x^\alpha \mathrm{d}x,$$

holds for all positive and measurable functions $f : (0,\ell) \to \mathbb{R}$ and all α such that

$$p \ge 1 \text{ and } \alpha < p-1 \quad \text{or} \quad p < 0 \text{ and } \alpha > p-1.$$

(b) Prove that the constant $\left(\frac{p}{p-1-\alpha}\right)^p$ is sharp in both cases.
[*Hint:* Use Lemma 2.3.2 and make the substitution

$$f(x) = h(x^{(p-\alpha-1)/p})x^{-(\alpha+1)/p}.]$$

5. (L.-E. Persson and N. Samko [472]) Let $0 \le \ell < \infty$ and $p \in \mathbb{R}\backslash\{0\}$.
(a) Prove that the following dual power weighted form of Hardy's inequality

$$\int_\ell^\infty \left(\frac{1}{x}\int_0^x f(t)\mathrm{d}t\right)^p x^\alpha \mathrm{d}x \le \left(\frac{p}{\alpha+1-p}\right)^p \int_\ell^\infty f^p(x)\left(1-\left(\frac{x}{\ell}\right)^{(p-\alpha-1)/p}\right) x^\alpha \mathrm{d}x$$

holds for all positive and measurable functions $f : [\ell,\infty) \to \mathbb{R}$ and all α such that

$$p \ge 1 \text{ and } \alpha > p-1 \text{ or } p < 0 \text{ and } \alpha < p-1.$$

(b) Prove that the constant $\left(\frac{p}{\alpha+1-p}\right)^p$ is sharp in both cases.

6. Notice that the inequality in Lemma 2.3.2 holds in the reversed direction if u is concave rather than convex. Formulate and prove the variants of Exercises 4 and 5 in the case $0 < p < 1$.

2.4 Building Blocks of Convex Functions. Applications

A powerful device to prove inequalities for convex functions is to take advantage of some approximation results. From this point of view, the class of

piecewise linear convex functions appears to be very important. Recall that a function $f\colon [a,b] \to \mathbb{R}$ is called *piecewise linear* if it is continuous and there exists a division $a = x_0 < \cdots < x_n = b$ such that the restriction of f to each partial interval $[x_k, x_{k+1}]$ is an affine function.

2.4.1 Lemma *Every continuous convex function $f : [a,b] \to \mathbb{R}$ is the uniform limit of a sequence of piecewise linear convex functions.*

In a similar manner, every increasing and convex function $f : [a,b] \to \mathbb{R}$ is the uniform limit of a sequence of piecewise linear, increasing, and convex functions.

It is worth mentioning that the continuous convex (and increasing) functions defined on compact intervals can be uniformly approximated by infinitely differentiable convex (and increasing) polynomials. See Popoviciu's approximation theorem, Theorem 2.6.5.

Proof. Due to the uniform continuity of f, the sequence of piecewise linear functions f_n obtained by joining the points

$$(a, f(a)), \left(a + \frac{b-a}{n}, f(a + \frac{b-a}{n})\right), ..., \left(a + n \cdot \frac{b-a}{n}, f(a + n \cdot \frac{b-a}{n})\right)$$

by linear segments is uniformly convergent to f. See [134], Theorem 6.8.7, p. 173. The fact that every piecewise linear function inscribed in a convex function is itself a convex function follows from Proposition 1.1.10 (h). ■

The structure of piecewise linear convex functions is surprisingly simple.

2.4.2 Lemma (K. Toda [566]) *Let $f\colon [a,b] \to \mathbb{R}$ be a piecewise linear convex function. Then f is the sum of an affine function and a linear combination, with positive coefficients, of translates of the absolute value function. In other words, f is of the form*

$$f(x) = \alpha x + \beta + \sum_{k=1}^{n} c_k |x - x_k| \tag{2.8}$$

for suitable $\alpha, \beta \in \mathbb{R}$ and suitable positive coefficients $c_1, \ldots, c_n$.

Here the absolute value functions $|x - x_k|$ can be replaced by functions of the form $(x - x_k)^+$ or $(x - x_k)^-$.

The piecewise linear increasing and convex functions can be represented as linear combinations, with positive coefficients, of functions of the form $(x - x_k)^+$.

Proof. Let $a = x_0 < \cdots < x_n = b$ be a division of $[a,b]$ such that the restriction of f to each partial interval $[x_k, x_{k+1}]$ is affine. If $\alpha x + \beta$ is the affine function whose restriction to $[x_0, x_1]$ coincides with $f|_{[x_0,x_1]}$, then it will be a support line for f and $f(x) - (\alpha x + \beta)$ will be an increasing convex function which vanishes on $[x_0, x_1]$. A moment's reflection shows the existence of a constant $c_1 \geq 0$ such

that $f(x) - (\alpha x + \beta) = c_1(x - x_1)^+$ on $[x_0, x_2]$. Repeating the argument we arrive at the representation

$$f(x) = \alpha x + \beta + \sum_{k=1}^{n-1} c_k (x - x_k)^+, \tag{2.9}$$

where all coefficients c_k are positive. The proof ends by replacing the translates of the positive part function by translates of the absolute value function. This is made possible by the formula $y^+ = (|y| + y)/2$. ∎

See Exercise 15, Section 1.1, for an alternative argument.

The representation formula (2.8) admits a generalization for all continuous convex functions on intervals. See L. Hörmander [268].

The above technique of approximation is instrumental in deriving many interesting results concerning the convex functions. A first example describes how convex functions relate the arithmetic means of the subfamilies of a given triplet of numbers.

2.4.3 Theorem (Popoviciu's inequality [488]) *If $f\colon I \to \mathbb{R}$ is a continuous function, then f is convex if and only if*

$$\frac{f(x) + f(y) + f(z)}{3} + f\Big(\frac{x+y+z}{3}\Big) \geq \frac{2}{3}\Big[f\Big(\frac{x+y}{2}\Big) + f\Big(\frac{y+z}{2}\Big) + f\Big(\frac{z+x}{2}\Big)\Big]$$

for all $x, y, z \in I$.

In the variant of strictly convex functions, the above inequality is strict except for $x = y = z$.

Proof. Necessity. This part does not make use of the continuity of f. According to Lemma 1.9.1 and Lemma 1.9.2, it suffices to consider the case where f is the absolute value function, that is, to show that

$$|x| + |y| + |z| + |x + y + z| \geq |x + y| + |y + z| + |z + x| \tag{2.10}$$

for all $x, y, z \in I$. This is an immediate consequence of the order properties of the real numbers. A second approach, based on the polynomial identity,

$$x^2 + y^2 + z^2 + (x + y + z)^2 = (x + y)^2 + (y + z)^2 + (z + x)^2$$

has the advantage to extend to the framework of Euclidean spaces. See Exercise 2 at the end of this section. Indeed,

$$\begin{aligned}
&(|x| + |y| + |z| + |x + y + z| - |x + y| - |y + z| - |z + x|) \\
&\qquad \times (|x| + |y| + |z| + |x + y + z|) \\
&= (|x| + |y| - |x + y|)\,(|z| + |x + y + z| - |x + y|) \\
&+ (|y| + |z| - |y + z|)\,(|x| + |x + y + z| - |y + z|) \\
&\qquad + (|z| + |x| - |z + x|)\,(|y| + |x + y + z| - |z + x|) \geq 0,
\end{aligned}$$

and the necessity part is done.

Sufficiency. Popoviciu's inequality (when applied for $y = z$) yields the following substitute for the condition of midpoint convexity:

$$\frac{1}{4}f(x) + \frac{3}{4}f\Big(\frac{x+2y}{3}\Big) \geq f\Big(\frac{x+y}{2}\Big) \quad \text{for all } x, y \in I. \tag{2.11}$$

The proof ends by following the argument in the proof of Theorem 1.1.8. ■

Two other proofs of Theorem 2.4.3 may be found in the first edition of this book. See [433], pp. 12 and 33. The above statement of Popoviciu's inequality is only a simplified version of a considerably more general result (also due to T. Popoviciu). See the Comments at the end of this chapter.

Another result whose proof can be considerably simplified by the piecewise linear approximation of convex functions is the following generalization of Jensen's inequality, due to J. F. Steffensen [550]. Unlike Jensen's inequality, it allows the use of negative weights.

2.4.4 Theorem (The Jensen–Steffensen inequality) *Suppose that $x_1, ..., x_n$ is a monotonic family of points in an interval $[a, b]$ and $w_1, ..., w_n$ are real weights such that*

$$\sum_{k=1}^{n} w_k = 1 \quad \text{and} \quad 0 \leq \sum_{k=1}^{m} w_k \leq \sum_{k=1}^{n} w_k \quad \text{for every } m \in \{1, ..., n\}. \tag{2.12}$$

Then every convex function f defined on $[a, b]$ verifies the inequality

$$f\left(\sum_{k=1}^{n} w_k x_k\right) \leq \sum_{k=1}^{n} w_k f(x_k).$$

Our argument for Theorem 2.4.4 combines the piecewise linear approximation of convex functions with a consequence of *Abel's partial summation formula* (also known as *Abel's transformation*).

2.4.5 Theorem (Abel's partial summation formula) *If $(a_k)_{k=1}^n$ and $(b_k)_{k=1}^n$ are two families of complex numbers, then*

$$\sum_{k=1}^{n} a_k b_k = \sum_{k=1}^{n-1} \left[(a_k - a_{k+1}) \left(\sum_{j=1}^{k} b_j \right) \right] + a_n \left(\sum_{j=1}^{n} b_j \right).$$

Abel [1] used his formula to derive a number of important results known today as the Abel criterion of convergence of signed series, the Abel theorem on power series, and the Abel summation method. In what follows we will be interested in another consequence of it:

2.4.6 Corollary (*The Abel–Steffensen inequality* [550]) *If* $x_1, x_2, ..., x_n$ *and* $y_1, y_2, ..., y_n$ *are two families of real numbers that verify one of the following two conditions:*

$$(a)\ x_1 \geq x_2 \geq \cdots \geq x_n \geq 0 \text{ and } \sum_{k=1}^{j} y_k \geq 0 \text{ for all } j \in \{1, 2, ..., n\},$$

$$(b)\ 0 \leq x_1 \leq x_2 \leq \cdots \leq x_n \text{ and } \sum_{k=j}^{n} y_k \geq 0 \text{ for all } j \in \{1, 2, ..., n\},$$

then

$$\sum_{k=1}^{n} x_k y_k \geq 0.$$

Therefore, if $x_1, x_2, ..., x_n$ *is a monotonic family and* $y_1, y_2, ..., y_n$ *is a family of real numbers such that*

$$0 \leq \sum_{k=1}^{j} y_k \leq \sum_{k=1}^{n} y_k,$$

for $j = 1, ..., n$, *then we have*

$$\left(\min_{1\leq k\leq n} x_k\right) \sum_{k=1}^{n} y_k \leq \sum_{k=1}^{n} x_k y_k \leq \left(\max_{1\leq k\leq n} x_k\right) \sum_{k=1}^{n} y_k.$$

Proof of the Jensen–Steffensen inequality. Taking into account Lemma 2.4.1 and Lemma 2.4.2, we may reduce ourselves to the case of absolute value function. This case can be settled as follows. Assuming the ordering $x_1 \leq \cdots \leq x_n$ (to make a choice), we infer that

$$0 \leq x_1^+ \leq \cdots \leq x_n^+ \text{ and } x_1^- \geq \cdots \geq x_n^- \geq 0.$$

According to Corollary 2.4.6,

$$\sum_{k=1}^{n} w_k x_k^+ \geq 0 \text{ and } \sum_{k=1}^{n} w_k x_k^- \geq 0,$$

which yields

$$\left|\sum_{k=1}^{n} w_k x_k\right| \leq \sum_{k=1}^{n} w_k \left|x_k\right|.$$

The proof is done. ■

The integral version of Jensen–Steffensen inequality can be established in the same manner, using integration by parts instead of Abel's partial summation formula.

2.4.7 Theorem (The integral version of Jensen–Steffensen inequality) *Suppose that $g : [a, b] \to \mathbb{R}$ is a monotone function and $w : [a, b] \to \mathbb{R}$ is an integrable function such that*

$$0 \leq \int_a^x w(t)\mathrm{d}t \leq \int_a^b w(t)\mathrm{d}t = 1 \quad \text{for every } x \in [a, b]. \tag{2.13}$$

Then every convex function f defined on an interval I that includes the range of g verifies the inequality

$$f\left(\int_a^b g(t)w(t)\mathrm{d}t\right) \leq \int_a^b f\left(g(t)\right) w(t)\mathrm{d}t.$$

Another application of Abel's partial summation formula is as follows.

2.4.8 Theorem (The discrete form of Hardy–Littlewood rearrangement inequality [247]) *Let $x_1, \ldots, x_n, y_1, \ldots, y_n$ be real numbers. Then*

$$\sum_{k=1}^n x_k^{\downarrow} y_{n-k+1}^{\downarrow} \leq \sum_{k=1}^n x_k y_k \leq \sum_{k=1}^n x_k^{\downarrow} y_k^{\downarrow},$$

where $z_1^{\downarrow} \geq \cdots \geq z_n^{\downarrow}$ denotes the decreasing rearrangement of a family $z_1, \ldots, z_n$ of real numbers.

Proof. Notice first that we may assume that $x_1 \geq \cdots \geq x_n \geq 0$. Then, apply Abel's partial summation formula. ∎

For a generalization of Theorem 2.4.8 see Exercise 6, Section 5.7. For more on rearrangements, see the Comments at the end of Chapter 6.

Exercises

1. Infer from Lemma 1.9.1 and Lemma 1.9.2 that the *running average* of every continuous convex function $f : [0, \infty) \to \mathbb{R}$,

$$F(x) = \frac{1}{x}\int_0^x f(t)\mathrm{d}t, \quad x > 0,$$

is convex too.

2. (The Hornich–Hlawka inequality) The inequality (2.10) that appears in the proof of Popoviciu's inequality can be extended to all inner product spaces H. For this, prove first Fréchet's identity [203],

$$\|x\|^2 + \|y\|^2 + \|z\|^2 + \|x + y + z\|^2 = \|x + y\|^2 + \|y + z\|^2 + \|z + x\|^2,$$

and infer from it the Hornich–Hlawka inequality,

$$\|x + y + z\| + \|x\| + \|y\| + \|z\| - \|x + y\| - \|y + z\| - \|z + x\| \geq 0.$$

Remark. For the history and ramifications of this inequality, see the Comments at the end of this chapter and the survey [362].

3. (An illustration of Popoviciu's inequality) Suppose that x_1, x_2, x_3 are positive numbers, not all equal. Prove that:
   (*a*) $27\prod_{i<j}(x_i + x_j)^2 > 64x_1x_2x_3(x_1 + x_2 + x_3)^3$;
   (*b*) $x_1^6 + x_2^6 + x_3^6 + 3x_1^2x_2^2x_3^2 > 2(x_1^3x_2^3 + x_2^3x_3^3 + x_3^3x_1^3)$.

4. Let $f\colon [0, 2\pi] \to \mathbb{R}$ be a convex function. Prove that

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f(t)\cos nt\,\mathrm{d}t \geq 0 \quad \text{for every } n \geq 1.$$

   [*Hint:* Use Corollary 1.4.7 by noticing that

$$\int_0^{2\pi} f(t)\cos t\,dt = \int_0^{\pi/2} [f(t) - f(\pi - t) - f(\pi + t) + f(2\pi - t)]\cos t\,dt.\ ]$$

   Remark. Exercise 4 admits a two-variable analogue based on Popoviciu's notion of (m, n) convexity. See C. P. Niculescu and I. Roventa [444] and B. Gavrea and I. Gavrea [215].

5. Infer the *AM-GM* inequality from Theorem 2.4.8 (the discrete form of Hardy–Littlewood–Pólya rearrangement inequality).

2.5 A Connection with Stochastic Orders

As was noticed in the preceding section, the two basic approximation results stated in Lemma 2.4.1 and Lemma 2.4.2 facilitate the proof of many important applications of convex analysis. Yet another application is presented in what follows.

In probability theory and statistics, the stochastic orders are used to formally define the concept that one random variable is "bigger" than another. One of the most relevant stochastic orders are the increasing convex order and the convex order, which are largely used in financial mathematics, risk management, insurance, actuarial science, etc. See, for example, the book of L. Rüschendorf [516].

2.5.1 Definition (Convex ordering) *Given two random variables X and Y we say that X is smaller then Y in convex order (abbreviated $X \leq_{cx} Y$) if*

$$\mathbb{E}\,f(X) \leq \mathbb{E}\,f(Y)$$

for all convex functions $f : \mathbb{R} \to \mathbb{R}$ such that the integrals defining the expectations exist.

We say that X is smaller then Y in increasing convex order (abbreviated $X \leq_{icx} Y$) if the above inequality is verified for all increasing convex functions $f : \mathbb{R} \to \mathbb{R}$ such that the integrals defining the expectations exist.

If $X \leq_{cx} Y$, then $\mathbb{E}\,X = \mathbb{E}\,Y$ and $\mathrm{var}(X) \leq \mathrm{var}(Y)$. In the first case, consider the functions $\pm x$, for the second relation, consider the function $f(x) = (x - \mathbb{E}\,X)^2$. For this reason, one can interpret $X \leq_{cx} Y$ as saying that X is *less variable* than Y.

Convex ordering can be suitable described via the *excess of loss function* π_X, defined as,

$$\pi_X(x) = \mathbb{E}\left((X - x)_+\right) = \int_x^\infty (X - t)\,\mathrm{d}F(t) = \int_x^\infty (1 - F(t))\mathrm{d}t. \tag{2.14}$$

2.5.2 Lemma *We have:*
(a) $X \leq_{icv} Y$ *if and only if* $\mathbb{E}\,((X - t)_+) \leq \mathbb{E}\,((Y - t)_+)$ *for all* $t \in \mathbb{R}$.
(b) *If* $X \leq_{icv} Y$ *and* $\mathbb{E}(X) = \mathbb{E}(Y)$, *then* $X \leq_{cv} Y$.

Proof. For (a), use the fact that the increasing convex functions defined on an interval $[a, b]$ can be uniformly approximated by linear combinations (with positive coefficients) of the functions $x \to (x-t)_+$; see Lemma 2.4.1 and Lemma 2.4.2. The assertion (b) is an easy consequence of (a). ∎

2.5.3 Theorem (The cut-criterion of S. Karlin and A. Novikoff [288]) *Let* X, Y *be two random variables such that* $\mathbb{E}\,X = \mathbb{E}\,Y$. *If the cumulative distribution functions* F_X *and* F_Y *of the random variables* X *and* Y, *respectively, cross exactly one time, that is, for some* $x_0 \in \mathbb{R}$,

$$F_X(x) \leq F_Y(x) \text{ if } x < x_0 \text{ and } F_X(x) \geq F_Y(x) \text{ if } x > x_0,$$

then $X \leq_{cx} Y$, *that is,*

$$\mathbb{E}\,f(X) \leq \mathbb{E}\,f(Y)$$

for all convex functions $f : \mathbb{R} \to \mathbb{R}$.

Proof. According to the formula (2.14),

$$\pi_Y(x) - \pi_X(x) = \int_x^\infty (F_X(t) - F_Y(t))\mathrm{d}t.$$

If $x \geq x_0$, then $F_X(t) \geq F_Y(t)$ for all $t > x$ and in this case $\pi_Y(x) - \pi_X(x) \geq 0$. Since

$$\lim_{x \to -\infty} (\pi_Y(x) - \pi_X(x)) = \mathbb{E}Y - \mathbb{E}X = 0,$$

it follows that for $x < x_0$ we have

$$\begin{aligned}\pi_Y(x) - \pi_X(x) &= \mathbb{E}Y - \mathbb{E}X + \int_{-\infty}^x (F_Y(t) - F_X(t))\mathrm{d}t \\ &= \int_{-\infty}^x (F_Y(t) - F_X(t))\mathrm{d}t \geq 0,\end{aligned}$$

taking into account our hypotheses. Therefore $\pi_Y(x) - \pi_X(x) \geq 0$ and according to Lemma 2.5.2 it follows that $X \leq_{cx} Y$. ∎

2.5.4 Remark *Consider the probability measures μ_X and μ_Y associated to the cumulative distribution functions F_X and F_Y via the formulas*

$$\mu_X((-\infty, x)) = F_X(x) \text{ and } \mu_Y((-\infty, x)) = F_Y(x)$$

for all $x \in \mathbb{R}$. As noticed T. Szostok [563], a careful inspection of the proof of Theorem 2.5.3 shows that if the measures μ_X and μ_Y are concentrated in the interval $[a, b]$, then the conclusion of this theorem works for all convex functions $f : [a, b] \to \mathbb{R}$.

The cut-criterion was rediscovered by J. Ohlin in 1969. It was extended by M. Denuit, C. Lefévre, and M. Shaked [158] to the case of multiple crossing points and higher order convex functions. The cut-criterion is a very useful tool for proving functional inequalities. See T. Rajba [495] and T. Szostok [563], as well as the next set of exercises.

Exercises

1. Let $a < b$ be two real numbers and X be a random variable taking values in $[a, b]$. Prove that

$$f(\mathbb{E}X) \leq \mathbb{E}f(X) \leq \frac{b - \mathbb{E}X}{b - a} f(a) + \frac{\mathbb{E}X - a}{b - a} f(b)$$

for every convex function $f : [a, b] \to \mathbb{R}$.

2. Let $f : [a, b] \to \mathbb{R}$ be a convex function and let the function $F : (\frac{1}{2}, 1] \to \mathbb{R}$ be given by

$$F(t) = \frac{1}{(2t - 1)(b - a)} \int_{ta+(1-t)b}^{(1-t)a+tb} f(x)dx.$$

Prove that F is increasing and compute its range.

3. Let $t_1, ..., t_n \in (0, 1)$ and $\alpha_1, ..., \alpha_n > 0$ be two families of numbers such that $\sum_{k=1}^{n} \alpha_k = 1$ and $\sum_{k=1}^{n} \alpha_k t_k = \frac{1}{2}$. Prove that

$$\sum_{n=1}^{k} \alpha_n f(t_n a + (1 - t_n)b) \leq \frac{f(a) + f(b)}{2}$$

for every convex function $f : [a, b] \to \mathbb{R}$.

4. (F. L. Nazarov and A. N. Podkorytov [407]) Let $(\mathcal{X}, \mu)$ be a measure space and f and g be two positive measurable functions such that their cumulative distribution functions F and G are real-valued. Suppose that $\int_{\mathcal{X}} (f^{s_0} - g^{s_0})\, \mathrm{d}\mu = 0$. Prove that $\int_{\mathcal{X}} (f^s - g^s)\, \mathrm{d}\mu = 0$ for all $s > s_0$.

Remark. Based on the result in Exercise 4, Nazarov and Podkorytov were able to provide a simple proof of the inequality

$$\int_{-\infty}^{\infty} \left| \frac{\sin \pi x}{\pi x} \right|^p \mathrm{d}x < \sqrt{\frac{2}{p}} \text{ for } p > 2,$$

(for $p = 2$, it turns into an identity), which lies at the basis of K. Ball's famous result on sections of the n-dimensional unit cube. See [36]. For other applications, see S V. Astashkin, K.V. Lykov, and M. Milman [27] (and references therein).

2.6 Higher order convexity

Higher order convexity was introduced by E. Hopf [259] and T. Popoviciu [483], [487], who defined it in terms of divided differences. Assuming a real-valued function f defined on a real interval I, its divided differences of order $0, 1, \ldots, n$ associated to a family $x_0, x_1, \ldots, x_n$ of $n+1$ distinct points are, respectively, defined by the formulas

$$\begin{aligned} [x_0; f] &= f(x_0) \\ [x_0, x_1; f] &= \frac{f(x_1) - f(x_0)}{x_1 - x_0} \\ &\ldots \\ [x_0, x_1, ..., x_n; f] &= \frac{[x_1, x_2, ..., x_n; f] - [x_0, x_1, ..., x_{n-1}; f]}{x_n - x_0}. \end{aligned}$$

Using mathematical induction, one can easily prove the formula

$$[x_0, x_1, ..., x_n; f] = \sum_{j=0}^{n} \frac{f(x_j)}{\prod_{k \neq j} (x_j - x_k)}, \tag{2.15}$$

which implies that the divided differences are invariant under the permutation of the points $x_0, x_1, ..., x_n$. As a consequence, we may always assume that $x_0 < x_1 < \cdots < x_n$.

A function f is called *n-convex* (respectively *n-concave*) if all divided differences $[x_0, x_1, \ldots, x_n; f]$ are positive (respectively negative). In particular,

- the convex functions of order 0 are precisely the positive functions;
- the convex functions of order 1 are the increasing functions;
- the convex functions of order 2 are nothing but the usual convex functions since in this case for all $x_0 < x_1 < x_2$ in I,

$$[x_0, x_1, x_2; f] = \frac{\frac{f(x_0)-f(x_1)}{x_0-x_1} - \frac{f(x_1)-f(x_2)}{x_1-x_2}}{x_0 - x_2} \geq 0,$$

that is, $(x_2 - x_0)\, f(x_1) \leq (x_2 - x_1)\, f(x_0) + (x_1 - x_0)\, f(x_2)$.

The description of convex functions of order $n \geq 3$ in terms of divided differences is rather intricate. For example, a function f is 3-*convex* if for every quadruple of elements $x_0 < x_1 < x_2 < x_3$ we have

$$[x_0, x_1, x_2, x_3; f] = \frac{f(x_0)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)} - \frac{f(x_1)}{(x_0 - x_1)(x_1 - x_2)(x_1 - x_3)} + \frac{f(x_2)}{(x_0 - x_2)(x_1 - x_2)(x_2 - x_3)} - \frac{f(x_3)}{(x_0 - x_3)(x_1 - x_3)(x_2 - x_3)} \geq 0,$$

equivalently,

$$(x_2 - x_0)(x_3 - x_0)(x_3 - x_2)f(x_1) + (x_1 - x_0)(x_2 - x_0)(x_2 - x_1)f(x_3) \geq (x_2 - x_1)(x_3 - x_1)(x_3 - x_2)f(x_0) + (x_1 - x_0)(x_3 - x_0)(x_3 - x_1)f(x_2).$$

When the points x_0, x_1, x_2, x_3 are equidistant, that is, when $x_1 = x_0 + h$, $x_2 = x_0 + 2h$, $x_3 = x_0 + 3h$ for some $h > 0$, the last inequality becomes

$$f(x_0 + 3h) - 3f(x_0 + 2h) + 3f(x_0 + h) - f(x_0) \geq 0,$$

equivalently,

$$f(x_0) + 3f\left(\frac{x_0 + 2x_3}{3}\right) \leq 3f\left(\frac{2x_0 + x_3}{3}\right) + f(x_3). \tag{2.16}$$

If f is n-times differentiable, then a repeated application of the mean value theorem yields the existence of a point $\xi \in (\min_k x_k, \max_k x_k)$ such that

$$[x_0, x_1, ..., x_n; f] = \frac{f^{(n)}(\xi)}{n!}.$$

As a consequence, one obtains the sufficiency part of the following practical criterion of n-convexity.

2.6.1 Lemma *Suppose that f is a continuous function defined on an interval I which is n-times differentiable on the interior of I. Then f is n-convex if and only if its nth derivative is positive.*

The necessity part is also immediate by using the standard formulas for derivatives via iterated differences,

$$f^{(n)}(x) = \lim_{h \to 0+} \frac{1}{h^n} \sum\nolimits_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(x + kh).$$

An important source of convex functions of higher order is provided by the Bernstein functions and the completely monotone functions. A function $f : (0, \infty) \to \mathbb{R}_+$ is a *Bernstein function* if it is infinitely differentiable and verifies the condition

$$(-1)^{n+1} f^{(n)}(x) \geq 0 \quad \text{for all } x > 0 \text{ and } n \geq 1;$$

the function f is *completely monotone* if instead

$$(-1)^n f^{(n)}(x) \geq 0 \quad \text{for all } x > 0 \text{ and } n \geq 0.$$

We adopt the convention to call a function $f : [0, \infty) \to \mathbb{R}_+$ a *Bernstein function* (respectively a *completely monotone function*) if it is the extension by continuity of a function of this type.

Every Bernstein function is $(2n+1)$-convex and every completely monotone function is $2n$-convex for every $n \geq 0$.

If $f : [0, \infty) \to \mathbb{R}_+$ is a Bernstein function, then so is $f - f(0)$; if f is a completely monotone function then $f(0) - f$ is a Bernstein function. An infinitely differentiable function $f : (0, \infty) \to \mathbb{R}_+$ is a Bernstein function if, and only if, its derivative is a completely monotone function.

Some simple examples of Bernstein functions are

$$x/(x+1),\ 1 - e^{-\alpha x} \text{ (for } \alpha > 0),$$
$$\log(1+x), (x-1)/\log x \text{ and } x^\alpha \text{(for } 0 < \alpha \leq 1).$$

The connection between the Bernstein functions and the operator monotone functions makes the objective of Theorem 7.2.12, Section 7.2.

A nice account of the Bernstein functions (as well as of the completely monotone functions) is offered by the monograph of Schilling, Song, and Vondraček [521]. According to Corollary 3.8, p. 28, in their monograph, the composition of every two Bernstein functions is always a Bernstein function (which is not the case for the class of 3-convex functions).

Notice also that the exponential function and the polynomials, all of whose coefficients are positive, are n-convex for every $n \geq 0$.

2.6.2 Remark *All polynomials of degree less than or equal to* $n-1$ $(n \geq 2)$ *are both* n*-convex and* n*-concave. These functions, together with the finite sums* $\sum_{i=1}^m c_i \left((x - a_i)_+\right)^{n-1}$ *with positive coefficients, represent the building blocks of any* n*-convex function. For* $n = 2$*, one recover the approximation of continuous convex functions by piecewise linear convex functions described in Section* 1.9*. See T. Popoviciu* [487]*, pp.* 29-30 (*and also* [76] *and* [565]).

Some more examples of 3-convex functions on $\mathbb{R}_+$ are

$$x^\alpha (\text{for } \alpha \in (0,1] \cup [2, \infty)),\ -x^2 + \sqrt{x},$$
$$\log x,\ -x \log x,\ \sinh,\ \cosh,\ \text{and}\ -\log(\Gamma(x)).$$

Also, the primitive of any continuous convex function is a 3-convex function.

The function $1 - (x-3) + \frac{(x-3)^3}{6}$ is continuous and 3-convex on $\mathbb{R}_+$ but not n-convex for any $n \in \{0, 1, 2\}$.

The extremal properties of n-convex functions differ from those of convex functions. For example, the maximum of two 3-convex functions is not necessarily a 3-convex function; see the case of the functions $-x$ and x. Also, an interior critical point of a 3-convex function is not necessarily a point of minimum. It

can be a point of inflection (the case of the cubic function x^3) or a point of maximum (the case of the function $-x^2 + \sqrt{x}$).

2.6.3 Remark (*Permanence properties*) *The continuous n-convex functions defined on an interval I constitute a convex cone in the vector space $C(I)$, consisting of all continuous functions on I. This cone is closed under convolution (provided it is well defined), but not under usual product.*

Every continuous function which is n-convex on the interior of I is n-convex on the whole interval.

The limit of a pointwise convergent sequence of n-convex functions is also an n-convex function.

As was noticed by T. Popoviciu [487], Section 24, p. 49, the property of n-convexity can be characterized in terms of higher order differences.

The *difference operator* Δ_h (of step size $h \geq 0$) associates to each function f defined on an interval I the function $\Delta_h f$ defined by

$$(\Delta_h f)(x) = f(x+h) - f(x),$$

for all x such that the right-hand side formula makes sense. Notice that no restrictions are necessary if $I = \mathbb{R}^+$ or $I = \mathbb{R}$.

The difference operators are linear and commute to each other,

$$\Delta_{h_1} \Delta_{h_2} = \Delta_{h_2} \Delta_{h_1}.$$

They also verify the following property of invariance under translation:

$$\Delta_h (f \circ T_a) = (\Delta_h f) \circ T_a,$$

where T_a is the translation defined by the formula $T_a(x) = x + a$.

2.6.4 Lemma *If $n \geq 1$ is an integer, then the following formula holds:*

$$\Delta_{h_1} \Delta_{h_2} \cdots \Delta_{h_n} f(x) = \sum_{\varepsilon_1, \ldots, \varepsilon_n \in \{0,1\}} (-1)^{n-(\varepsilon_1 + \cdots + \varepsilon_n)} f(x + \varepsilon_1 h_1 + \cdots + \varepsilon_n h_n).$$

Proof. Use mathematical induction in the definition of the operator Δ_h. ∎

By definition,

$$(\Delta_h)^0 f(x) = f(x),$$

while for $n \geq 1$, we infer from Lemma 2.6.4 that

$$\begin{aligned}(\Delta_h)^n f(x) &= \underbrace{\Delta_h \cdots \Delta_h}_{n \text{ times}} f(x) \\ &= \sum\nolimits_{k=0}^{n} (-1)^{n-k} \binom{n}{k} f(x + kh).\end{aligned}$$

Notice that every function f defined on an interval I of the form $[0, A]$ or $[0, \infty)$ verifies the formula:

$$(\Delta_h)^n f(x) = h^n[x, x+h, ..., x+nh; f],$$

at all points $x \in I$ and all steps $h > 0$ such that $x + nh \in I$. This implies that every n-convex function f defined on a such interval I verifies the inequality

$$(\Delta_h)^n f(x) \geq 0 \tag{2.17}$$

for all $x \in I$ and all $h > 0$ such that $x + nh \in I$.

The inequality (2.17) together with Bernstein's variant of the Weierstrass approximation theorem (see [134], Theorem 8.8.1, p. 256) yields the following shape-preserving approximation result:

2.6.5 Theorem (Popoviciu's approximation theorem [484]) *If a continuous function $f : [0, 1] \to \mathbb{R}$ is m-convex, then so are the Bernstein polynomials associated to it,*

$$B_n(f)(x) = \sum_{i=0}^{n} \binom{n}{i} x^i (1-x)^{n-i} f\left(\frac{i}{n}\right).$$

Moreover, by the well-known property of simultaneous uniform approximation of a function and its derivatives by the Bernstein polynomials and their derivatives, it follows that $B_n(f)$ and any derivative (of any order) of it converge uniformly to f and to its derivatives, correspondingly.

Using a change of variable, one can easily see that the approximation theorem extends to functions defined on compact intervals $[a, b]$ with $a < b$.

Proof. Using mathematical induction one can easily show that the derivatives of Bernstein's polynomials verify the formula:

$$B_n^{(k)}(f)(x)$$
$$= n(n-1)\cdots(n-k+1) \sum_{i=0}^{n-k} \underbrace{\Delta_{1/n} \cdots \Delta_{1/n}}_{k \text{ times}} f(j/n) \binom{n-k}{i} x^i (1-x)^{n-k-i}.$$

The proof ends by taking into account the formula (2.17) and Lemma 2.6.1. ∎

When combined with Remark 2.6.3, Theorem 2.6.5 implies that any result valid for smooth n-convex functions also works for all n-convex continuous functions.

When combined with Remark 2.6.3, Theorem 2.6.5 implies that any result valid for the smooth n-convex functions also works for all n-convex continuous functions.

2.6.6 Corollary *If $f : \mathbb{R}_+ \to [0, \infty)$ is a continuous 3-convex function which is also nondecreasing and concave, then the same properties hold for f^α if $\alpha \in (0, 1]$.*

Proof. According to Theorem 2.6.5, we may reduce the proof to the case where the involved function is of class C^3, and in this case the conclusion follows from Lemma 2.6.1. ■

2.6.7 Definition *A function f defined on an interval I of the form $[0, A]$ or $[0, \infty)$ has* positive differences of order $n \geq 1$ *if*

$$\Delta_{h_1} \Delta_{h_2} \cdots \Delta_{h_n} f(x) \geq 0$$

whenever the left-hand side is well defined. For convenience, we say that the function f has positive differences of order 0 *if $f \geq 0$.*

As was mentioned above, Popoviciu has characterized the continuous n-convex function by the positivity of their differences of order n. This is trivial for $n = 0$ and $n = 1$ but is a bit tricky for $n = 2$. See Exercise 1 at the end of this section. The next theorem combines Popoviciu's result with another characterization of n-convexity, due to Hopf ([259], p. 24), and Popoviciu ([483], p. 48), based on the existence of the derivatives of order $n - 2$.

2.6.8 Theorem *Suppose that $f : [0, A] \to \mathbb{R}$ is a continuous function and $n \geq 2$ is an integer. Then the following conditions are equivalent:*

(a) *f is n-convex;*
(b) *$\Delta_h^n f(x) \geq 0$ for all $x \in [0, A)$ and $h > 0$ such that $x + nh \leq A$;*
(c) *f has* positive differences of order n;
(d) *f is $(n-2)$-differentiable on $(0, A)$ and $f^{(n-2)}$ is a convex function.*

A detailed proof can be found in the book of M. Kuczma [311]; see Theorem 15.6.1, p. 440 and Theorem 15.8.4 p. 446. Notice that Kuczma makes use of another definition of n-convexity, precisely, $[x_1, ..., x_{n+2}; f] \geq 0$ for every family of distinct points. This means that the n-convex functions in the terminology of Kuczma correspond to the $(n+1)$-convex functions in the terminology of Hopf and Popoviciu (and vice versa).

Notice also that a simple proof for the implications (b)⇒(c)⇒(d) is made available by the paper of Boas and Widder [72], while the case $n = 3$ of Theorem 2.6.8 is presented in [361].

2.6.9 Corollary (The Hornich–Hlawka functional inequality) *Every continuous 3-convex function $f : [0, A] \to \mathbb{R}$ verifies the inequality*

$$f(x+t) + f(y+t) + f(z+t) + f(x+y+z+t) \geq f(x+y+t) + f(y+z+t) + f(x+z+t) + f(t)$$

whenever $x, y, z, t \in [0, A]$ such that $x + y + z + t \leq A$; if in addition $f(0) \geq 0$, then

$$f(x) + f(y) + f(z) + f(x+y+z) \geq f(x+y) + f(y+z) + f(z+x),$$

an inequality which is usually known as the Hornich–Hlawka functional inequality.

Proof. Notice that the inequality under attention is equivalent to

$$\Delta_x \Delta_y \Delta_z f(t) \geq 0.$$

■

As noticed H. S. Sendov and R. Zitikis [529], the Hornich–Hlawka functional inequality also works in the case of completely monotone functions (an example being e^{-x}, which is 3-concave but not 3-convex).

Usually, higher order convexity appears in contexts mixing various instances of n-convexity and it seems appropriate to use a notation such as

$$\mathcal{C}vx_{(0,-1,2,3)}([0,\infty)),$$

for the set of functions $f : [0,\infty) \to \mathbb{R}$ which are positive, decreasing, convex, and 3-convex and

$$\mathcal{C}vx_{(1,-2,3)}([0,\infty)),$$

for the set of real-valued functions $f : [0,\infty) \to \mathbb{R}$ which are increasing, concave, and 3-convex. So is the case of the following result (originally due to P. Ressel [502]) that deepens Corollary 2.6.9. We state it here following [361].

2.6.10 Theorem *If $f \in \mathcal{C}vx_{(1,-2,3)}([0,A])$, then the inequality*

$$f(|x|)+f(|y|)+f(|z|)+f(|x+y+z|) \geq f(|x+y|)+f(|y+z|)+f(|z+x|)+f(0)$$

holds for all $x,y,z \in [-A,A]$ with $|x|+|y|+|z| \leq A$.

Proof. Because the inequality under attention is invariant under permutation and to the symmetry $(x,y,z) \to (-x,-y,-z)$, it suffices to consider the following five cases:

(a) $x \geq y \geq z \geq 0$; (b) $x \geq y \geq 0 \geq z$ and $|z| \geq x+y$;
(c) $x \geq y \geq 0 \geq z$ and $x+y \geq |z| \geq x,y$;
(d) $x \geq y \geq 0 \geq z$ and $x \geq |z| \geq y$;
(e) $z \leq 0 \leq y \leq x$ and $x \geq y \geq |z|$.

Case (a) is covered by Corollary 2.6.9. Case (b) is a consequence of case (a).

In the case (c), we have $0 \leq x+y+z \leq |z| \leq x+y$, so by Corollary 1.4.7 it follows that

$$f(x+y)+f(0) \leq f(|z|)+f(x+y+z).$$

On the other hand, $|x+z| = |z|-x$, $|y+z| = |z|-y$ and $|x+y+z| = x+y+z$. Since f is nondecreasing, it follows that $f(|z|-x) \leq f(y)$ and $f(|z|-y) \leq f(|x|)$. Therefore

$$\begin{aligned} f(|x+y|)+f(|y+z|)+f(|z+x|)+f(0) &= f(x+y)+f(0)+f(|z|-y)+f(|z|-x) \\ &\leq f(x+y+z)+f(|z|)+f(x)+f(y) \\ &= f(|z|)+f(|x+y+z|)+f(|x|)+f(|y|), \end{aligned}$$

so also the case (c) is proved.

The cases (d) and (e) can be treated in the same manner. Notice that the property of 3-convexity is unnecessary in the cases (c), (d), and (e). ■

2.6.11 Corollary *If* $\alpha \in (0,1]$ *and* $f \in \mathcal{C}vx_{(1,-2,3)}([0,A])$*, then*

$$f^{\alpha}(|x|)+f^{\alpha}(|y|)+f^{\alpha}(|z|)+f^{\alpha}(|x+y+z|) \\ \geq f^{\alpha}(|x+y|)+f^{\alpha}(|y+z|)+f^{\alpha}(|z+x|)+f^{\alpha}(0)$$

for all $x,y,z \in [-A,A]$ *with* $|x|+|y|+|z| \leq A$.

This follows from Corollary 2.6.6 and Theorem 2.6.10.

One can extend Theorem 2.6.10 to the case of $n > 3$ variables via the following inductive scheme due to Vasić and Adamović (see [389], Theorem 2, p. 528).

2.6.12 Theorem *Suppose that* φ *is a real-valued function defined on a commutative additive semigroup* $\mathcal{S}$ *such that*

$$\sum_{k=1}^{3}\varphi(x_k)+\varphi\left(\sum_{k=1}^{3}x_k\right) \geqslant \sum_{1\leq i<j\leq 3}\varphi(x_i+x_j)$$

for all $x_1,x_2,x_3 \in \mathcal{S}$*. Then for each pair* (k,n) *of integers with* $2 \leq k < n$ *we also have*

$$\binom{n-2}{k-1}\sum_{k=1}^{n}\varphi(x_k)+\binom{n-2}{k-2}\varphi\left(\sum_{k=1}^{n}x_k\right) \geqslant \sum_{1\leq i_1<\ldots<i_k\leq n}\varphi\left(\sum_{j=1}^{k}x_{i_j}\right),$$

whenever $x_1,\ldots,x_n \in \mathcal{S}$.

We are thus led to the following generalization of Theorem 2.6.10:

2.6.13 Theorem *If* $f \in \mathcal{C}vx_{(1,-2,3)}([0,\infty])$*, then*

$$\binom{n-2}{k-1}\sum_{k=1}^{n}f(|x_k|)+\binom{n-2}{k-2}f\left(\left|\sum_{k=1}^{n}x_k\right|\right) \\ \geq \sum_{1\leq i_1<\ldots<i_k\leq n}f\left(\left|\sum_{j=1}^{k}x_{i_j}\right|\right)+(k-1)\binom{n-1}{k}f(0),$$

for all pairs $\{k,n\}$ *of integers with* $2 \leq k < n$ *and all strings* $x_1,\ldots,x_n$ *of real numbers.*

Proof. Apply Theorem 2.6.12 to $f(|\cdot|) - f(0)$, taking into account Theorem 2.6.10 and the formula

$$\binom{n-2}{k-1} + \binom{n-2}{k-2} = \binom{n-1}{k-1}.$$

■

Recent work by C. Schötz [523] and C. P. Niculescu [430] has drawn attention to a new class of functions generating functional inequalities: the class $\mathcal{C}vx_{(0,1,2,-3)}([0,\infty))$, of all functions f from $[0,\infty)$ into itself which are increasing, convex, and 3-concave. Some few examples of such functions vanishing at the origin are

$$\begin{array}{c} \left(1+\alpha x^2\right)^{1/2} - 1 \quad (\text{for } \alpha > 0), \\ x^\alpha \quad (\text{for } \alpha \in [1,2]), \quad x\log(x+1), \text{ and } \log(\cosh x). \end{array}$$

As noticed in [430], one can easily ascribe functional inequalities to the functions belonging to $\mathcal{C}vx_{(0,1,2,-3)}([0,\infty))$, using the following particular case of the Hardy–Littlewood–Pólya inequality of majorization.

2.6.14 Lemma (Truncated majorization) *Let $f : [0,\infty) \to \mathbb{R}$ be an increasing concave function and let $x_1, x_2, \ldots, x_n, y_1, y_2$ $(n \geq 2)$ be nonnegative numbers such that*

$$\max\{x_1, x_2, ..., x_n\} \leq \max\{y_1, y_2\}$$

and

$$x_1 + x_2 + \cdots + x_n \geq y_1 + y_2.$$

Then

$$f(x_1) + f(x_2) + \cdots + f(x_n) \geq f(y_1) + f(y_2) + (n-2)f(0).$$

Proof. Without loss of generality we may increase y_1 to $\tilde{y}_1$ and y_2 to $\tilde{y}_2$ so that

$$\sum_{k=1}^{n} x_k = \tilde{y}_1 + \tilde{y}_2$$

and also we may assume that $x_1 \geq x_2 \geq x_3 \geq \cdots \geq x_n$ and $\tilde{y}_1 \geq \tilde{y}_2$. Clearly, $\tilde{y}_1 \geq x_1$ and $\tilde{y}_1 + \tilde{y}_2 = \sum_{k=1}^{n} x_k \geq x_1 + \cdots + x_m$ for all $m \in \{1, ..., n\}$. Then the strings $\mathbf{x} = (x_1, x_2, x_3, ..., x_n)$ and $\mathbf{y} = (\tilde{y}_1, \tilde{y}_2, \underbrace{0, ..., 0}_{n-2 \text{ times}})$ fulfill the hypotheses of Theorem 6.1.3 and thus

$$\sum_{k=1}^{n} f(x_k) \geq f(z) + f(\tilde{y}_2) + (n-2)f(0).$$

The proof ends by noticing that $f(\tilde{y}_1) \geq f(y_1)$ and $f(\tilde{y}_2) \geq f(y_2)$ since f is nondecreasing. ■

A generalization of Lemma 2.6.14 makes the objective of Corollary 6.1.4, Section 6.1.

The applications of Lemma 2.6.14 often need the following ingredient:

2.6.15 Lemma *Suppose that $f : [0, \infty) \to \mathbb{R}$ is an increasing and continuous 3-concave function. Then the function $g(x) = f(x^\alpha)$ is increasing and concave for every $\alpha \in (0, 1/2]$.*

Proof. Combining Theorem 2.6.5 with Remark 2.6.3, we can reduce ourselves to the case where f is of class C^2. Since f is nondecreasing, it follows that $f' \geq 0$. According to Theorem 2.6.8, f' is a concave function, so by taking into consideration Exercise 4 (d) in Section 1.1, it results that $f'(x)/x$ is nonincreasing on $(0, \infty)$. The fact that g is increasing is clear. To prove that g is also concave it suffices to show that its derivative is decreasing. Indeed, g' can be represented as the product of two decreasing nonnegative functions,

$$g'(x) = \alpha x^{\alpha-1} f'(x^\alpha) = \alpha x^{2\alpha-1} \cdot \frac{f'(x^\alpha)}{x^\alpha},$$

and the proof is complete. ∎

Lemma 2.6.15 is not valid for $\alpha \in (1/2, \infty)$, a counterexample being provided by the function $f(x) = x^2$, $x \geq 0$.

We are now in a position to present a simple example how the above results lead to functional inequalities.

2.6.16 Example *Suppose that $f : [0, \infty) \to [0, \infty)$ is an increasing, convex, and 3-concave function such that $f(0) = 0$. Then, for all complex numbers a and b,*

$$f(|a|) + f(|b|) \leq 2f\left(\left|\frac{a-b}{2}\right|\right) + 2f\left(\left|\frac{a+b}{2}\right|\right). \tag{2.18}$$

In particular, for all $\alpha \in [1, 2]$,

$$|a|^\alpha + |b|^\alpha \leq 2\left|\frac{a-b}{2}\right|^\alpha + 2\left|\frac{a+b}{2}\right|^\alpha.$$

Indeed, according to Lemma 2.6.15, the function $g(x) = f(x^{1/2})$ is concave. It is also increasing, convex, and $g(0) = 0$. As a consequence, the inequality (2.18) follows from Lemma 2.6.14, when applied to the function g and to the elements

$$y_1 = |a|^2, \ y_2 = |b|^2,$$
$$x_1 = x_2 = \left|\frac{a-b}{2}\right|^2 \text{ and } x_3 = x_4 = \left|\frac{a+b}{2}\right|^2.$$

The fact that $y_1 + y_2 \leq x_1 + x_2 + x_3 + x_4$ is assured by the following special case of the parallelogram law: for all real numbers a and b,

$$|a|^2 + |b|^2 = 2\left|\frac{a-b}{2}\right|^2 + 2\left|\frac{a+b}{2}\right|^2.$$

Exercises

1. (A connection to Wright convexity [590]) Prove that a continuous function $f : I \to \mathbb{R}$ is convex if and only if the inequality

$$\Delta_a \Delta_b f(x) = f(a+b+x) - f(a+x) - f(b+x) + f(x) \geq 0$$

holds for all $x \in \operatorname{int} I$ and all $a, b \geq 0$ such that $x + a + b \in I$.
[Hint: Take into consideration Corollary 1.4.7.]

2. (Convexity from 4-convexity; see P. Nayar [406]) Suppose $\Phi : \mathbb{R} \to \mathbb{R}$ is a twice differentiable even function such that Φ'' is convex. Prove that for every real number s the function $\Psi(x) = \Phi(\sqrt{x}+s) + \Phi(\sqrt{x}-s)$ is convex on $\mathbb{R}_+$.

3. (The multiplicative form of the Hornich–Hlawka inequality) Prove that

$$(1+|x|)(1+|y|)(1+|z|)(1+|x+y+z|) \\ \geq (1+|x+y|)(1+|y+z|)(1+|z+x|)$$

for all $x, y, z \in \mathbb{R}$.

4. A continuous function f defined on an interval I is convex if and only if

$$\frac{1}{|J|}\int_J f(x)\mathrm{d}x \geq \frac{1}{|K|}\int_K f(x)\mathrm{d}x,$$

whenever $K \subset J$ are two compact subintervals of I with the same midpoint.

5. (C. Schötz [523]) Denote by $\mathcal{S}$ the set of all increasing, convex, and 3-concave functions defined on $[0, \infty)$ that vanish at the origin. Prove that every function $f \in \mathcal{S}$ verifies the following properties:
(a) f has a continuous derivative;
(b) $f\left(\sqrt{a+b}\right) \leq f\left(\sqrt{a}\right) + f\left(\sqrt{b}\right)$ for every $a, b \geq 0$.
(c) $f(a-b) + f(a+b) \leq 2f(a) + 2f(b)$ for every $a \geq b \geq 0$.

6. (The analogue of the Hermite–Hadamard theorem; M. Bessenyei and Zs. Páles [59]) Suppose that $f : I \to \mathbb{R}$ is a function continuous on the interval I and of class C^1 on the interior of I. Prove that the following assertions are equivalent:
(a) f is 3-convex;
(b) f verifies the inequality

$$f'\left(\frac{a+b}{2}\right) \leq \frac{f(b)-f(a)}{b-a},$$

for all points $a < b$ in $\operatorname{int} I$;

(c) f verifies the inequality

$$\frac{f(b)-f(a)}{b-a} \le \frac{1}{2}\left(\frac{f'(a)+f'(b)}{2} + f'\left(\frac{a+b}{2}\right)\right),$$

for all points $a < b$ in $\operatorname{int} I$.

2.7 Comments

The proof of the integral form of Jensen's inequality can be done using only techniques of measure theory. See L. Ambrosio, G. Da Prato, and A. Mennucci [18], pp. 55–56. The covariance form of Jensen's inequality (see Theorem 2.2.11) follows the paper of C. P. Niculescu [418].

A refinement of Jensen's inequality for "more convex" functions was proved by S. Abramovich, G. Jameson, and G. Sinnamon [3]. Call a function $\varphi\colon [0,\infty) \to \mathbb{R}$ *superquadratic* provided that for each $x \ge 0$ there exists a constant $C_x \in \mathbb{R}$ such that $\varphi(y) - \varphi(x) - \varphi(|y-x|) \ge C_x(y-x)$ for all $y \ge 0$. For example, if $\varphi\colon [0,\infty) \to \mathbb{R}$ is continuously differentiable, $\varphi(0) \le 0$ and either $-\varphi'$ is subadditive or $\varphi'(x)/x$ is increasing, then φ is superquadratic. Particularly, this is the case when $f(x) = x^p$ for $p \ge 2$, or $f(x) = x^2 \log x$. Moreover, every superquadratic positive function is convex. Their main result asserts that the inequality

$$\varphi\left(\int_X f(y)\,\mathrm{d}\mu(y)\right) \le \int_X \left[\varphi(f(x)) - \varphi\left(\left|f(x) - \int_X f(y)\,\mathrm{d}\mu(y)\right|\right)\right] \mathrm{d}\mu(x)$$

holds for all probability spaces (X, Ω, μ) and all positive μ-measurable functions f if and only if φ is superquadratic.

The prehistory of Hardy's inequality (until G. H. Hardy finally proved his inequality in 1925) is described in detail by A. Kufner, L. Maligranda, and L.-E. Persson in [313]. The further development is presented in [314], [315], and [456].

We can arrive at Hardy's inequality in many ways, for example, via mixed means. For a positive n-tuple $a = (a_1, \ldots, a_n)$, the *mixed arithmetic–geometric inequality* asserts that the arithmetic mean of the numbers

$$a_1, \sqrt{a_1 a_2}, \ldots, \sqrt[n]{a_1 a_2 \cdots a_n}$$

does not exceed the geometric mean of the numbers

$$a_1, \frac{a_1+a_2}{2}, \ldots, \frac{a_1+a_2+\cdots+a_n}{n}$$

(see K. Kedlaya [292]). As noted by B. Mond and J. E. Pečarić [393], the arithmetic and the geometric means can be replaced (in this order) by any pair (M_r, M_s) of power means with $r > s$. For $r = p > 1$ and $s = 1$, this gives us

$$\left[\frac{1}{n}\sum_{k=1}^{n}\left(\frac{a_1+a_2+\cdots+a_k}{k}\right)^p\right]^{1/p} \le \frac{1}{n}\sum_{k=1}^{n}\left(\frac{1}{k}\sum_{j=1}^{k} a_j^p\right)^{1/p}$$

so that $\sum_{k=1}^{n}((a_1+a_2+\cdots+a_k)/k)^p$ is less than or equal to

$$n^{1-p}\Big(\sum_{j=1}^{n}a_j^p\Big)\Big[\sum_{k=1}^{n}\Big(\frac{1}{k}\Big)^{1/p}\Big]^p \le \Big(\frac{p}{p-1}\Big)^p\Big(\sum_{j=1}^{n}a_j^p\Big),$$

since $\sum_0^n k^{-1/p} \le \frac{p}{p-1}n^{1-1/p}$. The integral case of this result is discussed by A. Čižmešija and J. E. Pečarić [135]. The limit case (known as Carleman's inequality), as well as its ramifications, has also received a great deal of attention in recent years. The reader may consult the papers by J. E. Pečarić and K. B. Stolarsky [465], J. Duncan and C. M. McGregor [168], M. Johansson, L.-E. Persson and A. Wedestig [278], S. Kaijser, L.-E. Persson and A. Öberg [285], and A. Čižmešija, J. E. Pečarić and L.-E. Persson [136].

Some interesting connections between the weighted inequalities of Hardy type and the spectral problems are mentioned by A. Kufner in his paper [312] and also in the Introduction to the book [315]. The following variant of Hardy's inequality is a crucial ingredient in the analysis of boundary behavior of self-adjoint, second-order, elliptic differential operators: *If* $f:[a,b]\to\mathbb{C}$ *is a continuously differentiable function with* $f(a)=f(b)=0$, *then*

$$\int_a^b \frac{|f(x)^2|}{4d(x)^2}\mathrm{d}x \le \int_a^b |f'(x)|^2\,\mathrm{d}x$$

where $d(x)=\min\{|x-a|,|x-b|\}$. See E. B. Davies [154], Lemma 1.5.1, p. 25. More applications of Hardy's inequality can be found in the books of A. Kufner, L.-E. Persson and N. Samko [315] and B. Opic and A. Kufner [456].

An account on the history, variations, and generalizations of the Chebyshev inequality can be found in the paper by D. S. Mitrinović and P. M. Vasić [390]. Other complements to Jensen's and Chebyshev's inequalities can be found in the papers by H. Heinig and L. Maligranda [248] and S. M. Malamud [350].

It is worth mentioning here the existence of a nonlinear framework for Jensen's inequality, initiated by B. Jessen [275] in 1931. Let $\mathcal{L}$ be a convex cone of real-valued functions defined on an abstract set S (that is, $\mathcal{L}$ is closed under addition and multiplication by positive scalars). We assume that $\mathcal{L}$ contains the constant function 1. A *normalized isotone sublinear functional* on $\mathcal{L}$ is any functional $A:\mathcal{L}\to\mathbb{R}$ such that

$$\begin{gathered}A(f+g)\le A(f)+A(g)\text{ for every }f,g\in\mathcal{L}\\ A(\lambda f)=\lambda A(f)\text{ for every }f\in\mathcal{L}\text{ and }\lambda\ge 0\\ f\le g\text{ in }\mathcal{L}\text{ implies }A(f)\le A(g)\\ A(\pm 1)=\pm 1.\end{gathered}$$

Some simple examples are as follows:

(a) $\mathcal{L}=c$, the space of convergent real sequences, and

$$A\left((a_n)_n\right)=\frac{1}{2}\left(a_0+\lim_{n\to\infty}a_n\right);$$

(b) $\mathcal{L} = C([a,b])$ and $A(f) = \frac{1}{b-a}\int_a^b f(t)\mathrm{d}t$ (the case that appears in Hermite–Hadamard inequality);

(c) $\mathcal{L} = C([a,b])$ and $A(f) = \sup_{x\in(a,b]} \frac{1}{x-a}\int_a^x f(t)\mathrm{d}t$;

(d) $\mathcal{L} = \mathcal{BUP}(\mathbb{R})$, the space of Bohr almost periodic functions defined on $\mathbb{R}$ (that is, the closure of the trigonometric polynomials with respect to the sup norm) and $A(f) = \lim_{T\to\infty} \frac{1}{T}\int_0^T f(t)\mathrm{d}t$.

In particular, B. Jessen [275] proved the following generalization of Jensen's inequality: *If* $A : \mathcal{L} \to \mathbb{R}$ *is a normalized isotone sublinear functional and* $\varphi : [a,b] \to \mathbb{R}$ *is a continuous convex function then, for every* $f \in \mathcal{L}$ *with* $\varphi \circ f \in \mathcal{L}$, *we have that* $A(f) \in [a,b]$ *and* $\varphi(A(f)) \le A(\varphi \circ f)$.

This fact was complemented by S. S. Dragomir, C. E. M. Pearce, and J. E. Pečarić [165], who proved the following generalization of Hermite–Hadamard inequality for isotone sublinear functionals:

$$A(\varphi \circ f) \le \frac{b\varphi(a) - a\varphi(b)}{b-a} + \frac{|\varphi(b) - \varphi(a)|}{b-a} A(\sigma f),$$

where $\sigma = \operatorname{sgn}(\varphi(b) - \varphi(a))$. See Exercise 8, Section 7.2, for more about the Hermite–Hadamard inequality.

Popoviciu's inequality [488], as stated in Theorem 2.4.3, was proved by him in a more general form that applies to all continuous convex functions $f\colon I \to \mathbb{R}$ and all finite families $x_1, \dots, x_n$ of $n \ge 2$ points with equal weights. However, his argument also covers the case of arbitrary weights $\lambda_1, \dots, \lambda_n > 0$, so that the general form of Popoviciu's inequality reads as follows:

$$\sum_{1\le i_1<\cdots<i_k\le n} (\lambda_{i_1} + \cdots + \lambda_{i_k}) f\Big(\frac{\lambda_{i_1}x_{i_1} + \cdots + \lambda_{i_k}x_{i_k}}{\lambda_{i_1} + \cdots + \lambda_{i_k}}\Big)$$
$$\le \binom{n-2}{k-2}\Big[\frac{n-k}{k-1}\sum_{i=1}^n \lambda_i f(x_i) + \Big(\sum_{i=1}^n \lambda_i\Big) f\Big(\frac{\lambda_1 x_1 + \cdots + \lambda_n x_n}{\lambda_1 + \cdots + \lambda_n}\Big)\Big].$$

Popoviciu's inequality was extended in various directions by M. Mihai and F.-C. Mitroi in [380] and C. P. Niculescu and I. Roventa [441]. An estimate from below of Popoviciu's inequality is available in [436] and [437]. The integral version of this inequality is discussed in [426] and [438]. A several variables analogue of Popoviciu's inequality is presented by M. Bencze, C. P. Niculescu and F. Popovici in their paper [48].

The proof of Jensen–Steffensen inequality in Section 1.8 is borrowed from C. P. Niculescu and M. M. Stănescu [448].

The Hornich–Hlawka inequality, in its original form, asserts that

$$\|\mathbf{x}\| + \|\mathbf{y}\| + \|\mathbf{z}\| + \|\mathbf{x}+\mathbf{y}+\mathbf{z}\| \ge \|\mathbf{x}+\mathbf{y}\| + \|\mathbf{y}+\mathbf{z}\| + \|\mathbf{z}+\mathbf{x}\| \tag{2.19}$$

for every triplet $\mathbf{x}, \mathbf{y}, \mathbf{z}$ of vectors in the Euclidean space $\mathbb{R}^N$ (and thus in any inner product space, real or complex). Its story started around 80 years ago,

when H. Hornich mentioned it as a consequence of his main result in [264]. He referred to E. Hlawka for a direct argument saying that *Für den Spezialfall* $m = 1$, $n = 2$ *hat Herr HLAWKA mir einen rein algebraischen Beweis angegeben* (in translation: For the special case $m = 1$, $n = 2$, Mr. Hlawka gave me one purely algebraic proof). Hlawka's argument is included in the Hint to Exercise 2, at the end of Section 1.9.

The Hornich–Hlawka inequality is not characteristic to inner product spaces! Indeed, by taking into account the inequality

$$|x| + |y| + |z| + |x + y + z| \geq |x + y| + |y + z| + |z + x| \quad \text{for all } x, y, z \in \mathbb{R},$$

which is an easy consequence of the ordering properties of $\mathbb{R}$, and replacing x, y, and z by elements of a real L^1 space one can easily deduce by integration that

$$\|\mathbf{x}\|_{L^1} + \|\mathbf{y}\|_{L^1} + \|\mathbf{z}\|_{L^1} + \|\mathbf{x} + \mathbf{y} + \mathbf{z}\|_{L^1} \geq \|\mathbf{x} + \mathbf{y}\|_{L^1} + \|\mathbf{y} + \mathbf{z}\|_{L^1} + \|\mathbf{z} + \mathbf{x}\|_{L^1}.$$

On the other hand, the Euclidean space $\mathbb{R}^N$ embeds linearly and isometrically into the Lebesgue space $L^1(S^{N-1})$, associated to the normalized Haar measure m on the unit sphere S^{N-1} of $\mathbb{R}^N$, invariant to rotations. The embedding $\Phi : \mathbb{R}^N \to L^1(S^{N-1})$ is defined via the formula:

$$\Phi(\mathbf{u}) = \langle \mathbf{u}, \cdot \rangle.$$

To check this, let $\mathbf{u}_0 \in S^{N-1}$ arbitrarily fixed and put

$$C = \|\Phi(\mathbf{u}_0)\|_{L^1} = \int_{S^{N-1}} |\langle \mathbf{u}_0, \mathbf{x} \rangle| \, \mathrm{d}m.$$

By rotational invariance, $C = \int_{S^{N-1}} |\langle \mathbf{u}, \mathbf{x} \rangle| \, \mathrm{d}m$ for all $\mathbf{u} \in S^{N-1}$, whence

$$\|\Phi(\mathbf{u})\|_{L^1} = \|\mathbf{u}\|_2 \quad \text{for all } \mathbf{u} \in \mathbb{R}^N.$$

In this manner, the original Hornich–Hlawka inequality is obtained by "transfer" from the L^1 spaces.

Notice that the Hornich–Hlawka inequality does not work in all Banach spaces. A counterexample is provided by $\mathbb{C}^2$, endowed with the sup norm, and the vectors $x = (1, -1)$, $y = (i, i)$, $z = (-i, 1)$.

The present state of the art concerning the Hornich–Hlawka inequality is the subject of the survey [362].



Chapter 3

Convex Sets in Real Linear Spaces

The natural domain for a convex function is a convex set. In this chapter, we review some basic facts, necessary for a deep understanding of the concept of convexity in real linear spaces. For reader's convenience, all results concerning the separation of convex sets in Banach spaces are stated in Section 2.2 with proofs covering only the particular (but important) case of Euclidean spaces. Full details in the general case are to be found in Appendix B.

3.1 Convex Sets

A subset C of a linear space E is said to be *convex* if it contains the *line segment*

$$[\mathbf{x}, \mathbf{y}] = \{(1-\lambda)\mathbf{x} + \lambda\mathbf{y} : \lambda \in [0,1]\},$$

connecting any of its points $\mathbf{x}$ and $\mathbf{y}$. As Figure 3.1 suggests, convexity is a weak form of rotundity (in the sense of M. M. Day [155]). Besides line segments, some other simple examples of convex sets in the Euclidean space $\mathbb{R}^N$ are the lines, the planes, the open disks (plus any part of their boundary), and the N-dimensional rectangles (Cartesian products of N nonempty intervals).

New examples from the old ones can be obtained by considering arbitrary intersections and/or the following two algebraic operations with sets:

$$A + B = \{\mathbf{x} + \mathbf{y} : \mathbf{x} \in A,\ \mathbf{y} \in B\}$$
$$\lambda A = \{\lambda\mathbf{x} : \mathbf{x} \in A\}$$

for $A, B \subset E$ and $\lambda \in \mathbb{R}$. See Figure 3.2. The addition of sets is also known as the *Minkowski addition*. Addition of sets is commutative and associative. One can prove easily that $\lambda A + \mu B$ is a convex set provided that A and B are convex and $\lambda, \mu \geq 0$.

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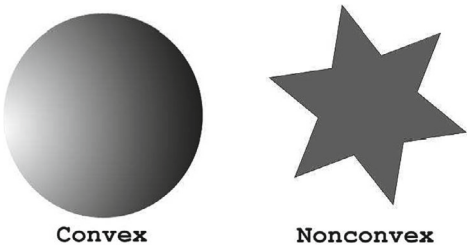


Figure 3.1: Convex and nonconvex planar sets.

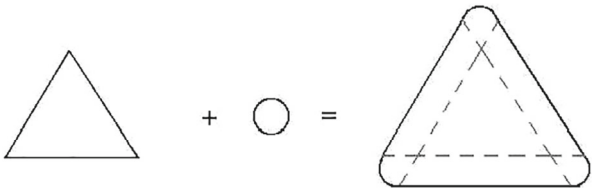


Figure 3.2: Minkowski addition of sets.

A subset A of E is said to be *affine* if it contains the whole line through any two of its points. Algebraically, this means that

$$\mathbf{x}, \mathbf{y} \in A \text{ and } \lambda \in \mathbb{R} \text{ imply } (1-\lambda)\mathbf{x} + \lambda\mathbf{y} \in A.$$

Clearly, any affine subset is also convex (but the converse is not true). It is important to notice that any affine subset A is just the translate of a (unique) linear subspace L (and all translations of a linear space represent affine sets). In fact, for every $\mathbf{a} \in A$, the translate

$$L = A - \mathbf{a}$$

is a linear space and it is clear that $A = L + \mathbf{a}$. For the uniqueness part, notice that if L and M are linear subspaces of E and $a, b \in E$ verify

$$L + \mathbf{a} = M + \mathbf{b},$$

then necessarily $L = M$ and $\mathbf{a} - \mathbf{b} \in L$. This remark allows us to introduce the concept of *dimension* for an affine set (as the dimension of the linear subspace of which it is a translate). Given a finite family $\mathbf{x}_1, \ldots, \mathbf{x}_n$ of points in E, an *affine combination* of them is any point of the form:

$$\mathbf{x} = \sum_{k=1}^{n} \lambda_k \mathbf{x}_k$$

, where $\lambda_1, \ldots, \lambda_n \in \mathbb{R}$, and $\sum_{k=1}^{n} \lambda_k = 1$. If in addition $\lambda_1, \ldots, \lambda_n \geq 0$, then $\mathbf{x}$ is called a *convex combination* of $\mathbf{x}_1, \ldots, \mathbf{x}_n$.

3.1.1 Lemma *A subset C of E is convex (respectively, affine) if and only if it contains every convex (respectively, affine) combination of points of C.*

Proof. The sufficiency part is clear, while the necessity part can be proved by mathematical induction. See the remark before Lemma 1.1.11. ■

Given a subset A of E, the intersection $\operatorname{conv}(A)$ of all convex subsets of E containing A is convex and thus it is the smallest set of this nature containing A. We call it the *convex hull* of A. By using Lemma 3.1.1, one can verify easily that $\operatorname{conv}(A)$ consists of all convex combinations of elements of A.

The affine variant of this construction yields the *affine hull* of A, denoted $\operatorname{aff}(A)$. As a consequence, we can introduce the concept of dimension for convex sets to be the dimension of their affine hulls.

A nice example of convex hull is offered by the *Gauss–Lucas theorem* on the distribution of the critical points of a polynomial: the roots $(\mu_k)_{k=1}^{n-1}$ of the derivative P' of any complex polynomial P of degree $n \geq 2$ lie in the smallest convex polygon containing the roots $(\lambda_j)_{j=1}^{n}$ of the polynomial P. Indeed assuming that w is a root of P' and $P(w) \neq 0$, we have

$$0 = \frac{P'(w)}{P(w)} = \sum_{k=1}^{n} \frac{1}{w - \lambda_k} = \sum_{k=1}^{n} \frac{\overline{w} - \overline{\lambda}_k}{|w - \lambda_k|^2},$$

whence

$$w = \sum_{k=1}^{n} \frac{1}{|w - \lambda_k|^2} \lambda_k \Bigg/ \sum_{k=1}^{n} \frac{1}{|w - \lambda_k|^2}.$$

If $(S_i)_{i \in \mathcal{I}}$ is a finite family of subsets of an N-dimensional linear space E, then the convex hull of their Minkowski addition equals the Minkowski addition of their convex hulls:

$$\operatorname{conv}\Big(\sum\nolimits_{i \in \mathcal{I}} S_i\Big) = \sum\nolimits_{i \in \mathcal{I}} \operatorname{conv}(S_i). \tag{3.1}$$

Surprisingly, this simple remark has deep consequences to the geometry of convex sets. See Theorems 3.1.3 and 3.1.8. The clue is provided by the following unifying lemma, used by R. M. Anderson in his course on Economic Theory, taught in Spring 2010 at Berkeley.

3.1.2 Lemma *Consider a finite family $(S_i)_{i \in \mathcal{I}}$ of nonempty subsets of $\mathbb{R}^N$. Then every $\mathbf{x} \in \operatorname{conv}\left(\sum_{i \in \mathcal{I}} S_i\right)$ admits a representation of the form*

$$\mathbf{x} = \sum_{i \in \mathcal{I}} \left(\sum_{1 \leq j \leq n_i} \lambda_{ij} \mathbf{x}_{ij} \right)$$

such that:

(a) $\sum_{i \in \mathcal{I}} n_i \leq |\mathcal{I}| + N$;
(b) $\mathbf{x}_{ij} \in S_i$ *and* $\lambda_{ij} > 0$ *for all* i, j;
(c) $\sum_{j=1}^{n_i} \lambda_{ij} = 1$ *for all* $i \in \mathcal{I}$.

Proof. According to formula (3.1), every point $\mathbf{x} \in \operatorname{conv}\left(\sum_{i\in\mathcal{I}} S_i\right)$ admits representations of the form $\mathbf{x} = \sum_{i\in\mathcal{I}} \mathbf{x}_i$ with $\mathbf{x}_i \in \operatorname{conv}(S_i)$ for all i. Therefore,

$$\mathbf{x} = \sum_{i\in\mathcal{I}} \left(\sum_{1\le j\le n_i} \lambda_{ij}\mathbf{x}_{ij} \right), \tag{3.2}$$

for suitable $\mathbf{x}_{ij} \in S_i$ and $\lambda_{ij} > 0$ with $\sum_{j=1}^{n_i} \lambda_{ij} = 1$. Clearly, one can choose such a representation for which $n = \sum_{i\in\mathcal{I}} n_i$ is minimal. If $n > |\mathcal{I}| + N$, then the vectors $\mathbf{x}_{ij} - \mathbf{x}_{i1}$ for $i \in \mathcal{I}$ and $j \in [2, n_i]$ are linearly dependent in $\mathbb{R}^N$. Then

$$\sum_{i\in\mathcal{I}} \sum_{2\le j\le n_i} c_{ij} \left(\mathbf{x}_{ij} - \mathbf{x}_{i1}\right) = 0$$

for some real coefficients, not all zero. Adding to Equation (3.2) the last equation multiplied by a real number λ we obtain

$$\mathbf{x} = \sum_{i\in\mathcal{I}} \sum_{1\le j\le n_i} \tilde{\lambda}_{ij}\mathbf{x}_{ij}, \tag{3.3}$$

where

$$\begin{aligned} \tilde{\lambda}_{ij} &= \lambda_{ij} + \lambda c_{ij} && \text{if } j \ge 2 \\ \tilde{\lambda}_{i1} &= \lambda_{i1} - \lambda \textstyle\sum_{2\le k\le n_i} c_{ik} && \text{if } j = 1. \end{aligned}$$

By a suitable choice of λ one can ensure that $\tilde{\lambda}_{ij} \ge 0$ for all indices i, j and that at least one coefficient $\tilde{\lambda}_{ij}$ is zero. The representation (3.3) eliminates one of x_{ij}, contrary to the minimality of n. Consequently $n \le |\mathcal{I}| + N$ and the proof is complete. ■

3.1.3 Theorem (Carathéodory's theorem) *Suppose that S is a subset of a linear space E and its convex hull* $\operatorname{conv}(S)$ *has dimension m. Then each point x of* $\operatorname{conv}(S)$ *is the convex combination of at most $m + 1$ points of S.*

Proof. Clearly, we may assume that $E = \mathbb{R}^m$. Then apply Lemma 3.1.2 for $\mathcal{I} = \{1\}$ and $S_1 = S$. ■

The sets of the form $C = \operatorname{conv}(\{\mathbf{x}_0, \ldots, \mathbf{x}_N\})$ are usually called *polytopes*. If $\mathbf{x}_1 - \mathbf{x}_0, \ldots, \mathbf{x}_N - \mathbf{x}_0$ are linearly independent, then C is called an *n-simplex* (with vertices $\mathbf{x}_0, \ldots, \mathbf{x}_N$). In this case, $\dim C = N$ and every point $\mathbf{x}$ of C has a unique representation $\mathbf{x} = \sum_{k=0}^{N} \lambda_k \mathbf{x}_k$, as a convex combination of vertices; the numbers $\lambda_0, \ldots, \lambda_N$ are called the *barycentric coordinates* of $\mathbf{x}$.

The *standard N-simplex* (or *unit N-simplex*) is the simplex Δ^N whose vertices are the elements of the canonical algebraic basis of $\mathbb{R}^{N+1}$, that is,

$$\Delta^N = \left\{ (\lambda_0, ..., \lambda_N) \in \mathbb{R}^{N+1} : \sum_{k=0}^{N} \lambda_k = 1 \text{ and } \lambda_k \ge 0 \text{ for all } k \right\}.$$

Given an arbitrary N-simplex C with vertices $(\mathbf{x}_0, \ldots, \mathbf{x}_N)$, the map

$$\omega : \Delta^N \to C, \quad \omega(\lambda_0, ..., \lambda_N) = \sum_{k=0}^{N} \lambda_k \mathbf{x}_k$$

is affine and bijective.

Notice that any polytope $\text{conv}(\{\mathbf{x}_0, \ldots, \mathbf{x}_N\})$ is a union of simplices whose vertices belong to $\{\mathbf{x}_0, \ldots, \mathbf{x}_N\}$.

3.1.4 Remark (Lagrange's barycentric identity) *Consider a finite system $\mathcal{S}$ of mass points $(\mathbf{x}_k, m_k)$ in $\mathbb{R}^{N+1}$, for $k = 1, ..., n$; $\mathbf{x}_k$ indicates position and m_k the mass. In mechanics and physics, one defines the* barycenter (*or* center of mass) *of the system by*

$$\text{bar}(\mathcal{S}) = \frac{\sum_{k=1}^n m_k \mathbf{x}_k}{\sum_{k=1}^n m_k}.$$

The mass point $(\text{bar}(\mathcal{S}), \sum_{k=1}^n m_k)$ *represents the* resultant *of the system $\mathcal{S}$. Notice that* $\text{bar}(\mathcal{S}) \in \text{conv}(\{\mathbf{x}_1, ..., \mathbf{x}_n\})$. *A practical way to determine the barycenter was found by J. L. Lagrange* [319]*, who proved the following identity:* For every family of points $\mathbf{x}, \mathbf{x}_1, ..., \mathbf{x}_n$ in $\mathbb{R}^N$ and every family of real weights $m_1, ..., m_n$ with $M = \sum_{k=1}^n m_k > 0$, we have

$$\sum_{k=1}^n m_k \|\mathbf{x} - \mathbf{x}_k\|^2 = M \left\| \mathbf{x} - \frac{1}{M} \sum_{k=1}^n m_k \mathbf{x}_k \right\|^2 + \frac{1}{M} \cdot \sum_{1 \le i < j \le n} m_i m_j \|\mathbf{x}_i - \mathbf{x}_j\|^2. \tag{3.4}$$

For the proof, use the formula $\|\mathbf{z}\|^2 = \langle \mathbf{z}, \mathbf{z} \rangle$. *The formula (3.4) yields the following variational definition of barycenter:* $\text{bar}(\mathcal{S})$ *is the unique point that minimizes the function* $\mathbf{x} \to \sum_{k=1}^n m_k \|\mathbf{x} - \mathbf{x}_k\|^2$*, that is,*

$$\text{bar}(\mathcal{S}) = \arg\min_{\mathbf{x} \in \mathbb{R}^N} \sum_{k=1}^n m_k \|\mathbf{x} - \mathbf{x}_k\|^2.$$

In Section 4.6, we will present the concept of barycenter associated to an arbitrary Borel probability measure.

Another important class of convex sets is the convex cones. A *convex cone* in a real linear space E is any subset C with the following two properties:

$$C + C \subset C$$
$$\lambda C \subset C \text{ for all } \lambda \ge 0.$$

The cone C is called *pointed* if $C \cap (-C) = \{0\}$. Here are some examples of pointed convex cones:

$\mathbb{R}^N_+ = \{(x_1, \ldots, x_N) \in \mathbb{R}^N : x_1, \ldots, x_N \ge 0\}$, the *positive orthant*;

$\left\{(x, y, z, t) \in \mathbb{R}^3 \times [0, \infty) : x^2 + y^2 + z^2 \le c^2 t^2\right\}$, the *future light cone* (c is the speed of light);

$\text{Sym}^+(N, \mathbb{R})$, the set of all *positive* (equivalently, positive semi-definite) matrices, that is, of real symmetric matrices A of $\text{M}_N(\mathbb{R})$ such that

$$\langle A\mathbf{x}, \mathbf{x} \rangle \ge 0 \quad \text{for all } \mathbf{x} \in \mathbb{R}^N.$$

An important source of pointed convex cones is provided by the theory of ordered Banach spaces, briefly summarized in Section 3.4. Two examples of convex cones which are not pointed are as follows:

$\mathbb{R}^N_{\geq} = \{(x_1, \ldots, x_N) \in \mathbb{R}^N : x_1 \geq \cdots \geq x_N\}$, the *monotone cone*

$\{(x, y, z) \in \mathbb{R}^3 : z \geq 0\}$, the *closed upper half-space.*

See the Comments section at the end of this chapter for a more comprehensive notion of cone.

So far we have not used any topology; only the linear properties of the space E have played a role. Suppose now that E is a linear normed space. The following two lemmas relate convexity and topology:

3.1.5 Lemma *If U is a convex set in a linear normed space, then its interior* $\operatorname{int} U$ *and its closure* $\overline{U}$ *are convex as well.*

Proof. For example, if $\mathbf{x},\ \mathbf{y} \in \operatorname{int} U$ and $\lambda \in (0, 1)$, then

$$\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} + \mathbf{u} = \lambda(\mathbf{x} + \mathbf{u}) + (1 - \lambda)(\mathbf{y} + \mathbf{u}) \in U$$

for all $\mathbf{u}$ in a suitable ball $B_\varepsilon(0)$. This shows that $\operatorname{int} U$ is a convex set. Now let $\mathbf{x}, \mathbf{y} \in \overline{U}$. Then there exist sequences $(\mathbf{x}_k)_k$ and $(\mathbf{y}_k)_k$ in U, converging to $\mathbf{x}$ and $\mathbf{y}$, respectively. This yields $\lambda\mathbf{x} + (1 - \lambda)\mathbf{y} = \lim_{k\to\infty}[\lambda\mathbf{x}_k + (1 - \lambda)\mathbf{y}_k] \in \overline{U}$ for all $\lambda \in [0, 1]$, that is, $\overline{U}$ is convex as well. ■

Notice that the affine sets in $\mathbb{R}^N$ are closed because finite-dimensional subspaces are always closed. Also, all cones mentioned above are closed.

As is well known, all norms on a finite-dimensional linear space are equivalent in the sense that they produce the same topology (see, e.g. R. Bhatia [63], p. 16). In the case of $\operatorname{Sym}(N, \mathbb{R})$, the natural norm is the Hilbertian norm inherited from $\mathbb{R}^{N\times N}$,

$$\|A\|_F = \left(\sum_{i,j=1}^{N} a_{ij}^2\right)^{1/2} \qquad \text{for } A = (a_{ij})_{i,j=1}^N, \tag{3.5}$$

usually known as the *Frobenius norm* (or the *Hilbert–Schmidt norm*). This norm is associated to the inner product

$$\langle A, B\rangle_F = \operatorname{trace}(B^T A),$$

where B^T denotes the transpose of B. Another important norm on $\mathrm{M}_N(\mathbb{R})$ is the *operator norm*,

$$\|A\| = \sup_{\|x\|\leq 1} \|A(x)\|,$$

the standard norm for the space of linear operators from $\mathbb{R}^N$ into itself. See Section 3.4 for more details.

The interior of $\mathbb{R}^N_+$ is the set

$$\mathbb{R}^N_{++} = \{(x_1, \ldots, x_N) \in \mathbb{R}^N : x_1, \ldots, x_N > 0\},$$

while the interior of $\mathrm{Sym}^{+}(N,\mathbb{R})$ is

$$\mathrm{Sym}^{++}(N,\mathbb{R}),$$

the set of all *strictly positive* (or *positive definite*) matrices; these are the symmetric matrices $A \in \mathrm{M}_N(\mathbb{R})$ such that

$$\langle A\mathbf{x},\mathbf{x}\rangle > 0 \quad \text{for all } \mathbf{x} \in \mathbb{R}^N,\ \mathbf{x} \neq 0.$$

Characteristic for a strictly positive matrix A is the existence of a constant $C > 0$ such that

$$\langle A\mathbf{x},\mathbf{x}\rangle \geq C\,\|\mathbf{x}\|^2 \quad \text{for all } \mathbf{x} \in \mathbb{R}^n;$$

to check this, put $C = \min\left\{\langle A\mathbf{x},\mathbf{x}\rangle : \mathbf{x} \in \mathbb{R}^N,\ \|\mathbf{x}\| = 1\right\}$ and take into account the Weierstrass extreme value theorem.

As is well known from linear algebra, every symmetric matrix has only real eigenvalues and is diagonalizable with respect to an orthonormal basis consisting of eigenvectors. Therefore, a symmetric matrix A belongs to $\mathrm{Sym}^{+}(N,\mathbb{R})$ (respectively to $\mathrm{Sym}^{++}(N,\mathbb{R})$) if and only if all its eigenvalues are positive (respectively strictly positive). See R. A. Horn and C. R. Johnson [263], Theorem 4.1.5, p. 171. We will come back to these facts in Section 3.5.

If $A \in \mathrm{Sym}^{++}(N,\mathbb{R})$, then the formula

$$\|\mathbf{x}\|_A = (\langle A\mathbf{x},\mathbf{x}\rangle)^{1/2}$$

defines a norm on $\mathbb{R}^N$, equivalent to the Euclidean norm. The associated closed balls,

$$Q = \{\mathbf{x} : \langle A(\mathbf{x}-\mathbf{v}),\mathbf{x}-\mathbf{v}\rangle \leq 1\},$$

represent *ellipsoids* centered at $\mathbf{v}$.

3.1.6 Lemma *If U is an open set in a normed linear space E, then its convex hull is open. If E is finite dimensional and K is a compact subset, then its convex hull is compact.*

Proof. For the first assertion, let $\mathbf{x} = \sum_{k=0}^{m} \lambda_k \mathbf{x}_k$ be a convex combination of elements of the open set U. Then

$$\mathbf{x} + \mathbf{u} = \sum_{k=0}^{m} \lambda_k(\mathbf{x}_k + \mathbf{u}) \quad \text{for all } \mathbf{u} \in E$$

and since U is open it follows that $\mathbf{x}_k + \mathbf{u} \in U$ for all k, provided that $\|\mathbf{u}\|$ is small enough. Consequently, $\mathbf{x}+\mathbf{u} \in \mathrm{conv}(U)$ for $\mathbf{u}$ in a ball $B_\varepsilon(\mathbf{0})$. We pass now to the second assertion. Clearly, we may assume that $E = \mathbb{R}^N$. Then, consider the map defined by

$$f(\lambda_0,\ldots,\lambda_N,\mathbf{x}_0,\ldots,\mathbf{x}_N) = \sum_{k=0}^{N} \lambda_k \mathbf{x}_k,$$

where $\lambda_0, \dots, \lambda_N \in [0,1]$, $\sum_{k=0}^{N} \lambda_k = 1$, and $\mathbf{x}_0, \dots, \mathbf{x}_N \in K$. Since f is continuous and its domain of definition $\Delta^N \times \underbrace{K \times \cdots \times K}_{n \text{ times}}$ is a compact set, then so is the range of f. According to Carathéodory's theorem, the range of f is precisely $\operatorname{conv}(K)$, and this ends the proof. ■

The union of a line and a point not on it is a closed set whose convex hull fails to be closed.

Notice also that the convex hull of a compact subset of an infinite-dimensional Banach space need not be compact. For example, consider the case of the Hilbert space ℓ^2 (of all square summable sequences of real numbers) and the compact set K consisting of elements $\mathbf{v}_n = \frac{1}{2^n}(\delta_{kn})_k$ for $n = 1, 2, 3, \dots$ and their limit, the sequence having all terms equal to zero. Then

$$\mathbf{v} = \sum_{n=1}^{\infty} \frac{1}{2^n} \mathbf{v}_n \in \overline{\operatorname{conv}(K)} \setminus \operatorname{conv}(K).$$

3.1.7 Lemma (Accessibility lemma) *Suppose that C is a convex set in a normed linear space E. If $\mathbf{x} \in \operatorname{int} C$ and $\mathbf{y} \in \overline{C}$, then the linear segment $[\mathbf{x}, \mathbf{y})$ is contained in* $\operatorname{int} C$.

Proof. Let $\varepsilon > 0$ such that $B_\varepsilon(\mathbf{x}) \subset C$. We have to show that $\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in \operatorname{int} C$ for all $\lambda \in (0,1)$. Since $\mathbf{y} \in \overline{C}$, there is a point $\mathbf{z} \in C$ such that $\|\mathbf{y} - \mathbf{z}\| < \varepsilon\lambda/(2-2\lambda)$. Suppose that $\mathbf{w} \in E$ and $\|\mathbf{w}\| < \varepsilon/2$. We have

$$\begin{aligned} \lambda\mathbf{x} + (1-\lambda)\mathbf{y} + \lambda\mathbf{w} &= \lambda\mathbf{x} + (1-\lambda)(\mathbf{y} - \mathbf{z}) + \lambda\mathbf{w} + (1-\lambda)\mathbf{z} \\ &= \lambda \left[\mathbf{x} + \mathbf{w} + (1-\lambda)\frac{\mathbf{y} - \mathbf{z}}{\lambda} \right] + (1-\lambda)\mathbf{z}. \end{aligned}$$

The point $\mathbf{w} + (1-\lambda)(\mathbf{y} - \mathbf{z})/\lambda$ belongs to $B_\varepsilon(\mathbf{0})$ since

$$\begin{aligned} \left\| \mathbf{w} + (1-\lambda)\frac{\mathbf{y} - \mathbf{z}}{\lambda} \right\| &\le \|\mathbf{w}\| + (1-\lambda)\|\mathbf{y} - \mathbf{z}\|/\lambda \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon. \end{aligned}$$

Hence $\mathbf{x} + \mathbf{w} + (1-\lambda)(\mathbf{y} - \mathbf{z})/\lambda \in B_\varepsilon(\mathbf{x}) \subset C$. As a consequence, $\lambda\mathbf{x} + (1-\lambda)\mathbf{y} + \lambda\mathbf{w} \in C$, that is, $\lambda\mathbf{x} + (1-\lambda)\mathbf{y} + \lambda B_{\varepsilon/2}(\mathbf{0}) \subset C$. Therefore, $\lambda\mathbf{x} + (1-\lambda)\mathbf{y} \in \operatorname{int} C$. ■

See Exercises 1, 6, and 12 for applications.

We next discuss an important feature of Minkowski addition of nonconvex sets in $\mathbb{R}^N$: approximate convexity of vector sums $\sum_{i \in \mathcal{I}} S_i$ of a large family of sets.

3.1.8 Theorem (Shapley–Folkman theorem, see [547]). *Consider a finite family $(S_i)_{i \in \mathcal{I}}$ of nonempty subsets of $\mathbb{R}^N$ with $|\mathcal{I}| > N$. If $\mathbf{x} \in \operatorname{conv}\left(\sum_{i \in \mathcal{I}} S_i\right)$, then there exists a subset J of $\mathcal{I}$, of cardinality at most N, such that*

$$\mathbf{x} \in \sum_{i \notin J} S_i + \operatorname{conv}\left(\sum_{i \in J} S_i \right).$$

Proof. Let $\mathbf{x} = \sum_{i\in\mathcal{I}} \left(\sum_{1\le j\le n_i} \lambda_{ij}\mathbf{x}_{ij} \right)$ be the decomposition given by Lemma 3.1.2. Because $\sum_{i\in\mathcal{I}} n_i \le |\mathcal{I}| + N$, we have $n_i = 1$ except for at most N values of i. If $n_i = 1$, then $\lambda_{i1} = 1$ and $\sum_{1\le j\le n_i} \lambda_{ij}\mathbf{x}_{ij} = \mathbf{x}_{i1} \in S_i$, which yields the conclusion of the theorem. ■

The Shapley–Folkman theorem can be connected with the following measure for the *deviation* of a nonempty compact subset S of $\mathbb{R}^N$ from its convex hull:

$$\rho(S) = \sup_{\mathbf{x}\in \mathrm{conv}(S)} \inf_{\mathbf{y}\in S} \|\mathbf{x} - \mathbf{y}\| .$$

In terms of the Pompeiu–Hausdorff distance between sets, this means

$$\rho(S) = d_H\left(S, \mathrm{conv}(S)\right).$$

Recall that the *Pompeiu–Hausdorff distance* between two nonempty subsets A and B (of a metric space $M = (M, d)$) is defined by the formula:

$$d_H(A, B) = \max\{\sup_{x\in A} \inf_{y\in B} d(x, y), \sup_{y\in B} \inf_{x\in A} d(x, y)\}.$$

A compact set S is convex if and only if $\rho(S) = 0$. It is immediate that

$$\mathrm{diam}\,(S) = \mathrm{diam}\,\mathrm{conv}\,(S) \ \text{ and } \rho(S) \le \mathrm{diam}\,(S).$$

3.1.9 Corollary *Let $(S_i)_{i\in\mathcal{I}}$ be a finite family of compact subsets of $\mathbb{R}^N$ and $L > 0$ a real number such that $\rho(S_i) \le L$ for all i. Then for any $\mathbf{x}$ in* conv $\left(\sum_{i\in\mathcal{I}} S_i\right)$ *there is $\mathbf{y} \in \sum_{i\in\mathcal{I}} S_i$ for which $\|\mathbf{x} - \mathbf{y}\| \le LN$.*

Proof. Given $\mathbf{x} \in \mathrm{conv}\left(\sum_{i\in\mathcal{I}} S_i\right)$, we infer from the Shapley–Folkman theorem the existence of a subset J of $\mathcal{I}$, of cardinality at most N, such that $\mathbf{x} = \sum_{i\in\mathcal{I}} \mathbf{x}_i$, where $\mathbf{x}_i \in S_i$ for $i \notin J$ and $\mathbf{x}_i \in \mathrm{conv}(S_i)$ for $i \in J$. According to the definition of the function ρ, for each $i \in J$ there is $\mathbf{y}_i \in S_i$ such that $\|\mathbf{x}_i - \mathbf{y}_i\| \le \rho\left(S_i\right)$. Put $\mathbf{y} = \sum_{i\in J} \mathbf{y}_i + \sum_{i\notin J} \mathbf{x}_i$. Then

$$\|\mathbf{x} - \mathbf{y}\| = \left\| \sum_{i\in J} (\mathbf{x}_i - \mathbf{y}_i) \right\| \le \sum_{i\in J} \|\mathbf{x}_i - \mathbf{y}_i\| \le \sum_{i\in J} \rho\left(S_i\right) \le NL$$

and the proof is done. ■

The estimate of Corollary 3.1.9 is independent of the size of $\mathcal{I}$ for $|\mathcal{I}| > N$. As a consequence, if $(S_n)_n$ is a sequence of compact subsets of $\mathbb{R}^N$ with $\sup_n \rho\left(S_n\right) < \infty$, then

$$d_H\left(\frac{1}{n}\left(S_1 + \cdots + S_n\right), \frac{1}{n}\,\mathrm{conv}(S_1 + \cdots + S_n)\right) \to 0$$

as $n \to \infty$. According to R. Schneider [522], p.142, the averaging of compact sets in the Minkowski sense is "asymptotically convexifying".

3.1.10 Remark (Relativization with respect to affine hull) *While working with a convex subset C of $\mathbb{R}^N$, the natural space containing it is often* $\operatorname{aff}(C)$, *not* $\mathbb{R}^N$, *which may be far too large. For example, if* $\dim C = k < n$, *then C has empty interior. We can talk more meaningfully about the topological notions of interior and boundary by using the notions of relative interior and relative boundary. If C is a convex subset of* $\mathbb{R}^N$, *the* relative interior *of C, denoted* $\operatorname{ri}(C)$, *is the interior of C relative to* $\operatorname{aff}(C)$. *That is, $a \in \operatorname{ri}(C)$ if and only if there is an $\varepsilon > 0$ such that $B_\varepsilon(a) \cap \operatorname{aff}(C) \subset C$. Every nonempty convex subset of* $\mathbb{R}^N$ *has a nonempty relative interior. The* relative boundary *of A, denoted* $\operatorname{rbd}(A)$, *is defined as* $\operatorname{rbd}(A) = \bar{A} \backslash \operatorname{ri}(A)$. *Many results concerning convex sets C in a finite-dimensional real space can be simplified by replacing E to a subspace F with* $\dim F = \dim C$ *(a fact that transforms* $\operatorname{ri}(C)$ *into* $\operatorname{int} C$*). Indeed, for every affine subset A of* $\mathbb{R}^N$ *with* $\dim A = n$, *there is a one-to-one affine transform of* $\mathbb{R}^N$ *into itself that maps A onto the subspace*

$$L = \left\{ \mathbf{x} \in \mathbb{R}^N : \mathbf{x} = (x_1, ..., x_N),\ x_{n+1} = \cdots = x_N = 0 \right\}.$$

For details, see R. T. Rockafellar [505], *Corollary* 1.6.1, *p.* 8. *More information on the relativization with respect to the affine hull is made available by Exercises* 4-7 *and Remark* 3.3.7.

3.1.11 Remark (Measurability) *A convex set in* $\mathbb{R}^N$ *is not necessarily a Borel set. Think of the open unit ball $B_1(\mathbf{0})$ to which it is added a non-Borel subset from its boundary. However, every convex set C in* $\mathbb{R}^N$ *is Lebesgue measurable. Indeed, since* $C = \bigcup_n (C \cap B_n(0))$, *we may reduce ourselves to the case where C is a convex and bounded set. If* $\operatorname{int} C = \emptyset$, *then C cannot contain any N-simplex and in this case C must lie in a subspace of dimension at most $N-1$, hence C is a Lebesgue null set (in particular, it is Lebesgue measurable). The argument in the case where* $\operatorname{int} C \neq \emptyset$ *follows from Exercise* 12.

Exercises

1. Suppose that C is a convex set in $\mathbb{R}^N$. Infer from the accessibility lemma (Lemma 3.1.7) that:
   (a) $\operatorname{int} C = \operatorname{int} \overline{C}$;
   (b) $\overline{C} = \overline{\operatorname{int} C}$ if the interior of C is nonempty.

2. Consider the disk $D = \{(x, y, z) : x^2 + y^2 \leq 1, z = 0\}$. Describe the relative interior and the relative boundary of D. Observe that in the standard topology D has empty interior and $\partial D = D$.

3. Prove that:
   (a) the convex hull of a closed set might be not closed;
   (b) if S is a bounded set, then $\operatorname{diam}(S) = \operatorname{diam}(\operatorname{conv}(S))$ and $\operatorname{conv}(\overline{S}) = \overline{\operatorname{conv}(S)}$.

4. Let $S = \{\mathbf{x}_0, \ldots, \mathbf{x}_m\}$ be a finite subset of $\mathbb{R}^N$. Prove that

$$\operatorname{ri}(\operatorname{conv} S) = \Big\{\sum_{k=0}^{m} \lambda_k \mathbf{x}_k : \lambda_k \in (0,1),\ \sum_{k=0}^{m} \lambda_k = 1\Big\}.$$

5. Suppose that A and B are convex subsets of $\mathbb{R}^N$, with $A \subset B$ and $\operatorname{aff}(A) = \operatorname{aff}(B)$. Prove that $\operatorname{ri}(A) \subset \operatorname{ri}(B)$.

6. (Extension of Accessibility Lemma 3.1.7) Suppose that A is a nonempty convex subset of $\mathbb{R}^N$, $\mathbf{x} \in \operatorname{ri}(A)$ and $\mathbf{y} \in \bar{A}$. Prove that

$$[\mathbf{x}, \mathbf{y}) = \{(1-\lambda)\mathbf{x} + \lambda\mathbf{y} : \lambda \in [0,1)\} \subset \operatorname{ri}(A).$$

   Infer that $\operatorname{ri}(A)$ and $\operatorname{int}(A)$ are both convex sets and every point of $\bar{A}$ can be approximated by points in $\operatorname{ri}(A)$. In particular, $\operatorname{ri}(A)$ is nonempty.

7. Suppose that C is a nonempty convex subset of $\mathbb{R}^N$. Prove that:

   (a) $\operatorname{ri}(C)$ and $\overline{C}$ have the same affine hull (and hence the same dimension);

   (b) $\operatorname{ri}(C) = \operatorname{ri}(\overline{C})$ and $\overline{C} = \overline{\operatorname{ri} C}$.

8. Prove that:

   (a) the map $\mathbf{x} \to \mathbf{x}/(1-\|\mathbf{x}\|^2)^{1/2}$ provides a homeomorphism between the open unit ball of the Euclidean space $\mathbb{R}^N$ and $\mathbb{R}^N$;

   (b) the map

$$(t_0, t_1, ..., t_N) \to \left(\log \frac{t_1}{t_0}, \log \frac{t_2}{t_0}, ..., \log \frac{t_N}{t_0}\right)$$

   provides a homeomorphism between the interior of the standard N-*simplex* Δ^N and $\mathbb{R}^N$.

   Remark. One can prove that all (nonempty) open convex subsets of $\mathbb{R}^N$ are homeomorphic to each other. For an argument, adapt the proof of Corollary 4.4.6.

9. A subset $S = \{\mathbf{x}_0, \ldots, \mathbf{x}_n\}$ of $\mathbb{R}^N$ is said to be *affinely independent* if the family $\{\mathbf{x}_1 - \mathbf{x}_0, \ldots, \mathbf{x}_n - \mathbf{x}_0\}$ is linearly independent. Prove that this means

$$\sum_{k=0}^{n} \lambda_k \mathbf{x}_k = 0 \text{ and } \sum_{k=0}^{n} \lambda_k = 0 \text{ imply } \lambda_k = 0 \text{ for all } k \in \{0, \ldots, n\}.$$

   Infer that an affinely independent set in $\mathbb{R}^N$ can have at most $N+1$ points.

10. (Radon's theorem) Prove that each finite set of at least $N+2$ points in $\mathbb{R}^N$ can be expressed as the union of two disjoint sets whose convex hulls have a common point.

[*Hint*: If $(\mathbf{x}_i)_{i=1}^n$ is a family of $n \geq N+2$ points in $\mathbb{R}^N$, then these points are affinely dependent, which yields real scalars λ_i, not all null, such that

$$\sum_{i=1}^{n} \lambda_i \mathbf{x}_i = 0 \quad \text{and} \quad \sum_{i=1}^{n} \lambda_i = 0.$$

We may assume (renumbering the indices) that $\lambda_1, ..., \lambda_m > 0$ and $\lambda_{m+1}, ..., \lambda_n \leq 0$. Then $\lambda = \sum_{i=1}^{m} \lambda_i > 0$ and

$$\mathbf{x} = \sum_{i=1}^{m} (\lambda_i/\lambda)\mathbf{x}_i = \sum_{i=m+1}^{n} (-\lambda_i/\lambda)\mathbf{x}_i$$

belongs to $\operatorname{conv}(\{\mathbf{x}_1, ..., \mathbf{x}_m\}) \cap \operatorname{conv}(\{\mathbf{x}_{m+1}, ..., \mathbf{x}_n\})$.]

11. (Helly's theorem) Let C_k be convex subsets of $\mathbb{R}^N$ for $1 \leq k \leq m$, where $m \geq N+1$. Suppose that every subcollection of at most $N+1$ sets has a nonempty intersection. Prove that the whole collection has a nonempty intersection.

    [*Hint:* Use induction on $m \geq N+1$. For $m = N+1$ things are clear. Suppose that $m \geq N+2$ and that the statement is true for families of $m-1$ sets. Choose $\mathbf{x}_i$ in the intersection of $C_1 \cap \cdots \cap C_{i-1} \cap \widehat{C}_i \cap C_{i+1} \cap \cdots \cap C_m$, where $\widehat{C}_i$ means that the set C_i is omitted. Then apply Radon's theorem to the family of $m \geq N+2$ points $\mathbf{x}_1, ..., \mathbf{x}_m$.]

    Remark. Helly's theorem also works for infinite families of subsets of $\mathbb{R}^N$ provided that they are convex and compact. An interesting reformulation of Helly's theorem in terms of convex inequalities is as follows: *Suppose there is given a system of inequalities*

    $$f_1(\mathbf{x}) < 0, ..., f_k(\mathbf{x}) < 0, \ f_{k+1}(\mathbf{x}) \leq 0, ..., f_m(\mathbf{x}) \leq 0,$$

    where the f_k are convex functions on $\mathbb{R}^N$, and the inequalities may be all strict or all weak. If every subsystem of $N+1$ or fewer inequalities has a solution in a given convex set C, then the entire system has a solution in C.

12. (Measurability of convex sets in $\mathbb{R}^N$) The aim of this exercise is to prove that every bounded convex subset $C \subset \mathbb{R}^N$ having interior points is Lebesgue measurable. Since $C = (C \cap \partial C) \cup \operatorname{int} C$, it suffices to prove that ∂C is a Lebesgue null set. Using the translation invariance of Lebesgue measure, we may assume that the origin 0 is interior to C.

    (a) Infer from the accessibility lemma (Lemma 3.1.7) that $\partial C \subset \frac{1}{1-\varepsilon} \operatorname{int} C$, for every $\varepsilon \in (0,1)$. Thus $\partial C \subset \left(\frac{1}{1-\varepsilon} \operatorname{int} C\right) \setminus \operatorname{int} C$.

    (b) Prove that $\operatorname{int} C \subset \frac{1}{1-\varepsilon} \operatorname{int} C$ for every $\varepsilon \in (0,1)$.

    (c) Prove that the Lebesgue measure of $\left(\frac{1}{1-\varepsilon} \operatorname{int} C\right) \setminus \operatorname{int} C$ is at most $\left(\frac{1}{1-\varepsilon}\right)^N \mathcal{L}(\operatorname{int} C) - \mathcal{L}(\operatorname{int} C)$ for every $\varepsilon \in (0,1)$.

(d) Conclude that ∂C is a Lebesgue null set.

13. A practical method for establishing convexity of a set C is to show that C is obtained from simple convex sets by operations that preserve convexity. Prove that images and inverse images of convex sets under affine functions of the form
$$f : \mathbb{R}^M \to \mathbb{R}^N, \quad f(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$$
(where A is an $N \times M$-dimensional matrix and $\mathbf{b} \in \mathbb{R}^N$) are convex sets. Applications: by scaling, translating, or projecting convex sets one obtains convex sets.

14. Extend the preceding exercise to the case of linear-fractional functions of the form
$$f : \{\mathbf{x} \in \mathbb{R}^M : \langle \mathbf{c}, \mathbf{x} \rangle + d > 0\} \to \mathbb{R}^N, \quad f(\mathbf{x}) = \frac{A\mathbf{x} + \mathbf{b}}{\langle \mathbf{c}, \mathbf{x} \rangle + d},$$
where A is an $N \times M$-dimensional matrix, $\mathbf{b} \in \mathbb{R}^N$, $\mathbf{c} \in \mathbb{R}^M$, and $d \in \mathbb{R}$.

3.2 The Orthogonal Projection

In any normed linear space E, one can speak about the *distance* from a point $\mathbf{x} \in E$ to a subset $A \subset E$. This is defined by the formula
$$d(\mathbf{x}, A) = \inf\{\|\mathbf{x} - \mathbf{a}\| : \mathbf{a} \in A\},$$
which represents a numerical indicator of how well $\mathbf{x}$ can be approximated by the elements of A. When $E = \mathbb{R}^3$ and A is the x-y-plane, the Pythagorean theorem shows that $d(\mathbf{x}, A)$ is precisely the distance between $\mathbf{x}$ and its orthogonal projection on that plane. This remark extends to the context of Hilbert spaces due to the presence of the *parallelogram law*,
$$\|\mathbf{x} - \mathbf{y}\|^2 + \|\mathbf{x} + \mathbf{y}\|^2 = 2\|\mathbf{x}\|^2 + 2\|\mathbf{y}\|^2.$$

3.2.1 Theorem *Let C be a nonempty closed convex subset of a Hilbert space H (particularly, of the Euclidean space $\mathbb{R}^N$). Then for every $\mathbf{x} \in H$ there is a unique point $P_C(\mathbf{x})$ of C such that*
$$d(\mathbf{x}, C) = \|\mathbf{x} - P_C(\mathbf{x})\|. \tag{3.6}$$
The point $P_C(\mathbf{x})$ is also characterized by the following variational inequality:
$$\langle \mathbf{x} - P_C(\mathbf{x}), \mathbf{y} - P_C(\mathbf{x}) \rangle \le 0 \quad \textit{for all } \mathbf{y} \in C. \tag{3.7}$$

We call $P_C(\mathbf{x})$ the *orthogonal projection* of $\mathbf{x}$ onto C (or the *nearest point* of C to $\mathbf{x}$). Accordingly, we call the map $P_C \colon \mathbf{x} \to P_C(\mathbf{x})$, from H into itself, the *orthogonal projection* associated to C. Clearly,
$$P_C(\mathbf{x}) \in C \quad \text{for every } \mathbf{x} \in H$$

and

$$P_C(\mathbf{x}) = \mathbf{x} \quad \text{if, and only if, } \mathbf{x} \in C.$$

In particular,

$$P_C^2 = P_C.$$

The terminology is motivated by the particular case of closed linear subspaces M of H. In their case, the variational inequality (3.7) shows that $\langle \mathbf{x} - P_M(\mathbf{x}), \mathbf{v}\rangle = 0$ for all $\mathbf{v} \in M$, that is, $\mathbf{x} - P_M(\mathbf{x})$ is orthogonal on M. Denoting by

$$M^{\perp} = \{\mathbf{y} \in H : \langle \mathbf{x}, \mathbf{y}\rangle = 0 \text{ for all } \mathbf{x} \in M\}$$

the *orthogonal complement* of M, one can easily verify that P_M is a linear self-adjoint projection and $I - P_M$ equals the orthogonal projection $P_{M^{\perp}}$. This fact is basic for the entire theory of orthogonal decompositions.

The orthogonal projection P_C is a *monotone* map in the sense that

$$\langle P_C(\mathbf{x}) - P_C(\mathbf{y}), \mathbf{x} - \mathbf{y}\rangle \geq 0 \quad \text{for all } \mathbf{x}, \mathbf{y} \in H. \tag{3.8}$$

This inequality follows by adding the inequalities $\|\mathbf{x} - P_C(\mathbf{x})\|^2 \leq \|\mathbf{x} - P_C(\mathbf{y})\|^2$ and $\|\mathbf{y} - P_C(\mathbf{y})\|^2 \leq \|\mathbf{y} - P_C(\mathbf{x})\|^2$ (and replacing the norms by the corresponding inner products). An immediate consequence of the inequality (3.8) is the property of P_C of being a Lipschitz map. Indeed,

$$\|P_C(\mathbf{x}) - P_C(\mathbf{y})\| \leq \|\mathbf{x} - \mathbf{y}\| \quad \text{for all } \mathbf{x}, \mathbf{y} \in H. \tag{3.9}$$

Proof of Theorem 3.2.1. The existence of $P_C(\mathbf{x})$ follows from the definition of the distance from a point to a set and the special geometry of the ambient space. In fact, any sequence $(\mathbf{y}_n)_n$ in C such that $\|\mathbf{x} - \mathbf{y}_n\| \to \alpha = d(\mathbf{x}, C)$ is a Cauchy sequence. This is a consequence of the following identity:

$$\|\mathbf{y}_m - \mathbf{y}_n\|^2 + 4\left\|\mathbf{x} - \frac{\mathbf{y}_m + \mathbf{y}_n}{2}\right\|^2 = 2(\|\mathbf{x} - y_m\|^2 + \|\mathbf{x} - y_n\|^2)$$

(motivated by the parallelogram law), and the definition of α as an infimum; notice that $\|\mathbf{x} - \frac{\mathbf{y}_m + \mathbf{y}_n}{2}\| \geq \alpha$, which forces $\limsup_{m,n\to\infty} \|\mathbf{y}_m - \mathbf{y}_n\|^2 = 0$. Since H is complete, there must exist a point $\mathbf{y} \in C$ at which $(\mathbf{y}_n)_n$ converges. Then necessarily $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{x}, C)$. The uniqueness of $\mathbf{y}$ with this property follows again from the parallelogram law. If $\mathbf{y}'$ is another point of C such that $d(\mathbf{x}, \mathbf{y}') = d(\mathbf{x}, C)$, then

$$\|\mathbf{y} - \mathbf{y}'\|^2 + 4\left\|\mathbf{x} - \frac{\mathbf{y} + \mathbf{y}'}{2}\right\|^2 = 2(\|\mathbf{x} - \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}'\|^2),$$

which gives us $\|\mathbf{y} - \mathbf{y}'\|^2 \leq 0$, a contradiction since it was assumed that the points $\mathbf{y}$ and $\mathbf{y}'$ are distinct. The implication (3.6) $\Rightarrow$ (3.7) follows from the fact that

$$\begin{aligned}\|\mathbf{x} - P_C(\mathbf{x})\|^2 &\leq \|\mathbf{x} - [(1-t)\,P_C(\mathbf{x}) + t\mathbf{y}]\,\|^2 = \|\mathbf{x} - P_C(\mathbf{x}) + t\,(P_C(\mathbf{x}) - \mathbf{y})\|^2 \\ &= \|\mathbf{x} - P_C(\mathbf{x})\|^2 + 2t\langle \mathbf{x} - P_C(\mathbf{x}), P_C(\mathbf{x}) - \mathbf{y}\rangle + t^2\,\|P_C(\mathbf{x}) - \mathbf{y}\|^2\end{aligned}$$

for all $t \in (0,1]$ and $\mathbf{y} \in C$. Indeed this condition can be restated as

$$2\langle \mathbf{x} - P_C(\mathbf{x}), P_C(\mathbf{x}) - \mathbf{y}\rangle + t\,\|P_C(\mathbf{x}) - \mathbf{y}\|^2 \geq 0$$

for all $t \in (0,1]$ and $\mathbf{y} \in C$ and we have to make $t \to 0$. For the implication (3.7) $\Rightarrow$ (3.6), notice that

$$0 \geq \langle \mathbf{x} - P_C(\mathbf{x}), \mathbf{y} - P_C(\mathbf{x})\rangle = \|\mathbf{x} - P_C(\mathbf{x})\|^2 - \langle \mathbf{x} - P_C(\mathbf{x}), \mathbf{x} - \mathbf{y}\rangle,$$

which yields (via the Cauchy–Bunyakovsky–Schwarz inequality),

$$\|\mathbf{x} - P_C(\mathbf{x})\| \cdot \|\mathbf{x} - \mathbf{y}\| \geq \langle \mathbf{x} - P_C(\mathbf{x}), \mathbf{x} - \mathbf{y}\rangle \geq \|\mathbf{x} - P_C(\mathbf{x})\|^2.$$

Thus $\|\mathbf{x} - \mathbf{y}\| \geq \|\mathbf{x} - P_C(\mathbf{x})\|$ for all $\mathbf{y} \in C$ and the proof is complete. ■

A simple geometric characterization of the projection onto a convex set (as well as an algorithm for finding the projection of a point on a simplex) can be found in the paper of A. Causa and F. Raciti [125].

It is important to reformulate Theorem 3.2.1 in the framework of approximation theory. Suppose that C is a nonempty closed subset in a real linear normed space E. The *set of best approximation* to $\mathbf{x} \in E$ from C is defined by

$$\mathcal{P}_C(\mathbf{x}) = \{\mathbf{z} \in C : d(\mathbf{x}, C) = \|\mathbf{x} - \mathbf{z}\|\};$$

We say that C is a *Chebyshev set* if $\mathcal{P}_C(\mathbf{x})$ is a singleton for all $\mathbf{x} \in E$, and a *proximinal set* if all the sets $\mathcal{P}_C(\mathbf{x})$ are nonempty. Theorem 3.2.1 asserts that all nonempty closed convex sets in a Hilbert space are Chebyshev sets. Is the converse true? One proves easily that every Chebyshev subset of a Banach space is necessarily (norm) closed. See Exercise 6. Convexity of such sets can be established only under additional assumptions.

3.2.2 Theorem (L. N. H. Bunt and T. S. Motzkin) *Every Chebyshev subset of* $\mathbb{R}^N$ *is convex.*

See R. Webster [582, pp. 362–365] for a proof based on Brouwer's fixed point theorem. Proofs based on the differentiability properties of the distance function $d_C\colon \mathbf{x} \to d(\mathbf{x}, C)$ are available in the paper by J.-B. Hiriart-Urruty [254], and in the monograph by L. Hörmander [268, pp. 62–63]. They are sketched in Exercises 5 and 6, Section 6.2.

V. Klee [300] proved in 1961 that all weakly closed Chebyshev subsets of a real Hilbert space are convex, but it is still unknown if the same works for all Chebyshev subsets. See [84], [254], and [578] for more information on this problem.

Hilbert spaces are not the only Banach spaces with the property that all nonempty closed convex subsets are Chebyshev sets. This is also true for all spaces $L^p(\mu)$, with $1 < p < \infty$ and, more generally, for all uniformly convex Banach spaces.

3.2.3 Definition *A Banach space is called* uniformly convex *if for every* $\varepsilon \in (0,2]$*, there exists* $\delta > 0$ *such that*

$$\|\mathbf{x}\| = \|\mathbf{y}\| = 1 \text{ and } \|\mathbf{x}-\mathbf{y}\| \geq \varepsilon \text{ imply } \left\|\frac{\mathbf{x}+\mathbf{y}}{2}\right\| \leq 1-\delta.$$

The condition $\|\mathbf{x}\| = \|\mathbf{y}\| = 1$ in the definition of uniformly convex spaces can be replaced by $\|\mathbf{x}\| \leq 1$ and $\|\mathbf{y}\| \leq 1$. See B. Beauzamy [45], Proposition 1, p. 190.

The uniform convexity of the spaces $L^p(\mu)$, with $1 < p < \infty$, follows from *Clarkson's inequalities* and the elementary fact that $(1-x)^{1/p} < 1 - x/p$ if $x \in (0,1)$ and $p > 1$. According to them, every pair of functions $u, v \in L^p(\mu)$ verifies

$$\left(\frac{\|u\|_{L^p}^p + \|v\|_{L^p}^p}{2}\right)^{1/p} \geq \left(\left\|\frac{u+v}{2}\right\|_{L^p}^q + \left\|\frac{u-v}{2}\right\|_{L^p}^q\right)^{1/q} \quad \text{for } p \in (1,2] \quad (3.10)$$

and

$$\frac{\|u\|_{L^p}^p + \|v\|_{L^p}^p}{2} \geq \left\|\frac{u+v}{2}\right\|_{L^p}^p + \left\|\frac{u-v}{2}\right\|_{L^p}^p \quad \text{for } p \in [2,\infty); \quad (3.11)$$

here $1/p + 1/q = 1$. For details, see Lemma A.2.4, Appendix A. Another simple proof is available in N. L. Carothers [119], pp. 117–119. Notice that only the case $1 < p \leq 2$ needs some effort because the case $2 \leq p < \infty$ follows from a combination between the triangle inequality and the convexity of the function x^p :

$$\begin{aligned}(|a+b|^p + |a-b|^p)^{1/p} &\leq \left(|a+b|^2 + |a-b|^2\right)^{1/2} \\ &= 2^{1/2}\left(|a|^2 + |b|^2\right)^{1/2} \leq 2^{1/2}2^{1/2-1/p}\left(|a|^p + |b|^p\right)^{1/p} \\ &\leq 2^{1-1/p}\left(|a|^p + |b|^p\right)^{1/p}.\end{aligned}$$

Clarkson's inequalities provide a substitute of the parallelogram law in the spaces $L^p(\mu)$, with $1 < p < \infty$, $p \neq 2$.

A couple of finer inequalities in L^p-spaces, the Hanner inequalities, make the object of Exercise 7, Section 4.5.

Every uniformly convex space is *strictly convex* in the sense that

$$\|\mathbf{x}\| = \|\mathbf{y}\| = 1 \text{ and } \left\|\frac{\mathbf{x}+\mathbf{y}}{2}\right\| = 1 \text{ imply } \mathbf{x} = \mathbf{y}.$$

See Exercise 8. The converse is true in the context of finite-dimensional Banach spaces and makes the object of Exercise 9. A condition equivalent to strict convexity is given in Exercise 7.

3.2.4 Theorem *Suppose that E is a uniformly convex Banach space and C is a nonempty closed convex subset of it. Then, for every $\mathbf{x} \in E$ there is a unique point $P_C(\mathbf{x})$ of C such that*

$$d(\mathbf{x}, C) = \|\mathbf{x} - P_C(\mathbf{x})\|.$$

In this context, the map $P_C : \mathbf{x} \to P_C(\mathbf{x})$ is called the *metric projection* of C.

Proof. The nontrivial case is where $\mathbf{x} \notin C$. Then $d = d(\mathbf{x}, C) > 0$ and there exists a sequence $(\mathbf{x}_n)_n$ of elements of C such that $\|\mathbf{x} - \mathbf{x}_n\| \to d$. Since C is convex,

$$d \leq \left\| \mathbf{x} - \frac{\mathbf{x}_m + \mathbf{x}_n}{2} \right\| \leq \frac{\|\mathbf{x} - \mathbf{x}_m\| + \|\mathbf{x} - \mathbf{x}_n\|}{2},$$

which yields $\left\| \frac{(\mathbf{x}-\mathbf{x}_m)+(\mathbf{x}-\mathbf{x}_n)}{2} \right\| \to d$ as $m, n \to \infty$. Then

$$\left\| \frac{\frac{\mathbf{x}-\mathbf{x}_m}{\|\mathbf{x}-\mathbf{x}_m\|} + \frac{\mathbf{x}-\mathbf{x}_n}{\|\mathbf{x}-\mathbf{x}_n\|}}{2} \right\| \to 1,$$

and taking into account the uniform convexity of E we infer that

$$\left\| \frac{\mathbf{x} - \mathbf{x}_m}{\|\mathbf{x} - \mathbf{x}_m\|} - \frac{\mathbf{x} - \mathbf{x}_n}{\|\mathbf{x} - \mathbf{x}_n\|} \right\| \to 0$$

as $m, n \to \infty$. Since

$$\begin{aligned} \|\mathbf{x}_m - \mathbf{x}_n\| &= \|(\mathbf{x} - \mathbf{x}_m) - (\mathbf{x} - \mathbf{x}_n)\| \\ &= \left\| \|\mathbf{x} - \mathbf{x}_m\| \frac{\mathbf{x} - \mathbf{x}_m}{\|\mathbf{x} - \mathbf{x}_m\|} - \|\mathbf{x} - \mathbf{x}_n\| \frac{\mathbf{x} - \mathbf{x}_n}{\|\mathbf{x} - \mathbf{x}_n\|} \right\| \\ &\leq \|\mathbf{x} - \mathbf{x}_m\| \left\| \frac{\mathbf{x} - \mathbf{x}_m}{\|\mathbf{x} - \mathbf{x}_m\|} - \frac{\mathbf{x} - \mathbf{x}_n}{\|\mathbf{x} - \mathbf{x}_n\|} \right\| + \left| \|\mathbf{x} - \mathbf{x}_m\| - \|\mathbf{x} - \mathbf{x}_n\| \right|, \end{aligned}$$

we conclude that $(\mathbf{x}_n)_n$ is a Cauchy sequence in E; its limit, say $\mathbf{z}$, verifies the condition $d = \|\mathbf{x} - \mathbf{z}\|$. The uniqueness of the point $\mathbf{z}$ of best approximation follows from the fact that E is strictly convex. ■

One can prove that all nonempty closed convex subsets of a reflexive Banach space are proximinal. See Corollary B.1.9, Appendix B. This fact extends Theorem 3.2.4 since every uniformly convex space is reflexive according to the Milman–Pettis theorem in Banach space theory. For a nice short argument of the Milman–Pettis theorem, see W. B. Johnson and J. Lindenstrauss [279], p. 31.

Exercises

1. Prove that:

   (a) the orthogonal projection of a vector $\mathbf{x} \in \mathbb{R}^N$ onto the closed unit ball of $\mathbb{R}^N$ is given by $P(\mathbf{x}) = \mathbf{x}$ if $\|\mathbf{x}\| \leq 1$ and $P(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|$ if $\|\mathbf{x}\| > 1$;

(b) the orthogonal projection of a vector $\mathbf{x} \in \mathbb{R}^N$ onto the affine set $C = \{\mathbf{x} : \langle \mathbf{x}, \mathbf{a} \rangle = b\}$ (with $\mathbf{a} \in \mathbb{R}^N \backslash \{0\}$ and $b \in \mathbb{R}$) is given by the formula $P_C(\mathbf{x}) = \mathbf{x} + (b - \langle \mathbf{x}, \mathbf{a} \rangle)\, \mathbf{a} / \|\mathbf{a}\|^2$;

(c) the orthogonal projection of a vector $\mathbf{x} \in \mathbb{R}^N$ onto the positive orthant $\mathbb{R}^N_+$ is found by replacing each strictly negative component of $\mathbf{x}$ with 0.

Remark. In connection with Exercise 1 (a), it is worth mentioning the following inequality due to J. L. Massera and J. J. Schäffer: if $\mathbf{x}$ and $\mathbf{y}$ are nonzero elements in a normed linear space, then

$$\left\| \frac{\mathbf{x}}{\|\mathbf{x}\|} - \frac{\mathbf{y}}{\|\mathbf{y}\|} \right\| \leq \frac{2}{\max\{\|\mathbf{x}\|, \|\mathbf{y}\|\}} \|\mathbf{x} - \mathbf{y}\|.$$

As was noticed by C. F. Dunkl and K. S. Williams, in the context of inner product space, the factor $2/\max\{\|\mathbf{x}\|, \|\mathbf{y}\|\}$ can be replaced by $2/(\|\mathbf{x}\| + \|\mathbf{y}\|)$. See J.-B. Hiriart-Urruty [255] for an elegant argument.

2. Find an explicit formula for the orthogonal projection $P_{\mathcal{E}}$ when $\mathcal{E}$ is the closed convex set in $\mathbb{R}^2$ bounded by the ellipse $x^2/a^2 + y^2/b^2 = 1$. Prove that for every point $A \notin \mathcal{E}$, the point $P_{\mathcal{E}}(A)$ is precisely the point X of the ellipse where the tangent to the ellipse is perpendicular to the line determined by A and X.

3. Let $\ell^\infty(2, \mathbb{R})$ be the space $\mathbb{R}^2$ endowed with the sup norm,

$$\|(x_1, x_2)\| = \sup\{|x_1|, |x_2|\},$$

and let C be the set of all vectors (x_1, x_2) such that $x_2 \geq x_1 \geq 0$. Prove that C is a nonconvex Chebyshev set.

4. Consider in $\mathbb{R}^2$ the nonconvex set $C = \{(x_1, x_2) : x_1^2 + x_2^2 \geq 1\}$. Prove that all points of $\mathbb{R}^2$, except the origin, admit a unique closest point in C.

5. (The nonexistence of metric projections in arbitrary Banach spaces) Consider the Banach space $C([0,1])$, of all continuous functions $f : [0,1] \to \mathbb{R}$, endowed with the sup norm. Prove that

$$C = \left\{ f \in C([0,1]) : \int_0^1 f(t)dt = 1,\ f(0) = 0,\ f(1) = 1 \right\}$$

is a closed convex cone in $C([0,1])$ and $d(0, C) = \inf\{\|f\| : f \in C\}$ is not attained.

6. Prove that every Chebyshev subset of a Banach space is necessarily closed.

7. Prove that a Banach space E is strictly convex if and only if whenever $\mathbf{x}$ and $\mathbf{y}$ are nonzero vectors in E, the equality $\|\mathbf{x} + \mathbf{y}\| = \|\mathbf{x}\| + \|\mathbf{y}\|$ implies $\mathbf{x} = c\mathbf{y}$ for some strictly positive number c.

[*Hint:* Suppose $\|\mathbf{x}+\mathbf{y}\| = \|\mathbf{x}\| + \|\mathbf{y}\|$, $0 < \|\mathbf{x}\| \leq 1$ and $\|\mathbf{y}\| = 1$. Then the functions $\varphi_1(t) = \|t\mathbf{x}+\mathbf{y}\|$ and $\varphi_2(t) = t\|\mathbf{x}\| + \|\mathbf{y}\|$ verify $\varphi_1(0) = \varphi_2(0)$ and $\varphi_1(1) = \varphi_2(1)$. Since φ_1 is convex and φ_2 is affine, they must be equal for all $t \geq 0$. In particular, $\left\| \frac{\mathbf{x}}{\|\mathbf{x}\|} + \mathbf{y} \right\| = 2$.]

8. Prove that every uniformly convex Banach space is also strictly convex.

9. Let E be a finite-dimensional strictly convex space. Prove that E is uniformly convex.

   [*Hint*: Use the fact that every bounded sequence of elements of E has a convergent subsequence.]

10. Prove that every real normed linear space H for which

$$\|x-y\|^2 + \|x+y\|^2 \leq 2\|x\|^2 + 2\|y\|^2$$

also verifies the parallelogram law (and thus it is isometric to an inner product space).

[*Hint*: Consider the change of variables $u = x+y$ and $v = x-y$.]

3.3 Hyperplanes and Separation Theorems in Euclidean Spaces

The notion of a hyperplane represents a natural generalization of the notion of a line in $\mathbb{R}^2$ or a plane in $\mathbb{R}^3$. Hyperplanes are useful to split the whole space into two pieces (called half-spaces). A *hyperplane* in a normed linear space E is any set of constancy of a nonzero linear functional. In other words, a hyperplane is a set of the form:

$$H = \{\mathbf{x} \in E : h(\mathbf{x}) = \alpha\}, \tag{3.12}$$

where $h\colon E \to \mathbb{R}$ is a suitable nonzero linear functional and α is a suitable scalar. In this case, the sets

$$\{\mathbf{x} \in E : h(\mathbf{x}) \leq \alpha\} \quad \text{and} \quad \{\mathbf{x} \in E : h(\mathbf{x}) \geq \alpha\}$$

are called the *half-spaces* determined by H. We say that H *separates* two sets U and V if they lie in opposite half-spaces (and *strictly separates* U and V if one set is contained in $\{\mathbf{x} \in E : h(\mathbf{x}) < \alpha\}$ and the other in $\{\mathbf{x} \in E : h(\mathbf{x}) \geq \alpha\}$). When the functional h which appears in the representation formula (3.12) is continuous (that is, when h belongs to the dual space E^*) we say that the corresponding hyperplane H is *closed*.

In the context of $\mathbb{R}^N$, all linear functionals are continuous and thus all hyperplanes are closed. In fact, any linear functional $h\colon \mathbb{R}^N \to \mathbb{R}$ has the form $h(\mathbf{x}) = \langle \mathbf{x}, \mathbf{z} \rangle$, for some $\mathbf{z} \in \mathbb{R}^N$, uniquely determined by h. This follows directly from the linearity of h and the representation of $\mathbb{R}^N$ with respect

to the canonical basis:

$$h(\mathbf{x}) = h\left(\sum_{k=1}^{N} x_k e_k\right) = \sum_{k=1}^{N} x_k h(e_k) = \langle \mathbf{x}, \mathbf{z} \rangle,$$

where $\mathbf{z} = \sum_{k=1}^{N} h(e_k) e_k$ is the *gradient* of h. Therefore, every hyperplane H in $\mathbb{R}^N$ is closed and has the form:

$$H = \{\mathbf{x} : \langle \mathbf{x}, \mathbf{z} \rangle = \alpha\}.$$

Some authors define the hyperplanes as the maximal proper affine subsets H of E. Here *proper* means different from E. One can prove that the hyperplanes are precisely the translates of codimension-1 linear subspaces, and this explains the equivalence of the two definitions. The following results on the separation of convex sets by closed hyperplanes are part of a much more general theory that will be presented in Appendix B:

3.3.1 Theorem (Separation theorem) *Let U and V be two convex sets in a normed linear space E, with $\operatorname{int} U \neq \emptyset$ and $V \cap \operatorname{int} U = \emptyset$. Then there exists a closed hyperplane that separates U and V.*

3.3.2 Theorem (Strong separation theorem) *Let K and C be two disjoint nonempty convex sets in a normed linear space E with K compact and C closed. Then there exists a closed hyperplane that separates strictly K and C.*

The special case of Theorem 3.3.2 when K is a singleton is known as the *basic separation theorem*. Important applications of this theorem can be found in Appendix B. See also Exercise 3, which deals with Farkas' theorem of the alternative (a result which historically played an important role in the development of mathematical optimization).

The proofs of Theorems 3.3.1 and 3.3.2 in the finite-dimensional case are sketched in Exercises 1 and 2.

Notice that not every pair of disjoint closed convex sets can be strictly separated. An example in $\mathbb{R}^2$ is offered by the sets $K = \{(x, 0) : x \in \mathbb{R}\}$ and $C = \{(x, y) : x \in \mathbb{R},\ y \geq \mathrm{e}^x\}$. This shows that it is essential that at least one of the sets K and C is compact.

Next we introduce the notion of a *supporting hyperplane* to a convex set A in a normed linear space E.

3.3.3 Definition *We say that the hyperplane H* supports *the set A at a point $\mathbf{a}$ in A if $\mathbf{a} \in H$ and A is contained in one of the half-spaces determined by H.*

Theorem 3.3.1 assures the existence of a supporting hyperplane to any convex set A at a boundary point, provided that A has nonempty interior. When $E = \mathbb{R}^N$, the existence of a supporting hyperplane of A at a boundary point $\mathbf{a}$ will mean the existence of a vector $z \in \mathbb{R}^N$ and of a real number α such that

$$\langle \mathbf{a}, \mathbf{z} \rangle = \alpha \quad \text{and} \quad \langle \mathbf{x}, \mathbf{z} \rangle \leq \alpha \quad \text{for all } \mathbf{x} \in A.$$

A direct argument for the existence of a supporting hyperplane in the finite-dimensional case is given in Exercise 6. We end this section with a discussion on the geometry of convex sets in finite-dimensional spaces.

3.3.4 Definition *Let U be a convex subset of a linear space E. A point $\mathbf{z}$ in U is an* extreme point *if it is not an interior point of any linear segment in U, that is, if there do not exist distinct points $\mathbf{x}, \mathbf{y} \in U$ and numbers $\lambda \in (0,1)$ such that*

$$\mathbf{z} = (1-\lambda)\mathbf{x} + \lambda\mathbf{y}.$$

The extreme points of a triangle are its vertices. More generally, every polytope $A = \text{conv}\{a_0, \ldots, a_m\}$ has finitely many extreme points, and they are among the points $a_0, \ldots, a_m$. All boundary points of a disk $\overline{D}_R(0) = \{(x,y) : x^2+y^2 \leq R^2\}$ are extreme points; this is an expression of the rotundity of disks. The closed upper half-plane $\mathbf{y} \geq 0$ in $\mathbb{R}^2$ has no extreme point. The extreme points are the landmarks of compact convex sets in $\mathbb{R}^N$:

3.3.5 Theorem (H. Minkowski) *Every nonempty convex and compact subset K of $\mathbb{R}^N$ is the convex hull of its extreme points.*

Proof. We use induction on the dimension m of K. If $m = 0$ or $m = 1$, that is, when K is a point or a closed segment, the above statement is obvious. Assume the theorem is true for all compact convex sets of dimension at most $m \leq N-1$. Consider now a compact convex set K whose dimension is $m+1$ and embed it into a linear subspace E of dimension $m+1$. If $\mathbf{z}$ is a boundary point of K, then we can choose a supporting hyperplane $H \subset E$ for K through $\mathbf{z}$. The set $K \cap H$ is compact and convex and its dimension is less than or equal to m. By the induction hypothesis, $\mathbf{z}$ is a convex combination of extreme points of $K \cap H$. Notice that any extreme point e of $K \cap H$ is also an extreme point of K. In fact, letting $H = \{t \in E : \varphi(t) = \alpha\}$, we may assume that K is included in the half-space $\varphi(t) \leq \alpha$. If $e = (1-\lambda)\mathbf{x} + \lambda\mathbf{y}$ with $\mathbf{x} \neq \mathbf{y}$ in K and $\lambda \in (0,1)$, then necessarily $\varphi(\mathbf{x}) = \varphi(\mathbf{y}) = \alpha$, that is, $\mathbf{x}$ and $\mathbf{y}$ should be in $K \cap H$, in contradiction with the choice of e. If $\mathbf{z}$ is an interior point of K, then each line through $\mathbf{z}$ intersects K in a segment whose endpoints belong to the boundary of K. Consequently, $\mathbf{z}$ is a convex combination of boundary points that in turn are convex combinations of extreme points. This ends the proof. ■

The result of Theorem 3.3.5 can be made more precise: *every point in a compact convex subset K of $\mathbb{R}^N$ is the convex combination of at most $N+1$ extreme points.* See Theorem 3.1.3.

The Krein–Milman theorem provides an extension of Minkowski's theorem to the framework of infinite-dimensional spaces. See Appendix B, Section B.3, for details.

Matrix theory related to probability and combinatorics emphasizes the importance of the *doubly stochastic* matrices. These are the square matrices $P = (p_{ij})_{i,j} \in \mathrm{M}_N(\mathbb{R})$ of positive real numbers, each of whose rows and columns sums to 1, that is,

$$\sum_{j=1}^{N} p_{ij} = 1 \text{ for } i = 1, ..., N \text{ and } \sum_{i=1}^{N} p_{ij} = 1 \text{ for } j = 1, ..., N.$$

The $N \times N$-dimensional doubly stochastic matrices constitute a compact convex set $\mathcal{B}_N$, known as the *Birkhoff polytope*. Since the conditions above form a set of $2N - 1$ independent linear constraints, the dimension of this polytope is $(N-1)^2 = N^2 - (2N-1)$. Among the elements of $\mathcal{B}_N$ we distinguish the *permutation* matrices, that is, the $N!$ matrices obtained by permuting the rows of the identity matrix. Any permutation matrix $P = (p_{ij})_{i,j=1}^N \in \mathrm{M}_N(\mathbb{R})$ is associated to a permutation π of the set $\{1, ..., N\}$ via the formula

$$p_{ij} = \delta_{i\pi(j)}.$$

3.3.6 Theorem (G. Birkhoff [67]) *The Birkhoff polytope $\mathcal{B}_N$ is the convex hull of permutation matrices.*

Proof. According to the Krein–Milman theorem, the proof of this result can be reduced to the fact that the permutation matrices are precisely the extreme points of $\mathcal{B}_N$. The fact that every permutation matrix is an extreme point of $\mathcal{B}_N$ is simple. See Exercise 9. To end the proof it remains to prove that every extreme point of $\mathcal{B}_N$ is a permutation matrix, that is, a matrix having exactly one entry $+1$ in each row and in each column and all other entries are zero. The following ingenious argument is due to C. Villani [575], p. 15.

Assume that $P = (p_{ij})_{i,j}$ is an element of $\mathcal{B}_N$ having an entry $p_{i_0 j_0}$ that lies in $(0, 1)$. Then choose distinct entries $p_{i_0 j_1}, p_{i_1 j_1}, p_{i_1 j_2}, p_{i_2 j_2}, ...$, that also lie in $(0, 1)$, and continue till $j_k = j_0$ or $i_k = i_0$. In the first case, we have an even number of entries and we can perturb these entries (and only them) to produce two new doubly stochastic matrices whose arithmetic mean is P. In the second case, leave $p_{i_0 j_0}$ out of the list and repeat the previous step with the other entries. In both cases, we conclude that P is not an extreme point. ■

The doubly stochastic matrices represent an important tool in the study of majorization. This will be presented in Chapter 6.

3.3.7 Remark (Relativization with respect to the affine hull and separation theorems) *One of the useful consequences of relative interior is that it allows us to relax interiority conditions in certain Euclidean space results. An easy application of the separation theorem yields the following fact: If C is finite-dimensional convex set in a finite-dimensional real Banach space E and $\mathbf{x} \in \mathrm{aff}(C) \setminus \mathrm{ri}(C)$, then there exists $x^* \in E^*$ such that $\sup x^*(C) \leq x^*(\mathbf{x})$ and x^* is nonconstant on C. Also, the separation theorem 3.3.1 admits the following "relative" variant: If C_1 and C_2 are convex subsets of $\mathbb{R}^N$ and $\mathrm{ri}\, C_1 \cap C_2 = \emptyset$, then there exists a nonzero vector $\mathbf{z} \in \mathbb{R}^N$ such that $\langle \mathbf{z}, \mathbf{x} \rangle \leq \langle \mathbf{z}, \mathbf{y} \rangle$ for all $\mathbf{x} \in C_1$ and all $\mathbf{y} \in C_2$.*

Exercises

1. Complete the following sketch of the proof of Theorem 3.3.2 in the case when $E = \mathbb{R}^N$:

   (a) First prove that the distance

   $$d = \inf\{\|\mathbf{x} - \mathbf{y}\| : \mathbf{x} \in K,\ \mathbf{y} \in C\}$$

   is attained for a pair $\mathbf{x}_0 \in K$ and $\mathbf{y}_0 \in C$, that is, $d = \|\mathbf{x}_0 - \mathbf{y}_0\|$.

   (b) Then notice that the hyperplane through $\mathbf{x}_0$, orthogonal to the linear segment $[\mathbf{x}_0, \mathbf{y}_0]$, determined by $\mathbf{x}_0$ and $\mathbf{y}_0$, has the equation $\langle \mathbf{y}_0 - \mathbf{x}_0, \mathbf{z} - \mathbf{x}_0 \rangle = 0$.

   (c) Fix arbitrarily a point $\mathbf{x} \in K$ and prove that $\langle \mathbf{y}_0 - \mathbf{x}_0, \mathbf{z} - \mathbf{x}_0 \rangle \leq 0$ for every point $\mathbf{z} \in [\mathbf{x}_0, \mathbf{x}]$ (and thus for every $\mathbf{z} \in K$). Conclude that every hyperplane through any point inside the segment $[\mathbf{x}_0, \mathbf{y}_0]$, orthogonal to this segment, separates strictly K and C.

2. Infer the finite-dimensional case of Theorem 3.3.1 from Theorem 3.3.2.

   [*Hint*: It suffices to assume that both sets U and V are closed. Then choose a point $\mathbf{x}_0 \in \operatorname{int} U$ and apply the preceding exercise to V and to the compact sets

   $$K_n = \{\mathbf{x}_0 + (1 - 1/n)(\mathbf{x} - \mathbf{x}_0) : \mathbf{x} \in U\} \cap \overline{B}_n(0)$$

   for $n \in \mathbb{N}^*$. This gives us a sequence of unit vectors $\mathbf{u}_n$ and numbers α_n such that $\langle \mathbf{u}_n, \mathbf{x} \rangle \leq \alpha_n$ for $\mathbf{x} \in K_n$ and $\langle \mathbf{u}_n, \mathbf{y} \rangle \geq \alpha_n$ for $\mathbf{y} \in V$. Since $(\mathbf{u}_n)_n$ and $(\alpha_n)_n$ are bounded, they admit converging subsequences, say to $\mathbf{u}$ and α, respectively. Conclude that $H = \{\mathbf{z} : \langle \mathbf{u}, \mathbf{z} \rangle = \alpha\}$ is the desired separation hyperplane.]

3. (Farkas' alternative theorem [189]) Suppose that A is an $n \times m$-dimensional real matrix and $\mathbf{b} \in \mathbb{R}^n$. Prove that precisely one of the following two alternatives is true:

   (F1) the linear system $A\mathbf{x} = \mathbf{b}$ admits a solution $\mathbf{x} \in \mathbb{R}^m_+$;

   (F2) there is $\mathbf{y} \in \mathbb{R}^n$ such that $A^*\mathbf{y} \geq \mathbf{0}$ and $\langle \mathbf{y}, \mathbf{b} \rangle < 0$.

   As an application, infer from Farkas' alternative theorem that the following system

   $$\begin{pmatrix} 4 & 1 & -5 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

   has a positive solution.

   [*Hint*: Notice first that (F1) and (F2) can't be true simultaneously. Thus, Farkas' result is equivalent to the following statement: there is a vector $\mathbf{y}$

such that $A^*\mathbf{y} \geq 0$ and $\langle \mathbf{y}, \mathbf{b} \rangle < 0$ if and only if b is not in the closed convex set $C = \left\{ A\mathbf{x} : \mathbf{x} \in \mathbb{R}^m_+ \right\}$. Then apply the strong separation theorem.]

Remark. Farkas' alternative theorem plays the same role in linear programming that the Fredholm alternative plays in linear algebra. See J. Franklin [202].

4. (Gordan's alternative theorem). Under the assumptions of Exercise 3, prove that exactly one of the following two alternatives is true:

   (G1) there is $\mathbf{x} \in \mathbb{R}^m$ such that $A\mathbf{x} < 0$;

   (G2) there is $\mathbf{y} \in \mathbb{R}^n_+$, $\mathbf{y} \neq 0$, such that $A^*\mathbf{y} = 0$.

   [*Hint:* Use Farkas' alternative theorem.]

5. (The spectrum of Markov matrices) A *Markov matrix* is a matrix $A \in \mathrm{M}_N(\mathbb{R})$ whose elements a_{ij} are positive and the sum of the elements in each column is one, that is,

$$a_{ij} \geq 0 \text{ and } \sum_{i=1}^{N} a_{ij} = 1 \text{ for all } j = 1, 2, ..., N.$$

   (a) Prove that all eigenvalues λ of a Markov matrix verify $|\lambda| \leq 1$.

   (b) Prove that if $A \in \mathrm{M}_N(\mathbb{R})$ is a Markov matrix, then 1 is an eigenvalue to which it corresponds to an eigenvector $v \in \mathbb{R}^N_+$.

   [*Hint:* (b) We have to prove the existence of a vector $\mathbf{x} \geq 0$ such that $(A - \mathrm{I})\mathbf{x} = 0$ and $\langle \mathbf{x}, \mathbf{e} \rangle = 1$, where $\mathbf{e} = (1, 1, ..., 1)$. Apply Farkas' alternative for the $(N + 1, N)$-dimensional matrix

$$\tilde{A} = \begin{pmatrix} A - \mathrm{I} \\ \mathbf{e} \end{pmatrix}$$

   and the vector $\tilde{\mathbf{b}} = (\mathbf{0}, 1) \in \mathbb{R}^{N+1}$.]

6. (The support theorem) Assume that $E = \mathbb{R}^N$ and $\mathbf{a}$ is a point in the relative boundary of the convex subset A of E. Prove that there exists a supporting hyperplane H to A at $\mathbf{a}$ which differs from $\mathrm{aff}(A)$.

   [*Hint*: We may assume that A is closed, by replacing A with $\overline{A}$. Choose a point $\mathbf{x}_0 \in S_1(\mathbf{a}) = \{\mathbf{x} : \|\mathbf{x} - \mathbf{a}\| = 1\}$ such that

$$d(\mathbf{x}_0, A) = \sup\{d(\mathbf{x}, A) : \mathbf{x} \in S_1(\mathbf{a})\},$$

   that is, $\mathbf{x}_0$ is the farthest point from A. Notice that $\mathbf{a}$ is the point of A closest to $\mathbf{x}_0$ and conclude that the hyperplane $H = \{\mathbf{z} : \langle \mathbf{x}_0 - \mathbf{a}, \mathbf{z} - \mathbf{a} \rangle = 0\}$ supports U at $\mathbf{a}$.]

7. Prove that every closed convex set in $\mathbb{R}^N$ is the intersection of closed half-spaces which contain it.

8. A set in $\mathbb{R}^N$ is a *polyhedral set* if it is a finite intersection of closed half-spaces. Prove that

   (a) Every compact polyhedral set is a polytope (the converse is also true).

   (b) Every polytope has finitely many extreme points.

   (c) $\mathrm{Sym}^+(2,\mathbb{R})$ is a closed convex cone (with interior $\mathrm{Sym}^{++}(2,\mathbb{R})$) but not a polyhedral set.

9. Prove that every permutation matrix is an extreme point of the Birkhoff polytope $\mathcal{B}_N$.

   [*Hint:* A permutation matrix has exactly one entry $+1$ in each row and in each column and all other entries are zero.]

10. Let C be a nonempty subset of a real Hilbert space H. The *polar* set of C is the set

$$C^\circ = \{\mathbf{y} \in H : \langle \mathbf{x}, \mathbf{y}\rangle \leq 1 \text{ for every } \mathbf{x} \in C\}.$$

    (a) Prove that C° is a closed convex set containing $\mathbf{0}$ and $C \subset D$ implies $D^\circ \subset C^\circ$.

    (b) (The bipolar theorem) Infer from the strong separation theorem 3.3.2 that the bipolar $C^{\circ\circ} = (C^\circ)^\circ$ of C verifies the formula:

$$C^{\circ\circ} = \overline{\mathrm{conv}}(C \cup \{\mathbf{0}\}).$$

    This implies that the polarity is an involution of the set of all closed and convex subsets of H that contain the origin.

    (c) Infer that the polar of the closed unit ball B of H is B itself and the polar of a linear subspace M is the orthogonal space $M^\perp$.

11. (The dual cone) Suppose that C is a cone in a Hilbert space H. The *dual cone* of C is the cone

$$C^* = \{\mathbf{y} \in H : \langle \mathbf{x}, \mathbf{y}\rangle \geq 0 \text{ for all } \mathbf{x} \in C\}.$$

    Prove that

    (a) The dual cone is equal to the negative of the polar cone.

    (b) $\mathbb{R}^N_+$ and $\mathrm{Sym}^+(N,\mathbb{R})$ are self-dual cones, that is, equal to their duals.

    (c) The dual of the monotone cone $\mathbb{R}^N_\geq = \{(x_1,\ldots,x_N) \in \mathbb{R}^N : x_1 \geq \cdots \geq x_N\}$ is the cone

$$\left(\mathbb{R}^N_\geq\right)^* = \left\{\mathbf{y} \in \mathbb{R}^N : \sum_{k=1}^{m} y_k \geq 0 \text{ for } m = 1,2,...,N-1, \text{ and } \sum_{k=1}^{N} y_k = 0\right\}.$$

    [*Hint:* (b) Use Abel's partial summation formula.]

12. Prove that the *norm cone* $C = \{(\mathbf{x}, t) \in \mathbb{R}^N \times [0, \infty) : \|\mathbf{x}\| \leq t\}$ is self-dual. What is happening if the natural norm on $\mathbb{R}^N$ (that is, the Euclidean norm) is replaced with an equivalent norm, for example, with $\|\mathbf{x}\|_p = \left(\sum_{k=1}^N |x_k|^p\right)^{1/p}$, for $p \in [1, \infty)$?

13. Suppose that C is a closed cone in a Hilbert space H. Prove that $P_{C^*}(\mathbf{x}) = \mathbf{x} - P_C(\mathbf{x})$ for all $\mathbf{x} \in H$ (which implies that P_C and P_{C^*} determine each other).

14. Prove that every nonempty compact and convex set K in a Hilbert space contains extreme points.

    [*Hint.* Consider the case of the points $\mathbf{z} \in K$ such that $\|\mathbf{z}\|^2 = \sup\{\|\mathbf{x}\|^2 : \mathbf{x} \in K\}$.]

3.4 Ordered Banach Spaces

An *ordered linear space* is any real linear space E endowed with a (partial) order relation $\leq$ such that for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in E$ and all numbers $\lambda \geq 0$ we have

$$\mathbf{x} \leq \mathbf{y} \text{ implies } \mathbf{x} + \mathbf{z} \leq \mathbf{y} + \mathbf{z}$$
$$\mathbf{x} \leq \mathbf{y} \text{ implies } \lambda\mathbf{x} \leq \lambda\mathbf{y}.$$

We write $\mathbf{x} < \mathbf{y}$ for $\mathbf{x} \leq \mathbf{y}$ and $\mathbf{x} \neq \mathbf{y}$; $\mathbf{y} \geq \mathbf{x}$ (respectively $\mathbf{y} > \mathbf{x}$) means $\mathbf{x} \leq \mathbf{y}$ ($\mathbf{x} < \mathbf{y}$). The set of positive elements,

$$E_+ = \{\mathbf{x} \in E : \mathbf{x} \geq 0\},$$

is a convex cone known as the *positive cone*. Necessarily, the positive cone is *pointed*, that is, $E_+ \cap (-E_+) = \{0\}$.

Conversely, every pointed convex cone C in a real linear space E makes it an ordered linear space with respect to the ordering

$$\mathbf{x} \leq \mathbf{y} \quad \text{if and only if } \mathbf{y} - \mathbf{x} \in C.$$

Consequently, for every real linear space E there is a one-to-one correspondence between the family of linear orderings and the family of pointed convex cones in E.

The natural order relation on the N-dimensional linear space $\mathbb{R}^N$ is that defined *coordinatewise* via the formula

$$\mathbf{x} \leq \mathbf{y} \text{ in } \mathbb{R}^N \text{ if and only if } x_k \leq y_k \text{ for all } k \in \{1, ..., N\}, \tag{3.13}$$

where $\mathbf{x} = (x_1, ..., x_N)$ and $\mathbf{y} = (y_1, ..., y_N)$. In this case, the positive cone is precisely the positive orthant.

Similarly, the sequence spaces c_0, c and ℓ^p for $1 \leq p \leq \infty$ become ordered linear spaces when endowed with the *coordinatewise ordering* defined as in formula (3.13), except that k runs over $\mathbb{N}$. The natural ordering in the case of function spaces such that

$$\begin{aligned} C(K) &= \{f : K \to \mathbb{R} : \ f \text{ continuous on the compact Hausdorff space } K\}, \\ C_b(X) &= \{f : X \to \mathbb{R} : \ f \text{ continuous and bounded on the metric space } X\}, \\ C_0(X) &= \{f : X \to \mathbb{R} : \ f \text{ continuous and null to infinity on the locally} \\ &\qquad \text{compact Hausdorff space } X\} \end{aligned}$$

is the *pointwise ordering*,

$$f \leq g \text{ if and only if } f(x) \leq g(x) \text{ for all } x.$$

For the Lebesgue spaces $L^p(\mu)$ ($p \in [1, \infty]$), the natural ordering is the *pointwise ordering modulo null sets*,

$$f \leq g \text{ if and only if } f(x) \leq g(x) \text{ for all } x \text{ except for a } \mu\text{-null set.}$$

If $\mathbf{x}$ and $\mathbf{y}$ are two vectors in an ordered linear space E satisfying $\mathbf{x} \leq \mathbf{y}$ then the *order interval* $[\mathbf{x}, \mathbf{y}]$ is the set defined by

$$[\mathbf{x}, \mathbf{y}] = \{\mathbf{z} \in E : \mathbf{x} \leq \mathbf{z} \leq \mathbf{y}\}$$

.

A subset C of an ordered linear space E is said to be *order convex* if for every pair of elements $\mathbf{x}, \mathbf{y} \in C$ with $\mathbf{x} \leq \mathbf{y}$ we have $[\mathbf{x}, \mathbf{y}] \subset C$. Clearly, every order interval is a convex set (though the converse is not true in general).

Suppose that E and F are two ordered linear spaces and C is an order convex subset of E. A function $f : C \to F$ is called *increasing* if $\mathbf{x} \leq \mathbf{y}$ implies $f(\mathbf{x}) \leq f(\mathbf{y})$ and *strictly increasing* if $\mathbf{x} < \mathbf{y}$ implies $f(\mathbf{x}) < f(\mathbf{y})$. Similarly, f is called *decreasing* if $\mathbf{x} \leq \mathbf{y}$ implies $f(\mathbf{x}) \geq f(\mathbf{y})$ and *strictly decreasing* if $\mathbf{x} < \mathbf{y}$ implies $f(\mathbf{x}) > f(\mathbf{y})$.

When $C = E$ and T is a linear operator, T is increasing if and only if T maps positive elements into positive elements (abbreviated, $T \geq 0$).

The concept of *ordered Banach space* is usually defined as any real ordered vector space E endowed with a complete norm such that the positive cone is closed in the norm topology. However, for the purposes of our book we will need a more restrictive concept of ordered Banach space. More precisely, we will consider *only* those ordered Banach spaces E which verify the following three properties:

(OBS1) the cone E_+ is *generating*, that is, $E = E_+ - E_+$;

(OBS2) the cone E_+ is closed;

(OBS3) the norm is *monotone on the positive cone,* that is,

$$0 \leq \mathbf{x} \leq \mathbf{y} \text{ implies } \|\mathbf{x}\| \leq \|\mathbf{y}\|.$$

An ordered linear space E such that every pair of elements $\mathbf{x}, \mathbf{y}$ admits a least upper bound $\sup\{\mathbf{x}, \mathbf{y}\}$ and a greatest lower bound $\inf\{\mathbf{x}, \mathbf{y}\}$ is called a *vector lattice*. In this case for each $x \in E$ we can define $x^+ = \sup\{\mathbf{x}, 0\}$ (the positive part of x), $\mathbf{x}^- = \sup\{-\mathbf{x}, 0\}$ (the negative part of $\mathbf{x}$) and $|\mathbf{x}| = \sup\{-\mathbf{x}, \mathbf{x}\}$ (the modulus of $\mathbf{x}$). We have $\mathbf{x} = \mathbf{x}^+ - \mathbf{x}^-$ and $|\mathbf{x}| = \mathbf{x}^+ + \mathbf{x}^-$.

A *Banach lattice* is any real Banach space E which is at the same time a vector lattice and verifies the condition

$$|x| \leq |y| \text{ implies } \|x\| \leq \|y\|.$$

All the above examples of ordered linear spaces are actually Banach lattices (with respect to their natural norms). A remark is necessary in the case of $\mathbb{R}^N$ because all norms on a finite-dimensional linear space are equivalent. See Bhatia [63], Theorem 13, p. 16. The natural norm on $\mathbb{R}^N$ is the Euclidean one, but sometimes it is more convenient to consider other norms such as

$$\|\mathbf{x}\|_p = \begin{cases} \left(\sum_{k=1}^N |x_k|^p\right)^{1/p} & \text{if } p \in [1, \infty) \\ \sup_{1 \leq k \leq N} |x_k| & \text{if } p = \infty. \end{cases}$$

To distinguish among these cases, we will make use of notations like $(\mathbb{R}^N, \|\cdot\|_p)$.

Two ordered Banach spaces E and F can be identified algebraically, isometrically, and in order if there exists a linear and bijective isometry $T : E \to F$ such that $\mathbf{x} \leq \mathbf{y}$ if and only if $T(\mathbf{x}) \leq T(\mathbf{y})$.

When endowed with the sup norm and the coordinatewise ordering, $\mathbb{R}^N$ can be identified algebraically, isometrically, and in order with the Banach lattice $C(\{1, ..., N\})$, where $\{1, ..., N\}$ carries the discrete topology.

The linear space $\mathrm{M}_N(\mathbb{R})$, endowed with the Hilbert–Schmidt norm and the order inherited from $\mathbb{R}^{N \times N}$, can be identified algebraically, isometrically, and in order with $\mathbb{R}^{N \times N}$. The positive cone is constituted by all matrices with positive coefficients.

If E is a Banach lattice, then its dual E^* is also a Banach lattice with respect to the cone

$$E_+^* = \{x^* \in E^* : x^*(\mathbf{x}) \geq 0 \quad \text{for every } \mathbf{x} \geq \mathbf{0}\}.$$

See Exercise 3.

The theory of Banach lattices can be covered from many sources. We recommend the monograph of H. H. Schaefer [519].

3.4.1 Remark *It is important to notice that all numerical inequalities of the form*

$$f(x_1, \ldots, x_n) \geq 0 \quad \textit{for } x_1, \ldots, x_n \geq 0, \tag{3.14}$$

where f is a continuous and positively homogeneous function of degree 1 *(that is, $f(\lambda x_1, \ldots, \lambda x_n) = \lambda f(x_1, \ldots, x_n)$ for $\lambda \geq 0$), extend to the context of Banach lattices, via a functional calculus invented by A. J. Yudin and J. L. Krivine. This allows us to replace the real variables of f by positive elements of a Banach lattice. See* [343, pp. 40–43]*. Particularly, this is the case of the AM–GM inequality, Rogers–Hölder inequality, and Minkowski inequality.*

A convenient way to emphasize the properties of ordered Banach spaces is presented by Davies in [153]. Davies calls an ordered Banach space E *regularly ordered* if

$$\|x\| = \inf\{\|y\| : y \in E, \ -y \le x \le y\} \quad \text{for all } x \in E.$$

This class of spaces brings together the Banach lattices and some other spaces which are not vector lattices, a simple example being presented in Exercise 1.

Another example of the same type is $\mathrm{Sym}(N, \mathbb{R})$, the ordered Banach space of all $N \times N$-dimensional symmetric matrices with real coefficients. This space is endowed with the Löwner order,

$$A \le B \text{ if and only if } \langle Ax, x\rangle \le \langle Bx, x\rangle \quad \text{for all } x \in H$$

and the norm $\|A\| = \sup\{\|A(x)\| : \|x\| \le 1\}$. Since

$$\|A\| = \sup\{|\langle Ax, x\rangle| : \|x\| \le 1\} \tag{3.15}$$

(see Bhatia [63], p. 120), it follows that

$$\pm A \le \|A\| I, \tag{3.16}$$

where I denotes the identity of $\mathbb{R}^N$. Therefore,

$$\|A\| = \inf\{\|B\| : -B \le A \le B\} \quad \text{for all } A \in \mathrm{Sym}(N, \mathbb{R}). \tag{3.17}$$

Notice also that for all $A, B \in \mathrm{Sym}(N, \mathbb{R})$ and $X \in M_N(\mathbb{R})$,

$$A \le B \text{ implies } X^*AX \le X^*BX. \tag{3.18}$$

3.4.2 Lemma *Every ordered Banach space can be renormed by an equivalent norm to become a regularly ordered Banach space.*

For details, see Namioka [403]. Some other useful properties of ordered Banach spaces are listed below.

3.4.3 Lemma *Suppose that E is a regularly ordered Banach space. Then:*

(a) There exists a constant $C > 0$ such that every element $x \in E$ admits a decomposition of the form $x = u - v$, where $u, v \in E_+$ and $\|u\|, \|v\| \le C\|x\|$.

(b) The dual space E^* of E, when endowed with the dual cone

$$E_+^* = \{x^* \in E^* : x^*(x) \ge 0 \text{ for all } x \in E_+\},$$

is a regularly ordered Banach space.

(c) $x \le y$ in E is equivalent to $x^*(x) \le x^*(y)$ for all $x^* \in E_+^*$.

(d) $\|x\| = \sup\{x^*(x) : x^* \in E_+^*, \ \|x^*\| \le 1\}$ for all $x \in E_+$.

Proof. The assertion (a) follows immediately from Lemma 3.4.2. For (b), see Davies [153], Lemma 2.4. The assertion (c) is an easy consequence of the Hahn–Banach separation theorem 3.3.2. The assertion (d) is also a consequence of the Hahn–Banach separation theorem; for full details, see [520], Theorem 4.3, p. 223. ■

The concept of strictly positive function admits a natural extension in the framework of ordered Banach spaces. Precisely, a positive element of an ordered Banach space E is said to be *strictly positive* provided that

$$x^*(x) > 0 \text{ for every nonzero functional } x^* \in E_+^*.$$

Strictly positive elements exist in every separable ordered Banach space E. For example, choose a sequence $(x_n)_n$ of elements of E, dense in the closed unit ball and consider their decomposition $x_n = u_n - v_n$ according to Lemma 3.4.3 (a). Then $x = \sum 2^{-n}(u_n + v_n)$ is a strictly positive element since any functional $x^* \in E_+^*$ vanishing at x will vanish on all elements x_n. By density, this implies that $x^* = 0$. Actually a stronger results holds, precisely, the set of all strictly positive elements of a separable ordered Banach space is dense into the positive cone. See [520], Theorem 7.6, p. 241.

An example of a Banach lattice without strictly positive elements is $C_0(X)$, provided that X is an uncountable set endowed with the discrete metric.

The spaces $C(K)$, $C_b(X)$, and $\mathcal{U}C_b(X)$ (separable or not) admit strictly positive elements with stronger properties, called order units. Recall that an element $u > 0$ of an ordered Banach space E is called an *order unit* if for each $x \in E$ there exists a real number $\lambda > 0$ such that $-\lambda u \leq x \leq \lambda u$. Necessarily, an order unit is a strictly positive element.

Another example of ordered Banach space with order unit is $\mathrm{Sym}(N, \mathbb{R})$, the unit being the identity I of $\mathbb{R}^N$. Together, these examples illustrate the importance of regularly ordered Banach spaces with an order unit u, whose norms coincide with the norms associated to the unit via the formula

$$\|x\|_u = \inf \{\lambda > 0 : -\lambda u \leq x \leq \lambda u\}.$$

The natural order unit of the vector lattice $\mathbb{R}^N$ is the vector $\mathbf{u}$, whose all components are equal to 1. The norm associated to this unit is the sup norm.

The following result (due to Gal and Niculescu [209]) explains the important role played by the Banach lattices $C(K)$ in the study of ordered Banach space-valued functions.

3.4.4 Theorem *Every ordered Banach space E can be represented as a linear subspace of a Banach lattice $C(K)$ via a linear and continuous map $\Phi : E \to C(K)$ such that $\Phi(x) \geq 0$ if and only if $x \geq 0$. If E has a strictly positive element e, then one can choose K such that $\Phi(e) = 1$, the unity of $C(K)$.*

Proof. According to the Banach–Alaoglu theorem (Theorem B.1.6, Appendix B), the set $K = \left\{x^* \in E_+^* : \|x^*\| \leq 1\right\}$ is compact relative to the weak-star topology. Taking into account the assertions (c) and (d) of Lemma 3.4.3 one can

easily conclude that E embeds into $C(K)$ via the positive linear isometry

$$\Phi : E \to C(K), \quad (\Phi(x))(x^*) = x^*(x).$$

For the second part of Theorem 3.4.4, notice that $\Phi(e)$ is an order unit for $C(K)$. Then the conclusion follows from a classical result due to Kadison [284], stating that every ordered real linear space with an order unit can be represented as a linear subspace of the space of continuous real-valued functions on a compact Hausdorff space via an order-preserving map that carries the order unit to the constant function 1. ■

Exercises

1. Prove that $\mathrm{Aff}([0,1])$, the linear space of all real-valued affine functions defined on $[0,1]$ is an ordered Banach space when endowed with the pointwise ordering and the sup norm. This is not a Banach lattice but the norm of every $f \in \mathrm{Aff}([0,1])$ verifies the formula

$$\|f\| = \inf\{\|g\| : g \in \mathrm{Aff}([0,1]),\ -g \le f \le g\}.$$

2. Suppose that E is a Banach lattice and $x^* : E \to \mathbb{R}$ is a positive linear functional, that is, a linear functional such that

$$\mathbf{x} \in E_+ \text{ implies } x^*(\mathbf{x}) \ge 0.$$

Prove that x^* is necessarily continuous and

$$\|x^*\| = \sup\{x^*(\mathbf{x}) : \mathbf{x} \in E_+,\ \|\mathbf{x}\| \le 1\}.$$

3. Suppose that E is a Banach lattice. Prove that E^* is also a Banach lattice, where the absolute value of a functional x^* verifies the formula

$$|x^*|(\mathbf{x}) = \sup\{x^*(\mathbf{y}) : \mathbf{y} \in E,\ |\mathbf{y}| \le \mathbf{x}\} \quad \text{for all } \mathbf{x} \ge 0.$$

Moreover, every family $(x^*_\alpha)_\alpha$ of elements of E^* which is bounded above by a functional $z^* \in E^*$ admits a least upper bound.

[*Hint.* Without loss of generality we may assume that the family $\mathcal{X} = (x^*_\alpha)_\alpha$ is upward directed (that is, every pair of elements is bounded above by an element of the family $\mathcal{X}$). For this, add to $\mathcal{X}$ the set of suprema of all finite subfamilies. Then $(\sup_\alpha x^*_\alpha)(\mathbf{x}) = \sup_\alpha x^*_\alpha(\mathbf{x})$ for all $\mathbf{x} \ge 0$.]

4. (a) Prove that a matrix $A \in \mathrm{M}_N(\mathbb{R})$ maps $\mathbb{R}^N_+$ into $\mathbb{R}^N_+$ if and only if all its coefficients are positive. These matrices constitute a generating pointed convex cone.

(b) Prove that the order induced by the aforementioned cone makes $\mathrm{M}_N(\mathbb{R})$ a Banach lattice when endowed with the Hilbert–Schmidt norm. Describe A^+, A^- and $|A|$ for $A \in M_N(\mathbb{R})$.

(c) Consider the case of the matrix $\begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$, to see that the spectrum of a matrix with strictly positive coefficients may contain negative eigenvalues.

5. (A nonlinear Perron-type theorem) Suppose that $A : \mathbb{R}^N \to \mathbb{R}^N$ is a continuous map such that the following three conditions are fulfilled:
   (a) $A(\mathbf{x}+\mathbf{y}) \geq A(\mathbf{x}) + A(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$;
   (b) $A(\lambda\mathbf{x}) = \lambda A(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^N$ and $\lambda \in \mathbb{R}_+$;
   (c) if $\mathbf{x} \in \mathbb{R}^N_+$, $\mathbf{x} \neq \mathbf{0}$, then $A(\mathbf{x}) > \mathbf{0}$.
   Prove that there is a strictly positive number λ and a vector $\mathbf{v} > \mathbf{0}$ such that $A\mathbf{v} = \lambda\mathbf{v}$.
   [*Hint:* The set
   $$S = \left\{\mu : \mu \in \mathbb{R}_+ \text{ and } A\mathbf{x} \geq \mu\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^N_+,\ \mathbf{x} \neq \mathbf{0}\right\}$$
   is nonempty and compact and one can choose λ as the maximum of S.]
   Remark. The classical Perron theorem asserts that if a matrix has strictly positive entries, then its largest eigenvalue is strictly positive, occurs with multiplicity 1, and the corresponding eigenvector can be chosen to have strictly positive components. See R. A. Horn and C. R. Johnson [263], Theorem 8.2.2, p. 496.

3.5 More on the Spectral Properties of Symmetric Matrices

An important result of linear algebra is the *spectral theorem* which asserts that every matrix $A \in \mathrm{Sym}(N, \mathbb{R})$ can be factorized as

$$A = UDU^{-1}, \tag{3.19}$$

where D is a diagonal matrix carrying the eigenvalues of A according to their multiplicity,

$$D = \begin{pmatrix} \lambda_1(A) & & 0 \\ & \ddots & \\ 0 & & \lambda_N(A) \end{pmatrix},$$

and $U \in \mathrm{M}_N(\mathbb{R})$ is an *orthogonal matrix*, that is, a real square matrix whose columns (and rows) are orthonormal vectors. Indeed, if U is an orthogonal matrix, then

$$\langle U\mathbf{x}, U\mathbf{y}\rangle = \langle \mathbf{x}, \mathbf{y}\rangle \quad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^N.$$

Notice that $U^{-1} = U^T$, due to the fact that $UU^T = U^TU = I$.

The proof of the spectral theorem is sketched in Exercise 1.

This result allows us to define the functions of a symmetric matrix $A \in \mathrm{Sym}(N, \mathbb{R})$ in a natural way. Given a function $f : \mathbb{R} \to \mathbb{R}$, we define $f(A)$ via the formula

$$f(A) = U \begin{pmatrix} f(\lambda_1(A)) & & 0 \\ & \ddots & \\ 0 & & f(\lambda_N(A)) \end{pmatrix} U^T.$$

3.5.1 Remark *One can easily verify that if f is a real polynomial, say $f(x) = \sum_{k=1}^{m} c_k x^k$, then*

$$f(A) = \sum_{k=1}^{m} c_k A^k. \tag{3.20}$$

Another important remark is the fact that $f(A)$ commutes with every symmetric matrix that commutes with A.

The main features of this construction are listed as follows:

$$\begin{cases} (f+g)(A) = f(A) + g(A) \\ (fg)(A) = f(A)g(A) = g(A)f(A) \\ f \geq 0 \text{ implies } f(A) \geq 0 \\ \|f(A)\| = \sup\{|f(\lambda)| : \lambda \in \sigma(A)\}. \end{cases} \tag{3.21}$$

The last property follows by combining Exercises 2, 3, and 4.

According to the last remark, one can define the *positive part* A^+, the *negative part* A, and the *modulus* $|A|$ for every symmetric matrix A. Clearly, $A = A^+ - A^-$, $A^+A^- = A^-A^+ = 0$, and $|A| = A^+ + A^-$, but in general it is not true that $|A+B| \leq |A| + |B|$. See Exercise 5 for an example.

If A is a positive matrix, then all its eigenvalues are positive. Every positive number has a unique positive square root. The analogous result for positive matrices also holds.

3.5.2 Theorem *For every matrix $A \in \mathrm{Sym}^+(N, \mathbb{R})$ there exists a unique $B \in \mathrm{Sym}^+(N, \mathbb{R})$ such that*

$$B^2 = A.$$

We will refer to B as the square root *of A and denote it $A^{1/2}$. This operator is a polynomial of A and thus it commutes with every operator that commutes with A.*

Proof. One can define B as

$$f(A) = U \begin{pmatrix} (\lambda_1(A))^{1/2} & & 0 \\ & \ddots & \\ 0 & & (\lambda_N(A))^{1/2} \end{pmatrix} U^{-1}.$$

For the uniqueness part, suppose that C is any positive matrix such that $C^2 = A$. Then

$$C^3 = CA = AC,$$

so C commutes with B. Let $\mathbf{x}$ be arbitrarily fixed in $\mathbb{R}^N$. Then the vector $\mathbf{y} = (B - C)\,x$ verifies

$$\langle B\mathbf{y}, \mathbf{y}\rangle + \langle C\mathbf{y}, \mathbf{y}\rangle = \langle (B+C)\mathbf{y}, \mathbf{y}\rangle = \langle \left(B^2 - C^2\right) x, \mathbf{y}\rangle = 0,$$

whence $\langle B\mathbf{y}, \mathbf{y}\rangle = \langle C\mathbf{y}, \mathbf{y}\rangle = 0$ (since B and C are positive). This implies

$$0 = \langle (B-C)\mathbf{y}, \mathbf{y}\rangle = \langle (B-C)^3\mathbf{x}, \mathbf{x}\rangle,$$

and taking into account that $\mathbf{x}$ was arbitrarily fixed we infer that $(B-C)^3 = 0$. Since $B - C$ is symmetric, we conclude that $B = C$. See Exercise 2. ■

3.5.3 Corollary *If $A, B \in \mathrm{Sym}^{++}(N, \mathbb{R})$, then $A \leq B$ if and only if $B^{-1} \leq A^{-1}$.*

Proof. If $B = I$, then from formula (3.18) we infer that

$$A^{-1/2} A A^{-1/2} \leq A^{-1/2} I A^{-1/2}$$

, that is, $I \leq A^{-1}$. In the general case,

$$A \leq B \;\Rightarrow\; B^{-1/2} A B^{-1/2} \leq I \;\Rightarrow I \leq B^{1/2} A^{-1} B^{1/2} \;\Rightarrow B^{-1} \leq A^{-1}.$$

■

Given an arbitrary matrix $A \in \mathrm{M}_N(\mathbb{R})$, the matrix $A^T A$ is always positive. We call the matrix

$$|A| = \left(A^T A\right)^{1/2} \tag{3.22}$$

the *modulus* of A since for a symmetric matrix, this definition of modulus agrees with that introduced before in this section. The reason for taking $A^T A$ rather than AA^T in (3.22) is that with this choice

$$\| |A| \, \mathbf{x} \| = \| A\mathbf{x} \| \quad \text{for all } \mathbf{x},$$

a fact that leads to the polar decomposition of a matrix.

The eigenvalues $s_1(A), s_2(A), ..., s_N(A)$ of $|A|$ counted by multiplicity are called the *singular values* of A.

3.5.4 Theorem (Polar decomposition of a matrix) *Every matrix $A \in \mathrm{M}_N(\mathbb{R})$ admits a decomposition of the form $A = U\,|A|$, where U is a suitable unitary matrix. If A is strictly positive, then U is uniquely determined by A.*

Proof. Notice first that $\|A\mathbf{x}\|^2 = \langle A\mathbf{x}, A\mathbf{x}\rangle = \langle A^T A\mathbf{x}, \mathbf{x}\rangle = \| |A| \, \mathbf{x}\|^2$, so $\ker A = \ker |A|$ and A and $|A|$ have the same rank n. Since $|A|$ is symmetric, there exists an orthonormal basis $\varphi_1, ..., \varphi_N$ of $\mathbb{R}^N$ consisting of eigenvectors of $|A|$ and $|A|\,\varphi_k = s_k(A)\,\varphi_k$ for all k. The vectors $\frac{1}{s_1(A)} A\varphi_1, ... \frac{1}{s_n(A)} A\varphi_n$ form an orthonormal family since

$$\begin{aligned} \langle \frac{1}{s_i(A)} A\varphi_i, \frac{1}{s_j(A)} A\varphi_j \rangle &= \frac{1}{s_i(A)\, s_j(A)} \langle A\varphi_i, A\varphi_j \rangle \\ &= \frac{s_i^2(A)}{s_i(A)\, s_j(A)} \langle \varphi_i, \varphi_j \rangle = \delta_{ij} \end{aligned}$$

for all $i, j \in \{1, ..., n\}$. If $n < N$ we complete this family with vectors $\psi_{n+1}, ..., \psi_N$ in $\ker A = \ker |A|$ to obtain an orthonormal basis of $\mathbb{R}^N$. The proof ends by choosing U as the matrix that acts on this new basis as follows:

$$U(\varphi_k) = \frac{1}{s_k(A)} A\varphi_k \text{ for } 1 \leq k \leq n \text{ and } U\varphi_k = \psi_k \text{ for } n+1 \leq k \leq N.$$

■

The polar decomposition easily yields the *singular value decomposition*: for every matrix $A \in \mathrm{M}_N(\mathbb{R})$ there exist two orthogonal matrices U and V such that

$$A = UDV^T,$$

where D is the diagonal matrix whose diagonal entries are the singular values of A. Equivalently, there exist two orthonormal families $(\mathbf{u}_k)_k$ and $(\mathbf{v}_k)_k$ such that

$$A = \sum_{k=1}^{N} s_k(A)\langle \cdot, \mathbf{u}_k\rangle \mathbf{v}_k.$$

An immediate consequence of the spectral theorem is the following result on the *simultaneous diagonalization* of commuting matrices.

3.5.5 Theorem *For every pair $A, B \in \mathrm{Sym}(N, \mathbb{R})$, of commuting matrices, there exists a basis of common eigenvectors that diagonalizes them both.*

Proof. A short argument consists in choosing an orthogonal matrix U that diagonalizes A as $A = UDU^T$. Then, denote $BD = U^T BU$ and compute the eigenvectors of BD, say $v_1, ..., v_N$. Finally, notice that the vectors $w_1, ..., w_N$, where $w_k = Uv_k$ are eigenvectors for both A and B. ■

It is also worth mentioning here the following result on simultaneous diagonalization of noncommuting matrices:

3.5.6 Lemma *If $A \in \mathrm{Sym}^{++}(n, \mathbb{R})$ and $B \in \mathrm{Sym}(n, \mathbb{R})$, then there exists a nonsingular matrix C such that $C^T AC = I$ and $C^T BC$ is diagonal.*

Proof. According to the spectral decomposition, one can find an orthogonal matrix C_1 such that $C_1^T AC_1$ is diagonal, and a strictly positive diagonal matrix D such that $D(C_1^T AC_1) = I$. There exists also an orthogonal matrix C_2 such that $C_2^T(D^{1/2}\left(C_1^T BC_1\right)D^{1/2})C_2$ is diagonal. Put $C = C_1 D^{1/2} C_2$. Then $C^T AC = I$ and $C^T BC$ is diagonal. ■

Much of the results presented in this section extend to the case of symmetric compact operators acting on a real Hilbert space. See the Comments at the end of this chapter.

Exercises

1. (Proof of the spectral theorem) Suppose that $A \in \mathrm{Sym}(N, \mathbb{R})$. Prove that:

   (a) all eigenvalues of $A \in \mathrm{Sym}(N, \mathbb{R})$ are real;

   (b) the eigenvectors corresponding to different eigenvalues are orthogonal;

   (c) If $\lambda_1(A), ..., \lambda_N(A)$ are the eigenvalues of A counted by multiplicity, then the space $\mathbb{R}^N$ admits an orthonormal basis consisting of eigenvectors $\mathbf{v}_1, ..., \mathbf{v}_N$ such that $A\mathbf{v}_k = \lambda_k(A)\mathbf{v}_k$ for all k.

(d) The assertion (c) proves the following representation of A :

$$A\mathbf{x} = \sum_{k=1}^{N} \lambda_k(A)\langle \mathbf{x}, \mathbf{v}_k\rangle \mathbf{v}_k \quad \text{for every } \mathbf{x} \in \mathbb{R}^N.$$

2. Let $A \in \mathrm{Sym}(N, \mathbb{R})$ and $f : \mathbb{R} \to \mathbb{R}$. Prove that $f(A) = 0$ if and only if $f = 0$ on $\sigma(A)$. Infer that every function of A is nothing but a polynomial of A and $\sigma(A) = \{\lambda\}$ if and only if $A = \lambda I$.

3. (Spectral mapping theorem for polynomials) Use the Fundamental Theorem of Algebra, to prove that for every matrix $A \in \mathrm{M}_N(\mathbb{C})$ and every polynomial P with complex coefficients,

$$\sigma(P(A) = P(\sigma(A)).$$

4. Prove that the norm of every matrix $A \in \mathrm{Sym}(N, \mathbb{R})$ verifies the formula

$$\|A\| = \sup\{|\lambda| : \lambda \in \sigma(A)\}.$$

5. The modulus of a matrix is not a genuine modulus because the inequality $|A + B| \leq |A| + |B|$ may fail. Here is an example in $\mathrm{M}_2(\mathbb{R})$, due to E. Nelson. Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Prove that $|(A + \mathrm{I}) + (B - \mathrm{I})| \not\leq |A + \mathrm{I}| + |B - \mathrm{I}|$.

6. (The trace functional) Suppose $A \in \mathrm{M}_N(\mathbb{R})$, $A = (a_{ij})_{i,j}$. By definition, the trace of A is the sum $\mathrm{trace}(A) = \sum_{k=1}^{N} a_{kk}$, of its diagonal elements. From the expression of the characteristic polynomial of A, we infer that $\mathrm{trace}(A)$ equals the sum of all eigenvalues of A, counted with their multiplicities.

(a) Prove that $\mathrm{trace} : \mathrm{M}_N(\mathbb{R}) \to \mathbb{R}$ is a linear map such that $\mathrm{trace}(A) \geq 0$ whenever $A \in \mathrm{Sym}^+(N, \mathbb{R})$ and $\mathrm{trace}(AB) = \mathrm{trace}(BA)$ for all $A, B \in \mathrm{M}_N(\mathbb{R})$. The inequality

$$\frac{1}{N}\,\mathrm{trace}(A) \geq (\det(A))^{1/N}$$

for $A \in \mathrm{Sym}^+(N, \mathbb{R})$ is a matrix variant of the *AM-GM* inequality.

(b) Prove that for every orthonormal basis $(\mathbf{u}_k)_k$ of $\mathbb{R}^N$ we have

$$\mathrm{trace}(A) = \sum_{k=1}^{N} \langle A\mathbf{u}_k, \mathbf{u}_k\rangle.$$

7. The Frobenius norm $\|\cdot\|_F$ on $\mathrm{M}_N(\mathbb{R})$ was introduced in Section 2.1 via the formula:

$$\|A\|_F = \left(\sum_{i,j=1}^{N} a_{ij}^2 \right)^{1/2},$$

where a_{ij} are the coefficients of A. This norm is associated to the inner product $\langle A, B \rangle_F = \mathrm{trace}(B^* A)$, where B^* denotes the transpose of B.

(a) Infer from the preceding exercise that

$$\|A\|_F = \left(\sum_{k=1}^{N} |\langle A\mathbf{u}_k, \mathbf{u}_k \rangle|^2 \right)^{1/2},$$

whenever $(\mathbf{u}_k)_k$ is an orthonormal basis of $\mathbb{R}^N$.

(b) Using the spectral decomposition, show that $\|A\|_F = \left(\sum_{k=1}^{N} \sigma_k^2 \right)^{1/2}$, where the σ_k's are the singular values of A, counted by their multiplicities.

(c) Prove that every linear functional f on $\mathrm{M}_N(\mathbb{R})$ can be represented uniquely as $f(X) = \mathrm{trace}(A^T X)$, for some $A \in \mathrm{M}_N(\mathbb{R})$.

3.6 Comments

The early history of the theorems of Carathéodory, Radon, and Helly, and many generalizations, ramifications, and analogues of these theorems forming an essential part of combinatorial convexity can be found in the survey article of L. Danzer, B. Grünbaum, and V. Klee [149].

The Shapley–Folkman theorem (due to L. S. Shapley and J. H. Folkman) was first published in R. Starr [547]. The unifying Lemma 2.1.2 appears in the notes prepared by R. M. Anderson for his students in economics (private communication).

Helly's theorem has many amazing consequences. Here is an example: Given s ($s \geq N+1$) points in the N-dimensional space such that every $N+1$ of them are contained in a ball of radius 1, then the whole family of points is contained in a ball of radius 1. This fact easily yields Jung's theorem: *If S is a finite set of points in the N-dimensional space, with all pairwise distances between them not exceeding* 1, *then S is contained in a ball of radius* $\sqrt{N/(2N+2)}$.

Some authors consider a much larger concept of cone. A subset C of a linear space is said to be a *linear cone* if

$$C = \mathbb{R}_{++} C.$$

If C is a linear cone, then C is a *convex linear cone* if and only if $C + C \subset C$. If C is a convex linear cone, then $C \cup \{0\}$ is a convex cone in the sense of the definition given in Section 2.1. If C is a convex set and $0 \in C$, then C is a convex cone if and only if $C + C \subset C$. The concept of convex linear cone includes

examples such as $\mathbb{R}^N_{++}$ and $\mathrm{Sym}^{++}(N,\mathbb{R})$ (that allow us to speak about *open convex cones*).

The Pompeiu–Hausdorff metric between two subsets of a metric space,

$$d_H(K_1,K_2)=\max\{\sup_{x\in K_1}\inf_{y\in K_2}d(x,y),\sup_{y\in K_2}\inf_{x\in K_1}d(x,y)\},$$

was introduced by D. Pompeiu in his Ph.D. thesis defended at Sorbonne in 1905 (and published in the same year in Annales de la Faculté de Sciences de Toulouse) and by F. Hausdorff in his book *Grundzüge der Mengenlehre*, first published in 1914. This metric plays an important role in image processing, face detection, convex geometry, etc. For example, one can show that the set $\mathcal{C}^N$, of all nonempty compact convex subsets of $\mathbb{R}^N$, can be made into a complete metric space via the Pompeiu–Hausdorff metric. For this, use the fact that

$$d_H(K_1,K_2)=\inf\{\delta>0:K_1\subset K_2+\delta B \text{ and } K_2\subset K_1+\delta B\},$$

where B denotes the open unit ball of $\mathbb{R}^N$. The main feature of the space $\mathcal{C}^N$ is described by the following important result due to W. Blaschke:

3.6.1 Theorem (The Blaschke selection theorem) *Every bounded sequence in $\mathcal{C}^N$ contains a convergent subsequence.*

Important applications of the Blaschke selection theorem (including a solution to the isoperimetric problem) can be found in the book of P. J. Kelly and M. L. Weiss [294].

Interestingly, the Blaschke selection theorem can be deduced from the well-known criterion of compactness due to Arzelà and Ascoli. Indeed, as it will be shown in Theorem 4.4.3 and Corollary 4.4.4, each nonempty compact convex subset K of the closed unit ball $\bar{B}$ of $\mathbb{R}^N$ can be associated uniquely to its support function,

$$h_K:\bar{B}\to\mathbb{R},\quad h_K(x)=\max_{k\in K}\langle x,k\rangle,$$

and this association provides an isometry between $\left\{K\in\mathcal{C}^N:K\subset\bar{B}\right\}$ and a subset $\mathcal{K}$ of $C(\bar{B})$. Clearly, $\mathcal{K}$ verifies the two requirements of the Arzelà–Ascoli criterion: boundedness and uniform equicontinuity.

Theorem 3.2.2 (about convexity of Chebyshev sets in $\mathbb{R}^N$) was proved independently by L. N. H. Bunt (as part of his Ph.D. thesis in 1934) and by T. S. Motzkin (1935). The present status of Klee's problem is discussed by J.-B. Hiriart-Urruty [254] and J. M. Borwein and J. Vanderwerff [90].

A substitute of orthogonal projection in the case of arbitrary Banach spaces is the following result due to F. Riesz [503], concerning the existence of *almost orthogonal elements*:

3.6.2 Theorem *Suppose that F is a proper closed subspace of the Banach space E. Then for every $\varepsilon>0$ there exists $\mathbf{x}_\varepsilon\in E$ such that $\|\mathbf{x}_\varepsilon\|=1$ and $d(\mathbf{x}_\varepsilon,F)\geq 1-\varepsilon$.*

3.6.3 Corollary *A normed linear space whose closed unit ball is compact is necessarily finite dimensional.*

See R. Bhatia [63], pp. 17–18, for details.

All results in Section 3.2 on hyperplanes and separation theorems in $\mathbb{R}^N$ are due to H. Minkowski [384]. Their extension to the general context of linear topological spaces is presented in Appendix B.

The critical role played by finite dimensionality in a number of important results on convex functions is discussed by J. M. Borwein and A. S. Lewis in [89, Chapter 9]. See also J. M. Borwein and J. Vanderwerff [90].

The comment in Remark 3.1.11 concerning the Lebesgue measurability of convex sets is generalized by the following result due to R. Lang [321]: every convex set A in $\mathbb{R}^N$ is measurable with respect to any σ-finite product measure $\mu = \otimes_{k=1}^N \mu_k$ defined on the Borel σ-algebra of $\mathbb{R}^N$.

Farkas' alternative theorem is due to J. Farkas [189], who published it in a paper laying the foundation of linear and nonlinear programming. It yields not only Gordan's theorem of alternative but also many others. See Section 4.9 of the book of C. L. Byrne [112].

The ordered linear space $\mathrm{Sym}(N,\mathbb{R})$ of all $N \times N$-dimensional symmetric matrices with real coefficients endowed with the *Löwner ordering*,

$$A \leq B \text{ if and only if } \langle A\mathbf{x},\mathbf{x}\rangle \leq \langle B\mathbf{x},\mathbf{x}\rangle \text{ for all } \mathbf{x} \in \mathbb{R}^N,$$

can be realized as an ordered Banach space in infinitely many ways, using the *Schatten trace norms* $\|\cdot\|_{S_p}$,

$$\|A\|_{S_p} = \begin{cases} \left(\sum_{k=1}^N |\lambda_k(A)|^p\right)^{1/p} & \text{if } p \in [1,\infty) \\ \sup_{1\leq k\leq N} |\lambda_k(A)| & \text{if } p = \infty. \end{cases}$$

Here $\lambda_1(A), ..., \lambda_N(A)$ is the list of the eigenvalues of A, repeated according to their multiplicities.

The space $\mathrm{Sym}(N,\mathbb{R})$ endowed with the norm $\|\cdot\|_{S_p}$ will be denoted $S_p(\mathbb{R}^N)$. We will refer to it as the *Schatten class* of index p.

It is worth noticing that the Schatten norm of index ∞ equals the *operator norm*,

$$\|A\| = \sup_{\|\mathbf{x}\|\leq 1} |\langle A\mathbf{x},\mathbf{x}\rangle|,$$

while the Schatten norm of index 2 is nothing but the Frobenius norm. See Exercise 7, Section 3.5.

All Schatten classes are ordered Banach spaces. This is a direct consequence of *Weyl's monotonicity principle* (see Corollary 6.4.3, Section 6.4), which asserts that

$$A \leq B \text{ in } \mathrm{Sym}(N,\mathbb{R}) \text{ implies } \lambda_i^{\downarrow}(A) \leq \lambda_i^{\downarrow}(B) \text{ for } i = 1, ..., N.$$

Here $\lambda_1^{\downarrow}(A) \geq \cdots \geq \lambda_N^{\downarrow}(A)$ and $\lambda_1^{\downarrow}(B) \geq \cdots \geq \lambda_N^{\downarrow}(B)$ are the eigenvalues of the matrices A and B in decreasing order and repeated according to their multiplicities. Indeed, according to this principle,

$$0 \le A \le B \text{ in } S_p(\mathbb{R}^N) \text{ implies } \|A\|_{S_p} \le \|B\|_{S_p}.$$

The infinite-dimensional analogue of the space $\mathrm{Sym}(n,\mathbb{R})$ is represented by the space $\mathcal{A}(H)$, of all symmetric operators acting on a real Hilbert space H. By the closed graph theorem, every operator $A \in \mathcal{A}(H)$ is bounded (that is, continuous).

If $A \in \mathcal{A}(H)$, then clearly $\langle A\mathbf{x},\mathbf{x}\rangle \in \mathbb{R}$ for all $\mathbf{x}$. Since the key formula

$$\|A\| = \sup_{\|\mathbf{x}\|\le 1} |\langle A\mathbf{x},\mathbf{x}\rangle| = \sup_{\|\mathbf{x}\|=1} |\langle A\mathbf{x},\mathbf{x}\rangle| \tag{3.23}$$

and its consequences (3.16) and (3.17) remain valid in this extended context, it follows that $\mathcal{A}(H)$ constitutes a regularly ordered Banach space when endowed with the operator norm and the positive cone

$$\mathcal{A}(H)_+ = \{A \in \mathcal{A}(H) : \langle A\mathbf{x},\mathbf{x}\rangle \ge 0 \quad \text{for all } \mathbf{x} \in H\}.$$

The interior of $\mathcal{A}(H)_+$ is the set of all strictly positive operators,

$$\mathcal{A}(H)_{++} = \{A \in \mathcal{A}(H) : \langle A\mathbf{x},\mathbf{x}\rangle > 0 \quad \text{for all } \mathbf{x} \in H\setminus\{0\}\}.$$

As in the finite-dimensional case, the spectrum $\sigma(A)$ of an operator $A \in \mathcal{A}(H)$ is a nonempty subset of the real line and A is positive if and only if $\sigma(A) \subset [0,\infty)$.

3.6.4 Theorem *Suppose that $A \in \mathcal{A}(H)$. Then its spectrum $\sigma(A)$ is included in the interval $[\omega_A,\Omega_A]$, where $\omega_A = \inf_{\|\mathbf{x}\|=1}\langle A\mathbf{x},\mathbf{x}\rangle$ and $\Omega_A = \sup_{\|\mathbf{x}\|=1}\langle A\mathbf{x},\mathbf{x}\rangle$. Moreover, ω_A and Ω_A belongs to $\sigma(A)$ and*

$$\|A\| = \sup_{\lambda\in\sigma(A)} |\lambda|. \tag{3.24}$$

Based on Theorem 3.6.4 one can easily extend Remark 3.5.1 to the case of symmetric operators $A \in \mathcal{A}(H)$. Precisely, to each such operator one can associate an application $\pi : C(\sigma(A)) \to \mathcal{A}(H)$ (known as the *Gelfand functional calculus*) which maps every real polynomial $P(t) = t^n + c_1 t^{n-1} + \cdots + c_n$ into the symmetric operator $P(A) = A^n + c_1 A^{n-1} + \cdots + c_n I$ and (using the Stone–Weierstrass Theorem of approximation), every continuous real function $f \in C(\sigma(A))$ into an operator $f(A) \in \mathcal{A}(H)$ such that

$$\begin{cases} (f+g)(A) = f(A) + g(A) \\ (fg)(A) = f(A)g(A) = g(A)f(A) \\ f \ge 0 \text{ implies } f(A) \ge 0 \\ \|f(A)\| = \sup\{|f(\lambda)| : \lambda \in \sigma(A)\}, \end{cases}$$

for all $f, g \in C(\sigma(A))$. See Ş. Strătilă and L. Zsidó [557], Theorem 2.6, p. 31.

The spectral decomposition of symmetric matrices admits an infinite-dimensional analogue in the case of compact self-adjoint operators (acting on complex Hilbert spaces). See R. Bhatia [63], p. 183, or M. Willem [588], p. 68.

The theory of Schatten classes (real or complex), associated to infinite-dimensional separable Hilbert spaces, is presented in the book of Simon [541].



Chapter 4

Convex Functions on a Normed Linear Space

Convex functions and their relatives are ubiquitous in a large variety of applications such as optimization theory, mass transportation, mathematical economics and geometric inequalities related to isoperimetric problems. This chapter is devoted to a succinct presentation of their theory in the context of *real* normed linear spaces, but most of the illustrations will refer to the Euclidean space $\mathbb{R}^N$, the matrix space $\mathrm{M}_N(\mathbb{R})$ of all $N \times N$-dimensional real matrices (endowed with the Hilbert-Schmidt norm or with the operator norm) and to the Lebesgue spaces $L^p\left(\mathbb{R}^N\right)$ with $p \in [1, \infty]$.

4.1 Generalities and Basic Examples

The notion of a convex function introduced in Chapter 1 can be extended verbatim to the framework of real-valued functions defined on arbitrary convex sets.

In what follows U will be a convex set in a real normed linear space E.

4.1.1 Definition *A function $f : U \to \mathbb{R}$ is said to be* convex *if*

$$f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) \leq (1-\lambda)f(\mathbf{x}) + \lambda f(\mathbf{y}) \tag{4.1}$$

for all $\mathbf{x}, \mathbf{y} \in U$ and $\lambda \in [0,1]$.

The function f is called strictly convex if the above inequality is strict whenever $\mathbf{x} \neq \mathbf{y}$ and $\lambda \in (0,1)$, and it is called strongly convex with parameter $C > 0$ (equivalently, C-strongly convex) if

$$f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) \leq (1-\lambda)f(\mathbf{x}) + \lambda f(\mathbf{y}) - \frac{C}{2}\lambda(1-\lambda)\left\|\mathbf{x}-\mathbf{y}\right\|^2 \tag{4.2}$$

for all $\mathbf{x}, \mathbf{y} \in U$ and $\lambda \in [0,1]$.

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It is worth mentioning that when U is a convex subset of a real inner product space, a function $f : U \to \mathbb{R}$ is strongly convex with parameter $C > 0$ if and only if the function $f - (C/2)\|\cdot\|^2$ is convex.

The related notions such as *concave function*, *strictly concave function*, *strongly concave function*, *affine function*, *midpoint convex/ midpoint concave function*, *quasiconvex/quasiconcave function*, *log-convex/ log-concave function* etc., can be introduced as in the one real variable case. To each of them one can ascribe a basic inequality like (4.1) and (4.2), that can be extended to the case of arbitrary finite convex combinations. We shall refer to these extensions as *discrete Jensen type inequalities*.

Clearly, strong convexity (concavity) implies strict convexity (concavity), that in turn implies convexity (concavity). On the other hand, log-convexity implies convexity, and this one implies quasiconvexity. Notice that every concave function is also quasiconcave and every strictly positive concave function is log-concave.

In the context of several variables, the property of convexity in the sense of Definition 4.1.1 is sometimes called *joint convexity* in order to distinguish it from the weaker property of *separate convexity* (that is, convexity with respect to each variable when the others are kept fixed). A simple example of a nonconvex function that is separately convex is provided by the polynomial xy. Separate convexity has a number of interesting features, for example, the supremum of such a function over a compact convex set equals the supremum over the boundary. See Exercise 7. A special case when separate convexity coincides with convexity makes the object of Exercise 2, Section 4.4.

As in the case of one real variable, the basic criterion of convexity is the second differentiability test. It will be discussed in Section 5.3, but some basic examples can be derived directly using the definition:

- $f(x_1, \ldots, x_N) = \sum_{k=1}^{N} \varphi(x_k)$ is (strictly) convex and continuous on $\mathbb{R}^N$, whenever φ is a (strictly) convex function on $\mathbb{R}$.
- The function $f(\mathbf{x}) = \sum_{k=1}^{n} c_k e^{\langle a_k, \mathbf{x}\rangle}$ is continuous and convex on $\mathbb{R}^N$, whenever $a_1, ..., a_n \in \mathbb{R}^N$ and $c_1, ..., c_n \geq 0$;
- The norm function $f(\mathbf{x}) = \|\mathbf{x}\|$ in any normed linear space E is continuous and convex. The same works for the *distance function* $d_U(\mathbf{x}) = d(\mathbf{x}, U)$, from a nonempty convex set U in E.
- As concerns the *distance function*, $d : E \times E \to \mathbb{R}$, $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$, this is continuous and jointly convex.
- The squared norm $f(\mathbf{x}) = \|\mathbf{x}\|^2$ in the Euclidean space $\mathbb{R}^N$ is strongly convex (with parameter $C = 2$). Renorming $\mathbb{R}^N$ with an equivalent norm, the squared function remains convex, but not necessarily strongly convex or strictly convex. See the case of sup norm.

It is worth noticing that the property of convexity is equivalent with convexity on each line segment included in the domain of definition:

4.1.2 Proposition *A function $f\colon U \to \mathbb{R}$ is convex (respectively strictly convex, strongly convex, log-convex, concave etc.) if and only if for every pair of distinct points $\mathbf{x}$ and $\mathbf{y}$ in U the function*

$$\varphi\colon [0,1] \to \mathbb{R}, \quad \varphi(t) = f((1-t)\mathbf{x} + t\mathbf{y})$$

is convex (respectively strictly convex, strongly convex, log-convex, concave etc.)

Letting $v = \mathbf{y} - \mathbf{x}$, one can reformulate Proposition 4.1.2 by asking the respective condition of convexity for the function

$$\psi(t) = f(\mathbf{x} + tv), \quad t \in \operatorname{dom} \psi = \{t \in \mathbb{R} : \mathbf{x} + tv \in U\},$$

whenever $\mathbf{x} \in U$ and $v \in E$ *(*the ambient linear space*).*

4.1.3 Corollary (The extension of Jensen's criterion of convexity) *Suppose that U is a convex subset of a real normed linear space. A continuous function $f\colon U \to \mathbb{R}$ is (strictly) convex if and only if it is (strictly) midpoint convex.*

Notice that continuity at the interior points of U (as well as the measurability of f) suffices as well. See Theorem 1.1.8 and the Comments at the end of Chapter 1. Moreover, as noticed K. Nikodem [452], for a real-valued function defined on an open convex subset of $\mathbb{R}^N$ the property of convexity is equivalent to midpoint convexity plus quasiconvexity.

Let us also mention the following result due to A. Aleman [11]: Suppose that f is a real-valued function defined on a closed convex subset C (of a real normed linear space). If f is lower semicontinuous and for each $x, y \in C$ there exists a λ in $(0,1)$ such that $f((1-\lambda)x + \lambda y) \le (1-\lambda)f(x) + \lambda f(y)$, then f is a convex function.

4.1.4 Example *The log-sum-exp function in $\mathbb{R}^N$ is a function of the form*

$$f(\mathbf{x}) = \log\left(\sum_{k=1}^{n} c_k e^{\langle a_k, \mathbf{x}\rangle}\right), \quad \mathbf{x} \in \mathbb{R}^N,$$

where the coefficients c_k are strictly positive and the vectors a_k belong to $\mathbb{R}^N$. This function is of class C^∞ and its convexity follows easily from Corollary 4.1.3 and the Cauchy–Bunyakovsky–Schwarz inequality.

4.1.5 Example (*Minkowski's inequality of order $p = 0$*) *According to Corollary 4.1.3, the proof of the property of concavity of the geometric mean $\left(\prod_{k=1}^n x_k\right)^{1/n}$ on $\mathbb{R}^n_+$ reduces to the fact that*

$$\left(\prod\nolimits_{k=1}^{n} (x_k + y_k)\right)^{1/n} \ge \left(\prod\nolimits_{k=1}^{n} x_k\right)^{1/n} + \left(\prod\nolimits_{k=1}^{n} y_k\right)^{1/n} \tag{4.3}$$

for every pair of distinct strings $x_1, \dots, x_n$ and $y_1, \dots, y_n \geq 0$. This is a consequence of the AM-GM inequality. Indeed, except in trivial cases when one or more elements are zero, we have

$$\frac{\left(\prod_{k=1}^n x_k\right)^{1/n} + \left(\prod_{k=1}^n y_k\right)^{1/n}}{\left(\prod_{k=1}^n (x_k + y_k)\right)^{1/n}} = \left(\prod_{k=1}^n \frac{x_k}{x_k + y_k}\right)^{1/n} + \left(\prod_{k=1}^n \frac{y_k}{x_k + y_k}\right)^{1/n}$$
$$\leq \frac{1}{n}\sum_{k=1}^n \frac{x_k}{x_k + y_k} + \frac{1}{n}\sum_{k=1}^n \frac{y_k}{x_k + y_k} = 1.$$

The inequality (4.3) has a matrix companion, Minkowski's determinant inequality, which asserts that the continuous function

$$f\colon \mathrm{Sym}^+(n, \mathbb{R}) \to \mathbb{R}, \quad f(A) = (\det A)^{1/n}$$

is concave. Using an approximation argument (see Exercise 10 (d)) we may restrict to the case of strictly positive matrices. Then the proof is immediate by taking into account that for every pair A, B of such matrices there exists a nonsingular matrix T such that T^*AT and T^*BT are both of diagonal form. See Lemma 3.5.6. The same argument together with the generalized AM-GM inequality yields

$$\det\left((1-\lambda)A + \lambda B\right) \geq (\det A)^{1-\lambda} (\det B)^{\lambda}$$

*for all $A, B \in \mathrm{Sym}^+(n, \mathbb{R})$ and $\lambda \in [0, 1]$. In turn, this implies that the log-*determinant function,

$$f\colon \mathrm{Sym}^{++}(n, \mathbb{R}) \to \mathbb{R}, \quad f(A) = \log(\det A)$$

is concave. Actually, the log-determinant function is strictly concave and this can be easily proved via Proposition 4.1.2. Indeed, for every $A, B \in \mathrm{Sym}^{++}(n, \mathbb{R})$ and $t \in \mathbb{R}$ with $A + tB \in \mathrm{Sym}^{++}(n, \mathbb{R})$, we have

$$\log\det(A + tB) = \log\det(A) + \log\det\left[I + tA^{-1/2}BA^{-1/2}\right]$$
$$= \log\det(A) + \sum_{k=1}^n \log\left(1 + t\lambda_k\right),$$

where $\lambda_1, ..., \lambda_n$ are the eigenvalues of $A^{-1/2}BA^{-1/2}$ (all strictly positive). Consequently, the function $\log(\det(A + tB))$ is the sum of a constant function and a family of strictly concave functions.

The log-determinant function is an archetype for the class of concave spectral functions. See Section 7.1.

4.1.6 Remark (Operations preserving convexity of functions) *All assertions of Proposition* 1.1.10 *remain valid in the context of several variables with only one exception, statement* (c), *where the connection between synchronicity and monotonicity in the same way should be eliminated. Some other techniques to construct convex functions from old ones are presented in Exercises 3, 5 and 11.*

In Section 1.1 we proved that every convex function defined on an open interval is continuous. This fact can be easily extended to the context of several variables, due to the following lemma.

4.1.7 Lemma *Every convex function f defined on an open convex set U in $\mathbb{R}^N$ is locally bounded (that is, each point $a \in U$ admits a neighborhood on which f is bounded).*

Proof. For $\mathbf{a} \in U$ arbitrarily fixed, choose a cube K in U, centered at $\mathbf{a}$, with vertices $\mathbf{v}_1, \ldots, \mathbf{v}_{2^N}$. Clearly, K is a neighborhood of $\mathbf{a}$. Every $\mathbf{x} \in K$ is a convex combination of vertices and thus

$$f(\mathbf{x}) = f\Big(\sum_{k=1}^{2^N} \lambda_k \mathbf{v}_k\Big) \leq M = \sup_{1 \leq k \leq 2^N} f(\mathbf{v}_k),$$

which shows that f is bounded above on K. By the symmetry of K, for every $\mathbf{x} \in K$ there is a $\mathbf{y} \in K$ such that $\mathbf{a} = (\mathbf{x}+\mathbf{y})/2$. Then $f(\mathbf{a}) \leq (f(\mathbf{x}) + f(\mathbf{y}))/2$, whence $f(\mathbf{x}) \geq 2f(\mathbf{a}) - f(\mathbf{y}) \geq 2f(\mathbf{a}) - M$, and the proof is complete. ■

4.1.8 Proposition *If f is a convex function defined on an open convex subset U of $\mathbb{R}^N$, then f is locally Lipschitz. In particular, f is continuous on U.*

According to a theorem due to Rademacher (see Appendix D), one can infer from Proposition 4.1.8 that every convex function on an open convex set U in $\mathbb{R}^N$ is almost everywhere differentiable. A direct proof of this property will be given in Section 5.2 (see Theorem 5.2.4).

Proof. According to the preceding lemma, given $\mathbf{a} \in U$, we may find a ball $B_{2r}(\mathbf{a}) \subset U$ on which f is bounded above, say by M. For $\mathbf{x} \neq \mathbf{y}$ in $B_r(\mathbf{a})$, put $\mathbf{z} = \mathbf{y} + (r/\alpha)(\mathbf{y} - \mathbf{x})$, where $\alpha = \|\mathbf{y} - \mathbf{x}\|$. Clearly, $\mathbf{z} \in B_{2r}(\mathbf{a})$. As

$$\mathbf{y} = \frac{r}{r+\alpha}\,\mathbf{x} + \frac{\alpha}{r+\alpha}\,\mathbf{z},$$

from the convexity of f we infer that

$$f(\mathbf{y}) \leq \frac{r}{r+\alpha} f(\mathbf{x}) + \frac{\alpha}{r+\alpha} f(\mathbf{z}).$$

Then

$$\begin{aligned} f(\mathbf{y}) - f(\mathbf{x}) &\leq \frac{\alpha}{r+\alpha}[f(\mathbf{z}) - f(\mathbf{x})] \\ &\leq \frac{\alpha}{r}\,[f(\mathbf{z}) - f(\mathbf{x})] \leq \frac{2M}{r}\,\|\mathbf{y} - \mathbf{x}\| \end{aligned}$$

and the proof ends by interchanging the roles of $\mathbf{x}$ and $\mathbf{y}$. ■

4.1.9 Corollary *Let f be a convex function defined on a convex set A in $\mathbb{R}^N$. Then f is Lipschitz on each compact convex subset of* $\mathrm{ri}(A)$. *In particular, f is continuous on* $\mathrm{ri}(A)$.

Proof. Clearly, we may assume that $\operatorname{aff}(A) = \mathbb{R}^N$. In this case, $\operatorname{ri}(A) = \operatorname{int}(A)$ and Proposition 4.1.8 applies. ∎

4.1.10 Corollary *Let the convex functions $f_n : \mathbb{R}^N \to \mathbb{R}$ converge pointwise for $n \to \infty$ to $f : \mathbb{R}^N \to \mathbb{R}$. Then f is convex and, for each compact set K, the convergence of f_n to f is uniform on K.*

Proof. The convexity of f is clear. As concerns the second assertion, choose $R > 0$ such that $K \subset B_R(\mathbf{0})$. The function $g(\mathbf{x}) = \sup_n f_n(\mathbf{x})$ is real-valued and convex so, by Proposition 4.1.8, it must be continuous on the compact ball $\overline{B}_{2R}(\mathbf{0})$. Hence g is bounded above on this ball and an inspection of the argument of Proposition 4.1.8 shows that g is Lipschitz on $B_R(\mathbf{0})$. As a consequence, there is some $L > 0$ (independent of n) such that $\|f_n(\mathbf{x}) - f_n(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in B_R(\mathbf{0})$. By passing to the limit as $n \to \infty$ we infer that f also verifies this inequality. To end the proof, consider a compact set K in $\mathbb{R}^N$ and a strictly positive number ε arbitrarily fixed. Since K is compact, it admits a finite covering with balls $B_\varepsilon(\mathbf{x}_1), ..., B_\varepsilon(\mathbf{x}_m)$. Then each $\mathbf{x} \in K$ belongs to a ball $B_\varepsilon(\mathbf{x}_{k(\varepsilon)})$ and this fact yields

$$\begin{aligned} |f_n(\mathbf{x}) - f(\mathbf{x})| &\leq \left|f_n(\mathbf{x}) - f_n(\mathbf{x}_{k(\varepsilon)})\right| + \left|f_n(\mathbf{x}_{k(\varepsilon)}) - f(\mathbf{x}_{k(\varepsilon)})\right| + \left|f(\mathbf{x}_{k(\varepsilon)}) - f(\mathbf{x})\right| \\ &\leq L\left\|\mathbf{x} - \mathbf{x}_{k(\varepsilon)}\right\| + \left|f_n(\mathbf{x}_{k(\varepsilon)}) - f(\mathbf{x}_{k(\varepsilon)})\right| + L\left\|\mathbf{x}_{k(\varepsilon)} - \mathbf{x}\right\| \\ &\leq 2L\varepsilon + \max_{1\leq k\leq m} |f_n(\mathbf{x}_k) - f(\mathbf{x}_k)|\,. \end{aligned}$$

Due to the pointwise convergence of the functions f_n to f, there is a rank N_ε such that $\max_{1\leq k\leq m} |f_n(\mathbf{x}_k) - f(\mathbf{x}_k)| < \varepsilon$ for all $n \geq N_\varepsilon$. Therefore

$$|f_n(\mathbf{x}) - f(\mathbf{x})| \leq (2L + 1)\varepsilon$$

for all $\mathbf{x} \in K$ and $n \geq N_\varepsilon$. The proof is complete. ∎

Another instance of uniform convergence is mentioned in Exercise 14.

The infinite-dimensional analogue of Proposition 4.1.8 is as follows:

4.1.11 Proposition (Local Lipschitz continuity via convexity) *Let f be a convex function on an open convex set U in a normed linear space. If f is bounded above in a neighborhood of one point of U, then f is locally Lipschitz on U. In particular, f is a continuous function.*

The proof is similar to that of Proposition 4.1.8, with the difference that the role of Lemma 4.1.7 is taken by the following lemma:

4.1.12 Lemma *Let f be a convex function on an open convex set U in a normed linear space. If f is bounded above in a neighborhood of one point of U, then f is locally bounded on U.*

Proof. Suppose that f is bounded above by M on a ball $B_r(\mathbf{a})$. Let $\mathbf{x} \in U$ and choose $\rho > 1$ such that $\mathbf{z} = \mathbf{a} + \rho(\mathbf{x} - \mathbf{a}) \in U$. If $\lambda = 1/\rho$, then

$$V = \{\mathbf{v} : \mathbf{v} = (1 - \lambda)\mathbf{y} + \lambda\mathbf{z},\ \mathbf{y} \in B_r(\mathbf{a})\}$$

equals the ball with center $\mathbf{x} = (1-\lambda)\mathbf{a} + \lambda\mathbf{z}$ and radius $(1-\lambda)r$. Moreover, for $\mathbf{v} \in V$ we have

$$f(\mathbf{v}) \leq (1-\lambda)f(\mathbf{y}) + \lambda f(\mathbf{z}) \leq (1-\lambda)M + \lambda f(\mathbf{z}).$$

To show that f is bounded below in the same neighborhood V, choose arbitrarily $\mathbf{v} \in V$ and notice that $2\mathbf{x} - \mathbf{v} \in V$. Consequently, $f(\mathbf{x}) \leq f(\mathbf{v})/2 + f(2\mathbf{x} - \mathbf{v})/2$, which yields $f(\mathbf{v}) \geq 2f(\mathbf{x}) - f(2\mathbf{x} - \mathbf{v}) \geq 2f(\mathbf{x}) - M$. The proof is complete. ∎

4.1.13 Remark *On every infinite-dimensional normed linear space E there are discontinuous linear functions (and a fortiori discontinuous convex functions). Indeed, in this case E contains an infinite set $\{\mathbf{x}_n : n \in \mathrm{N}\}$ of linearly independent elements. One can assume that $\|\mathbf{x}_n\| = 1$ for all indices n (since normalizing the vectors does not affect the linear independence). According to the Axiom of Choice, this family can be completed to a normalized algebraic basis $\left\{\mathbf{x}_n : n \in \widetilde{\mathrm{N}}\right\}$ of E (that is, to a maximal family of linearly independent norm-1 elements). See* [252]*, Theorem* 3.19*, p.* 18*. The functional $L : \left\{\mathbf{x}_n : n \in \widetilde{\mathrm{N}}\right\} \to \mathbb{R}$ defined by*

$$L(\mathbf{x}_n) = \begin{cases} n & \text{if } n \in \mathrm{N} \\ 0 & \text{if } n \in \widetilde{\mathrm{N}} \backslash \mathrm{N} \end{cases}$$

admits a unique extension by linearity to $E = \operatorname{span}\left\{\mathbf{x}_n : n \in \widetilde{\mathrm{N}}\right\}$ and this extension is discontinuous because it is not bounded on the unit ball.

4.1.14 Remark *As noticed J. M. Borwein, S. Fitzpatrick and J. A Vanderwerff* [88]*, a Banach space E is finite dimensional if (and only if) every continuous convex function on E is bounded on bounded sets. A similar result holds when every continuous convex function defined on the unit open ball of E is Lipschitz. See D. Popa* [481].

Exercises

1. Prove that the general form of an affine function $f\colon \mathbb{R}^N \to \mathbb{R}$ is $f(\mathbf{x}) = \langle \mathbf{x}, \mathbf{u} \rangle + a$, where $\mathbf{u} \in \mathbb{R}^N$ and $a \in \mathbb{R}$. This fact can be extended to all affine functions $g : \mathbb{R}^N \to \mathbb{R}^M$. Precisely, $g\left((1-\lambda)\mathbf{x} + \lambda\mathbf{y}\right) = (1-\lambda)\, g(\mathbf{x}) + \lambda g(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$ and $\lambda \in [0,1]$ if and only if $g(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ for some $M \times N$-dimensional real matrix A and some $\mathbf{b} \in \mathbb{R}^M$.

2. The quadratic function associated to a symmetric matrix $A \in \operatorname{Sym}(N, \mathbb{R})$, $A \neq 0$, is $f(\mathbf{x}) = \langle A\mathbf{x}, \mathbf{x} \rangle$ for $\mathbf{x} \in \mathbb{R}^N$. Prove that f is convex if and only if $A \in \operatorname{Sym}^+(N, \mathbb{R})$.

   Extend this result, by considering a real Banach space E and a continuous bilinear map $B : E \times E \to \mathbb{R}$. Then $B(\mathbf{x}, \mathbf{x})$ is a convex function on E if and only if $B(\mathbf{x}, \mathbf{x}) \geq 0$ for all $\mathbf{x}$.

   [*Hint:* Use Corollary 4.1.3.]

3. (Convex monotone superposition) Let $f(\mathbf{x}) = (f_1(\mathbf{x}), ..., f_M(\mathbf{x}))$ be a vector-valued function on $\mathbb{R}^N$ with convex components f_k, and assume that F is a convex function on $\mathbb{R}^M$ which is monotone increasing (that is, $\mathbf{u} \leq \mathbf{v}$ in $\mathbb{R}^M$ implies $F(\mathbf{u}) \leq F(\mathbf{v})$). Prove that their superposition $\varphi(\mathbf{x}) = F(f(\mathbf{x}))$ is convex on $\mathbb{R}^N$. This result easily yields the convexity of the log-sum-exp function and of the powers $\|\cdot\|^\alpha$ of any norm for $\alpha \geq 1$.

4. Let f be a real-valued function defined on a convex set A in $\mathbb{R}^N$. Prove that f is convex if and only if

$$f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) < (1-\lambda)\alpha + \lambda\beta$$

whenever $0 < \lambda < 1$, $f(\mathbf{x}) < \alpha$ and $f(\mathbf{y}) < \beta$.

5. (Partial minimization) Suppose that $f : \mathbb{R}^M \times \mathbb{R}^N \to \mathbb{R}$, $f = f(\mathbf{x}, \mathbf{y})$ is jointly convex and $\inf_{\mathbf{y}\in\mathbb{R}^N} f(\mathbf{x}, \mathbf{y}) > -\infty$ for all $\mathbf{x} \in \mathbb{R}^M$. Infer from the preceding exercise that the function $g(\mathbf{x}) = \inf_{\mathbf{y}\in\mathbb{R}^N} f(\mathbf{x}, \mathbf{y})$ is convex on $\mathbb{R}^M$. Notice that the quasiconvex variant of this assertion also works.

6. Prove that the functions $f(x_1, \ldots, x_N) = \max\{x_1, \ldots, x_N\}$ and

$$g(x_1, \ldots, x_N) = \max\{x_1, \ldots, x_N\} - \min\{x_1, \ldots, x_N\}$$

are convex on $\mathbb{R}^N$.

7. (Separately convex functions). A function of several variables is called *separately convex* if it is convex with respect to each variable (when the others are kept fixed). Prove that the supremum of any such function over a compact convex set equals the supremum over the boundary. Apply this result in the case of the function

$$f(\mathbf{x}, \mathbf{y}) = \sum_{1\leq i\leq j\leq N} (x_i y_j - x_j y_i)^2,$$

defined for pair of vectors $\mathbf{x}, \mathbf{y}$ with $x_1, ..., x_N \in [m_1, M_1]$ and $y_1, ..., y_N \in [m_2, M_2]$.

8. Find the maximum of the function

$$f(a, b, c) = \left[3\left(a^5 + b^7 \sin\frac{\pi a}{2} + c\right) - 2(bc + ca + ab)\right]$$

for $a, b, c \in [0, 1]$.

[*Hint*: Notice that $f(a, b, c) \leq \sup[3(a + b + c) - 2(bc + ca + ab)] = 4.$]

9. It was shown in Example 4.1.5 that

$$(\det(A + B))^{1/n} \geq (\det A)^{1/n} + (\det B)^{1/n}$$

for all $A, B \in \mathrm{Sym}^+(n, \mathbb{R})$. Prove that when $A, B \in \mathrm{Sym}^{++}(n, \mathbb{R})$, equality holds if and only if there is a positive constant c so that $B = cA$. In particular, if equality holds and $\det A = \det B$, then $A = B$.

10. (An alternative approach of the log-concavity of the function det)
    (a) Prove that every matrix $A \in \mathrm{Sym}^{++}(n,\mathbb{R})$ verifies the formula
    $$\int_{\mathbb{R}^n} \mathrm{e}^{-\langle A\mathbf{x},\mathbf{x}\rangle}\,\mathrm{d}\mathbf{x} = \pi^{n/2}/\sqrt{\det A};$$
    there is no loss of generality assuming that A is diagonal.
    (b) Infer from the Rogers–Hölder inequality that
    $$\int_{\mathbb{R}^n} \mathrm{e}^{-\langle [\alpha A+(1-\alpha)B]\mathbf{x},\mathbf{x}\rangle}\,\mathrm{d}\mathbf{x} \le \left(\int_{\mathbb{R}^n} \mathrm{e}^{-\langle A\mathbf{x},\mathbf{x}\rangle}\,\mathrm{d}\mathbf{x}\right)^{\alpha}\left(\int_{\mathbb{R}^n} \mathrm{e}^{-\langle B\mathbf{x},\mathbf{x}\rangle}\,\mathrm{d}\mathbf{x}\right)^{1-\alpha},$$
    for every $A, B \in \mathrm{Sym}^{++}(n,\mathbb{R})$ and every $\alpha \in (0,1)$.
    (c) Conclude that $\det(\alpha A + (1-\alpha)B) \ge (\det A)^{\alpha}(\det B)^{1-\alpha}$.
    (d) Extend this conclusion for $A, B \in \mathrm{Sym}^{+}(n,\mathbb{R})$, using perturbations of the form $A+\varepsilon \mathrm{I}$ and $B+\varepsilon \mathrm{I}$ with $\varepsilon > 0$ arbitrarily small.

11. Prove that the function det is neither convex nor concave on $\mathrm{Sym}^{+}(N,\mathbb{R})$, as soon as $N \ge 2$.

12. (a) Prove that the limit of a pointwise converging sequence of (quasi)convex functions defined on the same convex set U is a (quasi)convex function.
    (b) Let $(f_\alpha)_\alpha$ be a family of (quasi)convex functions defined on the same convex set U, such that $f(\mathbf{x}) = \sup_\alpha f_\alpha(\mathbf{x}) < \infty$ for all $\mathbf{x} \in U$. Prove that f is (quasi)convex.
    (c) Suppose that $(f_\alpha)_\alpha$ is a family of (quasi)concave functions defined on the same convex set U, such that $f(\mathbf{x}) = \inf_\alpha f_\alpha(\mathbf{x}) > -\infty$ for all $\mathbf{x} \in U$. Prove that f is (quasi)concave.

13. Prove that if $g : \mathbb{R}^N \to \mathbb{R}$ is quasiconvex and $h : \mathbb{R} \to \mathbb{R}$ is increasing, then $f = h \circ g$ is quasiconvex.

14. Let $f : \mathbb{R}^N \to \mathbb{R}$ be a convex function. Prove that there is a constant $C > 0$ depending only on N, such that
    $$\sup_{\mathbf{x}\in\overline{B}_{r/2}(\mathbf{a})} |f(\mathbf{x})| \le \frac{C}{\mathrm{Vol}(B_r(\mathbf{a}))}\int_{B_r(\mathbf{a})} |f(y)|\,\mathrm{d}y,$$
    for every $r > 0$ and $\mathbf{a} \in \mathbb{R}^N$. As a consequence, the convergence in the L^1 norm of every sequence of convex and integrable functions defined on $\mathbb{R}^N$ implies uniform convergence on the compact sets.
    [*Hint:* We have $f(\mathbf{x}) \le (f(\mathbf{x}+\mathbf{y}) + f(\mathbf{x}-\mathbf{y}))/2$ for all $\mathbf{x} \in K = \overline{B}_{r/2}(\mathbf{a})$ and all $\mathbf{y}$ with $\|\mathbf{y}\| < r/2$; integrating this inequality over $B_{r/2}(\mathbf{0})$, infer that $f(\mathbf{x}) \le A = \big(\mathrm{Vol}B_{r/2}(\mathbf{0})\big)^{-1}\int_{B_r(\mathbf{a})} |f(\mathbf{y})|\,\mathrm{d}\mathbf{y}$. A lower bound for $f(\mathbf{x})$ results by integrating the inequality $f(\mathbf{x}) \ge 2f(\mathbf{x}+\mathbf{y}) - f(\mathbf{x}+2\mathbf{y}) \ge f(\mathbf{x}+\mathbf{y}) - A$ for $\mathbf{y} \in B_{r/2}(\mathbf{0})$.]

15. (Convexity via substitution) Consider the function

$$f(x_1,\dots,x_N)=\sum_{k=1}^{m}a_k x_1^{r_{1k}}\cdots x_N^{r_{Nk}},\quad (x_1,\dots,x_N)\in\mathbb{R}^N_{++},$$

where $a_k>0$ and $r_{ij}\in\mathbb{R}$. Prove that $g(y_1,\dots,y_N)=\log f(\mathrm{e}^{y_1},\dots,\mathrm{e}^{y_N})$ is convex on $\mathbb{R}^N$.

16. (A topological property of convex functions) Let $f:\mathbb{R}^N\to\mathbb{R}$ be a convex function such that $f(x_0)<0$ for some $x_0\in\mathbb{R}^N$. Prove that:
(a) $\overline{\{x:f(x)<0\}}=\{x:f(x)\le 0\}$;
(b) $\operatorname{int}\{x:f(x)\le 0\}=\{x:f(x)<0\}$;
(c) $\operatorname{rbd}\operatorname{int}\{x:f(x)\le 0\}=\{x:f(x)=0\}$.
The existence of x_0 is crucial for these equivalences. See the case of the function $f(x)=\min\left\{0,x^2-1\right\}$, $x\in\mathbb{R}$.
[*Hint*: See Exercise 7, Section 3.1.]

4.2 Extended Real-valued Convex Functions

The concepts of convex function and convex set can be related to each other in many ways. Of special interest is the characterization of convex functions in terms of their epigraphs.

As in the precedent section, U denotes a convex subset of a real normed linear space E.

By definition, the *epigraph* of a function $f\colon U\to\mathbb{R}$ is the set

$$\operatorname{epi}(f)=\{(\mathbf{x},\alpha):\mathbf{x}\in U,\ \alpha\in\mathbb{R}\text{ and }f(\mathbf{x})\le\alpha\}.$$

4.2.1 Theorem *A function $f\colon U\to\mathbb{R}$ is convex if and only if* $\operatorname{epi}(f)$ *is convex as a subset of $E\times\mathbb{R}$.*

The proof is straightforward and we omit the details.

For concave functions $f:U\to\mathbb{R}$, it is the hypograph

$$\operatorname{hyp}(f)=\{(\mathbf{x},\alpha):\mathbf{x}\in U,\ \alpha\in\mathbb{R}\text{ and }\alpha\le f(\mathbf{x})\}$$

rather than the epigraph that is a convex subset of $E\times\mathbb{R}$.

As we already noticed, the discrete case of Jensen's inequality (as well as its argument) extends verbatim to the case of convex functions of a vector variable. Theorem 4.2.1 offers an alternative simple argument.

4.2.2 Proposition (The discrete case of Jensen's inequality) *If $f\colon U\to\mathbb{R}$ is a convex function and $\sum_{k=1}^{n}\lambda_k\mathbf{x}_k$ is a convex combination of points of U, then*

$$f\left(\sum_{k=1}^{n}\lambda_k\mathbf{x}_k\right)\le\sum_{k=1}^{n}\lambda_k f(\mathbf{x}_k).$$

For concave functions, the inequality works in the reversed direction.

Proof. Since f is convex, its epigraph is a convex set, so considering n points $(\mathbf{x}_1, f(\mathbf{x}_1)), ..., (\mathbf{x}_n, f(\mathbf{x}_n))$ in epi(f), any convex combination of them,

$$\left(\sum_{k=1}^{n} \lambda_k \mathbf{x}_k, \sum_{k=1}^{n} \lambda_k f(\mathbf{x}_k) \right),$$

also belongs to epi(f). By the definition of the epigraph, this is equivalent to Jensen's inequality. ■

4.2.3 Corollary *If $f\colon U \to \mathbb{R}$ is a convex function and $\mathbf{x}_1, ..., \mathbf{x}_n$ are points in U, then*

$$f(\mathbf{x}) \leq \max\{f(\mathbf{x}_1), ..., f(\mathbf{x}_n)\} \quad \textit{for every } \mathbf{x} \in \operatorname{conv}(\{\mathbf{x}_1, ..., \mathbf{x}_n\}).$$

As a consequence, the function f attains its supremum on conv $(\{\mathbf{x}_1, ..., \mathbf{x}_n\})$ *at one of the points $\mathbf{x}_1, ..., \mathbf{x}_n$.*

The property stated in Corollary 4.2.3 characterizes the *quasiconvex functions*, that is, the functions $f\colon U \to \mathbb{R}$ such that

$$f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) \leq \max\{f(\mathbf{x}), f(\mathbf{y})\} \tag{4.4}$$

for all $\mathbf{x}, \mathbf{y} \in U$ and all $\lambda \in [0,1]$. This corollary represents the analogue of Jensen's discrete inequality in the context of quasiconvex functions.

Given a function $f\colon U \to \mathbb{R}$ and a real number α, the *level set* of height α of f is defined as

$$\operatorname{level}_\alpha(f) = \{\mathbf{x} \in U : f(\mathbf{x}) \leq \alpha\}.$$

4.2.4 Lemma *The quasiconvex functions are precisely the functions whose level sets are convex.*

Reversing the inequality sign in (4.4) and replacing max by min, we arrive at the concept of *quasiconcave function.* Every convex (concave) function is also quasiconvex (quasiconcave). The converse works for positively homogeneous functions of degree 1, taking positive values. See Section 4.4, Lemma 4.4.2. The function $f(x_1, x_2) = x_1 x_2$ is quasiconcave on $\mathbb{R}^2_+$ but not concave.

In applications, it is often useful to consider extended real-valued functions, defined on the whole ambient real linear space E.

4.2.5 Definition *An extended real-value function $f\colon E \to \overline{\mathbb{R}}$ is said to be* convex *if its epigraph,*

$$\operatorname{epi}(f) = \{(\mathbf{x}, \alpha) : \mathbf{x} \in E,\ \alpha \in \mathbb{R} \textit{ and } f(\mathbf{x}) \leq \alpha\},$$

is a convex subset of $E \times \mathbb{R}$.

A characterization of these functions via inequalities is offered by Exercise 2. In what follows we shall *only* deal with *proper convex* functions, that is, with those functions $f : E \to \mathbb{R} \cup \{\infty\}$ which are not identically ∞ and verify the inequality

$$f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) \leq (1-\lambda)f(\mathbf{x}) + \lambda f(\mathbf{y}),$$

for all $\mathbf{x}, \mathbf{y} \in E$ and all $\lambda \in (0,1)$. See Exercise 4 for an explanation why $-\infty$ is excluded from the range of a proper convex function.

A function $f\colon E \to \mathbb{R} \cup \{-\infty\}$ is said to be a *proper concave function* if $-f$ is proper convex.

The related notions of proper quasiconvex/quasiconcave functions can be introduced in a similar way.

The *effective domain* of a proper convex function $f : E \to \mathbb{R} \cup \{\infty\}$ is the set

$$\operatorname{dom} f = \{\mathbf{x} : f(\mathbf{x}) < \infty\}.$$

Since $\mathbf{x} \in \operatorname{dom} f$ is equivalent to the existence of some $\alpha \in \mathbb{R}$ such that $(\mathbf{x}, \alpha) \in \operatorname{epi}(f)$, the effective domain of a convex function is a convex set.

The restriction of a proper convex function to its effective domain is a convex function in the sense of Definition 4.1.1.

If U is a convex subset of a linear space E, then every convex function $f\colon U \to \mathbb{R}$ extends to a proper convex function $\widetilde{f}$ on E, letting $\widetilde{f}(\mathbf{x}) = \infty$ for $\mathbf{x} \in E \backslash U$. A basic example is provided by the indicator function. The *indicator function* of a *nonempty* subset A of E is defined by the formula

$$\iota_A(\mathbf{x}) = \begin{cases} 0 & \text{if } x \in A, \\ \infty & \text{if } x \in E \backslash A. \end{cases}$$

Clearly, A is convex if and only if ι_A is a proper convex function.

A discussion of the topological nature of level sets needs the framework of lower semicontinuity.

4.2.6 Definition *An extended real-valued function f defined on a Hausdorff topological space X is called* lower semicontinuous *if*

$$f(\mathbf{x}) \leq \liminf_{\mathbf{y} \to \mathbf{x}} f(\mathbf{y}) \quad \textit{for all } \mathbf{x} \in X.$$

In the same framework, a function g is called *upper semicontinuous* if $-g$ is lower semicontinuous.

When X is a metrizable space (in particular, a normed linear space), one can formulate the condition of lower semicontinuity in terms of sequences:

$$\mathbf{x}_n \to \mathbf{x} \text{ in } X \text{ implies } f(\mathbf{x}) \leq \liminf_{n \to \infty} f(\mathbf{x}_n).$$

Lower semicontinuity is related to the topological nature of the epigraph. Assuming E a Banach space, the epigraph of any extended real-valued function

f defined on a subset of E is endowed with the topology of subspace of the product Banach space $E \times \mathbb{R}$, associated to the norm

$$\|(\mathbf{x}, \alpha)\| = \left(\|\mathbf{x}\|^2 + |\alpha|^2\right)^{1/2}.$$

As in the case of the space $\mathbb{R}^N$, this norm is equivalent to each of the norms $\|(\mathbf{x}, \alpha)\|_p = (\|\mathbf{x}\|^p + |\alpha|^p)^{1/p}$ (for $1 \leq p < \infty$) and $\|(\mathbf{x}, \alpha)\|_\infty = \max\{\|\mathbf{x}\|, |\alpha|\}$, since they produce the same topology. The dual space of $E \times \mathbb{R}$ is $E^* \times \mathbb{R}$, the duality being defined by the formula

$$\langle (\mathbf{x}, \alpha), (x^*, \beta) \rangle = x^*(\mathbf{x}) + \alpha\beta.$$

4.2.7 Theorem *For a function $f : E \to \mathbb{R} \cup \{\infty\}$ the following statements are equivalent:*

(a) *The epigraph of f is closed in $E \times \mathbb{R}$, that is, for every sequence of points $(\mathbf{x}_n, y_n) \in \operatorname{epi}(f)$ converging to some point $(\mathbf{x}, y) \in E \times \mathbb{R}$ we have $(\mathbf{x}, y) \in \operatorname{epi}(f)$;*

(b) *Every level set of f is closed;*

(c) *f is lower semicontinuous on E.*

This result motivates the alternative terminology of *closed functions* for lower semicontinuous functions.

Proof. (a) $\Rightarrow$ (b) Let α be any real number. If $\operatorname{level}_\alpha(f)$ is the empty set, then it is a closed set. If $\operatorname{level}_\alpha(f)$ is nonempty, then choose $(\mathbf{x}_n)_n$ a sequence of points of $\operatorname{level}_\alpha(f)$ such that $\mathbf{x}_n \to \mathbf{x}$ in E. We have $f(\mathbf{x}_n) \leq \alpha$ for all n, whence $(\mathbf{x}_n, \alpha) \in \operatorname{epi}(f)$ for all n. Since $(\mathbf{x}_n, \alpha) \to (\mathbf{x}, \alpha)$ and the epigraph of f is closed, it follows that $(\mathbf{x}, \alpha) \in \operatorname{epi}(f)$, hence $\mathbf{x} \in \operatorname{level}_\alpha(f)$.

(b) $\Rightarrow$ (c) We reason by contradiction. If f were not lower semicontinuous, then it would exist $\mathbf{x} \in E$ and a sequence $(\mathbf{x}_n)_n$ such that $\mathbf{x}_n \to \mathbf{x}$ in E and

$$f(\mathbf{x}) > \liminf_{n\to\infty} f(\mathbf{x}_n).$$

This yields a number c and a subsequence $(\mathbf{x}_{k(n)})_n$ such that $f(\mathbf{x}) > c \geq f(\mathbf{x}_{k(n)})$. Then $\mathbf{x}_{k(n)} \in \operatorname{level}_c(f)$ for all n. Since $\mathbf{x}_{k(n)} \to \mathbf{x}$ and every level set of f is closed it follows that $\mathbf{x} \in \operatorname{level}_c(f)$. Hence $f(\mathbf{x}) \leq c$, a contradiction.

(c) $\Rightarrow$ (a) Assume that $\operatorname{epi}(f)$ is not closed. This implies the existence of a sequence of elements $(\mathbf{x}_n, \mathbf{y}_n) \in \operatorname{epi}(f)$ such that $(\mathbf{x}_n, \mathbf{y}_n) \to (\mathbf{x}, \mathbf{y})$ in E and $(\mathbf{x}, \mathbf{y}) \notin \operatorname{epi}(f)$. Since $f(\mathbf{x}_n) \leq \mathbf{y}_n$ for all n, we infer that

$$\liminf_{n\to\infty} f(\mathbf{x}_n) \leq \lim_{n\to\infty} \mathbf{y}_n = \mathbf{y}.$$

On the other hand, $\mathbf{y} < f(\mathbf{x})$ because $(\mathbf{x}, \mathbf{y}) \notin \operatorname{epi}(f)$. Hence $\liminf_{n\to\infty} f(\mathbf{x}_n) < f(\mathbf{x})$, a fact that contradicts the lower semicontinuity of f. In conclusion, $\operatorname{epi}(f)$ must be a closed set. ■

4.2.8 Corollary *The pointwise supremum of any family $(f_i)_{i\in I}$ of lower semicontinuous convex functions defined on a normed linear space E is a function of the same nature.*

Proof. Indeed, the function $f(\mathbf{x}) = \sup\{f_i(\mathbf{x}) : i \in I\}$, verifies the equality

$$\operatorname{epi}(f) = \cap_{i\in I}\operatorname{epi}(f_i).$$

■

If the effective domain of a proper convex function f is closed and f is continuous relative to dom f, then f is lower semicontinuous. However, f can be lower semicontinuous under more general circumstances. The following function,

$$\varphi(x,y) = \begin{cases} y^2/2x & \text{if } x > 0, \\ \alpha & \text{if } x = y = 0, \\ \infty & \text{otherwise,} \end{cases}$$

is illustrative on what can happen at the boundary points of the effective domain. In fact, f is a proper convex function for each $\alpha \in [0,\infty]$. All points of its effective domain are points of continuity except the origin, where the limit does not exist. However the function φ is lower semicontinuous for $\alpha = 0$.

The possibility of modifying the values of a proper convex function at the boundary of its effective domain to become lower semicontinuous is discussed in Exercise 10.

4.2.9 Theorem *Every lower semicontinuous proper convex function f defined on a Banach space is continuous at every point of $U = \operatorname{int}(\operatorname{dom} f)$.*

Notice that if $\mathbf{a} \in \operatorname{dom} f$ is a point of continuity, then f is bounded above on a neighborhood of $\mathbf{a}$ (see Exercise 6) and this fact implies that $\mathbf{a} \in \operatorname{int}(\operatorname{dom} f)$.

Proof. According to Proposition 4.1.11, it suffices to prove that f is bounded above in the neighborhood of one point in U. We will use the fact that every nonempty open subset of a complete metric space is a Baire space. See J. Munkres [402], Lemma 48.4, p. 297. In our case, U is the union of the closed sets $U_n = \{\mathbf{x} \in U : f(\mathbf{x}) \le n\}$ for $n \in \mathbb{N}$. Consequently there is a natural number n such that $\operatorname{int} U_n \neq \emptyset$ and the proof is done. ■

A function f as in the statement of Theorem 4.2.9 can be continuous at certain boundary points a of dom f (as a function taking values in the metric space $\mathbb{R} \cup \{\infty\}$). See the case of the function

$$f(x) = \begin{cases} 1/x & \text{for } x > 0 \\ \infty & \text{for } x \in (-\infty, 0] \end{cases}$$

and the point $a = 0$.

Notice that if f is a continuous convex function defined on a nonempty closed convex subset C of E and if we extend f to be ∞ on $E\backslash C$, then the resulting function $\tilde{f}$ is a lower semicontinuous proper convex function. The function $\tilde{f}$ is

continuous at $x \in C$ if and only if $x \in \operatorname{int} C$. Thus, if $\operatorname{int} C = \emptyset$, then $\tilde{f}$ is not continuous at any point of $C = \operatorname{dom} \tilde{f}$.

The standard notation in convex analysis for the set of all lower semicontinuous proper convex functions $f : E \to \mathbb{R} \cup \{\infty\}$ is $\Gamma(E)$. This set occupies a central role in the theory of duality of convex functions. See Chapter 8.

4.2.10 Remark *According to a theorem due to A. Brøndsted and R.T. Rockafellar, if $f : E \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper convex function, then the domain of ∂f is dense in* dom f. *For details see R. R. Phelps* [473], *Theorem* 3.17, *p.* 48.

4.2.11 Remark *Every separately convex proper function $f : \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ is locally Lipschitz, hence continuous on* int(dom f). *See B. Dacorogna* [146], *Theorem* 2.31, *p.* 47.

Exercises

1. Prove that the multiplicative inverse of a strictly positive convex function is a quasiconcave function, while the multiplicative inverse of a strictly positive concave function is a convex function.

2. Prove that $f : E \to \overline{\mathbb{R}}$ is a convex function if and only if

$$f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) \leq (1-\lambda)\alpha + \lambda\beta$$

whenever $f(\mathbf{x}) < \alpha$, $f(\mathbf{y}) < \beta$ and $\lambda \in (0,1)$. An important consequence is the fact that for any nonempty convex subset C of $E \times \mathbb{R}$, the function

$$f(\mathbf{x}) = \inf\{\alpha : (\mathbf{x}, \alpha) \in C \text{ for some} \mathbf{x} \in E\}$$

is convex.

3. Let C be a set in a Banach space. Prove that the indicator ι_C of C is lower semicontinuous if and only if C is closed.

4. Prove that if a lower semicontinuous convex function takes the value $-\infty$, then it is nowhere finite.

5. The Laplace transform of a Borel probability measure μ defined on $\mathbb{R}^N$ is defined by the formula

$$\mathcal{L}(\mu)(\xi) = \int_{\mathbb{R}^N} \mathrm{e}^{\langle \mathbf{x}, \xi \rangle} \mathrm{d}\mu(\mathbf{x}).$$

Prove that $\mathcal{L}(\mu)$ is a lower semicontinuous proper convex function on $\mathbb{R}^N$.

[*Hint:* Lower semicontinuity follows from Fatou's lemma. See [134].]

6. Suppose $f : E \to (-\infty, \infty]$ is a proper convex function and $\mathbf{a} \in \operatorname{dom} f$. Prove that the following assertions are equivalent:

   (a) f is Lipschitz on some neighborhood of $\mathbf{a}$;

   (b) f is continuous at $\mathbf{a}$;

   (c) f is bounded on some neighborhood of $\mathbf{a}$;

   (d) f is bounded above on some neighborhood of $\mathbf{a}$.

7. The *closure* $\operatorname{cl} f$ of a proper function $f : E \to \mathbb{R} \cup \{\infty\}$ is the largest lower semicontinuous minorant of f. Prove that $\operatorname{epi}(\operatorname{cl} f) = \overline{\operatorname{epi}(f)}$ and

$$(\operatorname{cl} f)(\mathbf{x}) = \liminf_{\mathbf{y} \to \mathbf{x}} f(\mathbf{y}).$$

8. Compute the closure of the function f which is 0 on the open unit disc in $\mathbb{R}^2$, arbitrary on its boundary and infinite at all points $\mathbf{x} \in \mathbb{R}^2$ with $\|\mathbf{x}\| > 1$.

9. Prove that the closure of every proper convex function is convex too.

10. (Construction of lower semicontinuous proper convex functions) Let $f : E \to \mathbb{R} \cup \{\infty\}$ be a proper convex function whose effective domain $U = \operatorname{dom} f$ is an open set. Assuming f is continuous on U, prove that

$$(\operatorname{cl} f)(\mathbf{x}) = \begin{cases} f(\mathbf{x}) & \text{if } x \in U \\ \liminf\limits_{\mathbf{y} \to \mathbf{x}} f(\mathbf{y}) & \text{if } x \in \partial U \\ \infty & \text{if } x \in E \backslash \overline{U}. \end{cases}$$

11. Let D be a bounded closed convex subset of the real plane. Prove that D can be always represented as

$$D = \{(x, y) : f(x) \leq y \leq g(x),\ x \in [a, b]\}$$

    for suitable functions $f\colon [a, b] \to \mathbb{R}$ convex, and $g\colon [a, b] \to \mathbb{R}$ concave.

    Infer that the boundary of D is smooth except possibly for an enumerable subset.

12. Prove that $-\log|x - y|$ is a lower semicontinuous proper convex function on $\mathbb{R}^2$.

13. (The convexification of a function) By definition, the *convex hull* of a function $f : E \to \overline{\mathbb{R}}$ is the largest convex minorant $\operatorname{conv}(f)$ of f. Suppose that $f : E \to \mathbb{R} \cup \{+\infty\}$ is a proper function for which there exists a continuous affine function $h : E \to \mathbb{R}$ such that $h \leq f$. Prove that:

    (a) The convex hull of f equals the pointwise supremum of all continuous affine minorants of f and $\operatorname{conv}(f) = \operatorname{cl} f$.

    (b) $\operatorname{epi}(\operatorname{conv}(f)) = \operatorname{conv}(\operatorname{epi}(f))$.

(c) $\operatorname{conv}(f)$ verifies the formula

$$(\operatorname{conv}(f))(\mathbf{x}) = \inf\{t : (\mathbf{x}, t) \in \operatorname{conv}(\operatorname{epi}(f))\}$$
$$= \inf\left\{\sum_{k=1}^{n} \lambda_k f(\mathbf{x}_k) : \mathbf{x} = \sum_{k=1}^{n} \lambda_k \mathbf{x}_k \text{ convex representation}\right\}.$$

(d) The functions f and $\operatorname{conv}(f)$ have the same global minimum (if any).

Remark. The points x for which $f(\mathbf{x}) = (\operatorname{conv}(f))(\mathbf{x})$ are called *points of convexity*. According to Exercise 12, Jensen's discrete inequality remains valid for any probability measure with barycenter at a point of convexity (though the function f under attention might not be convex). See also the comments after Exercise 10, Section 2.2.

4.3 The Subdifferential

A deep consequence of the characterization of convex functions through their epigraphs (see Theorem 4.2.1 above) is the existence of affine supports for convex functions.

In what follows E denotes a real linear normed space and E^* denotes its dual space.

4.3.1 Theorem *Let $f : E \to \mathbb{R} \cup \{\infty\}$ be a lower semicontinuous proper convex function. Then, for every point $\mathbf{a} \in \operatorname{int}(\operatorname{dom} f)$ there is a continuous linear functional $x^* \in E^*$ such that*

$$f(\mathbf{x}) \geq f(\mathbf{a}) + x^*(\mathbf{x} - \mathbf{a}) \quad \textit{for all } \mathbf{x} \in E.$$

The affine minorant $h(\mathbf{x}) = f(\mathbf{a}) + x^*(\mathbf{x} - \mathbf{a})$ is said to be a *support* of f at $\mathbf{a}$. The terminology is motivated by the fact that the set $\{(\mathbf{x}, h(\mathbf{x})) : \mathbf{x} \in E\}$ is a hyperplane supporting the convex set epi f at the point $(\mathbf{a}, f(\mathbf{a}))$.

Proof. By Theorem 4.2.1 and Theorem 4.2.7, the epigraph of f is a closed convex set in $E \times \mathbb{R}$. It is also nonempty because f is proper. If $\mathbf{a} \in \operatorname{int}(\operatorname{dom} f)$, then $\mathbf{a}$ is a point of continuity of f; see Theorem 4.2.9. This implies that $(\mathbf{a}, f(\mathbf{a}) + \varepsilon) \in \operatorname{int}(\operatorname{epi}(f))$ for every $\varepsilon > 0$ and $(\mathbf{a}, f(\mathbf{a}))$ is a boundary point of epi(f). An appeal to the separation Theorem 3.3.1 yields the existence of a nonzero element in the dual of $E \times \mathbb{R}$ (that is, of a pair $(y^*, \lambda) \in E^* \times \mathbb{R}$ with $(y^*, \lambda) \neq (0, 0)$) such that

$$y^*(\mathbf{x}) + \lambda r \leq y^*(\mathbf{a}) + \lambda f(\mathbf{a})$$

for all $(\mathbf{x}, r) \in \operatorname{epi} f$. Since $\mathbf{a} \in \operatorname{int}(\operatorname{dom} f)$, it follows that $\lambda \neq 0$. On the other hand, letting $r \to \infty$ we infer that $\lambda \leq 0$. Therefore $\lambda < 0$, which yields

$$f(\mathbf{x}) \geq f(\mathbf{a}) + \left(\frac{-1}{\lambda} y^*\right)(\mathbf{x} - \mathbf{a}).$$

for all $\mathbf{x} \in E$. The proof ends by choosing $x^* = -y^*/\lambda$. ■

It is worth noticing that the affine minorants allow the reconstruction of lower semicontinuous proper convex functions.

4.3.2 Theorem *Every lower semicontinuous proper convex function f (from E into $\mathbb{R} \cup \{\infty\}$) is the pointwise supremum of the family of its continuous and affine minorants.*

Proof. We will show that for every $\mathbf{x}_0 \in E$ and every $\alpha \in \mathbb{R}$ such that $\alpha < f(\mathbf{x}_0)$ there exists a continuous affine minorant h of f for which $\alpha \leq h(\mathbf{x}_0) \leq f(\mathbf{x}_0)$. The details are similar to those used in the proof of Theorem 4.3.1. Since $\operatorname{epi}(f)$ is a closed convex set in $E \times \mathbb{R}$ and $(\mathbf{x}_0, \alpha) \notin \operatorname{epi}(f)$, the separation Theorem 3.3.2 yields the existence of an element $(x^*, \lambda) \in E^* \times \mathbb{R}$ and of a number $\varepsilon > 0$ such that

$$x^*(\mathbf{x}) + \lambda r < x^*(\mathbf{x}_0) + \lambda\alpha - \varepsilon \tag{4.5}$$

whenever $(\mathbf{x}, r) \in \operatorname{epi}(f)$ (that is, for all $x \in \operatorname{dom} f$ and $r \geq f(\mathbf{x})$).

If $(\mathbf{x}_0, f(\mathbf{x}_0)) \in \operatorname{epi}(f)$, then the inequality (4.5) imposes $\lambda < 0$ and in this case

$$f(\mathbf{x}) \geq h(\mathbf{x}) = \left(\frac{1}{-\lambda}\right) x^*(\mathbf{x}) + \alpha + \frac{1}{\lambda}\left(x^*(\mathbf{x}_0) - \varepsilon\right)$$

for all $\mathbf{x} \in \operatorname{dom} f$. Clearly, the function h is continuous and affine, $h(\mathbf{x}_0) > \alpha$ and $f(\mathbf{x}) \geq h(\mathbf{x})$ for every $\mathbf{x} \in E$.

If $f(\mathbf{x}_0) = \infty$ and $\lambda \neq 0$, letting $r \to \infty$ in (4.5) we infer that $\lambda < 0$ and the proof ends as in the preceding case.

If $f(\mathbf{x}_0) = \infty$ and $\lambda = 0$, then the inequality (4.5) shows that the continuous affine function

$$p(\mathbf{x}) = x^*(\mathbf{x}) - x^*(\mathbf{x}_0) + \varepsilon$$

is strictly negative at all $\mathbf{x} \in \operatorname{dom} f$.

Since $\operatorname{dom} f$ is nonempty, the discussion above yields the existence of a continuous affine minorant q for f. Therefore

$$h_n(\mathbf{x}) = q(\mathbf{x}) + n\left[x^*(\mathbf{x}) - x^*(\mathbf{x}_0) + \varepsilon\right],$$

for n large enough, provides an example of continuous affine minorant which takes at x_0 a value greater than α. The proof is complete. ■

Theorem 4.3.1 imposes the following definition:

4.3.3 Definition *A functional $x^* \in E^*$ is said to be a* subgradient *of the function f at the point $\mathbf{a} \in \operatorname{dom} f$ if*

$$f(\mathbf{x}) \geq f(\mathbf{a}) + x^*(\mathbf{x} - \mathbf{a}) \quad \textit{for all } \mathbf{x}; \tag{4.6}$$

we will refer to this inequality as the subgradient inequality.

The set of all subgradients of f at the point a is called the *subdifferential* of f at $\mathbf{a}$ and is denoted by $\partial f(\mathbf{a})$. The set-valued map $\partial f : \mathbf{a} \to \partial f(\mathbf{a})$ is called the *subdifferential* of f. In general, $\partial f(\mathbf{a})$ may be empty, a singleton or it may consist of more points.

If $\partial f(\mathbf{a})$ is not empty, then f is said to be *subdifferentiable* at $\mathbf{a}$.

We denote by $\mathrm{dom}\partial f$ the *domain* of ∂f, that is, the set of all points $\mathbf{a} \in \mathrm{dom}\ f$ such that $\partial f(\mathbf{a}) \neq \emptyset$. According to Theorem 4.3.1,

$$\mathrm{int}\,(\mathrm{dom}\ f) \subset \mathrm{dom}\partial f \subset \mathrm{dom}\ f. \tag{4.7}$$

Both inclusions can be strict. Also, $\mathrm{dom}\partial f$ is not necessarily a convex set. See Exercise 3.

When E is a real Hilbert space, the subgradients $\mathbf{x}^*$ can be viewed as vectors in E and the subgradient inequality becomes

$$f(\mathbf{x}) \geq f(\mathbf{a}) + \langle \mathbf{x}^*, \mathbf{x} - \mathbf{a} \rangle \quad \text{for all } \mathbf{x}.$$

4.3.4 Remark *When E is finite dimensional, Theorem 4.3.1 works for all $\mathbf{a} \in \mathrm{ri}\,(\mathrm{dom}\ f)$. See Remark 3.3.7. As a consequence, in this case the inclusions (4.7) can be relaxed as follows:*

$$\mathrm{ri}\,(\mathrm{dom}\ f) \subset \mathrm{dom}\ \partial f \subset \mathrm{dom}\ f.$$

4.3.5 Remark *The subdifferential at a point $\mathbf{a}$ is always a weak-star closed convex set. Recall that the weak-star topology of $E^\star$ is the topology of pointwise convergence. See Appendix B.*

If U is an open convex set in the Banach space E and $f : U \to \mathbb{R}$ is a continuous convex function, then the map $\mathbf{x} \to \partial f(\mathbf{x})$ is locally bounded. Indeed, according to Proposition 4.1.11, for each $x \in U$ there are an open ball $B_r(\mathbf{x})$ and a positive number L such that

$$|f(\mathbf{u}) - f(\mathbf{v})| \leq L\,\|\mathbf{u} - \mathbf{v}\|$$

for all $\mathbf{u}, \mathbf{v} \in B_r(\mathbf{x})$. Fix $\mathbf{y} \in B_r(\mathbf{x})$. Then $B_r(\mathbf{x})$ includes a closed ball $\overline{B}_\varepsilon(\mathbf{y})$ centered at $\mathbf{y}$, which yields

$$x^*(\mathbf{v}) \leq f(\mathbf{y} + \mathbf{v}) - f(\mathbf{y}) \leq L\,\|\mathbf{v}\|$$

for all $x^* \in \partial f(\mathbf{y})$ and all $\mathbf{v} \in E$ with $\|\mathbf{v}\| \leq \varepsilon$. Therefore $\|x^*\| \leq L$ for all

$$x^* \in \bigcup\nolimits_{\mathbf{y} \in B_r(\mathbf{x})} \partial f(\mathbf{y}).$$

The convex functions are the *only* functions that can be subdifferentiable at all points in their effective domains.

4.3.6 Theorem *Let E be a normed linear space and let $f : E \to \mathbb{R} \cup \{\infty\}$ be a function not identically ∞ that is subdifferentiable at any point $\mathbf{x} \in \mathrm{dom}\ f$. Then f is convex and lower semicontinuous.*

Proof. Let $\mathbf{x} \in \operatorname{dom} f$ arbitrarily fixed and $x^* \in \partial f(\mathbf{x})$. Then for every two points $\mathbf{x}_1$ and $\mathbf{x}_2$ in $\operatorname{dom} f$ we have

$$f(\mathbf{x}_1) \geq f(\mathbf{x}) + x^*(\mathbf{x}_1 - \mathbf{x}) \quad \text{and} \quad f(\mathbf{x}_2) \geq f(\mathbf{x}) + x^*(\mathbf{x}_2 - \mathbf{x}).$$

Multiplying the first inequality by $\lambda \in (0,1)$ and the second one by $1-\lambda$, their sum yields

$$\lambda f(\mathbf{x}_1) + (1-\lambda) f(\mathbf{x}_2) \geq f(\mathbf{x}) + x^* \left(\lambda \mathbf{x}_1 + (1-\lambda)\mathbf{x}_2 - \mathbf{x}\right)$$

and the convexity of f follows by choosing $\mathbf{x} = \lambda \mathbf{x}_1 + (1-\lambda)\mathbf{x}_2$.

To prove that f is also lower semicontinuous, suppose that $\mathbf{x}_n \to \mathbf{x}$ in E and $x^* \in \partial f(\mathbf{x})$. Then $f(\mathbf{x}_n) \geq f(\mathbf{x}) + x^*(\mathbf{x}_n - \mathbf{x})$ for all n, which implies $\liminf_{n\to\infty} f(\mathbf{x}_n) \geq f(\mathbf{x})$. Hence f is also lower semicontinuous. ∎

In Section 5.1 we will discuss the connection between subdifferentiability and directional derivatives as well as the calculus rules with subdifferentials.

The *subdifferential* of a lower semicontinuous proper convex function $f : \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ is defined as the set-valued map ∂f which associates to each $\mathbf{x} \in \mathbb{R}^N$ the subset $\partial f(\mathbf{x}) \subset \mathbb{R}^N$. Equivalently, ∂f may be seen as a graph in $\mathbb{R}^N \times \mathbb{R}^N$.

Given a set-valued map $u\colon \mathbb{R}^N \to \mathcal{P}(\mathbb{R}^N)$, we define

domain of u, $\operatorname{dom} u = \{\mathbf{x} : u(\mathbf{x}) \neq \emptyset\}$;

graph of u, $\operatorname{graph} u = \{(\mathbf{x}, \mathbf{y}) : \mathbf{x} \in \operatorname{dom} u \text{ and } \mathbf{y} \in u(\mathbf{x})\}$;

inverse of u, $u^{-1}(\mathbf{y}) = \{\mathbf{x} : \mathbf{y} \in u(\mathbf{x})\}$.

If $u, v\colon \mathbb{R}^N \to \mathcal{P}(\mathbb{R}^N)$ are set-valued maps, we write $u \subset v$, if the graph of u is contained in the graph of v.

4.3.7 Definition *A set-valued map* $u\colon \mathbb{R}^N \to \mathcal{P}(\mathbb{R}^N)$ *is said to be* monotone *if it verifies*

$$\langle \mathbf{x}_1 - \mathbf{x}_2, \mathbf{y}_1 - \mathbf{y}_2 \rangle \geq 0$$

for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^N$ *and all* $\mathbf{y}_1 \in u(\mathbf{x}_1)$, $\mathbf{y}_2 \in u(\mathbf{x}_2)$. *A monotone function* u *is called* maximal monotone *when it is maximal with respect to inclusion in the class of monotone functions, that is, if the following implication holds:*

$$v \supset u \text{ and } v \text{ monotone} \implies v = u.$$

The subdifferential ∂f of every lower semicontinuous proper convex function $f\colon \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ is monotone. This is a straightforward consequence of the subgradient inequality (4.6). As will be shown in Theorem 4.3.12, the subdifferential ∂f is actually a maximal monotone map.

Zorn's lemma easily yields the fact that every monotone set-valued map u admits a maximal monotone extension $\widetilde{u}$.

4.3.8 Remark *The graph of any maximal monotone map* $u\colon \mathbb{R}^N \to \mathcal{P}(\mathbb{R}^N)$ *is closed and thus it verifies the following conditions of upper semicontinuity:*

$$\mathbf{x}_k \to \mathbf{x},\ \mathbf{y}_k \to \mathbf{y},\ \text{and}\ \mathbf{y}_k \in u(\mathbf{x}_k)\ \text{for all}\ k \in \mathbb{N} \implies \mathbf{y} \in u(\mathbf{x}).$$

We shall prove the existence of a one-to-one correspondence between the graphs of maximal monotone maps and graphs of nonexpansive functions. Recall that a function $h\colon \mathbb{R}^N \to \mathbb{R}^N$ is called *nonexpansive* if its Lipschitz constant verifies $\mathrm{Lip}(h) \le 1$.

We shall need the following result concerning the extension of Lipschitz functions:

4.3.9 Theorem (M. D. Kirszbraun) *Suppose that* A *is a subset of* $\mathbb{R}^N$ *and* $f\colon A \to \mathbb{R}^m$ *is a Lipschitz function. Then there exists a Lipschitz function* $\widetilde{f}\colon \mathbb{R}^N \to \mathbb{R}^m$ *such that* $\widetilde{f} = f$ *on* A *and* $\mathrm{Lip}(\widetilde{f}) = \mathrm{Lip}(f)$. *Moreover, we may choose* $\widetilde{f}$ *convex, when* A *and* f *are also convex.*

Proof. When $m = 1$, we may choose

$$\widetilde{f}(\mathbf{x}) = \inf_{\mathbf{y}\in A} \{f(\mathbf{y}) + \mathrm{Lip}(f) \cdot \|\mathbf{x} - \mathbf{y}\|\}.$$

In the general case, a direct application of this remark at the level of components of f leads to an extension $\widetilde{f}$ with $\mathrm{Lip}(\widetilde{f}) \le \sqrt{m}\,\mathrm{Lip}(f)$. The existence of an extension with the same Lipschitz constant is described in [191, Section 2.10.43, p. 201]. ■

The aforementioned correspondence between graphs is realized by the *Cayley transform*, that is, by the linear isometry

$$\Phi\colon \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}^N \times \mathbb{R}^N, \quad \Phi(\mathbf{x}, \mathbf{y}) = \frac{1}{\sqrt{2}}(\mathbf{x} + \mathbf{y}, -\mathbf{x} + \mathbf{y}).$$

When $N = 1$, the Cayley transform represents a clockwise rotation of angle $\pi/4$. The precise statement of this correspondence is as follows:

4.3.10 Theorem (*G. Minty* [385]) *Let* $u\colon \mathbb{R}^N \to \mathcal{P}(\mathbb{R}^N)$ *be a maximal monotone map. Then* $J = (I + u)^{-1}$ *is defined on the whole* $\mathbb{R}^N$ *and* $\Phi(\text{graph } u)$ *is the graph of a nonexpansive function* $v\colon \mathbb{R}^N \to \mathbb{R}^N$, *given by*

$$v(\mathbf{x}) = \mathbf{x} - \sqrt{2}\,(I + u)^{-1}(\sqrt{2}\,\mathbf{x}). \tag{4.8}$$

Conversely, if $v\colon \mathbb{R}^N \to \mathbb{R}^N$ *is a nonexpansive function, then the inverse image of graph* v *under* Φ *is the graph of a maximal monotone map on* $\mathbb{R}^N$.

Proof. Let u be a monotone map and let v be the set-valued function whose graph is $\Phi(\text{graph } u)$. We shall show that v is nonexpansive in its domain (and thus single-valued). In fact, given $\mathbf{x} \in \mathbb{R}^N$ we have

$$\mathbf{y} \in v(\mathbf{x}) \quad \text{if and only if} \quad \frac{\mathbf{x} + \mathbf{y}}{\sqrt{2}} \in u\Big(\frac{\mathbf{x} - \mathbf{y}}{\sqrt{2}}\Big) \tag{4.9}$$

and this yields $\mathbf{y} \in \mathbf{x} - \sqrt{2}\,(I+u)^{-1}(\sqrt{2}\mathbf{x})$ for all $\mathbf{y} \in v(\mathbf{x})$. Now, if $\mathbf{x}_k \in \mathbb{R}^N$ and $\mathbf{y}_k \in v(\mathbf{x}_k)$ for $k = 1, 2$, then we infer from (4.9) that

$$\langle (\mathbf{x}_1 - \mathbf{y}_1) - (\mathbf{x}_2 - \mathbf{y}_2), (\mathbf{x}_1 + \mathbf{y}_1) - (\mathbf{x}_2 + \mathbf{y}_2) \rangle \geq 0,$$

hence $\|\mathbf{y}_1 - \mathbf{y}_2\|^2 \leq \|\mathbf{x}_1 - \mathbf{x}_2\|^2$. This shows that v is indeed nonexpansive. The same argument shows that Φ^{-1} maps graphs of nonexpansive functions into graphs of monotone functions. Assuming that u is the maximal monotone, we shall show that the domain of v is $\mathbb{R}^N$. In fact, if the contrary was true, we could apply Theorem 4.3.9 to extend v to a nonexpansive function $\widetilde{v}$ defined on the whole $\mathbb{R}^N$, and then $\Phi^{-1}(\mathrm{graph}\widetilde{v})$ provides a monotone extension of u, which contradicts the maximality of u. ■

4.3.11 Corollary *Let $u\colon \mathbb{R}^N \to \mathcal{P}(\mathbb{R}^N)$ be a maximal monotone map. Then $J = (I+u)^{-1}$ is a nonexpansive map of $\mathbb{R}^N$ into itself.*

Proof. It is easy to see that $I + u$ (and thus $(I+u)^{-1}$) is monotone. By Theorem 4.3.10, the maximality of u yields the surjectivity of $I + u$, hence $\mathrm{dom}(I+u)^{-1} = \mathbb{R}^N$. In order to prove that $(I+u)^{-1}$ is also a nonexpansive function, let us consider points $\mathbf{x}_k \in \mathbb{R}^N$ and $\mathbf{y}_k \in u(\mathbf{x}_k)$ (for $k = 1, 2$). Then

$$\begin{aligned} \|\mathbf{x}_1 - \mathbf{x}_2\|^2 &\leq \langle \mathbf{x}_1 - \mathbf{x}_2, \mathbf{x}_1 - \mathbf{x}_2 + \mathbf{y}_1 - \mathbf{y}_2 \rangle \\ &\leq \|\mathbf{x}_1 - \mathbf{x}_2\| \cdot \|\mathbf{x}_1 + \mathbf{y}_1 - (\mathbf{x}_2 + \mathbf{y}_2)\|, \end{aligned} \tag{4.10}$$

which yields $\|\mathbf{x}_1 - \mathbf{x}_2\| \leq \|(\mathbf{x}_1 + \mathbf{y}_1) - (\mathbf{x}_2 + \mathbf{y}_2)\|$. Particularly, if $\mathbf{x}_1 + \mathbf{y}_1 = \mathbf{x}_2 + \mathbf{y}_2$, then $\mathbf{x}_1 = \mathbf{x}_2$, and this shows that $(I+u)^{-1}$ is single-valued. Consequently, $(I+u)^{-1}(\mathbf{x}_k + \mathbf{y}_k) = \mathbf{x}_k$ for $k \in \{1, 2\}$ and thus (4.10) yields the nonexpansivity of $(I+u)^{-1}$. ■

An important class of maximal monotone maps is provided by the subdifferentials of convex functions.

4.3.12 Theorem *If $f\colon \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper convex function, then ∂f is a maximal monotone map.*

Proof. We already noticed that ∂f is a monotone map. According to Theorem 4.3.10, the maximality of ∂f is equivalent to the surjectivity of $\partial f + I$. A simple argument based on the optimality of convex functions follows from Corollary 5.4.5. ■

An example of a maximal monotone set-valued function is the *duality map* of a Banach space E

$$J(\mathbf{x}) = \left\{ x^* \in E^* : x^*(\mathbf{x}) = \|\mathbf{x}\|^2 = \|x^*\|^2 \right\};$$

indeed, $J(\mathbf{x}) = \partial f(\mathbf{x})$, where f is the function defined by $f(\mathbf{x}) = \|\mathbf{x}\|^2/2$ for $\mathbf{x} \in E$.

Let E be one of the spaces $L^p(\mathbb{R}^N)$ with $p \in (1, \infty)$, and let A be a maximal monotone operator from E to E^*. Let $f : E \to \mathbb{R} \cup \{\infty\}$ be a lower semicontinuous proper convex function such that at least one of the following conditions are verified:

$$\mathrm{dom} A \cap \mathrm{int}\,(\mathrm{dom}\ f) \neq \emptyset \text{ or } \mathrm{dom}\ f \cap \mathrm{int}\,(\mathrm{dom} A) \neq \emptyset.$$

Then $A + \partial f$ is a maximal monotone operator. The details are quite involving, but this example is important in the theory of partial differential equations. See V. Barbu and T. Precupanu [37], Theorem 2.4, pp. 124-128.

The subdifferential of a lower semicontinuous proper convex function on $\mathbb{R}^N$ plays yet another special property that will be discussed in what follows.

4.3.13 Definition *A* cyclically monotone *map is any set-valued function u (from $\mathbb{R}^N$ into $\mathcal{P}(\mathbb{R}^N)$) such that*

$$\langle \mathbf{y}_1, \mathbf{x}_2 - \mathbf{x}_1 \rangle + \langle \mathbf{y}_2, \mathbf{x}_3 - \mathbf{x}_2 \rangle + \cdots + \langle \mathbf{y}_m, \mathbf{x}_1 - \mathbf{x}_m \rangle \leq 0$$

for all finite families of points $(\mathbf{x}_k, \mathbf{y}_k) \in \mathbb{R}^N \times \mathbb{R}^N$ such that $\mathbf{y}_k \in u(\mathbf{x}_k)$ for $k \in \{1, \ldots, m\}$.

By the subgradient inequality (4.6), the subdifferential of a lower semicontinuous proper convex function on $\mathbb{R}^N$ is cyclically monotone. The converse also holds, as is shown by the following integrability result:

4.3.14 Theorem (R. T. Rockafellar [505, pp. 238–239]) *For any cyclically monotone map $u\colon \mathbb{R}^N \to \mathcal{P}(\mathbb{R}^N)$ there exists a lower semicontinuous proper convex function $f : \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ such that $u \subset \partial f$.*

Proof. Put $\Gamma = \left\{(\mathbf{x}, \mathbf{y}) \in \mathbb{R}^N \times \mathbb{R}^N : \mathbf{y} \in u(\mathbf{x})\right\}$ and chose arbitrarily a point $(\mathbf{x}_0, \mathbf{y}_0) \in \Gamma$. We define the function f by the formula

$$f(\mathbf{x}) = \sup \left\{ \langle \mathbf{y}_0, \mathbf{x}_1 - \mathbf{x}_0 \rangle + \langle \mathbf{y}_1, \mathbf{x}_2 - \mathbf{x}_1 \rangle + \cdots + \langle \mathbf{y}_m, \mathbf{x} - \mathbf{x}_m \rangle \right\},$$

where the supremum is taken over all finite families of points $(\mathbf{x}_1, \mathbf{y}_1), ..., (\mathbf{x}_m, \mathbf{y}_m)$ in Γ. Since f is the supremum of a family of continuous affine functions, it is a lower semicontinuous and convex. Moreover, $f(\mathbf{x}_0) = 0$ due to the property of cyclic monotonicity of u. Hence f is also a proper function. It remains to prove that $\Gamma \subset \partial f$, that is, for every $(\mathbf{x}, \mathbf{y}) \in \Gamma$ we have

$$f(\mathbf{z}) \geq f(\mathbf{x}) + \langle \mathbf{y}, \mathbf{z} - \mathbf{x} \rangle$$

whenever $\mathbf{z} \in \mathbb{R}^N$. For this, it suffices to show that for all $\alpha < f(\mathbf{x})$ and $\mathbf{z} \in \mathbb{R}^N$ we have

$$f(\mathbf{z}) \geq \alpha + \langle \mathbf{y}, \mathbf{z} - \mathbf{x} \rangle.$$

Because of the definition of $f(\mathbf{x})$, the fact that $\alpha < f(\mathbf{x})$ implies the existence of a finite family $(\mathbf{x}_1, \mathbf{y}_1), ..., (\mathbf{x}_m, \mathbf{y}_m)$ of points in Γ such that

$$\alpha \leq \langle \mathbf{y}_0, \mathbf{x}_1 - \mathbf{x}_0 \rangle + \langle \mathbf{y}_1, \mathbf{x}_2 - \mathbf{x}_1 \rangle + \cdots + \langle \mathbf{y}_m, \mathbf{x} - \mathbf{x}_m \rangle.$$

Then

$$\alpha + \langle \mathbf{y}, \mathbf{z} - \mathbf{x} \rangle \leq \langle \mathbf{y}_0, \mathbf{x}_1 - \mathbf{x}_0 \rangle + \langle \mathbf{y}_1, \mathbf{x}_2 - \mathbf{x}_1 \rangle + \cdots + \langle \mathbf{y}_m, \mathbf{x} - \mathbf{x}_m \rangle + \langle \mathbf{y}, \mathbf{z} - \mathbf{x} \rangle \leq f(x)$$

and the proof is done. ■

Exercises

1. Consider the norm function $f(\mathbf{x}) = \|\mathbf{x}\|$ in an arbitrary real Banach space E. Prove that:

$$\partial f(\mathbf{x}) = \begin{cases} \{x^* \in E^* : \|x^*\| \leq 1\} & \text{if } \mathbf{x} = 0 \\ \{x^* \in E^* : \|x^*\| = 1, x^*(\mathbf{x}) = \|\mathbf{x}\|\} & \text{if} \mathbf{x} \neq 0. \end{cases}$$

   Compare to the case of absolute value on the real line.

2. Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a convex function. Show that the subdifferential of the function $F(x, y) = f(x) - y$ is given by the formula

$$\partial F(x, y) = \{(\lambda, -1) : \lambda \in \partial f(x)\}.$$

3. (Domain of the subdifferential) Consider the function f defined on $\mathbb{R}^2$ as follows:

$$f(x) = \begin{cases} \max\{1 - \sqrt{x}, |y|\} & \text{if } x \geq 0 \\ \infty & \text{if } x < 0. \end{cases}$$

   Prove that f is a convex function and that the domain of ∂f is

$$\{(x, y) : x \geq 0\} \setminus \{(0, y) : |y| < 1)\},$$

   which is not a convex set.

4. Suppose that $f : \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper convex function. Prove that:

   (a) $\partial f(\mathbf{a})$ is a closed convex set for every $\mathbf{a} \in \text{dom } f$;

   (b) if $\mathbf{a}$ belongs to the relative interior of dom f, then $\partial f(\mathbf{a})$ is a nonempty compact set;

   (c) the graph of the subdifferential is closed in the sense that $\mathbf{x}_n \to \mathbf{x}$ in dom f and $\mathbf{y}_n \to \mathbf{y}$ in $\mathbb{R}^N$ with $\mathbf{y}_n \in \partial f(\mathbf{x}_n)$ for all n imply that $\mathbf{y} \in \partial f(\mathbf{x})$.

   [*Hint:* (b) It suffices to prove that $\partial f(\mathbf{a})$ is a bounded set. Use Corollary 4.1.9.]

5. Suppose that E is a Banach space and $f : E \to \mathbb{R}$ is a convex Lipschitz function. Adapt the argument of Remark 4.3.5 to prove that $\partial f(A)$ is bounded, whenever A is a bounded subset of E.

6. Let f be a continuous convex function on a Banach space E. Show that f is bounded on the bounded subsets of E if and only if ∂f is bounded on the bounded subsets of E.

7. (Subdifferentiabily and strict convexity) Let U be an open convex subset of $\mathbb{R}^N$ and let $f : U \to \mathbb{R}$ be a convex function. Prove that f is strictly convex if, and only if, it verifies one of the following equivalent conditions:

   (a) $\mathbf{x}, \mathbf{y} \in U$, $\mathbf{x} \neq \mathbf{y}$ and $\mathbf{z} \in \partial f(\mathbf{x})$ implies $f(\mathbf{y}) - f(\mathbf{x}) > \langle \mathbf{y} - \mathbf{x}, \mathbf{z} \rangle$;

   (b) $\mathbf{x}, \mathbf{y} \in U$, $\partial f(\mathbf{x}) \cap \partial f(\mathbf{y}) \neq \emptyset$ implies $\mathbf{x} = y$.

8. Consider the function $f(\mathbf{x}) = -\left(1 - \|\mathbf{x}\|^2\right)^{1/2}$ for $\mathbf{x}$ in the closed unit ball $\overline{B}$ of $\mathbb{R}^N$ and $f(\mathbf{x}) = \infty$ on $\mathbb{R}^N \backslash \overline{B}$. Prove that f is convex and $\partial f(\mathbf{x}) = \emptyset$ at all points $\mathbf{x}$ with $\|\mathbf{x}\| = 1$.

9. (I. Popescu [482]) Suppose that $f : \mathbb{R}^N_+ \to \mathbb{R}$ is a convex function with the property that $f(0) = 0$ and there exists $C \geq 0$ such that

$$f(\mathbf{x}) \geq -C \|\mathbf{x}\|^2 \quad \text{for all} \mathbf{x} \in \mathbb{R}^N_+.$$

   Prove that $f(\mathbf{x}) \geq 0$ everywhere on his domain.

4.4 Positively Homogeneous Convex Functions

A function f defined on a convex cone C is said to be *positively homogeneous* (of degree 1) if

$$f(\lambda \mathbf{x}) = \lambda f(\mathbf{x}) \quad \text{for all } \mathbf{x} \in C \text{ and } \lambda > 0.$$

Two simple examples of positively homogeneous functions defined on a real normed linear space E are the norm function $\|\cdot\|$ and the function $\mathbf{x} \to |x^*(\mathbf{x})|$, associated to a continuous linear functional $x^* \in E^*$.

Convexity of positively homogeneous functions can be related to other important properties like subadditivity and sublinearity. Recall that a function $p\colon E \to \mathbb{R}$ is called *subadditive* if $p(\mathbf{x}+\mathbf{y}) \leq p(\mathbf{x})+p(\mathbf{y})$ for all $\mathbf{x}, y \in E$ and *superadditive* if $-p$ is subadditive; p is called *sublinear* if it is both positively homogeneous and subadditive. A sublinear function p is a *seminorm* if $p(\lambda \mathbf{x}) = |\lambda| p(\mathbf{x})$ for all $\lambda \in \mathbb{R}$. A seminorm p is a *norm* if

$$p(\mathbf{x}) = 0 \implies \mathbf{x} = 0.$$

Seminorms are instrumental in the theory of locally convex spaces.

4.4.1 Lemma *If f is a positively homogeneous function defined on a convex cone C, then f is convex (concave) if and only if f is subadditive (superadditive).*

Proof. Suppose that f is convex and $\mathbf{x}, \mathbf{y} \in C$. Then

$$\frac{1}{2} f(\mathbf{x}+\mathbf{y}) = f\Big(\frac{\mathbf{x}+\mathbf{y}}{2}\Big) \leq \frac{1}{2}\left(f(\mathbf{x}) + f(\mathbf{y})\right)$$

and so $f(\mathbf{x}+y) \leq f(\mathbf{x}) + f(y)$. Conversely, suppose that f is subadditive. Then

$$f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) \leq f((1-\lambda)\mathbf{x}) + f(\lambda\mathbf{y}) = (1-\lambda)f(\mathbf{x}) + \lambda f(\mathbf{y})$$

for all $\mathbf{x}, \mathbf{y} \in C$ and $\lambda \in (0,1)$, which shows that f is convex. ■

For concave functions f defined on $[0,\infty)$, the property of subadditivity is equivalent to the fact that $f(0) \geq 0$. See Section 1.1, Exercise 4. This does not work in higher dimensions. For example, as noticed in Section 4.1, the geometric mean of $N \geq 2$ variables is a positive, concave and superadditive function.

4.4.2 Lemma *Let f be a positively homogeneous function defined on a convex cone C and taking positive values. If the level set $\{\mathbf{x} \in C : f(\mathbf{x}) \leq 1\}$ is convex, then f is a convex function.*

In other words, every positively homogeneous, quasiconvex and positive function is convex.

Proof. According to Lemma 4.4.1, it suffices to show that f is subadditive. For that, let $\mathbf{x}, \mathbf{y} \in C$ and choose scalars α and β such that $\alpha > f(\mathbf{x})$, and $\beta > f(\mathbf{y})$. Since f is positive and positively homogeneous, $f(\mathbf{x}/\alpha) \leq 1$ and $f(\mathbf{y}/\beta) \leq 1$. Thus $\mathbf{x}/\alpha$ and $\mathbf{y}/\beta$ both lie in the level set of f at height 1. The assumed convexity of this level set shows that

$$\frac{1}{\alpha+\beta} f(\mathbf{x}+\mathbf{y}) = f\Big(\frac{\mathbf{x}+\mathbf{y}}{\alpha+\beta}\Big) = f\Big(\frac{\alpha}{\alpha+\beta}\cdot\frac{\mathbf{x}}{\alpha} + \frac{\beta}{\alpha+\beta}\cdot\frac{\mathbf{y}}{\beta}\Big) \leq 1,$$

that is, $f(\mathbf{x}+\mathbf{y}) \leq \alpha+\beta$ whenever $\alpha > f(\mathbf{x})$, $\beta > f(\mathbf{y})$. Hence $f(\mathbf{x}+\mathbf{y}) \leq f(\mathbf{x}) + f(\mathbf{y})$, which shows that f is subadditive. ■

An example of how the last lemma yields the convexity of some functions is as follows. Let $p \geq 1$ and consider the function f given on the positive orthant $\mathbb{R}^N_+$ by the formula

$$f(x_1, \ldots, x_N) = (x_1^p + \cdots + x_N^p)^{1/p}.$$

Clearly, f is positive and positively homogeneous, and f^p is convex as a sum of convex functions. Hence the level set

$$\{\mathbf{x} \in X : f(\mathbf{x}) \leq 1\} = \{\mathbf{x} \in X : f^p(\mathbf{x}) \leq 1\}$$

is convex and this implies that f is a convex function. By Lemma 4.4.1 we conclude that f is subadditive, a fact which is equivalent with the Minkowski inequality.

We finish this section by introducing and investigating two important examples of positively homogeneous functions that can be attached to certain convex sets.

The *support function* of a nonempty convex set C in a real normed linear space E is defined by the formula

$$h_C(u^*) = \sup_{\mathbf{x}\in C} u^*(\mathbf{x}), \quad u^* \in E^*.$$

This is a lower semicontinuous sublinear function, taking values in $\mathbb{R}\cup\{\infty\}$. The domain of a support function is a convex cone; the domain is the entire space E if and only if the set C is bounded.

When $E = \mathbb{R}^N$, the dual space can be identified with E, so

$$h_C(\mathbf{u}) = \sup_{\mathbf{x}\in C}\langle \mathbf{x}, \mathbf{u}\rangle, \quad \mathbf{u} \in \mathbb{R}^N.$$

Moreover, if $\|\mathbf{u}\| = 1$, then the sets $H_\alpha = \{\mathbf{x} \in \mathbb{R}^N : \langle \mathbf{x}, \mathbf{u}\rangle = \alpha\}$ represent parallel hyperplanes, each having $\mathbf{u}$ as a normal vector; $\alpha = h_C(\mathbf{u})$ is the value for which H_α supports C, and C is contained in the half-space H_α^-.

The support function of a singleton $C = \{\mathbf{a}\}$ in $\mathbb{R}^N$ is $h_C(\mathbf{u}) = \langle \mathbf{u}, \mathbf{a}\rangle$, while in the case of the closed unit ball, $h_C(\mathbf{u}) = \|\mathbf{u}\|$.

The support functions and Minkowski operations interact in a very simple way, precisely, for every compact nonempty convex sets K and L in $\mathbb{R}^N$ and every $\alpha, \beta \geq 0$ we have

$$h_{\alpha K+\beta L} = \alpha h_K + \beta h_L.$$

More on the correspondence between convex sets and sublinear functions is offered by the following result:

4.4.3 Theorem (L. Hörmander [267])

(a) *The support function h_C of a nonempty compact convex set C in $\mathbb{R}^N$ is positively homogeneous and convex. Moreover,*

$$C = \{\mathbf{x} \in \mathbb{R}^N : \langle \mathbf{x}, \mathbf{u}\rangle \leq h_C(\mathbf{u}) \text{ for every } \mathbf{u} \in \mathbb{R}^N\},$$

which shows that C is the intersection of all half-spaces that contain it.

(b) *Conversely, let $h\colon \mathbb{R}^N \to \mathbb{R}$ be a positively homogeneous convex function. Then the set*

$$C = \{\mathbf{x} \in \mathbb{R}^N : \langle \mathbf{x}, \mathbf{u}\rangle \leq h(\mathbf{u}) \quad \text{for every } \mathbf{u} \in \mathbb{R}^N\}$$

is nonempty, compact, convex and its support function is h.

Proof. (a) Clearly, $C \subset \{\mathbf{x} \in \mathbb{R}^N : \langle \mathbf{x}, \mathbf{u}\rangle \leq h_C(\mathbf{u}) \text{ for every } \mathbf{u} \in \mathbb{R}^N\}$. If a vector $\mathbf{z}$ verifies $\langle \mathbf{z}, \mathbf{u}\rangle \leq h_C(\mathbf{u})$ for every $\mathbf{u} \in \mathbb{R}^N$ and $\mathbf{z} \notin C$, then by the basic separation theorem we infer the existence of a vector $\mathbf{u}$ such that $h_C(\mathbf{u}) = \sup\{\langle \mathbf{x}, \mathbf{u}\rangle : \mathbf{x} \in C\} < \langle \mathbf{z}, \mathbf{u}\rangle$, a contradiction. Therefore $\mathbf{z} \in C$.

(b) By Theorem 4.3.1, $\partial h(\mathbf{v}) \neq \emptyset$ for every $\mathbf{v} \in \mathbb{R}^N$. If $\mathbf{x} \in \partial h(\mathbf{v})$, then $h(\mathbf{u}) \geq h(\mathbf{v}) + \langle \mathbf{u} - \mathbf{v}, \mathbf{x}\rangle$ for all $\mathbf{u}$. Replacing $\mathbf{u}$ by $\lambda\mathbf{u}$ (for $\lambda > 0$ arbitrarily fixed) and taking into account that h is positively homogeneous, we infer that $h(\mathbf{u}) \geq h(\mathbf{v})/\lambda + \langle \mathbf{u} - v/\lambda, \mathbf{x}\rangle$, whence, letting $\lambda \to \infty$, we infer that $h(\mathbf{u}) \geq \langle \mathbf{u}, \mathbf{x}\rangle$

for all $\mathbf{u} \in \mathbb{R}^N$. Therefore C contains the union of all subdifferentials $\partial f(\mathbf{v})$. In particular, it is a nonempty set. C is also a closed convex set being an intersection of closed subspaces.

Let $\mathbf{e}_1, ..., \mathbf{e}_N$ be the natural basis of $\mathbb{R}^N$. For each $\mathbf{x} \in C$ we have $-h(-\mathbf{e}_k) \leq -\langle \mathbf{x}, -\mathbf{e}_k \rangle = \langle \mathbf{x}, \mathbf{e}_k \rangle \leq h(\mathbf{e}_k)$ for all k, which yields the boundedness of C. Thus C is a compact set.

Clearly $h_C \leq h$. For the other inequality, choose arbitrarily $\mathbf{u} \in \mathbb{R}^N$ and $\mathbf{z} \in \partial h(\mathbf{u})$ and notice that $0 = h(\mathbf{0}) \geq h(\mathbf{u}) + \langle \mathbf{0} - \mathbf{u}, \mathbf{z} \rangle$, that is, $h(\mathbf{u}) \leq \langle \mathbf{u}, \mathbf{z} \rangle$. Since $\mathbf{z} \in C$ we conclude that $h(\mathbf{u}) \leq h_C(\mathbf{u})$. ∎

4.4.4 Corollary *If two nonempty, compact and convex sets in $\mathbb{R}^N$ have the same support, then they coincide.*

See Exercise 8 for an alternative proof.

Let C be a nonempty convex set in a normed linear space E. The *gauge function* (also called the *Minkowski functional*) associated to C is the function

$$\gamma_C : E \to \mathbb{R} \cup \{\infty\}, \quad \gamma_C(\mathbf{x}) = \inf\{\lambda > 0 : \mathbf{x} \in \lambda C\},$$

with the convention that $\gamma_C(\mathbf{x}) = \infty$ if $\mathbf{x} \in \lambda C$ for no $\lambda > 0$.

4.4.5 Proposition *If the convex set C is closed and contains the origin, then:*

(a) *the gauge function γ_C is a lower semicontinuous, sublinear and positive function*;

(b) $C = \{\mathbf{x} \in E : \gamma_C(\mathbf{x}) \leq 1\}$;

(c) *γ_C is real valued and continuous if the origin lies in the interior of C.*

Proof. Clearly, $\gamma_C \geq 0$ and $\gamma_C(0) = 0$. The positive homogeneity of γ_C results as follows:

$$\begin{aligned}
\gamma_C(\lambda \mathbf{x}) &= \inf\{\mu > 0 : \lambda \mathbf{x} \in \mu C\} \\
&= \inf\{\mu > 0 : \mathbf{x} \in \lambda^{-1} \mu C\} \\
&= \lambda \inf\{\lambda^{-1} \mu > 0 : \mathbf{x} \in \lambda^{-1} \mu C\} \\
&= \lambda \inf\{\tau > 0 : \mathbf{x} \in \tau C\} = \lambda \gamma_C(\mathbf{x}),
\end{aligned}$$

for all $\lambda > 0$ and $\mathbf{x} \in \mathrm{dom}\gamma_C$. We have

$$\begin{aligned}
\{\mathbf{x} \in E : \gamma_C(\mathbf{x}) \leq 1\} &= \{\mathbf{x} \in E : \inf\{\lambda > 0 : \lambda^{-1} \mathbf{x} \in C\} \leq 1\} \\
&= \{\mathbf{x} \in E : \lambda^{-1} \mathbf{x} \in C \text{ for all} \lambda > 1\} \\
&= \bigcap\nolimits_{\lambda > 1} \lambda C = C,
\end{aligned}$$

whence, according to Lemma 4.4.2, it follows that γ_c is a convex function. Taking into account Lemma 4.4.1, we infer that γ_C is actually sublinear. On the other hand, due to the positive homogeneity of γ_C, we have $\{\mathbf{x} \in E : \gamma_C(\mathbf{x}) \leq r\} = rC$ for all $r > 0$, a fact which implies that γ_C is lower semicontinuous.

If 0 is interior to C, then C includes a ball $\overline{B}_\varepsilon(0)$ for some $\varepsilon > 0$. Then for every point $\mathbf{x} \in E$, $\mathbf{x} \neq 0$, we have

$$\gamma_C(\mathbf{x}) = \gamma_C\left(\frac{\|\mathbf{x}\|}{\varepsilon} \cdot \frac{\varepsilon \mathbf{x}}{\|\mathbf{x}\|}\right) = \frac{\|\mathbf{x}\|}{\varepsilon}\gamma_C\left(\frac{\varepsilon \mathbf{x}}{\|\mathbf{x}\|}\right) \leq \frac{1}{\varepsilon}\|\mathbf{x}\| < \infty,$$

so γ_C is finite everywhere. ■

4.4.6 Corollary *If C is a closed, bounded and convex subset of a real normed linear space E and $\operatorname{int} C \neq \emptyset$, then C is homeomorphic to the closed unit ball $B = \{\mathbf{x} \in E : \|\mathbf{x}\| \leq 1\}$.*

Proof. Via a translation, we may assume that $0 \in \operatorname{int} C$. Then there is $r > 0$ such that C contains all $\mathbf{x} \in E$ with $\|\mathbf{x}\| \leq r$, whence $\gamma_C(\mathbf{x}) \leq \|\mathbf{x}\|/r$ for all $\mathbf{x} \in E$. Since C is bounded, there is $R > 0$ such that $\|\mathbf{x}\| \leq R$ for all $\mathbf{x} \in C$. Therefore, if $\mathbf{x} \in \lambda C$, then $\|\mathbf{x}/\lambda\| \leq R$, whence $\gamma_C(\mathbf{x}) \geq \|\mathbf{x}\|/R$ for all $\mathbf{x} \in E$.

An immediate consequence is the continuity of the gauge function γ_C. Indeed,

$$|\gamma_C(\mathbf{x}) - \gamma_C(\mathbf{y})| \leq \max\{\gamma_C(\mathbf{x}-\mathbf{y}), \gamma_C(\mathbf{y}-\mathbf{x})\} \leq \frac{1}{r}\|\mathbf{x}-\mathbf{y}\|$$

for all $\mathbf{x}, y \in E$. The proof ends by noticing that the map

$$T : E \to E, \quad T(\mathbf{x}) = \begin{cases} \gamma_C(\mathbf{x})\frac{\mathbf{x}}{\|\mathbf{x}\|} & \text{if } \mathbf{x} \neq 0 \\ 0 & \text{if } \mathbf{x} = 0 \end{cases}$$

is continuous, with continuous inverse and $T(C) = B$. ■

We shall show in Section 8.1 that the support function and the gauge function are related through the concept of Legendre-Fenchel duality. A particular case makes the object of Exercise 6.

Exercises

1. Let $p : E \to \mathbb{R}$ be a sublinear function on a normed linear space. Prove that p is continuous if and only if there is $M > 0$ such that $p(\mathbf{x}) \leq M\|\mathbf{x}\|$ for all $\mathbf{x}$.

2. (B. Dacorogna [146]) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be separately convex and positively homogeneous of degree one. Prove that f is convex. Give an example to show that this result does not work for functions of three (or more) variables.

3. Let f be a positive function defined on a cone C such that for some $\alpha > 0$ it verifies the condition $f(\lambda\mathbf{x}) = \lambda^\alpha f(\mathbf{x})$ for all $\lambda \geq 0$ and $\mathbf{x} \in C$ (that is, f is positively homogeneous of degree α).

   (a) Assume $\alpha \geq 1$. Prove that f is convex if and only if it is quasiconvex. Infer the convexity of the function $f(x_1, \ldots, x_N) = \left(\sum_{k=1}^N x_k^\alpha\right)^p$ for all $p \geq 1/\alpha$.

   (b) If $\alpha < 1$, then prove that f is concave if and only if it is quasiconcave.

4. The *Cobb–Douglas function* is defined as

$$f(\mathbf{x}) = Ax_1^{r_1} \cdots x_N^{r_n}, \quad (x_1, \ldots, x_n) \in \mathbb{R}^N_{++},$$

where $A > 0$ and all exponents r_k are strictly positive.

(a) Prove that $\log f$ is a concave function and infer that $f = e^{\log f}$ is quasiconcave.

(b) Prove that f is concave if and only if $\sum_{k=1}^N r_k \leq 1$.

5. The *Leontief function* is defined by $f(\mathbf{x}) = \left(\min\left\{\frac{x_k}{\alpha_k} : 1 \leq k \leq N\right\}\right)^\alpha$ for $\mathbf{x} \in \mathbb{R}^N_{++}$, where $\alpha > 0$ and all parameters α_k are strictly positive. Prove that this function is quasiconcave and that it is concave if and only if $\alpha \in (0, 1]$.

6. (Polar duality) Let C be a closed convex set containing the origin. Prove that:

(a) The Minkowski functional of C is the support function of the polar set C°, and the Minkowski functional of C° is the support function of C, that is,

$$p_C = h_{C^\circ} \text{ and } p_{C^\circ} = h_C.$$

(b) C° is bounded if and only if $0 \in \operatorname{int} C$ (so, by the bipolar theorem, C is bounded if and only if $0 \in \operatorname{int} C^\circ$). Infer that the Minkowski functional of C is real-valued if $0 \in \operatorname{int} C$.

7. Find the support and the gauge function of each of the following sets:

(a) $\{\mathbf{x} \in \mathbb{R}^N : \sup\{|x_1|, \ldots, |x_N|\} \leq 1\}$;

(b) $\{\mathbf{x} \in \mathbb{R}^N : |x_1| + \cdots + |x_N| \leq 1\}$;

(c) $\{\mathbf{x} \in \mathbb{R}^N : x_1^2 + \cdots + x_N^2 \leq 1\}$.

8. Suppose that C_1 and C_2 are two nonempty compact convex subsets in $\mathbb{R}^N$. Prove that

$$C_1 \subset C_2 \text{ if and only if } h_{C_1} \leq h_{C_2}.$$

This shows that $C_1 = C_2$ if and only if $h_{C_1} = h_{C_2}$.

9. If C is a nonempty convex set in $\mathbb{R}^N$, then prove that

$$\operatorname{ri}(C) = \{\mathbf{x} : \langle \mathbf{x}, \mathbf{v} \rangle < h_C(\mathbf{v}) \text{ for all } \mathbf{v} \neq 0\}.$$

10. (A converse of Proposition 4.4.5). Prove that any positively homogeneous, subadditive and continuous function $p : E \to [0, \infty)$ is a gauge function.

4.5 Inequalities Associated to Perspective Functions

Many basic inequalities can be easily derived via the following functional operation generating new convex functions from old ones.

Assuming C a convex cone in a linear space, the *perspective* of a function $f : C \to \mathbb{R}$ is the positively homogeneous function

$$\widetilde{f} : C \times (0, \infty) \to \mathbb{R}, \quad \widetilde{f}(\mathbf{x}, t) = tf\left(\frac{\mathbf{x}}{t}\right).$$

4.5.1 Lemma *The perspective of a convex/concave function is a function of the same nature.*

Proof. Indeed, assuming (to make a choice) that f is convex, then for all $(\mathbf{x}, s), (\mathbf{y}, t) \in C \times (0, \infty)$ and $\lambda \in [0, 1]$ we have

$$\begin{aligned} f\left(\frac{(1-\lambda)\mathbf{x} + \lambda\mathbf{y}}{(1-\lambda)s + \lambda t}\right) &= f\left(\frac{(1-\lambda)s}{(1-\lambda)s + \lambda t} \cdot \frac{\mathbf{x}}{s} + \frac{\lambda t}{(1-\lambda)s + \lambda t} \cdot \frac{\mathbf{y}}{t}\right) \\ &\leq \frac{(1-\lambda)s}{(1-\lambda)s + \lambda t} f\left(\frac{\mathbf{x}}{s}\right) + \frac{\lambda t}{(1-\lambda)s + \lambda t} f\left(\frac{\mathbf{y}}{t}\right), \end{aligned} \tag{4.11}$$

that is,

$$\widetilde{f}((1-\lambda)\mathbf{x} + \lambda\mathbf{y}, (1-\lambda)s + \lambda t) \leq (1-\lambda)\widetilde{f}(\mathbf{x}, s) + \lambda\widetilde{f}(\mathbf{y}, t). \tag{4.12}$$

■

4.5.2 Corollary *The following functions are convex on the indicated domains:*

(a) $x \log x - x \log t$ *on* $(0, \infty) \times (0, \infty)$;

(b) $\|\mathbf{x}\|^2 / t$ *on* $\mathbb{R}^N \times (0, \infty)$;

(c) $(\langle \mathbf{c}, \mathbf{x} \rangle + d)\, f\left(\frac{A\mathbf{x}+b}{\langle \mathbf{c}, \mathbf{x} \rangle + d}\right)$ *on* $\left\{\mathbf{x} : \langle \mathbf{c}, \mathbf{x} \rangle + d > 0 \text{ and } \frac{A\mathbf{x}+b}{\langle \mathbf{c}, \mathbf{x} \rangle + d} \in \operatorname{dom} f\right\}$,

where $A \in \mathrm{M}_N(\mathbb{R})$, $b \in \mathbb{R}^N$ *and* f *is a convex function defined on a convex set in* $\mathbb{R}^N$.

The convexity of the function in case (a) easily yields the convexity of the Kullback–Leibler divergence.

According to Lemma 4.4.1, the perspective of a convex (respectively concave) function is subadditive (respectively superadditive). Exercise 1 addresses the case of strictly concave functions and indicates at this basis short proofs for some classical inequalities.

Deeper results can be obtained via a generalization of Jensen's inequality that works for positively homogeneous continuous convex functions $\Phi \colon \mathbb{R}^2_+ \to \mathbb{R}$. Since

$$\Phi(u, v) = \begin{cases} v\Phi(u/v, 1) & \text{if } u \geq 0 \text{ and } v > 0 \\ u\Phi(1, 0) & \text{if } u \geq 0 \text{ and } v = 0, \end{cases}$$

it follows that Φ is the perspective function associated to the continuous convex function $\varphi(u) = \Phi(u, 1)$, $u \geq 0$.

Since Φ is continuous, $\Phi(0, 0) = 0$. Notice also that

$$\sup_{u>0} \varphi'_+(u) = \lim_{u\to\infty} \frac{\varphi(u)}{u} = \Phi(1, 0). \tag{4.13}$$

The details are left as Exercise 4. According to Theorem 2.1.5, the function φ is the pointwise supremum of a family of affine functions, precisely,

$$\varphi(u) = \sup\{a + bu : (a, b) \in G\}, \text{ for all } u \geq 0,$$

where $G = \{(\varphi(s) - s\varphi'_+(s), \varphi'_+(s)) : s \in (0, \infty)\}$. Therefore for $u, v \in \mathbb{R}^2_+$ with $v > 0$ we have

$$\Phi(u, v) = \sup\{av + bu : (a, b) \in G\}, \tag{4.14}$$

but the last equality also works when $v = 0$, due to formula (4.13).

The next result extends the basic ingredient used by O. Hanner [236] to prove his famous inequalities. See Exercise 7.

4.5.3 Theorem (T.-C. Lim [337]) *Let $\Phi\colon \mathbb{R}^2_+ \to \mathbb{R}$ be a positively homogeneous continuous convex function. Then for every measure space (X, Σ, μ) and every μ-integrable function $f\colon X \to \mathbb{R}^2_+$ we have the inequality*

$$\Phi\left(\int_X f\, d\mu\right) \leq \int_X \Phi \circ f \, \mathrm{d}\mu. \tag{4.15}$$

If Φ is concave, then the inequality holds in the reverse way.

Proof. Put $f = (f_1, f_2)$. Since Φ is continuous, the inverse image of every Borel set is a Borel set, so $\Phi \circ f$ is measurable. On the other hand, $\Phi \circ f$ is majorized in absolute value by a multiple of $f_1 + f_2$. Indeed, for all x such that $f_1(x) + f_2(x) \neq 0$ we have

$$\begin{aligned} |(\Phi \circ f)(x)| &= |\Phi(f_1(x), f_2(x))| \\ &= (f_1(x) + f_2(x)) \left|\Phi\left(\frac{f(x)}{f_1(x) + f_2(x)}, \frac{f_2(x)}{f_1(x) + f_2(x)}\right)\right| \\ &\leq \max_{0\leq x\leq 1} |\Phi(x, 1 - x)|\,(f_1(x) + f_2(x)), \end{aligned}$$

while for all other values of x we have $\Phi(f_1(x), f_2(x)) = \Phi(0, 0) = 0$. Therefore $\Phi \circ f$ is an integrable function. According to formula (4.14),

$$\begin{aligned} \int_X (\Phi \circ f)(x)\, \mathrm{d}\mu &= \int_X \sup_{(a,b)\in G} (af_2 + bf_1)\, \mathrm{d}\mu \\ &\geq \sup_{(a,b)\in G} \left(a \int_X f_2 \, \mathrm{d}\mu + b \int_X f_1 \, \mathrm{d}\mu\right) = \Phi\left(\int_X f \, \mathrm{d}\mu\right). \end{aligned}$$

■

See Exercise 5 for a converse. Also, the role of $\mathbb{R}^2_+$ can be taken by every convex cone in $\mathbb{R}^N$ $(N \geq 2)$. Indeed, the only change is to use Theorem 4.3.2 instead of Theorem 2.1.5.

In applications we shall need the following remark to derive the convexity of the function Φ.

4.5.4 Remark *If $\Phi : \mathbb{R}^2_+ \to \mathbb{R}$ is a positively homogeneous continuous function, then its convexity is equivalent to the convexity of the restriction of Φ to $(0,\infty)\times(0,\infty)$, which in turn is equivalent to the convexity of the function $\varphi(u) = \Phi(u,1)$ on $(0,\infty)$.*

4.5.5 Theorem (Minkowski's inequality) *For $p \in (-\infty,0)\cup[1,\infty)$ and $f,g \in L^p(\mu)$ we have*

$$\|f+g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}, \tag{4.16}$$

while for $0 < p < 1$ the inequality works in the reverse sense,

$$\|f+g\|_{L^p} \geq \|f\|_{L^p} + \|g\|_{L^p}. \tag{4.17}$$

If f is not 0 almost everywhere, then we have equality if and only if $g = \lambda f$ almost everywhere, for some $\lambda \geq 0$.

Proof. Apply Theorem 4.5.3 for $f(x) = (|u(x)|^p, |v(x)|^p)$ and

$$\Phi(u,v) = (u^{1/p} + v^{1/p})^p \quad (p \in \mathbb{R},\ p \neq 0)$$

The convexity of Φ follows from that of $\varphi(u) = (1+u^{1/p})^p$ for $u > 0$. Using the second derivative test we obtain that φ is strictly convex for $0 < p < 1$ and strictly concave for $p \in (-\infty,0)\cup(1,\infty)$. ∎

There is an integral Minkowski type inequality even in the case $p = 0$. In fact, letting $p \to 0+$ in (4.17), and taking into account Exercise 1 (d) in Section 2.2, we obtain the following result:

4.5.6 Theorem (Minkowski's inequality for $p = 0$) *Assume that (Ω,Σ,μ) is a finite measure space. For every $f,g \in L^1(\mu)$, $f,g \geq 0$, we have*

$$\exp\left(\frac{1}{\mu(X)}\int_X \log(f(x)+g(x))\,\mathrm{d}\mu\right) \geq \exp\left(\frac{1}{\mu(X)}\int_X \log f(x)\,\mathrm{d}\mu\right) + \exp\left(\frac{1}{\mu(X)}\int_X \log g(x)\,\mathrm{d}\mu\right).$$

Exercises

1. Prove that the functions $xy/(x+y)$ and $(xy+yz+zx)/(x+y+z)$ are concave for $x,y,z > 0$. For more examples of rational concave functions, see Appendix C.

2. (E. K. Godunova [222] and G. J. Woeginger [589]) Consider the perspective function $\widetilde{f}(x,y) = yf(x/y)$, associated to a strictly concave function $f : [0,\infty) \to \mathbb{R}$. Prove that

$$\sum_{k=1}^{n} \tilde{f}\,(x_k, y_k) \leq \tilde{f}\left(\sum_{k=1}^{n} x_k, \sum_{k=1}^{n} y_k\right)$$

whenever $x_1, ..., x_n$ and $y_1, ..., y_n$ are strictly positive numbers. The inequality works in the reverse way if f is strictly convex. In either case the equality holds if and only if there exists a number $t > 0$ such that $x_k = ty_k$ for all k.

This result encompasses many classical inequalities such as those of Cauchy, Rogers-Hölder, Minkowski and Gibbs (for $f(x) = x^{1/2}$, $x^{1/p}$, $(x^{1/p} + 1)^p$ and $\log x$ respectively).

3. (E. K. Godunova [222]) Suppose that $M(x,y)$ is a concave function of two real variables, increasing with x for constant y and decreasing with y for constant x. Prove that:

(a) $M(f(x), g(y))$ is a concave function whenever $f : U \to \mathbb{R}$ is concave and $g : V \to \mathbb{R}$ is convex.

(b) A similar result holds if the words "convex" and "concave" are interchanged.

If $g(y) = y$ for all y, then the condition that M decreases with y for constant x is not necessary. Infer that the function $y \log f(x) - y \log y$ is concave on $U \times (0,\infty)$, whenever $f : U \to \mathbb{R}$ is a strictly positive concave function.

4. Prove the formula (4.13), that is, the fact $\sup_{u>0} \varphi'_+(u) = \lim_{u\to\infty} \frac{\varphi(u)}{u} = \Phi(1,0)$.

5. Prove the following converse of Theorem 4.5.3: If Φ is continuous and the inequality (4.15) holds for every μ-integrable function $f\colon X \to \mathbb{R}^2_+$ for which $\Phi \circ f$ is also μ-integrable, then Φ is positively homogeneous and convex.

6. (More on the Rogers-Hölder inequality) Extend the result of Exercise 3, Section 1.2, to the general context of measure spaces.

7. (Hanner's inequalities [236]) If $f, g \in L^p(\mu)$ and $2 \leq p < \infty$, then

$$\|f+g\|_{L^p}^p + \|f-g\|_{L^p}^p \leq (\|f\|_{L^p} + \|g\|_{L^p})^p + \big|\|f\|_{L^p} - \|g\|_{L^p}\big|^p,$$

equivalently (by making the replacements $f \to f+g$ and $g \to f-g$),

$$(\|f+g\|_{L^p} + \|f-g\|_{L^p})^p + \big|\|f+g\|_{L^p} - \|f-g\|_{L^p}\big|^p \geq 2^p(\|f\|_{L^p}^p + \|g\|_{L^p}^p).$$

In case $1 < p \leq 2$, the above inequalities are reversed.

[*Hint*: Apply Theorem 4.5.3 for $f(x) = (|u(x)|^p, |v(x)|^p)$ and $\Phi(u,v) = (u^{1/p} + v^{1/p})^p + |u^{1/p} - v^{1/p}|^p$. The function Φ is convex for $1 < p \leq 2$ and concave for $2 \leq p < \infty$. To check this, notice that the second derivative of the function $\varphi(u) = \Phi(u,1) = (u^{1/p}+1)^p + |u^{1/p} - 1|^p$ is

$$\varphi''(u) = \frac{1-p}{p} x^{-(p+1)/p} \left[\left(x^{-1/p} + 1 \right)^{p-2} - \left| x^{-1/p} - 1 \right|^{p-2} \right] .]$$

Remark. The *modulus of convexity* of a Banach space E is the function $\delta_E : (0,2] \to [0,1]$ defined by

$$\delta_E(\varepsilon) = \inf\{1 - \left\| \frac{\mathbf{x}+\mathbf{y}}{2} \right\| : \|\mathbf{x}\| \leq 1, \|\mathbf{y}\| \leq 1 \text{ and } \|\mathbf{x}-\mathbf{y}\| \geq \varepsilon\}.$$

Hanner's inequalities provide exact values for the moduli of convexity in the case of L^p-spaces. Precisely,

$$\left(1 - \delta_{L^p}(\varepsilon) + \frac{\varepsilon}{2}\right)^p + \left|1 - \delta_{L^p}(\varepsilon) - \frac{\varepsilon}{2}\right|^p = 2 \text{ for } p \in (1,2)$$

and $\delta_{L^p}(\varepsilon) = 1 - (1 - (\varepsilon/2)^p)^{1/p}$ for p in $[2,\infty)$. As a consequence, $\delta_{L^p}(\varepsilon) > 0$ for all $\varepsilon \in (0,2]$, which shows that the L^p-spaces with $p \in (1,\infty)$ are uniformly convex.

8. (P. Maréchal [359]) Notice that the function $f(x_1, ..., x_N) = \prod_{k=1}^N x_k^{-\alpha_k}$ is log-convex on $\mathbb{R}^N_{++}$ whenever $\alpha_1, ..., \alpha_N > 0$. Infer that the function

$$g(x_1, ..., x_N, z) = |z|^\gamma \prod_{k=1}^N x_k^{-\alpha_k}$$

is convex on $\mathbb{R}^N_{++} \times \mathbb{R}$ provided that $\gamma \geq 1 + \sum_{k=1}^N \alpha_k$.

[*Hint:* In the case of function g, use the properties of perspective functions.]

9. (The contractive version of Jensen's inequality) Suppose that C is a convex set that contains the origin and $f : C \to \mathbb{R} \cup \{\infty\}$ is a proper convex function such that $f(0) \leq 0$. Prove that

$$f\left(\sum_{k=1}^n \lambda_k x_k\right) \leq \sum_{k=1}^n \lambda_k f(x_k)$$

for all finite families $\{x_1, ..., x_n\} \subset C$ and $\{\lambda_1, ..., \lambda_n\} \subset [0,1]$ with $\sum_{k=1}^n \lambda_k \leq 1$. Infer that

$$\sum_{k=1}^n \lambda_k x_k \geq \prod_{k=1}^n x_k^{\lambda_k}$$

for all $x_1, ..., x_n > 0$ and all $\lambda_1, ..., \lambda_n \in [0,1]$ with $\sum_{k=1}^n \lambda_k \leq 1$.

10. (A generalization of the concept of perspective function due to P. Maréchal [359]) Let C be a cone in a linear space E, and let K be a convex set in a linear space F. The perspective function $f \triangle g$, associated to a pair of functions $f : C \to \mathbb{R}$ and $g : K \to (0, \infty)$ is defined by the formula

$$(f \triangle g)(\mathbf{x}, \mathbf{y}) = g(\mathbf{y}) f\left(\frac{\mathbf{x}}{g(\mathbf{y})}\right) \quad \text{for } (\mathbf{x}, \mathbf{y}) \in C \times K.$$

Prove that:

(a) $f \triangle g$ is convex provided that f is a convex function such that $f(\mathbf{0}) \leq 0$ and g is concave;

(b) $f \triangle g$ is concave when f and g are concave and $f(\mathbf{0}) \geq 0$.

A noncommutative generalization of the notion of perspective function is described in Section 7.2.

11. (The closure of the perspective function) Assuming C is a convex cone in a Hilbert space H, the perspective of a convex function $f : C \to \mathbb{R}$ was defined in Section 3.5 as $\widetilde{f} : C \times (0, \infty) \to \mathbb{R}$, $\widetilde{f}(\mathbf{x}, t) = tf(\mathbf{x}/t)$. This does not necessarily produce a lower semicontinuous function. Prove that the lower semicontinuous envelope of the perspective function of f is

$$\widetilde{f} : H \times \mathbb{R} \to \mathbb{R}, \quad \widetilde{f}(\mathbf{x}, t) = \begin{cases} tf(\mathbf{x}/t) & \text{if } t > 0 \\ (\mathrm{rec} f)(\mathbf{0}) & \text{if} t = 0 \\ \infty & \text{otherwise.} \end{cases}$$

Notice that $\tilde{f}$ is proper, convex, and positively homogeneous.

4.6 Jensen's Integral Inequality in the Context of Several Variables

Given an arbitrary measure space (Ω, Σ, μ), the integrability of a vector-valued function $f : \Omega \to \mathbb{R}^N$, $f = (f_1, ..., f_N)$, means the integrability of its components, so that its integral is defined via the formula

$$\int_\Omega f(x) \mathrm{d}\mu(x) = \left(\int_\Omega f_1(x) \mathrm{d}\mu(x), ..., \int_\Omega f_N(x) \mathrm{d}\mu(x) \right).$$

Moreover, for every linear functional L on $\mathbb{R}^N$ we have

$$L\left(\int_\Omega f(x) \mathrm{d}\mu(x) \right) = \int_\Omega L(f(x)) \mathrm{d}\mu(x).$$

As in the scalar case, we call the integral of a function f with respect to a probability measure, the *expectation* (or *expected value,* or *arithmetic mean*) of f and we will denote it also by $E(f)$.

In this section we are especially interested in the case where Ω is a convex subset of $\mathbb{R}^N$. According to Remark 3.1.11, the convex subsets C of $\mathbb{R}^N$ are not necessarily Borel sets if $N \geq 2$. Therefore, in order to ensure that the inclusion of C into $\mathbb{R}^N$ is measurable, we will endow C with its *relative Borel algebra*,

$$\mathcal{B}(C) = \left\{ B \cap C : B = \text{Borel subset of } \mathbb{R}^N \right\},$$

and we will call every probability measure $\mu : \mathcal{B}(C) \to [0,1]$ a *Borel probability measure* on C. The concept of barycenter is attached to all Borel probability measures on C that belong to the class

$$\mathcal{P}^1(C) = \left\{ \mu : \mu \text{ Borel probability measure on} C \text{ and } \int_C \|\mathbf{x}\| \, \mathrm{d}\mu(\mathbf{x}) < \infty \right\}.$$

This class includes all Borel probability measures if C is bounded.

4.6.1 Definition *The barycenter of a probability measure $\mu \in \mathcal{P}^1(C)$ is the point*

$$bar(\mu) = \int_C \mathbf{x} \mathrm{d}\mu(\mathbf{x}).$$

Clearly,

$$x^*(\mathrm{bar}(\mu)) = \int_C x^*(\mathbf{x}) \mathrm{d}\mu(\mathbf{x})$$

for all linear functionals x^* on $\mathbb{R}^N$. This fact serves as a model for introducing the concept of barycenter for any Borel probability measure defined on an arbitrary compact convex set. See Chapter 9.2.

The barycenter of the discrete probability measure $\lambda = \sum_{k=1}^n \lambda_k \delta_{\mathbf{x}_k}$ concentrated at the points $\mathbf{x}_1, ..., \mathbf{x}_n \in C$ is

$$\mathrm{bar}(\lambda) = \int_{\mathbb{R}^N} \mathbf{x} \mathrm{d}\lambda(\mathbf{x}) = \sum_{k=1}^n \lambda_k \mathbf{x}_k,$$

which clearly belongs to the convex set C. The membership of the barycenter to the convex set on which the probability measure is defined is valid in all finite-dimensional spaces.

4.6.2 Proposition (H. Rubin and O. Wesler [513]) *If C is a nonempty convex set in $\mathbb{R}^N$, then it contains the barycenter of every probability measure $\mu \in \mathcal{P}^1(C)$.*

Proof. By mathematical induction with respect to $n = \dim C$. For $n = 0$, C consists of one single point, $\mathbf{x}_0$, which implies that $\mathcal{P}^1(C) = \{\delta_{\mathbf{x}_0}\}$. Clearly, $\mathrm{bar}(\delta_{\mathbf{x}_0}) = \mathbf{x}_0$.

Suppose now that $\dim C = n \leq N$ and the statement holds for all convex sets of dimension m with $m < n$. Since C is nonempty, its relative interior is nonempty, and we can assume without loss of generality that $\mathbf{0} \in \mathrm{ri}(C)$. In this

case $\mathrm{aff}(C)$ coincides with $L = \mathrm{span}(C)$ and $\mathbf{0}$ is an interior point of C relative to the ambient space L.

We have to prove that the barycenter of every probability measure $\mu \in \mathcal{P}^1(C)$ lies in C. For this, notice first that $\mathrm{bar}(\mu) \in L$. Indeed, if $\mathrm{bar}(\mu) \notin L$, then there would exist a linear functional $x^* : \mathbb{R}^N \to \mathbb{R}$ such that $x^*(\mathrm{bar}(\mu)) > 0$ and $x^*|_L = 0$. But this is impossible because

$$0 = x^*(\mathrm{bar}(\mu)) - x^*\left(\int_C \mathbf{x}\mathrm{d}\mu(\mathbf{x})\right) = x^*(\mathrm{bar}(\mu)) - \int_C x^*(\mathbf{x})\mathrm{d}\mu(\mathbf{x}) = x^*(\mathrm{bar}(\mu)).$$

Therefore $\mathrm{bar}(\mu) \in L$. Now consider the case $\mathrm{bar}(\mu) \in L \backslash C$. According to the Separation Theorem 3.3.2. there exists a nonzero linear functional $x^* : L \to \mathbb{R}$ such that

$$x^*(\mathrm{bar}(\mu)) > \sup\{x^*(\mathbf{c}) : \mathbf{c} \in C\}.$$

This functional can be extended to a linear functional on $\mathbb{R}^N$ (also denoted x^*) by letting $x^*(\mathbf{x}) = 0$ for $\mathbf{x}$ in the orthogonal complement of L with respect to $\mathbb{R}^N$. We have

$$\begin{aligned}\int_C [x^*(\mathrm{bar}(\mu)) - x^*(\mathbf{x})]\,\mathrm{d}\mu(\mathbf{x}) &= x^*(\mathrm{bar}(\mu)) - \int_C x^*(\mathbf{x})\mathrm{d}\mu(\mathbf{x}) \\ &= x^*(\mathrm{bar}(\mu)) - x^*\left(\int_C \mathbf{x}\mathrm{d}\mu(\mathbf{x})\right) = 0,\end{aligned}$$

and since the function under square brackets is positive, it follows that μ is concentrated on the set $C_0 = C \cap H$, where $H = \{\mathbf{x} \in L : x^*(\mathbf{x}) = x^*(\mathrm{bar}(\mu))\}$ is a hyperplane in L. Thus $\dim C_0 \leq n-1$ and $\mu_0 = \mu|_{C_0} \in \mathcal{P}^1(C_0)$. According to our induction assumption, $\mathrm{bar}(\mu) = \mathrm{bar}(\mu_0) \in C_0 \subset C$, and this contradiction completes the proof. ■

A special method to construct new probability measure spaces from old ones is described in what follows.

Let (Ω_1, Σ_1) and (Ω_2, Σ_2) be two measurable spaces and let $g : \Omega_1 \to \Omega_2$ be a measurable map (that is, $A \in \Sigma_2$ implies $g^{-1}(A) \in \Sigma_1$). Then, for every probability measure $\mu_1 : \Sigma_1 \to [0,1]$, the formula

$$\mu_2(A) = \mu_1(g^{-1}(A)), \quad A \in \Sigma_2,$$

defines a probability measure μ_2 on Σ_2 called the *push-forward measure* (or the *image* of μ_1 through g). Usually, the measure μ_2 is denoted $g\#\mu_1$. The main feature of this measure is the following *change-of-variables formula,*

$$\int_{\Omega_2} f(y)\mathrm{d}\,(g\#\mu_1)\,(y) = \int_{\Omega_1} f(g(x))\mathrm{d}\mu_1(x), \tag{4.18}$$

valid for all μ_2-integrable functions f.

When $\Omega_2 = \mathbb{R}^N$, $\Sigma_2 = \mathcal{B}\left(\mathbb{R}^N\right)$, $f =$ the identity of $\mathbb{R}^N$, and $g \in L^1(\mu_1)$, this formula yields the existence of the barycenter of $g\#\mu_1$ and also the fact that

$$\mathrm{bar}(g\#\mu_1) = \int_{\Omega_1} g(x)\mathrm{d}\mu_1(x) = E(g). \tag{4.19}$$

When f is an arbitrary convex functions, the formulas (4.18) and (4.19) are connected by Jensen's inequality:

4.6.3 Theorem (The integral form of Jensen's inequality) *Let (Ω, Σ, μ) be a probability space and let $g\colon \Omega \to \mathbb{R}^N$ be a μ-integrable function. If f is a lower semicontinuous proper convex function whose effective domain C includes the image of g, then $E(g) \in C$ and*

$$f(E(g)) \leq \int_\Omega f(g(x))\mathrm{d}\mu(x). \tag{4.20}$$

Notice that the right-hand side integral always exists but it might be ∞ if the μ-integrability of $f \circ g$ is not expressly asked. When both functions g and $f \circ g$ are μ-integrable, then the above inequality becomes

$$f(E(g)) \leq E(f \circ g).$$

Moreover, if in addition f is strictly convex and $\partial f\,(E(g)) \neq \emptyset$, then this inequality becomes an equality if and only if g is constant μ-almost everywhere.

Proof. The membership of $E(g)$ to C follows from Proposition 4.6.2 and the fact that $E(g) = \mathrm{bar}(g\#\mu)$.

The function f is $\mathcal{B}(C)$-measurable since the level sets of f are relatively closed subsets of C. As a consequence, the function $f \circ g$ is μ-measurable (and the same is true for $(f \circ g)^+$ and $(f \circ g)^-$).

According to Theorem 4.3.2, for every $\varepsilon > 0$ there is a continuous affine function $A(\mathbf{y}) = \langle \mathbf{y}, \mathbf{v}\rangle + a$ on $\mathbb{R}^N$ such that

$$A(\mathbf{y}) \leq f(\mathbf{y}) \text{ for all } \mathbf{y} \in C \text{ and } A(E(g)) > f(E(g)) - \varepsilon.$$

Then

$$f(g(x)) \geq A(g(x)) = \langle g(x), \mathbf{v}\rangle + a \quad \text{for all } x \in \Omega, \tag{4.21}$$

whence we infer that $(f(g(x)))^-$ is a μ-integrable function (being dominated by the μ-integrable function $\|g(x)\|\,\|\mathbf{v}\| + a$). Since every positive measurable function admits an integral (possibly $+\infty$) it follows that the integral in the right-hand side of inequality (4.20) exists. This inequality holds because $\varepsilon > 0$ was arbitrarily chosen and

$$\begin{aligned}\int_\Omega f(g(x))\,\mathrm{d}\mu(x) &\geq \int_\Omega A(g(x))\,\mathrm{d}\mu(x)\\ &= A\left(\int_\Omega g(x)\mathrm{d}\mu\right) = A(E(g)) > f(E(g)) - \varepsilon.\end{aligned}$$

The last assertion, concerning the equality case when f is strictly convex, is a consequence of the inequality

$$f(\mathbf{y}) > f(E(g)) + \langle \mathbf{y} - E(g), \mathbf{z}\rangle, \quad \text{for all } \mathbf{y} \in C, \mathbf{y} \neq E(g),$$

where $\mathbf{z} \in \partial f(E(g))$. This makes the equality $E(f \circ g) = f(E(g))$ possible if, and only if, $g = E(g)$ μ-almost everywhere. ■

The condition $\partial f(E(g)) \neq \emptyset$ in Theorem 4.6.3 is fulfilled when $E(g) \in \operatorname{ri}(C)$. See Remark 4.3.4.

4.6.4 Corollary *If f is a lower semicontinuous proper convex function defined on a convex subset C of $\mathbb{R}^N$ and $\mu \in \mathcal{P}^1(C)$, then $bar(\mu) \in C$ and*

$$f(bar(\mu)) \leq \int_C f \mathrm{d}\mu.$$

Theorem 4.6.3 and Corollary 4.6.4 are actually equivalent (due to the technique of push-forward measure).

The derivation of the discrete case of Jensen's inequality from Theorem 4.6.3 can be done in two ways:

(a) by considering the probability measure space (Ω, Σ, μ), where

$$\Omega = \{1, ..., n\}, \ \Sigma = \mathcal{P}(\Omega) \text{ and } \mu = \sum_{k=1}^{n} \lambda_k \delta_k, \tag{4.22}$$

and the function $g : \{1, ..., n\} \to \mathbb{R}$ defined by $g(k) = \mathbf{x}_k$ for all k. In this case the inequality $f(E(g)) \leq E(f \circ g)$ becomes

$$f\left(\sum_{k=1}^{n} \lambda_k \mathbf{x}_k\right) \leq \sum_{k=1}^{n} \lambda_k f(\mathbf{x}_k).$$

(b) by considering the probability measure space $(\mathbb{R}^N, \mathcal{B}(\mathbb{R}^N), \lambda)$, corresponding to the discrete probability measure $\lambda = \sum_{k=1}^{n} \lambda_k \delta_{\mathbf{x}_k}$ concentrated at the points $\mathbf{x}_1, ..., \mathbf{x}_n \in \mathbb{R}^N$ and choosing as g the identity of $\mathbb{R}^N$. In this variant, Theorem 2.2.3 yields the same inequality, written as

$$f(\mathrm{bar}(\lambda)) \leq E(f).$$

Surprisingly, Jensen's integral inequality can be deduced from the discrete case via an approximation argument. See Lemma 9.2.6.

Exercises

1. (Convex series closed sets) A subset C of a real Banach space E is called convex series closed if every point $\mathbf{x} \in E$ that can be represented as the sum of a convergent series $\sum_{k=1}^{\infty} \lambda_k \mathbf{x}_k$ with $\lambda_k \geq 0$, $\sum_{k=1}^{\infty} \lambda_k = 1$ and $\mathbf{x}_k \in C$, must be in C. Prove that all open convex subsets (as well as all closed convex subsets) of E are convex series closed.

   [*Hint:* Use the separation theorems.]

2. (Jensen's inequality for series) Suppose that $f : \mathbb{R}^N \to \mathbb{R}$ is a convex function, $(\mathbf{x}_n)_n$ is a sequence in $\mathbb{R}^N$ and $(\lambda_n)_n$ is a sequence in $[0,1]$ such that $\sum_{n=1}^{\infty} \lambda_n = 1$ and the series $\sum_{n=1}^{\infty} \lambda_n \mathbf{x}_n$ is absolutely convergent. Prove that the series $\sum_{n=1}^{\infty} \lambda_n f(\mathbf{x}_n)$ is either convergent or has infinite sum, and in both cases,

$$f\left(\sum_{n=1}^{\infty} \lambda_n \mathbf{x}_n\right) \leq \sum_{n=1}^{\infty} \lambda_n f(\mathbf{x}_n).$$

3. (An estimate of the Jensen gap) Let (Ω, Σ, μ) be a probability measure space and let $g : \Omega \to \mathbb{R}^N$, $g = (g_1, ..., g_N)$, be a function such that $\|g(\cdot)\| \in L^2(\mu)$. Consider also an open convex set $U \subset \mathbb{R}^N$ that includes the image of g and a convex function $f \in C^2(U)$ such that

$$\alpha \|\mathbf{v}\|^2 \leq d^2 f(\mathbf{x})(\mathbf{v}, \mathbf{v}) \leq \beta \|\mathbf{v}\|^2 \quad \text{for all } \mathbf{v} \in \mathbb{R}^N,$$

for suitable strictly positive constants α and β. Prove that

$$\alpha \|\mathrm{var}(g)\|_{HS} \leq E(f \circ g) - f(E(g)) \leq \beta \|\mathrm{var}(g)\|_{HS}.$$

Here $\mathrm{cov}(g) = (\mathrm{cov}(g_i, g_j))_{i,j}$ denotes the *covariance matrix* of g.

[*Hint*: Apply Taylor's formula.]

4.7 The Prékopa–Leindler Inequality

The aim of this section is to present some results related to the geometric theory of convexity.

4.7.1 Theorem (Prékopa–Leindler inequality) *Let $0 < \lambda < 1$ and let f, g, and h be positive integrable functions on $\mathbb{R}^N$ satisfying*

$$h((1-\lambda)\mathbf{u} + \lambda \mathbf{v}) \geq f(\mathbf{u})^{1-\lambda} g(\mathbf{v})^{\lambda}$$

for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^N$. Then

$$\int_{\mathbb{R}^N} h(\mathbf{x})\, \mathrm{d}\mathbf{x} \geq \left(\int_{\mathbb{R}^N} f(\mathbf{x})\, \mathrm{d}\mathbf{x}\right)^{1-\lambda} \left(\int_{\mathbb{R}^N} g(\mathbf{x})\, \mathrm{d}\mathbf{x}\right)^{\lambda}.$$

The equality condition for this inequality can be found in a paper by S. Dubuc [167].

The Prékopa–Leindler inequality is just the particular case of a much more general result, the Borell–Brascamp–Lieb inequality. See Theorem 4.7.6. Before entering into details, we mention several immediate consequences of the Prékopa–Leindler inequality.

4.7.2 Corollary (Preservation of log-concavity by integration) *If*

$$F : \mathbb{R}^M \times \mathbb{R}^N \to [0, \infty), \quad F = F(\mathbf{x}, \mathbf{y}),$$

is an integrable log*-concave function, then the function*

$$\varphi(\mathbf{x}) = \int_{\mathbb{R}^N} F(\mathbf{x}, \mathbf{y}) \mathrm{d}\mathbf{y}, \quad \mathbf{x} \in \mathbb{R}^M,$$

is also log*-concave.*

Proof. Fix arbitrarily $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^M$ and $\lambda \in (0, 1)$. The functions $h(\mathbf{y}) = F((1-\lambda)\mathbf{x}_1 + \lambda\mathbf{x}_2, \mathbf{y})$, $f(\mathbf{y}) = F(\mathbf{x}_1, \mathbf{y})$ and $g(\mathbf{y}) = F(\mathbf{x}_2, \mathbf{y})$ verify the hypotheses of Theorem 4.7.1, which implies that

$$\begin{aligned} \varphi\left((1-\lambda)\mathbf{x}_1 + \lambda\mathbf{x}_2\right) &= \int_{\mathbb{R}^N} F((1-\lambda)\mathbf{x}_1 + \lambda\mathbf{x}_2, \mathbf{y}) \mathrm{d}\mathbf{y} \\ &\geq \left(\int_{\mathbb{R}^N} F(\mathbf{x}_1, \mathbf{y}) \mathrm{d}\mathbf{y}\right)^{1-\lambda} \left(\int_{\mathbb{R}^N} F(\mathbf{x}_2, \mathbf{y}) \mathrm{d}\mathbf{y}\right)^{\lambda} \\ &= \left(\varphi(\mathbf{x}_1)\right)^{1-\lambda} \left(\varphi(\mathbf{x}_2)\right)^{\lambda}. \end{aligned}$$

Therefore φ is log-concave. ■

4.7.3 Corollary (Preservation of log-concavity by convolution product) *The convolution product of two* log*-concave functions defined on* $\mathbb{R}^N$ *is also a* log*-concave function.*

Proof. Indeed, if $f_1, f_2 : \mathbb{R}^N \to [0, \infty)$ are log-concave, so is the function $F(\mathbf{x}, \mathbf{y}) = f_1(\mathbf{x} - \mathbf{y}) f_2(\mathbf{y})$ on $\mathbb{R}^N \times \mathbb{R}^N$. The proof ends by taking into account Corollary 4.7.2. ■

It is worth noticing that the Prékopa–Leindler inequality represents the functional analogue of another important inequality related to the isoperimetric problem in Euclidean spaces:

4.7.4 Theorem (Lusternik's general Brunn–Minkowski inequality) *Let* $s, t > 0$ *and let* A *and* B *be two nonempty bounded measurable sets in* $\mathbb{R}^N$ *such that* $sA + tB$ *is also measurable. Then*

$$\mathrm{Vol}_N(sA + tB)^{1/N} \geq s\mathrm{Vol}_N(A)^{1/N} + t\mathrm{Vol}_N(B)^{1/N}.$$

Here Vol_N *denotes the* N*-dimensional Lebesgue measure.*

Proof. Since the Lebesgue measure Vol_N is positively homogeneous of degree N (that is, $\mathrm{Vol}_N(\alpha A) = \alpha^N \mathrm{Vol}_N(A)$ for every Borel set A and every $\alpha \geq 0$), we may restrict to the case where $s = 1 - \lambda$ and $t = \lambda$ for some $\lambda \in (0, 1)$. Then we

apply the Prékopa–Leindler inequality for $f = \chi_A$, $g = \chi_B$ and $h = \chi_{(1-\lambda)A+\lambda B}$, which yields

$$\begin{aligned}\mathrm{Vol}_N((1-\lambda)A+\lambda B) &= \int_{\mathbb{R}^N} \chi_{(1-\lambda)A+\lambda B}(\mathbf{x})\,\mathrm{d}x \\ &\geq \left(\int_{\mathbb{R}^N} \chi_A(\mathbf{x})\,\mathrm{d}\mathbf{x}\right)^{1-\lambda}\left(\int_{\mathbb{R}^N} \chi_B(\mathbf{x})\,\mathrm{d}\mathbf{x}\right)^{\lambda} \\ &= \mathrm{Vol}_N(A)^{1-\lambda}\mathrm{Vol}_N(B)^{\lambda}.\end{aligned}$$

Applying this inequality for A replaced by $\mathrm{Vol}_N(A)^{-1/N}A$, B replaced by $\mathrm{Vol}_N(B)^{-1/N}B$, and λ replaced by

$$\frac{\lambda \mathrm{Vol}_N(B)^{1/N}}{(1-\lambda)\mathrm{Vol}_N(A)^{1/N}+\lambda\mathrm{Vol}_N(B)^{1/N}},$$

we obtain

$$\mathrm{Vol}_N((1-\lambda)A+\lambda B)^{1/N} \geq (1-\lambda)\mathrm{Vol}_N(A)^{1/N}+\lambda\mathrm{Vol}_N(B)^{1/N},$$

which ends the proof. ■

The hypothesis on the measurability of $sA + tB$ cannot be deduced from the measurability of A and B. A counterexample can be found in a paper by W. Sierpiński [536].

The classical Brunn–Minkowski inequality represents the particular case of Theorem 4.7.4 for convex bodies. A *convex body* is understood as a compact convex set in $\mathbb{R}^N$, with nonempty interior. In this case the measurability of the sets $sX + tY$ is automatic.

4.7.5 Theorem (The Brunn–Minkowski inequality) *Let $\lambda \in (0,1)$ and let K and L be two convex bodies in $\mathbb{R}^N$. Then*

$$\mathrm{Vol}_N((1-\lambda)K+\lambda L)^{1/N} \geq (1-\lambda)\mathrm{Vol}_N(K)^{1/N}+\lambda\mathrm{Vol}_N(L)^{1/N}.$$

Equality holds precisely when K and L are equal up to translation and dilation.

The connection of the Brunn–Minkowski inequality with the isoperimetric problems is briefly described in the Comments at the end of this chapter.

The Prékopa–Leindler inequality represents the case $p = 0$ of the following general result:

4.7.6 Theorem (The Borell–Brascamp–Lieb inequality) *Suppose that $0 < \lambda < 1$, $-1/N \leq p \leq \infty$, and f, g, and h are positive integrable functions on $\mathbb{R}^N$ satisfying*

$$h((1-\lambda)\mathbf{x}+\lambda\mathbf{y}) \geq M_p(f(\mathbf{x}), g(\mathbf{y}); 1-\lambda, \lambda),$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$. Then

$$\int_{\mathbb{R}^N} h(\mathbf{x})\,\mathrm{d}\mathbf{x} \geq M_{p/(Np+1)}\left(\int_{\mathbb{R}^N} f(\mathbf{x})\,\mathrm{d}\mathbf{x}, \int_{\mathbb{R}^N} g(\mathbf{x})\,\mathrm{d}\mathbf{x}; 1-\lambda, \lambda\right).$$

Here $p/(Np+1)$ means $-\infty$, if $p = -1/N$, and $1/N$, if $p = \infty$.

Proof. We start with the case $N = 1$. Without loss of generality, we may assume that

$$\int_{\mathbb{R}} f(x)\,\mathrm{d}x = A > 0 \quad \text{and} \quad \int_{\mathbb{R}} g(x)\,\mathrm{d}x = B > 0.$$

We define $u, v\colon [0,1] \to \overline{\mathbb{R}}$ such that $u(t)$ and $v(t)$ are the smallest numbers satisfying

$$\frac{1}{A}\int_{-\infty}^{u(t)} f(x)\,\mathrm{d}x = \frac{1}{B}\int_{-\infty}^{v(t)} g(x)\,\mathrm{d}x = t.$$

Clearly, the two functions are increasing and thus they are differentiable almost everywhere. This yields

$$\frac{f(u(t))u'(t)}{A} = \frac{g(v(t))v'(t)}{B} = 1 \quad \text{almost everywhere,}$$

so that $w(t) = (1-\lambda)u(t) + \lambda v(t)$ verifies

$$\begin{aligned} w'(t) &= (1-\lambda)u'(t) + \lambda v'(t) \\ &= (1-\lambda)\frac{A}{f(u(t))} + \lambda\frac{B}{g(v(t))} \end{aligned}$$

at every t with $f(u(t)) > 0$ and $g(v(t)) > 0$. Or,

$$\int_{\mathbb{R}} h(x)\,\mathrm{d}x \geq \int_0^1 h(w(t))w'(t)\,\mathrm{d}t$$

and the last inequality can be continued as

$$\begin{aligned} &\geq \int_0^1 M_p\big(f(u(t)), g(v(t)); 1-\lambda, \lambda\big) M_1\Big(\frac{A}{f(u(t))}, \frac{B}{g(v(t))}; 1-\lambda, \lambda\Big)\,\mathrm{d}t \\ &\geq \int_0^1 M_{p/(p+1)}(A, B; 1-\lambda, \lambda)\,\mathrm{d}t \\ &= M_{p/(p+1)}(A, B; 1-\lambda, \lambda), \end{aligned}$$

by a generalization of the discrete Rogers–Hölder inequality (provided by Exercise 3 (a), Section 1.2, for $p+1 \geq 0$ and $q = 1$.

The general case follows by induction. Suppose that it is true for all natural numbers less than N. For each $s \in \mathbb{R}$, attach to f, g, and h section functions f_s, g_s, and h_s, following the model

$$f_s\colon \mathbb{R}^{N-1} \to \mathbb{R}, \quad f_s(\mathbf{z}) = f(\mathbf{z}, s).$$

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{N-1}$, let $a, b \in \mathbb{R}$ and put $c = (1-\lambda)a + \lambda b$. Then

$$\begin{aligned} h_c((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) &= h((1-\lambda)\mathbf{x} + \lambda\mathbf{y}, (1-\lambda)a + \lambda b) \\ &= h((1-\lambda)(\mathbf{x}, a) + \lambda(\mathbf{y}, b)) \\ &\geq M_p(f(\mathbf{x}, a), g(\mathbf{y}, b); 1-\lambda, \lambda) \\ &= M_p(f_a(\mathbf{x}), g_b(\mathbf{y}); 1-\lambda, \lambda) \end{aligned}$$

and thus, by our inductive hypothesis,

$$\int_{\mathbb{R}^{N-1}} h_c(\mathbf{x})\,\mathrm{d}\mathbf{x} \geq M_{p/((N-1)p+1)}\left(\int_{\mathbb{R}^{N-1}} f_a(\mathbf{x})\,\mathrm{d}\mathbf{x}, \int_{\mathbb{R}^{N-1}} g_b(\mathbf{x})\,\mathrm{d}\mathbf{x}; 1-\lambda, \lambda\right).$$

Letting

$$H(c) = \int_{\mathbb{R}^{N-1}} h_c(\mathbf{x})\,\mathrm{d}x, \quad F(a) = \int_{\mathbb{R}^{N-1}} f_a(\mathbf{x})\,\mathrm{d}\mathbf{x}, \quad G(b) = \int_{\mathbb{R}^{N-1}} g_b(\mathbf{x})\,\mathrm{d}\mathbf{x},$$

we have

$$H(c) = H((1-\lambda)a + \lambda b) \geq M_r(F(a), G(b); 1-\lambda, \lambda),$$

where $r = p/((N-1)p+1)$, so by Fubini's theorem and our inductive hypothesis we conclude that

$$\begin{aligned}\int_{\mathbb{R}^N} h(\mathbf{x})\,\mathrm{d}\mathbf{x} &\geq \int_{\mathbb{R}}\int_{\mathbb{R}^{N-1}} h_c(\mathbf{z})\,\mathrm{d}\mathbf{z}\,\mathrm{d}c = \int_{\mathbb{R}} H(c)\,\mathrm{d}c \\ &\geq M_{r/(r+1)}\left(\int_{\mathbb{R}} F(a)\,\mathrm{d}a, \int_{\mathbb{R}} G(b)\,\mathrm{d}b; 1-\lambda, \lambda\right) \\ &= M_{p/(Np+1)}\left(\int_{\mathbb{R}^N} f(\mathbf{x})\,\mathrm{d}\mathbf{x}, \int_{\mathbb{R}^N} g(\mathbf{x})\,\mathrm{d}\mathbf{x}; 1-\lambda, \lambda\right).\end{aligned}$$

■

The above argument in the proof of Theorem 4.7.6 goes back to R. Henstock and A. M. Macbeath [250] (when $N = 1$) and illustrates a powerful tool of convex analysis: the *Brenier map*. See the Comments at the end of this chapter. Basically the same argument (plus some computation that makes the objective of Exercise 10, Section 8.1) led F. Barthe [40], [41] to a simplified approach of the best constants in some famous inequalities like the Young convolution inequality and the reverse Young convolution inequality.

Exercises

1. Settle the equality case in the Brunn–Minkowski inequality (as stated in Theorem 4.7.5).

2. (Log-concavity of marginals) Let $f = f(\mathbf{x}, \mathbf{y})$ be an integrable log-concave function defined on an open convex subset Ω of $\mathbb{R}^m \times \mathbb{R}^n$. For each $\mathbf{x}$ in the orthogonal projection $\mathrm{pr}_1\,\Omega$, of Ω onto $\mathbb{R}^m$, put

$$F(\mathbf{x}) = \int_{\Omega(\mathbf{x})} f(\mathbf{x}, \mathbf{y})\,\mathrm{d}\mathbf{y},$$

where $\Omega(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^n : (\mathbf{x}, \mathbf{y}) \in \Omega\}$. Infer from the Prékopa–Leindler inequality that the function $F(\mathbf{x})$ is log-concave on $\mathrm{pr}_1\,\Omega$.

[*Hint*: Suppose that $\mathbf{x}_k \in \mathrm{pr}_1\,\Omega$ and $\mathbf{y}_k \in \Omega(\mathbf{x}_k)$ for $k = 1, 2$ and $\lambda \in (0, 1)$. Then

$$\Omega((1-\lambda)\mathbf{x}_1 + \lambda\mathbf{x}_2) \supset (1-\lambda)\Omega(\mathbf{x}_1) + \lambda\Omega(\mathbf{x}_2)$$

and

$$f((1-\lambda)\mathbf{x}_1+\lambda\mathbf{x}_2,(1-\lambda)\mathbf{y}_1+\lambda\mathbf{y}_2)\geq f(\mathbf{x}_1,\mathbf{y}_1)^{1-\lambda}f(\mathbf{x}_2,\mathbf{y}_2)^{\lambda}. \quad]$$

3. Consider a convex body K in $\mathbb{R}^3$ and a vector $\mathbf{v}\in\mathbb{R}^3$ with $\|\mathbf{v}\|\neq 0$. Prove that the function

$$A(t)=\sqrt{Area(K\cap\{\mathbf{x}:\langle\mathbf{x},\mathbf{v}\rangle=t\})}$$

is concave on the interval consisting of all t for which

$$K\cap\{\mathbf{x}:\langle\mathbf{x},\mathbf{v}\rangle=t\}\neq\emptyset.$$

In other words, the square root of the area of the cross section of K by parallel hyperplanes is a concave function.

4. (The essential form of the Prékopa–Leindler inequality; see H. J. Brascamp and E. H. Lieb [96]) Let $f,g\in L^1(\mathbb{R}^N)$ be two positive functions and let $\lambda\in(0,1)$. The function

$$S(\mathbf{x})=\operatorname*{esssup}_y f\Big(\frac{\mathbf{x}-\mathbf{y}}{1-\lambda}\Big)^{1-\lambda}g\Big(\frac{\mathbf{y}}{\lambda}\Big)^{\lambda}$$

is measurable since

$$S(\mathbf{x})=\sup_n\int_{\mathbb{R}^N}f\Big(\frac{\mathbf{x}-\mathbf{y}}{1-\lambda}\Big)^{1-\lambda}g\Big(\frac{\mathbf{y}}{\lambda}\Big)^{\lambda}\varphi_n(\mathbf{y})\,d\mathbf{y}$$

for every sequence $(\varphi_n)_n$, dense in the unit ball of $L^1(\mathbb{R}^N)$. Prove that

$$\|S(\mathbf{x})\|_{L^1}\geq\|f\|_{L^1}^{1-\lambda}\,\|g\|_{L^1}^{\lambda} \tag{4.23}$$

and derive from this result the classical Prékopa–Leindler inequality.

Remark. As was noticed in [96], the essential form of the Prékopa–Leindler inequality represents the limiting case as $r\to 0$ of the following reverse Young convolution inequality with sharp constants: *Let* $0<p,q,r\leq 1$ *with* $1/p+1/q=1+1/r$, *and let* $f\in L^p(\mathbb{R}^N)$ *and* $g\in L^q(\mathbb{R}^N)$ *be positive functions. Then*

$$\|f*g\|_{L^r}\geq C(p,q,r,N)\|f\|_{L^p}\,\|g\|_{L^q}.$$

5. A positive regular measure μ defined on the (Lebesgue) measurable subsets of $\mathbb{R}^N$ is called M_p-*concave* (for some $p\in\overline{\mathbb{R}}$) if

$$\mu((1-\lambda)X+\lambda Y)\geq M_p(\mu(X),\mu(Y);1-\lambda,\lambda)$$

for all measurable sets X and Y in $\mathbb{R}^N$ and all $\lambda\in(0,1)$ such that the set $(1-\lambda)X+\lambda Y$ is measurable. When $p=0$, an M_p-concave measure is

also called log-*concave*. By the Prékopa–Leindler inequality, the Lebesgue measure is $M_{1/N}$-concave. Suppose that $-1/N \leq p \leq \infty$, and let f be a positive integrable function which is M_p-concave on an open convex set C in $\mathbb{R}^N$. Prove that the measure $\mu(X) = \int_{C\cap X} f(\mathbf{x})\,\mathrm{d}\mathbf{x}$ is $M_{p/(Np+1)}$-concave. Infer that the standard *Gaussian measure* in $\mathbb{R}^N$,

$$\mathrm{d}\gamma_N = (2\pi)^{-N/2}\ \mathrm{e}^{-\|\mathbf{x}\|^2/2}\mathrm{d}\mathbf{x},$$

is log-concave.

6. (S. Dancs and B. Uhrin [148]) Extend Theorem 4.7.6 by replacing the Lebesgue measure by an M_q-concave measure, for some $-\infty \leq q \leq 1/N$.

4.8 Convex Functions Acting on Ordered Banach Spaces

The notion of convex function (as well as its relatives) admits a verbatim generalization to the context of functions taking values in an ordered Banach space. For example, if E is a real linear space, U is a convex subset of E and F is an ordered Banach space then a function $f : U \to F$ is called *convex* if for all $\mathbf{x}, \mathbf{y} \in U$ and $\lambda \in [0, 1]$,

$$f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) \leq (1-\lambda)f(\mathbf{x}) + \lambda f(\mathbf{y}),$$

equivalently,

$$(1-\lambda)f(\mathbf{x}) + \lambda f(\mathbf{y}) - f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) \in F_+;$$

f is called *strictly convex* if the above inequality is strict for $\mathbf{x} \neq \mathbf{y}$ and $\lambda \in (0, 1)$.

When $F = \mathbb{R}^N$, the function f is (strictly) convex if and only if all its components $f_k = \mathrm{pr}_k \circ f$ are (strictly) convex functions. According to Lemma 3.4.3 (c), this remark extends for any ordered Banach space F as follows:

4.8.1 Theorem *If U is a convex subset of a real linear space E and F is an ordered Banach space, then a function $f : U \to F$ is (strictly) convex if and only if $x^* \circ f$ is (strictly) convex, whenever $x^* \in F_+^*$, $x^* \neq 0$.*

An immediate consequence is the fact that many important results such as those stated in Proposition 4.1.2 (the equivalence of convexity with convexity along linear segments), Corollary 4.1.3 (the extension of Jensen's criterion of convexity) and Proposition 4.2.2 (the discrete case of Jensen's inequality) can be extended to this vector-valued context.

Notice also that every convex function with values in an ordered Banach space can be turned into a convex function with values in a space $C(K)$ due to Theorem 3.4.4.

An important class of real-valued convex functions is that of sublinear operators. Suppose that E and F are ordered vector spaces. A map $P : E \to F$ is

called a *sublinear operator* if it is subadditive and positively homogeneous, that is,

$$P(x+y) \leq P(x) + P(y) \text{ and } P(\lambda x) = \lambda P(x)$$

for all $x, y \in E$ and $\lambda \geq 0$. Every sublinear operator is convex and a vector-valued convex function is a sublinear operator if and only if it is positively homogeneous.

Every sublinear operator $T : E \to F$ verifies the conditions $T(0) = 0$ and

$$-T(-f) \leq T(f) \quad \text{for all } f \in E.$$

As a consequence, every sublinear and increasing operator T maps positive elements into positive elements, that is,

$$Tf \geq 0 \quad \text{for all } f \geq 0. \tag{4.24}$$

The property (4.24) is equivalent to increasing monotonicity in the case of linear operators.

4.8.2 Example *Suppose that E is a Banach lattice, F is an ordered Banach space and $A : E \to F$ is a positive linear operator. Then the formula*

$$P : E \to F, \quad P(x) = A\left(x^{+}\right), \quad x \in E$$

defines a monotonically increasing and continuous sublinear operator.

4.8.3 Example *The formula*

$$T : \ell^{\infty} \to \ell^{\infty}, \quad T((x_n)_n) = \left(\limsup_{n\to\infty} x_n\right) \cdot 1;$$

defines an example of monotonically increasing and continuous sublinear operator such that $P(1) = 1$; here 1 denotes the sequence with all components equal to unity.

4.8.4 Example *Real analysis and harmonic analysis offer numerous examples of sublinear and monotone operators. Assuming that Ω is an open subset of $\mathbb{R}^N$ and $p > 1$, we may consider the* signed Hardy–Littlewood maximal operator *M_Ω, acting from the Lebesgue space $L^p(\Omega)$ into itself via the formula*

$$\left(M_\Omega f\right)(x) = \sup \frac{1}{\mathrm{Vol}(B_r(x))} \int_{B_r(x)} f(y)\mathrm{d}y, \tag{4.25}$$

where the supremum is taken over all radii $r > 0$ with $0 < r < \mathrm{dist}(x, \partial\Omega)$. Clearly, M_Ω is sublinear monotone increasing. M_Ω is unital if Ω is bounded.

An important source of sublinear operators is provided by the theory of Choquet's integral. See [207], [208] and [209], where the reader will find several generalizations of Korovkin's approximation theorem.

The *norm* of a continuous sublinear operator $T : E \to F$ can be introduced as in the linear case, via the formula

$$\begin{aligned}\|T\| &= \inf\{\lambda > 0 : \|T(f)\| \leq \lambda \|f\| \text{ for all } f \in E\} \\ &= \sup\{\|T(f)\| : f \in E, \|f\| \leq 1\}.\end{aligned}$$

An arbitrary sublinear operator may be discontinuous, but when it is continuous, it is Lipschitz continuous. More precisely, if $T : E \to F$ is a continuous sublinear operator, then

$$\|T(f) - T(g)\| \leq 2\|T\| \|f - g\| \quad \text{for all } f \in E.$$

As shows the following theorem, in the context of Banach lattices, the norm of every sublinear and monotone operator equals its Lipschitz constant.

4.8.5 Theorem (See [208]) *Suppose that E and F are two Banach lattices. Then every sublinear and increasing operator $T : E \to F$ verifies the inequality*

$$|T(f) - T(g)| \leq T(|f - g|) \quad \textit{for all } f, g \in E$$

and thus it is Lipschitz continuous with Lipschitz constant equals to $\|T\|$, that is,

$$\|T(f) - T(g)\| \leq \|T\| \|f - g\| \quad \textit{for all } f, g \in E.$$

Proof. Notice first that for all $f, g \in E$ we have $f \leq g + |f - g|$, whence $T(f) - T(g) \leq T(|f - g|)$ due to the monotonicity and subadditivity of T. Interchanging the role of f and g we infer that $-(T(f) - T(g)) \leq T(|f - g|)$. Therefore

$$|T(f) - T(g)| = \sup\{T(f) - T(g), -(T(f) - T(g))\} \leq T(|f - g|). \tag{4.26}$$

Now the continuity of T can be established by reductio ad absurdum. Indeed, if T were not continuous, then would exist a sequence $(x_n)_n$ of elements of E such that $\|f_n\| \leq 1/(n2^n)$ and $\|T(f_n)\| \geq n$. Taking into account the inequality $|T(f_n)| \leq T(|f_n|)$, and replacing each x_n by $|f_n|$ if necessary, one may restrict ourselves to the case where all elements f_n belong to E_+. Then the series $\sum f_n$ is absolutely converging, with sum $f \geq 0$. Since T is monotone,

$$T(f) \geq T(f_n)$$

which implies $\|T(f)\| \geq \|T(f_n)\| \geq n$ for all positive integers n, a contradiction.

Once the continuity of T was established, we infer from (4.26) that

$$\|T(f) - T(g)\| \leq \|T(|f - g|)\| \leq \|T\| \|f - g\|,$$

which ends the proof. ■

Theorem 4.8.5 extends the classical result of M. G. Krein concerning the continuity of positive linear functionals. See [2] for historical details.

The subgradients theory presented in Sections 4.3 and 5.1 admits a natural extension to the case of vector-valued convex functions.

Suppose that C is a convex subset of a Banach space E and F is an order complete Banach space (that is, an ordered Banach space such that every nonempty and bounded from the above subset of F has a least upper bound). Some few examples of order complete Banach spaces are $\mathbb{R}^N$, all Lebesgue spaces $L^p(\mu)$ $(1 \leq p \leq \infty)$ and the sequence space c_0. See the monograph of H. H. Schaefer [519].

Given a function $f : C \to F$ and a point $x_0 \in C$ we call *subgradient of f at x_0* every continuous linear operator $T : E \to F$ such that

$$f(x) \geq f(x_0) + T(x - x_0) \quad \text{for all } x \in C. \tag{4.27}$$

The set $\partial_{x_0} f$, of all such subgradients, is referred to as the *subdifferential of f at x_0*.

The following result is due to Valadier [572], p. 68, and Zowe [603], p. 286.

4.8.6 Theorem *If C is a convex subset of a Banach space E and F is an order complete Banach space, then the subdifferential of every continuous convex function $f : C \to F$ at an interior point of C is a nonempty convex set.*

The converse of Theorem 4.8.6 also works: the only continuous functions $f : C \to F$ with the property that $\partial_{x_0} f \neq \emptyset$ at any interior point x_0 of C are the convex ones. To check this, adapt the argument used in the proof of Theorem 4.3.6.

4.8.7 Corollary *Under the hypotheses of Theorem 4.8.6, every continuous convex function $f : C \to F$ is the upper envelope of its continuous affine minorants.*

Valadier's argument for the non-emptiness of the subdifferential is based on the existence of the directional derivatives

$$f'(x_0; h) = \lim_{t \to 0+} \frac{f(x_0 + th) - f(x_0)}{t}$$

and the fact that a linear operator $T : E \to F$ is a subgradient of f at x_0 if and only if $T(h) \leq f'(x_0; h)$ for every $h \in E$. He assumed F to be an order complete Banach lattice, a fact relaxed by Zowe who used the following generalization of the analytical form of the Hahn–Banach Theorem:

4.8.8 Theorem (The Generalized Hahn–Banach Extension Theorem) *Let f be a convex function defined on the real vector space E and taking values in an order complete vector space F. If H is a linear subspace of E and $T_H : H \to F$ is a linear operator satisfying $T_H(x) \leq f(x)$ for all $x \in H$, then there exists a linear operator $T : E \to F$ which extends T_H and $T(x) \leq f(x)$ for all $x \in E$.*

See Zowe [603], Theorem 2.1, p. 284, for a simple argument.

4.8.9 Corollary (The topological version of Theorem 4.8.8) *Let f be a convex function defined on the Banach space E and taking values in an order complete Banach space F. Suppose in addition that f is continuous at the origin and $f(0) = 0$. If H is a linear subspace of E and $T_H : H \to F$ is a linear operator satisfying the inequality $T_H(x) \leq f(x)$ for all $x \in H$, then there exists a continuous linear operator $T : E \to F$ which extends T_H and verifies the inequality*

$$T(x) \leq f(x) \quad \text{for all } x \in E. \tag{4.28}$$

Proof. Indeed, the existence of a linear operator $T : E \to F$ which extends T_H and verifies the inequality (4.28) follows from Theorem 4.8.8. Since T is linear, its continuity is equivalent to the continuity at the origin. Taking into account that

$$-f(-x) \leq T(x) \leq f(x) \quad \text{for all } x \in E,$$

we infer that for every sequence $x_n \to 0$ we have $-f(-x_n) \leq T(x_n) \leq f(x_n)$ and thus the fact that $T(x_n) \to 0$ is a consequence of the squeeze theorem. ■

For $H = \{0\}$ and f replaced by $f(x + x_0) - f(x_0)$, where x_0 is arbitrarily fixed in E, Theorem 4.8.8 yields the following particular case of Theorem 4.8.6:

4.8.10 Corollary *Let f be a convex function defined on the Banach space E and taking values in an order complete Banach space F. Then $\partial_{x_0} f \neq \emptyset$ at every point x_0 of continuity of f.*

It is worth noticing that the hypothesis concerning the order completeness of the codomain F is not only sufficient for the validity of Theorem 4.8.8, but it is also necessary (in the class of ordered vector spaces). See G. Buskes [111] for a survey on this challenging result due to W. E. Bonnice, R. J. Silvermann and Ting On To.

Exercises

1. Prove that the function $\Phi : A \to \log \operatorname{trace}(e^A)$ is convex on $\operatorname{Sym}(n, \mathbb{R})$.

   [*Hint*: Notice that the convexity of Φ is equivalent to the convexity of the function $f(t) = \log \operatorname{trace}(e^{A+tB})$ on $\mathbb{R}$, for all $A, B \in \operatorname{Sym}(n, \mathbb{R})$.]

2. Suppose that C is an open convex subset of a Banach space E, F is an order complete Banach space, and $f : C \to F$ is a continuous convex function. Prove that for every pair of points x and y in C and every operators $T \in \partial_x f$ and $S \in \partial_y f$ we have

$$T(y - x) \leq \Phi(y) - \Phi(x) \leq S(y - x).$$

3. (A characterization of the monotonicity of continuous convex functions) Suppose that E is an ordered Banach space, F is an order complete Banach space, and $f : E \to F$ is a continuous convex function. Prove that the following statements are equivalent:

(a) $x \le y$ in E implies $f(x) \le f(y)$;

(b) the subdifferential of f at any point $x_0 \in E$ consists of positive linear operators;

(c) for each $x_0 \in E$, there exists a positive linear operator $T \in \partial_{x_0} f$.

4. Suppose that E is an ordered Banach space and F is an order complete Banach space. Then a continuous convex function $f : E_+ \to F$ is increasing if (and only if) for every $x_0 \in E_+$, there exists a positive linear operator $T \in \mathcal{L}(E, F)$ such that

$$f(x) - f(x_0) \ge T(x - x_0) \quad \text{for all } x \ge x_0.$$

4.9 Comments

The theory of convex functions of one real variable can be generalized to several variables in many different ways. A long time ago, P. Montel [394] pointed out the alternative of subharmonic functions. The question of when the superharmonic functions are concave is discussed by B. Kawohl [290]. Nowadays, many other alternatives are known. An authoritative monograph on this subject has been published by L. Hörmander [268].

The first modern exposition on convexity in $\mathbb{R}^N$ was written by W. Fenchel [194]. He used the framework of lower semicontinuous proper convex functions to provide a valuable extension of the classical theory. At the beginning of the 1960s, R. T. Rockafellar and J.-J. Moreau initiated a systematic study of the field of convex analysis, that paved the way for many deep applications in convex optimization, variational analysis, partial differential equations, mathematical economics, etc.

The notion of perspective function has a long history with deep roots in statistics. The term "perspective function" was coined by C. Lemaréchal ca. 1987–1988, and first appeared in print in [256], Section IV.2.2. An account on the present-day research based on this notion is available in the paper of P. L. Combettes [141].

The continuous convex functions defined on polytopes have a special boundary behavior. If K is a polytope in $\mathbb{R}^N$, then every bounded convex function defined on the relative interior of K has a unique extension to a continuous convex function on K. D. Gale, V. Klee and R. T. Rockafellar [210] have noticed that this property actually characterizes the polytopes among the convex sets in $\mathbb{R}^N$.

Support functions, originally defined by H. Minkowski in the case of bounded convex sets, have been studied for general convex sets in $\mathbb{R}^N$ by W. Fenchel [193], [194] and, in infinite-dimensional spaces, by L. Hörmander [267].

The Brunn–Minkowski inequality is among the most important results in the theory of convex bodies. R. J. Gardner [213], [214] wrote a nice survey about it, including historical remarks and a description of links to other inequalities and various extensions. Nowadays, a dozen of Brunn–Minkowski type inequalities are

already known for the first eigenvalue of the Laplacian, the Newton capacity, the torsional rigidity, etc. See the paper of A. Colesanti [140] and references therein.

The Brunn–Minkowski inequality was primarily motivated by the isoperimetric problem in Euclidean spaces. The clue is provided by the formula

$$S = \frac{\mathrm{d}V}{\mathrm{d}R},$$

that relates the volume V of a ball $B_R(0)$ in $\mathbb{R}^3$ and the area S of its surface. This led H. Minkowski to define the *surface area* of a convex body K in $\mathbb{R}^N$ by the formula

$$S_{N-1}(K) = \lim_{\varepsilon \to 0+} \frac{\mathrm{Vol}_N(K + \varepsilon B) - \mathrm{Vol}_N(K)}{\varepsilon},$$

where B denotes the closed unit ball of $\mathbb{R}^N$. The agreement of this definition with the usual definition of the surface of a smooth surface is discussed in books by H. Federer [191] and Y. D. Burago and V. A. Zalgaller [107].

4.9.1 Theorem (The isoperimetric inequality for convex bodies in $\mathbb{R}^N$) *Let K be a convex body in $\mathbb{R}^N$ and let B denote the closed unit ball of this space. Then*

$$\Big(\frac{\mathrm{Vol}_N(K)}{\mathrm{Vol}_N(B)}\Big)^{1/N} \le \Big(\frac{S_{N-1}(K)}{S_{N-1}(B)}\Big)^{1/(N-1)},$$

with equality if and only if K is a ball.

For $N = 2$, this means that the area A of a compact convex domain in the plane, bounded by a curve of length L, verifies the inequality $4\pi A \le L^2$, the equality taking place only for discs.

Proof. In fact, by the Brunn–Minkowski inequality,

$$\begin{aligned} S_{N-1}(K) &= \lim_{\varepsilon \to 0+} \frac{\mathrm{Vol}_N(K + \varepsilon B) - \mathrm{Vol}_N(K)}{\varepsilon} \\ &\ge \lim_{\varepsilon \to 0+} \frac{(\mathrm{Vol}_N(K)^{1/N} + \varepsilon \mathrm{Vol}_N(B)^{1/N})^N - \mathrm{Vol}_N(K)}{\varepsilon} \\ &= N \mathrm{Vol}_N(K)^{(N-1)/N} \mathrm{Vol}(B)^{1/N}, \end{aligned}$$

and it remains to notice that $S_{N-1}(B) = N\mathrm{Vol}_N(B)$. ∎

A survey of the convexity properties of solutions of partial differential equations may be found in the notes of B. Kawohl [291]. We shall mention here one application of the Prékopa–Leindler inequality, which refers to the diffusion equation

$$\frac{\partial u}{\partial t} = \frac{1}{2}\,\Delta u - V(\mathbf{x})u \quad \text{for } (\mathbf{x}, t) \in \Omega \times (0, \infty)$$

with zero Dirichlet boundary conditions (that is, $\lim_{\mathbf{x} \to \partial\Omega} u(\mathbf{x}, t) = 0$ for each t). Here Ω is an open convex set in $\mathbb{R}^N$, $u : \Omega \times (0, \infty) \to \mathbb{R}$, $u = u(\mathbf{x}, t)$, is the unknown function, and V is a positive continuous function defined on Ω.

When $\Omega = \mathbb{R}^N$ and $V = 0$, the fundamental solution is given by formula $f(\mathbf{x}, \mathbf{y}, t) = (2\pi t)^{-n/2} \mathrm{e}^{-\|\mathbf{x}-\mathbf{y}\|^2/2t}$, which is log-concave on $\mathbb{R}^N \times \mathbb{R}^N$ for all $t > 0$. H. J. Brascamp and E. H. Lieb [96] have proved, based on the Prékopa–Leindler inequality, that in general the fundamental solution $f(\mathbf{x}, \mathbf{y}, t)$ of the above Dirichlet problem is log-concave on $\Omega \times \Omega$, whenever V is a convex function. The idea is to show that $f(\mathbf{x}, \mathbf{y}, t)$ is a pointwise limit of convolutions of log-concave functions. Later on, Ch. Borell [79] considered potentials that depend on a parameter, and this fact led him to more general results and a Brownian motion treatment of the Brunn–Minkowski inequality.

The Borell–Brascamp–Lieb inequality was first stated and proved in full generality by Ch. Borell [78] and, independently, by H. J. Brascamp and E. H. Lieb [96].

Last but not least, let us mention the existence of an extensive literature devoted to the theory of the vector-valued convex functions. See J. M. Borwein [81], A. G. Kusraev and S. S. Kutateladze [316], S. S. Kutateladze [317], N. S. Papageorgiou [460] and the references therein.



Chapter 5

Differentiable Convex Functions

There are two basic properties of convex functions that make them so widely used in theoretical and applied mathematics:

The maximum is attained at a boundary point.

Any local minimum is a global one. Moreover, a strictly convex function admits at most one minimum.

Differentiability provides the basic tools to derive from these properties of the convex functions a series of powerful tools for handling numerous optimization problems that appear in economics, engineering, operations research, machine learning, etc. So are the conditions of optimality and the numerical algorithms for computing the minimum/maximum of a given function.

5.1 Directional Derivatives

In what follows E denotes a real Banach space and U is a nonempty open subset of it.

The following notion is fundamental to the study of differentiability of convex functions.

5.1.1 Definition *Let $f : E \to \mathbb{R} \cup \{\infty\}$ be a proper convex function. The right-hand directional derivative of f at $\mathbf{a} \in \operatorname{int}(\operatorname{dom} f)$ relative to $\mathbf{v} \in E$ is defined via the formula*

$$f'_+(\mathbf{a}\,;\mathbf{v}) = \lim_{t \to 0+} \frac{f(\mathbf{a}+t\mathbf{v}) - f(\mathbf{a})}{t}.$$

The main features of the notion of right-hand directional derivative are summarized in the following lemma.

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C. P. Niculescu and L.-E. Persson, *Convex Functions and Their Applications*, CMS/CAIMS Books in Mathematics 14, https://doi.org/10.1007/978-3-031-71967-7_5

5.1.2 Lemma *Suppose $f : E \to \mathbb{R} \cup \{\infty\}$ is a proper convex function and $\mathbf{a} \in \operatorname{int}(\operatorname{dom} f)$. Then:*

(a) *the right-hand directional derivative $f'_+(\mathbf{a}\,;\mathbf{v})$ is a real-valued positively homogeneous sublinear function of $\mathbf{v} \in E$;*

(b) $f'_+(\mathbf{a}\,;\mathbf{0}) = 0$ *and*

$$f'_+(\mathbf{a}\,;\mathbf{v}) \geq -f'_+(\mathbf{a}\,;-\mathbf{v}) \quad \text{for all } \mathbf{v} \in E;$$

(c) *if $\mathbf{a}$ is a point of continuity for f, then the directional derivative $f'_+(\mathbf{a}\,;\mathbf{v})$ is a continuous function in $\mathbf{v}$.*

Proof. (a) & (b) Since $\mathbf{a} \in \operatorname{int}(\operatorname{dom} f)$, there is $\varepsilon > 0$ such that the function $t \to f(\mathbf{a}+t\mathbf{v})$ is convex on an interval $[-\varepsilon, \varepsilon]$ whenever $\mathbf{v} \in E$. According to the three chords inequality (1.18), the ratio $[f(\mathbf{a}+t\mathbf{v}) - f(\mathbf{a})]/t$ is an increasing function of t, which yields

$$\frac{f(\mathbf{a}-\varepsilon\mathbf{v}) - f(\mathbf{a})}{-\varepsilon} \leq -\frac{f(\mathbf{a}-t\mathbf{v}) - f(\mathbf{a})}{t} \leq \frac{f(\mathbf{a}+t\mathbf{v}) - f(\mathbf{a})}{t} \leq \frac{f(\mathbf{a}+\varepsilon\mathbf{v}) - f(\mathbf{a})}{\varepsilon}$$

for all $t \in (0, \varepsilon]$. This easily implies the existence and finiteness of $f'_+(\mathbf{a}\,;\mathbf{v})$ and $f'_+(\mathbf{a}\,;-\mathbf{v})$ and also the inequality $-f'_+(\mathbf{a}\,;-\mathbf{v}) \leq f'_+(\mathbf{a}\,;\mathbf{v})$. Notice that

$$f'_+(\mathbf{a}\,;\mathbf{v}) = \inf_{t>0} \frac{f(\mathbf{a}+t\mathbf{v}) - f(\mathbf{a})}{t} \tag{5.1}$$

and $f'_+(\mathbf{a}\,;\mathbf{0}) = 0$. For the positive homogeneity of $f'_+(\mathbf{a}\,;\cdot)$ notice that

$$\begin{aligned} f'_+(\mathbf{a}\,;\alpha\mathbf{v}) &= \lim_{t\to 0+} \frac{f(\mathbf{a}+t\alpha\mathbf{v}) - f(\mathbf{a})}{t} = \alpha \lim_{t\to 0+} \frac{f(\mathbf{a}+\alpha t\mathbf{v}) - f(\mathbf{a})}{\alpha t} \\ &= \alpha \lim_{s\to 0+} \frac{f(\mathbf{a}+s\mathbf{v}) - f(\mathbf{a})}{s} = \alpha\ f'(\mathbf{a}\,;\mathbf{v}) \end{aligned}$$

for all $\alpha > 0$. The fact that $f'_+(\mathbf{a}\,;\cdot)$ is subadditive (and thus sublinear) is left as an exercise. See Exercise 1.

(c) If $\mathbf{a}$ is a point of continuity for f, then f is Lipschitz in a ball $\overline{B}_R(\mathbf{a})$. See Proposition 4.1.11. Therefore, there is $M > 0$ such that for all $\mathbf{v} \in E$, $\mathbf{v} \neq 0$,

$$\left| \frac{f(\mathbf{a}+t\mathbf{v}) - f(\mathbf{a})}{t} \right| \leq M \|\mathbf{v}\|$$

for all $t \in (0, R/\|v\|)$. This implies $f'_+(\mathbf{a}\,;\mathbf{v}) \leq M\|\mathbf{v}\|$ for all $\mathbf{v} \in E$ and the continuity of $f'_+(\mathbf{a}\,;\mathbf{v})$ follows. See Exercise 1, Section 4.4. ■

Next, we will discuss the connection between subgradients and right-hand directional derivatives.

5.1.3 Lemma *Suppose that $f : E \to \mathbb{R} \cup \{\infty\}$ is a proper convex function and $\mathbf{a} \in \operatorname{int}(\operatorname{dom} f)$. Then $x^* \in E^*$ is a subgradient of f at $\mathbf{a}$ if and only if*

$$x^*(\mathbf{v}) \leq f'_+(\mathbf{a}\,;\mathbf{v}) \quad \text{for all } \mathbf{v} \in E.$$

Proof. If $x^* \in \partial f(\mathbf{a})$, then for every $\mathbf{v} \in E$ and every $t > 0$ we have

$$x^*(\mathbf{v}) \leq \frac{f(\mathbf{a}+t\mathbf{v}) - f(\mathbf{a})}{t},$$

whence, by taking limits as $t \to 0+$, we infer that $x^*(\mathbf{v}) \leq f'_+(\mathbf{a}\,;\mathbf{v})$.

Conversely, if $x^*(\mathbf{v}) \leq f'_+(\mathbf{a}\,;\mathbf{v})$ for all $\mathbf{v} \in E$, then the formula (5.1) yields

$$x^*(\mathbf{v}) \leq f'_+(\mathbf{a}\,;\mathbf{v}) \leq \frac{f(\mathbf{a}+t\mathbf{v}) - f(\mathbf{a})}{t} \quad \text{for all } \mathbf{v} \in E \text{ and } t > 0.$$

In particular, for $\mathbf{v} = \mathbf{x} - \mathbf{a}$ and $t = 1$, we obtain $x^*(\mathbf{x} - \mathbf{a}) \leq f(\mathbf{x}) - f(\mathbf{a})$, that is, $x^* \in \partial f(\mathbf{a})$. The proof is complete. ■

5.1.4 Theorem (Moreau's max formula) *If $f : E \to \mathbb{R} \cup \{\infty\}$ is a proper convex function and a is a point of continuity for f, then $\partial f(\mathbf{a})$ is nonempty and*

$$f'_+(\mathbf{a}\,;\mathbf{v}) = \max\{x^*(\mathbf{v}) : x^* \in \partial f(\mathbf{a})\}.$$

In other words, $f'_+(\mathbf{a}\,;\cdot)$ is the support function of $\partial f(\mathbf{a})$.

Proof. According to Lemma 5.1.3, we have to prove that for $\mathbf{v} \in E$ arbitrarily fixed there is a subgradient $x^* \in E^*$ such that $x^*(\mathbf{v}) = f'_+(\mathbf{a}\,;\mathbf{v})$. This follows from the Hahn–Banach theorem (see Theorem B.1.1 in Appendix B). Indeed, by Lemma 5.1.2 (c), the sublinear function $\mathbf{x} \to f'_+(\mathbf{a}\,;\mathbf{x})$ is continuous, and an appeal to Corollary B.1.2 in Appendix B shows the existence of a linear functional $x^* \colon E \to \mathbb{R}$ such that $x^*(\mathbf{v}) = f'_+(\mathbf{a}\,;\mathbf{v})$ and $x^*(\mathbf{x}) \leq f'_+(\mathbf{a}\,;\mathbf{x})$ for all $\mathbf{x}$ in E. It remains to prove the continuity of x^* at the origin. We have $\limsup_{\mathbf{x}\to 0} x^*(\mathbf{x}) \leq \lim_{\mathbf{x}\to 0} f'_+(\mathbf{a}\,;\mathbf{x}) = 0$ and

$$\limsup_{\mathbf{x}\to 0} x^*(-\mathbf{x}) = -\liminf_{\mathbf{x}\to 0} x^*(\mathbf{x}) \leq \lim_{\mathbf{x}\to 0} f'_+(\mathbf{a}\,;-\mathbf{x}) = 0,$$

so

$$0 \leq \liminf_{\mathbf{x}\to 0} x^*(\mathbf{x}) \leq \limsup_{\mathbf{x}\to 0} x^*(\mathbf{x}) \leq 0,$$

that is, $\lim_{\mathbf{x}\to 0} x^*(\mathbf{x}) = 0$. ■

In connection with Moreau's max formula notice that for every nonempty convex compact C in $\mathbb{R}^N$ we have

$$\partial h_C(0) = C \quad \text{and} \quad (h_C)'_+(0;\cdot) = h_C.$$

Some applications of Moreau's max formula are in order.

5.1.5 Theorem (Subdifferential of the max-function) *Suppose that $f_1, \ldots, f_n$ are real-valued convex functions on $\mathbb{R}^N$ and denote*

$$f = \max\{f_1, \ldots, f_n\}.$$

For $\mathbf{a} \in \mathbb{R}^N$, let $J(\mathbf{a}) = \{j : f_j(\mathbf{a}) = f(\mathbf{a})\}$ be the active index set. Then

$$\partial f(\mathbf{a}) = conv\left(\cup_{j\in J(\mathbf{a})} \partial f_j(\mathbf{a})\right).$$

More general results, as well as applications to the other subdifferential calculus rules, can be found in the paper of A. Hantoute, M. A. López, and C. Zălinescu [242].

Proof. The inclusion $\supset$ follows from the fact that $\partial f(\mathbf{a})$ is a closed convex set and $\partial f(\mathbf{a}) \supset \partial f_j(\mathbf{a})$ for all $j \in J(\mathbf{a})$. Since the functions f_k are continuous, there is $\varepsilon > 0$ such that $f_k(\mathbf{x}) < f(\mathbf{x})$ for all $k \notin J(\mathbf{a})$ and $\mathbf{x} \in B_\varepsilon(\mathbf{a})$. Then for every $\mathbf{v} \in \mathbb{R}^N \backslash \{\mathbf{0}\}$ arbitrarily fixed and $0 < t < \varepsilon / \|v\|$ we have $f(\mathbf{a} + t\mathbf{v}) = \max_{j \in J(\mathbf{a})} f_j(\mathbf{a} + t\mathbf{v})$. Therefore

$$\begin{aligned} f'_+(\mathbf{a}; \mathbf{v}) &= \lim_{t \to 0+} \frac{f(\mathbf{a} + t\mathbf{v}) - f(\mathbf{a})}{t} = \lim_{t \to 0+} \max_{j \in J(\mathbf{a})} \frac{f_j(\mathbf{a} + t\mathbf{v}) - f(\mathbf{a})}{t} \\ &= \lim_{t \to 0+} \max_{j \in J(\mathbf{a})} \frac{f_j(\mathbf{a} + t\mathbf{v}) - f_j(\mathbf{a})}{t} = \max_{j \in J(a)} f'_j(\mathbf{a}, \mathbf{v}) \\ &= \max_{j \in J(\mathbf{a})} \left\{ \langle \mathbf{v}, x_j^* \rangle : x_j^* \in \partial f_j(\mathbf{a}) \right\} \\ &= \max \left\{ \langle \mathbf{v}, x^* \rangle : x^* \in \bigcup\nolimits_{j \in J(\mathbf{a})} \partial f_j(\mathbf{a}) \right\} \\ &= \max \left\{ \langle \mathbf{v}, x^* \rangle : x^* \in \operatorname{conv} \bigcup\nolimits_{j \in J(\mathbf{a})} \partial f_j(\mathbf{a}) \right\}, \end{aligned}$$

which shows that the function $\mathbf{v} \to f'_+(\mathbf{x}; \mathbf{v})$ is the support of the convex set $C = \operatorname{conv}\left(\cup_{j \in J(\mathbf{x})} \partial f_j(\mathbf{x})\right)$. By Theorem 5.1.4, this function is also the support of $\partial f(\mathbf{x})$, so by Theorem 4.4.3 we conclude that these two sets coincide. ■

An application of Theorem 5.1.5 is indicated in Exercise 3.

5.1.6 Theorem (Subdifferential sum rule) *Suppose that f_1 and f_2 are two real-valued convex functions defined on $\mathbb{R}^N$ and $t_1, t_2 > 0$. Then, for every $\mathbf{x} \in \mathbb{R}^N$,*

$$\partial(t_1 f_1 + t_2 f_2)(\mathbf{x}) = t_1 \partial f_1(\mathbf{x}) + t_2 \partial f_2(\mathbf{x}).$$

In the general setting of proper convex functions, only the inclusion $\supset$ works. The equality needs additional assumptions, for example, $\operatorname{ri}(\operatorname{dom} f_1) \cap \operatorname{ri}(\operatorname{dom} f_2) \neq \emptyset$. A proof based on the Fenchel–Moreau duality is available in [505], Theorem 23.8, p. 223.

Proof. Indeed, $t_1 \partial f_1(\mathbf{x}) + t_2 \partial f_2(\mathbf{x})$ is a compact convex set whose support function is $t_1 f'_1(\mathbf{x}; \cdot) + t_2 f'_2(\mathbf{x}; \cdot)$. On the other hand, the support function of $\partial(t_1 f_1 + t_2 f_2)(\mathbf{x})$ is

$$(t_1 f_1 + t_2 f_2)'(\mathbf{x}; \cdot) = t_1 f'_1(\mathbf{x}; \cdot) + t_2 f'_2(\mathbf{x}; \cdot).$$

Therefore the two compact convex sets $t_1 \partial f_1(\mathbf{x}) + t_2 \partial f_2(\mathbf{x})$ and $\partial(t_1 f_1 + t_2 f_2)(\mathbf{x})$ have the same support. Hence, according to Corollary 4.4.4, they coincide. ■

5.1.7 Theorem (Subdifferential of a composition) *Let f be a real-valued convex function on $\mathbb{R}^N$ and let A be a linear transformation from $\mathbb{R}^M$ to $\mathbb{R}^N$. Then, for every $\mathbf{x} \in \mathbb{R}^M$,*

$$\partial(f \circ A)(\mathbf{x}) = A^* \partial f(A\mathbf{x}).$$

In the general setting of proper convex functions, only the inclusion $\partial(f \circ A)(\mathbf{x}) \supset A^*\partial f(A\mathbf{x})$ works. The equality needs additional assumptions, for example, it works when the range of A contains a point of $\mathrm{ri}(\mathrm{dom}\ f)$. See [505], Theorem 23.9, p. 225, for a proof based on Fenchel–Moreau duality.

Proof. Using the definition of directional derivatives and Moreau's max formula, one can show that $(f \circ A)'(\mathbf{x}; \mathbf{v}) = f'(A\mathbf{x}, A\mathbf{v})$. Thus the support function of the compact convex set $\partial(f \circ A)(\mathbf{x})$ verifies the formula

$$h_{\partial(f\circ A)(\mathbf{x})}(\mathbf{v}) = f'(A\mathbf{x}, A\mathbf{v}).$$

On the other hand,

$$h_{A^*\partial f(A\mathbf{x})}(\mathbf{v}) = \sup_{\mathbf{y}\in\partial f(A\mathbf{x})} \langle A^*\mathbf{y}, \mathbf{v}\rangle = \sup_{\mathbf{y}\in\partial f(A\mathbf{x})} \langle \mathbf{y}, A\mathbf{v}\rangle = f'(A\mathbf{x}, A\mathbf{v}).$$

Therefore the compact convex sets $\partial(f \circ A)(\mathbf{x})$ and $A^*\partial f(A\mathbf{x})$ have the same support. Hence, according to Corollary 4.4.4, they coincide. ■

Subdifferentiability can be seen as a generalization of Gâteaux differentiability.

5.1.8 Definition *Suppose that E and F are two real Banach spaces and U is an open subset of E. A function $f : U \to F$ is said to be Gâteaux differentiable at a point $\mathbf{a} \in U$ if the directional derivatives*

$$f'(\mathbf{a}; \mathbf{v}) = \lim_{t\to 0} \frac{f(\mathbf{a} + t\mathbf{v}) - f(\mathbf{a})}{t}.$$

exist for every $\mathbf{v} \in E$ and define a continuous linear operator $f'(\mathbf{a})\colon \mathbf{v} \to f'(\mathbf{a}; \mathbf{v})$ from E to F (called the Gâteaux derivative of f at $\mathbf{a}$ in the direction $\mathbf{v}$).

In the case of real-valued functions f defined on $\mathbb{R}^N$, the Gâteaux differentiability implies the existence of all partial derivatives

$$\frac{\partial f}{\partial x_k}(\mathbf{a}) = f'(\mathbf{a})(\mathbf{e}_k) \quad \text{for } k = 1, ..., N,$$

where $\mathbf{e}_1 = (1, 0, ..., 0), ..., \mathbf{e}_N = (0, 0, ..., 1)$ represent the natural basis of $\mathbb{R}^N$. Thus

$$f'(\mathbf{a})(\mathbf{v}) = \langle \nabla f(\mathbf{a}), \mathbf{v}\rangle \quad \text{for all } \mathbf{v} \in \mathbb{R}^N,$$

where

$$\nabla f(\mathbf{a}) = \sum_{k=1}^{n} \frac{\partial f}{\partial x_k}(\mathbf{a})\mathbf{e}_k$$

denotes the *gradient* of f at $\mathbf{a}$.

In the case of a proper convex function $f : E \to \mathbb{R} \cup \{\infty\}$, the definition of Gâteaux differentiability applies to all points $\mathbf{a} \in \mathrm{int}\,(\mathrm{dom}\ f)$ (which corresponds to the choice $U = \mathrm{int}\,(\mathrm{dom}\ f)$ and $F = \mathbb{R}$ in Definition 5.1.8).

According to Lemma 5.1.2, the function f is Gâteaux differentiable at a point $\mathbf{a} \in \operatorname{int}(\operatorname{dom} f)$, of continuity for f, if and only if

$$f'_+(\mathbf{a}\,;\mathbf{v}) = -f'_+(\mathbf{a}\,;-\mathbf{v}) \quad \text{for all } \mathbf{v} \in E.$$

Indeed, a sublinear function p is linear if and only if $p(-\mathbf{x}) = -p(\mathbf{x})$ for all x. The condition of continuity is automatically fulfilled when E is finite dimensional. See Proposition 4.1.8.

5.1.9 Proposition *If a proper convex function $f : E \to \mathbb{R} \cup \{\infty\}$ is* Gâteaux differentiable *at a point $a \in \operatorname{int}(\operatorname{dom} f)$, then*

$$\partial f(\mathbf{a}) = \{f'(\mathbf{a})\}$$

and the subgradient inequality becomes

$$f(\mathbf{x}) \geq f(\mathbf{a}) + f'(\mathbf{a})\,(\mathbf{x} - \mathbf{a}) \quad \textit{for all } \mathbf{x} \in E.$$

Proof. Clearly, $f'(\mathbf{a}) \in \partial f(\mathbf{a})$. If $x^* \in \partial f(\mathbf{a})$, then $x^*(\mathbf{v}) \leq f'(\mathbf{a})(\mathbf{v})$ for all $\mathbf{v} \in E$. Replacing $\mathbf{v}$ by $-\mathbf{v}$, we infer the converse inequality. Therefore $x^* = f'(\mathbf{a})$. ■

Gâteaux differentiability is equivalent to the uniqueness of the support function.

5.1.10 Corollary *Let $f : E \to \mathbb{R} \cup \{\infty\}$ be a proper convex function which is continuous at a point $\mathbf{a} \in \operatorname{int}(\operatorname{dom} f)$. Then f is Gâteaux differentiable at $\mathbf{a}$ if and only if f has a unique support at $\mathbf{a}$.*

See Exercise 5 for an alternative formulation.

Proof. Suppose that $\partial f(\mathbf{a}) = \{x^*\}$. Then, by Moreau's max formula, $f'_+(\mathbf{a};\mathbf{v}) = x^*(\mathbf{v})$ and $f'_+(\mathbf{a};-\mathbf{v}) = -x^*(\mathbf{v})$ for all $\mathbf{v} \in E$. According to the comments after Definition 5.1.8, this implies that f is Gâteaux differentiable at a. The other implication follows from Proposition 5.1.9. ■

If E is finite dimensional, then the condition on continuity in Corollary 5.1.10 is automatic. See Proposition 4.1.8. In infinite dimensions things are different and without the continuity hypothesis the assertion of Corollary 5.1.10 may fail. An example is provided by the indicator function ι_C of the *Hilbert cube*,

$$C = \left\{(x_n)_{n=1}^{\infty} \in \ell^2 : |x_n| \leq 1/n \text{ for all } n\right\}.$$

The function ι_C is lower semicontinuous because C is a weakly compact convex set and all such sets are also closed in the norm topology. See Exercise 7. Since $\operatorname{int} C = \emptyset$, the function ι_C is nowhere continuous. The subdifferential of ι_C at the origin of ℓ^2 is a singleton, but ι_C is nowhere Gâteaux differentiable.

As we already noticed, the subdifferentiability is related to the existence of directional derivatives. However, while a derivative is a local property, the subgradient definition (4.3.3) describes a global property. An illustration of this fact is the following theorem that plays an important role in optimization theory.

5.1.11 Theorem (Subdifferential at optimality) *Let E be a Banach space and let $f : E \to \mathbb{R} \cup \{\infty\}$ be a proper convex function. Then a point $\mathbf{a} \in \operatorname{dom} f$ is a global minimizer of f if and only if f is subdifferentiable at $\mathbf{a}$ and*

$$\mathbf{0} \in \partial f(\mathbf{a}).$$

If $\mathbf{a} \in \operatorname{int}(\operatorname{dom} f)$ and f is differentiable at $\mathbf{a}$, then the condition $\mathbf{0} \in \partial f(\mathbf{a})$ reduces to $f'(\mathbf{a}) = 0$.

The proof of Theorem 5.1.11 is straightforward.

Theorem 5.1.11 represents the analogue of *Fermat's rule* for finding points of extrema in the context of convex functions.

Important. In the general framework of differentiable functions a critical point is not necessarily a point of extremum. However, when $f : E \to \mathbb{R}$ is differentiable and convex, the condition $f'(\mathbf{a}) = 0$ is equivalent to the fact that $\mathbf{a}$ is a global minimizer!

A detailed discussion on the extremal properties of convex functions makes the objective of Section 5.4.

Exercises

1. (Subadditivity of the directional derivatives) Suppose that f is a convex function (on an open convex set U in a normed linear space E). For $\mathbf{a} \in U$, $\mathbf{u}, \mathbf{v} \in E$ and $t > 0$ small enough, show that

$$\frac{f(\mathbf{a} + t(\mathbf{u} + \mathbf{v})) - f(\mathbf{a})}{t} \leq \frac{f(\mathbf{a} + 2t\mathbf{u}) - f(\mathbf{a})}{2t} + \frac{f(\mathbf{a} + 2t\mathbf{v}) - f(\mathbf{a})}{2t}$$

   and conclude that $f'_+(\mathbf{a}\,; \mathbf{u} + \mathbf{v}) \leq f'_+(\mathbf{a}\,; \mathbf{u}) + f'_+(\mathbf{a}\,; \mathbf{v})$.

2. Let $f, g : \mathbb{R} \to \mathbb{R}$ be two convex functions. Infer from Theorem 5.1.6 that the subdifferential of the function $h(x, y) = f(x) + g(y)$ is given by the formula $\partial h(x, y) = \partial f(x) \times \partial g(y)$.

3. Consider the space $\mathbb{R}^N$ endowed with the norm $\|\mathbf{x}\|_1 = \sum_{n=1}^N |x_n|$. Prove that the subdifferential of the function $f(\mathbf{x}) = \|\mathbf{x}\|_1$ is

$$\partial f(\mathbf{x}) = \left\{ \sigma \in \mathbb{R}^N : \sup_{1 \leq n \leq N} |\sigma_n| \leq 1, \sum_{n=1}^N \sigma_n x_n = \|\mathbf{x}\|_1 \right\}.$$

   [*Hint:* Apply Theorem 5.1.5, taking into account that f can be expressed as the maximum of $2N$ linear and differentiable functions,

$$f(\mathbf{x}) = \max \left\{ \langle \mathbf{x}, \sigma \rangle = \sum\nolimits_{n=1}^N \sigma_n x_n : \sigma_n \in \{-1, 1\} \text{ for all } n \right\},$$

   and the subgradient of each map $\mathbf{x} \to \langle \mathbf{x}, \sigma \rangle$ is σ.]

4. Prove that every lower semicontinuous proper convex function $f : E \to (-\infty, \infty]$ which is and Gâteaux differentiable at a point $\mathbf{a}$, is continuous at that point.

   [*Hint*: See Exercise 6 at the end of Section 4.2.]

5. (Unique subgradients) Suppose that $f : E \to (-\infty, \infty]$ is a proper convex function, continuous at a point a. Prove that f is Gâteaux differentiable at a if and only if $\partial f(a)$ is a singleton.

6. (Mean value theorem) Let $f : U \to \mathbb{R}$ be a convex function defined on an open convex subset U of the Banach space E and let $\mathbf{x}$ and $\mathbf{y}$ be two distinct points in E. Prove that there is a number $t \in (0, 1)$ such that

$$f(\mathbf{x}) - f(\mathbf{y}) \in \langle \mathbf{x} - \mathbf{y}, \partial f\left((1-t)\mathbf{x} + t\mathbf{y}\right)\rangle.$$

   [*Hint:* Consider a local extremum for the convex function $\varphi : [0, 1] \to \mathbb{R}$ defined by $\varphi(t) = f((1-t)\mathbf{x} + t\mathbf{y})$.]

   Remark. The mean value theorem easily yields the following characterization of the Lipschitz property: a convex function $f : U \to \mathbb{R}$ is Lipschitz with $\mathrm{Lip}(f) = K$ if and only if $\|x^*\| \leq K$ for all $x^* \in \partial f(\mathbf{x})$ and $\mathbf{x} \in U$.

7. Prove that the Hilbert cube is a weakly compact convex subset of ℓ^2.

   [*Hint:* Since ℓ^2 is a reflexive Banach space, it suffices to prove that the Hilbert cube is a closed, bounded, and convex set. See the Banach–Alaoglu theorem (Theorem B.1.6, Appendix B).]

8. Find convex functions $f, g : \mathbb{R} \to \mathbb{R} \cup \{\infty\}$ with $\partial f(0) + \partial g(0) \neq \partial(f+g)(0)$.

9. Give an example of convex function $g : \mathbb{R}^2 \to \mathbb{R} \cup \{\infty\}$ and of a linear map $A : \mathbb{R} \to \mathbb{R}^2$ such that $\partial(g \circ A)(0) \neq A^* \partial g(0)$.

5.2 Fréchet Differentiability of Convex Functions

A stronger condition than Gâteaux differentiability is Fréchet differentiability, which is crucial for some basic results in nonlinear analysis such as the local inversion theorem and the implicit function theorem.

5.2.1 Definition *Suppose that E and F are two real Banach spaces and f is a function defined on an open subset U of E and taking values in F. We say that f is* Fréchet differentiable (*or simply* differentiable) *at a point $\mathbf{a} \in U$ if there exists a continuous linear operator $df(\mathbf{a}) \in L(E, F)$ such that*

$$\lim_{\mathbf{x} \to \mathbf{a}} \frac{\|f(\mathbf{x}) - f(\mathbf{a}) - df(\mathbf{a})(\mathbf{x} - \mathbf{a})\|}{\|\mathbf{x} - \mathbf{a}\|} = 0.$$

Equivalently, using Landau's little-o notation,

$$f(\mathbf{a} + \mathbf{v}) = f(\mathbf{a}) + df(\mathbf{a})(\mathbf{v}) + o(\mathbf{v}), \tag{5.2}$$

which means the existence of a ball $B_\varepsilon(\mathbf{0}) \subset E$ *centered at* $\mathbf{0}$ *and a function* $\omega : B_\varepsilon(\mathbf{0}) \to \mathbb{R}$ *such that* $\omega(\mathbf{0}) = \lim_{\mathbf{v}\to\mathbf{0}} \omega(\mathbf{v}) = 0$ *and*

$$f(\mathbf{a}+\mathbf{v}) = f(\mathbf{a}) + df(\mathbf{a})(\mathbf{v}) + \omega(\mathbf{v})\,\|\mathbf{v}\|$$

for all $\mathbf{v} \in B_\varepsilon(\mathbf{0})$.

The operator $df(\mathbf{a})$, when it exists, is unique. It is called the *Fréchet derivative* of f at $\mathbf{a}$.

Notice the following analogue of formula (5.2) in the case of Gâteaux differentiability:

$$f(\mathbf{a}+\varepsilon\mathbf{v}) = f(\mathbf{a}) + \varepsilon f'(\mathbf{a})(\mathbf{v}) + o(\varepsilon). \tag{5.3}$$

Clearly, Fréchet differentiability implies Gâteaux differentiability and also the equality of the respective derivatives,

$$f'(\mathbf{a}) = df(\mathbf{a}).$$

5.2.2 Remark *It is worth noticing that the definition of Fréchet differentiability at a point extends verbatim to the case of functions defined on a closed subset* $C \subset \mathbb{R}^N$ *having some special geometric properties such as the "cone condition", which means that every point* $\mathbf{a} \in C$ *is the vertex of a finite right circular cone with nonempty interior. So is the case of every compact N-dimensional interval* $[a_1, b_1] \times \cdots \times [a_N, b_N]$ *and of the upper half-space* $x_N \geq 0$ *in* $\mathbb{R}^N$. *The cone condition plays an important role in the theory of Sobolev spaces. See the book of R. A. Adams and J. J. F. Fournier* [8]. *However, except in the case of Exercise 3 (b), this book is devoted entirely to the case of differentiable functions defined on open sets.*

A simple test of Fréchet differentiability in the case of functions defined on open sets in $\mathbb{R}^N$ is the membership to the class C^1 (that is, the existence and continuity of all partial derivatives of first order). This result, as well as many more on Fréchet differentiability, can be found in the book by R. Coleman [139], Theorem 2.2, p. 45.

Remarkably, the Gâteaux and Fréchet differentiability coincide in the case of convex functions defined on open subsets of $\mathbb{R}^N$. Even more, every Fréchet differentiable function is continuously differentiable. See Exercise 3 (a).

5.2.3 Theorem *Suppose that a convex function* f *defined on an open convex set* U *in* $\mathbb{R}^N$ *possesses all its partial derivatives* $\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_N}$ *at some point* $\mathbf{a} \in U$. *Then* f *is Fréchet differentiable at* $\mathbf{a}$.

Actually much more can be proved: the set U_0, of all points of differentiability of f is dense in U, $U \backslash U_0$ is a set of measure zero and of first category, and the function $\nabla f : U_0 \to \mathbb{R}$ is continuous in U_0. See M. Kuczma [311], Theorem 7.9.1, p. 191.

Proof. Since U is open, there is an $r > 0$ such that $B_r(\mathbf{a}) \subset U$. We have to prove that the function

$$g(\mathbf{u}) = f(\mathbf{a}+\mathbf{u}) - f(\mathbf{a}) - \sum_{k=1}^{N} \frac{\partial f}{\partial x_k}(\mathbf{a})u_k,$$

defined for all vectors $\mathbf{u} = (u_1, \ldots, u_N)$ with $\|\mathbf{u}\| < r$, verifies

$$\lim_{\|\mathbf{u}\|\to\mathbf{0}} g(\mathbf{u})/\|\mathbf{u}\| = 0.$$

Clearly, the function g is convex. Then

$$0 = g(\mathbf{0}) = g\Big(\frac{\mathbf{u}+(-\mathbf{u})}{2}\Big) \le \frac{1}{2}\,(g(\mathbf{u}) + g(-\mathbf{u})),$$

which yields $g(\mathbf{u}) \ge -g(-\mathbf{u})$. On the other hand, for each $\mathbf{u}$ with $N\|\mathbf{u}\| < r$, we have

$$\begin{aligned} g(\mathbf{u}) &= g\Big(\frac{1}{N}\sum_{k=1}^{N} Nu_k\mathbf{e}_k\Big) \le \frac{1}{N}\sum_{k=1}^{N} g(Nu_k\mathbf{e}_k) \\ &= \sum_{\{k:u_k\neq 0\}} u_k\,\frac{g(Nu_k\mathbf{e}_k)}{Nu_k} \le \|\mathbf{u}\| \sum_{\{k:u_k\neq 0\}} \Big|\frac{g(Nu_k\mathbf{e}_k)}{Nu_k}\Big|, \end{aligned}$$

where $\mathbf{e}_1, ..., \mathbf{e}_N$ denotes the natural basis of $\mathbb{R}^N$. Similarly,

$$g(-\mathbf{u}) \le \|\mathbf{u}\| \sum_{\{k:u_k\neq 0\}} \Big|\frac{g(-Nu_k\mathbf{e}_k)}{Nu_k}\Big|.$$

Then

$$-\|\mathbf{u}\| \sum_{\{k:u_k\neq 0\}} \Big|\frac{g(-Nu_k\mathbf{e}_k)}{Nu_k}\Big| \le -g(-\mathbf{u}) \le g(\mathbf{u}) \le \|\mathbf{u}\| \sum_{\{k:u_k\neq 0\}} \Big|\frac{g(Nu_k\mathbf{e}_k)}{Nu_k}\Big|,$$

and to complete the proof it remains to notice that from the definition of the partial derivative we have $g(Nu_k\mathbf{e}_k)/(Nu_k) \to 0$ as $u_k \to 0$. ∎

The analogue of Theorem 5.2.3 for differentiability of higher order does not work. See Exercise 7, Section 5.3 and the Comments at the end of this chapter.

In the context of several variables, the set of points where a convex function is not differentiable is negligible (though it can be uncountable). This fact can be seen as a consequence of Rademacher's theorem on almost everywhere differentiability of Lipschitz functions, but we prefer to present here a direct argument.

5.2.4 Theorem *Suppose that f is a convex function on an open subset U of $\mathbb{R}^N$. Then f is differentiable almost everywhere in U.*

Proof. Consider first the case when U is also bounded. According to Theorem 5.2.3 we must show that each of the sets

$$E_k = \left\{ \mathbf{x} \in U : \frac{\partial f}{\partial x_k}(\mathbf{x}) \text{ does not exist} \right\}, \quad k = 1, ..., N,$$

is a set of zero Lebesgue measure. The measurability of E_k is a consequence of the fact that the limit of a pointwise converging sequence of measurable functions is measurable too. In fact, the formula

$$f'_+(\mathbf{x}, \mathbf{e}_k) = \lim_{j \to \infty} \frac{f(\mathbf{x} + \mathbf{e}_k/j) - f(\mathbf{x})}{1/j}$$

motivates the measurability of the one-sided directional derivative $f'_+(\mathbf{x}, \mathbf{e}_k)$ and a similar argument applies for $f'_-(\mathbf{x}, \mathbf{e}_k)$. Consequently the set

$$E_k = \{\mathbf{x} \in U : f'_+(\mathbf{x}, \mathbf{e}_k) - f'_-(\mathbf{x}, \mathbf{e}_k) > 0\}$$

is measurable. Being bounded, it is also integrable. By Fubini's theorem,

$$\begin{aligned} \mathcal{L}(E_k) &= \int_{\mathbb{R}^N} \chi_{E_k} \, \mathrm{d}\mathbf{x} \\ &= \int_{\mathbb{R}} \cdots \left(\int_{\mathbb{R}} \chi_{E_k} \, \mathrm{d}x_k \right) \mathrm{d}x_1 \cdots \mathrm{d}x_{k-1} \, \mathrm{d}x_{k+1} \cdots \mathrm{d}x_N \end{aligned}$$

and the interior integral is zero since f is convex as a function of x_k (and thus differentiable except at a countable set of points). If U is arbitrary, the argument above shows that all the sets $E_k \cap B_n(\mathbf{0})$ are negligible. Or, $E_k = \bigcup_{n=1}^{\infty} (E_k \cap B_n(\mathbf{0}))$ and a countable union of negligible sets is negligible too. ∎

The function $f(x, y) = \sup\{x, 0\}$ is convex on $\mathbb{R}^2$ and nondifferentiable at the points of the y-axis (which constitutes an uncountable set).

The coincidence of Gâteaux and Fréchet differentiability for convex functions is no longer true in the context of infinite-dimensional spaces. See Exercises 3-6.

An example of a real-valued continuous convex function which is not Gâteaux differentiable at any point is provided by Exercise 6. However, according to Theorem 4.3.1, such a function is everywhere subdifferentiable.

5.2.5 Theorem *Let E be a Banach space such that for each continuous convex function $f : E \to \mathbb{R}$, every point of Gâteaux differentiability is also a point of Fréchet differentiability. Then E is finite dimensional.*

The proof depends on a deep result in Banach space theory:

5.2.6 Theorem (The Josefson–Nissenzweig theorem [280], [455]) *If E is a Banach space such that*

$$x_n^\star \to 0 \text{ in the weak-star topology of } E^\star \quad \text{implies} \quad \|x_n^\star\| \to 0,$$

then E is finite dimensional.

Proof. (of Theorem 5.2.5). Consider a sequence $(x_n^*)_n$ of functionals in the unit ball of E^* and a sequence $(\alpha_n)_n$ of real numbers such that $\alpha_n \downarrow 0$. Then the function

$$f(\mathbf{x}) = \sup_n [\langle \mathbf{x}, x_n^* \rangle - \alpha_n]$$

is convex and continuous and, moreover,

$$f \text{ is Gâteaux differentiable at } 0 \iff x_n^*(\mathbf{x}) \to 0 \text{ for all } x \in E$$
$$f \text{ is Fréchet differentiable at } 0 \iff \|x_n^*\| \to 0.$$

The proof ends by applying the Josefson–Nissenzweig Theorem 5.2.6. ■

5.2.7 Remark *A result due to S. Mazur asserts that if E is a separable Banach space and f is a continuous convex function defined on a convex open subset U of E, then the set of points where f is Gâteaux differentiable is always a dense subset of U (even more, it is the intersection of a sequence of dense open subsets). See R. R. Phelps [473], Theorem 1.20, p. 12. A similar result in the case of Fréchet differentiability is known to work in the framework of reflexive Banach spaces. See Theorem 5.7.1 in the Comments section at the end of this chapter.*

We next discuss an important geometric property of L^p-spaces with $1 < p < \infty$, the smoothness of the unit sphere.

5.2.8 Definition *A Banach space E is called* smooth *if its norm is Gâteaux differentiable at every nonzero point.*

Smoothness and strict convexity are related by duality. If E^* is strictly convex (respectively, smooth), then E is smooth (respectively, strictly convex). The converse is true for reflexive spaces, but not in general.

5.2.9 Proposition *Let (Ω, Σ, μ) be a measure space. Then all Lebesgue spaces $L^p(\mu)$ with $1 < p < \infty$ are smooth and the Gâteaux derivative of the norm function*

$$\mathrm{N}(u) = \|u\|_p = \left(\int_\Omega |u(x)|^p \, \mathrm{d}\mu \right)^{1/p}$$

at a point $u \in L^p(\mu)$, $u \neq 0$, is

$$\mathrm{N}'(u)(v) = \|u\|^{1-p} \int_\Omega |u(x)|^{p-2} u(x) v(x) \mathrm{d}\mu \quad \text{for all } v \in L^p(\mu).$$

Proof. We prove first that the function $G(u) = \|u\|_p^p$ is Gâteaux differentiable on $L^p(\mu)$ and

$$G'(u)(v) = p \int_\Omega |u(x)|^{p-2} u(x) v(x) \mathrm{d}\mu \quad \text{for all } v.$$

Indeed, for every two real numbers a, b we have

$$\lim_{t\to 0} \frac{|a+tb|^p - |a|^p}{t} = p\,|a|^{p-2}\,ab; \tag{5.4}$$

the case $a = 0$ is not excluded because the function $|a|^{p-2}\,a$ has the limit 0 at $a = 0$ when $p > 1$. The function $|a + tb|^p$ is convex (being the composition of the convex and increasing function x^p on $[0,\infty)$ and the convex function $|a + tb|$). Therefore

$$|a|^p - |a-b|^p \le \frac{|a+tb|^p - |a|^p}{t} \le |a+b|^p - |a|^p\,, \tag{5.5}$$

according to the three chords inequality. The double estimate (5.5) yields the integrability of the function

$$\frac{|u(x)+tv(x)|^p - |u(x)|^p}{t}.$$

Taking into account the formula (5.4) and Lebesgue's dominated convergence theorem we infer that

$$G'(u)(v) = \lim_{t\to 0} \frac{\|u+tv\|^p - \|u\|^p}{t} = p\int_\Omega |u(x)|^{p-2}\,u(x)v(x)\mathrm{d}\mu.$$

Hence, for all $u, v \in L^p(\mu)$, $u \neq 0$,

$$\begin{aligned} \mathrm{N}'(u)(v) &= \frac{d}{d\varepsilon} G^{1/p}(u+\varepsilon v)\Big|_{\varepsilon=0} = G^{(1-p)/p}(u)\int_\Omega |u(x)|^{p-2}\,u(x)v(x)\mathrm{d}\mu \\ &= \|u\|^{1-p}\int_\Omega |u(x)|^{p-2}\,u(x)v(x)\mathrm{d}\mu \end{aligned}$$

and the proof is complete. ■

If one considers the squared norm $\mathrm{N}^2(\mathbf{x}) = \|\mathbf{x}\|^2$ in a real Hilbert space H, then the identity

$$\|\mathbf{x}+\mathbf{v}\|^2 = \|\mathbf{x}\|^2 + 2\langle \mathbf{x}, \mathbf{v}\rangle + \|\mathbf{v}\|^2$$

easily yields that this function is Fréchet differentiable everywhere and its Fréchet derivative verifies the formula

$$d\mathrm{N}^2(\mathbf{x})(\mathbf{v}) = 2\langle \mathbf{x}, \mathbf{v}\rangle \quad \text{ for all } \mathbf{x}, \mathbf{v} \in H.$$

The Fréchet differentiability of the norm function $\mathrm{N}(\mathbf{x}) = \|\mathbf{x}\| = \sqrt{\mathrm{N}^2(\mathbf{x})}$ for all $\mathbf{x} \in H$, $\mathbf{x} \neq 0$, results via the chain rule. We have

$$d\mathrm{N}(\mathbf{x})(\mathbf{v}) = \langle \mathbf{v}, \frac{\mathbf{x}}{\|\mathbf{x}\|}\rangle \quad \text{ whenever } \mathbf{x}, \mathbf{v} \in H,\ \mathbf{x} \neq 0.$$

One can prove that all L^p-norms are Fréchet differentiable at every nonzero point of $L^p(\mu)$ when $1 < p < \infty$. See K. Sundaresan and S. Swaminathan [559].

Exercises

1. Prove the following formulas:

$$\nabla_{\mathbf{x}}\langle A\mathbf{x},\mathbf{x}\rangle = 2A\mathbf{x} \quad (A \in \mathrm{Sym}(N,\mathbb{R}),\ \mathbf{x} \in \mathbb{R}^N),$$
$$\nabla_A \mathrm{trace}(BA) = B^T \quad (A, B \in M_N(\mathbb{R})),$$
$$\nabla_A \log\det(A) = (A^{-1})^T \quad (A \in \mathrm{Sym}^{++}(N,\mathbb{R})).$$

2. Consider a Lipschitz function f defined on an open set in $\mathbb{R}^N$ and taking values in a Banach space E. Prove that f is Gâteaux differentiable if and only if it is Fréchet differentiable.

3. (a) Suppose that U is an open convex subset of $\mathbb{R}^N$ and $f : U \to \mathbb{R}$ is a differentiable convex function. Prove that f is actually a continuously differentiable function.

   (b) The above property does not apply to the differentiable convex functions defined on closed convex sets. Consider the case of the function $f(x,y) = x^2 e^{-y/x^2}$ for $(x,y) \in (0,\infty)\times\mathbb{R}_+$ and $f(0,y) = 0$ for $y \in \mathbb{R}_+$.

4. Consider the space ℓ^1, of all absolutely summing sequences of real numbers, endowed with its natural norm, $f(\mathbf{x}) = \|\mathbf{x}\|_1 = \sum_{n=1}^{\infty} |x_n|$. The dual of this space is ℓ^∞, the space of all bounded sequences $\sigma = (\sigma_n)_n$ of real numbers, endowed with the sup norm, $\|\sigma\|_\infty = \sup|\sigma_n|$. See R. Bhatia [63], p. 50. Prove that the norm in ℓ^1 is Gâteaux differentiable only at the points $\mathbf{x} = (x_n)_n$ in ℓ^1 whose all entries x_n are nonzero.

5. Prove that the norm $\|\cdot\|_1$ in ℓ^1 is not Fréchet differentiable at any point.

6. (The case of uncountable index sets) Consider the space $\ell^1(A)$, of all absolutely summing families $\mathbf{x} = (x_a)_{a\in A}$ of real numbers, endowed with the norm

$$\|\mathbf{x}\|_1 = \sup_{F\subset A,\ F \text{ finite}} \sum_{\alpha\in F} |x_a|\,.$$

   Prove that the function $\|\cdot\|_1 : \ell^1(A) \to \mathbb{R}$ is not Gâteaux differentiable at any point if A is uncountable.

7. Prove that the norm function on $C([0,1])$,

$$\|x\| = \sup\{|x(t)| : t \in [0,1]\},$$

   is not Fréchet differentiable at any point, but it is Gâteaux differentiable at those points x of the unit sphere for which $|x(t_0)| = 1$ is attained for only one value t_0.

8. (An example due to M. Sova; see [219]) Consider the maps $f : L^1([0,\pi]) \to \mathbb{R}$, $f(x) = \int_0^\pi \sin(x(t))\mathrm{d}t$, and $g : L^2([0,\pi]) \to \mathbb{R}$, $g(x) = \int_0^\pi \sin(x(t))\mathrm{d}t$. Prove that:

   (a) f is everywhere Gâteaux differentiable, with $f'(x) = \cos x$, but nowhere Fréchet differentiable;

   (b) g is Fréchet differentiable everywhere.

9. Infer from the Josefson–Nissenzweig Theorem 5.2.6 that any Banach space E on which all continuous convex functions are bounded on bounded subsets is finite dimensional.

   [*Hint*: Consider a sequence $(x_n^*)_n$ of norm-1 functionals in E^* and the convex function $f(\mathbf{x}) = \sum_n n(|x_n^*(\mathbf{x})| - 1/2)^+$. Then notice that f is finite and continuous if and only if $(x_n^*)_n$ is weak-star convergent to 0.]

10. (Convergence of sequences of differentiable convex functions) Let U be an open convex set in $\mathbb{R}^N$ and let f_n, $f : U \to \mathbb{R}$ $(n \in \mathbb{N}^*)$ be differentiable convex functions. Assuming that $f_n \to f$ pointwise, prove that

$$\lim_{n\to\infty} \nabla f_n(x) = \nabla f(x), \text{ uniformly on the compact subsets of } U.$$

   [*Hint*: It suffices to prove that the partial derivatives of f_n converge uniformly to the corresponding partial derivatives of f, on every compact subset of U. Use Exercise 6, Section 4.3.]

5.3 Differential Criteria of Convexity

The following characterization of convexity within the class of Gâteaux differentiable functions combines Theorem 4.3.6 and Proposition 5.1.9:

5.3.1 Theorem (First derivative test of convexity) *Suppose that U is an open convex set in a real Banach space E. Then a Gâteaux differentiable function $f : U \to \mathbb{R}$ is convex if and only if*

$$f(\mathbf{x}) \geq f(\mathbf{a}) + f'(\mathbf{a})(\mathbf{x} - \mathbf{a}) \quad \text{for every } \mathbf{x}, \mathbf{a} \in U. \tag{5.6}$$

The function f is strictly convex if and only if the inequality (5.6) is strict for $\mathbf{x} \neq \mathbf{a}$.

The case of strongly convex functions will be detailed in Section 5.5.

On intervals, a differentiable function is convex if and only if its derivative is increasing. The higher dimensional analogue of this result is due to I. R. Kachurovskii [283] and illustrates the concept of monotone map.

5.3.2 Definition *Suppose that C is a nonempty set in a real Banach space E. A map $V : C \to E^*$ is called monotone if*

$$(V(\mathbf{x}) - V(\mathbf{y}))(\mathbf{x} - \mathbf{y}) \geq 0$$

for all $\mathbf{x}, \mathbf{y} \in C$ and strictly monotone if this inequality is strict for $\mathbf{x} \neq \mathbf{y}$.

This definition is motivated by the case where $E = \mathbb{R}$ and C is an interval in $\mathbb{R}$. In this case the monotonicity of a function $V : C \to \mathbb{R}$ reduces to the fact that $(V(x) - V(y)) \cdot (x - y)) \geq 0$, equivalently,

$$x \leq y \text{ implies } V(x) \leq V(y).$$

5.3.3 Theorem *Suppose that f is Gâteaux differentiable on the open convex set U in a real Banach space E. Then f is (strictly) convex if and only if its Gâteaux differential is a (strictly) monotone map from U to E^*.*

Proof. If f is convex, then for $\mathbf{x}$ and $\mathbf{y}$ in U and $0 < t < 1$ we have

$$\frac{f(\mathbf{y} + t(\mathbf{x} - \mathbf{y})) - f(\mathbf{y})}{t} \leq f(\mathbf{x}) - f(\mathbf{y}),$$

so by letting $t \to 0+$ we obtain that $f'(\mathbf{y})(\mathbf{x}-\mathbf{y}) \leq f(\mathbf{x})-f(\mathbf{y})$. Interchanging $\mathbf{x}$ and $\mathbf{y}$, we also have $f'(\mathbf{x})(\mathbf{y}-\mathbf{x}) \leq f(\mathbf{y})-f(\mathbf{x})$. By adding these two inequalities we obtain the monotonicity of f', that is,

$$(f'(\mathbf{x}) - f'(\mathbf{y}))(\mathbf{x} - \mathbf{y}) \geq 0 \quad \text{for all } \mathbf{x}, \mathbf{y} \in U. \tag{5.7}$$

Suppose now that (5.7) holds. Let $\mathbf{x}, \mathbf{y} \in U$ and consider the function $g(\lambda) = f((1 - \lambda)\mathbf{x} + \lambda\mathbf{y})$, $\lambda \in [0, 1]$. One can easily verify that

$$\lambda_1 \leq \lambda_2 \text{ implies } g'(\lambda_1) \leq g'(\lambda_2),$$

which shows that g is convex. Then

$$\begin{aligned} f((1 - \lambda)\mathbf{x} + \lambda\mathbf{y}) &= g(\lambda) = g(\lambda \cdot 1 + (1 - \lambda) \cdot 0) \\ &\leq \lambda g(1) + (1 - \lambda)g(0) = (1 - \lambda)f(\mathbf{x}) + \lambda f(\mathbf{y}), \end{aligned}$$

which proves that f is a convex function. The case of strictly convex functions can be treated in the same manner. ■

When the ambient space is a real Hilbert space (in particular, a Euclidean space), the inequality (5.7) defining the monotonicity of the differential becomes

$$\langle \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq 0. \tag{5.8}$$

The analogue of Theorem 5.3.3 in the context of subdifferentiability was presented in Section 4.3.

Our next goal is an important characterization of convexity under the presence of twice Gâteaux differentiability. For convenience, we recall here some basic facts.

Given a Gâteaux differentiable function $f : U \to \mathbb{R}$, the limit

$$f''(\mathbf{a}\,; \mathbf{v}, \mathbf{w}) = \lim_{\lambda \to 0} \frac{f'(\mathbf{a} + \lambda\mathbf{w})(\mathbf{v}) - f'(\mathbf{a})(\mathbf{v})}{\lambda}$$

(when it exists in $\mathbb{R}$) is called the *second directional derivative* of f at the point $\mathbf{a}$ in the directions $\mathbf{v}$ and $\mathbf{w}$. We say that f is *twice Gâteaux differentiable* at

$\mathbf{a}$ if the second directional derivatives $f''(\mathbf{a}\,;\mathbf{v},\mathbf{w})$ exist for all vectors $\mathbf{v},\mathbf{w}$ in the ambient Banach space E and the map $f''(\mathbf{a}) : (\mathbf{v},\mathbf{w}) \to f''(\mathbf{a}\,;\mathbf{v},\mathbf{w})$ is a continuous bilinear form on $E \times E$; in this case, $f''(\mathbf{a})$ represents the *second Gâteaux differential* of f at $\mathbf{a}$.

When $E = \mathbb{R}^N$, the twice Gâteaux differentiability simply means the existence of the *Hessian matrix* of f,

$$\nabla^2 f(\mathbf{a}) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{a}) \right)_{i,j=1}^{N}$$

at all points $\mathbf{a} \in U$. We have

$$f''(\mathbf{a})(\mathbf{v},\mathbf{w}) = \langle \nabla^2 f(\mathbf{a})\mathbf{v},\mathbf{w}\rangle \quad \text{for all } \mathbf{a} \in U \text{ and } \mathbf{v},\mathbf{w} \in \mathbb{R}^N. \tag{5.9}$$

The Hessian matrix $\nabla^2 f(\mathbf{a})$ is symmetric when all second order derivatives of f are continuous at $\mathbf{a}$.

Some authors prefer to denote the Hessian matrix as $H_f(\mathbf{a})$ or $\mathrm{Hess}_{\mathbf{a}} f$.

The twice Fréchet differentiability of a function $f\colon U \to \mathbb{R}$ means the differentiability of both f and $\mathrm{d}f$. The *second Fréchet differential* of a function $f\colon U \to \mathbb{R}$ at $\mathbf{a} \in U$ is defined as the differential at $\mathbf{a}$ of the first differential, that is,

$$\mathrm{d}^2 f(\mathbf{a}) = \mathrm{d}\left(\mathrm{d}f(\mathbf{x})\right)(\mathbf{a}).$$

This imposes the existence of the first differential at least on a neighborhood of $\mathbf{a}$. The Fréchet differential of second order belongs to the space $L(E, L(E,\mathbb{R}))$. This space can be identified with the space $\mathcal{B}(E \times E, \mathbb{R})$ (of all continuous bilinear maps from $E \times E$ to $\mathbb{R}$) via the linear isomorphism

$$\Phi : L(E, L(E,\mathbb{R})) \to \mathcal{B}(E \times E, \mathbb{R}),$$

that associates to each operator $T \in L(E, L(E,\mathbb{R}))$ the bilinear map $B = \Phi(T)$ given by $B(\mathbf{v},\mathbf{w}) = (T(\mathbf{v}))(\mathbf{w})$.

5.3.4 Remark *All functions defined on open sets in $\mathbb{R}^N$ that admit continuous partial derivatives up to second order (that is, all functions of class C^2) are twice Fréchet differentiable. See R. Coleman* [139]*, Proposition* 4.9*, p.* 99.

The Hessian matrix of every twice Fréchet differentiable function defined on an open subset of $\mathbb{R}^N$ is symmetric. See J. Dieudonné [160]*, Theorem* 8.12.2*, p.* 180.

5.3.5 Lemma *If $f\colon U \to \mathbb{R}$ is twice Fréchet differentiable, then it is also twice Gâteaux differentiable and*

$$\mathrm{d}^2 f(\mathbf{a})(\mathbf{v},\mathbf{w}) = f''(\mathbf{a}\,;\mathbf{v},\mathbf{w}) \quad \textit{for all } \mathbf{a} \in U \textit{ and } \mathbf{v},\mathbf{w} \in E.$$

Our next goal is to establish the analogue of Taylor's formula in the context of Gâteaux differentiability and to infer from it an important characterization of convexity under the presence of twice Gâteaux differentiability.

5.3.6 Theorem (Taylor's formula) *If f is twice Gâteaux differentiable at all points of the linear segment $[\mathbf{a}, \mathbf{a}+\mathbf{v}]$ relative to the pair $(\mathbf{v}, \mathbf{v})$, then there exists a number $\theta \in (0,1)$ such that*

$$f(\mathbf{a}+\mathbf{v}) = f(\mathbf{a}) + f'(\mathbf{a}\,;\mathbf{v}) + \frac{1}{2} f''(\mathbf{a}+\theta\mathbf{v}\,;\mathbf{v},\mathbf{v}). \tag{5.10}$$

Proof. Consider the function $g(t) = f(\mathbf{a}+t\mathbf{v})$, for $t \in [0,1]$. Its first derivative is

$$\begin{aligned} g'(t) &= \lim_{\varepsilon \to 0} \frac{g(t+\varepsilon) - g(t)}{\varepsilon} \\ &= \lim_{\varepsilon \to 0} \frac{f(\mathbf{a}+t\mathbf{v}+\varepsilon\mathbf{v}) - f(\mathbf{a}+t\mathbf{v})}{\varepsilon} = f'(\mathbf{a}+t\mathbf{v}\,;\mathbf{v}) \end{aligned}$$

and similarly we infer that $g''(t) = f''(\mathbf{a}+t\mathbf{v}\,;\mathbf{v},\mathbf{v})$. By usual Taylor's formula we infer the existence of a number $\theta \in (0,1)$ such that

$$g(1) = g(0) + g'(0) + \frac{1}{2} g''(\theta),$$

which in turn yields the formula (5.10). ∎

5.3.7 Corollary (The second derivative test of convexity) *Suppose that f is twice Gâteaux differentiable on an open convex subset U of the real Banach space E.*

(a) *If*

$$f''(\mathbf{x}\,;\mathbf{v},\mathbf{v}) \geq 0 \quad \text{for all } \mathbf{x} \in U,\ \mathbf{v} \in E, \tag{5.11}$$

then f is convex on U.

(b) *If the above inequality is strict for $\mathbf{v} \neq \mathbf{0}$, then f is strictly convex.*

Proof. Combine Taylor's formula (5.10) and Theorem 5.3.1. ∎

A practical second derivative test of convexity in the context of finite dimensional Banach spaces is as follows.

5.3.8 Corollary *Suppose that f is a twice Fréchet differentiable function defined on an open convex subset U of $\mathbb{R}^N$ (for example, this is the case if $f \in C^2(U)$). Then:*

(a) f is convex if and only if its Hessian matrix $\nabla^2 f(\mathbf{a})$ is positive at all points $\mathbf{a}$ of U (equivalently, if and only if all eigenvalues of $\nabla^2 f(\mathbf{a})$ are positive, whenever $\mathbf{a}$ belongs to U)

(b) f is strictly convex if its Hessian matrix $\nabla^2 f(\mathbf{a})$ is strictly positive at all points of U (equivalently, if all eigenvalues of $\nabla^2 f(\mathbf{a})$ are strictly positive, whenever $\mathbf{a}$ belongs to U).

Notice that in general it is not true that the Hessian matrix of a strictly convex function is strictly positive.

Proof. According to Corollary 5.3.7, the positivity (strict positivity) of the Hessian matrix of f at all points of U guarantees the convexity (strict convexity) of f. It remains to prove that the convexity of f implies that $\langle \nabla^2 f(\mathbf{a})\mathbf{v}, \mathbf{v}\rangle \geq 0$ for all $\mathbf{a} \in U$ and $\mathbf{v} \in \mathbb{R}^N$. For this, recall that convexity is equivalent to convexity along all lines. See Proposition 4.1.2. Therefore, for all $\mathbf{a} \in U$ and $\mathbf{v} \in \mathbb{R}^N$ the function $g(t) = f(\mathbf{a} + t\mathbf{v})$ is convex and of class C^2 on a suitable open interval centered at 0. According to Corollary 1.4.9, this happens if

$$g''(t) = \langle \nabla^2 f(\mathbf{a} + t\mathbf{v})\mathbf{v}, \mathbf{v}\rangle \geq 0$$

for all t around 0, in particular for $t = 0$. ■

In connection with Corollary 5.3.8 it is important to recall here the following result from linear algebra:

5.3.9 Theorem (Sylvester's criterion of positivity of a symmetric matrix) *Suppose that $A \in \mathrm{Sym}(N, \mathbb{R})$. Then:*

(a) *$A > 0$ if and only if all the leading principal minors of A are strictly positive;*

(b) *$A \geq 0$ if and only if all principal minors of A are positive.*

For details, see the book of C. D. Meyer [378], p. 559 and p. 566. Another simple proof of the assertion (a) makes the objective of Exercise 3, Section 6.4.

Simple examples show that the strict positivity of the Hessian matrix is a sufficient condition but not necessary for the strict convexity of a function. See Exercise 7.

In practice, handling second-order differentials is not always very comfortable and the methods exposed in sections 3.1–3.3 for recognizing convexity are much more convenient.

5.3.10 Example *For $A \in \mathrm{Sym}^{++}(N, \mathbb{R})$ and $\mathbf{b} \in \mathbb{R}^N$, the function*

$$F(\mathbf{x}) = \frac{1}{2}\langle A\mathbf{x}, \mathbf{x}\rangle - \langle \mathbf{x}, \mathbf{b}\rangle$$

is Fréchet differentiable of class C^∞ and its first two differentials are

$$\mathrm{d}F(\mathbf{x})\mathbf{v} = \langle A\mathbf{x}, \mathbf{v}\rangle - \langle \mathbf{v}, \mathbf{b}\rangle$$
$$\mathrm{d}^2F(\mathbf{x})(\mathbf{v}, \mathbf{v}) = \langle A\mathbf{v}, \mathbf{v}\rangle.$$

This implies that F is strictly convex (and even strongly convex, according to Theorem 5.5.2).

5.3.11 Example *According to Example 4.1.4, the following particular variant of the log-sum-exp function,*

$$f(x) = \log\left(\sum_{k=1}^{N} e^{x_k}\right)$$

is convex on $\mathbb{R}^N$. *An alternative argument based on Corollary* 5.3.8 *starts by noticing that this function has continuous partial derivatives of order* 2. *Its Hessian matrix* $\nabla^2 f(\mathbf{x})$ *equals*

$$\frac{1}{\left(\sum_{k=1}^N e^{x_k}\right)^2}\left[\left(\sum_{k=1}^N e^{x_k}\right)\operatorname{diag}\left(e^{x_1},...,e^{x_N}\right)-\begin{pmatrix} e^{x_1}\\ \vdots\\ e^{x_N}\end{pmatrix}\left(e^{x_1},...,e^{x_N}\right)\right],$$

so by taking into account the Cauchy–Bunyakovsky–Schwarz inequality we infer that

$$\left(\sum_{k=1}^N e^{x_k}\right)^2\cdot\langle\nabla^2 f(\mathbf{x})\mathbf{z},\mathbf{z}\rangle=\left(\sum_{k=1}^N e^{x_k}z_k^2\right)\left(\sum_{k=1}^N e^{x_k}\right)-\left(\sum_{k=1}^N e^{x_k}z_k\right)^2\geq 0,$$

for all $\mathbf{z}\in\mathbb{R}^N$. *Consequently the log-sum-exp function* f *is convex* (*and not strictly convex*).

5.3.12 Example *According to Example* 4.1.5, *the function* $f(A)=(\det A)^{1/N}$ *is concave on* $\operatorname{Sym}^+(N,\mathbb{R})$. *Since* f *is continuous and* $\operatorname{Sym}^{++}(N,\mathbb{R})$ *is a convex open set dense in* $\operatorname{Sym}^+(N,\mathbb{R})$, *one can deduce this property of* f *by applying Corollary* 5.3.8 *to its restriction to* $\operatorname{Sym}^{++}(N,\mathbb{R})$. *We have*

$$\mathrm{d}f(A)(X)=\frac{1}{N}(\det A)^{1/N}\operatorname{trace}(A^{-1}X)$$

and

$$\mathrm{d}^2 f(A)(X,Y)=\frac{1}{N^2}(\det A)^{1/N}\operatorname{trace}(A^{-1}X)\operatorname{trace}(A^{-1}Y)-\frac{1}{N}(\det A)^{1/N}\operatorname{trace}(A^{-1}XA^{-1}Y).$$

See J. R. Magnus and H. Neudecker [349] *for these (and many other) formulas from the matrix differential calculus. Therefore*

$$\begin{aligned}\mathrm{d}^2 f(A)(X,X)&=\frac{1}{N^2}(\det A)^{1/N}\left(\operatorname{trace}(A^{-1}X)\right)^2\\&\quad-\frac{1}{N}(\det A)^{1/N}\operatorname{trace}(A^{-1}XA^{-1}X)\\&=\frac{1}{N^2}(\det A)^{1/N}\left[\left(\operatorname{trace}(A^{-1/2}XA^{-1/2})\right)^2-N\operatorname{trace}\left((A^{-1/2}XA^{-1/2})^2\right)\right]\end{aligned}$$

and the fact that $\mathrm{d}^2 f(A)(X,X)\leq 0$ *for all* $A\in\operatorname{Sym}^{++}(N,\mathbb{R})$ *and* $X\in\operatorname{Sym}(N,\mathbb{R})$ *is now a consequence of the Cauchy–Bunyakovsky–Schwarz inequality applied to the inner product*

$$\langle A,B\rangle=\operatorname{trace}(AB),\qquad \text{for } A,B\in\operatorname{Sym}(N,\mathbb{R}).$$

Surprisingly, things are simpler in the case of the differentiable function $g(A) = \log\det(A)$ *(defined on* $\mathrm{Sym}^{++}(N,\mathbb{R})$*). Indeed, in this case*

$$\mathrm{d}g(A) = A^{-1} \quad \textit{and} \quad \mathrm{d}^2 g(A)(X,Y) = -\mathrm{trace}(A^{-1}XA^{-1}Y),$$

which yields $\mathrm{d}^2 g(A)(X,X) \le 0$ *for all* $X \in \mathrm{Sym}(N,\mathbb{R})$ *(and thus the fact that* g *is concave).*

5.3.13 Example *One can prove that the following function defined on* $\mathbb{R}^N_{++}$ *is convex:*

$$f(x_1,\dots,x_N) = \sum_{i=1}^{N} \frac{1}{x_i} - \sum_{1\le i<j\le N} \frac{1}{x_i+x_j} + \sum_{1\le i<j<k\le N} \frac{1}{x_i+x_j+x_k} - \cdots + (-1)^{N-1}\frac{1}{x_1+x_2+\cdots+x_N},$$

See J. M. Borwein, D. Bailey, and R. Girgensohn [85]*, p.* 36*, for details and history of this example. For* $N = 2$*, or* $N = 3$*, the Hessian method works well, but for large* N *this method is certainly not helpful.*

5.3.14 Remark *Differential criteria apply to functions defined on open sets. In the context of convex functions, they can be applied to the relative interior of the domain, taking into account the following simple consequence of the Accessibility Lemma (see Exercise* 6*, Section* 3.1*) : Suppose that* C *is a convex set in a Banach space* E *and* $f : C \to \mathbb{R}$ *is a continuous function. If* f *is convex on* $\mathrm{ri}(C)$*, then* f *is convex on* C*.*

Exercises

1. Prove the strict convexity on $\mathbb{R}^N$ of the function

$$f(x_1,\dots,x_n) = \sum_{1\le i<j\le N} c_{ij}(x_i - x_j)^2,$$

where the coefficients c_{ij} are strictly positive.

2. Consider the open set $A = \{(x,y,z) \in \mathbb{R}^3 : x,y > 0, xy > z^2\}$. Prove that A is convex and the function

$$f\colon A \to \mathbb{R}, \quad f(x,y,z) = \frac{1}{xy - z^2}$$

is strictly convex. Then, infer the inequality

$$\frac{8}{(x_1+x_2)(y_1+y_2) - (z_1+z_2)^2} < \frac{1}{x_1y_1 - z_1^2} + \frac{1}{x_2y_2 - z_2^2},$$

which works for every pair of distinct points (x_1,y_1,z_1) and (x_2,y_2,z_2) of the set A.

[*Hint*: Compute the Hessian of f.]

3. Use calculus to prove that the function

$$\Pi\colon (0,\infty)^n \to \mathbb{R}, \quad \Pi(x_1,\dots,x_n) = x_1^{\alpha_1}\cdots x_n^{\alpha_n}$$

is concave if $\alpha_1,...,\alpha_n > 0$ and $\alpha_1 + \cdots + \alpha_n \leq 1$. The corresponding Jensen's inequality provides a generalization of Minkowski's inequality for $p = 0$.

[*Hint*: Denote $H(x) = \nabla^2\Pi(\mathbf{x})/\Pi(\mathbf{x})$ and notice that for every vector $\mathbf{v} \in \mathbb{R}^n$ we have

$$\langle H(\mathbf{x})\mathbf{v},\mathbf{v}\rangle = \left(\sum_{j=1}^n \frac{\alpha_j v_j}{x_j}\right)^2 - \sum_{j=1}^n \frac{1}{\alpha_j}\left(\frac{\alpha_j v_j}{x_j}\right)^2 \leq 0,$$

according to the Cauchy–Bunyakovsky–Schwarz inequality.]

4. Prove that the function $f(x,y) = y^2/(1-|x|)$ is convex and bounded on the open unit disc $D_1(0) = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$.

[*Hint*: The function f can be represented as the maximum of two convex functions, $y^2/(1-x)$ and $y^2/(1+x)$.]

5. Let f be a continuous function on the closure of an open convex subset D of $\mathbb{R}^N$ and quasiconvex on the interior of D. Prove that f is quasiconvex on $\overline{D}$.

6. Suppose that f is continuously differentiable on an open convex subset D of $\mathbb{R}^N$. Prove that f is quasiconvex if, and only if,

$$f(\mathbf{y}) \leq f(\mathbf{x}) \text{ implies } \langle \nabla f(\mathbf{x}), \mathbf{y}-\mathbf{x}\rangle \leq 0.$$

Infer that the function $g(x,y) = -2x^2 - 2xy$ is quasiconvex on $\mathbb{R}^2_+$.

7. (J. Stoer and C. Witzgall [552], p. 152) There is no analogue of Theorem 5.2.3 for the second-order partial derivatives. Consider the function

$$f(x,y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2} + 13x^2 + 13y^2 & \text{if } (x,y) \in \mathbb{R}^2 \setminus \{(0,0)\} \\ 0 & \text{if } x = y = 0. \end{cases}$$

(a) Prove that f is convex.

(b) Prove that all second-order partial derivatives of f exist, but $\frac{\partial^2 f}{\partial x \partial y}(0,0)$ differs from $\frac{\partial^2 f}{\partial y \partial x}(0,0)$. According to Remark 5.3.4, f is not twice differentiable, despite the fact that its Hessian matrix exists.

5.4 Extrema of Convex Functions

Convex functions exhibit a series of nice properties concerning their extreme values, which make them successful in convex optimization. So is the following result.

5.4.1 Theorem *Suppose that E is a real Banach space E and $f\colon E \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper convex function. If a point $\mathbf{a}$ is a local minimizer for f, then it is also a global minimizer. In other words, if*

$$f(\mathbf{x}) \geq f(\mathbf{a}) \quad \textit{in a neighborhood of } \mathbf{a},$$

then

$$f(\mathbf{x}) \geq f(\mathbf{a}) \quad \textit{for all } \mathbf{x} \in E.$$

Moreover, the set of global minimizers of f is a convex set. If f is strictly convex, then this set is either empty or a singleton.

It is worth noticing that, according to Theorem 5.1.11, if f is a differentiable convex function on intdom f, then every point $\mathbf{a} \in$ intdom f that verifies the condition $f'(\mathbf{a}) = 0$ is a global minimizer.

Proof. Let $r > 0$ be such that $\mathbf{a}$ is a minimizer of $f|_{B_r(\mathbf{a}) \cap \text{dom } f}$ and let $\mathbf{x} \in$ dom f. Since dom f is convex, the line segment $[\mathbf{a}, \mathbf{x}]$ is contained in dom f and for some $\varepsilon \in (0, 1)$, the point $(1-\varepsilon)\mathbf{a} + \varepsilon\mathbf{x} = \mathbf{a} + \varepsilon(\mathbf{x} - \mathbf{a})$ must be in $B_r(\mathbf{a}) \cap$ dom f. By the convexity of f,

$$\begin{aligned} f(\mathbf{a}) &\leq f(\mathbf{a} + \varepsilon(\mathbf{x} - \mathbf{a})) = f((1-\varepsilon)\mathbf{a} + \varepsilon\mathbf{x}) \\ &\leq (1-\varepsilon)f(\mathbf{a}) + \varepsilon f(\mathbf{x}), \end{aligned} \tag{5.12}$$

which yields $f(\mathbf{a}) \leq f(\mathbf{x})$. Hence a is a global minimizer. If f is strictly convex in a neighborhood of $\mathbf{a}$, then the last inequality in (5.12) is strict and the conclusion becomes $f(\mathbf{x}) > f(\mathbf{a})$ for all $\mathbf{x} \in$ dom f, $\mathbf{x} \neq \mathbf{a}$. The convexity of the set of global minimizers of f is immediate. ■

It is customary in optimization theory to denote the problem of minimizing a function f (as in Theorem 5.4.1) by the symbol

$$\min_{\mathbf{x} \in E} f(\mathbf{x}) \tag{5.13}$$

and the fact that $\mathbf{a}$ is a (global) minimizer by

$$\mathbf{a} \in \arg\min_{\mathbf{x} \in E} f(\mathbf{x}).$$

The problem of minimizing f over a convex subset C of E, denoted

$$\min_{\mathbf{x} \in C} f(\mathbf{x}), \tag{5.14}$$

is called a problem of *constrained optimization* (to distinguish it from the problem (5.13), of *unconstrained optimization*).

Typically C is the set where a number of inequalities of the form

$$g_k(\mathbf{x}) \leq b_k, \quad k = 1, ..., m,$$

associated to appropriate convex functions, are fulfilled. See, for example, the case of linear programming, discussed in Section 8.4.

Using the indicator function, one can easily rewrite every problem of constrained optimization (5.14) into a problem of unconstrained minimization as follows,

$$\min_{\mathbf{x}\in E} \left(f(\mathbf{x}) + \iota_C(\mathbf{x})\right).$$

An archetype for the existence of a global minimum is the following classical result:

5.4.2 Theorem (K. Weierstrass) *Assume that $f\colon \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper function whose level sets are bounded. Then f has a global minimum.*

Proof. Choose arbitrarily $\bar{\mathbf{x}} \in \operatorname{dom} f$ and notice that

$$m = \inf\left\{f(\mathbf{x}) : \mathbf{x} \in \mathbb{R}^N\right\} = \inf\left\{f(\mathbf{x}) : \mathbf{x} \in \operatorname{level}_{f(\bar{\mathbf{x}})(f)}(f)\right\}.$$

According to our hypotheses, the level set $\operatorname{level}_{f(\bar{\mathbf{x}})}(f)$ is bounded and closed in $\mathbb{R}^N$ and thus compact. By the definition of m, there exists a sequence $(\mathbf{x}_n)_n$ in $\operatorname{level}_{f(\bar{\mathbf{x}})}(f)$ such that $f(\mathbf{x}_n) \to m$. Passing to a subsequence (if necessary) we may assume that $(\mathbf{x}_n)_n$ is convergent to a point $\mathbf{a}$. Necessarily, $\mathbf{a} \in \operatorname{level}_{f(\bar{\mathbf{x}})}(f)$. Since f is lower semicontinuous, we have

$$f(\mathbf{a}) \leq \liminf_{n\to\infty} f(\mathbf{x}_n) = m,$$

which shows that f has a global minimum at $\mathbf{a}$. ■ The condition of having bounded level sets is a growth condition.

5.4.3 Lemma *Let E be a Banach space and $f : E \to \mathbb{R} \cup \{\infty\}$ be a lower semicontinuous proper convex function. Then, the following conditions are equivalent:*

(a) *all levels sets of f are bounded;*

(b) *the function f is coercive in the sense that*

$$\lim_{\|\mathbf{x}\|\to\infty} f(\mathbf{x}) = \infty; \tag{5.15}$$

(c) *there exist $\alpha > 0$ and $\beta \in \mathbb{R}$ such that $f(\mathbf{x}) \geq \alpha \|\mathbf{x}\| + \beta$ for all $\mathbf{x} \in E$;*

(d) *we have*

$$\liminf_{\|\mathbf{x}\|\to\infty} \frac{f(\mathbf{x})}{\|\mathbf{x}\|} > 0.$$

Proof. The only nontrivial implication is (b) ⇒ (c) This will be done noticing that we may assume (replacing f by a translate of it) that $\mathbf{0} \in \operatorname{dom} f$ and $f(\mathbf{0}) = 0$. Since f is coercive, there is a number $R > 0$ such that $f(\mathbf{x}) \geq 1$ if $\|\mathbf{x}\| \geq R$. Then for all such $\mathbf{x}$ we have

$$1 \leq f\left(\frac{R\mathbf{x}}{\|\mathbf{x}\|}\right) \leq \frac{R}{\|\mathbf{x}\|} f(\mathbf{x}) + \left(1 - \frac{R}{\|\mathbf{x}\|}\right) f(\mathbf{0}) = \frac{R}{\|\mathbf{x}\|} f(\mathbf{x}),$$

whence $f(\mathbf{x}) \geq \|\mathbf{x}\| / R$. Since f is lower semicontinuous at the origin, the set $\{\mathbf{x} : f(\mathbf{x}) > -1\}$ is open and thus there is $\varepsilon \in (0, R)$ such that $\|\mathbf{x}\| \leq \varepsilon$ implies $f(\mathbf{x}) > -1$. Then for every $\mathbf{x}$ with $\|\mathbf{x}\| \leq R$ we have

$$-1 < f\left(\frac{\varepsilon}{R}\mathbf{x}\right) \leq \frac{\varepsilon}{R} f(\mathbf{x}) + \left(1 - \frac{\varepsilon}{R}\right) f(\mathbf{0}) = \frac{\varepsilon}{R} f(\mathbf{x}).$$

Therefore $f(\mathbf{x}) \geq \|\mathbf{x}\| / R - 1 - R/\varepsilon$ for all $\mathbf{x}$, which shows that assertion (c) works for $\alpha = 1/R$ and $\beta = -1 - R/\varepsilon$. ■

Combining Lemma 5.4.3 with Theorem 5.4.1 and Theorem 5.4.2 one obtains the following important consequences concerning the minimization of convex functions defined on an Euclidean space.

5.4.4 Corollary *Every lower semicontinuous, proper, convex and coercive function $f : \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ has a global minimizer.*

5.4.5 Corollary *If $f \colon \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper convex function, then for every $\mathbf{y} \in \mathbb{R}^N$ there is $\mathbf{a} \in \mathbb{R}^N$ such that $\mathbf{y} \in (I + \partial f)(\mathbf{a})$.*

Proof. The function

$$g \colon \mathbf{x} \to f(\mathbf{x}) + \frac{1}{2}\|\mathbf{x}\|^2 - \langle \mathbf{x}, \mathbf{y} \rangle$$

is lower semicontinuous, strictly convex, and coercive, so it admits a unique minimizer $\mathbf{x} = \mathbf{a}$. Then $\mathbf{0} \in \partial g(\mathbf{a})$, whence $\mathbf{y} \in (\partial f + I)(\mathbf{a})$ since $\partial(f + \|\cdot\|^2/2) = \partial f + I$. ■

A nice consequence of Theorem 5.4.2 and Lemma 5.4.3 is the existence of "approximate" critical points:

5.4.6 Proposition *If $f : \mathbb{R}^N \to \mathbb{R}$ is differentiable and bounded below, then for every $\varepsilon > 0$ there exist points $\mathbf{x}_\varepsilon$ such that $\|\nabla f(\mathbf{x}_\varepsilon)\| \leq \varepsilon$.*

Proof. The function $f + \varepsilon\|\mathbf{x}\|$ is coercive, so by Theorem 5.4.2 it has a global minimizer $\mathbf{x}_\varepsilon$. We will show that $\mathbf{v} = \nabla f(\mathbf{x}_\varepsilon)$ verifies $\|\mathbf{v}\| \leq \varepsilon$. Indeed, assuming that $\|\mathbf{v}\| > \varepsilon$, the formula

$$\lim_{t \to 0+} \frac{f(\mathbf{x}_\varepsilon - t\mathbf{v}) - f(\mathbf{x}_\varepsilon)}{t} = -\langle \nabla f(\mathbf{x}_\varepsilon), \mathbf{v} \rangle = -\|\mathbf{v}\|^2 < -\varepsilon\|\mathbf{v}\|$$

implies for $t > 0$ small enough that

$$\begin{aligned} -t\varepsilon \|\mathbf{v}\| &> f(\mathbf{x}_\varepsilon - t\mathbf{v}) - f(\mathbf{x}_\varepsilon) \\ &= [f(\mathbf{x}_\varepsilon - t\mathbf{v}) + \varepsilon \|\mathbf{x}_\varepsilon - tv\|] - [f(\mathbf{x}_\varepsilon) + \varepsilon \|\mathbf{x}_\varepsilon\|] \\ &+ \varepsilon [\|\mathbf{x}_\varepsilon\| - \|\mathbf{x}_\varepsilon - t\mathbf{v}\|] \\ &> -\varepsilon t \|\mathbf{v}\|, \end{aligned}$$

which is a contradiction. Therefore, $\|\mathbf{v}\| \leq \varepsilon$ and the proof is complete. ■

Given a finite family of vectors $\mathbf{v}_0, ..., \mathbf{v}_m$ in $\mathbb{R}^N$, their convex hull may contain the origin or not. Using the separation Theorem 3.3.2, this dichotomy can be reformulated as follows: *Either*

$$\textit{there exist } \lambda_0, ..., \lambda_m \geq 0 \textit{ such that } \sum_{k=0}^{m} \lambda_k = 1 \textit{ and } \sum_{k=0}^{m} \lambda_k \mathbf{v}_k = 0, \tag{5.16}$$

or

$$\textit{there exists } \mathbf{x} \in \mathbb{R}^N \textit{ such that } \langle \mathbf{v}_k, \mathbf{x} \rangle < 0 \quad \textit{for } k = 0, ..., m. \tag{5.17}$$

Taking into account Proposition 5.4.6, one obtains the following surprising consequence:

5.4.7 Corollary *If* $\mathbf{v}, \mathbf{v}_0, ..., \mathbf{v}_m \in \mathbb{R}^N$, *then* $\mathbf{v} \in conv\{\mathbf{v}_0, ..., \mathbf{v}_m\}$ *if and only if the function*

$$f(\mathbf{x}) = \log \left(\sum_{k=0}^{m} \mathrm{e}^{\langle \mathbf{v}_k - \mathbf{v}, \mathbf{x} \rangle} \right)$$

is bounded below on $\mathbb{R}^N$.

Proof. Replacing $\mathbf{v}_0, ..., \mathbf{v}_m$ by $\mathbf{v}_0 - \mathbf{v}, ..., \mathbf{v}_m - \mathbf{v}$, we may restrict ourselves to the case $\mathbf{v} = 0$. If $\mathbf{0} \in \operatorname{conv}\{\mathbf{v}_0, ..., \mathbf{v}_m\}$, then the alternative (5.17) fails and clearly the function f is bounded below.

Conversely, if the function f is bounded below, then Proposition 5.4.6 applies, which yields the existence of a sequence $(\mathbf{x}_n)_n$ in $\mathbb{R}^N$ such that $\|\nabla f(\mathbf{x}_n)\| \leq 1/n$ for all n. We have

$$\nabla f(\mathbf{x}_n) = \sum_{k=0}^{m} \lambda_{nk} \mathbf{v}_k,$$

where the scalars

$$\lambda_{nk} = \mathrm{e}^{\langle \mathbf{v}_k, \mathbf{x}_n \rangle} / \sum_{k=0}^{m} \mathrm{e}^{\langle \mathbf{v}_k, \mathbf{x}_n \rangle}$$

are positive and verify $\sum_{k=0}^{m} \lambda_{nk} = 1$. The sequences $(\lambda_{n0})_n, ..., (\lambda_{nm})_n$ are bounded so, by the Bolzano–Weierstrass theorem, we may assume that the limits $\lim_{n\to\infty} \lambda_{nk} = \lambda_k$ exist in $\mathbb{R}$ for $k = 0, ..., m$. This provides a solution for the system (5.16) and the proof is complete. ■

Corollary 5.4.7 can be used to prove Birkhoff's theorem 3.3.6. See the book of J. M. Borwein and Q. J. Zhu [92], Theorem 2.4.10, pp. 25–26.

A stronger condition than coercivity is *supercoercivity*, which means that

$$\lim_{\|\mathbf{x}\|\to\infty} \frac{f(\mathbf{x})}{\|\mathbf{x}\|} = \infty. \tag{5.18}$$

5.4.8 Proposition *Every lower semicontinuous, proper, strongly convex function f defined on a real Hilbert space H is supercoercive* (*and thus has exactly one minimizer in H*).

Proof. Indeed, if f is strongly convex with parameter $\alpha > 0$, then $g(\mathbf{x}) = f(\mathbf{x}) - \frac{\alpha}{2}\|\mathbf{x}\|^2$ is a lower semicontinuous, proper convex function. According to Theorem 4.3.1, there is $\mathbf{x}_0 \in H$ such that $\partial g(\mathbf{x}_0) \neq \emptyset$. Therefore, for $\mathbf{z}$ arbitrarily fixed in $\partial g(\mathbf{x}_0)$, we have

$$f(\mathbf{x}) \geq f(\mathbf{x}_0) + \frac{\alpha}{2}\left(\|\mathbf{x}\|^2 - \|\mathbf{x}_0\|^2\right) + \langle \mathbf{x} - \mathbf{x}_0, \mathbf{z}\rangle,$$

whenever $\mathbf{x} \in H$. By the Cauchy–Bunyakovsky–Schwarz inequality, $\langle \mathbf{x}-\mathbf{x}_0, \mathbf{z}\rangle \geq -\|\mathbf{x}\|\,\|\mathbf{z}\| - \langle \mathbf{x}_0, \mathbf{z}\rangle$, and the fact that $\lim_{\|\mathbf{x}\|\to\infty} f(\mathbf{x})/\|\mathbf{x}\| = \infty$ is now clear. ■

The converse of Proposition 5.4.8 is not true. For example, the function $f(x) = \left(x^2 - 1\right)^+$ is convex, continuous and supercoercive on $\mathbb{R}$, but it is not strongly convex.

If $A \in \mathrm{Sym}^{++}(N, \mathbb{R})$ and $\mathbf{b} \in \mathbb{R}^N$, then the function

$$F(\mathbf{x}) = \frac{1}{2}\langle A\mathbf{x}, \mathbf{x}\rangle - \langle \mathbf{x}, \mathbf{b}\rangle$$

is strongly convex and differentiable, of class C^∞ on $\mathbb{R}^N$. By Proposition 5.4.8, it is also supercoercive and admits a (unique) global minimizer $\bar{\mathbf{x}}$. Necessarily,

$$\nabla F(\bar{\mathbf{x}}) = A\bar{\mathbf{x}} - \mathbf{b} = 0,$$

and the subgradient inequality assures the sufficiency of this condition. Therefore

$$\bar{\mathbf{x}} = \arg\min_{\mathbf{x}\in\mathbb{R}^N} \left[\frac{1}{2}\langle A\mathbf{x}, \mathbf{x}\rangle - \langle \mathbf{x}, \mathbf{b}\rangle\right] \iff A\bar{\mathbf{x}} = \mathbf{b}.$$

The above reasoning suggests the valuable idea to solve certain systems of equations by finding the minimum of suitable functionals. By putting this idea in the general framework of Hilbert spaces, it constitutes the basis of a very useful tool in partial differential equations. See Appendix E.

5.4.9 Remark (Barycenters as minimizers) *Suppose that C is a convex set in $\mathbb{R}^N$. The barycenter of a probability measure $\mu \in \mathcal{P}^1(C)$ was defined in Section* 4.6 *as the point*

$$\mathrm{bar}(\mu) = \int_C \mathbf{x}\mathrm{d}\mu(\mathbf{x}).$$

We will show that the barycenter can also be characterized as the unique minimizer of the strongly convex continuous function

$$F(\mathbf{x}) = F_{\mathbf{y}}(\mathbf{x}) = \int_C \left[\|\mathbf{x}-\mathbf{u}\|^2 - \|\mathbf{y}-\mathbf{u}\|^2 \right] \mathrm{d}\mu(\mathbf{u}), \quad \mathbf{x} \in \mathbb{R}^N, \tag{5.19}$$

where $\mathbf{y} \in \mathbb{R}^N$ *is an arbitrarily fixed parameter.*

Notice first that F *is well defined. Indeed, the integrand is continuous and its absolute value is majorized by a* μ*-integrable function since*

$$\begin{aligned} \left| \|\mathbf{x}-\mathbf{u}\|^2 - \|\mathbf{y}-\mathbf{u}\|^2 \right| &\le |\|\mathbf{x}-\mathbf{u}\| - \|\mathbf{y}-\mathbf{u}\|| \, |\|\mathbf{x}-\mathbf{u}\| + \|\mathbf{y}-\mathbf{u}\|| \\ &\le \|\mathbf{x}-\mathbf{y}\| \left(\|\mathbf{x}-\mathbf{u}\| + \|\mathbf{y}-u\| \right) \\ &\le \|\mathbf{x}-\mathbf{y}\| \left(\|\mathbf{x}\| + \|\mathbf{y}\| + 2\|\mathbf{u}\| \right). \end{aligned}$$

The function $F(\mathbf{x})$ *is continuous, being Lipschitz on every bounded ball.*

Let $\mathbf{x}', \mathbf{x}'' \in C$ *and* $\lambda \in [0,1]$*. Since the function* $\mathbf{u} \to \|\mathbf{x}-\mathbf{u}\|^2$ *is* 1*-strongly convex, an easy computation yields*

$$\begin{aligned} F((1-\lambda)\mathbf{x}' + \lambda\mathbf{x}'') &= \textstyle\int_C \left[\|(1-\lambda)\mathbf{x}' + \lambda\mathbf{x}'' - \mathbf{u}\|^2 - \|\mathbf{y}-\mathbf{u}\|^2 \right] \mathrm{d}\mu(\mathbf{u}) \\ &\le (1-\lambda) \textstyle\int_C \left[\|\mathbf{x}'-\mathbf{u}\|^2 - \|\mathbf{y}-\mathbf{u}\|^2 \right] \mathrm{d}\mu(\mathbf{u}) \\ &\quad + \lambda \textstyle\int_C \left[\|\mathbf{x}''-u\|^2 - \|\mathbf{y}-\mathbf{u}\|^2 \right] \mathrm{d}\mu(\mathbf{u}) \\ &\quad - \tfrac{1}{2}\lambda(1-\lambda) \|\mathbf{x}'-\mathbf{x}''\|^2 \\ &= (1-\lambda)F(\mathbf{x}') + \lambda F(\mathbf{x}'') - \tfrac{1}{2}\lambda(1-\lambda) \|\mathbf{x}'-\mathbf{x}''\|^2, \end{aligned}$$

so $F(\mathbf{x})$ *is also* 1*-strongly convex. By Proposition* 5.4.8*, the function* $F(\mathbf{x})$ *admits a unique minimizer, say* $\mathbf{z}_\mu$*. Since*

$$\begin{aligned} F'(\mathbf{x})(\mathbf{v}) &= \lim_{t\to 0} \frac{F(\mathbf{x}+t\mathbf{v}) - F(\mathbf{x})}{t} = 2\int_C \langle \mathbf{x}-\mathbf{u}, \mathbf{v}\rangle \mathrm{d}\mu(\mathbf{u}) \\ &= 2\langle \mathbf{x} - \mathrm{bar}(\mu), \mathbf{v}\rangle, \end{aligned}$$

for all $\mathbf{x}$ *and* $\mathbf{v}$ *in* $\mathbb{R}^N$*, we infer that* F *is differentiable and* $\nabla F(\mathbf{x}) = \mathbf{x} - \mathrm{bar}(\mu)$*. According to Theorem* 5.1.11*,* $\mathbf{z}_\mu$ *is a solution of the equation* $\nabla F(\mathbf{x}) = 0$*, whence*

$$\mathbf{z}_\mu = \mathrm{bar}(\mu).$$

The discussion above shows that $F_{\mathrm{bar}(\mu)}(\mathbf{x}) \ge 0$*, which in turn yields*

$$\int_C \|\mathbf{x}-\mathbf{u}\|^2 \mathrm{d}\mu(\mathbf{u}) \ge \int_C \|\mathrm{bar}(\mu)-\mathbf{u}\|^2 \mathrm{d}\mu(\mathbf{u})$$

for all $\mathbf{x} \in \mathbb{R}^N$*. When* μ *represents a mass distribution and* $\|\mathbf{x}-\mathbf{u}\|^2$ *represents the transportation cost per unit of mass from* $\mathbf{u}$ *to* $\mathbf{x}$*, then the last inequality*

asserts that the transportation cost for moving the whole mass at a point $\mathbf{x}$ *is minimal when* $\mathbf{x} = \text{bar}(\mu)$.

Another simple example illustrating how supercoercivity yields the existence of solutions of certain nonlinear equations is offered by Exercise 1.

The main ingredient of Theorem 5.4.2 is the fact that all bounded and closed sets in $\mathbb{R}^N$ are compact sets. This does not work in the context of infinite-dimensional Banach spaces, but the reflexive Banach spaces exhibit a valuable substitute, the sequential weak compactness of their bounded closed and convex subsets. See Appendix E for applications to calculus of variations.

Some of the results above concerning the minimization of convex functions can be extended to the case of quasiconvex functions defined on reflexive Banach spaces. See Exercise 7.

Convex functions attain their maxima at the relative boundary:

5.4.10 Theorem (The maximum principle) *Let* U *be a convex subset of* $\mathbb{R}^N$. *If a convex function* $f\colon U \to \mathbb{R}$ *attains its maximum on* U *at a point* $\mathbf{a}$ *from the relative interior of* U, *then* f *is constant on* U.

Proof. We should prove that $f(\mathbf{x}) = f(\mathbf{a})$ for $\mathbf{x} \in U$, $\mathbf{x} \neq \mathbf{a}$. Since $\mathbf{a} \in \text{ri}U$, there exists a point $\mathbf{x}' \in U$ such that $\mathbf{a}$ is an interior point of the segment $[\mathbf{x}, \mathbf{x}']$, that is, $\mathbf{a} = (1-\lambda)\mathbf{x} + \lambda\mathbf{x}'$ for some $\lambda \in (0,1)$. From the definition of convexity,

$$f(\mathbf{a}) \leq (1-\lambda)f(\mathbf{x}) + \lambda f(\mathbf{x}').$$

Since both $f(\mathbf{x})$ and $f(\mathbf{x}')$ do not exceed $f(\mathbf{a})$ and both weights $1-\lambda$ and λ are strictly positive, the last inequality forces $f(\mathbf{a}) = f(\mathbf{x}) = f(\mathbf{x}')$. The proof is complete. ■

A generalization of Theorem 5.4.10, known as the Bauer maximum principle, makes the object of Corollary B.3.4, Appendix B.

We end this section with an important consequence of Minkowski's Theorem 3.3.5.

5.4.11 Theorem *If* f *is a continuous convex function on a compact convex subset* K *of* $\mathbb{R}^N$, *then* f *attains a global maximum at an extreme point.*

Proof. Assume that f attains its global maximum at $\mathbf{a} \in K$. By Theorem 3.3.5, the point $\mathbf{a}$ can be represented as a convex combination of extreme points, say $\mathbf{a} = \sum_{k=1}^{m} \lambda_k \mathbf{e}_k$. Then $f(\mathbf{a}) \leq \sum_{k=1}^{m} \lambda_k f(\mathbf{e}_k) \leq \sup_k f(\mathbf{e}_k)$, which forces $f(\mathbf{a}) = f(\mathbf{e}_k)$ for some k. ■

For functions defined on N-dimensional intervals $[a_1, b_1] \times \cdots \times [a_N, b_N]$ in $\mathbb{R}^N$, Theorem 5.4.11 extends to the case of continuous functions which are convex in each variable (when the others are kept fixed). This fact can be proved by one-variable means (taking into account that convexity is equivalent to convexity on segment lines).

In the infinite-dimensional setting, it is difficult to state fairly general results on maximum attaining. Besides, the deep results of Banach space theory appear

to be crucial in answering questions which at first glance may look simple. Here is an example. By the Alaoglu–Bourbaki theorem (see Theorem B.1.6, Appendix B) it follows that each continuous linear functional on a reflexive Banach space E achieves its norm on the unit ball. Surprisingly, these are the *only* Banach spaces for which the norm-attaining phenomenon occurs. This was proved by R. C. James [272].

Exercises

1. (Surjectivity of derivatives) Suppose that $f : \mathbb{R}^N \to \mathbb{R}$ is a Fréchet differentiable supercoercive function. Prove that its differential $df : \mathbb{R}^N \to \mathbb{R}^N$, defined by $df(\mathbf{x})(\mathbf{v}) = \langle \nabla f(\mathbf{x}), \mathbf{v} \rangle$ for all $\mathbf{v} \in \mathbb{R}^N$, is onto.

   [*Hint:* For $\mathbf{z} \in \mathbb{R}^N$ arbitrarily fixed, the function $g(\mathbf{x}) = f(\mathbf{x}) - \langle \mathbf{x}, \mathbf{z} \rangle$ admits a global minimum.]

2. (First order condition of minimization) Let C be a nonempty convex subset of $\mathbb{R}^N$ and let $f : C \to \mathbb{R}$ be a convex and differentiable function. Prove that $\bar{\mathbf{x}}$ is a minimizer of f on C if and only if

   $$\langle \nabla f(\bar{\mathbf{x}}), \mathbf{x} - \bar{\mathbf{x}} \rangle \geq 0 \text{ for all } \mathbf{x} \in C.$$

   Infer that the orthogonal projection $P_C(\mathbf{x})$ of a point $\mathbf{x} \in \mathbb{R}^N$ onto a nonempty closed convex set C in $\mathbb{R}^N$ verifies the condition

   $$\langle \mathbf{x} - P_C(\mathbf{x}), \mathbf{y} - P_C(\mathbf{x}) \rangle \leq 0 \quad \text{for all } \mathbf{y} \in C.$$

3. Suppose that $f : E \to \mathbb{R}$ is convex and C is a nonempty subset of E. Prove that

   $$\sup_{\mathbf{x} \in C} f(\mathbf{x}) = \sup_{\mathbf{x} \in \operatorname{conv}(C)} f(\mathbf{x}).$$

4. Suppose that E is a smooth Banach space. Prove that the Gâteaux differential of the norm function $\mathrm{N}(\mathbf{x}) = \|\mathbf{x}\|$ at $\mathbf{x} \neq 0$ is a norm-1 functional that verifies the formula

   $$\mathrm{N}'(\mathbf{x})(\mathbf{x}) = \|\mathbf{x}\| = \sup \{x^*(\mathbf{x}) : x^* \in E^*, \, \|x^*\| = 1\}.$$

5. Suppose that E is a uniformly convex Banach space and $x^* \in E^*$, $x^* \neq 0$. Prove that x^* attains its norm at an element $\mathbf{u} \in E$ with $\|\mathbf{u}\| = 1$.

   Remark. According to the preceding exercise, the element $\mathbf{u}$ is unique if E is also a smooth space.

6. (J. J. Moreau) Let H be a Hilbert space, $\mathbf{z}$ a point in H and $f : H \to (-\infty, \infty]$ a lower semicontinuous proper convex function. Prove that the function $F(\mathbf{x}) = \frac{1}{2} \|x - z\|^2 + f(\mathbf{x})$ has a strict minimum.

   Remark. The unique point $\mathbf{x} = \operatorname{prox}_f(\mathbf{z})$ where F attains its minimum is known as the *proximal point to* $\mathbf{z}$, *relative to* f. For f the indicator function of a closed convex set C, $\operatorname{prox}_f(\mathbf{z})$ is just the projection of $\mathbf{z}$ onto C.

7. (Minimum of quasiconvex functions) Let E be a real Banach space.

   (a) Prove that every lower semicontinuous quasiconvex proper function $f : E \to \mathbb{R} \cup \{\infty\}$ is also (sequentially) weakly lower semicontinuous, that is,

   $$\mathbf{u}_n \to \mathbf{u} \text{ weakly in } E \text{ implies } f(\mathbf{u}) \leq \liminf_{n\to\infty} f(\mathbf{u}_n).$$

   (b) Suppose in addition that E is an L^p-space (or, more generally, a reflexive Banach space) and that $f : E \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous, quasiconvex, coercive, and proper function. Prove that f attains its infimum.

   (c) A function $f : E \to \mathbb{R} \cup \{\infty\}$ is said to be *semistrictly quasiconvex* if it is quasiconvex and for every $\lambda \in (0,1)$ and every $\mathbf{x}_0, \mathbf{x}_1 \in E$ with $f(\mathbf{x}_0) \neq f(\mathbf{x}_1)$ we have $f((1-\lambda)\mathbf{x}_0 + \lambda\mathbf{x}_1) < \max\{f(\mathbf{x}_0), f(\mathbf{x}_1)\}$. Prove that every superposition $g \circ f$ of a strictly increasing function $g : \mathbb{R} \to \mathbb{R}$ and a strictly convex function $f : E \to \mathbb{R}$ is semistrictly quasiconvex. Indicate concrete examples when $E = \mathbb{R}^N$.

   (d) Prove that for every semistrictly quasiconvex function any local minimizer is also a global minimizer.

   (e) Give an example of quasiconvex function on $\mathbb{R}$ for which not every local minimizer is a global minimizer.

   [*Hint:* For (a), see Corollary B.2.6, Appendix B. For (b), adapt the argument of Theorem 5.4.2 by using the Eberlein–Šmulyan compactness theorem (Theorem B.1.8, Appendix B).]

   The following set of exercises concerns the constrained optimization.

8. Suppose that C is a convex subset of $\mathbb{R}^N$ and $\mathbf{a} \in C$. The *tangent cone* to C at $\mathbf{a}$ is the closed cone $T_C(\mathbf{a}) = \overline{\mathbb{R}_+(C-\mathbf{a})}$, and the *normal cone* to C at $\mathbf{a}$ is the closed cone $N_C(\mathbf{a}) = \{\mathbf{v} \in \mathbb{R}^N : \langle \mathbf{v}, \mathbf{x}-\mathbf{a}\rangle \leq 0 \text{ for all } \mathbf{x} \in C\}$.

   (a) Compute $T_C(\mathbf{a})$ and $N_C(\mathbf{a})$ when C is the unit disc.

   (b) Prove that the polar set of $T_C(\mathbf{a})$ is $N_C(\mathbf{a})$ (and vice-versa).

9. Let $f\colon \mathbb{R}^N \to \mathbb{R}$ be a convex function and $C = \{\mathbf{x} : f(\mathbf{x}) \leq 0\}$. Assume that there exists a point $\mathbf{x}$ such that $f(\mathbf{x}) < 0$. Prove that

   $$T_C(\mathbf{a}) = \{\mathbf{v} : f'(\mathbf{a};\mathbf{v}) \leq 0\} \quad \text{and} \quad N_C(\mathbf{a}) = \mathbb{R}_+ \partial f(\mathbf{a})$$

   for all $\mathbf{a} \in \mathbb{R}^N$ such that $f(\mathbf{a}) = 0$.

10. (Self-dual cones) Suppose that C is one of the following cones: $\mathbb{R}^N$, $\mathrm{Sym}^+(n,\mathbb{R})$ or $\{\mathbf{x} \in \mathbb{R}^N : x_1^2 \geq x_2^2 + \cdots + x_n^2\}$. Prove that $N_C(\mathbf{0}) = -C$ in all these cases.

11. Suppose that C is a convex subset of $\mathbb{R}^N$ and that $f\colon \mathbb{R}^N \to \mathbb{R}$ is a convex function. Prove that the following assertions are equivalent for $\mathbf{a} \in C$:

    (a) $\mathbf{a}$ is a minimizer for $f|_C$;

(b) $f'(\mathbf{x};\mathbf{v}) \geq 0$ for all $\mathbf{v} \in T_C(\mathbf{a})$;

(c) $\mathbf{0} \in \partial f(\mathbf{a}) + N_C(\mathbf{a})$.

5.5 Special Classes of Functions and the Gradient Descent Algorithm

Several variants of convexity play prominent roles in convex optimization, calculus of variations, isoperimetric inequalities, mass transportation, etc. A way to bring it all together is to consider the class of perturbed convex functions.

Let $\omega : [0,\infty) \to \mathbb{R}$ be a function referred to as a *modulus of perturbation.*

A real-valued function f defined on a convex subset U of $\mathbb{R}^N$ is called a *perturbed convex function with modulus* ω (abbreviated, ω*-convex function*) if it verifies an estimate of the form

$$f((1-\lambda)\mathbf{x}+\lambda\mathbf{y}) \leq (1-\lambda)f(\mathbf{x}) + \lambda f(\mathbf{y}) - \lambda(1-\lambda)\omega\left(\|\mathbf{x}-\mathbf{y}\|\right), \tag{5.20}$$

for all $\mathbf{x},\mathbf{y}$ in U and $\lambda \in (0,1)$. The usual convex functions represent the particular case when ω is identically 0. Every ω-convex function associated to a modulus $\omega \geq 0$ is necessarily convex. When ω is allowed to take negative values, there are ω-convex functions which are not convex. See the case of semiconvex functions, described below.

The ω-convex functions whose moduli ω are strictly positive except at the origin (where $\omega(0) = 0$) are usually called *uniformly convex.* In their case the inequality (5.20) is strict whenever $\mathbf{x} \neq \mathbf{y}$ and $\lambda \in (0,1)$. For more information see D. Azé and J.-P. Penot [34], J. M. Borwein et al. [87], [91] and C. Zălinescu [597], [598].

By changing f to $-f$ one obtains the notions of ω*-concave function* and *uniformly concave function.*

For $\omega(t) = (\alpha/2)t^2$, $\alpha > 0$, we retrieve the case of α*-strongly convex* functions (that is, strongly convex with parameter α). See Definition 4.1.1. As was noticed in Section 4.1, a function f is α-strongly convex if and only if the function $f - (\alpha/2)\|\cdot\|^2$ is still convex.

The case when $\omega(t) = -(\alpha/2)t^2$ $(\alpha > 0)$ is that of α*-semiconvex functions* (that is, semiconvex with parameter $\alpha > 0$). A function f is α*-semiconvex* if and only if it becomes convex after the addition of $(\alpha/2)\|\cdot\|^2$, equivalently, if it verifies the inequality

$$f((1-\lambda)\mathbf{x}+\lambda\mathbf{y}) \leq (1-\lambda)f(\mathbf{x}) + \lambda f(\mathbf{y}) + \frac{\alpha}{2}\lambda(1-\lambda)\|\mathbf{x}-\mathbf{y}\|^2, \tag{5.21}$$

for all $\mathbf{x},\mathbf{y}$ in U and $\lambda \in (0,1)$. By changing f to $-f$ one obtains the notion of α*-semiconcave function.*

Under the presence of differentiability, each of the above classes of functions generates specific gradient inequalities.

5.5.1 Lemma *Suppose that U is an open convex subset of $\mathbb{R}^N$ and $f : U \to \mathbb{R}$ is a Gâteaux differentiable ω-convex function. Then*

$$f(\mathbf{x}) \geq f(\mathbf{a}) + \langle \nabla f(\mathbf{a}), \mathbf{x} - \mathbf{a} \rangle + \omega\left(\|\mathbf{x} - \mathbf{a}\|\right)$$

for all points $\mathbf{a} \in U$ and $\mathbf{x} \in U$.

Proof. Indeed, according to the estimate (5.20),

$$f(\mathbf{x}) - f(\mathbf{a}) \geq \frac{f(\mathbf{a} + \lambda(\mathbf{x} - \mathbf{a})) - f(\mathbf{a})}{\lambda} + (1 - \lambda)\omega\left(\|\mathbf{x} - \mathbf{a}\|\right)$$

for all $\lambda \in (0, 1]$, and the conclusion follows by passing to the limit as $\lambda \to 0^+$. ■

The next result collects several useful characterizations of strongly convex functions (that can be easily turned into characterizations of semiconvex functions).

5.5.2 Theorem *Suppose that f is a differentiable function defined on an open convex set U in $\mathbb{R}^N$ and α is a strictly positive number. Then the following assertions are equivalent:*

(a) f is strongly convex with parameter $\alpha > 0$;
(b) $f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\alpha}{2} \|\mathbf{x} - \mathbf{y}\|^2$ for all $\mathbf{x}, \mathbf{y} \in U$;
(c) $\langle \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq \alpha \|\mathbf{x} - \mathbf{y}\|^2$ for all $\mathbf{x}, \mathbf{y} \in U$.

If f is a twice continuously differentiable function, then f is α-strongly convex if and only if

$$\langle \nabla^2 f(\mathbf{x})(\mathbf{v}), \mathbf{v} \rangle \geq \alpha \|\mathbf{v}\|^2 \qquad \text{for all } \mathbf{x} \in U \text{ and } \mathbf{v} \in \mathbb{R}^N.$$

Proof. The implication (a) $\Rightarrow$ (b) follows from Lemma 5.5.1.

(b) $\Rightarrow$ (c) Add side by side the inequality that makes the objective of assertion (b) and the inequality obtained by interchanging the roles of $\mathbf{x}$ and $\mathbf{y}$. For the implication (c) $\Rightarrow$ (a), notice that

$$\begin{aligned}\langle (\nabla f(\mathbf{x}) - \alpha \mathbf{x}) - (\nabla f(\mathbf{y}) - \alpha \mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \\ = \langle \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle - \alpha \|\mathbf{x} - \mathbf{y}\|^2 \geq 0,\end{aligned}$$

which, according to Theorem 5.3.3, implies that $f - \frac{\alpha}{2} \|\cdot\|^2$ is convex.

When f is twice continuously differentiable, the fact that $\nabla^2 f \geq \alpha I$ implies the assertions (b) (and thus the α-strong convexity of f) is a straightforward consequence of Taylor's formula,

$$f(\mathbf{y}) = f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{1}{2} \langle \nabla^2 f(\mathbf{z})(\mathbf{y} - \mathbf{x}), \mathbf{y} - \mathbf{x} \rangle,$$

where $\mathbf{z}$ is a suitable point on the line segment of endpoints $\mathbf{x}$ and $\mathbf{y}$. Finally, apply Corollary 5.3.8 to the convex function $f - \frac{\alpha}{2} \|\cdot\|^2$. ■

Geometrically, the assertion (b) of Theorem 5.5.2, shows that a Gâteaux differentiable function f is strongly convex if and only if at each point $\mathbf{x}_0$ it is bounded from below by the quadratic convex function,

$$g : \mathbf{x} \to f(\mathbf{x}_0) + \langle \nabla f(\mathbf{x}_0), \mathbf{x} - \mathbf{x}_0 \rangle + \frac{\alpha}{2} \|\mathbf{x} - \mathbf{x}_0\|^2 ;$$

notice that $g(\mathbf{x}_0) = f(\mathbf{x}_0)$ and $\nabla g(\mathbf{x}_0) = \nabla f(\mathbf{x}_0)$.

5.5.3 Corollary (The strong convexity of the negative Shannon entropy) *Consider the function*

$$f : [0,1]^N \to \mathbb{R}, \quad f(\mathbf{x}) = \sum_{k=1}^{N} x_k \log x_k,$$

which is continuous on $[0,1]^N$ *and of class* C^∞ *on* $(0,1)^N$. *Notice that* $f(\mathbf{x}) = -H(\mathbf{x})$, *where* H *denotes the Shannon entropy.*

Proof. According to Theorem 5.5.2 and Pinsker's inequality (1.12) we have

$$\begin{aligned} f(\mathbf{y}) - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle &= \sum_{k=1}^{N} y_k \log \frac{y_k}{x_k} + \sum_{k=1}^{N} (x_k - y_k) \\ &\geq \frac{1}{2} \left(\sum_{k=1}^{N} |x_k - y_k| \right)^2 \geq \frac{1}{2} \left(\sum_{k=1}^{N} |x_k - y_k|^2 \right)^2, \end{aligned}$$

so the negative Shanon entropy is $1/2$-strongly convex. ■

5.5.4 Corollary (*The Polyak-Łojasiewick inequality*) *If* f *is a differentiable and* α*-strongly convex function defined on an open convex set* U *in* $\mathbb{R}^N$, *then*

$$f(\mathbf{y}) \geq f(\mathbf{x}) - \frac{1}{2\alpha} \|\nabla f(\mathbf{x})\|^2 \quad \text{for all } \mathbf{x}, \mathbf{y} \in U.$$

Proof. According to Theorem 5.5.2 (b),

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\alpha}{2} \|\mathbf{x} - \mathbf{y}\|^2 \quad \text{for all } \mathbf{x}, \mathbf{y} \in U.$$

The right-hand side is a quadratic function in $\mathbf{y}$ and attains its minimum at the point where its gradient $\nabla f(\mathbf{x}) + \alpha(\mathbf{y} - \mathbf{x})$ vanishes, that is, at $\mathbf{y} = \mathbf{x} - (1/\alpha)\nabla f(\mathbf{x})$. Therefore

$$\begin{aligned} f(\mathbf{y}) &\geq f(\mathbf{x}) - \frac{1}{\alpha} \langle \nabla f(\mathbf{x}), \nabla f(\mathbf{x}) \rangle + \frac{\alpha}{2} \left\| \frac{1}{\alpha} \nabla f(\mathbf{x}) \right\|^2 \\ &= f(\mathbf{x}) - \frac{1}{2\alpha} \|\nabla f(\mathbf{x})\|^2 \end{aligned}$$

for all $\mathbf{x}, \mathbf{y} \in U$. ■

The Polyak-Łojasiewick inequality, also known as the *PL inequality*, is instrumental in outlining the rate of convergence in some optimization algorithms (such as gradient descent for smooth convex functions). Some of its consequences are listed below.

5.5.5 Remark *Suppose $f : U \to \mathbb{R}$ is a function that verifies the hypotheses of Corollary 5.5.4. Then:*

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{1}{2\alpha} \parallel \nabla f(\mathbf{y}) - \nabla f(\mathbf{x}) \parallel^2, \tag{5.22}$$

and

$$\langle \nabla f(\mathbf{x}) - \nabla f(y), \mathbf{x} - \mathbf{y} \rangle \leq \frac{1}{\alpha} \parallel \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}) \parallel^2 \tag{5.23}$$

for all $\mathbf{x}, \mathbf{y} \in U$.

For the first inequality, consider the auxiliary function $f_{\mathbf{x}}(\mathbf{z}) = f(\mathbf{z}) - \langle \nabla f(\mathbf{x}), \mathbf{z} \rangle$. This function is α-strongly convex since

$$\langle \nabla f_{\mathbf{x}}(\mathbf{z}_1) - \nabla f_{\mathbf{x}}(\mathbf{z}_2), \mathbf{z}_1 - \boldsymbol{z}_2 \rangle = \langle \nabla f(\mathbf{z}_1) - \nabla f(\mathbf{z}_2), \mathbf{z}_1 - \mathbf{z}_2 \rangle \geq \alpha \left\| \mathbf{z}_1 - \mathbf{z}_2 \right\|^2$$

for all $\mathbf{z}_1, \mathbf{z}_2 \in U$. Then apply the PL inequality to the function $f_{\mathbf{x}}(\mathbf{z})$ to get

$$f_{\mathbf{x}}(\mathbf{y}) \geq f_{\mathbf{x}}(\mathbf{y}) - \frac{1}{2\alpha} \left\| \nabla f_{\mathbf{x}}(\mathbf{y}) \right\|^2,$$

equivalently, $f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{x} \rangle \geq f(\mathbf{y}) - \langle \nabla f(\mathbf{x}), \mathbf{y} \rangle - \frac{1}{2\alpha} \left\| \nabla f(\mathbf{y}) - \nabla f(\mathbf{x}) \right\|^2$. Rearranging, one obtains the inequality (5.22). Adding this inequality with the inequality obtained from it by interchanging the roles of $\mathbf{x}$ and $\mathbf{y}$ we arrive at the inequality (5.23).

Another class of functions that plays an important role in optimization is that of strongly smooth functions.

5.5.6 Definition *A continuously differentiable function $f : U \to \mathbb{R}$ defined on an open convex subset of $\mathbb{R}^N$ is said to be L-strongly smooth if its gradient is L-Lipschitz, that is,*

$$\left\| \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}) \right\| \leq L \left\| \mathbf{x} - \mathbf{y} \right\| \qquad \text{for all } \mathbf{x}, \mathbf{y} \in U.$$

Notice that *every differentiable and α-strongly convex function is also α-strongly smooth.* This is a consequence of Theorem 5.5.2 (c), when combined with the Cauchy–Bunyakovsky–Schwarz inequality.

If an L-strongly smooth function f is of class C^2, then $\left\| \nabla^2 f(\mathbf{x}) \right\| \leq L$.

The connection between L-strong smoothness, convexity, and semiconvexity is outlined by the following theorem.

5.5.7 Theorem *Suppose that U is an open convex subset of $\mathbb{R}^N$ and $f : U \to \mathbb{R}$ is an L-strongly smooth function. Then*

$$\left| (1-\lambda) f(\mathbf{x}) + \lambda f(\mathbf{y}) - f((1-\lambda)\mathbf{x} + \lambda \mathbf{y}) \right| \leq \lambda \, (1-\lambda) \, L \left\| \mathbf{x} - \mathbf{y} \right\|^2$$

for all $\mathbf{x}, \mathbf{y} \in U$ *and* $\lambda \in [0,1]$, *which implies that* $f + \frac{L}{2}\|\cdot\|^2$ *is a convex function and* $f - \frac{L}{2}\|\cdot\|^2$ *is a concave function.*

Proof. Indeed, if $\mathbf{x}, \mathbf{y} \in U$ and $\lambda \in [0,1]$, then

$$\begin{aligned}(1-\lambda)f(\mathbf{x}) + \lambda f(\mathbf{y}) - f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) \\ = (1-\lambda)\left[f(\mathbf{x}) - f((1-\lambda)\mathbf{x} + \lambda\mathbf{y})\right] + \lambda\left[f(\mathbf{y}) - f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}\right] \\ = -\lambda(1-\lambda)\langle\nabla f(\xi_1), \mathbf{y} - \mathbf{x}\rangle + \lambda(1-\lambda)\langle\nabla f(\xi_2), \mathbf{y} - \mathbf{x}\rangle \\ = \lambda(1-\lambda)\langle\nabla f(\xi_2) - \nabla f(\xi_1), \mathbf{y} - \mathbf{x}\rangle,\end{aligned}$$

for suitable points ξ_1 and ξ_2 belonging to the line segment joining $\mathbf{x}$ and $\mathbf{y}$. Using the mean value theorem, this implies the inequality

$$|(1-\lambda)f(\mathbf{x}) + \lambda f(\mathbf{y}) - f((1-\lambda)\mathbf{x} + \lambda\mathbf{y})| \leq \lambda(1-\lambda)L\|\mathbf{x} - \mathbf{y}\|^2,$$

from which we infer that

$$\begin{aligned}\left(f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) + \frac{L}{2}\|(1-\lambda)\mathbf{x} + \lambda\mathbf{y}\|^2\right) \\ - (1-\lambda)\left(f(\mathbf{x}) + \frac{L}{2}\|\mathbf{x}\|^2\right) - \lambda\left(f(\mathbf{y}) + \frac{L}{2}\|\mathbf{y}\|^2\right) \\ = f((1-\lambda)\mathbf{x} + \lambda\mathbf{y}) - (1-\lambda)f(\mathbf{x}) - \lambda f(\mathbf{y}) \\ + \frac{L}{2}\left[\|(1-\lambda)\mathbf{x} + \lambda\mathbf{y}\|^2 - (1-\lambda)\|\mathbf{x}\|^2 - \lambda\|\mathbf{y}\|^2\right] \\ \leq \lambda(1-\lambda)L\|\mathbf{x} - \mathbf{y}\|^2 - \lambda(1-\lambda)\frac{L}{2}\|\mathbf{x} - \mathbf{y}\|^2 \\ = \lambda(1-\lambda)\frac{L}{2}\|\mathbf{x} - \mathbf{y}\|^2,\end{aligned}$$

that is, the convexity of the function $f + \frac{L}{2}\|\cdot\|^2$, according to Definition 4.1.1.

The proof of the fact that $f - \frac{L}{2}\|\cdot\|^2$ is concave is similar. ■

5.5.8 Remark *As noticed P. Cannarsa and C. Sinestrari* [114]*, Corollary* 3.3.8*, p.* 61*, every function* $f : U \to \mathbb{R}$ *which is simultaneously semiconvex and semiconcave with parameter* $\alpha > 0$*, is also* α*-strongly smooth.*

5.5.9 Lemma *Every* L*-strongly smooth function* $f : U \to \mathbb{R}$ *verifies the following variant of the gradient inequality:*

$$f(\mathbf{y}) - f(\mathbf{x}) \leq \langle\nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x}\rangle + \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|^2, \quad \mathbf{x}, \mathbf{y} \in U. \tag{5.24}$$

As a consequence,

$$\langle\nabla f(\mathbf{x}) - \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y}\rangle \leq L\|\mathbf{y} - \mathbf{x}\|^2, \quad \mathbf{x}, \mathbf{y} \in U. \tag{5.25}$$

Proof. Indeed,

$$\begin{aligned}
&|f(\mathbf{y}) - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x}\rangle| \\
&\qquad = \left| \int_0^1 \langle \nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})), \mathbf{y} - \mathbf{x}\rangle \mathrm{d}t - \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x}\rangle \right| \\
&\qquad\quad \leq \int_0^1 |\langle \nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) - \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x}\rangle| \, \mathrm{d}t \\
&\qquad , \leq \int_0^1 \|\nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) - \nabla f(\mathbf{x})\| \cdot \|\mathbf{y} - \mathbf{x}\| \, \mathrm{d}t \\
&\qquad\qquad\qquad\qquad \leq L \left(\int_0^1 t \mathrm{d}t \right) \|\mathbf{y} - \mathbf{x}\|^2 = \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2 ,
\end{aligned}$$

whence the inequality (5.24) holds. The proof of the fact that this inequality implies the inequality (5.25) is straightforward. ■

5.5.10 Corollary *If f is simultaneously convex and L-strongly smooth, then*

$$0 \leq f(\mathbf{y}) - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x}\rangle \leq \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2 \tag{5.26}$$

and

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x}\rangle + \frac{1}{2L} \|\nabla f(\mathbf{y}) - \nabla f(\mathbf{x})\|^2 \tag{5.27}$$

for all $\mathbf{x}, \mathbf{y} \in U$.

Proof. The two-sided inequality (5.26) follows from the convexity of f and Lemma 5.5.9. For the inequality (5.27), consider the auxiliary function $f_{\mathbf{x}}(\mathbf{z}) = f(\mathbf{z}) - \langle \nabla f(\mathbf{x}), \mathbf{z}\rangle$. Since f is convex, it verifies the inequality $f(\mathbf{z}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{z} - \mathbf{x}\rangle$, so the function $f_{\mathbf{x}}$ attains its minimum at $\mathbf{z} = \mathbf{x}$. By inequality (5.25) we have

$$\langle \nabla f_{\mathbf{x}}(\mathbf{z}_1) - \nabla f_{\mathbf{x}}(\mathbf{z}_2), \mathbf{z}_1 - \mathbf{z}_2\rangle \leq L \|\mathbf{z}_1 - \mathbf{z}_2\|^2$$

for all $\mathbf{z}_1, \mathbf{z}_2 \in U$, which implies

$$f_{\mathbf{x}}(\mathbf{z}) \leq f_{\mathbf{x}}(\mathbf{y}) + \langle \nabla f_{\mathbf{x}}(\mathbf{y}), \mathbf{z} - \mathbf{y}\rangle + \frac{L}{2} \|\mathbf{z} - \mathbf{y}\|^2 .$$

Taking minimization over $\mathbf{z}$, one obtains the inequality (5.27). ■

In Section 8.2 we will show that the α-strongly convex functions and the $(1/\alpha)$-strongly smooth functions are related via the Legendre-Fenchel duality.

Gradient descent Gradient descent is a very popular optimization algorithm used to minimize some differentiable convex functions $f : \mathbb{R}^N \to \mathbb{R}$. In its simplest variant, one starts from some initial point $\mathbf{x}_0$ and proceeds by applying the following iterative formula

$$\mathbf{x}_{n+1} = \mathbf{x}_n - \gamma \nabla f(\mathbf{x}_n).$$

Here $\gamma > 0$ represents the step size (also known as the *learning rate* in the machine learning literature). If f is convex and decays at an appropriate rate, then it is guaranteed that as $k \to \infty$, $\mathbf{x}_n \to \mathbf{x}^*$, where $\mathbf{x}^*$ is the optimal point of f, that is, the point where f attains its optimal value $p^* = f(\mathbf{x}^*) = \inf_{\mathbf{x}} f(\mathbf{x})$.

Since $\langle \nabla f(\mathbf{x}_n), \mathbf{x}_{n+1} - \mathbf{x}_n \rangle = -(1/\gamma) \|\mathbf{x}_{n+1} - \mathbf{x}_n\|^2 < 0$ (except when $\mathbf{x}_n$ is optimal), the convexity of f yields the fact that

$$f(\mathbf{x}_{n+1}) < f(\mathbf{x}_n),$$

which motivates the name of this algorithm as gradient descent.

5.5.11 Theorem *Let $f : \mathbb{R}^N \to \mathbb{R}$ be an L-strongly smooth and convex function. Then the iterates of the gradient descent with $\gamma = 1/L$ verify*

$$f(\mathbf{x}_n) - f(\mathbf{x}^*) \leq \frac{2L \|\mathbf{x}_0 - \mathbf{x}^*\|^2}{n} \quad \text{for } n \geq 1.$$

Proof. We will first prove that $(\|\mathbf{x}_k - \mathbf{x}^*\|)_k$ is a decreasing sequence. For this, we need the inequality

$$\langle \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq \frac{1}{L} \| \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}) \|^2 \text{ for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^N, \quad (5.28)$$

which follows easily by adding the inequality (5.27) with the similar inequality obtained by interchanging the roles of $\mathbf{x}$ and $\mathbf{y}$. We have, according to this inequality,

$$\begin{aligned} \|\mathbf{x}_{k+1} - \mathbf{x}^*\|^2 &= \left\| \mathbf{x}_k - \frac{1}{L} \nabla f(\mathbf{x}_k) - \mathbf{x}^* \right\|^2 \\ &= \|\mathbf{x}_k - \mathbf{x}^*\|^2 - \frac{2}{L} \langle \nabla f(\mathbf{x}_k), \mathbf{x}_k - \mathbf{x}^* \rangle + \frac{1}{L^2} \|\nabla f(\mathbf{x}_k)\|^2 \\ &\leq \|\mathbf{x}_k - \mathbf{x}^*\|^2 - \frac{1}{L^2} \|\nabla f(\mathbf{x}_k)\|^2, \\ &\leq \|\mathbf{x}_k - \mathbf{x}^*\|^2, \end{aligned}$$

which proves the decreasing monotonicity of the sequence $(\|\mathbf{x}_k - \mathbf{x}^*\|)_k$.

Using the inequality (5.26) in Corollary 5.5.10, we infer that

$$f\left(\mathbf{x}_k - \frac{1}{L} \nabla f(\mathbf{x}_k)\right) - f(\mathbf{x}_k) \leq -\frac{1}{2L} \|\nabla f(\mathbf{x}_k)\|^2,$$

that is,

$$f(\mathbf{x}_{k+1}) - f(\mathbf{x}_k) \leq -\frac{1}{2L} \|\nabla f(\mathbf{x}_k)\|^2.$$

Therefore

$$\varepsilon_{k+1} - \varepsilon_k \leq -\frac{1}{2L} \|\nabla f(\mathbf{x}_k)\|^2, \quad (5.29)$$

where $\varepsilon_k = f(\mathbf{x}_k) - f(\mathbf{x}^*)$ for all k.

According to the gradient inequality for convex functions and the Cauchy–Bunyakovsky–Schwarz inequality,

$$\varepsilon_k \leq \langle \nabla f(\mathbf{x}_k), \mathbf{x}_k - \mathbf{x}^* \rangle \leq \|\mathbf{x}_k - \mathbf{x}^*\| \, \|\nabla f(\mathbf{x}_k)\|,$$

whence $\|\nabla f(\mathbf{x}_k)\| \geq \varepsilon_k / \|\mathbf{x}_0 - \mathbf{x}^*\|$, due to the fact that $(\|\mathbf{x}_k - \mathbf{x}^*\|)_k$ is a decreasing sequence. Combining this fact with (5.29), one obtains

$$\varepsilon_{k+1} \leq \varepsilon_k - \frac{\varepsilon_k^2}{2L \|\mathbf{x}_0 - \mathbf{x}^*\|^2},$$

that is,

$$\varepsilon_{k+1} + \frac{\varepsilon_k^2}{2L \|\mathbf{x}_0 - \mathbf{x}^*\|^2} \leq \varepsilon_k.$$

Therefore

$$\frac{1}{2L \|\mathbf{x}_0 - \mathbf{x}^*\|^2} \cdot \frac{\varepsilon_k}{\varepsilon_{k+1}} + \frac{1}{\varepsilon_k} \leq \frac{1}{\varepsilon_{k+1}},$$

which yields

$$\frac{1}{\varepsilon_{k+1}} - \frac{1}{\varepsilon_k} \geq \frac{1}{2L \|\mathbf{x}_0 - \mathbf{x}^*\|^2}$$

since the sequence $(\varepsilon_k)_k$ is decreasing. Then

$$\sum_{k=0}^{n-1} \left(\frac{1}{\varepsilon_{k+1}} - \frac{1}{\varepsilon_k} \right) = \frac{1}{\varepsilon_n} - \frac{1}{\varepsilon_0} \geq \frac{n}{2L \|\mathbf{x}_0 - \mathbf{x}^*\|^2}$$

According to the first assertion of Corollary 5.5.10, $\varepsilon_0 \leq \frac{L}{2} \|\mathbf{x}_0 - \mathbf{x}^*\|^2$, so $\varepsilon_n \leq \frac{2L}{n+4} \|\mathbf{x}_0 - \mathbf{x}^*\|^2$ and the proof is done. ■

An application of the gradient descent to dynamical systems is presented in the Comments section, at the end of this chapter.

There are numerous books available on optimization algorithms, covering various aspects and levels of complexity. A nice introduction is offered by the books of S. Bubeck [103] and S. Shalev-Shwartz and S. Ben-David [531] (focusing on machine learning) and Ch. L. Byrne [112] (focusing on the fundamental problems of constrained and unconstrained optimization).

Exercises

1. Let $f : U \to \mathbb{R}$ be a convex function defined on an open and convex subset U of $\mathbb{R}^N$ and α is a strictly positive parameter. Prove that the following conditions are equivalent:

   (a) f is α-strongly convex;

   (b) $f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \mathbf{s}, \mathbf{y} - \mathbf{x} \rangle + \frac{\alpha}{2} \|\mathbf{x} - \mathbf{y}\|^2$ for all $\mathbf{x}, \mathbf{y} \in U$ and $\mathbf{s} \in \partial f(\mathbf{x})$;

   (c) $f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \mathbf{s}, \mathbf{y} - \mathbf{x} \rangle + \frac{\alpha}{2} \|\mathbf{x} - \mathbf{y}\|^2$ for all $\mathbf{x}, \mathbf{y} \in U$ and some $\mathbf{s} \in \partial f(\mathbf{x})$.

   [*Hint*: Use the fact that f is α-strongly convex if and only if the function $g = f - \frac{\alpha}{2} \|\cdot\|^2$ is convex if and only if $\partial g(\mathbf{x}) \neq \emptyset$ at each $\mathbf{x} \in U$.]

2. (Azagra et al [32]) Suppose that $f : \mathbb{R}^N \to \mathbb{R}$ is a α-strongly convex function and $\varphi \in C^2\left(\mathbb{R}^N\right)$ is a function such that

$$N \max_{1\leq i,j\leq N} \max_{\mathbf{x}\in\mathbb{R}^N} \left| \frac{\partial^2 \varphi}{\partial x_i \partial x_j}(\mathbf{x}) \right| \leq \alpha - \tilde{\alpha}$$

for some constant $\tilde{\alpha} \in (0, \alpha)$. Prove that $f + \varphi$ is a $\tilde{\alpha}$-strongly convex function.

[*Hint*: Notice that $\left|\langle \nabla^2 \varphi(\mathbf{x})\mathbf{v}, \mathbf{v}\rangle\right| \leq (\alpha - \tilde{\alpha}) \|\mathbf{v}\|^2$ for all $\mathbf{x}, \mathbf{v} \in \mathbb{R}^N$.]

3. Suppose that f is a Gâteaux differentiable convex function defined on an open convex set U in $\mathbb{R}^N$ and L is a strictly positive number. Prove that the following assertions are equivalent:

(a) f is strongly L-smooth;

(b) $\frac{L}{2} \|\cdot\|^2 - f$ is convex;

(c) $\langle \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y}\rangle \leq L \|\mathbf{y} - \mathbf{x}\|^2$ for all $\mathbf{x}, \mathbf{y} \in U$.

5.6 Supermodularity and Monotonicity of the Differentials

In the case of differentiable functions of one real variable the property of (strict) convexity is equivalent to the (strict) increasing of its derivative. See Theorem 1.4.3. However, this is no longer true in the case of $n \geq 2$ variables. Here is an example borrowed from [429].

5.6.1 Example (A difference between the differentiable convex functions of one real variable and those of several variables) *The twice continuously differentiable function*

$$f(x, y) = -2\,(xy)^{1/2}, \quad (x, y) \in \mathbb{R}^2_{++},$$

is convex due to the fact that its Hessian,

$$\nabla^2 f(x, y) = \frac{1}{2} \begin{pmatrix} x^{-3/2} y^{1/2} & -x^{-1/2} y^{-1/2} \\ -x^{-1/2} y^{-1/2} & x^{1/2} y^{-3/2} \end{pmatrix},$$

is a positive semidefinite matrix. However, unlike the case of convex functions of one real variable, the differential of f,

$$df : \mathbb{R}^2_{++} \to \mathbb{R}^2, \quad df(x, y) = -(x^{-1/2} y^{1/2}, x^{1/2} y^{-1/2}),$$

is not increasing. Indeed, $(1, 1) < (2, 1)$ in $\mathbb{R}^2_{++}$ while

$$df(1, 1) = -\,(1, 1) \text{ and } df(2, 1) = -(1/\sqrt{2}, \sqrt{2})$$

are not even comparable.

On the other hand, a simple example of a nonconvex differentiable function whose differential is isotone is provided by the function

$$H(x, y) = (2x - 1)(2y - 1), \quad (x, y) \in \mathbb{R}^2.$$

As noticed Amann (see [17], Proposition 3.2, p. 184), the Gâteaux differentiability offers a convenient way to recognize the order properties of functions acting on ordered Banach spaces: the positivity of the differential. In the case of convex functions his result can be stated as follows:

5.6.2 Lemma *Suppose that E and F are two ordered Banach spaces, C is a convex subset of E with nonempty interior and $f : C \to F$ is a convex function, continuous on C and Gâteaux differentiable on* int C. *Then f is increasing on C if and only if $f'(\mathbf{a}) \geq 0$ for all $\mathbf{a} \in$ int C.*

Proof. The "only if" part follows immediately from the definition of the Gâteaux derivative. For the other implication, notice that the gradient inequality

$$f(\mathbf{x}) - f(\mathbf{a}) \geq f'(\mathbf{a})(\mathbf{x} - \mathbf{a})$$

shows that f is increasing on int C if $f'(\mathbf{a}) \geq 0$ for all $\mathbf{a} \in$ int C. As concerns the monotonicity on C, this follows by an approximation argument. Indeed, suppose that $\mathbf{x}, \mathbf{y} \in C$ and $\mathbf{x} \leq \mathbf{y}$. For $\mathbf{x}_0 \in$ int U arbitrarily fixed and $t \in [0, 1)$, both elements $\mathbf{u}_t = \mathbf{x}_0 + t(\mathbf{x} - \mathbf{x}_0)$ and $\mathbf{v}_t = \mathbf{x}_0 + t(\mathbf{y} - \mathbf{x}_0)$ belong to int C. See Lemma 2.1.7. Clearly, $\mathbf{u}_t \leq \mathbf{v}_t$. Moreover, $\mathbf{u}_t \to \mathbf{x}$ and $\mathbf{v}_t \to \mathbf{y}$ as $t \to 1$. Passing to the limit in the inequality $f(\mathbf{u}_t) \leq f(\mathbf{v}_t)$ and using the continuity, we conclude that $f(\mathbf{x}) \leq f(\mathbf{y})$. ■

5.6.3 Remark *If the ordered Banach space E has finite dimension, then the statement of Lemma 5.6.2 remains valid by replacing the interior of C by the relative interior of C. See Exercise 6, Section* 3.1.

Using the aforementioned result of H. Amann, one can easily prove the following criterion of monotonicity of the differentials:

5.6.4 Lemma *Suppose that U is an open convex subset of the Banach lattice $\mathbb{R}^N$ and $f : U \to \mathbb{R}$ is a twice differentiable function. Then df is increasing on U if (and only if) all partial derivatives of second order of f are positive.*

When f is also convex, the increasing monotonicity of f' is equivalent to the condition that all mixed derivatives $\frac{\partial^2 f}{\partial x_i \partial x_j}$ are nonnegative.

Three examples of differentiable functions which are isotone and/or admit isotone differentials are presented in what follows.

5.6.5 Example *The following variant of the negative entropy function,*

$$E(\mathbf{x}) = \sum_{k=1}^{N} x_k \log x_k, \quad \mathbf{x} = (x_1, ..., x_N),$$

is C^∞-differentiable on $\mathbb{R}^N_{++}$ and strongly convex on any compact subset K of $\mathbb{R}^N_{++}$ because its Hessian matrix $H(\mathbf{x}) = \operatorname{diag}\left(\frac{1}{x_1}, ..., \frac{1}{x_N}\right)$ verifies the inequality

$$\langle H(\mathbf{x})\mathbf{v}, \mathbf{v}\rangle \geq \frac{1}{\max\{x_1 + \cdots + x_N : \mathbf{x} \in K\}} \|\mathbf{v}\|^2$$

for all $\mathbf{x} \in K$ *and* $\mathbf{v} \in \mathbb{R}^N$; *see Theorem* 5.5.2. *Clearly, the function* E *is increasing on the subset* U *of* $\mathbb{R}^N_{++}$ *constituted by all vectors* $\mathbf{x}$ *whose all components are greater than or equal to* $1/e$. *The differential of* E *is the map* $dE : \mathbb{R}^N_{++} \to \mathbb{R}^N$ *given by the formula*

$$dE(\mathbf{x})\mathbf{v} = \sum_{k=1}^{N} (1 + \log x_k) v_k, \quad \mathbf{v} \in \mathbb{R}^N,$$

so that, according to Lemma 5.6.4, $\mathbf{x} \leq \mathbf{y}$ *in* U *implies* $dE(\mathbf{x}) \leq dE(\mathbf{y})$.

5.6.6 Example *Let* $p \in (1, \infty)$. *The function*

$$\Phi : L^p(\mathbb{R}) \to \mathbb{R}, \quad \Phi(f) = \|f\|_p^p = \int_{\mathbb{R}} |f|^p \, \mathrm{d}t$$

is convex being the composition of the increasing and convex function $t \to t^p$ $(t \in \mathbb{R}_+)$ *with the convex function* $f \to \|f\|_p$. *According to Proposition* 5.2.9, *the function* Φ *is differentiable and its differential is defined by the formula*

$$d\Phi(f)(h) = p \int_{\mathbb{R}} h \, |f|^{p-2} f \mathrm{d}t \quad \text{for all } f, h \in L^p(\mathbb{R}).$$

Clearly, $0 \leq f \leq g$ *in* $L^p(\mathbb{R})$ *implies* $\Phi(f) \leq \Phi(g)$, *which agrees with the fact that* $d\Phi(f) \geq 0$ *for all* $f \geq 0$. *See Lemma* 5.6.2. *It is worth mentioning another nice property of the function* Φ,

$$0 \leq f \leq g \text{ in } L^p(\mathbb{R}) \text{ implies } d\Phi(f) \leq d\Phi(g).$$

5.6.7 Example *This example is inspired by Malamud's approach of majorization for sequences of vectors with non-equal numbers of entries. See* [353], *Proposition* 2.16, *where he considered the composite convex functions of the form*

$$\Phi(\mathbf{x}) = f(\langle \mathbf{x}, \mathbf{w} \rangle), \quad \mathbf{x} \in \mathbb{R}^N,$$

where f *is any convex function defined on* $\mathbb{R}$ *and* $\mathbf{w} \in \mathbb{R}^N$. *Clearly,* Φ *is increasing when* f *is increasing and* $\mathbf{w} \in \mathbb{R}^N_+$.

The function Φ *is differentiable and its differential is increasing if* f *is differentiable and* $\mathbf{w} \in \mathbb{R}^N_+$. *Indeed, in this case the differential of* Φ *is given by the formula*

$$d\Phi(\mathbf{x})\mathbf{v} = f'(\langle \mathbf{x}, \mathbf{v} \rangle) \langle \mathbf{v}, \mathbf{w} \rangle, \quad \mathbf{x}, \mathbf{v} \in \mathbb{R}^N,$$

which implies that $d\Phi(\mathbf{x}) \leq d\Phi(\mathbf{y})$ *whenever* $\mathbf{x} \leq \mathbf{y}$ *in* $\mathbb{R}^N$.

One can characterize the monotonicity of the differential of a convex function defined on an open N-box in $\mathbb{R}^N$ by using the concept of supermodularity (common in comparative statistics and combinatorial optimization).

Recall that an N-box in $\mathbb{R}^N$ is nothing but a Cartesian product $U = \prod_{k=1}^N I_k$ of N real intervals I_k; the box is compact (respectively open) if each of the intervals I_k has this property. In the compact case, some of the intervals I_k may reduce to singletons.

Clearly, every N-box is an order convex set.

5.6.8 Definition *A real-valued function f defined on an N-box U is called* supermodular *if*

$$f\left(\sup\{\mathbf{x},\mathbf{y}\}\right)+f\left(\inf\{\mathbf{x},\mathbf{y}\}\right)\geq f(\mathbf{x})+f(\mathbf{y})$$

whenever $\mathbf{x},\mathbf{y}\in U$; f is called $2D$-supermodular *if the above inequality holds for all $\mathbf{x},\mathbf{y}\in U$ situated in a plane parallel to one of the coordinate planes.*

Notice that the function f is $2D$-*supermodular* if

$$\begin{aligned}\Delta(f;B_{ij})=f(u_1,...,v_i,...,v_j,...,u_N)-f(u_1,...,v_i,...,w_j,...,u_N)\\ -f(u_1,...,w_i,...,v_j,...,u_N)+f(u_1,...,w_i,...,w_j,...,u_N)\geq 0.\end{aligned}$$

for every nondegenerate 2-dimensional box

$$B_{ij}=\{u_1\}\times\cdots\times[v_i,w_i]\times\cdots\times\{u_k\}\times\cdots\times[v_j,w_j]\times\cdots\times\{u_N\},\quad 1\leq i<j\leq N,$$

included in U and parallel to one of the planes of coordinates is positive.

In what follows we will refer to $\Delta(f;B_{ij})$ as the *increment* of f over the 2-dimensional box B_{ij}.

The notions of supermodularity and $2D$-supermodularity coincide in the case of functions defined on the Cartesian product of two intervals.

Two simple examples of supermodular functions on $\mathbb{R}^2$ are x^2+y^2 and xy.

Since any convex function $F:\mathbb{R}\to\mathbb{R}$ verifies the inequality

$$\begin{vmatrix}1 & x & F(x)\\ 1 & y & F(y)\\ 1 & z & F(z)\end{vmatrix}=(z-y)F(x)-(z-x)F(y)+(y-x)F(z)\geq 0$$

for all $x<y<z$ (see the introduction to Section 1.4), one can attach to it a supermodular function on $\mathbb{R}^2$ via the formula $f(x,y)=-F(x-y)$. In a similar way, if $F:[0,\infty)\to\mathbb{R}$ is a convex function and $\lambda>0$, then $f(x,y)=F(\lambda\,(x+y))$ is a supermodular function on $[0,\infty)^2$.

Probability and statistics provide yet another major source of supermodular functions: the functions that couple multivariate distribution functions to their one-dimensional marginal distribution functions, called copulas. See R. B. Nelsen [409] for details.

S. G. Gal and C. P. Niculescu [206] have described the concept of $2D$-supermodularity (under the name of 2-box monotonicity) by its relationship with the concept of $(2,2)$-convex function in the sense of T. Popoviciu [483], [487].

The analogue of the $2D$-supermodularity for functions f defined on an interval $[a,b]$ is the property of *equal increasing increments* (which is equivalent to convexity):

$$f(x+z)-f(x)\leq f(y+z)-f(y)$$

whenever $x\leq y$, $z>0$ and $x,y,y+z\in[a,b]$. See Remark 1.4.1 and Corollary 1.4.7.

A characterization of $2D$-supermodularity in the presence of twice differentiability is as follows.

5.6.9 Theorem *Suppose that U is an open N-box of the Banach lattice $\mathbb{R}^N$ and $f : U \to \mathbb{R}$ is a twice continuously differentiable convex function. Then f is $2D$-supermodular if (and only if) all mixed derivatives $\frac{\partial^2 f}{\partial x_i \partial x_j}$ are positive.*

Proof. Notice the identity

$$\begin{aligned}\int_{v_i}^{w_i}\int_{v_j}^{w_j}\frac{\partial^2 f}{\partial x_i \partial x_j}(u_1,...,x_i,...,x_j,...,u_N)\mathrm{d}x_i\mathrm{d}x_j \\ = f(u_1,...,v_i,...,v_j,...,u_N) - f(u_1,...,v_i,...,w_j,...,u_N) \\ - f(u_1,...,w_i,...,v_j,...,u_N) + f(u_1,...,w_i,...,w_j,...,u_N),\end{aligned}$$

which holds for all pairs of indices $i < j$ and all pairs of points in U satisfying

$$(u_1,...,v_i,...,v_j,...,u_N) \leq (u_1,...,w_i,...,w_j,...,u_N).$$

Since f is of class C^2, the integral in the left-hand side is strictly positive only when the integrand is strictly positive on an open N-box included in U. ■

Combining Lemma 5.6.4 with Theorem 5.6.9 one can easily infer that for the convex functions of class C^2 the property of increasing monotonicity is equivalent to the property of $2D$-supermodularity. This equivalence continues to work in the context of functions of class C^1.

5.6.10 Theorem *Suppose that U is an open N-box of the Banach lattice $\mathbb{R}^N$ and $f : U \to \mathbb{R}$ is a continuously differentiable convex function. Then df is increasing on U if (and only if) f is $2D$-supermodular.*

Proof. The fact that df is increasing is equivalent to

$$\frac{\partial f}{\partial x_k}(u_1,...,u_N) \leq \frac{\partial f}{\partial x_k}(v_1,...,v_N) \tag{5.30}$$

for all pairs of points $\mathbf{u} = (u_1,...,u_N) \leq \mathbf{v} = (v_1,...,v_N)$ in C and all indices $k \in \{1,...,N\}$.

Since f is differentiable and convex in each variable, we have

$$\frac{\partial f}{\partial x_k}(u_1,...,u_{k-1},x_k,u_{k+1},...,u_N) \leq \frac{\partial f}{\partial x_k}(u_1,...,u_{k-1},y_k,u_{k+1},...,u_N)$$

whenever $(u_1,...,u_{k-1},x_k,u_{k+1},...,u_N) \leq (u_1,...,u_{k-1},y_k,u_{k+1},...,u_N)$ in C.

Using the identity

$$\begin{aligned}\int_{x_j}^{y_j}\left(\frac{\partial f}{\partial x_j}(u_1,...,y_i,...,t,...,u_N) - \frac{\partial f}{\partial x_j}(u_1,...,x_i,...,t,...,u_N)\right)\mathrm{d}t \\ = f(u_1,...,y_i,...,y_j,...,u_N) - f(u_1,...,y_i,...,x_j,...,u_N) \\ - f(u_1,...,x_i,...,y_j,...,u_N) + f(u_1,...,x_i,...,x_j,...,u_N),\end{aligned}$$

which works for every nondegenerate 2-dimensional box

$$\{u_1\} \times \cdots \times [x_i, y_i] \times \cdots \times \{u_k\} \times \cdots \times [x_j, y_j] \times \cdots \times \{u_N\},$$

included in C, one can easily infer that the $2D$-supermodularity is equivalent with the monotonicity of each partial derivative $\frac{\partial f}{\partial x_k}$ in each variable different from the kth, when the others are kept fixed. By mathematical induction one can put together all these facts to obtain the inequalities (5.30). ■

The reader can easily verify that the following two nondifferentiable functions,

$$\min\{x_1, x_2\} \text{ and } \max\{x_1 + x_2 - 1, 0\}, \quad x_1, x_2 \in [0, 1],$$

are $2D$-supermodular. They are known in the theory of copulas as the *Fréchet-Hoeffding bounds*. See [409]. The second function is convex but the first one is not convex. This fact combined with Example 5.6.1 shows that the notions of convexity and $2D$-supermodularity are independent in dimensions $N \geq 2$.

The increment of the log-sum-exp function over the box

$$[0, 1] \times [0, 1] \times \{0\} \times \cdots \times \{0\}$$

equals

$$\log N - 2\log(e + N - 1) + \log(2e + N - 2) < 0$$

so that, this function is not $2D$-supermodular. According to Theorem 5.6.10, the differential of the log-sum-exp function is not increasing.

We end this section by showing that every $2D$-supermodular convex function can be approximated by indefinitely differentiable and strongly convex functions with increasing differentials.

5.6.11 Theorem *Suppose that f is a $2D$-supermodular convex function defined on an open N-box U included in $\mathbb{R}^N$. Then on every compact box $K \subset U$, f is the uniform limit of a sequence of infinitely differentiable strongly convex functions with increasing differentials.*

When $U \subset \mathbb{R}^N_{++}$ and the function f is also increasing, then the approximants of f are of the same nature.

Proof. We use the usual convolution based smooth approximation. Let $\varepsilon > 0$ be arbitrarily fixed. Then the function $g = f + (\varepsilon/2)\|\cdot\|^2$ is $2D$-supermodular and ε-strongly convex. Besides,

$$\|g(\mathbf{x}) - f(\mathbf{x})\| \leq (\varepsilon/2) \sup\left\{\|\mathbf{x}\|^2 : \mathbf{x} \in K\right\} \quad \text{for all } \mathbf{x} \in K.$$

According to the method of mollifiers (see Theorem 5.7.4), the convolution

$$(g * \varphi)(\mathbf{x}) = \int_{\mathbb{R}^N} g(\mathbf{x} - \mathbf{y})\varphi(\mathbf{y})\mathrm{d}\mathbf{y},$$

of g with any infinitely differentiable function $\varphi : \mathbb{R}^N \to [0, \infty)$ such that $\varphi = 0$ on $\mathbb{R}^N \backslash K$ and $\int_{\mathbb{R}^N} \varphi(\mathbf{y})\mathrm{d}\mathbf{y} = 1$, is an infinitely differentiable function

that provides a regularization of g since $g * \varphi \to g$ uniformly on K as the support of φ shrinks to $\{0\}$. An easy computation shows that $g * \varphi$ is also a $2D$-supermodular and ε-strongly convex function. Indeed, the increment of g over a 2-dimensional box B_{ij} verifies the inequality

$$\Delta(g * \varphi; B_{ij}) = \int_{\mathbb{R}^N} \Delta(g(\mathbf{x} - \mathbf{y}); B_{ij})\varphi(\mathbf{y})\mathrm{d}\mathbf{y} \geq \mathbf{0}$$

and for all $\mathbf{u}, \mathbf{v} \in U$ and $\lambda \in [0, 1]$ we have

$$\begin{aligned}(g * \varphi)((1-\lambda)\mathbf{u}+\lambda\mathbf{v}) &= \int_{\mathbb{R}^N} g((1-\lambda)(\mathbf{u}-\mathbf{y})+\lambda(\mathbf{v}-\mathbf{y}))\varphi(\mathbf{y})\mathrm{d}\mathbf{y} \\ &\leq \int_{\mathbb{R}^N} \left[(1-\lambda)\, g(\mathbf{u}-\mathbf{y})+\lambda g(\mathbf{v}-\mathbf{y})) - \frac{\varepsilon}{2}\lambda(1-\lambda)\left\|\mathbf{u}-\mathbf{v}\right\|^2\right]\varphi(\mathbf{y})\mathrm{d}\mathbf{y} \\ &= (1-\lambda)\,(g*\varphi)(\mathbf{u}) + \lambda\,(g*\varphi)(\mathbf{v}) - \frac{\varepsilon}{2}\lambda(1-\lambda)\left\|\mathbf{u}-\mathbf{v}\right\|^2.\end{aligned}$$

Then the conclusion of Theorem 5.6.11 follows from Theorem 5.6.10. ■

Exercises

1. (Concavity and order preservation). Suppose C is a convex set that contains the origin and $f : C \to \mathbb{R}$ is a concave function such that $f(0) \geq 0$.

   (a) Prove that $f(t\mathbf{x}) \geq tf(\mathbf{x})$ for all $\mathbf{x} \in C$ and $t \in [0, 1]$.

   (b) Infer that every positive concave function $f : \mathbb{R}^N_+ \to \mathbb{R}$ is increasing, that is,

   $$\mathbf{x} \leq \mathbf{y} \text{ (coordinatewise) implies } f(\mathbf{x}) \leq f(\mathbf{y}).$$

   (c) Example 4.1.5 exhibits the case of several variables concave functions which are both positive, superadditive, and increasing. Prove that actually every positive and superadditive function defined on $\mathbb{R}^N_+$ is increasing. Show by an example that not every positive and concave function is superadditive.

2. Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a convex function, A is an $N \times N$-dimensional matrix with real coefficients and $\mathbf{w} \in \mathbb{R}^N$. Show that $f(\langle A(\mathbf{x}), \mathbf{w}\rangle)$ is a $2D$-submodular and convex function. In particular, so is $f(x_1 + \cdots + x_N)$.

3. If $f_1, ..., f_N : \mathbb{R} \to \mathbb{R}$ are convex functions, prove that $f_1(x_1) + \cdots + f_N(x_N)$ is $2D$-supermodular and convex.

4. If $f_1, ..., f_N : \mathbb{R}_+ \to \mathbb{R}_+$ are differentiable and increasing functions, prove that $f_1(x_1) \cdots f_N(x_N)$ is $2D$-supermodular and increasing. In particular, so is the case of the functions $x_1 \cdots x_N$.

5. Let $\varphi : \mathbb{R}^2 \to \mathbb{R}$ be a supermodular increasing function and let f be an increasing and convex function from $\mathbb{R}$ to $\mathbb{R}$.

(a) Show that the composition $f \circ \varphi$ is increasing and supermodular. When $\varphi(s,t) = s + t$, shows that this is also true if f is monotone decreasing.

(b) Extend the assertion (a) to the case when $\varphi : \mathbb{R}^N \to \mathbb{R}$ is an increasing $2D$-supermodular function.

6. (A generalization of the discrete form of Hardy–Littlewood rearrangement inequality) Suppose that I is an interval and $\Phi : I \times I \to \mathbb{R}$ is a supermodular function. Prove that for all elements $x_1, \dots, x_n, y_1, \dots, y_n$ in I,

$$\sum_{k=1}^{n} \Phi(x_k^{\downarrow}, y_k^{\uparrow}) \leq \sum_{k=1}^{n} \Phi(x_k, y_k) \leq \sum_{k=1}^{n} \Phi(x_k^{\downarrow}, y_k^{\downarrow})$$

where $x_1^{\downarrow} \geq \cdots \geq x_n^{\downarrow}$ represents the decreasing rearrangement of the family $x_1, \dots, x_n$, and $x_1^{\uparrow} \leq \cdots \leq x_n^{\uparrow}$ represents the increasing rearrangement of this family.

For more general results see F. Brock [102] (and the references therein).

5.7 Comments

The almost everywhere differentiability of convex functions defined on $\mathbb{R}^N$ is generalized by a famous result due to H. Rademacher concerning locally Lipschitz functions. See Theorem D.1.1, Appendix D.

As we noticed in Section 5.3, the second derivative test of convexity needs some good experience with matrix differential calculus when applied to functions whose variables are the minors of a given matrix. The Matrix Cookbook of K. B. Petersen and M. S. Pedersen, might be helpful in this context.

A Banach space E is said to be a *strong differentiability space* if, for each convex open set U in E and each continuous convex function $f : U \to \mathbb{R}$, the set of points of Fréchet differentiability of f contains a dense G_δ subset of U. A characterization of such spaces is given by the following result:

5.7.1 Theorem (E. Asplund [26], I. Namioka and R. R. Phelps [404]) *A Banach space E is a strong differentiability space if and only if the dual of every separable subspace of E is separable.*

In particular, every reflexive Banach space is a strong differentiability space. See R. R. Phelps [473] for details and a survey on the differentiability properties of convex functions on a Banach space.

A survey on the problem of renorming Banach spaces to improve their smoothness properties is available in the book by M. M. Day [155].

The convex functions can be characterized in terms of distributional derivatives:

5.7.2 Theorem (G. Alberti and L. Ambrosio [9]) *If Ω is an open convex subset of $\mathbb{R}^n$, and $f : \Omega \to \mathbb{R}$ is a convex function, then Df is monotone, and*

$D^2 f$ is a positive and symmetric (matrix-valued and locally bounded) measure. Conversely, if f is locally integrable and $D^2 f$ is a positive (matrix-valued) distribution on Ω, then f agrees almost everywhere on Ω with a convex function g such that $\Omega \subset \operatorname{dom} g$.

If $f\colon \mathbb{R}^N \to \mathbb{R}$ is a convex function, then it admits distributional derivatives of second order. More precisely, for all $i, j \in \{1, \dots, N\}$ there exist signed Radon measures μ_{ij} (with $\mu_{ij} = \mu_{ji}$) such that

$$\int_{\mathbb{R}^N} f(\mathbf{x}) \frac{\partial^2 \varphi}{\partial x_i \partial x_j}(\mathbf{x})\, \mathrm{d}\mathbf{x} = \int_{\mathbb{R}^N} \varphi(\mathbf{x}) \mathrm{d}\mu_{ij} \quad \text{for every } \varphi \in C_c^2(\mathbb{R}^N).$$

In addition, the measures μ_{ii} are positive. For details, see L. C. Evans and R. Gariepy [184], Theorem 6.8, p. 271. Yu. G. Reshetnyak [501] proved that a locally integrable function f defined on an open subset of $\mathbb{R}^N$ is equal almost everywhere to a convex function if and only if for every $\xi_1, \dots, \xi_N \in \mathbb{R}$, $\sum_{i,j=1}^N \xi_i \xi_j \frac{\partial^2 [f]}{\partial x_i \partial x_j}$ is a positive Radon measure; here $[f]$ represents the distribution associated to f.

In Appendix D it will be proved (following the argument of M. G. Crandall, H. Ishii and P.-L. Lions [144]), the following theorem of A. D. Alexandrov [12], concerning the twice differentiability of convex functions defined on $\mathbb{R}^N$:

5.7.3 Theorem *Every convex function f on $\mathbb{R}^N$ is twice differentiable almost everywhere in the following sense: f is twice differentiable at $\mathbf{a}$, with Alexandrov Hessian $\nabla^2 f(\mathbf{a})$ in $\operatorname{Sym}^+(N, \mathbb{R})$, if $\nabla f(\mathbf{a})$ exists, and if for every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$\|\mathbf{x} - \mathbf{a}\| < \delta \text{ implies } \sup_{\mathbf{y} \in \partial f(\mathbf{x})} \|\mathbf{y} - \nabla f(\mathbf{a}) - \nabla^2 f(\mathbf{a})(\mathbf{x} - \mathbf{a})\| \le \varepsilon \|\mathbf{x} - \mathbf{a}\|.$$

Moreover, if $\mathbf{a}$ is such a point, then

$$\lim_{\mathbf{h} \to 0} \frac{f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) - \langle \nabla f(\mathbf{a}), \mathbf{h} \rangle - \frac{1}{2} \langle \nabla^2 f(\mathbf{a}) \mathbf{h}, \mathbf{h} \rangle}{\|\mathbf{h}\|^2} = 0.$$

Convolution by smooth functions provides a powerful technique for approximating locally integrable functions by C^∞ functions. Particularly, this applies to the convex functions.

Let φ be a *mollifier*, that is, a positive function in $C_c^\infty(\mathbb{R}^N)$ such that

$$\int_{\mathbb{R}^N} \varphi \, \mathrm{d}\mathbf{x} = 1 \quad \text{and} \quad \operatorname{supp} \varphi \subset \overline{B}_1(\mathbf{0}).$$

The standard example of such a function is

$$\varphi(\mathbf{x}) = \begin{cases} C \exp(-1/(1 - \|\mathbf{x}\|^2)) & \text{if } \|\mathbf{x}\| < 1, \\ 0 & \text{if } \|\mathbf{x}\| \ge 1, \end{cases}$$

where C is chosen such that $\int_{\mathbb{R}^N} \varphi \, d\mathbf{x} = 1$. Each mollifier φ gives rise to an one-parameter family of positive functions

$$\varphi_\varepsilon(\mathbf{x}) = \frac{1}{\varepsilon^N} \varphi\left(\frac{\mathbf{x}}{\varepsilon}\right), \quad \varepsilon > 0$$

with similar properties:

$$\varphi_\varepsilon \in C_c^\infty(\mathbb{R}^N), \quad \text{supp}\varphi_\varepsilon \subset \overline{B}_\varepsilon(\mathbf{0}) \quad \text{and} \quad \int_{\mathbb{R}^N} \varphi_\varepsilon \, d\mathbf{x} = 1.$$

The following result is standard and its proof is available from many sources. See, for example [184], Theorem 4.1, p. 146.

5.7.4 Theorem *Suppose that $f \in L^1_{loc}(\mathbb{R}^N)$ and $(\varphi_\varepsilon)_{\varepsilon>0}$ is the one-parameter family of functions associated to a mollifier φ. Then:*

(a) *The functions $f_\varepsilon = \varphi_\varepsilon * f$ belong to $C^\infty(\mathbb{R}^N)$ and*

$$D^\alpha f_\varepsilon = D^\alpha \varphi_\varepsilon * f \quad \text{for every multi-index } \alpha \in \mathbb{N}^N;$$

(b) *$f_\varepsilon(\mathbf{x}) \to f(\mathbf{x})$ whenever $\mathbf{x}$ is a point of continuity of f. If f is continuous on an open subset U, then f_ε converges uniformly to f on each compact subset of U;*

(c) *If $f \in L^p(\mathbb{R}^N)$ (for some $p \in [1, \infty)$), then $f_\varepsilon \in L^p(\mathbb{R}^N)$, $\|f_\varepsilon\|_{L^p} \leq \|f\|_{L^p}$ and $\lim_{\varepsilon \to 0} \|f_\varepsilon - f\|_{L^p} = 0$;*

(d) *If f is a (strongly) convex function on an open convex subset U of $\mathbb{R}^N$, then f_ε is (strongly) convex too.*

In Section 8.5 we will discuss the approximation of convex functions via infimal convolution, that provides convex and C^1 smooth approximants. The following result due to D. Azagra and C. Mudarra [33], shows that the uniform approximation of continuous convex functions in Banach spaces is possible only in some special cases.

5.7.5 Theorem *For a separable Banach space E, the following statements are equivalent:*

(a) *E^* is separable;*

(b) *For every $U \subset E$ open and convex, every continuous convex function $f : U \to \mathbb{R}$ and every $\varepsilon > 0$, there exists a convex function $g \in C^1(U)$ such that $f - \varepsilon \leq g \leq f$ on U.*

Recently, D. Azagra, M. Drake and P. Hajłasz [32] proved that for every strongly convex function $f : \mathbb{R}^N \to \mathbb{R}$ and every $\varepsilon > 0$ there is a strongly convex function $g \in C^2(\mathbb{R}^N)$ such that $\mathcal{L}(\{\mathbf{x} : f(\mathbf{x}) \neq g(\mathbf{x})\}) < \varepsilon$ and $\|f - g\|_\infty < \varepsilon$.

In Section 5.4 we noticed the role played by lower semicontinuity in the problem of finding the minimum of a convex function defined on $\mathbb{R}^N$. For the applications of the calculus of variations to solving partial differentiable equations it is necessary to consider also the variant of sequential lower semicontinuity.

See Appendix E. The ubiquity of convexity in this connection was noticed by a number of mathematicians starting with L. Tonelli. For simplicity, we will consider here only the case of functionals of the form

$$I = I(u) = \int_\Omega f(u(\mathbf{x}))d\mathbf{x}, \quad u \in L^p(\Omega),$$

where Ω is a bounded open subset of $\mathbb{R}^n$ with smooth boundary, $p \in (1, \infty)$ and $f : \mathbb{R} \to \mathbb{R}$ is a continuous function bounded below.

5.7.6 Theorem (L. Tonelli) *The functional I is sequentially weakly lower semicontinuous, that is*

$$u_n \to u \text{ weakly in } L^p(\Omega) \text{ implies } I(u) \leq \liminf_{n\to\infty} I(u_n),$$

if, and only if, the function f is convex.

For details and much more general results (involving integrands of the form $f = f(\mathbf{x}, u(\mathbf{x}), \nabla u(\mathbf{x}))$, see the books of L. C. Evans [182] and B. Dacorogna [146]. A proof of Tonelli's theorem is also available in the book of M. Renardy and R. C. Rogers [499], pp. 347-350.

Our presentation of strongly smooth functions in Section 5.5 borrowed some ideas from the survey of X. Zhou [600].

The problem of solving a system of equations $F(\mathbf{x}) = 0$, where $F : U \to R^N$, $F = (F_1, ..., F_N)$, is the gradient of a convex function $f : U \to \mathbb{R}$ of class C^1, defined on an open convex subset of $\mathbb{R}^N$, reduces to the problem of finding the minimizers of f. Systems of this form are quite restrictive. Indeed, F is the gradient of a differentiable function $f : D \to \mathbb{R}$ if and only if $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$ for all pairs of distinct indices i and j. See J. M. Ortega and W. C. Rheinboldt [457], Theorem 4.1.6, pp. 95-96. However, as they noticed, there is always a simple way of converting the problem of solving the system $F(\mathbf{x}) = 0$ into aminimization problem even if F is not a gradient.

The quasiconvex functions (introduced by B. De Finetti around 1949) play many properties similar to those of convex functions. For example, any lower semicontinuous quasiconvex function is weakly lower semicontinuous. Thus, if the ambient space is reflexive and if moreover the function is coercive, it attains its infimum. See Exercise 6, Section 5.4. Meanwhile, there are serious differences. The sums of quasiconvex functions may not be quasiconvex. In fact, as observed by J.-P. Crouzeix, a real-valued function f defined on a normed linear space E is convex if, and only if, for each $x^* \in E^*$ the function $f + x^*$ is quasiconvex. The quasiconvex functions defined on open convex sets in $\mathbb{R}^N$ may be continuous only almost everywhere and the existence of the one-sided partial derivatives takes place only almost everywhere.

A systematic treatment of the subject of quasiconvex functions can be found in the books of M. Avriel, W. E. Diewert, S. Schaible and I. Zang [31], and of

A. Cambini and L. Martein [113]. See also the surveys by H. J. Greenberg and W. P. Pierskalla [225], and J.-P. Penot [468].

The theory of differentiable convex functions offers a large generalization of the Kullback–Leibler divergence: the Bregman divergence. Given a strictly convex function $\varphi : C \to \mathbb{R}$ defined on a closed convex set $C \subset \mathbb{R}^N$ that is of class C^1 on int C, the corresponding *Bregman divergence* is given by the formula

$$D_\varphi(\mathbf{v}, \mathbf{u}) = \varphi(\mathbf{v}) - \varphi(\mathbf{u}) - \langle \nabla\varphi(\mathbf{u}), \mathbf{v} - \mathbf{u}\rangle \quad \text{for } \mathbf{v} \in \mathbb{R}, \mathbf{u} \in \text{int } C;$$

in other words, the Bregman divergence measures the difference between the function $\varphi(\mathbf{v})$ and its first-order Taylor approximation $\varphi(\mathbf{u}) - \langle \nabla\varphi(\mathbf{u}), \mathbf{v} - \mathbf{u}\rangle$.

In general, the Bregman divergences are positive, strictly convex in the first argument and zero if and only if $\mathbf{x} = \mathbf{y}$. However, it is not necessarily symmetric in its arguments and also it may fail the triangle inequality.

For example, the strictly convex functions

$$\begin{array}{ll} \varphi_1(\mathbf{v}) = \|\mathbf{v}\|_2^2 & \varphi_2(\mathbf{v}) = -\sum_{k=1}^N \log v_k \\ \varphi_3(\mathbf{v}) = \sum_{k=1}^N v_k \log v_k & \varphi_4(\mathbf{v}) = \langle A\mathbf{v}, \mathbf{v}\rangle, \end{array}$$

(where A is a positive definite matrix), generate the Bregman divergences

$$\begin{array}{ll} D_{\varphi_1}(\mathbf{v}, \mathbf{u}) = \|\mathbf{v} - \mathbf{u}\|_2^2 & D_{\varphi_2}(\mathbf{v}, \mathbf{u}) = \sum_{k=1}^N \left[\frac{v_k}{u_k} - \log\left(\frac{v_k}{u_k}\right) - 1\right] \\ D_{\varphi_3}(\mathbf{v}, \mathbf{u}) = \sum_{k=1}^N v_k \log\left(\frac{v_k}{u_k}\right) & D_{\varphi_4}(\mathbf{v}, \mathbf{u}) = \langle A(\mathbf{v} - \mathbf{u}), \mathbf{v} - \mathbf{u}\rangle. \end{array}$$

Notice that $D_{\varphi_2}(v, u)$ (the so called Itakura–Saito divergence) is not a metric (being nonsymmetric). The Kullback–Leibler divergence $D_{\varphi_3}(\mathbf{v}, \mathbf{u})$ is associated to the negative Shannon entropy φ_3.

Bregman divergences are useful in connection with the miror descent, an iterative optimization algorithm for finding a local minimum of a differentiable function. See S. Bubeck [103] and Ch. L. Byrne [112] for an introduction.

The following application of the gradient descent to dynamical systems was mentioned by R. I. Boţ in a conference presented at the Transylvania University, Braşov, June 13, 2024. Let H be a real Hilbert space and consider the optimization problem

$$\min_{\mathbf{x} \in H} f(x),$$

where $f : H \to \mathbb{R}$ is an L-strongly smooth convex function such that $\arg\min_H f$ is assumed to be nonempty. Then the first-order dynamical system

$$\left\{ \begin{array}{rcl} \dot{x}(t) & = & -\nabla f(x(t)) \\ x(0) & = & x_0 \in H \end{array} \right.$$

admits a unique bounded trajectory $x : [0, \infty) \to H$ such that $\lim_{t\to\infty} \dot{x}(t) = 0$ and $x(t)$ converges weakly to a minimizer of f as $t \to \infty$. Moreover,

$$f(x(t)) - \min_H f = O(1/t) \quad \text{as } t \to \infty.$$

Explicit time discretization of this system leads to the gradient method

$$\frac{x_{k+1} - x_k}{\gamma} = -\nabla f(x_k), \quad k \geq 0,$$

equivalently,

$$x_{k+1} = x_k - \gamma \nabla f(x_k), \quad k \geq 0.$$

If $\gamma \in (0, \frac{1}{L}]$, then $(x_k)_k$ converges weakly to a minimizer of f and

$$f(x_k) - \min_H f = O(1/t) \quad \text{as } t \to \infty.$$

The details can be found in the paper of H. Attouch, R. I. Boţ, and D.-K. Nguyen [29], which considers the more general case of the perturbed dynamical systems.

It is worth mentioning here the paper of A. Chambolle and Ch. Dossal [127], which focuses on the following fast gradient algorithm.

Given $x_0 = x_1 \in H$, $0 < \gamma \leq 1/L$ and, for $\alpha \geq 3$, $t_k = \frac{k+\alpha-2}{\alpha-1}$ for $k \geq 1$, set for every $k \geq 1$,

$$y_k = x_k + \frac{t_k - 1}{t_{k+1}}(x_k - x_{k-1})$$
$$x_{k+1} = y_k - \gamma f(y_k).$$

Then $f(x_k) - \min_H f = O(1/k^2)$ as $k \to \infty$. Moreover, if $\alpha > 3$, then

$$f(x_k) - \min_H f = o(1/k^2) \quad \text{as } k \to \infty$$

and $(x_k)_k$ converges weakly to a minimizer of f.

Chapter 6

Convexity and Majorization

This chapter is aimed to offer a glimpse on the majorization theory and the beautiful inequalities associated to it. Introduced by G. H. Hardy, J. E. Littlewood and G. Pólya [246] in 1929, and popularized by their celebrated book on *Inequalities* [247], the relation of majorization has attracted along the time a big deal of attention not only from the mathematicians, but also from people working in various other fields such as statistics, economics, physics, signal processing, data mining, etc. Part of this research activity is summarized in the 900 pages of the recent book by A. W. Marshall, I. Olkin, and B. Arnold [363].

The discussion in this chapter is limited to some very basic aspects. However, due to the importance of the subject of majorization, we will come back to it in Chapter 9, by presenting a series of integral inequalities involving signed measures rather than positive measures.

6.1 The Hardy–Littlewood–Pólya Theory of Majorization

The main problem under attention in this section is to find necessary and sufficient conditions under which two families of real numbers, $x_1 \geq \cdots \geq x_N$ and $y_1 \geq \cdots \geq y_N$, accompanied by a family $(p_k)_{k=1}^N$ of positive weights, verify the inequality

$$\sum_{k=1}^{N} p_k f(x_k) \leq \sum_{k=1}^{N} p_k f(y_k) \tag{6.1}$$

for every real-valued convex function f defined on an interval that contains all numbers of x_k and y_k.

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C. P. Niculescu and L.-E. Persson, *Convex Functions and Their Applications*, CMS/CAIMS Books in Mathematics 14, https://doi.org/10.1007/978-3-031-71967-7_6

Notice that this problem is related to Jensen's inequality, which occurs in the particular case where

$$\sum_{k=1}^{N} p_k = 1 \text{ and } x_1 = \cdots = x_N = \sum_{k=1}^{N} p_k x_k.$$

Taking into account that the identity function and its opposite are convex functions, we infer that the inequality (6.1) imposes the equality

$$\sum_{k=1}^{N} p_k x_k = \sum_{k=1}^{N} p_k y_k. \tag{6.2}$$

Further, using the convexity of the functions $f = (x - y_k)^+$, we obtain

$$\begin{aligned} p_1 x_1 + \cdots + p_k x_k - (p_1 + \cdots + p_k) y_k &\le \sum_{j=1}^{N} f(x_j) \\ &\le \sum_{j=1}^{N} f(y_j) \le p_1 y_1 + \cdots + p_k y_k - (p_1 + \cdots + p_k) y_k, \end{aligned}$$

which provides a new set of necessary conditions:

$$\sum_{k=1}^{m} p_k x_k \le \sum_{k=1}^{m} p_k y_k \text{ for all } m \in \{1, ..., N-1\}. \tag{6.3}$$

Surprisingly, the conditions (6.2) and (6.3) are also sufficient for solving the aforementioned problem, even in the case of real weights. This fact, known as *Fuchs' generalization of the Hardy–Littlewood–Pólya inequality of majorization*, can be stated as follows:

6.1.1 Theorem *If $(x_k)_{k=1}^N$ and $(y_k)_{k=1}^N$ are two families of real numbers directed downwards,*

$$x_1 \ge \cdots \ge x_N \text{ and } y_1 \ge \cdots \ge y_N,$$

and $(p_k)_k$ is a family of real weights which fulfill the conditions (6.2) and (6.3), then

$$\sum_{k=1}^{N} p_k f(x_k) \le \sum_{k=1}^{N} p_k f(y_k) \tag{6.4}$$

for every convex function f whose domain of definition is an interval that contains all numbers x_k and y_k. Moreover, when the two families of real numbers $(x_k)_k$ and $(y_k)_k$ are directed upwards, the inequality (6.4) works in the reverse direction.

Proof. Without loss of generality, we may assume that $x_k \neq y_k$ for all indices k. Then, according to Abel's partial summation formula (see Theorem 2.4.5), we have

$$\begin{aligned}\sum_{k=1}^{N} p_k f(y_k) - \sum_{k=1}^{N} p_k f(x_k) &= \sum_{k=1}^{N} \left[p_k \, (y_k - x_k) \, \frac{f(y_k) - f(x_k)}{y_k - x_k} \right] \\ &= \sum_{k=1}^{N-1} \Big(\frac{f(y_k) - f(x_k)}{y_k - x_k} - \frac{f(y_{k+1}) - f(x_{k+1})}{y_{k+1} - x_{k+1}} \Big) \Big(\sum_{i=1}^{k} p_i y_i - \sum_{i=1}^{k} p_i x_i \Big) \\ &\qquad + \frac{f(y_N) - f(x_N)}{y_N - x_N} \Big(\sum_{i=1}^{N} p_i y_i - \sum_{i=1}^{N} p_i x_i \Big) \\ &= \sum_{k=1}^{N-1} \Big(\frac{f(y_k) - f(x_k)}{y_k - x_k} - \frac{f(y_{k+1}) - f(x_{k+1})}{y_{k+1} - x_{k+1}} \Big) \Big(\sum_{i=1}^{k} p_i y_i - \sum_{i=1}^{k} p_i x_i \Big) \geq 0\end{aligned}$$

due to our hypotheses (6.2) and (6.3) and the three chords inequality. See Remark 1.4.1.

The proof ends by noticing that in the case where $x_1 \leq \cdots \leq x_N$ and $y_1 \leq \cdots \leq y_N$, the three chords inequality implies

$$\frac{f(y_k) - f(x_k)}{y_k - x_k} - \frac{f(y_{k+1}) - f(x_{k+1})}{y_{k+1} - x_{k+1}} \leq 0.$$

■

An inspection of the argument of Theorem 6.1.1, easily yields *Fuchs' generalization of the Tomić-Weyl inequality of majorization*:

6.1.2 Theorem *If $(x_k)_{k=1}^N$ and $(y_k)_{k=1}^N$ are two families of real numbers directed downwards,*

$$x_1 \geq \cdots \geq x_N \text{ and } y_1 \geq \cdots \geq y_N,$$

and $(p_k)_k$ is a family of real weights which fulfill the inequalities

$$\sum_{k=1}^{m} p_k x_k \leq \sum_{k=1}^{m} p_k y_k \quad \text{for all } m \in \{1, ..., N\},$$

then

$$\sum_{k=1}^{N} p_k f(x_k) \leq \sum_{k=1}^{N} p_k f(y_k)$$

for every convex and increasing function f whose domain of definition is an interval that contains all numbers x_k and y_k. Moreover, when the two families of real numbers $(x_k)_k$ and $(y_k)_k$ are directed upwards, the last inequality works in the reverse direction.

G. H. Hardy, J. E. Littlewood and G. Pólya [246], [247] (as well as M. Tomić [567] and H. Weyl [584]) have considered only the unweighted case of their inequalities of majorization, that is, the case where all weights equal the unity. We recall them here by removing the unnecessary assumption on the continuity of the functions involved and noting that the monotonicity of only one of the two families of numbers is sufficient.

6.1.3 Theorem (The Hardy–Littlewood–Pólya inequality of majorization) *Let f be a convex function defined on an interval I and let $\mathbf{x} = (x_k)_{k=1}^N$ and $\mathbf{y} = (y_k)_{k=1}^n$ be two families of numbers in I such that*

$$\sum_{k=1}^{m} x_k \leq \sum_{k=1}^{m} y_k \quad \text{for } m = 1, \ldots, N-1 \tag{6.5}$$

and

$$\sum_{k=1}^{N} x_k = \sum_{k=1}^{N} y_k. \tag{6.6}$$

If $x_1 \geq \cdots \geq x_N$, then

$$\sum_{k=1}^{N} f(x_k) \leq \sum_{k=1}^{N} f(y_k), \tag{6.7}$$

while if $y_1 \leq \cdots \leq y_N$, then the last inequality works in the reverse direction.

Proof. Clearly, when $x_1 \geq \cdots \geq x_N$ and the hypotheses (6.5) and (6.6) hold, they will continue to work when $\mathbf{y}$ is replaced by the vector obtained from $\mathbf{y}$ by rearranging its components in decreasing order. Thus the conclusion is a direct consequence of Theorem 6.1.1.

Alternatively, one can avoid the use of Theorem 6.1.1, by taking into account the property of subdifferentiability of convex functions. For this, rearrange $\mathbf{y}$ as above and notice that we may assume that $x_k \neq y_k$ for all k. According to the hypotheses (6.5) and (6.6), it follows that $x_1 < y_1$ and $x_N > y_N$, so all points x_k are interior to I. By Theorem 2.1.2, we infer that

$$\begin{aligned}
\sum_{k=1}^{N} f(y_k) - \sum_{k=1}^{N} f(x_k) &= \sum_{k=1}^{N} \left(f(y_k) - f(x_k)\right) \geq \sum_{k=1}^{N} f'_+(x_k)\,(y_k - x_k) \\
&= \sum_{m=1}^{N-1} \left(f'_+(x_m) - f'_+(x_{m+1})\right) \left(\sum_{k=1}^{m} (y_k - x_k)\right) + f'_+(x_N) \sum_{k=1}^{N} (y_k - x_k) \\
&= \sum_{m=1}^{N-1} \left(f'_+(x_m) - f'_+(x_{m+1})\right) \left(\sum_{k=1}^{m} (y_k - x_k)\right)
\end{aligned}$$

and the proof ends by noticing that the right derivative f'_+ of the convex function f is increasing on the interior of I. See Theorem 1.4.2.

The case when $y_1 \leq \cdots \leq y_N$ can be treated in a similar way, by replacing $\mathbf{x}$ by the vector obtained from it by rearranging its components in increasing order. ■

See [429] for an order-theoretical generalization of the Hardy–Littlewood–Pólya inequality. A sample is offered by Exercise 5 in Section 6.5.

An immediate consequence of Theorem 6.1.3 is as follows:

6.1.4 Corollary (Truncated majorization) *Let $f : [0, \infty) \to \mathbb{R}$ be an increasing concave function and let $x_1, x_2, \ldots, x_n, y_1, y_2, ..., y_m$ $(2 \leq m \leq n)$ be nonnegative numbers such that*

$$\begin{gathered}\max\{x_1, x_2, ..., x_n\} \leq \max\{y_1, y_2, ..., y_m\},\\ \max\{x_{i_1} + x_{i_2} : i_1 \neq i_2\} \leq \max\{y_{j_1} + y_{j_2} : j_1 \neq j_2\}\\ \ldots\\ \max\left\{\sum_{p=1}^{k} x_{i_p} : i_r \neq i_s\right\} \leq \max\left\{\sum_{p=1}^{k} y_{j_p} : j_r \neq j_s\right\}\end{gathered}$$

for $k \leq m$ and

$$\sum_{i=1}^{n} x_i \geq \sum_{j=1}^{m} y_j.$$

Then

$$\sum_{i=1}^{n} f(x_i) \geq \sum_{j=1}^{m} f(y_j) + (n-m)f(0).$$

For the proof, adapt the argument of Lemma 2.6.14, Section 2.6.

The argument of Theorem 6.1.3 also yields the following result:

6.1.5 Theorem (The Tomić-Weyl inequality of weak majorization [567], [584]). *Let f be a convex and increasing function defined on an interval I and let $\mathbf{x} = (x_k)_{k=1}^N$ and $\mathbf{y} = (y_k)_{k=1}^N$ be two families of numbers in I such that*

$$\sum_{k=1}^{m} x_k \leq \sum_{k=1}^{m} y_k \quad \text{for } m = 1, \ldots, N.$$

If $x_1 \geq \cdots \geq x_N$, then

$$\sum_{k=1}^{N} f(x_k) \leq \sum_{k=1}^{N} f(y_k),$$

while if $y_1 \leq \cdots \leq y_N$, then the last inequality works in the reverse direction.

See Exercise 8 for a generalization of the Tomić-Weyl inequality.

We are now in a position to indicate the precise definition of the notion of majorization.

Given a vector $\mathbf{x} = (x_1, \dots, x_N) \in \mathbb{R}^N$, we denote by $\mathbf{x}^{\downarrow} = (x_1^{\downarrow}, \dots, x_N^{\downarrow})$ the vector obtained from $\mathbf{x}$ by rearranging its components in decreasing order,

$$x_1^{\downarrow} \geq \cdots \geq x_N^{\downarrow}.$$

In a similar way we may introduce the vector $\mathbf{x}^{\uparrow} = (x_1^{\uparrow}, \dots, x_N^{\uparrow})$, obtained from $\mathbf{x}$ by rearranging its components in increasing order.

6.1.6 Definition *Given two vectors* $\mathbf{x}$ *and* $\mathbf{y}$ *in* $\mathbb{R}^N$*, we say that* $\mathbf{x}$ *is* weakly majorized *by* $\mathbf{y}$ (*denoted* $\mathbf{x} \prec_{HLPw} \mathbf{y}$) *if*

$$\sum_{i=1}^{k} x_i^{\downarrow} \leq \sum_{i=1}^{k} y_i^{\downarrow} \quad \text{for } k = 1, \dots, N$$

and that $\mathbf{x}$ *is* majorized *by* $\mathbf{y}$ (*denoted* $\mathbf{x} \prec_{HLP} \mathbf{y}$) *if in addition*

$$\sum_{i=1}^{N} x_i^{\downarrow} = \sum_{i=1}^{N} y_i^{\downarrow}.$$

When considering the similar relations for vectors rearranged in increasing order, one obtains respectively the corresponding relations denoted $\mathbf{x} \prec^*_{HLPw} \mathbf{y}$ *and* $\mathbf{x} \prec^*_{HLP} \mathbf{y}$.

Notice that $\prec_{HLP}$ provides a partial order on $\mathbb{R}^N$ and the initial order of the components in the vectors does not play any role in terms of majorization.

As it will be shown in Theorem 6.1.12, the fact that $\mathbf{x} \prec_{HLP} \mathbf{y}$ means geometrically that the components of $\mathbf{x}$ spread out less than those of $\mathbf{y}$.

In information theory, the relation $\mathbf{p} \prec_{HLP} \mathbf{q}$ (for $\mathbf{p}$ and $\mathbf{q}$ probability distributions on N outcomes) implies that $\mathbf{p}$ is more *disordered* than $\mathbf{q}$. Indeed, according to Theorem 6.1.3,

$$H(\mathbf{p}) = -\sum_{k=1}^{N} p_k \log_2 p_k \geq H(\mathbf{q}) = -\sum_{k=1}^{N} q_k \log_2 q_k,$$

that is, the Shannon entropy of $\mathbf{q}$ does not exceed the Shannon entropy of $\mathbf{p}$. Since the converse does not work (that is, in general the inequality $H(\mathbf{p}) \geq H(\mathbf{q})$ does not imply $\mathbf{p} \prec_{HLP} \mathbf{q}$), this suggests that the majorization theory could offer stronger criteria for the measurement of disorder in a system than the entropic inequalities. Indeed, this is the case, and the details can be found in the survey of M. A. Nielsen [450] and the monograph of M. A. Nielsen and I. L. Chuang [451].

6.1.7 Remark *According to the discussion at the beginning of this section, if* $\mathbf{x}$ *and* $\mathbf{y}$ *are two vectors in* $\mathbb{R}^N$*, for which the inequality (6.7) holds for every convex* (*respectively convex and increasing*) *function defined on an interval including the components of* $\mathbf{x}$ *and* $\mathbf{y}$*, then* $\mathbf{x} \prec_{HLP} \mathbf{y}$ (*respectively* $\mathbf{x} \prec_{HLPw} \mathbf{y}$).

6.1.8 Remark *The fact that the Tomić-Weyl inequality of weak majorization was obtained as a consequence of the Hardy–Littlewood–Pólya inequality of majorization is not an accident. Indeed, as was noticed by G. Pólya* [477]*, if*

$$(x_1, \dots, x_N) \prec_{HLPw} (y_1, \dots, y_N),$$

then there exist real numbers x_{N+1} *and* y_{N+1} *such that*

$$(x_1, \dots, x_N, x_{N+1}) \prec_{HLP} (y_1, \dots, y_N, y_{N+1}).$$

To check this, choose

$$x_{N+1} = \min\{x_1, \dots, x_N, y_1, \dots, y_N\} \text{ and } y_{N+1} = \sum_{k=1}^{N+1} x_k - \sum_{k=1}^{N} y_k.$$

W. Arveson and R. Kadison have found another reduction of weak majorization to majorization. Let $x_1 \geq \cdots \geq x_N$ *and* $y_1 \geq \cdots \geq y_N$ *be two decreasing sequences of positive real numbers such that*

$$x_1 + \cdots + x_k \leq y_1 + \cdots + y_k \quad \text{for } k = 1, ..., N.$$

Then there is a decreasing sequence $\bar{y}_1 \geq \cdots \geq \bar{y}_N$ *such that* $0 \leq \bar{y}_k \leq y_k$ *for all* k *and*

$$(x_1, \dots, x_N) \prec_{HLP} (\bar{y}_1, \dots, \bar{y}_N).$$

See [25]*, Lemma* 4.3.

The aforementioned book [247] of Hardy, Littlewood and Pólya also includes a description of the relation of majorization by averaging means, based on the doubly stochastic matrices. Recall that a matrix $P \in \mathrm{M}_N(\mathbb{R})$ is doubly stochastic if P has positive entries and each row and each column sums to unity. A special class of doubly stochastic matrices is that of *T-transformations*. They have the form

$$T = \lambda I + (1 - \lambda)Q$$

where $0 \leq \lambda \leq 1$ and Q is a permutation mapping which interchanges two coordinates, that is,

$$Tx = \big(x_1, \dots, x_{j-1}, \lambda x_j + (1-\lambda)x_k, x_{j+1}, \dots, x_{k-1}, \\ \lambda x_k + (1-\lambda)x_j, x_{k+1}, \dots, x_N\big).$$

6.1.9 Theorem *Let* $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$*. Then the following assertions are equivalent:*

(a) $\mathbf{x} \prec_{HLP} \mathbf{y}$;

(b) $\mathbf{x} = P\mathbf{y}$ *for a suitable doubly stochastic matrix* $P \in \mathrm{M}_N(\mathbb{R})$;

(c) $\mathbf{x}$ *can be obtained from* $\mathbf{y}$ *by successive applications of finitely many T-transformations.*

The implication (b) $\Rightarrow$ (a) is due to I. Schur [526] and constituted the starting point for this theorem.

Proof. (c) $\Rightarrow$ (b) Since T-transformations are doubly stochastic, the product of T-transformations is a doubly stochastic transformation.

The implication (b) $\Rightarrow$ (a) is a consequence of Remark 6.1.7. Assume that $P = (p_{jk})_{j,k=1}^N$ and consider an arbitrary convex function f defined on an interval including the components of $\mathbf{x}$ and $\mathbf{y}$. Since $x_k = \sum_j y_j p_{jk}$, and $\sum_j p_{jk} = 1$ for all indices k, it follows from Jensen's inequality that

$$f(x_k) \leq \sum_{j=1}^N p_{jk} f(y_j).$$

Then

$$\sum_{k=1}^N f(x_k) \leq \sum_{k=1}^N \left(\sum_{j=1}^N p_{jk} f(y_j) \right) = \sum_{j=1}^N \left(\sum_{k=1}^N p_{jk} f(y_j) \right) = \sum_{j=1}^N f(y_j)$$

and it remains to apply Remark 6.1.7.

(a) $\Rightarrow$ (c) Let $\mathbf{x}$ and $\mathbf{y}$ be two distinct vectors in $\mathbb{R}^N$ such that $\mathbf{x} \prec_{HLP} \mathbf{y}$. Since permutations are T-transformations, we may assume that their components verify the conditions

$$x_1 \geq x_2 \geq \cdots \geq x_N \quad \text{and} \quad y_1 \geq y_2 \geq \cdots \geq y_N.$$

Let j be the largest index such that $x_j < y_j$ and let k be the smallest index such that $k > j$ and $x_k > y_k$. The existence of such a pair of indices is motivated by the fact that the largest index i with $x_i \neq y_i$ verifies $x_i > y_i$. Then

$$y_j > x_j \geq x_k > y_k.$$

Put $\varepsilon = \min\{y_j - x_j, x_k - y_k\}$, $\lambda = 1 - \varepsilon/(y_j - y_k)$ and

$$\mathbf{y}^* = (y_1, \ldots, y_{j-1}, y_j - \varepsilon, y_{j+1}, \ldots, y_{k-1}, y_k + \varepsilon, y_{k+1}, \ldots, y_N).$$

Clearly, $\lambda \in (0, 1)$. Denoting by Q the permutation matrix which interchanges the components of order j and k, we see that $\mathbf{y}^* = T\mathbf{y}$ for the representation

$$T = \lambda I + (1 - \lambda)Q.$$

According to the implication $(b) \Rightarrow (a)$, it follows that $\mathbf{y}^* \prec_{HLP} \mathbf{y}$. On the other hand, $\mathbf{x} \prec_{HLP} \mathbf{y}^*$. In fact,

$$\begin{aligned}
\sum_{r=1}^s y_r^* = \sum_{r=1}^s y_r \geq \sum_{r=1}^s x_r \quad & \text{for } s = 1, \ldots, j-1, \\
y_j^* \geq x_j \quad \text{and} \quad y_r^* = y_r \quad & \text{for } r = j+1, \ldots, k-1, \\
\sum_{r=1}^s y_r^* = \sum_{r=1}^s y_r \geq \sum_{r=1}^s x_r \quad & \text{for } s = k+1, \ldots, N
\end{aligned}$$

and

$$\sum_{r=1}^{N} y_r^* = \sum_{r=1}^{N} y_r = \sum_{r=1}^{N} x_r.$$

Letting $d(\mathbf{u}, \mathbf{v})$ be the number of indices r such that $u_r \neq v_r$, it is clear that $d(\mathbf{x}, \mathbf{y}^*) \leq d(\mathbf{x}, \mathbf{y}) - 1$, so by repeating the above algorithm (at most) $N - 1$ times, we arrive at $\mathbf{x}$. ■

The Hardy–Littlewood–Pólya inequality of majorization admits a generalization (due to S. Sherman [533]) to the case where the vectors $\mathbf{x}$ and $\mathbf{y}$ are not necessarily in the same vector space. This will be presented in Section 6.8, in the more general context of vector majorization. For functions defined on intervals, it reduces to the following result.

6.1.10 Theorem *Suppose that I is an interval of $\mathbb{R}$ and consider the vectors $\mathbf{a} = (a_1, ..., a_m) \in \mathbb{R}_+^m$, $\mathbf{b} = (b_1, ..., b_n) \in \mathbb{R}_+^n$, $\mathbf{x} = (x_1, ..., x_m) \in I^m$ and $\mathbf{y} = (y_1, ..., y_n) \in I^n$. Then*

$$\sum\nolimits_{i=1}^{n} b_i f(y_i) \leq \sum\nolimits_{j=1}^{m} a_j f(x_j)$$

for every convex function $f : I \to \mathbb{R}$ if and only if there exists an $n \times m$ dimensional matrix $S = (s_{ij})_{i,j}$ with positive entries such that

$$\mathbf{y} = S\mathbf{x}, \quad \mathbf{a} = S^T\mathbf{b}$$

and

$$\sum\nolimits_{j=1}^{m} s_{ij} = 1 \text{ for every } i.$$

An elegant proof of this result has been presented by G. Bennett [52], Lemma 6.1, p. 894.

The next two results need some preparation concerning the action of the permutation group on the Euclidean space.

The permutation group of order N is the group $\Pi(N)$ of all bijective functions from $\{1, ..., N\}$ onto itself. This group acts on $\mathbb{R}^N$ via the map $\Psi : \Pi(N) \times \mathbb{R}^N \to \mathbb{R}^N$, defined by the formula

$$\Psi(\pi, \mathbf{x}) = \pi\mathbf{x} = (x_{\pi(1)}, ..., x_{\pi(N)}).$$

The orbits of this action, that is, the sets of the form $\mathcal{O}(\mathbf{x}) = \{\pi\mathbf{x} : \pi \in \Pi(N)\}$, play an important role in majorization theory.

6.1.11 Definition *A subset C of $\mathbb{R}^N$ is called* invariant under permutations *(or $\Pi(N)$-invariant)* if $\pi\mathbf{x} \in C$ *whenever* $\pi \in \Pi(N)$ *and* $\mathbf{x} \in C$. *Accordingly, a function F defined on a $\Pi(N)$-invariant subset C is called* $\Pi(N)$-invariant *(or invariant under permutations)* if $F(\pi\mathbf{x}) = F(\mathbf{x})$ *whenever* $\pi \in \Pi(N)$ *and* $\mathbf{x} \in C$.

For example, all elementary symmetric functions (as well as all norms of index $p \in [1, \infty]$) are invariant under permutations. Moreover, to every convex function $\varphi : \mathbb{R}^N \to \mathbb{R}$ one can attach a convex function $\varphi_\Pi : \mathbb{R}^N \to \mathbb{R}$ invariant under permutations via the formula

$$\varphi_\Pi(\mathbf{x}) = \sum_{\pi \in \Pi(N)} \varphi(\pi \mathbf{x}).$$

A geometric insight into majorization was revealed by R. Rado, who noticed that $\mathbf{x} \prec_{HLP} \mathbf{y}$ means that the components of $\mathbf{x}$ spread out less than those of $\mathbf{y}$ in the sense that $\mathbf{x}$ lies in the convex hull of the $N!$ permutations of y.

6.1.12 Theorem (R. Rado [493]) $\mathbf{x} \prec_{HLP} \mathbf{y}$ *in* $\mathbb{R}^N$ *if and only if* $\mathbf{x}$ *belongs to the convex hull of the* $N!$ *permutations of* $\mathbf{y}$. *Therefore,*

$$\{\mathbf{x} : \mathbf{x} \prec_{HLP} \mathbf{y}\} = \operatorname{conv}\{\pi \mathbf{y} : \pi \in \Pi(N)\}.$$

Proof. According to Theorem 6.1.9, if $\mathbf{x} \prec_{HLP} \mathbf{y}$, then $\mathbf{x} = P\mathbf{y}$ for some doubly stochastic matrix. Taking into account Birkhoff's Theorem 3.3.6, P can be represented as a convex combination $P = \sum_\pi \lambda_\pi P_\pi$ of the $N!$ permutation matrices P_π. Then

$$\mathbf{x} = \sum_\pi \lambda_\pi P_\pi(\mathbf{y}) \in \operatorname{conv}\{P_\pi(\mathbf{y}) : \pi \in \Pi(N)\}.$$

Conversely, if $\mathbf{x} \in \operatorname{conv}\{P_\pi(\mathbf{y}) : \pi \in \Pi(N)\}$, then $\mathbf{x}$ admits a convex representation of the form $\mathbf{x} = \sum_\pi \lambda_\pi P_\pi(\mathbf{y})$, whence $\mathbf{x} = \left(\sum_\pi \lambda_\pi P_\pi\right)(\mathbf{y})$. ■

Remarkably, the relation of majorization gives rise to inequalities of the type (6.7) not only for the continuous convex functions of the form

$$F(x_1, ..., x_N) = \sum_{k=1}^N f(x_k),$$

but also for all quasiconvex functions $F(x_1, ..., x_N)$ which are invariant under the action of the permutation group.

6.1.13 Theorem (I. Schur [526]) *If* C *is a convex set in* $\mathbb{R}^N$ *invariant under permutations and* $F : C \to \mathbb{R}$ *is a quasiconvex function invariant under permutations, then*

$$\mathbf{x} \prec_{HLP} \mathbf{y} \text{ implies } F(\mathbf{x}) \leq F(\mathbf{y}).$$

Proof. Indeed, according to Theorem 6.1.12,

$$\begin{aligned} F(\mathbf{x}) &\leq \sup\{F(\mathbf{u}) : \mathbf{u} \in \operatorname{conv}\{\pi \mathbf{y} : \pi \in \Pi(N)\}\} \\ &= \sup\{F(\pi \mathbf{y}) : \pi \in \Pi(N)\} = F(\mathbf{y}). \end{aligned}$$

■

Two simple examples of quasiconvex functions invariant under permutations that are not convex are

$$\log\left(\sum\nolimits_{k=1}^{N} x_k\right) \text{ and } \frac{x_1^\alpha + \cdots + x_N^\alpha}{\sqrt[N]{x_1 \cdots x_N}} \quad \text{for } \alpha > 0 \text{ and } x_1, \ldots, x_N > 0.$$

For more examples, notice that if f and g are two functions defined on a convex set $C \subset \mathbb{R}^N$ such that f is positive and convex and g is strictly positive and concave, then f/g is quasiconvex.

An illustration of Theorem 6.1.13 is offered by the following result due to R. F. Muirhead [400]:

6.1.14 Theorem (Muirhead's inequality) *If* $\mathbf{x}$ *and* $\mathbf{y}$ *are two vectors in* $\mathbb{R}^N$ *such that* $\mathbf{x} \prec_{HLP} \mathbf{y}$ *and* $\alpha_1, \ldots, \alpha_N$ *are strictly positive numbers, then*

$$\sum_{\pi \in \Pi(N)} \alpha_{\pi(1)}^{x_1} \cdots \alpha_{\pi(N)}^{x_N} \leq \sum_{\pi \in \Pi(N)} \alpha_{\pi(1)}^{y_1} \cdots \alpha_{\pi(N)}^{y_N}, \tag{6.8}$$

the sum being taken over all permutations π *of the set* $\{1, \ldots, N\}$.

Actually, Muirhead has considered only the case where $\mathbf{x}$ and $\mathbf{y}$ have positive integer components; the extension to the case of real components is due to G. H. Hardy, J. E. Littlewood and G. Pólya [247].

Proof. Put $\mathbf{w} = (\log a_1, ..., \log a_N)$. Then we have to prove that

$$\sum_{\pi \in \Pi(N)} \mathrm{e}^{\langle \mathbf{x}, \pi \mathbf{w} \rangle} \leq \sum_{\pi \in \Pi(N)} \mathrm{e}^{\langle \mathbf{y}, \pi \mathbf{w} \rangle}.$$

This follows from Theorem 6.1.13 because the function $\mathbf{u} \to \sum_{\pi \in \Pi(N)} \mathrm{e}^{\langle \mathbf{u}, \pi \mathbf{w} \rangle}$ is convex and invariant under permutations. ■

The converse of Theorem 6.1.14 also works: *If the inequality* (6.8) *is valid for all* $\alpha_1, \ldots, \alpha_N > 0$, *then* $\mathbf{x} \prec_{HLP} \mathbf{y}$. Indeed, the case where $\alpha_1 = \cdots = \alpha_N > 0$ gives us

$$\alpha_1^{\sum_{k=1}^N x_k} \leq \alpha_1^{\sum_{k=1}^N y_k},$$

so that $\sum_{k=1}^N x_k = \sum_{k=1}^N y_k$ since $\alpha_1 > 0$ is arbitrary. Denote by $\mathcal{P}$ the set of all subsets of $\{1, \ldots, N\}$ of size k and take $\alpha_1 = \cdots = \alpha_k > 1$ and $\alpha_{k+1} = \cdots = \alpha_N = 1$. By our hypotheses,

$$\sum_{S \in \mathcal{P}} \alpha_1^{\sum_{i \in S} x_i} \leq \sum_{S \in \mathcal{P}} \alpha_1^{\sum_{i \in S} y_i}.$$

If $\sum_{j=1}^k x_j^\downarrow > \sum_{j=1}^k y_j^\downarrow$, this leads to a contradiction for α_1 large enough. Thus $\mathbf{x} \prec_{HLP} \mathbf{y}$.

For applications of Muirhead's inequality, see Exercises 3 and 5.

Exercises

1. Infer from the Hardy–Littlewood–Pólya inequality of majorization the following inequality due to M. Petrović: if f is a convex function defined on $[0,\infty)$, then
$$\sum_{k=1}^{N} f(x_k) \le f\left(\sum_{k=1}^{N} x_k\right) + (N-1)f(0)$$
for all $x_1,\dots,x_N \ge 0$. As an application show that
$$\sum_{1\le i<j\le N} x_i x_j \left(x_i^2 + x_j^2\right) \le \frac{1}{8}\left(\sum_{k=1}^{N} x_k\right)^4,$$
whenever $x_1, x_2, ..., x_N \ge 0$ with $N \ge 2$.

2. (R. Bellman) Suppose that $a_1 \ge a_2 \ge \cdots \ge a_N > 0$ and f is a convex function defined on $[0, a_1]$ such that $\left((-1)^N + 1\right) f(0) \le 0$. Prove that
$$\sum_{k=1}^{N} (-1)^{k-1} f(a_k) \ge f\Big(\sum_{k=1}^{N} (-1)^{k-1} a_k\Big).$$

3. Use the relation $(1/N, 1/N, \dots, 1/N) \prec_{HLP} (1, 0, \dots, 0)$ to infer the *AM-GM* inequality from the Muirhead inequality of majorization.

4. Prove that a positive matrix $P \in \mathrm{M}_N(\mathbb{R})$ is doubly stochastic if, and only if, $P\mathbf{x} \prec_{HLP} \mathbf{x}$ for every $\mathbf{x} \in \mathbb{R}^N$.

5. Use Muirhead's inequality of majorization to prove that every completely monotonic function f verifies *Fink's inequalities* [196],
$$(-1)^{nk}(f^{(k)}(x))^n \le (-1)^{nk}(f^{(n)}(x))^k (f(x))^{n-k}$$
for all $x > 0$ and all integers n, k with $n \ge k \ge 0$.

   [*Hint:* See the integral representation of completely monotonic functions provided by the formula (1.15), Section 1.3.]

   Remark. A. M. Fink [196] also noticed that this inequality yields similar inequalities for moments: if μ is a positive Borel measure that admits moments of all orders,
$$\mu_n = \int_0^1 x^n \mathrm{d}\mu(x), \quad n = 0, 1, 2, ...,$$
then $\prod_{k=1}^{N} \mu_{\alpha_k} \le \prod_{k=1}^{N} \mu_{\beta_k}$, provided that $(\alpha_1, ..., \alpha_N) \prec_{HLP} (\beta_1, ..., \beta_N)$.

6. (A rearrangement-free characterization of majorization). Adapt the proof of the Hardy–Littlewood–Pólya inequality of majorization to show that $(x_1, \dots, x_N) \prec_{HLP} (y_1, \dots, y_N)$ if and only if $\sum_{k=1}^{N} x_k = \sum_{k=1}^{N} y_k$ and
$$\sum_{k=1}^{N} (x_k - t)^+ \le \sum_{k=1}^{N} (y_k - t)^+ \text{ for all } t \in \mathbb{R}.$$

Here the positive part can be replaced by the absolute value function.

7. (Q. J. Zhu; see [92]) Prove that $(x_1, \ldots, x_N) \prec_{HLP} (y_1, \ldots, y_N)$ if and only if for every $(z_1, ..., z_N) \in \mathbb{R}^N$ we have

$$\sum_{k=1}^{N} x_k^{\downarrow} z_k^{\downarrow} \leq \sum_{k=1}^{N} y_k^{\downarrow} z_k^{\downarrow}.$$

[*Hint:* Use Abel's partial summation formula.]

8. (A generalization of the Tomić-Weyl inequality of majorization [430]) Suppose that $f : [0, \infty) \to \mathbb{R}$ is a continuous function. Prove that the following two assertions are equivalent:

   (a) f is an increasing convex function such that $f(0) = 0$;

   (b) for all integer numbers $2 \leq m \leq n$ and all strings $x_1, \ldots, x_n$ and $y_1, \ldots, y_m$ of positive numbers such that

$$\max\left\{\sum_{p=1}^{k} x_{i_p} : i_r \neq i_s\right\} \leq \max\left\{\sum_{p=1}^{k} y_{j_p} : j_r \neq j_s\right\}$$

   for $k = 1, ..., m$ and

$$\sum_{k=1}^{n} x_k \leq \sum_{k=1}^{m} y_k,$$

   we have

$$\sum_{k=1}^{n} f(x_k) \leq \sum_{k=1}^{m} f(y_k).$$

   The assertion (b) in the variant where $m = n \geq 2$, is equivalent to the fact that f is increasing and convex. See Exercise 4 in Section 6.5, for an application leading to a matrix analog of Popoviciu's inequality.

9. (S. Karlin and A. Novikoff [288]) Let $n \geq 0$ be an integer number and consider a finite real sequence $\{a_0, ..., a_n\}$. Prove that the conditions

$$\sum_{k=0}^{n} a_k = 0$$
$$\sum_{k=0}^{n} k a_k = 0$$
$$\sum_{j=0}^{k} \sum_{i=0}^{j} a_i \geq 0 \text{ for } k = 0, ..., n$$

   are necessary and sufficient in order that

$$\sum_{k=0}^{n} a_k \varphi(k) \geq 0$$

   for every convex sequence $\varphi(0), ..., \varphi(n)$.

10. (S V. Astashkin, K.V. Lykov, and M. Milman [27]) Suppose that $\Phi : [0,\infty) \to \mathbb{R}$ is an increasing function and $\Psi : [0,\infty) \to \mathbb{R}$ is an increasing convex function. Prove that for every pair of positive and decreasing sequences $(x_n)_n$ and $(y_n)_n$, for which there exists n_0 such that $x_n \geq y_n$ if $n \leq n_0$ and $x_n \leq y_n$ if $n > n_0$ we have

$$\sum_{k=1}^{\infty} \Phi(|x_k|) \geq \sum_{k=1}^{\infty} \Phi(|y_k|)$$

and

$$\sum_{k=1}^{\infty} \Psi\left(\Phi(|x_k|)\right) \geq \sum_{k=1}^{\infty} \Psi\left(\Phi(|y_k|)\right).$$

6.2 Two Applications to Linear Algebra

A well known result in linear algebra asserts that the trace of a matrix equals the sum of its eigenvalues. What more can be said about the possible diagonal entries of the real symmetric matrices having a fixed set of eigenvalues? The answer to this problem is given by the following result:

6.2.1 Theorem (The Schur–Horn theorem) *Suppose that*

$$\mathbf{d} = (d_1, ..., d_N) \text{ and } \lambda = (\lambda_1, ..., \lambda_N)$$

are two vectors in $\mathbb{R}^N$. *Then there is a real symmetric matrix with diagonal entries* $d_1, ..., d_N$ *and eigenvalues* $\lambda_1, ..., \lambda_N$ *if and only if* $\mathbf{d} \prec_{HLP} \lambda$.

Schur's contribution was the striking remark concerning the implication of majorization to this matter.

6.2.2 Lemma (I. Schur [526]) *Let* $A \in \mathrm{M}_N(\mathbb{R})$ *be a self-adjoint matrix with diagonal elements* $a_{11}, \ldots, a_{NN}$ *and eigenvalues* $\lambda_1, \ldots, \lambda_N$. *Then*

$$(a_{11}, \ldots, a_{NN}) \prec_{HLP} (\lambda_1, \ldots, \lambda_N).$$

Proof. By the spectral decomposition theorem, $A = UDU^*$, where $U = (u_{kj})_{k,j}$ is orthogonal and D is diagonal, with diagonal entries $\lambda_1, \ldots, \lambda_N$. The diagonal elements of A are

$$a_{kk} = \langle A\mathbf{e}_k, \mathbf{e}_k \rangle = \sum_{j=1}^{N} \lambda_j u_{kj}^2 = \sum_{j=1}^{N} p_{kj}\lambda_j,$$

where $p_{kj} = u_{kj}^2$; as usually, $\mathbf{e}_1, ..., \mathbf{e}_N$ denote the natural basis of $\mathbb{R}^N$. Since U is orthogonal, the matrix $P = (p_{kj})_{k,j}$ is doubly stochastic and Theorem 6.1.9 applies. ■

Since the function log is concave, from Theorem 6.1.3 and Lemma 6.2.2 we infer the following inequality:

6.2.3 Corollary (Hadamard's determinant inequality) *If A is an $N\times N$-dimensional positive matrix with diagonal elements $a_{11},\dots,a_{NN}$ and eigenvalues $\lambda_1,\dots,\lambda_N$, then*

$$\prod_{k=1}^{N} a_{kk} \geq \prod_{k=1}^{N} \lambda_k.$$

See Exercise 1 for a more general result.

A. Horn [260] has proved a converse to Lemma 6.2.2, which led to the statement of Theorem 6.2.1:

6.2.4 Lemma *If $\mathbf{x}$ and $\mathbf{y}$ are two vectors in $\mathbb{R}^N$ such that $\mathbf{x} \prec_{HLP} \mathbf{y}$, then there exists a symmetric matrix A such that the entries of $\mathbf{x}$ are the diagonal elements of A and the entries of $\mathbf{y}$ are the eigenvalues of A.*

Proof. We follow here the argument of W. Arveson and R. Kadison [25].

Step 1: If $A = (a_{ij})_{i,j} \in \mathrm{M}_N(\mathbb{R})$ is a symmetric matrix with diagonal $\mathbf{d}$, then for every T-transform T there exists a unitary matrix U such that UAU^* has diagonal $T\mathbf{d}$.

Indeed, suppose that $T = (1-\cos^2\theta)I + \left(\sin^2\theta\right) P_\pi$, where π is a permutation that interchanges i_0 and j_0. Then the matrix $U = (u_{ij})_{i,j}$, obtained by modifying four entries of the identity matrix as follows,

$$\begin{aligned} u_{i_0 i_0} &= \mathrm{i}\sin\theta, \quad u_{i_0 j_0} = -\cos\theta, \\ u_{j_0 i_0} &= \mathrm{i}\cos\theta, \quad u_{j_0 j_0} = \sin\theta, \end{aligned}$$

is unitary and a straightforward computation shows that the diagonal of UAU^* is equal to $(1-\cos^2\theta)d + \left(\sin^2\theta\right) P_\pi(d)$.

Step 2: Let $\Lambda = \mathrm{Diag}(\mathbf{y})$. By Theorem 6.1.9, $\mathbf{x}$ can be obtained from $\mathbf{y}$ by successive applications of finitely many T -transformations,

$$\mathbf{x} = T_m T_{m-1} \cdots T_1 \mathbf{y}.$$

By Step 1, there is a unitary matrix U_1 such that $U_1 \Lambda U_1^*$ has diagonal $T_1\mathbf{y}$. Similarly, there is a unitary matrix U_2 such that $U_2 \left(U_1 \Lambda U_1^*\right) U_2^*$ has diagonal $T_2(T_1\mathbf{y})$. Iterating this argument, we obtain a self-adjoint matrix

$$A = \left(U_m U_{m-1} \cdots U_1\right) \Lambda \left(U_m U_{m-1} \cdots U_1\right)^*$$

whose diagonal elements are the entries of $\mathbf{x}$ and the eigenvalues are the entries of $\mathbf{y}$. ■

The Schur–Horn theorem (when combined with Theorem 6.1.12) offers an illustration of Kostant's convexity theorem. See [305]. Indeed, if we fix an N-tuple α of real numbers and denote by $\mathcal{O}_\alpha$ the set of all symmetric matrices in $\mathrm{M}_N(\mathbb{R})$ with eigenvalues α, then the range of the map $\Phi\colon \mathcal{O}_\alpha \to \mathbb{R}^N$, that takes a matrix to its diagonal, is a convex polyhedron (whose vertices are the $N!$ permutations of α). See M. F. Atiyah [28] for a large generalization and a surprising link between mechanics, Lie group theory, and spectra of matrices.

Our next goal is to investigate how the eigenvalues of a matrix are related to its singular numbers. For this we need the following multiplicative analogue of the Hardy–Littlewood–Pólya inequality of majorization:

6.2.5 Theorem *Suppose that $x_1 \geq x_2 \geq \cdots \geq x_n$ and $y_1 \geq y_2 \geq \cdots \geq y_n$ are two families of numbers in a subinterval I of $(0, \infty)$ such that*

$$\begin{aligned} x_1 &\leq y_1 \\ x_1 x_2 &\leq y_1 y_2 \\ &\cdots \\ x_1 x_2 \cdots x_{n-1} &\leq y_1 y_2 \cdots y_{n-1} \\ x_1 x_2 \cdots x_n &= y_1 y_2 \cdots y_n. \end{aligned}$$

Then

$$f(x_1) + f(x_2) + \cdots + f(x_n) \leq f(y_1) + f(y_2) + \cdots + f(y_n)$$

for every function $f\colon I \to \mathbb{R}$ such that $f(e^x)$ is increasing and convex.

Proof. Notice that $(\log x_1, \ldots \log x_n) \prec_{HLP} (\log y_1, \ldots \log y_n)$ and apply the Hardy–Littlewood–Pólya inequality of majorization to the increasing convex function $f(e^x)$. ■

It is worth mentioning that if $f\colon I \to \mathbb{R}$ is increasing and convex then so is $f(e^x)$.

Weyl's inequality for singular numbers stated below, gives us the basic example of a pair of sequences satisfying the hypothesis of Theorem 6.2.5; the fact that all examples come this way was noted by A. Horn [261].

6.2.6 Theorem (Weyl's inequality for singular numbers; H. Weyl [584]) *Consider a matrix $A \in M_N(\mathbb{C})$ having the eigenvalues $\lambda_1, \ldots, \lambda_N$ and the singular numbers $s_1, \ldots, s_N$, and assume that they are rearranged such that $|\lambda_1| \geq \cdots \geq |\lambda_N|$, and $s_1 \geq \cdots \geq s_N$. Then:*

$$\prod_{k=1}^{m} |\lambda_k| \leq \prod_{k=1}^{m} s_k \quad \text{for } m = 1, \ldots, N \quad \text{and} \quad \prod_{k=1}^{N} |\lambda_k| = \prod_{k=1}^{N} s_k.$$

Proof. The fact that $|\lambda_1| \leq \|A\| = s_1$ is clear. The other inequalities follow by applying this argument to the antisymmetric tensor powers $\wedge^k A$. See R. Bhatia [61], for details. As concerns the equality $\prod_{k=1}^{N} |\lambda_k| = \prod_{k=1}^{N} s_k$, notice that each side equals $(\det(A^* A))^{1/2}$. ■

An immediate consequence of Theorem 6.2.6, also due to H. Weyl [584], is as follows:

6.2.7 Corollary *Let $A \in M_N(\mathbb{C})$ be a matrix having the eigenvalues $\lambda_1, \ldots, \lambda_N$ and the singular numbers $s_1, \ldots, s_N$, listed such that $|\lambda_1| \geq \cdots \geq |\lambda_N|$ and $s_1 \geq \cdots \geq s_N$. Then*

$$\sum_{k=1}^{m} \varphi\left(|\lambda_k|\right) \leq \sum_{k=1}^{m} \varphi(s_k)$$

for every number $m \in \{1, ..., N\}$ *and every function* $\varphi : \mathbb{R}_+ \to \mathbb{R}_+$ *such that* $\varphi(e^x)$ *is increasing and convex.*

In particular,

$$\sum_{k=1}^{m} |\lambda_k|^p \leq \sum_{k=1}^{m} s_k^p \quad \textit{for all } p > 0 \textit{ and } m \in \{1, ..., N\}.$$

Proof. Combine Weyl's inequality for singular numbers with Theorem 6.1.5 (the Tomić-Weyl inequality of weak majorization). ■

Exercises

1. (Hadamard's determinant inequality) *Suppose that* $A = (a_{jk})_{j,k}$ *is an* $N \times N$*-dimensional complex matrix. Prove that*

$$|\det A| \leq \left(\prod_{j=1}^{N} \sum_{k=1}^{N} |a_{jk}|^2 \right)^{1/2}.$$

[*Hint:* Use Corollary 6.2.3 and the fact that $|\det A|^2 = \det(AA^*)$.]

2. (I. Schur [524]) Consider a matrix $A = (a_{ij})_{i,j=1}^{N}$, whose eigenvalues are $\lambda_1, \ldots, \lambda_N$. Prove that

$$\sum_{k=1}^{N} |\lambda_k|^2 \leq \sum_{j,k=1}^{N} |a_{jk}|^2 = \|A\|_F^2 .$$

3. Apply the result of the preceding exercise to derive the *AM-GM* inequality.

[*Hint*: Write down an $N \times N$-dimensional matrix whose nonzero entries are $x_1, \ldots, x_N > 0$ and whose characteristic polynomial is $x^N - \prod_{k=1}^{N} x_k$.]

4. Infer from Theorem 6.2.5 and Theorem 6.2.6 that for every $A \in M_N(\mathbb{C})$ and every integer $n \geq 1$, one has

$$|\operatorname{trace} A^n| \leq \operatorname{trace} |A|^n ;$$

here $|A| = (A^*A)^{1/2}$ denotes the modulus of A.

6.3 Schur Convexity

According to Theorem 6.1.13, the quasiconvex functions $f : \mathbb{R}^N \to \mathbb{R}$ invariant under permutations are isotonic with respect to the relation of majorization, that is,

$$\mathbf{x} \prec_{HLP} \mathbf{y} \text{ implies } f(\mathbf{x}) \leq f(\mathbf{y}). \tag{6.9}$$

Simple examples (such as $f(x_1, x_2) = -x_1 x_2$ on $\mathbb{R}^2$) show that the implication (6.9) holds true beyond the framework of quasiconvex functions invariant under permutations. This led I. Schur [526] to initiate a systematic study of the functions that verify the property (6.9).

6.3.1 Definition *A function $f : C \to \mathbb{R}$ defined on a set invariant under permutations is called Schur convex if*

$$\mathbf{x} \prec_{HLP} \mathbf{y} \text{ implies } f(\mathbf{x}) \leq f(\mathbf{y});$$

if in addition $f(\mathbf{x}) < f(\mathbf{y})$ whenever $\mathbf{x} \prec_{HLP} \mathbf{y}$ but $\mathbf{x}$ is not a permutation of $\mathbf{y}$, then f is said to be strictly Schur convex.

We call the function f (strictly) Schur concave if $-f$ is (strictly) Schur convex.

Every Schur convex (as well as every Schur concave) function defined on a set C invariant under permutations is a function invariant under permutations. This is a consequence of the fact that $\mathbf{x} \prec_{HLP} P_\pi(\mathbf{x})$ and $P_\pi(\mathbf{x}) \prec_{HLP} \mathbf{x}$ for every vector $\mathbf{x} \in \mathbb{R}^N$ and every permutation matrix $P_\pi \in \mathrm{M}_N(\mathbb{R})$.

The Schur convex functions encompass a large variety of examples.

6.3.2 Example *If $H : \mathbb{R}^N \to \mathbb{R}$ is a function (strictly) increasing in each variable and $f_1, ..., f_N$ are (strictly) Schur convex functions on $\mathbb{R}^n$, then the function $h(\mathbf{x}) = H(f_1(\mathbf{x}), ..., f_N(\mathbf{x}))$ is (strictly) Schur convex on $\mathbb{R}^n$. Some particular examples of strictly Schur convex functions are*

$$\max\{x_1, ..., x_N\} \text{ and } \log\left(\sum\nolimits_{k=1}^{N} x_k^2\right) \text{ on } \mathbb{R}^N;$$

$$-\prod\nolimits_{k=1}^{N} x_k \text{ on } (0, \infty)^N.$$

6.3.3 Theorem (The Schur-Ostrowski criterion of Schur convexity) *Let I be a nonempty open interval. A differentiable function $f : I^N \to \mathbb{R}$ is Schur convex if and only if it fulfills the following two conditions:*

(a) *f is invariant under permutations;*

(b) *for every $\mathbf{x} \in I^N$ and $i, j \in \{1, \ldots, N\}$ we have*

$$(x_i - x_j)\left(\frac{\partial f}{\partial x_i}(\mathbf{x}) - \frac{\partial f}{\partial x_j}(\mathbf{x})\right) \geq 0.$$

Proof. Necessity. For (a), see the comments after Definition 6.3.1. This reduces the verification of (b) to the case where $i = 1$ and $j = 2$. Fix arbitrarily $\mathbf{x} \in I^N$ and choose $\varepsilon > 0$ sufficiently small such that

$$\mathbf{x}(t) = ((1-t)x_1 + tx_2, tx_1 + (1-t)x_2, x_3, ..., x_N) \in D \tag{6.10}$$

for $t \in (0, \varepsilon]$. Then $\mathbf{x}(t) \prec_{HLP} \mathbf{x}$, which yields $f(\mathbf{x}(t)) \leq f(\mathbf{x})$. Therefore

$$0 \geq \lim_{t \to 0} \frac{f(\mathbf{x}(t)) - f(\mathbf{x})}{t} = \left.\frac{\mathrm{d}f(\mathbf{x}(t))}{\mathrm{d}t}\right|_{t=0} = -(x_1 - x_2)\left(\frac{\partial f}{\partial x_1}(\mathbf{x}) - \frac{\partial f}{\partial x_2}(\mathbf{x})\right).$$

Sufficiency. We have to prove that $\mathbf{y} \prec_{HLP} \mathbf{x}$ implies $f(\mathbf{y}) \leq f(\mathbf{x})$. According to Theorem 6.1.9, it suffices to consider the case where

$$\mathbf{y} = ((1-s)\,x_1 + sx_2, sx_1 + (1-s)x_2, x_3, ..., x_N).$$

for some $s \in [0, 1/2]$. Consider $x(t)$ as in formula (6.10). Then

$$\begin{aligned} f(\mathbf{y}) - f(\mathbf{x}) &= \int_0^s \frac{\mathrm{d}}{\mathrm{d}t} f(\mathbf{x}(t))\mathrm{d}t \\ &= -\int_0^s (x_1 - x_2)\left(\frac{\partial f}{\partial x_1}(\mathbf{x}(t)) - \frac{\partial f}{\partial x_2}(\mathbf{x}(t))\right)\mathrm{d}t \\ &= -\int_0^s \frac{\mathrm{pr}_1\,\mathbf{x}(t) - \mathrm{pr}_2\,\mathbf{x}(t)}{1-2t}\left(\frac{\partial f}{\partial x_1}(\mathbf{x}(t)) - \frac{\partial f}{\partial x_2}(\mathbf{x}(t))\right)\mathrm{d}t, \end{aligned}$$

where pr_k denotes the projection to the $k-$th coordinate. According to condition (b), $f(\mathbf{y}) - f(\mathbf{x}) \leq 0$, and the proof is complete. ■

The elementary symmetric functions of N variables are defined by the formulas

$$\begin{aligned} e_0(x_1, x_2, \ldots, x_N) &= 1 \\ e_1(x_1, x_2, \ldots, x_N) &= x_1 + x_2 + \cdots + x_N \\ &\vdots \\ e_k(x_1, x_2, \ldots, x_N) &= \sum_{1 \leq i_1 < \cdots < i_k \leq N} x_{i_1} \cdots x_{i_k} \\ &\vdots \\ e_N(x_1, x_2, \ldots, x_N) &= x_1 x_2 \cdots x_N. \end{aligned}$$

Clearly, they are invariant under permutations. A small computation shows that

$$\frac{\partial}{\partial x_i} e_k(x_1, \ldots, x_N) = e_{k-1}(x_1, ..., \widehat{x_i}, \ldots, x_N)$$

and

$$\frac{\partial}{\partial x_i} e_k(x_1, \ldots, x_N) - \frac{\partial}{\partial x_j} e_k(x_1, \ldots, x_N) = -(x_i - x_j) e_{k-2}(x_1, ..., \widehat{x_i}, \ldots, \widehat{x_j}, ..., x_N),$$

where the cup indicates the omission of the coordinate underneath. This leads us to the following consequence of Theorem 6.3.3:

6.3.4 Corollary (I. Schur) *All elementary symmetric functions* $e_k(x_1, x_2, \ldots, x_N)$ *of* N *variables are Schur concave on* $\mathbb{R}_+^N$.

Since

$$\left(\frac{x_1 + \cdots + x_N}{N}, \ldots, \frac{x_1 + \cdots + x_N}{N}\right) \prec_{HLP} (x_1, \ldots, x_N)$$

for every $(x_1, \ldots, x_N) \in \mathbb{R}_+^N$, we infer from Corollary 6.3.4 that

$$e_k(x_1, x_2, \ldots, x_N) = \sum_{1 \leq i_1 < \cdots < i_k \leq N} x_{i_1} \cdots x_{i_k} \leq \binom{N}{k}\left(\frac{x_1 + \cdots + x_N}{N}\right)^k;$$

for $k = N$ we retrieve the *AM-GM* inequality. A nice application of these inequalities is mentioned in Exercise 7.

More information about the elementary symmetric functions can be found in Appendix C.

6.3.5 Remark *The relation of majorization is closely related to the duality of cones, more precisely, to the fact that the dual cone of the monotone cone* $\mathbb{R}^N_{\geq} = \{(x_1, \ldots, x_N) \in \mathbb{R}^N : x_1 \geq \cdots \geq x_N\}$ *is the cone*

$$\left(\mathbb{R}^N_{\geq}\right)^* = \left\{ \mathbf{y} \in \mathbb{R}^N : \sum_{k=1}^{m} y_k \geq 0 \text{ for } m = 1, 2, ..., N-1, \text{ and } \sum_{k=1}^{N} y_k = 0 \right\}.$$

See Exercise 11, Section 3.3. Indeed, if $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N_{\geq}$, *then*

$$\mathbf{x} \prec_{HLP} \mathbf{y} \quad \text{if, and only if,} \quad \mathbf{y} - \mathbf{x} \in \left(\mathbb{R}^N_{\geq}\right)^*.$$

This leads to the more general concept of majorization with respect to a convex cone C *in a real vector space* V,

$$\mathbf{x} \prec_C \mathbf{y} \quad \text{if, and only if,} \quad \mathbf{y} - \mathbf{x} \in C^*,$$

and, implicitly, to a generalization of Schur convexity. In the case of self-dual cones C *(like* $\mathbb{R}^N_+$ *and* $\mathrm{Sym}^+(n, \mathbb{R})$*), the corresponding concept of Schur convexity coincides with that of a function which is monotone increasing on* C.

Exercises

1. Suppose that $f : [0, \infty) \to (0, \infty)$ is a log-convex function. Prove that $f(x) + f(y)$ and $f(x)f(y)$ are Schur convex functions on $[0, \infty) \times [0, \infty)$. Infer that for every $0 < \varepsilon < x \leq y$, we have
$$f(x)f(y) \leq f(x - \varepsilon)f(y + \varepsilon) \leq f(0)f(x + y).$$

2. Prove that the area of a triangle is a Schur concave function of its sides. Also, the radius of the circumscribed circle of a triangle is a Schur convex function of its sides.

3. Suppose that f is a real-valued convex function defined on an interval I.
   (a) Prove that the function
   $$F : I^N \to \mathbb{R}^N, \quad F(x_1, ..., x_N) = (f(x_1), ..., f(x_N))$$
   verifies the condition
   $$\mathbf{x} \prec_{HLP} \mathbf{y} \text{ implies } F(\mathbf{x}) \prec_{HLPw} F(\mathbf{y}).$$
   (b) Suppose that f is convex and increasing. Prove that in this case $\mathbf{x} \prec_{HLP} \mathbf{y}$ implies $F(\mathbf{x}) \prec_{HLP} F(\mathbf{y})$.
   (c) Illustrate with examples the assertions (a) and (b).

4. Consider the Laplace transform of a Borel probability measure μ on $\mathbb{R}$,

$$\mathcal{L}(s) = \int_{\mathbb{R}^N} \mathrm{e}^{sx} \mathrm{d}\mu(x).$$

Prove that the function $f(s_1, ..., s_N) = \prod_{k=1}^{N} \mathcal{L}(s_k)$ is Schur convex on $[0, \infty)^N$.

5. (D. B. Hunter [269]) The complete homogeneous symmetric polynomials of N variables are defined by the formulas

$$\begin{aligned} h_0(x_1, .., x_N) &= 0 \\ h_d(x_1, \ldots, x_N) &= \sum_{1 \le i_1 \le \cdots \le i_d \le N} x_{i_1} \cdots x_{i_d} \quad \text{for } d \ge 1. \end{aligned}$$

Prove that all such polynomials of even degree d are Schur convex and positive on $\mathbb{R}^N$.

[*Hint:* Use mathematical induction and the identity

$$(x_i - x_j)\left[\frac{\partial h_d}{\partial x_i}(x_1, \ldots, x_N) - \frac{\partial h_d}{\partial x_j}(x_1, \ldots, x_N)\right] = (x_i - x_j)^2 h_{d-2}(x_1, \ldots, x_N).\ ]$$

6. (A. W. Marshall, I. Olkin and B. C. Arnold [363]) Suppose that φ and ψ are Schur concave functions defined on $\mathbb{R}^N$. Prove that their convolution product

$$(\varphi * \psi)(\mathbf{x}) = \int_{\mathbb{R}^N} \varphi(\mathbf{x} - \mathbf{y}) \psi(y) \mathrm{d}\mathbf{y}$$

is Schur concave (whenever the integral exists).

7. (The birthday inequality) Let the n-tuple $\mathbf{p} = (p_1, ..., p_n)$ be any probability vector of birthdays for the $n = 365$ days of the year. Given $\mathbf{p}$, let $P_k(\mathbf{p})$ be the probability of at least one match among k people. Prove that

$$P_k(\mathbf{p}) \ge 1 - \frac{n(n-1)\cdots(n-k+1)}{n^k},$$

along the following steps:

(a) Let $Q_k(\mathbf{p}) = 1 - P_k(\mathbf{p})$ be the probability of the complementary event that all k birthdays are different. Show that $Q_k(\mathbf{p}) = \sum p_{i_1} \cdots p_{i_k}$, where the sum is taken over all strings $i_1, ..., i_k$ of distinct integers in $\{1, ..., n\}$.

(b) Prove that $Q_k(\mathbf{p}) = k! e_k(p_1, ..., p_n)$.

(c) Compute $P_k(\mathbf{u})$, where $\mathbf{u} = (1/n, ..., 1/n)$ is the uniform distribution of the birthdays.

(d) Conclude the proof by using Corollary 6.3.4.

8. (D. E. Wulbert) Suppose that f is a continuous function defined on an interval I. Prove that the symmetric function

$$F(x,y) = \frac{1}{y-x}\int_x^y f(t)dt \text{ for } x,y \in I,\ x \neq y$$
$$F(x,y) = f(x) \text{ for } x = y \in I$$

is convex if and only if the function f is convex.

9. (An example of Schur convex like function due to V. Senderov) Let $f : [0,\infty)^N \to \mathbb{R}$ be a function such that

$$f(s/N, ..., s/N) \leq f(x_1, \ldots, x_N) \leq f(s, 0, ..., 0)$$

whenever $s \geq 0$ and $x_1 + \cdots + x_N = s$. Every Schur convex function f verifies this property, but the converse is not true in general. Illustrate this fact using the polynomial function $4x_1^3 + 4x_2^3 + 4x_3^3 + 15x_1x_2x_3$.

6.4 Eigenvalue Inequalities

According to the spectral decomposition, for every matrix $A \in \mathrm{Sym}(N,\mathbb{R})$ there is an orthonormal basis $(\mathbf{u}_k)_k$ of $\mathbb{R}^N$ consisting of eigenvectors of A. Assuming $A\mathbf{u}_k = \lambda_k(A)\mathbf{u}_k$ for each index k, this fact gives rise to a representation of the form

$$A = \sum_{k=1}^{N} \lambda_k(A)\langle \cdot, \mathbf{u}_k\rangle \mathbf{u}_k. \tag{6.11}$$

In what follows the downwards/upwards rearrangements of the eigenvalues $\lambda_k(A)$ will be denoted respectively by $\lambda_k^{\downarrow}(A)$ and $\lambda_k^{\uparrow}(A)$. Of a special interest is the *eigenvalues map,*

$$\Lambda : \mathrm{Sym}(N,\mathbb{R}) \to \mathbb{R}^N, \quad \Lambda(A) = \lambda^{\downarrow}(A) = \left(\lambda_1^{\downarrow}(A), ..., \lambda_N^{\downarrow}(A)\right).$$

The representation (6.11) easily yields the equalities

$$\lambda_1^{\downarrow}(A) = \max_{\|\mathbf{x}\|=1} \langle A\mathbf{x}, \mathbf{x}\rangle \tag{6.12}$$

and

$$\lambda_N^{\downarrow}(A) = \min_{\|\mathbf{x}\|=1} \langle A\mathbf{x}, \mathbf{x}\rangle. \tag{6.13}$$

Therefore

$$\{\langle A\mathbf{x}, \mathbf{x}\rangle : \mathbf{x} \in \mathbb{R}^N, \|\mathbf{x}\| = 1\} = [\lambda_N^{\downarrow}(A), \lambda_1^{\downarrow}(A)], \tag{6.14}$$

due to the fact that the function $\mathbf{x} \to \langle A\mathbf{x}, \mathbf{x}\rangle$ is continuous and the unit sphere is compact and connected.

The equalities (6.12) and (6.13) also imply the following result:

6.4.1 Proposition *If A and B belong to* $\mathrm{Sym}(N,\mathbb{R})$, *then*

$$\lambda_1^{\downarrow}(A+B) \le \lambda_1^{\downarrow}(A) + \lambda_1^{\downarrow}(B)$$

and

$$\lambda_N^{\downarrow}(A+B) \ge \lambda_N^{\downarrow}(A) + \lambda_N^{\downarrow}(B).$$

Thus the function $\lambda_1^{\downarrow}(A)$ is positively homogeneous and subadditive (in particular, convex), while the function $\lambda_N^{\downarrow}(A)$ is positively homogeneous and superadditive (in particular, concave). These two conclusions are equivalent since

$$\lambda_k^{\downarrow}(-A) = -\lambda_{N-k+1}^{\downarrow}(A) = -\lambda_k^{\uparrow}(A). \tag{6.15}$$

The formulas (6.12) and (6.13) admit analogues for all indices k:

6.4.2 Theorem (Courant-Fischer minimax principle) *If $A \in \mathrm{Sym}(N,\mathbb{R})$, then its eigenvalues can be computed by the formulas*

$$\lambda_k^{\downarrow}(A) = \max_{\substack{V\subset\mathbb{R}^N \\ \dim V = k}} \min_{\substack{\mathbf{x}\in V \\ \|\mathbf{x}\|=1}} \langle A\mathbf{x},\mathbf{x}\rangle = \min_{\substack{V\subset\mathbb{R}^N \\ \dim V = N-k+1}} \max_{\substack{\mathbf{x}\in V \\ \|\mathbf{x}\|=1}} \langle A\mathbf{x},\mathbf{x}\rangle.$$

Equivalently, taking into account (6.15),

$$\lambda_k^{\uparrow}(A) = \min_{\substack{V\subset\mathbb{R}^N \\ \dim V = k}} \max_{\substack{\mathbf{x}\in V \\ \|\mathbf{x}\|=1}} \langle A\mathbf{x},\mathbf{x}\rangle = \max_{\substack{V\subset\mathbb{R}^N \\ \dim V = N-k+1}} \min_{\substack{\mathbf{x}\in V \\ \|\mathbf{x}\|=1}} \langle A\mathbf{x},\mathbf{x}\rangle.$$

Proof. Let $\mathbf{u}_1,\dots,\mathbf{u}_N$ be the orthonormal basis which appears in the spectral representation (6.11) of A. The vector space $W = \mathrm{span}\{\mathbf{u}_k,\mathbf{u}_{k+1},\dots,\mathbf{u}_N\}$ is $(N-k+1)$-dimensional and thus every k-dimensional vector subspace $V \subset \mathbb{R}^N$ will contain a point $\mathbf{z} \in W \cap V$ with $\|\mathbf{z}\| = 1$. According to (6.14),

$$\langle A\mathbf{z},\mathbf{z}\rangle \in [\lambda_N^{\downarrow}(A), \lambda_k^{\downarrow}(A)],$$

from which it follows that

$$\min_{\substack{\mathbf{x}\in V \\ \|\mathbf{x}\|=1}} \langle A\mathbf{x},\mathbf{x}\rangle \le \lambda_k^{\downarrow}(A).$$

Finally, note that equality occurs for $V = \mathrm{span}\{\mathbf{u}_1,\dots,\mathbf{u}_k\}$. ■

6.4.3 Corollary (Weyl's monotonicity principle [583]) *If A and B belong to* $\mathrm{Sym}(N,\mathbb{R})$ *and $A \le B$, then $\lambda_k^{\downarrow}(A) \le \lambda_k^{\downarrow}(B)$ for every index k.*

If $A \in \mathrm{Sym}(N,\mathbb{R})$ and P_W is the orthogonal projection of $\mathbb{R}^N$ onto the m-dimensional vector subspace W, we call

$$B = P_W A P_W$$

the *compression* of A to W. For example, the principal submatrix obtained from A by deleting the last row and last column is a compression of A, corresponding to the linear span W of the first $N-1$ vectors of the canonical basis of $\mathbb{R}^N$.

6.4.4 Corollary (Cauchy's interlace theorem) *Let $A \in \mathrm{Sym}(N,\mathbb{R})$ be a symmetric matrix and let $B = P_W A P_W$ be a compression of it. If $m = \dim W$, then*

$$\lambda_k^{\uparrow}(A) \le \lambda_k^{\uparrow}(B) \le \lambda_{N-m+k}^{\uparrow}(A) \quad \textit{for every } k \in \{1,...,m\}. \tag{6.16}$$

Proof. Denote by $\mathbf{w}_1,...,\mathbf{w}_m$ an orthonormal basis of W consisting of eigenvectors of B such that $B\mathbf{w}_i = \lambda_i^{\uparrow}(B)\mathbf{w}_i$ for every i. Put $W_k = \mathrm{span}\{w_1,...,w_k\}$. According to the Courant-Fischer minimax theorem,

$$\begin{aligned}\lambda_k^{\uparrow}(B) &= \max_{\mathbf{x}\in W_k, \|\mathbf{x}\|=1} \langle B\mathbf{x},\mathbf{x}\rangle = \max_{\mathbf{x}\in W_k, \|\mathbf{x}\|=1} \langle AP_W\mathbf{x}, P_W\mathbf{x}\rangle \\ &= \max_{\mathbf{x}\in W_k, \|\mathbf{x}\|=1} \langle A\mathbf{x},\mathbf{x}\rangle \ge \lambda_k^{\uparrow}(A).\end{aligned}$$

In a similar way, by considering the space $\widetilde{W_k} = \mathrm{span}\{w_k,...,w_m\}$,

$$\begin{aligned}\lambda_k^{\uparrow}(B) &= \min_{\mathbf{x}\in \widetilde{W}_k, \|\mathbf{x}\|=1} \langle B\mathbf{x},\mathbf{x}\rangle = \min_{\mathbf{x}\in \widetilde{W}_k, \|\mathbf{x}\|=1} \langle AP_W\mathbf{x}, P_W\mathbf{x}\rangle \\ &= \min_{\mathbf{x}\in \widetilde{W}_k, \|\mathbf{x}\|=1} \langle A\mathbf{x},\mathbf{x}\rangle \le \lambda_{N-m+k}^{\uparrow}(A).\end{aligned}$$

■

One can prove (see [263]) that the converse of Corollary 6.4.4 is also true, that is, for any two sequences $(\lambda_k)_{k=1}^N$ and $(\mu_k)_{k=1}^{N-1}$ of real numbers, satisfying

$$\lambda_1 \le \mu_1 \le \lambda_2 \le \mu_2 \le ... \le \lambda_{N-1} \le \mu_{N-1} \le \lambda_N,$$

there exists a matrix $A \in \mathrm{Sym}(N,\mathbb{R})$ such that $\sigma(A) = (\lambda_k)_{k=1}^N$ and $\sigma(A_{N-1}) = (\mu_k)_{k=1}^{N-1}$, where A_{N-1} is the principal submatrix obtained from A by deleting the last row and column. We say that such a matrix A solves the inverse spectral problem for these sequences.

6.4.5 Theorem (Weyl's inequalities) *If A and B belong to $\mathrm{Sym}(N,\mathbb{R})$, then*

$$\lambda_{i+j-1}^{\downarrow}(A+B) \le \lambda_i^{\downarrow}(A) + \lambda_j^{\downarrow}(B) \quad \textit{if } i+j-1 \le N.$$

Proof. Suppose that A, B and $A+B$ have the spectral representations

$$A = \sum_{k=1}^N \lambda_k^{\downarrow}(A)\langle\cdot,\mathbf{u}_k\rangle\mathbf{u}_k, \quad B = \sum_{k=1}^N \lambda_k^{\downarrow}(B)\langle\cdot,\mathbf{v}_k\rangle\mathbf{v}_k, \quad A+B = \sum_{k=1}^N \lambda_k^{\downarrow}(C)\langle\cdot,\mathbf{w}_k\rangle\mathbf{w}_k.$$

Since

$$\dim \mathrm{span}\{\mathbf{u}_i,\dots,\mathbf{u}_N\} + \dim \mathrm{span}\{\mathbf{v}_j,\dots,\mathbf{v}_N\} + \dim \mathrm{span}\{\mathbf{w}_1,\dots,\mathbf{w}_{i+j-1}\}$$

equals $(N-i+1)+(N-j+1)+(i+j-1) = 2N+1$, the spaces

$$\mathrm{span}\{\mathbf{u}_i,\dots,\mathbf{u}_N\}, \mathrm{span}\{\mathbf{v}_j,\dots,\mathbf{v}_N\} \text{ and } \mathrm{span}\{\mathbf{w}_1,\dots,\mathbf{w}_{i+j-1}\}$$

must have in common a vector $\mathbf{x}$ with $\|\mathbf{x}\| = 1$. Then, according to (6.12) and (6.13),

$$\langle A\mathbf{x}, \mathbf{x}\rangle \leq \lambda_i^{\downarrow}(A), \quad \langle B\mathbf{x}, \mathbf{x}\rangle \leq \lambda_j^{\downarrow}(B), \quad \langle (A+B)\mathbf{x}, \mathbf{x}\rangle \geq \lambda_{i+j-1}^{\downarrow}(C)$$

and the proof is complete. ■

6.4.6 Corollary *The following inequalities hold:*

$$\lambda_i^{\downarrow}(A) + \lambda_N^{\downarrow}(B) \leq \lambda_i^{\downarrow}(A+B) \leq \lambda_i^{\downarrow}(A) + \lambda_1^{\downarrow}(B) \quad \textit{for } i = 1, ..., N.$$

6.4.7 Theorem (Weyl's perturbation theorem) *For every pair of matrices A, B in* $\mathrm{Sym}(N, \mathbb{R})$*, we have*

$$\max_{1\leq k\leq n} |\lambda_k(A) - \lambda_k(B)| \leq \|A - B\|.$$

In particular, the eigenvalues $\lambda_k(A)$ are continuous functions of A.

Proof. In fact, for every symmetric matrix A we have

$$\|A\| = \sup_{\|\mathbf{x}\|=1} |\langle A\mathbf{x}, \mathbf{x}\rangle| = \max\{|\lambda_1^{\downarrow}(A)|, |\lambda_n^{\downarrow}(A)|\}.$$

Consequently, by applying Corollary 6.4.6 to A and $B - A$, we get

$$\lambda_k(A) - \|B - A\| \leq \lambda_k(B) \leq \lambda_k(A) + \|B - A\|.$$

■

Weyl's perturbation theorem has applications to numerical analysis.

According to Lemma 6.2.2, if A is a symmetric matrix with diagonal elements $a_{11}, \ldots, a_{NN}$ and eigenvalues $\lambda_1, \ldots, \lambda_N$, then

$$(a_{11}, \ldots, a_{NN}) \prec_{HLP} (\lambda_1, \ldots, \lambda_N).$$

Since the spectrum is invariant under unitary equivalence of matrices, this result yields *Ky Fan's maximum principle*:

$$\sum_{k=1}^{r} \lambda_k^{\downarrow}(A) = \max_{\substack{(\mathbf{x}_k)_{k=1}^{r} \\ \text{orthonormal} \\ \text{family}}} \sum_{k=1}^{r} \langle A\mathbf{x}_k, \mathbf{x}_k\rangle \quad \text{for } r = 1, \ldots, N. \tag{6.17}$$

As a consequence, the sums $\sum_{k=1}^{r} \lambda_k^{\downarrow}(A)$ are *convex* functions of A and the following set of inequalities (known as *Ky Fan's inequalities*) hold:

$$\sum_{k=1}^{r} \lambda_k^{\downarrow}(A+B) \leq \sum_{k=1}^{r} \lambda_k^{\downarrow}(A) + \sum_{k=1}^{r} \lambda_k^{\downarrow}(B), \quad \text{for } r = 1, \ldots, N. \tag{6.18}$$

In other words,

$$\lambda^{\downarrow}(A+B) \prec_{HLP} \lambda^{\downarrow}(A) + \lambda^{\downarrow}(B). \tag{6.19}$$

The complementary inequality,

$$\lambda^{\downarrow}(A) + \lambda^{\uparrow}(B) \prec_{HLP} \lambda(A+B), \tag{6.20}$$

also works and it was proved in an equivalent form by V. B. Lidskii [335] and later by H. Wielandt [587]:

6.4.8 Theorem (Lidskii–Wielandt inequalities) *Let A, B, C be three $N \times N$-dimensional symmetric matrices such that $C = A + B$. Then, for every $1 \leq r \leq N$ and every $1 \leq i_1 < \cdots < i_r \leq N$, we have the inequalities*

$$\sum_{k=1}^{r} \lambda_{i_k}^{\downarrow}(C) \leq \sum_{k=1}^{r} \lambda_{i_k}^{\downarrow}(A) + \sum_{k=1}^{r} \lambda_k^{\downarrow}(B)$$

as well as the corresponding inequalities obtained by interchanging A and B.

Proof. (C. K. Li and R. Mathias [333]) We have to prove the inequality

$$\sum_{k=1}^{r} \left[\lambda_{i_k}^{\downarrow}(A+B) - \lambda_{i_k}^{\downarrow}(A)\right] \leq \sum_{k=1}^{r} \lambda_k^{\downarrow}(B). \tag{6.21}$$

Without loss of generality we may assume that $\lambda_r^{\downarrow}(B) = 0$; for this, replace B by $B - \lambda_r^{\downarrow}(B) \cdot \mathrm{I}$. Let $B = B^+ - B^-$ be the canonical decomposition of B into the positive and negative part. Since $B \leq B^+$, Weyl's monotonicity principle yields $\lambda_{i_k}^{\downarrow}(A+B) \leq \lambda_{i_k}^{\downarrow}(A+B^+)$, so that the left-hand side of (6.21) is majorized by

$$\sum_{k=1}^{r} [\lambda_{i_k}^{\downarrow}(A+B^+) - \lambda_{i_k}^{\downarrow}(A)],$$

which in turn is less than or equal to

$$\sum_{k=1}^{n} [\lambda_k^{\downarrow}(A+B^+) - \lambda_k^{\downarrow}(A)] = \mathrm{trace}(B^+).$$

Or, $\mathrm{trace}(B^+) = \sum_{k=1}^{r} \lambda_k^{\downarrow}(B)$ since $\lambda_r^{\downarrow}(B) = 0$. ■

The following result is known as von Neumann's trace inequality:

6.4.9 Theorem (von Neumann [413]) *Any matrices $A, B \in \mathrm{M}_N(\mathbb{R})$ verify the inequality*

$$|\mathrm{trace}(A^*B)| \leq \langle s^{\downarrow}(A), s^{\downarrow}(B) \rangle.$$

Here $s^{\downarrow}(A) = (s_1^{\downarrow}(A), s_2^{\downarrow}(A), ..., s_N^{\downarrow}(A))$ and similarly for $s^{\downarrow}(B)$.

The proof needs the following generalization of the Hardy–Littlewood–Pólya rearrangement inequality.

6.4.10 Lemma *Suppose that* $\mathbf{x} = (x_1, ..., x_N)$ *and* $\mathbf{y} = (y_1, ..., y_N)$ *are two vectors in* $\mathbb{R}^N$ *such that* $x_1 \geq \cdots \geq x_N \geq 0$ *and* $y_1 \geq \cdots \geq y_N \geq 0$. *Then, for every doubly stochastic matrix* $D \in \mathrm{M}_N(\mathbb{R})$, *we have the inequality*

$$\langle D\mathbf{x}, \mathbf{y}\rangle \leq \langle \mathbf{x}, \mathbf{y}\rangle.$$

Proof. This is an immediate consequence of the Hardy–Littlewood–Pólya rearrangement inequality 2.4.8 and of the Birkhoff Theorem 3.3.6. ■

Proof of Theorem 6.4.9. According to the singular value decomposition theorem,

$$A = USV^* \text{ and } B = XTY^*$$

where S and T are diagonal matrices and U, V, X, Y are orthogonal matrices. Then

$$\begin{aligned}\operatorname{trace}(A^*B) &= \operatorname{trace}(VSU^*XTY^*) = \operatorname{trace}(Y^*VSU^*XT)\\ &= \operatorname{trace}(Q^*SPT),\end{aligned}$$

where $P = U^*X = (p_{ij})_{i,j}$ and $Q = V^*Y = (q_{ij})_{i,j}$ are orthogonal matrices. Thus the matrices $(p_{ij}^2)_{i,j}$ and $(q_{ij}^2)_{i,j}$ are doubly stochastic and we have

$$\begin{aligned}\operatorname{trace}(A^*B) &= \operatorname{trace}\left((SQ)^*\, PT\right)\\ &= \sum_{i,j=1}^{N} s_i(A)s_j(B)q_{ij}p_{ij}\\ &\leq \frac{1}{2}\sum_{i,j=1}^{N} s_i(A)s_j(B)q_{ij}^2 + \frac{1}{2}\sum_{i,j=1}^{N} s_i(A)s_j(B)p_{ij}^2\\ &\leq \sum_{k=1}^{N} s_k^{\downarrow}(A)s_k^{\downarrow}(B),\end{aligned}$$

the last inequality being implied by Lemma 6.4.10. ■

6.4.11 Corollary *All matrices* $A, B \in \mathrm{Sym}(N, \mathbb{R})$ *verify the inequality*

$$\operatorname{trace}(AB) \leq \langle \lambda^{\downarrow}(A), \lambda^{\downarrow}(B)\rangle.$$

Equality occurs if, and only if, there exists an orthogonal matrix V *such that* $V^*AV = \operatorname{Diag}\lambda^{\downarrow}(A)$ *and* $V^*BV = \operatorname{Diag}\lambda^{\downarrow}(B)$.

Proof. Choose scalars α and β such $A + \alpha\mathrm{I} \geq 0$ and $B + \beta\mathrm{I} \geq 0$ and apply Theorem 6.4.9 taking into account that $\lambda^{\downarrow}(C) = s^{\downarrow}(C)$ if $C \geq 0$. A simple proof of the equality case is available in [331], Theorem 2.2. ■

Most of the above results extend easily to the framework of compact self-adjoint operators. A good starting point in this direction is offered by the book of B. Simon [541].

Exercises

1. Recall Ky Fan's maximum principle (6.17),

$$\sum_{k=1}^{r} \lambda_k^{\downarrow}(A) = \max_P \operatorname{trace}(AP) \quad \text{for } r = 1, \ldots, N,$$

where the maximum is taken over all orthogonal projections P, having r-dimensional range. Prove that this principle is equivalent to Schur's Lemma 6.2.2.

2. Consider a symmetric matrix $A \in \operatorname{Sym}(N, \mathbb{R})$ and its principal submatrix B, obtained from A by deleting the last row and column of A. Prove that

$$(\lambda_1^{\downarrow}(B), ..., \lambda_{N-1}^{\downarrow}(B), \sum\nolimits_{i=1}^{N} \lambda_1^{\downarrow}(A) - \sum\nolimits_{i=1}^{N-1} \lambda_1^{\downarrow}(B)) \prec_{HLP} (\lambda_1^{\downarrow}(A), ..., \lambda_N^{\downarrow}(A)).$$

3. Infer from Cauchy's interlace theorem that a real symmetric matrix A is strictly positive if and only if the determinants of all its leading minors are strictly positive.

4. Consider two interlaced sequences $a_1 \geq b_1 \geq a_2 \geq \cdots \geq b_{N-1} \geq a_N$. Prove that

$$(b_1, ..., b_{N-1}, \sum\nolimits_{i=1}^{N} a_i - \sum\nolimits_{j=1}^{N-1} b_j) \prec_{HLP} (a_1, ..., a_{N-1}, a_N)\,.$$

5. (A converse to Weyl's monotonicity principle 6.4.3) Suppose that $A, B \in \operatorname{Sym}^+(N, \mathbb{R})$ and $\lambda_k^{\downarrow}(A) \leq \lambda_k^{\downarrow}(B)$ for all $k \in \{1, ..., N\}$. Prove that $B \geq UAU^T$ for some orthogonal matrix U.

6. (Ky Fan) Consider a matrix $A \in \mathrm{M}_N(\mathbb{C})$ and put $\operatorname{Re} A = (A + A^*)/2$. Infer from Ky Fan's maximum principle (6.17) that

$$(\operatorname{Re} \lambda_1(A), ..., \operatorname{Re} \lambda_N(A)) \prec_{HLP} (\lambda_1(\operatorname{Re} A), ..., \lambda_N(\operatorname{Re} A)).$$

6.5 Convexity Versus Positive Differences

In Section 2.6 we noticed that the higher order convexity of functions defined on intervals can be characterized in terms of positive differences. The difference operators

$$\Delta_h f : x \to \Delta_h f(x) = f(x + h) - f(x)$$

are well defined for any function f defined on a (pointed) convex cone $\mathcal{C}$ (in a real linear space E). Such a function is said to be a *function with positive differences of order* n $(n \geq 1)$ if

$$\Delta_{h_1} \Delta_{h_2} \cdots \Delta_{h_n} f(x) = \sum_{\varepsilon_1, ..., \varepsilon_n \in \{0,1\}} (-1)^{n - (\varepsilon_1 + \cdots + \varepsilon_n)} f(x + \varepsilon_1 h_1 + \cdots + \varepsilon_n h_n) \geq 0$$

for all $x, h_1, \ldots, h_n \in \mathcal{C}$.

For convenience, we say that f has *positive differences of order* 0 if $f \geq 0$.

The functions $f : \mathcal{C} \to \mathbb{R}$ which verify the condition $f(0) \leq 0$ and have positive differences of second order, that is,

$$\Delta_x \Delta_y f(z) = f(\mathbf{x}+\mathbf{y}+\mathbf{z}) + f(\mathbf{z}) - f(\mathbf{x}+\mathbf{z}) - f(\mathbf{y}+\mathbf{z}) \geq 0$$

for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{C}$, are usually called *strongly superadditive*; f is called *strongly subadditive* if $-f$ is strongly superadditive. Notice that every strongly superadditive function is also superadditive.

An important example of function which admits positive differences of any order and vanishes at the origin is the determinant function when restricted to $\mathrm{Sym}^+(N, \mathbb{R})$. Notice that this function is *neither convex nor concave*.

Clearly, $\det(A) \geq 0$ for all $A \in \mathrm{Sym}^+(N, \mathbb{R})$. According to Weyl's monotonicity principle (see Corollary 6.4.3),

$$(\Delta_A \det)(X) = \det(X + A) - \det(X) \geq 0,$$

for all $A, X \in \mathrm{Sym}^+(N, \mathbb{R})$, that is, the function det admits positive differences of first order. As noticed F. Zhang (see [599], Section 7.2, Exercise 36), the function det also admits positive differences of second order:

6.5.1 Lemma *The restriction to* $\mathrm{Sym}^+(N, \mathbb{R})$ *of the function* det *is strongly superadditive, equivalently, for all* $A, B, X \in \mathrm{Sym}^+(N, \mathbb{R})$,

$$(\Delta_A (\Delta_B \det))(X) = \det(A+B+X) - \det(A+X) - \det(B+X) + \det(X) \geq 0.$$

Proof. Clearly, it suffices to prove the lemma for X replaced by $X + \varepsilon I$ for any $\varepsilon > 0$ and then to pass to the limit as $\varepsilon \to 0$.

Since $X + \varepsilon I$ is invertible and the function det is multiplicative, we may reduce ourselves to the case where $X = I$, that is, to the formula

$$\det(I + A + B) - \det(I + A) - \det(I + B) + 1 \geq 0. \tag{6.22}$$

Since

$$\det(I + X) = \prod_{k+1}^{N} (1 + \lambda_k^{\downarrow}(X)) = 1 + \sum_{j=1}^{N} e_j(\lambda_1^{\downarrow}(X), \ldots, \lambda_1^{\downarrow}(X))$$

for all $X \in \mathrm{Sym}(N, \mathbb{R})$, the proof of (6.22) reduces to the proof of inequalities

$$e_j(\lambda_1^{\downarrow}(A+B), \ldots, \lambda_1^{\downarrow}(A+B)) \geq e_j(\lambda_1^{\downarrow}(A), \ldots, \lambda_1^{\downarrow}(A)) + e_j(\lambda_1^{\downarrow}(B), \ldots, \lambda_1^{\downarrow}(B)),$$

which follows from Ky Fan's inequality (6.19) combined with the property of Schur concavity of the elementary symmetric functions (see Corollary 6.3.4). ∎

The fact that the restriction of det to $\mathrm{Sym}^+(N, \mathbb{R})$ admits positive differences of any order was established by C. P. Niculescu and S. Sra [446]. See Exercise 2 at the end of this section.

Notice that not every convex function of several variables admits positive differences of second order. See the case of the function $R(x, y) = -\sqrt{xy}$ defined on $\mathbb{R}^2_+$. Indeed,

$$\left(\Delta_{(1/4,1/3)} \left(\Delta_{(1/4,2/3)} R\right)\right)(0,0) = -\sqrt{\frac{1}{2}} + \sqrt{\frac{1}{12}} + \sqrt{\frac{1}{6}} < 0.$$

A nice exception is offered by the case of differentiable convex functions $f : \mathbb{R}^N_{++} \to \mathbb{R}$ with increasing differentials. See Exercise 7.

Exercises

1. Prove that every function of the form
$$\Phi(\mathbf{x}) = f\left(\langle \mathbf{x}, \mathbf{w}\rangle\right), \quad \mathbf{x} \in \mathbb{R}^N_+,$$
associated to a continuous n-convex function $f : \mathbb{R}_+ \to \mathbb{R}$ and a vector $\mathbf{w} \in \mathbb{R}^N_+$, has positive differences of order n.

2. (C. P. Niculescu and S. Sra [446]) Prove that the restriction of det to $\mathrm{Sym}^+(N, \mathbb{R})$ is a function with positive differences of every order $n \geq 0$.

3. (A determinant inequality due to M. Lin [339]) Suppose that $A, B, C \in \mathrm{Sym}^+(N, \mathbb{R})$. Prove the following matrix analogue of the Hornich-Hlawka inequality:
$$\det(A+B+C)+\det A+\det B+\det C \geq \det(A+B)+\det(B+C)+\det(C+A).$$
(a) Show that the inequality is equivalent to its particular case where $C = I$.
(b) Prove the inequality in the case where A and B are diagonal matrices, that is,
$$\prod_{i=1}^N (a_i + b_i + 1) + \prod_{i=1}^N a_i + \prod_{i=1}^N b_i + 1 \geq \prod_{i=1}^N (a_i + b_i) + \prod_{i=1}^N (1 + a_i) + \prod_{i=1}^N (1 + b_i)$$
(c) Prove that the function $\prod_{i=1}^N (1 + x_i) - \prod_{i=1}^N x_i$ is Schur concave on $\mathbb{R}^N_+$.
(d) Notice that
$$\det(A + B + I) - \det(A + B) = \prod_{i=1}^N (1 + \lambda_i(A + B)) - \prod_{i=1}^N \lambda_i(A + B)$$
and conclude the proof using (in order) Ky Fan's inequality (6.19), the assertion (b) and then the assertion (a).

4. (C. P. Niculescu [430]) Suppose that $f : [0, \infty) \to \mathbb{R}$ is a nondecreasing and convex function. Infer from Exercise 8, Section 6.1, that
$$f(\det(A + B + C)) + f(\det C) \geq f(\det(A + C)) + f(\det(B + C)),$$
for all positive semidefinite matrices A, B and C. Then prove that this inequality yields the following matrix analog of Popoviciu's inequality,
$$\frac{f(\det A) + f(\det B) + f(\det C)}{3} + f(\det(A + B + C)) \geq \frac{2}{3}\left(f(\det(A + B)) + f(\det(B + C)) + f(\det(A + C))\right).$$

5. (An order-theoretical generalization of the Hardy–Littlewood–Pólya inequality from [429]) Suppose that U is an open convex subset of the ordered Banach space $\mathbb{R}^N$ and $f : U \to \mathbb{R}$ is a differentiable convex function whose differential is increasing. Consider also a probability distribution $\{\lambda_1, \dots \lambda_n\}$ and two families $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ and $\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$ of elements in U such that

$$\mathbf{x}_1 \geq \cdots \geq \mathbf{x}_n$$

$$\sum_{k=1}^{j} \lambda_k \mathbf{x}_k \leq \sum_{k=1}^{j} \lambda_k \mathbf{y}_k \quad \text{for } j \in \{1, \dots, n-1\}$$

and

$$\sum_{k=1}^{n} \lambda_k \mathbf{x}_k = \sum_{k=1}^{n} \lambda_k \mathbf{y}_k.$$

Prove that

$$\sum_{k=1}^{n} \lambda_k \Phi(\mathbf{x}_k) \leq \sum_{k=1}^{n} \lambda_k \Phi(\mathbf{y}_k).$$

[*Hint:* Adapt the argument in the proof of Theorem 6.1.3.]

6. Show that the result of Exercise 5 also works for the $2D$-supermodular convex functions defined on an open N-box in $\mathbb{R}^N$.

7. Suppose that $f : \mathbb{R}^N_+ \to \mathbb{R}$ is a differentiable convex function whose differential is increasing. Infer from Exercise 5 that f has positive differences of second order.

8. (The extension of Szegö and Bellman inequalities) Suppose that E and F are two ordered Banach spaces, U is an open convex subset of E that contains the origin and $\Phi : U \to F$ is a differentiable convex function whose differential is isotone. Prove that for every finite family $\mathbf{x}_1 \geq \mathbf{x}_2 \geq \cdots \geq \mathbf{x}_n \geq 0$ of points in U we have

$$\left(1 - \sum_{k=1}^{n} (-1)^{k+1}\right) \Phi(0) + \sum_{k=1}^{n} (-1)^{k+1} \Phi(\mathbf{x}_k) \geq \Phi\left(\sum_{k=1}^{n} (-1)^{k+1} \mathbf{x}_k\right).$$

[*Hint:* Consider separately the cases where n is odd or even.]

9. (The extension of an inequality due to K. L. Chung; see [429]) Let f be an increasing, differentiable and convex function defined on $\mathbb{R}$. If $A_1, A_2, ..., A_n$ and $B_1, B_2, ..., B_n$ are two families of matrices in $\mathrm{Sym}(N, \mathbb{R})$ such that

$$A_1 \geq A_2 \geq \cdots \geq A_n \geq 0 \text{ and } \sum_{k=1}^{j} A_k \leq \sum_{k=1}^{j} B_k \text{ for } j \in \{1, 2, ..., n\},$$

prove that

$$\sum_{k=1}^{n} \operatorname{Trace} f(A_k) \leq \sum_{k=1}^{n} \operatorname{Trace} f(B_k).$$

6.6 Horn's Inequalities

The following problem was raised by H. Weyl [583] in 1912: *Let A,B and C be Hermitian $N \times N$ matrices and denote the string of eigenvalues of A by α, where*

$$\alpha : \quad \alpha_1 \geq \cdots \geq \alpha_N,$$

and similarly write β and γ for the spectra of B and C. What α, β and γ can be when $C = A + B$?

There is one obvious condition, namely that the trace of C is the sum of the traces of A and B:

$$\sum_{k=1}^{N} \gamma_k = \sum_{k=1}^{N} \alpha_k + \sum_{k=1}^{N} \beta_k. \tag{6.23}$$

H. Weyl was able to indicate supplementary additional conditions in terms of linear inequalities on the possible eigenvalues. They were presented in Section 6.4.

Weyl's problem was studied extensively by A. Horn [262] who solved it for small N and proposed a complete set of necessary inequalities to accompany (6.23) for $N \geq 5$. Horn's inequalities have the form

$$\sum_{k \in K} \gamma_k \leq \sum_{i \in I} \alpha_i + \sum_{j \in J} \beta_j, \tag{6.24}$$

where

$$I = \{i_1, \ldots, i_r\}, \quad J = \{j_1, \ldots, j_r\}, \quad K = \{k_1, \ldots, k_r\}$$

are subsets of $\{1, \ldots, N\}$ with the same cardinality $r \in \{1, \ldots, N-1\}$ in a certain finite set T_r^N. Let us call such triplets (I, J, K) *admissible*. When $r = 1$, the condition of admissibility is

$$i_1 + j_1 = k_1 + 1.$$

If $r > 1$, this condition is

$$\sum_{i \in I} i + \sum_{j \in J} j = \sum_{k \in K} k + \binom{r+1}{2}$$

and, for all $1 \leq p \leq r-1$ and all $(U, V, W) \in T_p^r$,

$$\sum_{u \in U} i_u + \sum_{v \in V} j_v = \sum_{w \in W} k_w + \binom{p+1}{2}.$$

Notice that Horn's inequalities are defined by an inductive procedure.

Horn's Conjecture. A triplet (α, β, γ) of elements of $\mathbb{R}^N_{\geq}$ displays the eigenvalues of three symmetric matrices $A, B, C \in \mathrm{M}_N(\mathbb{R})$ with $C = A + B$, if and only if the trace equality (6.23) and Horn's inequalities (6.24) hold for every (I, J, K) in T_r^N and every $r < N$.

Nowadays this conjecture is a theorem due to works by A. A. Klyachko [301] and A. Knutson and T. Tao [303]. It appeals to advanced facts from algebraic geometry and representation theory (beyond the goal of this book). A thorough introduction to the mathematical world of Horn's conjecture is offered by the paper of R. Bhatia [62]. A more technical description of the work of Klyachko, Knutson and Tao can be found in the paper of W. Fulton [205].

Just to get a flavor of what Horn's conjecture is about we will detail here the proof in the case of 2×2 symmetric matrices. Precisely, we will prove that for all families of real numbers $\alpha_1 \geq \alpha_2$, $\beta_1 \geq \beta_2$, $\gamma_1 \geq \gamma_2$, which verify Weyl's inequalities,

$$\gamma_1 \leq \alpha_1 + \beta_1, \quad \gamma_2 \leq \alpha_2 + \beta_1, \quad \gamma_2 \leq \alpha_1 + \beta_2,$$

and the trace formula (6.23),

$$\gamma_1 + \gamma_2 = \alpha_1 + \alpha_2 + \beta_1 + \beta_2,$$

there exist symmetric matrices $A, B, C \in \mathrm{M}_2(\mathbb{R})$ with $C = A + B$, $\sigma(A) = \{\alpha_1, \alpha_2\}$, $\sigma(B) = \{\beta_1, \beta_2\}$ and $\sigma(C) = \{\gamma_1, \gamma_2\}$.

Assume, for the sake of simplicity, that the spectra of A and B are respectively $\alpha = (4, 2)$ and $\beta = (2, -2)$. Then the conditions above may be read as

$$\gamma_1 + \gamma_2 = 6, \quad \gamma_1 \geq \gamma_2 \tag{6.25}$$

$$\gamma_1 \leq 6, \quad \gamma_2 \leq 2. \tag{6.26}$$

This shows that γ has the form $\gamma = (6 - a, a)$, with $0 \leq a \leq 2$; clearly, $\gamma_1 \geq \gamma_2$. We shall prove that every pair $(6 - a, a)$ with $0 \leq a \leq 2$ can be the spectrum of a sum $A + B$.

In fact, the relations (6.25) and (6.26) lead us to consider (in the plane $0\gamma_1\gamma_2$) the line segment XY, where $X = (6, 0)$ and $Y = (4, 2)$. Starting with the matrices

$$A = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$$

and

$$R_\theta^\star \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} R_\theta,$$

where

$$R_\theta = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix},$$

we should remark that the spectrum $(\lambda_1^{\downarrow}(C_\theta), \lambda_2^{\downarrow}(C_\theta))$ of the matrix

$$C_\theta = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} + R_\theta^\star \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} R_\theta$$

lies on the line segment XY for all $\theta \in [0, \pi/2]$. In fact, since the eigenvalues of a matrix are continuous functions on the entries of that matrix, the map

$$\theta \to (\lambda_1^{\downarrow}(C_\theta), \lambda_2^{\downarrow}(C_\theta))$$

is continuous. The trace formula shows that the image of this map is a subset of the line $\gamma_1 + \gamma_2 = 6$. The point X corresponds to $\theta = 0$, and Y corresponds to $\theta = \pi/2$. Since the image should be a line segment, we conclude that each point of the linear segment XY represents the spectrum of a matrix C_θ with $\theta \in [0, \pi/2]$.

The list of inequalities involved in the 3×3-dimensional case of Horn's conjecture is considerably larger and consists of 13 items:

- *Weyl's inequalities*,

$$\begin{aligned} &\gamma_1 \le \alpha_1 + \beta_1 \quad \gamma_2 \le \alpha_1 + \beta_2 \quad \gamma_2 \le \alpha_2 + \beta_1 \\ &\gamma_3 \le \alpha_1 + \beta_3 \quad \gamma_3 \le \alpha_3 + \beta_1 \quad \gamma_3 \le \alpha_2 + \beta_2; \end{aligned}$$

- *Ky Fan's inequality*,

$$\gamma_1 + \gamma_2 \le \alpha_1 + \alpha_2 + \beta_1 + \beta_2;$$

- *Lidskii–Wielandt inequalities* (taking into account the symmetric role of A and B),

$$\begin{aligned} \gamma_1 + \gamma_3 &\le \alpha_1 + \alpha_3 + \beta_1 + \beta_2 \\ \gamma_2 + \gamma_3 &\le \alpha_2 + \alpha_3 + \beta_1 + \beta_2 \\ \gamma_1 + \gamma_3 &\le \alpha_1 + \alpha_2 + \beta_1 + \beta_3 \\ \gamma_2 + \gamma_3 &\le \alpha_1 + \alpha_2 + \beta_2 + \beta_3; \end{aligned}$$

- *Horn's inequality*,

$$\gamma_2 + \gamma_3 \le \alpha_1 + \alpha_3 + \beta_1 + \beta_3; \tag{6.27}$$

- *trace identity*,

$$\gamma_1 + \gamma_2 + \gamma_3 = \alpha_1 + \alpha_2 + \alpha_3 + \beta_1 + \beta_2 + \beta_3.$$

Horn's inequality (6.27) follows from (6.20), which in the case $n = 3$ may be read as

$$(\alpha_1 + \beta_3, \alpha_2 + \beta_2, \alpha_3 + \beta_1) \prec_{HLP} (\gamma_1, \gamma_2, \gamma_3).$$

The aforementioned set of 13 relations provides necessary and sufficient conditions for the existence of three symmetric matrices $A, B, C \in \mathrm{M}_3(\mathbb{R})$, with $C = A + B$, and spectra equal respectively to

$$\alpha_1 \ge \alpha_2 \ge \alpha_3; \quad \beta_1 \ge \beta_2 \ge \beta_3; \quad \gamma_1 \ge \gamma_2 \ge \gamma_3.$$

The proof is similar to that in the case $n = 2$.

For larger n, things become much more intricate. For example, for $n = 7$, Horn's list includes 2062 inequalities, not all of them independent.

The multiplicative companion to Horn's inequalities is a by-product of the solution of Horn's conjecture. See W. Fulton [205].

6.6.1 Theorem *Let $\alpha_1 \geq \cdots \geq \alpha_N$, $\beta_1 \geq \cdots \geq \beta_N$, $\gamma_1 \geq \cdots \geq \gamma_N$, be strings of positive real numbers. Then there exist matrices A and B with singular numbers $s_k(A) = \alpha_k$, $s_k(B) = \beta_k$, $s_k(AB) = \gamma_k$, if and only if*

$$\prod_{k\in K} \gamma_k \leq \prod_{i\in I} \alpha_i \prod_{j\in J} \beta_j$$

for all admissible triplets (I, J, K).

6.7 The Way of Hyperbolic Polynomials

Motivated by the theory of hyperbolic partial differential equations, L. Gårding [216], [217] has developed the theory of hyperbolic polynomials, that connects in an unexpected way algebra with convex analysis.

In what follows, by a polynomial on a finite-dimensional real vector space V, we will mean any real-valued functions on V that can be represented as a finite sum of finite products of linear functionals. Assuming that $\dim V = N$, and considering a vector basis $\mathbf{v}_1, ..., \mathbf{v}_N$ of V and its dual vector basis $\mathbf{v}_1^*, ..., \mathbf{v}_N^*$ of V^*, it follows that every such polynomial $p(\mathbf{x})$ is nothing but a usual polynomial in the coefficients of $\mathbf{x}$.

If p is a nonconstant polynomial on V and m is a strictly positive integer, then we say that p is *homogeneous* of degree m if $p(t\mathbf{x}) = t^m p(\mathbf{x})$ for all $t \in \mathbb{R}$ and every $\mathbf{x} \in V$.

6.7.1 Definition (L. Gårding [216]) *A homogeneous polynomial p is called* hyperbolic *in the direction $\mathbf{d} \in V$ if $p(\mathbf{d}) > 0$ and the one real-variable polynomial function $t \to P(\mathbf{x} - t\mathbf{d})$ has only real roots for all vectors $\mathbf{x}$.*

Some simple examples of hyperbolic polynomials are as follows.

The polynomial $p(x_1, ..., x_N) = x_1 \cdots x_N$ is hyperbolic in the direction $\mathbf{d} = (1, ..., 1)$; the roots of $p(\mathbf{x} - t\mathbf{d})$ are exactly $x_1, x_2, ..., x_N$. This polynomial is actually hyperbolic in every direction $\mathbf{d} \in \mathbb{R}^N_{++}$ such that $p(\mathbf{d}) > 0$.

The polynomial $p(x_0, x_1, ..., x_N) = x_0^2 - x_1^2 - \cdots - x_N^2$ is hyperbolic in the direction $\mathbf{d} = (1, 0, ..., 0)$; the roots of $p(\mathbf{x} - t\mathbf{d})$ are $x_0 \pm \sqrt{x_1^2 + \cdots + x_N^2}$. It is also hyperbolic in every direction $\mathbf{d}$ for which $p(\mathbf{d}) > 0$.

The function det, when restricted to $\mathrm{Sym}(N, \mathbb{R})$, can be considered as a polynomial in the entries above the diagonal. The polynomial det is hyperbolic in the direction $\mathbf{d} = \mathrm{I}$, the identity matrix. For each $X \in \mathrm{Sym}(N, \mathbb{R})$, the roots of $\det(X - t\mathrm{I})$ are precisely the eigenvalues of X; det is also hyperbolic in every direction $\mathbf{d} \in \mathrm{Sym}^{++}(N, \mathbb{R})$.

Now, consider the polynomial $p(\mathbf{x}) = \det(x_1A_1 + \cdots + x_NA_N)$, where $A_1, ..., A_N \in \mathrm{Sym}^{++}(N, \mathbb{R})$. Then $p(x)$ is hyperbolic in the direction $\mathbf{d} = (1, 0, ..., 0)$ because

$$p(\mathbf{x} - t\mathbf{d}) = \det\left(\sum_{k=1}^{n} x_kA_k - tA_1\right) = \det A_1 \det\left(\sum_{k=1}^{n} x_kA_1^{-1/2}A_kA_1^{-1/2} - t\mathrm{I}\right).$$

6.7.2 Remark *The Helton–Vinnikov theorem* [249] (*previously a conjecture due to P. Lax*) *asserts that every hyperbolic polynomial $p(x_1, x_2, x_3)$ in three variables possesses a representation of the form*

$$p(x_1, x_2, x_3) = \det(x_1A_1 + x_2A_2 + x_3A_3),$$

where A_1, A_2, A_3 are real symmetric matrices.

A simple way to generate new hyperbolic polynomials from old ones is by differentiation:

6.7.3 Proposition (L. Gårding [217]) *Suppose that p is a hyperbolic polynomial in the direction $\mathbf{d}$ and p is homogeneous of degree $m > 1$. Then $Q(\mathbf{x}) = \langle \nabla p(\mathbf{x}), \mathbf{d}\rangle$ is also a hyperbolic polynomial in the direction $\mathbf{d}$.*

Proof. Since $Q(\mathbf{x} + t\mathbf{d}) = \langle \nabla p(\mathbf{x} + t\mathbf{d}), d\rangle = \frac{\mathrm{d}}{\mathrm{d}t}p(\mathbf{x} + t\mathbf{d})$, we infer from Rolle's theorem that the polynomial $Q(\mathbf{x}+t\mathbf{d})$ has $m-1$ real roots, separating the roots of $p(\mathbf{x} + t\mathbf{d})$. On the other hand, by differentiating the identity $p(t\mathbf{d}) = t^m p(\mathbf{d})$ we obtain $\langle \nabla p(t\mathbf{d}), \mathbf{d}\rangle = mt^{m-1}p(\mathbf{d}) > 0$ for $t \neq 0$, whence $Q(\mathbf{d}) > 0$. ■

6.7.4 Corollary *All elementary symmetric polynomials $e_k(x_1, ..., x_n)$ of strictly positive degree are hyperbolic in the direction $\mathbf{d} = (1, ..., 1)$.*

Proof. Indeed, it was already noticed that $e_n(x_1, ..., x_n) = x_1 \cdots x_n$ is hyperbolic in the direction $\mathbf{d}$. Then $e_{n-1}(\mathbf{x}) = \langle \nabla e_n(\mathbf{x}), \mathbf{d}\rangle$ is also hyperbolic by Proposition 6.7.3 and this argument can be iterated $n - 1$ steps. ■

By applying Proposition 6.7.3 to the polynomial $p(X) = \det X$ and the direction $\mathbf{d} = \mathrm{I}$, one obtains a new family of hyperbolic polynomials defined on $\mathrm{Sym}(N, \mathbb{R})$,

$$\sigma_k(X) = e_k(\lambda_1(X), ..., \lambda_N(X)), \quad k = 1, ..., N,$$

where $\lambda_1(X), ..., \lambda_N(X)$ are the eigenvalues of X. This example can be generalized.

If $p(\mathbf{x})$ is a hyperbolic polynomial in the direction $\mathbf{d}$, of degree m, then for each $\mathbf{x} \in V$ we have

$$p(\mathbf{x} + t\mathbf{d}) = p(\mathbf{d}) \prod_{k=1}^{m} (t + \lambda_k(\mathbf{x})),$$

where $\lambda_1(\mathbf{x}), ..., \lambda_m(\mathbf{x})$ are the so called $\mathbf{d}$-*eigenvalues* of $\mathbf{x}$. The terminology is motivated by the case of the polynomial $p(X) = \det X$ and the direction $\mathbf{d} = \mathrm{I}$. These functions are continuous (and dependent on p and $\mathbf{d}$). The continuity

follows from the continuous dependence of the roots of the polynomial of its coefficients. See Q. I. Rahman and G. Schmeisser [494], Theorem 1.3.1, p. 10.

Notice that

$$p(\mathbf{x}) = p(\mathbf{d}) \prod_{k=1}^{m} \lambda_k(\mathbf{x}). \tag{6.28}$$

The following immediate result implies that the $\mathbf{d}$-eigenvalues of $\mathbf{x}$ are also positively homogeneous.

6.7.5 Lemma *Denote by* $\lambda_k^{\downarrow}(\mathbf{x})$ *the* k*-th largest* $\mathbf{d}$*-eigenvalue of* $\mathbf{x}$. *Then for all* $s, t \in \mathbb{R}$ *we have*

$$\lambda_k^{\downarrow}(s\mathbf{x} + t\mathbf{d}) = \begin{cases} s\lambda_k^{\downarrow}(\mathbf{x}) + t & \text{if } s \geq 0 \\ s\lambda_{m-k+1}^{\downarrow}(\mathbf{x}) + t & \text{if } s < 0. \end{cases}$$

The *eigenvalues map* associated to the family of $\mathbf{d}$-eigenvalues of $\mathbf{x}$ is defined as

$$\Lambda : V \to \mathbb{R}^m, \quad \Lambda(\mathbf{x}) = \left(\lambda_1^{\downarrow}(\mathbf{x}), ..., \lambda_m^{\downarrow}(\mathbf{x})\right).$$

See Exercise 2 for an important result concerning this map.

As will be shown in what follows, the $\mathbf{d}$-eigenvalues share many of the properties of the eigenvalues of symmetric matrices.

The *hyperbolicity cone* of p is the set

$$\begin{aligned} \Gamma_{p,\mathbf{d}} &= \{\mathbf{x} \in V : p(\mathbf{x} - t\mathbf{d}) \neq 0, \text{ for all } t \leq 0\} \\ &= \{\mathbf{x} \in V : \lambda_k(\mathbf{x}) > 0, \text{ for } k = 1, ..., m\}. \end{aligned}$$

The hyperbolicity cone $\Gamma_{p,\mathbf{d}}$ associated with the polynomial $x_1 \cdots x_n$ and the direction $\mathbf{d} = (1, ..., 1)$ is the open positive orthant $\mathbb{R}^N_{++}$. In the case of the polynomial det and the direction $\mathbf{d} = \mathrm{I}$, the hyperbolicity cone is $\mathrm{Sym}^{++}(n, \mathbb{R})$.

The hyperbolicity cone plays an important role in emphasizing the convexity properties of hyperbolic polynomials.

6.7.6 Theorem (L. Gårding [216], [217]) *Suppose that* $p(\mathbf{x})$ *is a hyperbolic polynomial in the direction* $\mathbf{d}$, *of degree* m. *Then:*

(a) *The hyperbolicity cone* $\Gamma_{p,\mathbf{d}}$ *is open and convex and* $\mathbf{d} \in \Gamma_{p,\mathbf{d}}$;

(b) *The polynomial* p *is hyperbolic in every direction* $\mathbf{e} \in \Gamma_{p,\mathbf{d}}$ *and* $\Gamma_{p,d} = \Gamma_{p,\mathbf{e}}$;

(c) $p(\mathbf{x} + \mathbf{y}) \geq p(\mathbf{x})$ *whenever* $\mathbf{y} \in \Gamma_{p,\mathbf{d}}$;

(d) $\Gamma_{p,\mathbf{d}} = \left\{\mathbf{x} \in V : \lambda_m^{\downarrow}(\mathbf{x}) > 0\right\}$ *and* $\overline{\Gamma}_{p,\mathbf{d}} = \left\{\mathbf{x} \in V : \lambda_m^{\downarrow}(\mathbf{x}) \geq 0\right\}$;

(e) $p(\mathbf{x})^{1/m}$ *is a concave function on* $\Gamma_{p,\mathbf{d}}$, *which vanishes on its boundary;*

(f) *The largest eigenvalue* $\lambda_1^{\downarrow}(\mathbf{x})$ *is a convex function* (*while* $\lambda_m^{\downarrow}(\mathbf{x})$ *is a concave function*).

An important ingredient in the proof of Theorem 6.7.6 is the following lemma.

6.7.7 Lemma (J. Renegar [500]) *The hyperbolicity cone is the connected component of the open set* $\{\mathbf{x} : p(\mathbf{x}) \neq 0\}$, *containing* $\mathbf{d}$.

Proof. Let S denote the connected component containing $\mathbf{d}$. Since $\mathbf{x}$ has 0 as an eigenvalue only if $p(\mathbf{x}) = 0$, and since $\mathbf{d} \in \Gamma_{p,\mathbf{d}}$, it follows from the continuity of the eigenvalues that $S \subset \Gamma_{p,\mathbf{d}}$. For the other inclusion, consider $\mathbf{x} \in \Gamma_{p,\mathbf{d}}$ and let L be the line segment with endpoints $\mathbf{x}$ and $\mathbf{d}$. For sufficiently large $t^* > 0$, all $\mathbf{y} \in L$ satisfy $p(\mathbf{y} + t^*\mathbf{d}) > 0$. Also, since $\mathbf{x}, \mathbf{d} \in \Gamma_{p,\mathbf{d}}$, we know that $\mathbf{x} + t\mathbf{d}$ and $\mathbf{d} + t\mathbf{d}$ belong to $\Gamma_{p,\mathbf{d}}$ for $t \geq 0$, which implies that $p(\mathbf{x} + t\mathbf{d}) > 0$ and $p(\mathbf{d} + t\mathbf{d}) > 0$ for $t \geq 0$. Thus, the segments $\{\mathbf{x} + t\mathbf{d} : 0 \leq t \leq t^*\}$, $\{\mathbf{y} + t^*\mathbf{d} : \mathbf{y} \in L\}$, and $\{\mathbf{d} + t\mathbf{d} : 0 \leq t \leq t^*\}$ form a path from $\mathbf{x}$ to $\mathbf{d}$ on which p remains strictly positive. This shows that $\mathbf{x} \in S$ and the proof of the inclusion $\Gamma_{p,\mathbf{d}} \subset S$ is done. ■

Proof of Theorem 6.7.6. (a) According to Lemma 6.7.7, the hyperbolicity cone $\Gamma_{p,\mathbf{d}}$ is open and contains $\mathbf{d}$. This cone is also convex. To show this, let $\mathbf{v}, \mathbf{w} \in \Gamma_{p,\mathbf{d}}$. As was noticed in the proof of Lemma 6.7.7, the line segment joining $\mathbf{v}$ and $\mathbf{d}$ lies inside $\Gamma_{p,\mathbf{d}}$. Now, if $\mathbf{e}$ is an arbitrary point in $\Gamma_{p,\mathbf{d}}$, we have $\Gamma_{p,\mathbf{d}} = \Gamma_{p,\mathbf{e}}$ because of the maximality of a connected component. Therefore $\mathbf{v} \in \Gamma_{p,\mathbf{e}}$ and the line segment joining $\mathbf{v}$ and $\mathbf{w}$ lies inside $\Gamma_{p,\mathbf{w}} = \Gamma_{p,\mathbf{e}}$ $(= \Gamma_{p,\mathbf{d}})$, which ends the proof.

(b) If $\mathbf{e} \in \Gamma_{p,\mathbf{d}}$, then $p(\mathbf{e}) > 0$. See formula (6.28). We will show that for every $\mathbf{x} \in V$, the polynomial $t \to p(\mathbf{x} + t\mathbf{e})$ has only real roots. Let $\alpha > 0$ and $s \in \mathbb{R}$ be two parameters, and consider the polynomial $t \to p(s\mathbf{x} + t\mathbf{e} + \alpha\mathrm{i}\mathbf{d})$. We claim that if $s \geq 0$, then this polynomial has only roots with strictly negative imaginary part. Clearly, this is true for $s = 0$ since $t \to p(t\mathbf{e} + \alpha\mathrm{i}\mathbf{d})$ cannot have a root at $t = 0$, since $p(\alpha\mathrm{i}\mathbf{d}) = (\alpha\mathrm{i})^m p(\mathbf{d}) = 0$. If $t \neq 0$, then $p(t\mathbf{e} + \alpha\mathrm{i}\mathbf{d}) = 0$ if, and only if, $p(\mathbf{e} + \alpha t^{-1}\mathrm{i}\mathbf{d}) = 0$ which implies $\alpha t^{-1}\mathrm{i} < 0$, and thus $t = r\mathrm{i}$ for some $r < 0$.

If for some $s > 0$ the polynomial $t \to p(s\mathbf{x} + t\mathbf{e} + \alpha\mathrm{i}\mathbf{d})$ would have a root in the upper half-plane, then there must exists s^* for which $t \to p(s^*\mathbf{x} + t\mathbf{e} + \alpha\mathrm{i}\mathbf{d})$ has a real root t^*, that is, $p(s^*\mathbf{x} + t^*\mathbf{e} + \alpha\mathrm{i}\mathbf{d}) = 0$. However, this contradicts the hyperbolicity of p, since $s^*\mathbf{x} + t^*\mathbf{e} \in V$. Thus, for all $s \geq 0$, the roots of $t \to p(s\mathbf{x} + t\mathbf{e} + \alpha\mathrm{i}\mathbf{d})$ have strictly negative imaginary parts.

The conclusion above was true for every $\alpha > 0$. Letting $\alpha \to 0$, by continuity of the roots we have that the polynomial $t \to p(s\mathbf{x} + t\mathbf{e})$ must also have only roots with negative imaginary parts. However, since it is a polynomial with real coefficients (and therefore its roots always appear in complex conjugate pairs), then all the roots must actually be real. Taking now $s = 1$, we have that $t \to p(\mathbf{x} + t\mathbf{e})$ has real roots for all $\mathbf{x}$.

The assertion (c) is a consequence of assertion (b) and Lemma 6.7.5.

(d) The proof is left to the reader as Exercise 7.

(e) We follow the ingenious argument of L. Hörmander [268], p. 63. According to Proposition 4.1.2, it suffices to prove that if $\mathbf{x} \in \Gamma$ and $\mathbf{y} \in V$, then the function $\varphi(t) = p^{1/m}(\mathbf{x} + t\mathbf{y})$ is concave on $\operatorname{dom}\varphi = \{t \in \mathbb{R} : \mathbf{x} + t\mathbf{y} \in \Gamma\}$. We know that

$$p(\mathbf{y} + t\mathbf{x}) = p(\mathbf{x}) \prod_{k=1}^{m} (t - r_i)$$

for suitable real numbers r_i, whence

$$p(\mathbf{x}+t\mathbf{y}) = t^m p(\mathbf{y}+\mathbf{x}/m) = p(\mathbf{x})\prod_{k=1}^{m}(1-tr_i);$$

we have $1-tr_i > 0$ since $\mathbf{x}+t\mathbf{y} \in \Gamma$. Consider the function $f(t) = \log p(\mathbf{x}+t\mathbf{y})$. Then

$$f'(t) = -\sum_{k=1}^{m}\frac{r_i}{1-tr_i}, \quad f''(t) = -\sum_{k=1}^{m}\frac{r_i^2}{(1-tr_i)^2},$$

and

$$\begin{aligned} m^2 \mathrm{e}^{-f(t)/m}\frac{\mathrm{d}^2}{\mathrm{d}t^2}\mathrm{e}^{f(t)/m} &= f'(t)^2 + mf''(t) \\ &= \left(\sum_{k=1}^{m}\frac{r_i}{1-tr_i}\right)^2 - m\sum_{k=1}^{m}\frac{r_i^2}{(1-tr_i)^2} \le 0, \end{aligned}$$

according to the Cauchy–Bunyakovsky–Schwarz inequality. This proves that the function $t \to p^{1/m}(\mathbf{x}+t\mathbf{y})$ is concave on $\Gamma_{p,\mathbf{d}}$. The fact that p vanishes on the boundary of $\Gamma_{p,\mathbf{d}}$ follows from assertion (d).

(f) Notice that $\lambda_m^{\downarrow}(\mathbf{x}) \ge \alpha$ if, and only if, $\lambda_m^{\downarrow}(\mathbf{x}-\alpha\mathbf{d}) \ge 0$, whence

$$\left\{\mathbf{x} : \lambda_m^{\downarrow}(\mathbf{x}) \ge \alpha\right\} = \alpha\mathbf{d} + \Gamma_{p,\mathbf{d}}.$$

According to assertion (a) this is a convex set. Since the function $\lambda_m^{\downarrow}(\mathbf{x})$ is positively homogeneous, we conclude from Lemma 4.4.2 that $\lambda_m^{\downarrow}(\mathbf{x})$ is also a concave function. ■

The following theorem extends the results stated in Section 6.6.

6.7.8 Theorem (D. Serre [530]) *Horn's inequalities remain valid for the eigenvalues of every hyperbolic polynomial.*

Exercises

1. (New hyperbolic polynomials from old ones)

   (a) Suppose that $p : V \to \mathbb{R}$ is a hyperbolic polynomial in the direction $\mathbf{d} \in V$ and W is a vector subspace of V that contains $\mathbf{d}$. Prove that $p_{|W}$ is also hyperbolic (in the same direction). Moreover $\Gamma_{p_{|W},\mathbf{d}} = \Gamma_{p,\mathbf{d}} \cap W$.

   (b) Let p and q be two hyperbolic polynomials in the direction $\mathbf{d}$. Prove that pq is also hyperbolic in the direction $\mathbf{d}$ and $\Gamma_{pq,\mathbf{d}} = \Gamma_{p,\mathbf{d}} \cap \Gamma_{q,\mathbf{d}}$.

   (c) Let p be a hyperbolic polynomial in the direction $\mathbf{d}$, of degree $m \ge 1$, and assume that $p(\mathbf{x}+t\mathbf{d}) = \sum_{k=0}^{m} p_k(\mathbf{x})t^k$. Prove that all coefficients $p_k(\mathbf{x})$ are hyperbolic polynomials in the direction $\mathbf{d}$.

   [*Hint:* For (c), use the formula

   $$p_k(\mathbf{x}) = \frac{1}{k!}\,\frac{d^k}{dt^k}p(\mathbf{x}+t\mathbf{d})\bigg|_{t=0} = \langle \nabla^k p(\mathbf{x}), \mathbf{d}\rangle. \ ]$$

2. (H. H. Bauschke, O. Güler, A. S. Lewis and H. S. Sendov [44]) Infer from Exercise 1 (c) the following new construction for hyperbolic polynomials. Suppose that q is a homogeneous symmetric polynomial of degree m on $\mathbb{R}^N$, hyperbolic in the direction $\mathbf{e} = (1, 1, ..., 1)$, with eigenvalue map Φ and let Λ be the eigenvalue map of a hyperbolic polynomial of degree m in the direction $\mathbf{d}$. Prove that $q \circ \Lambda$ is also a hyperbolic polynomial of degree m in the direction $\mathbf{d}$, with eigenvalue map $\Phi \circ \Lambda$.

3. Let $1 \le k \le N$ and $\mathbf{e} = (1, 1, ..., 1) \in \mathbb{R}^N$. Prove that the polynomial

$$q(\mathbf{u}) = \prod_{1 \le i_1 < \cdots < i_k \le N} \sum_{j=1}^{k} u_{i_j}$$

is hyperbolic, of degree $\binom{N}{k}$, in the direction $\mathbf{e}$, and its eigenvalues are the functions

$$\lambda_{i_1,...,i_k}(\mathbf{u}) = \frac{1}{k} \sum_{j=1}^{k} u_{i_j} \quad \text{for } 1 \le i_1 < \cdots < i_k \le N.$$

Notice that the largest eigenvalue is the arithmetic mean of the k largest components of $\mathbf{u}$.

4. Infer from Exercises 2 and 3 and Theorem 6.7.6 (f) that the sums of the first k largest eigenvalues of a hyperbolic polynomial of degree m

$$\sigma_k(\mathbf{x}) = \lambda_1^{\downarrow}(\mathbf{x}) + \cdots + \lambda_k^{\downarrow}(\mathbf{x}) \quad \text{for } k = 1, ..., m,$$

are positively homogeneous and sublinear (in particular, convex). Prove that $\sigma_m(\mathbf{x})$ is actually a linear function.

5. Let $\mathbf{w} \in \mathbb{R}^m$ be a vector such that $w_1 \ge w_2 \ge \cdots \ge w_m$ and let Λ be the eigenvalues map of a hyperbolic polynomial of degree m. Infer from Exercise 4 that the function $\mathbf{x} \to \langle \mathbf{w}, \Lambda(\mathbf{x}) \rangle$ is positively homogeneous and sublinear (in particular, convex). This extends the fact that the largest eigenvalue is positively homogeneous and sublinear.

[*Hint:* According to Abel's partial summation formula,

$$\langle \mathbf{w}, \Lambda(\mathbf{x}) \rangle = \sum_{k=1}^{m} w_k \lambda_k(\mathbf{x}) = w_m \sigma_m(\mathbf{x}) + \sum_{k=1}^{m-1} (w_k - w_{k+1}) \lambda_k(\mathbf{x}). \ ]$$

6. Prove Lemma 6.7.5.

7. Prove assertion (d) of Theorem 6.7.6, that is, the formulas

$$\Gamma_{p,\mathbf{d}} = \left\{ \mathbf{x} \in V : \lambda_m^{\downarrow}(\mathbf{x}) > 0 \right\} \text{ and } \overline{\Gamma}_{p,\mathbf{d}} = \left\{ \mathbf{x} \in V : \lambda_m^{\downarrow}(\mathbf{x}) \ge 0 \right\}.$$

8. A polynomial $q \in \mathbb{R}[x_1, ..., x_N]$ of degree $m \geq 1$ is called *real stable* if for all $\mathbf{x} \in \mathbb{R}^N$ and $\mathbf{d} \in \mathbb{R}^N_{++}$, the polynomial $q(\mathbf{x} + t\mathbf{d}) \in \mathbb{R}[t]$ has only real roots. Prove that q is real stable if, and only if, its homogenization

$$q_h(x_0, x_1, ...x_N) = x_0^m q\left(\frac{x_1}{x_0}, ..., \frac{x_N}{x_0}\right)$$

is hyperbolic in the direction $(0, \mathbf{d})$ for all $\mathbf{d} \in \mathbb{R}^N_{++}$.

6.8 Vector Majorization in $\mathbb{R}^N$

The usual relation of majorization, described in Section 6.1 as a relation between strings of real numbers can be easily generalized as a relation between strings of weighted vectors in $\mathbb{R}^N$. This was done by S. Sherman [533], inspired by the equivalence of conditions (a) and (b) in Theorem 6.1.9.

6.8.1 Definition *The relation of majorization*

$$(\mathbf{x}_1, ..., \mathbf{x}_m; \lambda_1, ..., \lambda_m) \prec_{Sh} (\mathbf{y}_1, ..., \mathbf{y}_n; \mu_1, ..., \mu_n) \tag{6.29}$$

between two strings of weighted points in $\mathbb{R}^N$ *is defined by asking the existence of an* $m \times n$*-dimensional matrix* $A = (a_{ij})_{i,j}$ *such that*

$$a_{ij} \geq 0 \quad \textit{for } (i,j) \in \{1, ..., m\} \times \{1, ..., n\} \tag{6.30}$$

$$\sum_{j=1}^{n} a_{ij} = 1 \quad \textit{for } i = 1, ..., m \tag{6.31}$$

$$\mu_j = \sum_{i=1}^{m} a_{ij}\lambda_i \quad \textit{for } j = 1, ..., n \tag{6.32}$$

and

$$\mathbf{x}_i = \sum_{j=1}^{n} a_{ij}\mathbf{y}_j \quad \textit{for } i = 1, ..., m. \tag{6.33}$$

We assume that all weights λ_i and μ_j belong to $[0, 1]$ and

$$\sum_{i=1}^{m} \lambda_i = \sum_{j=1}^{n} \mu_j = 1.$$

The matrices verifying the conditions (6.30) and (6.31) are called *stochastic on rows*. When $m = n$ and all weights λ_i and μ_j are equal to each other, then the condition (6.32) assures the *stochasticity on columns,* so in that case we deal with doubly stochastic matrices.

6.8.2 Remark *The relation of majorization introduced by Definition 6.8.1 can be restated as a relation between probability measures, letting*

$$\sum_{i=1}^{m} \lambda_i \delta_{\mathbf{x}_i} \prec_{Sh} \sum_{j=1}^{n} \mu_j \delta_{\mathbf{y}_j},$$

if the conditions (6.30) - (6.33) *hold. In this context, the condition* (6.33) *means that*

$$\mathbf{x}_i = \text{bar}\left(\sum_{j=1}^{n} a_{ij}\delta_{\mathbf{y}_j}\right) \quad \text{for } i = 1, ..., m,$$

a fact that allows easily to define the relation of majorization between probability measures not only in $\mathbb{R}^N$*, but also in any space where the notion of barycenter of a probability measure makes sense. Notice also that the conditions* (6.31) *and* (6.33) *imply that*

$$\mathbf{x}_1, ..., \mathbf{x}_m \in \text{conv}\{\mathbf{y}_1, ..., \mathbf{y}_n\}.$$

The following theorem provides a large extension of the Hardy–Littlewood–Pólya inequality of majorization:

6.8.3 Theorem (S. Sherman [533]) *Suppose that* $\sum_{i=1}^{m} \lambda_i \delta_{\mathbf{x}_i}$ *and* $\sum_{j=1}^{n} \mu_j \delta_{\mathbf{y}_j}$ *are two Borel probability measures on* $\mathbb{R}^N$*. Then the following assertions are equivalent:*

(a) $\sum_{i=1}^{m} \lambda_i \delta_{\mathbf{x}_i} \prec_{Sh} \sum_{j=1}^{n} \mu_j \delta_{\mathbf{y}_j}$;

(b) $\mathbf{x}_1, ..., \mathbf{x}_m \in \text{conv}\{\mathbf{y}_1, ..., \mathbf{y}_n\}$ *and every continuous convex function* f *defined on* $\text{conv}\{\mathbf{y}_1, ..., \mathbf{y}_n\}$ *(or to a larger convex subset of* $\mathbb{R}^N$*) verifies the inequality*

$$\sum_{i=1}^{m} \lambda_i f(\mathbf{x}_i) \leq \sum_{j=1}^{n} \mu_j f(\mathbf{y}_j). \tag{6.34}$$

Proof. The implication (a) ⇒ (b) is a consequence of Remark 6.8.2. Indeed, assuming the existence of an $m \times n$-dimensional matrix $A = (a_{ij})_{i,j}$ which verifies the conditions (6.30) - (6.33), we infer from Jensen's inequality that

$$f(\mathbf{x}_i) \leq \sum_{j=1}^{n} a_{ij} f(\mathbf{y}_j)$$

for every $i \in \{1, ..., m\}$. Multiplying each side by λ_i and then summing up over i we conclude that

$$\sum_{i=1}^{m} \lambda_i f(\mathbf{x}_i) \leq \sum_{i=1}^{m} \left(\lambda_i \sum_{j=1}^{n} a_{ij} f(\mathbf{y}_j)\right) = \sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij}\lambda_i\right) f(\mathbf{y}_j) = \sum_{j=1}^{n} \mu_j f(\mathbf{y}_j).$$

The proof of the implication (b) ⇒ (a) follows the argument of J. Borcea [77], pp. 3212-3214.

The starting remark is the non-emptiness of the set $\mathcal{M}$, of $m \times n$-dimensional matrices $A = (a_{ij})_{i,j}$ that are stochastic on rows and verify the condition (6.33); this is a consequence of the fact that $\mathbf{x}_1, ..., \mathbf{x}_m \in \text{conv}\{\mathbf{y}_1, ..., \mathbf{y}_n\}$. Thus, the proof of the implication (b) ⇒ (a) reduces to the existence of a matrix $A \in \mathcal{M}$ such that

$$\mu = \lambda \cdot A,$$

where $\mu = (\mu_1, ..., \mu_n)$ and $\lambda = (\lambda_1, ..., \lambda_m)$. The set $\mathcal{M}$ is a closed convex set and the same is true for

$$\lambda\mathcal{M} = \{\lambda \cdot A : A \in \mathcal{M}\}.$$

By the basic separation theorem, if $\mu \notin \lambda\mathcal{M}$, then there exist $c \in \mathbb{R}$ and a $(r_1, ..., r_n) \in \mathbb{R}^n$ such that

$$\sum_{j=1}^{n} r_j \nu_j < c < \sum_{j=1}^{n} r_j \mu_j \tag{6.35}$$

for all $(\nu_1, ..., \nu_n) \in \lambda\mathcal{M}$. Consider the function $F : \operatorname{conv}\{\mathbf{y}_1, ..., \mathbf{y}_n\} \to \mathbb{R}$ defined by

$$F(\mathbf{y}) = \max \sum_{j=1}^{n} r_j c_j,$$

where the maximum is taken over the nonempty compact convex set

$$C(\mathbf{y}) = \left\{ (c_1, ..., c_n) : c_1, ..., c_n \geq 0, \sum_{j=1}^{n} c_j = 1 \text{ and } \mathbf{y} = \sum_{j=1}^{n} c_j \mathbf{y}_j \right\}.$$

One can show easily that this is a continuous concave function.

Since $C(\mathbf{y})$ is compact, we may choose $(c_1(\mathbf{y}), ..., c_n(\mathbf{y}))$ in $C(\mathbf{y})$ such that $F(\mathbf{y}) = \sum_{j=1}^{n} r_j c_j(\mathbf{y})$. Since $c_j(\mathbf{y}_i) = \delta_{ij}$, we have $F(\mathbf{y}_j) \geq r_j$ for all j, hence

$$\sum_{j=1}^{n} \mu_j F(\mathbf{y}_j) \geq \sum_{j=1}^{n} r_j \mu_j > c, \tag{6.36}$$

according to the right inequality in (6.35). On the other hand, since

$$(c_1(x_i), ..., c_n(x_i)) \in \mathcal{M}$$

for every index i, we infer from the left inequality in (6.35) that

$$c \geq \sum_{j=1}^{n} r_j \left(\sum_{i=1}^{m} \lambda_i c_j(\mathbf{x}_i) \right) = \sum_{i=1}^{m} \lambda_i \left(\sum_{j=1}^{n} r_j c_j(\mathbf{x}_i) \right) = \sum_{i=1}^{m} \lambda_i F(\mathbf{x}_i). \tag{6.37}$$

Combining (6.36) and (6.37) we get

$$-\sum_{i=1}^{m} \lambda_i F(\mathbf{x}_i) > -\sum_{j=1}^{n} \mu_j F(\mathbf{y}_j),$$

a fact that contradicts our hypothesis (b) because the function $-F$ is convex and continuous. Consequently $\mu \in \lambda\mathcal{M}$ and the proof is complete. ■

Concerning the assertion (b) of Theorem 6.8.3, let us notice that if the inequality (6.34) holds for all continuous convex functions defined on convex

sets containing all points $\mathbf{x}_1, ..., \mathbf{x}_m, \mathbf{y}_1, ..., \mathbf{y}_n$, then necessarily $\mathbf{x}_1, ..., \mathbf{x}_m \in \operatorname{conv}\{\mathbf{y}_1, ..., \mathbf{y}_n\}$. To check this, see the case of the function

$$f(\mathbf{x}) = d(\mathbf{x}, \operatorname{conv}\{\mathbf{y}_1, ..., \mathbf{y}_n\}).$$

We next present a connection between the majorization theory and the distribution of roots of a polynomial.

The Gauss–Lucas theorem asserts that the roots $(\mu_k)_{k=1}^{n-1}$ of the derivative P' of any complex polynomial $P \in \mathbb{C}[z]$ of degree $n \geq 2$ lie in the smallest convex polygon containing the roots $(\lambda_j)_{j=1}^{n}$ of the polynomial P. See Section 3.1. Letting $\mu_n = (1/n)\sum_{k=1}^{n} \lambda_k$, this theorem yields the existence of a stochastic on rows matrix S such that $\mu = S\lambda$, where $\mu = (\mu_1, ..., \mu_n)$ and $\lambda = (\lambda_1, ..., \lambda_n)$. See Theorem 6.1.9 and Theorem 6.1.12. S. M. Malamud [353] and, independently, R. Pereira [469], were able to prove stronger results concerning the location of the roots of a polynomial and of its derivative. In particular, they turned into a theorem a conjecture raised by N. G. de Bruijn and T. A. Springer in 1947:

6.8.4 Theorem *For any convex function $f : \mathbb{C} \to \mathbb{R}$ and any polynomial P of degree $n \geq 2$,*

$$\frac{1}{n-1}\sum_{k=1}^{n-1} f(\mu_k) \leq \frac{1}{n}\sum_{k=1}^{n} f(\lambda_k),$$

where $(\lambda_k)_{k=1}^{n}$ and $(\mu_k)_{k=1}^{n-1}$ are respectively the roots of P and P'.

The proof of Theorem 6.8.4 needs some preliminaries relating the pair of roots $(\lambda_k)_{k=1}^{n}$ and $(\mu_k)_{k=1}^{n-1}$ with the spectrum of a normal matrix and the spectrum of a compression of it. Our argument follows closely that of J. Borcea [77], pp. 3222-3223.

Recall that a matrix $A \in \mathrm{M}_n(\mathbb{C})$ is called *normal* if it commutes with its adjoint, that is, $AA^* = A^*A$. Since

$$A = \frac{A + A^*}{2} + \mathrm{i}\frac{A - A^*}{2\mathrm{i}}$$

and the self-adjoint matrices $(A + A^*)/2$ and $-\mathrm{i}(A - A^*)/2$ commute, the spectral decomposition of self-adjoint matrices easily yields the existence of a unitary matrix U such that UAU^* is diagonal.

Choose a normal matrix $A \in \mathrm{M}_n(\mathbb{C})$ whose characteristic polynomial is P (as in Theorem 6.8.4) and denote by $(\mathbf{v}_k)_{k=1}^{n}$ an orthonormal basis of $H = \mathbb{C}^n$ consisting of eigenvectors of A such that $A\mathbf{v}_k = \lambda_k \mathbf{v}_k$ for all k. Let

$$\mathbf{w}_n = (1/\sqrt{n})\sum\nolimits_{k=1}^{n} \mathbf{v}_k$$

and let π be the projection onto the linear subspace X, orthogonal to w_n. According to Schur's unitary triangularization theorem (see R. A. Horn and C. R. Johnson [263], Theorem 2.3.1, p. 79), there exists an orthonormal basis $(\mathbf{w}_k)_{k=1}^{n-1}$ of X which triangularizes the operator

$$B = \pi A \pi|_X,$$

the compression of A to X. Since $\|\mathbf{w}_n\| = 1$, the family $\{\mathbf{w}_1, ..., \mathbf{w}_{n-1}, \mathbf{w}_n\}$ constitutes an orthonormal basis of $\mathbb{C}^n$.

6.8.5 Lemma *We have*

$$\langle (A - z\mathrm{I}_H)^{-1} \mathbf{w}_n, \mathbf{w}_n \rangle = \frac{\det(B - z\mathrm{I}_X)}{\det(A - z\mathrm{I}_H)} = -\frac{P'(z)}{nP(z)},$$

for every $z \in \mathbb{C} \setminus \sigma(A)$.

Proof. The matrix representation of B in the basis $(\mathbf{w}_k)_{k=1}^{n-1}$ of X is given by the $(n-1) \times (n-1)$-dimensional principal minor in the matrix of representation of A in the basis $(\mathbf{w}_k)_{k=1}^{n}$ of H. The $n \times n$ coefficient of the matrix representing $(A - z\mathrm{I}_H)^{-1}$ in this basis is

$$\langle (A - z\mathrm{I}_H)^{-1} \mathbf{w}_n, \mathbf{w}_n \rangle = \langle \sum_{k=1}^{n} \frac{\langle \mathbf{w}_n, \mathbf{v}_k \rangle}{\lambda_k - z} \mathbf{v}_k, \sum_{k=1}^{n} \langle \mathbf{w}_n, \mathbf{v}_k \rangle \mathbf{v}_k \rangle = -\frac{P'(z)}{nP(z)}$$

for every $z \in \mathbb{C} \setminus \sigma(A)$. On the other hand, by Cramer's rule, this coefficient equals the cofactor of the $n \times n$ coefficient of $A - z\mathrm{I}_H$ in the same basis. Thus

$$\langle (A - z\mathrm{I}_H)^{-1} \mathbf{w}_n, \mathbf{w}_n \rangle = \frac{\det(B - z\mathrm{I}_X)}{\det(A - z\mathrm{I}_H)}$$

for every $z \in \mathbb{C} \setminus \sigma(A)$ and the proof is complete. ■

Proof of Theorem 6.8.4. According to Lemma 6.8.5, the roots μ_k are the same with the eigenvalues of the compression $B = \pi A \pi|_X$ as defined in the comments preceding this lemma. As was noticed above $\{w_1, ..., w_{n-1}\}$ is an orthonormal basis of X which triangularizes the operator $B = \pi A \pi|_X$ (that is, the compression of A to X). We have

$$\mu_j = \langle A\mathbf{w}_j, \mathbf{w}_j \rangle = \sum\nolimits_{i=1}^{n} \lambda_i \left| \langle \mathbf{w}_j, \mathbf{v}_i \rangle \right|^2.$$

Let S denote the $n-1$ by n matrix with coefficients $s_{ij} = \left| \langle \mathbf{w}_i, \mathbf{v}_j \rangle \right|^2$. By Parseval's theorem,

$$\sum\nolimits_{j=1}^{n} s_{ij} = \sum\nolimits_{j=1}^{n} \left| \langle \mathbf{w}_i, \mathbf{v}_j \rangle \right|^2 = \|\mathbf{w}_i\|^2 = 1$$

for $i \in \{1, ..., n-1\}$, and

$$\sum\nolimits_{i=1}^{n-1} s_{ij} = \sum\nolimits_{i=1}^{n-1} \left| \langle \mathbf{w}_i, \mathbf{v}_j \rangle \right|^2 = \|\mathbf{v}_j\|^2 - \left| \langle \mathbf{v}_j, \mathbf{w}_n \rangle \right|^2 = 1 - 1/n,$$

for $j \in \{1, ..., n-1\}$. This shows that S is a rectangular row stochastic matrix whose columns sum to $(n-1)/n$. Its existence shows that

$$\frac{1}{n-1} \sum_{k=1}^{n-1} \delta_{\mu_k} \prec_{Sh} \frac{1}{n} \sum_{k=1}^{n} \delta_{\lambda_k},$$

so the conclusion of Theorem 6.8.4 follows now from Sherman's majorization Theorem 6.8.3. ■

Other results connecting the roots of a polynomial with the roots of its derivative are mentioned in the Comments section at the end of this chapter.

Exercises

1. Let $A \in \mathrm{M}_n(\mathbb{C})$ be a normal matrix with diagonal elements $a_{11}, \ldots, a_{nn}$ and eigenvalues $\lambda_1, \ldots, \lambda_n$. Prove that there exists a doubly stochastic matrix $S \in \mathrm{M}_n(\mathbb{R})$ such that $\mathbf{a} = S\lambda$, where $\mathbf{a} = (a_{kk})_{k=1}^n$ and $\lambda = (\lambda_k)_{k=1}^n$.

   [*Hint:* Adapt the argument used in the proof of Schur's lemma 6.2.2.]

2. Suppose that $(\lambda_k)_{k=1}^n$ and $(\mu_k)_{k=1}^{n-1}$ are respectively the roots of a polynomial P and of its derivative P' and $n \geq 2$. Assuming that they are all positive, prove that

$$\left(\prod\nolimits_{k=1}^{n} \lambda_k\right)^{1/n} \leq \left(\prod\nolimits_{k=1}^{n-1} \mu_k\right)^{1/n-1}.$$

   This result can be easily generalized within the class of elementary symmetric functions, taking into account that $e_k^{1/k}(x_1, x_2, \ldots, x_n)$ is a concave function. See Section C.1, Appendix C.

3. The Legendre polynomial of order $n \geq 0$ is defined by the formula $P_n(x) = \frac{1}{n!2^n}\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left(x^2-1\right)^n$. Prove that its roots $x_1 < \cdots < x_n$ verify the formula

$$\frac{1}{n}\sum\nolimits_{k=1}^{n} f(x_k) \leq \frac{f(-1)+f(1)}{2}$$

   for all convex functions defined on $[-1, 1]$ and all $n \geq 2$.

4. Consider two interlace sequences

$$\lambda_1 \leq \mu_1 \leq \lambda_2 \leq \cdots \leq \mu_{n-1} \leq \lambda_n.$$

   For example, the roots of successive Legendre polynomials interlace (and the same works in the case of Hermite and Laguerre polynomials). See G. Andrews et al. [21].

   (a) Prove the Markov–Krein identity,

$$\sum_{k=1}^{n} \frac{p_k}{z-\lambda_k} = \frac{(z-\mu_1)\cdots(z-\mu_{n-1})}{(z-\lambda_1)\cdots(z-\lambda_n)},$$

   where $p_1, ..., p_n > 0$ are suitable weights with $\sum_{k=1}^n p_k = 1$.

   (b) Let $\alpha, \beta \in \mathbb{C}$, $\alpha \neq 0$. Prove that

$$\sum_{k=1}^{n-1} f(\alpha\mu_k + \beta) \leq \sum_{k=1}^{n}(1-p_k) f(\alpha\lambda_k + \beta),$$

   for every convex function $f : \mathbb{C} \to \mathbb{R}$.

6.9 Comments

In their celebrated book on *Inequalities*, G. H. Hardy, J. E. Littlewood and G. Pólya [247] proved a number of important characterizations of convex functions based on a reflexive and transitive relation defined on $\mathbb{R}^N$, called by them *majorization*. Their approach nicely incorporated previous work done by R. F. Muirhead, H. Dalton, I. Schur and J. Karamata. Around the 1950s, motivated by the subject of comparison of experiments, researchers like D. Blackwell, S. Sherman, C. Stein, V. Strassen, P. Cartier, J. M. G. Fell, and P. A. Meyer added significant results that connected the majorization theory with statistical decision theory. See the treatise of E. Torgersen [569]. Choquet's theory [132], [474], on the representation of points of a convex compact set as barycenters of probability measures supported by their extreme points enlarged the fields of applications to quantum statistical mechanics. Nowadays, the theory of majorization has become an important tool in numerous other applications to economics, combinatorics, signal processing, etc.

The authoritative monograph on majorization theory is that by A. W. Marshall, I. Olkin and B. C. Arnold [363]. The reader will find there almost all significative results which were published before 2011.

The integral version of the Hardy–Littlewood–Pólya inequality of majorization is based on the concept of decreasing rearrangement.

Given a σ-finite measure space (X, μ) and a measurable function φ from X into $\overline{\mathbb{R}}$ that takes real values μ-almost everywhere, we define the *complementary distribution function* of φ by the formula

$$\lambda_\varphi(y) = \mu(\{x \in X : \varphi(x) > y\}) \quad \text{for } y \in \mathbb{R},$$

and the *decreasing rearrangement* (or, sometimes, decreasing nonsymmetric rearrangement) of φ by the formula

$$\varphi^\downarrow(t) = \inf\{y : \lambda_\varphi(y) \le t\} \quad \text{for } t \in [0, \mu(X)].$$

One can easily prove that φ and $\varphi^\downarrow$ have the same complementary distribution function. As a consequence,

$$\|\varphi\|_{L^p(\mu)} = \left\|\varphi^\downarrow\right\|_{L^p([0,\infty))} \text{ for all } p > 0.$$

Notice also the *Hardy–Littlewood rearrangement inequality*,

$$\int_X |\varphi(x)\psi(x)|\, \mathrm{d}\mu \le \int_X \varphi^\downarrow(t)\psi^\downarrow(t)\mathrm{d}t.$$

See C. Bennett and R. C. Sharpley [51] for details. For more recent results, see A. Burchard and H. Hajaiej [109] (and references therein).

Decreasing rearrangements make possible the following integral variant of the majorization theorems of Hardy, Littlewood, Pólya and Schur:

6.9.1 Theorem *If $\varphi \in L^1(X,\mu)$ and $\psi \in L^1(Y,\nu)$, where $\alpha = \mu(X) = \nu(Y) < \infty$, then the following assertions are equivalent:*

(a) For every convex function $f : \mathbb{R} \to \mathbb{R}$,

$$\int_X f(\varphi(x))\,\mathrm{d}\mu \le \int_Y f(\psi(y))\,\mathrm{d}\nu;$$

(b) $\varphi \prec_{HLP} \psi$ in the sense that

$$\int_0^s \varphi^{\downarrow}(t)\,\mathrm{d}t \le \int_0^s \psi^{\downarrow}(t)\,\mathrm{d}t \quad \text{for all } s \in [0,\alpha).$$

and

$$\int_0^\alpha \varphi^{\downarrow}(t)\,\mathrm{d}t \le \int_0^\alpha \psi^{\downarrow}(t)\,\mathrm{d}t;$$

(c) $\varphi = S(\psi)$ for some doubly stochastic operator $S : L^1(Y,\nu) \to L^1(X,\mu)$, that is, for a positive linear operator S such that $S(1) = 1$ and

$$\int_X S(\xi)\,\mathrm{d}\mu = \int_Y \xi\,\mathrm{d}\nu \quad \text{for all } \xi \in L^1(Y,\nu).$$

See K.-M. Chong [130], Theorem 2.5, for the equivalence (a) $\Leftrightarrow$ (b), and P. W. Day [156], Theorem 4.9, for the equivalence (b) $\Leftrightarrow$ (c).

Applications of the theory of decreasing rearrangements to mathematical risk analysis can be found in the book of L. Rüschendorf [516].

E. H. Lieb and M. Loss [336] offer a thorough presentation of the theory of symmetric-decreasing rearrangements and its applications to problems in mathematical physics. For more applications, see the lecture notes of A. Burchard [108].

Majorization is sometimes related to inequality measurement, but in a series of problems like tax progression and income redistribution, Condorcet jury theorems, "fair representation" in parliaments etc. the use of *Lorenz curves* seems to be more appropriate. See B. C. Arnold [22]. Suppose that X is a positive random variable with $0 < E(X) < \infty$. The Lorenz curve associated to it is parametrized by the formula

$$L_X(u) = \frac{1}{E(X)} \int_0^u F_X^{-1}(t)\mathrm{d}t \quad \text{for } u \in [0,1],$$

where F_X^{-1} represents the generalized inverse of the cumulative distribution function F_X. Clearly, the function L_X is continuous, increasing and convex and moreover, $L_X(0) = 0$ and $L_X(1) = 1$. In risk analysis it is customary to define

$$Y \ge_L X \text{ if and only if } L_Y \le L_X,$$

meaning that Y is more unequal (or more spread out, or more variable) than X in the Lorenz sense.

The quick review of Horn's inequalities as presented in Section 6.6 was inspired by a beautiful paper of R. Bhatia [62].

The results in Section 6.8 concerning the connection between the roots of a polynomial and the roots of its derivative represent the starting point for a sequence of much better inequalities based on other variants of majorization.

6.9.2 Definition *Let* $\mathbf{x} = \{x_1, ..., x_m\}$ *and* $\mathbf{y} = \{y_1, ..., y_n\}$ *be two strings of vectors in* $\mathbb{R}^N$, *with* $m \leq n$. *The doubly stochastic majorization,*

$$\mathbf{x} \prec_{ds} \mathbf{y},$$

means the existence of elements $x_{m+1}, ..., x_n$ *and of a doubly stochastic matrix* $S \in \mathrm{M}_n(\mathbb{R})$ *such that* $\widetilde{\mathbf{x}} = \{x_1, ..., x_n\} = S\mathbf{y}$. *The special case when* S *can be chosen of the form* $S = (|u_{ij}|^2)_{i,j}$, *for a suitable unitary matrix* $U = (u_{ij})_{i,j}$ *will be denoted*

$$\mathbf{x} \prec_{uds} \mathbf{y}.$$

Given a vector $(\lambda_j)_{j=1}^n$ in $\mathbb{C}^n$ and a number $k \in \{1, ..., n\}$ one can attach to them the following element of $\mathbb{C}^{\binom{n}{k}}$:

$$C_k\left((\lambda_j)_{j=1}^n\right) = \{\lambda_{i_1} \cdots \lambda_{i_k} : 1 \leq i_1 < \cdots < i_k \leq n\}.$$

6.9.3 Theorem (S. M. Malamud [353]) *Let* $\{\lambda_1, ..., \lambda_n\}$ *and* $\{\mu_1, ..., \mu_{n-1}\}$ *be two families of complex numbers. Then, for the existence of a normal matrix* A *such that* $\sigma(A) = (\lambda_j)_{j=1}^n$ *and* $\sigma(A_{n-1}) = (\mu_k)_{k=1}^{n-1}$ (*where* A_{n-1} *denotes the principal submatrix of* A *obtained by deleting the last row and last column*) *it is necessary that the condition*

$$C_k\left((\mu_j - w)_{j=1}^{n-1}\right) \prec_{uds} C_k\left((\lambda_j - w)_{j=1}^n\right)$$

is fulfilled for any complex number w *and any* $k \in \{1, ..., n-1\}$ *and it is sufficient to be fulfilled for* $k = n - 1$ *and all* $w \in \{\lambda_1, ..., \lambda_n\}$.

This leads to the following improvement of Theorem 6.8.4, also due to S. M. Malamud [353]:

6.9.4 Theorem *Let* $f : \mathbb{C} \to \mathbb{R}$ *be a convex function and* α *a complex number. Then for any polynomial* P *of degree* $n \geq 2$ *and any integer* $k \in \{1, ..., n-1\}$ *we have*

$$\frac{1}{\binom{n-1}{k}} \sum_{1 \leq i_1 < \cdots < i_k \leq n} f(\prod_{j=1}^k \mu_{i_j} - \alpha)$$
$$\leq \frac{1}{\binom{n}{k}} \sum_{1 \leq i_1 < \cdots < i_k \leq n} f(\prod_{j=1}^k \lambda_{i_j} - \alpha),$$

where $(\lambda_k)_{k=1}^n$ *and* $(\mu_k)_{k=1}^{n-1}$ *are respectively the roots of* P *and* P'.

Similar results were obtained by J. Borcea [77], who discussed the distribution of the equilibrium points of logarithmic potentials of the form

$$U(z) = \sum_k a_k \log \left| 1 - \frac{z}{z_k} \right| .$$



Chapter 7

Convexity in Spaces of Matrices

In this chapter, we investigate three subjects concerning the convexity of functions defined on a space of matrices (or just on a convex subset of it). The first one is devoted to the convex spectral functions, that is, to the convex functions $F : \mathrm{Sym}(n, \mathbb{R}) \to \mathbb{R}$ whose values $F(A)$ depend only on the spectrum of A. The main result concerns their description as superpositions $f \circ \Lambda$ between convex functions $f : \mathbb{R}^n \to \mathbb{R}$ invariant under permutations and the eigenvalues map Λ.

The second subject, the matrix convexity, is aimed to clarify the conditions under which the functional calculus associates to a convex function $f : \mathbb{R} \to \mathbb{R}$ a convex function on $\mathrm{Sym}(n, \mathbb{R})$.

The third subject concerns the hyperbolic geometry of $\mathrm{Sym}^{++}(n, \mathbb{R})$ and the geodesic convexity structure associated to it. Being at the frontier between differential geometry, linear algebra and spectral theory, a presentation of this subject with full details would be quite involving, so our option is to offer the reader just a glimpse of what is going on in spaces with a curved geometry.

7.1 Convex Spectral Functions

The structure of convex functions under the presence of symmetry can be surprisingly simple. For example, this is the case of functions f defined on the space $\mathrm{Sym}(n, \mathbb{R})$ and invariant under the action of the *orthogonal group*

$$\mathrm{O}(n) = \{U \in \mathrm{M}_n(\mathbb{R}) : U^T U = UU^T = I\},$$

the elements of $\mathrm{O}(n)$ being the orthogonal matrices.

The group $\mathrm{O}(n)$ acts on $\mathrm{Sym}(n, \mathbb{R})$ via the map

$$\Phi : \mathrm{O}(n) \times \mathrm{Sym}(n, \mathbb{R}) \to \mathrm{Sym}(n, \mathbb{R}), \quad \Phi(U, A) = U^T AU.$$

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A subset $\mathcal{K}$ of $\mathrm{Sym}(n,\mathbb{R})$ is called $\mathrm{O}(n)$-*invariant* if

$$A \in \mathcal{K} \text{ implies } U^T A U \in \mathcal{K} \text{ for all } U \in \mathrm{O}(n),$$

while a function F defined on an $\mathrm{O}(n)$-invariant subset of $\mathrm{Sym}(n,\mathbb{R})$ is called $\mathrm{O}(n)$-*invariant* if

$$F(A) = F(U^T A U) \text{ for all } U \in \mathrm{O}(n).$$

Some simple examples of $\mathrm{O}(n)$-invariant convex functions defined on $\mathrm{Sym}(n,\mathbb{R})$ are trace and the largest eigenvalue $\lambda_1^{\downarrow}(\cdot)$. The same is true for their restrictions to $\mathrm{Sym}^{+}(n,\mathbb{R})$, $\mathrm{Sym}^{++}(n,\mathbb{R})$, $\{A \in \mathrm{Sym}(n,\mathbb{R}) : \mathrm{trace}\, A = 0\}$ (as well as to any other $\mathrm{O}(n)$-invariant convex subset of $\mathrm{Sym}(n,\mathbb{R})$).

According to the spectral decomposition theorem, the orbit of any symmetric matrix A,

$$\mathcal{O}(A) = \left\{U^T A U : U \in \mathrm{O}(n)\right\},$$

contains all diagonal matrices carrying the various permutations of eigenvalues of A with their multiplicities. As a consequence, the values $F(A)$ of every $\mathrm{O}(n)$-invariant function F depend only on the spectrum of the matrix A. That's why the $\mathrm{O}(n)$-invariant functions are also called *spectral functions.*

The spectral functions can be identified with the functions of n real variables, invariant under the permutation of coordinates. If $F : \mathcal{K} \to \mathbb{R}$ is a spectral function, then $\mathcal{C} = \{\mathbf{x} \in \mathbb{R}^n : \mathrm{diag}(\mathbf{x}) \in \mathcal{K}\}$ is a subset of $\mathbb{R}^n$ invariant under permutations and the function

$$f : \mathcal{C} \to \mathbb{R}, \quad f(\mathbf{x}) = F(\mathrm{diag}(\mathbf{x})),$$

called *the symmetric function associated to* F, is invariant under permutations.

Conversely, every function $f : \mathcal{C} \to \mathbb{R}$, invariant under permutations, represents the symmetric function associated to a spectral function, more precisely to the function

$$F : \mathcal{K} \to \mathbb{R}, \quad F = f \circ \Lambda,$$

where $\mathcal{K} = \{A \in \mathrm{Sym}(n,\mathbb{R}) : \Lambda(A) \in \mathcal{C}\}$ and $\Lambda : \mathrm{Sym}(n,\mathbb{R}) \to \mathbb{R}^n$ is the eigenvalues map that associates to each symmetric matrix A the string $\lambda^{\downarrow}(A)$, of its eigenvalues with multiplicities listed downward. It is a simple exercise that $\mathcal{C}$ is a convex set when $\mathcal{K}$ is convex (and vice-versa).

As was shown in Example 4.1.5, the convexity of the spectral function $-\log\det(A)$ on $\mathrm{Sym}^{++}(n,\mathbb{R})$ can be derived from the convexity of its associated symmetric function, $-\log(x_1 \cdots x_n)$ on $\mathbb{R}^n_{++}$. Remarkably, this fact works for any spectral function.

7.1.1 Theorem (C. Davis [151]) *A spectral function $F : \mathcal{K} \to \mathbb{R}$ is convex if and only if its associated symmetric function $f : \mathcal{C} \to \mathbb{R}$ is convex.*

This result remains valid in the framework of log-convex functions, concave functions, log-concave functions, quasiconvex functions, etc.

Proof. If F is convex, then for every $\mathbf{x}, \mathbf{y} \in \mathcal{C}$ and $\lambda \in [0,1]$ we have

$$\begin{aligned} f\left((1-\lambda)\mathbf{x}+\lambda\mathbf{y}\right) &= F\left(\operatorname{diag}\left((1-\lambda)\mathbf{x}+\lambda\mathbf{y}\right)\right) \\ &= F\left((1-\lambda)\,\operatorname{diag}(\mathbf{x})+\lambda\,\operatorname{diag}(\mathbf{y})\right) \\ &\leq (1-\lambda)F\left(\operatorname{diag}(\mathbf{x})\right)+\lambda F\left(\operatorname{diag}(\mathbf{y})\right) \\ &= (1-\lambda)f(\mathbf{x})+\lambda f\left(\mathbf{y}\right), \end{aligned}$$

so that f is convex too.

Suppose now that f is convex and fix arbitrarily A, B in $\mathcal{K}$ and λ in $[0,1]$. Since F is $\mathrm{O}(n)$-invariant, we may assume that $C=(1-\lambda)A+\lambda B$ is diagonal. Denote by A_C and B_C the matrices having the same main diagonal as A and B respectively, but whose off-diagonal elements are zero. Then $C=(1-\lambda)A_C+\lambda B_C$ and the fact that f is convex implies

$$F(C) \leq (1-\lambda)F(A_C)+\lambda F\left(B_C\right).$$

We next show that $F(A_C) \leq F(A)$ and $F\left(B_C\right) \leq F(B)$, a fact that will complete the proof.

Let $\mathbf{e}_1, ..., \mathbf{e}_n$ be the natural basis of $\mathbb{R}^n$ and let $\mathbf{u}_1, ..., \mathbf{u}_n$ be an orthonormal basis of it consisting of eigenvectors of A. The eigenvalues α_i of A can be counted so that $A\mathbf{u}_i = \alpha_i \mathbf{u}_i$ for every i. Then, the diagonal elements of A and A_C are given by the formula

$$a_{ii} = \langle A\mathbf{e}_i, \mathbf{e}_i\rangle = \langle \sum_{j=1}^{n} \alpha_j \langle \mathbf{e}_i, \mathbf{u}_j\rangle \mathbf{u}_j, \mathbf{e}_i\rangle = \sum_{j=1}^{n} \alpha_j \left|\langle \mathbf{e}_i, \mathbf{u}_j\rangle\right|^2.$$

According to the Pythagorean theorem, the matrix $D=(D_{ij})_{i,j}$ with coefficients $D_{ij} = \left|\langle \mathbf{e}_i, \mathbf{u}_j\rangle\right|^2$ is doubly stochastic. Taking into account Birkhoff's Theorem 3.3.6, we infer that D is a convex combination $D=\sum_{\pi\in\Pi(n)} d_\pi P^{(\pi)}$ of the $n!$ permutation matrices $P^{(\pi)}$ with coefficients

$$P_{ij}^{(\pi)} = \begin{cases} 1 & \text{if } j=\pi(i) \\ 0 & \text{otherwise.} \end{cases}$$

Notice that a permutation matrix is a doubly stochastic matrix whose elements are either 0 or 1. Then

$$\begin{aligned} F(A_C) &= f\left((a_{11}, ..., a_{nn})\right) \\ &= f\left(\sum_{j=1}^{n} D_{1,j}\alpha_j, ..., \sum_{j=1}^{n} D_{n,j}\alpha_j\right) \\ &= f\left(\sum_{\pi} d_\pi \sum_{j=1}^{n} P_{1,j}^{(\pi)}\alpha_j, ..., \sum_{\pi} d_\pi \sum_{j=1}^{n} P_{n,j}^{(\pi)}\alpha_j\right) \end{aligned}$$

which is bounded from above by

$$\sum_{\pi} d_\pi f\left(\sum_{j=1}^{n} P_{1,j}^{(\pi)}\alpha_j, ..., \sum_{j=1}^{n} P_{n,j}^{(\pi)}\alpha_j\right)$$
$$= \sum_{\pi} d_\pi f\left(\alpha_{\pi(1)}, ..., \alpha_{\pi(n)}\right) = f\left(\alpha_1, ..., \alpha_n\right) = F(A).$$

In the same manner one can show that $F(B_C) \leq F(B)$, which completes the proof. ■

Theorem 7.1.1 has a large range of applications. For example, it allows us the transfer some important results from the information theory to the quantum information theory, by considering their noncommutative analogs.

In quantum information theory the role of the set

$$\text{Prob}\,(n) = \left\{(\lambda_1, ..., \lambda_n) \in [0,1]^n : \sum_{k=1}^{n} \lambda_k = 1\right\}$$

of all probability distributions of order n, is taken by the set

$$\mathbf{S}_n = \left\{A \in \text{Sym}^+(n, \mathbb{R}) : \text{trace}(A) = 1\right\},$$

of the so called *density matrices*. This is a compact and convex subset of $\text{Sym}(n, \mathbb{R})$, so, according to Minkowski's Theorem 3.3.5, every density matrix is a convex combination of pure states, that is, of the extreme points of the set $\mathbf{S}_n$. One can show that the pure states are nothing but the operators $\rho = \langle \cdot, \mathbf{v}\rangle\mathbf{v}$ with $\mathbf{v} \in \mathbb{R}^n$, $\|\mathbf{v}\| = 1$ (that is, the rank one orthogonal projections on $\mathbb{R}^n$). See S. Strătilă and L. Zsidó [557].

Notice also that the set $\mathbf{S}_n$ is $\text{O}(n)$ invariant. The map

$$S : \mathbf{S}_n \to \mathbb{R}, \quad S(\rho) = -\,\text{trace}(\rho \log \rho),$$

known as the *von Neumann entropy*, is also $\text{O}(n)$ invariant. Indeed, according to the spectral theorem from linear algebra, every $\rho \in \mathbf{S}_n$ admits the decomposition $\rho = U\,\text{diag}(\lambda_1, ..., \lambda_n)U^T$, where $\lambda_1, ..., \lambda_n$ is the spectrum of ρ (each eigenvalue being counted with its multiplicity) and U is a suitable orthogonal matrix. Then

$$S(\rho) = -\sum_{k=1}^{n} \lambda_k \log \lambda_k$$

and for every orthogonal matrix V we have $S(V\rho V^T) = S(\rho)$ because $V\rho V^T$ and ρ have the same family of eigenvalues (counted with their multiplicities). Notice also the connection with the Shannon entropy,

$$S(\rho) = H(\lambda_1, ..., \lambda_n),$$

so the von Neumann entropy is the noncommutative analogue of the Shannon entropy.

According to Theorem 7.1.1, it follows that the von Neumann entropy is a strictly concave function. Actually, taking into account Corollary 5.5.3, this is a 1/2-strongly convex function

Since the von Neumann entropy is continuous on $\mathbf{S}_n$ and of class C^∞ on $\operatorname{int}\mathbf{S}_n$, we may consider the Bregman function associated to it,

$$\begin{aligned} D_S(\sigma||\rho) &= S(\sigma) - S(\rho) - dS(\rho)(\sigma - \rho) \\ &= S(\sigma) - S(\rho) + \langle \log\rho + I, \sigma - \rho\rangle_{HS} \\ &= -\operatorname{trace}(\sigma\log\sigma) + \operatorname{trace}(\rho\log\rho) + \operatorname{trace}(\sigma\log\rho - \rho\log\rho + \sigma - \rho) \\ &= \operatorname{trace}\sigma(\log\sigma - \log\rho) \end{aligned}$$

for all $\sigma \in \mathbf{S}_n$ and $\rho \in \operatorname{int}\mathbf{S}_n$. In quantum information theory this is known as the *quantum relative entropy* of the pair (σ, ρ) and usually denoted $S(\sigma||\rho)$. Thus

$$S(\sigma||\rho) = \operatorname{trace}\sigma(\log\sigma - \log\rho)$$

and this formula can be extended to any pair (σ, ρ) of matrix densities by adopting the conventions made in Remark 1.2.7. The above considerations show that

$$S(\sigma||\rho) \geq 0$$

and that the equality holds if and only if $\rho = \sigma$

If the densities ρ and σ commute, then from Theorem 3.5.5, on simultaneous diagonalization of symmetric matrices, we infer that $\rho = SD_1S^T$ and $\sigma = SD_2S^T$, where S is an orthogonal matrix, $D_1 = \operatorname{diag}(\lambda_1, ..., \lambda_n)$ and $D_2 = \operatorname{diag}(\mu_1, ..., \mu_n)$. Thus, in the commutative case, the quantum relative entropy $S(\rho||\sigma)$ coincides with the Kulback–Leibler divergence from $(\lambda_1, ..., \lambda_n)$ to $(\mu_1, ..., \mu_n)$,

$$S(\rho||\sigma) = \sum_{k=1}^{n} \lambda_k \log\frac{\lambda_k}{\mu_k}.$$

We will show in the next section that the quantum relative entropy is a jointly convex function of its arguments. See Corollary 7.2.11.

We continue with another application of Theorem 7.1.1.

7.1.2 Corollary (The noncommutative analog of the log-sum-exp function). *The function* $F(A) = \log\operatorname{trace}\left(\mathrm{e}^A\right)$ *is convex on the space* $\operatorname{Sym}(n, \mathbb{R})$.

Proof. Indeed, if $U \in \mathrm{O}(n)$, then

$$\begin{aligned} e^{UAU^T} &= I + \frac{1}{1!}\left(UAU^T\right) + \frac{1}{2!}\left(UAU^T\right)^2 + \cdots \\ &= U\left(I + \frac{1}{1!}A + \frac{1}{2!}A^2 + \cdots\right)U^T = Ue^AU^T, \end{aligned}$$

so F is a spectral function. The symmetric function associated to F is the convex function $f(x) = \log\left(\sum_{k=1}^n e^{x_k}\right)$, and Theorem 7.1.1 applies. ∎

The negative von Neumann entropy and the noncommutative analog of the log-sum-exp function are related by the Legendre-Fenchel duality. See Theorem 8.2.7.

The next application of Theorem 7.1.1 is related to the mechanics of hyperelastic materials.

7.1.3 Corollary (S. Lehmich, P. Neff, and J. Lankeit [328]) *Let h be a twice continuously differentiable real function defined on the interval $(0,\infty)$. Then the function $\Phi(A) = h(\det A)$ is convex on $\mathrm{Sym}^{++}(n,\mathbb{R})$ for $n \geq 2$, if and only if*

$$nth''(t) + (n-1)h'(t) \geq 0 \text{ and } h'(t) \leq 0 \text{ for all } t > 0. \tag{7.1}$$

Proof. The following short argument is due to M. Šilhavý [539]. According to Theorem 7.1.1, the function $\Phi(A)$ is convex if and only if the function $\varphi(\lambda) = h(\lambda_1 \cdots \lambda_n)$ is convex on $(0,\infty)^n$. Since φ is twice Fréchet differentiable, the second derivative test of convexity applies. Letting $t = \lambda_1 \cdots \lambda_n$, a simple computation shows that for all $\lambda = (\lambda_1, ..., \lambda_n) \in (0,\infty)^n$ and $\mathbf{x} = (x_1, ..., x_n) \in \mathbb{R}^n$ we have

$$\begin{aligned} \mathrm{d}^2\varphi(\lambda)(\mathbf{x},\mathbf{x}) &= \sum_{i,j=1}^{n} \frac{\partial^2 \varphi}{\partial\lambda_i \partial\lambda_j}(\lambda_1, ..., \lambda_n) x_i x_j \\ &= \left(t^2 h''(t) + t h'(t)\right) \left(\sum_{i=1}^{n} \frac{x_i}{\lambda_i}\right)^2 - t h'(t) \sum_{i=1}^{n} \left(\frac{x_i}{\lambda_i}\right)^2. \end{aligned}$$

The function φ is convex if and only if $\mathrm{d}^2\varphi(\lambda)(\mathbf{x},\mathbf{x}) \geq 0$ for all λ and $\mathbf{x}$. The necessity of the conditions (7.1) follows by considering separately the following two particular cases

a) $x_i/\lambda_i = 1$ for $i = 1, ..., n$;
b) $x_1/\lambda_1 = 1$, $x_2/\lambda_2 = -1$ and $x_i/\lambda_i = 0$ for $i \geq 3$.

As concerns their sufficiency, one has to remark that due to the Cauchy–Bunyakovsky–Schwarz inequality, we have

$$\mathrm{d}^2\varphi(\lambda)(\mathbf{x},\mathbf{x}) \geq \frac{t}{n}\left(nth''(t) + (n-1)h'(t)\right) \left(\sum_{i=1}^{n} \frac{x_i}{\lambda_i}\right)^2 \geq 0$$

for all λ and $\mathbf{x}$. ∎

As in the case of convex functions, it is sometimes convenient to extend a spectral function to the whole space ($\mathrm{Sym}(n,\mathbb{R})$ or $\mathbb{R}^n$) by assigning ∞ at each point outside the invariant set representing the effective domain of definition. In particular, this is the case when dealing with their subdifferentiability. See Exercises 4 and 5 at the end of this section.

The following result, due to H. H. Bauschke, O. Güler, A. S. Lewis and H. S. Sendov [44], extends Theorem 7.1.1 to the context of hyperbolic polynomials.

7.1.4 Theorem *If $f : \mathbb{R}^m \to \mathbb{R}\cup\{\infty\}$ is convex and symmetric and Λ is the eigenvalues map of a hyperbolic polynomial of degree m defined on a finite-dimensional vector space V, then $f \circ \Lambda$ is also convex on V.*

Proof. If $\mathbf{x}$ and $\mathbf{y}$ are vectors in V and $\alpha \in [0,1]$, then $\mathbf{u} = (1-\alpha)\Lambda(\mathbf{x}) + \alpha\Lambda(\mathbf{y})$ and $\mathbf{v} = \Lambda\left((1-\alpha)\mathbf{x} + \alpha\mathbf{y}\right)$ both belong to the monotone cone $\mathbb{R}^m_{\geq}$. According to Exercise 5 in Section 6.7, if $\mathbf{w}$ is an arbitrarily fixed vector in $\mathbb{R}^m_{\geq}$, then the function $\mathbf{x} \to \langle \mathbf{w}, \Lambda(\mathbf{x})\rangle$ is convex on V. Hence

$$\langle \mathbf{w}, \Lambda((1-\alpha)\mathbf{x} + \alpha\mathbf{y})\rangle \leq (1-\alpha)\langle \mathbf{w}, \Lambda(\mathbf{x})\rangle + \alpha\langle \mathbf{w}, \Lambda(\mathbf{y})\rangle,$$

equivalently,

$$\langle \mathbf{w}, \mathbf{u} - \mathbf{v}\rangle \geq 0.$$

As $\mathbf{w}$ was arbitrarily fixed in $\mathbb{R}^m_{\geq}$, the last inequality shows that $\mathbf{u}-\mathbf{v} \in \left(\mathbb{R}^m_{\geq}\right)^*$. Under these circumstances we infer from Remark 6.3.5 that $\mathbf{v} \prec_{HLP} \mathbf{u}$. Then $f(\mathbf{u}) \geq f(\mathbf{v})$, by Theorem 6.1.13, which means that

$$f\left(\Lambda\left((1-\alpha)\mathbf{x} + \alpha\mathbf{y}\right)\right) \leq f\left((1-\alpha)\Lambda(\mathbf{x}) + \alpha\Lambda(\mathbf{y})\right).$$

The proof ends by noticing that $f\left((1-\alpha)\Lambda(\mathbf{x}) + \alpha\Lambda(\mathbf{y})\right) \leq (1-\alpha)\Lambda(\mathbf{x}) + \alpha\Lambda(\mathbf{y})$, due to the property of convexity of f. ∎

Applying Theorem 7.1.4 to the function $f(x_1, ..., x_m) = \max\{x_k : 1 \leq k \leq m\}$, one retrieves the fact that the largest eigenvalue of a hyperbolic polynomial is a convex function on V. See Theorem 6.7.6 (f).

Exercises

1. Prove the convexity of the following functions:
   (a) $F(A) = A^2$, for $A \in \mathrm{Sym}(n, \mathbb{R})$;
   (b) the partial sums

$$A \to \sum_{k=1}^{m} \lambda_k^{\downarrow}(A) = \sup_{J\subset\{1,\ldots,n\},\ |J|=m} \sum_{k\in J} \lambda_k,$$

   where $A \in \mathrm{Sym}(n, \mathbb{R})$ and $1 \leq m \leq n$;
   (c) $-\operatorname{trace}\left(A^{1/2}\right)$, for $A \in \mathrm{Sym}^+(n, \mathbb{R})$;
   (d) $\operatorname{trace}\left(A^{-1}\right)$, for $A \in \mathrm{Sym}^{++}(n, \mathbb{R})$.

2. Prove that $|\operatorname{trace} A| \leq \operatorname{trace}|A|$ for all $A \in \mathrm{Sym}(n, \mathbb{R})$.

3. Let $F : \mathrm{Sym}(n, \mathbb{R}) \to \mathbb{R}\cup\{\infty\}$ be an $\mathrm{O}(n)$-invariant function. Prove that f is a lower semicontinuous proper convex function if, and only if, $F \circ \mathrm{diag}$ is likewise on $\mathbb{R}^n$.

4. Prove that the differentiability of a convex spectral function F is characterized by that of $f = F \circ \mathrm{diag}$. More precisely, if $F : \mathrm{Sym}(n, \mathbb{R}) \to \mathbb{R}\cup\{\infty\}$ is an $\mathrm{O}(n)$-invariant lower semicontinuous proper convex function, then F is Gâteaux differentiable at A if and only if f is Gâteaux differentiable at $\Lambda(A)$.

5. Suppose that $F : \mathrm{Sym}(n,\mathbb{R}) \to \mathbb{R}\cup\{\infty\}$ is an $\mathrm{O}(n)$-invariant lower semi-continuous proper convex function. Prove that $B \in \partial F(A)$ if and only if A and B admit simultaneous spectral decomposition,

$$U(\mathrm{diag}(x))U^T = A \text{ and } U(\mathrm{diag}(y))U^T = B,$$

for some U with $U^TU = I$ and some $y \in \partial f(x)$.

6. Since the function det is concave and strictly positive on $\mathrm{Sym}^{++}(n,\mathbb{R})$, its reciprocal $1/\det$ is convex, so $h(t) = 1/t$ is an example of function for which Theorem 7.1.3 applies. Indicate other examples.

7. Prove that the von Neumann entropy verifies $0 \le S(\rho) \le \log n$ for all $\rho \in \mathbf{S}_n$. There is equality on the left if and only if ρ is a pure state and there is equality on the right if and only if $\rho = (1/n)I$.

8. (Spectral order) Given two symmetric matrices $A, B \in \mathrm{Sym}(n,\mathbb{R})$, we say that A is majorized by B with respect to the spectral order (abbreviated, $A \prec_{sp} B$) if $\Lambda(A) \prec_{HLP} \Lambda(B)$. Prove that the following assertions are equivalent:

   (a) $A \prec_{sp} B$;

   (b) there exist positive numbers λ_k and orthogonal matrices U_k such that

$$\sum_{k=1}^{n} \lambda_k = 1 \text{ and } A = \sum_{k=1}^{n} \lambda_k U_k^T B U_k;$$

   (c) there exists a doubly stochastic operator $\Phi : \mathrm{M}_n(\mathbb{R}) \to \mathrm{M}_n(\mathbb{R})$ such that $A = \Phi(B)$, that is, a linear operator that verifies the following three properties:

$$\Phi(I) = I,\ \Phi(\mathrm{Sym}^+(n,\mathbb{R})) \subset \mathrm{Sym}^+(n,\mathbb{R}) \\ \text{and } \mathrm{trace}\,\Phi(A) = \mathrm{trace}\,A \text{ for all } A.$$

   (d) For every convex function $f : \mathbb{R} \to \mathbb{R}$, we have $\mathrm{trace}(f(A)) \le \mathrm{trace}(f(B))$.

9. Suppose that $F : \mathrm{Sym}(n,\mathbb{R}) \to \mathbb{R}$ is an $\mathrm{O}(n)$-invariant convex function. Prove that $F(A) \le F(B)$ whenever $A \prec_{sp} B$. Infer that

$$\rho, \sigma \in \mathbf{S}_n \text{ and } \rho \prec_{sp} \sigma \text{ implies } S(\rho) \ge S(\sigma).$$

10. Prove that the rank function is quasiconcave on $\mathrm{Sym}^+(n,\mathbb{R})$ and its convex hull is the trace function.

7.2 Matrix Convexity

According to the functional calculus with continuous functions one can associate to each symmetric operator $A \in \mathcal{A}(H)$ (from the real Hilbert space H into itself) an application $\pi : C(\sigma(A)) \to \mathcal{A}(H)$ that transforms every real polynomial $P(t) = t^n + c_1 t^{n-1} + \cdots + c_n$ into the symmetric operator $P(A) = A^n + c_1 A^{n-1} + \cdots + c_n I$ and, in general, every continuous real function $f \in C(\sigma(A))$ into an operator $f(A) \in \mathcal{A}(H)$ such that

$$\begin{cases} (f+g)(A) = f(A) + g(A) \\ (fg)(A) = f(A)g(A) \\ f \geq 0 \text{ implies } f(A) \geq 0 \\ \|f(A)\| = \sup\{|f(\lambda)| : \lambda \in \sigma(A)\}, \end{cases}$$

for all $f, g \in C(\sigma(A))$. See Section 3.5 for the case of symmetric real matrices and the Comments at the end of Chapter 3 for the general case. In the case of matrices, the assumption of the continuity of f is automatically fulfilled because the spectrum of any matrix is a discrete set.

If $A, B \in \mathcal{A}(H)$ are two operators whose spectra are included in the compact interval $[m, M]$, then $mI \leq A \leq MI$ and $mI \leq B \leq MI$, so that $mI \leq (1-\lambda)A + \lambda B \leq MI$. See Theorem 3.6.4. Therefore, for every nonempty real interval J, the set $\mathcal{A}_J(H)$, of all symmetric operators A with $\sigma(A) \subset J$, constitutes a convex set in $\mathcal{A}(H)$.

7.2.1 Definition (F. Kraus [308]) *A function $f : J \to \mathbb{R}$ is said to be n-matrix convex if*

$$f((1-\lambda)A + \lambda B) \leq (1-\lambda)f(A) + \lambda f(B) \tag{7.2}$$

for every $\lambda \in [0,1]$ and every pair of matrices A, $B \in \mathrm{Sym}(n, \mathbb{R})$ with spectra in J. If the above inequality holds for every $n \geq 1$, then f is called matrix convex (or operator convex).

The concepts of strict n-matrix convexity and strict matrix convexity refer to the case where the inequality (7.2) is strict when $\lambda \in (0,1)$ and $A \neq B$.

The matrix companions to concave functions are defined by reversing the inequality sign in (7.2).

7.2.2 Example *The function $f(t) = 1/t$ is strictly matrix convex on $(0, \infty)$: for each integer number $n \geq 1$, the induced function*

$$f : \mathrm{Sym}^{++}(n, \mathbb{R}) \to \mathrm{Sym}^{++}(n, \mathbb{R}), \quad f(A) = A^{-1},$$

is strictly convex, that is,

$$((1-\lambda)A + \lambda B)^{-1} < (1-\lambda)A^{-1} + \lambda B^{-1}$$

for all $A, B \in \mathrm{Sym}^{++}(n, \mathbb{R})$ with $A \neq B$, and $\lambda \in (0,1)$. For $B = I$ and $A \neq I$, this reduces to the inequality

$$((1-\lambda)A + \lambda I)^{-1} < (1-\lambda)A^{-1} + \lambda I,$$

which easily follows by simple algebraic manipulations. To obtain from here the general case we have to replace A by $A^{-1/2}BA^{-1/2}$.

7.2.3 Example *The preceding example can be used to prove the strict matrix convexity of the function $-\log$ on $(0,\infty)$. Indeed, using the representation formula*

$$-\log x = \int_0^\infty \left(\frac{1}{x+t} - \frac{1}{1+t}\right) \mathrm{d}t,$$

we infer that for every $Z \in \mathrm{Sym}^{++}(n,\mathbb{R})$,

$$-\log Z = \int_0^\infty \left[(Z+tI)^{-1} - (I+tI)^{-1}\right] \mathrm{d}t,$$

whence

$$\begin{aligned}
-\log\left((1-\lambda)A+\lambda B\right) &= \int_0^\infty \left[((1-\lambda)A+\lambda B+tI)^{-1} - (I+tI)^{-1}\right] \mathrm{d}t \\
&= \int_0^\infty \left[\left((1-\lambda)(A+tI)+\lambda\,(B+tI)\right)^{-1} - (I+tI)^{-1}\right] \mathrm{d}t \\
&< (1-\lambda)\int_0^\infty \left[(A+tI)^{-1} - (I+tI)^{-1}\right] \mathrm{d}t \\
&+ \lambda \int_0^\infty \left[(B+tI)^{-1} - (I+tI)^{-1}\right] \mathrm{d}t \\
&= -(1-\lambda)\log A - \lambda \log B,
\end{aligned}$$

for all $A, B \in \mathrm{Sym}^{++}(n,\mathbb{R})$, $A \neq B$, and $\lambda \in (0,1)$.

The matrix concave functions can be characterized in terms of matrix monotonicity.

7.2.4 Definition (K. Löwner [346]) *A function $f : J \to \mathbb{R}$ is called n-matrix monotone if for every pair of matrices A, $B \in Sym(n,\mathbb{R})$ with spectra in J, the following implication holds:*

$$A \leq B \text{ implies } f(A) \leq f(B).$$

The function f is called matrix monotone if the above inequality holds for every integer $n \geq 1$.

One can prove (using an approximation argument) that if a function is matrix monotone, then $A \leq B$ also implies $f(A) \leq f(B)$ for all symmetric operators defined on infinite-dimensional Hilbert spaces. See Exercise 1. This explains why the matrix monotone functions are also called *operator monotone* functions.

The matrix convexity/monotonicity of a function $f : J \to \mathbb{R}$ implies the convexity/monotone increasing of f, but the converse is not true.

As noticed K. Löwner, the theory of matrix monotone functions can benefit from the complex analysis by replacing the above framework with that of self-adjoint matrices acting on complex Hilbert spaces. See F. Hansen [237] for a thorough presentation of this theory. For more applications, the reader may consult the books of R. Bhatia [61], F. Hiai and D. Petz [253] and B. Simon [543]. However, this is outside the goals of our book and we will keep working in the real framework.

According to Examples 7.2.2 and 7.2.3, the functions $-1/t$ and log are matrix concave on $(0,\infty)$. They are also matrix monotone, since $A \leq B$ in $\mathrm{Sym}^{++}(n,\mathbb{R})$ is equivalent to $-A^{-1} \leq -B^{-1}$ (due to Corollary 3.5.3).

The following two results due to F. Hansen and G. K. Pedersen [239] describe the general relationship between the matrix monotone functions and the matrix concave functions defined on the interval $(0,\infty)$.

7.2.5 Theorem (a) *Every matrix $2n$-monotone function $f:(0,\infty)\to\mathbb{R}$ is also matrix n-concave. In particular, f is continuous.*

(b) *Every matrix monotone function defined on $(0,\infty)$ is also matrix concave.*

Proof. We follow the argument of F. Hansen [237], based on the handling of block matrices. Let A_1 and A_2 be two elements of $\mathrm{Sym}^{++}(n,\mathbb{R})$ and $s\in[0,1]$. Then, consider the orthogonal block matrix V of order $2n\times 2n$ given by

$$V=\begin{pmatrix} s^{1/2}I & -(1-s)^{1/2}I \\ (1-s)^{1/2}I & s^{1/2}I \end{pmatrix}.$$

Clearly

$$V^T\begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}V=\begin{pmatrix} sA_1+(1-s)A_2 & s^{1/2}(1-s)^{1/2}(A_2-A_1) \\ s^{1/2}(1-s)^{1/2}(A_2-A_1) & (1-s)A_1+sA_2 \end{pmatrix}.$$

Denote $D=-s^{1/2}(1-s)^{1/2}(A_2-A_1)$ and notice that for every $\varepsilon>0$ we have

$$\begin{pmatrix} sA_1+(1-s)A_2+\varepsilon I & 0 \\ 0 & 2\lambda I \end{pmatrix}-V^T\begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}V\geq\begin{pmatrix} \varepsilon I & D \\ D & \lambda I \end{pmatrix}$$

provided $\lambda\geq(1-s)A_1+sA_2$. Since the last block matrix is positive (semidefine) for $\lambda\geq\|D\|^2/\varepsilon$, we infer that

$$V^T\begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}V\leq\begin{pmatrix} sA_1+(1-s)A_2+\varepsilon I & 0 \\ 0 & 2\lambda I \end{pmatrix}$$

for λ large enough. Therefore for any such λ,

$$f\left(V^T\begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}V\right)\leq f\left(\begin{pmatrix} sA_1+(1-s)A_2+\varepsilon I & 0 \\ 0 & 2\lambda I \end{pmatrix}\right),$$

due to the $2n$-matrix monotonicity of f. Since

$$f\left(V^T\begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}V\right) = V^T f\left(\begin{pmatrix} f(A_1) & 0 \\ 0 & f(A_2) \end{pmatrix}\right)V$$
$$= \begin{pmatrix} sf(A_1)+(1-s)f(A_2) & s^{1/2}(1-s)^{1/2}(f(A_2)-f(A_1)) \\ s^{1/2}(1-s)^{1/2}(f(A_2)-f(A_1)) & (1-s)f(A_1)+sf(A_2) \end{pmatrix},$$

we obtain

$$sf(A_1)+(1-s)f(A_2) \le f(sA_1+(1-s)A_2+\varepsilon I). \tag{7.3}$$

Since f is monotone increasing, the limit

$$f^+(t) = \lim_{s\to 0+} f(t+\varepsilon)$$

is well defined for every $t>0$, so using the inequality (7.3) we infer that

$$sf^+(t_1)+(1-s)f^+(t_2) \le sf(t_1+\varepsilon)+(1-s)f(t_2+\varepsilon) \le f(st_1+(1-s)t_2+2\varepsilon)$$

for all $t_1, t_2, \varepsilon > 0$. Passing to the limit as $\varepsilon \to 0+$, we obtain

$$sf^+(t_1)+(1-s)f^+(t_2) \le f^+(st_1+(1-s)t_2),$$

hence f^+ is concave and thus continuous. Since f is monotone increasing,

$$f^+(t-\varepsilon) \le f(t) \le f^+(t+\varepsilon) \quad \text{for } t>0 \text{ and } \varepsilon \in (0,t),$$

whence $f = f^+$ due to the continuity of f^+. The proof ends by letting ε tend to 0 in inequality (7.3) to obtain

$$sf(A_1)+(1-s)f(A_2) \le f(sA_1+(1-s)A_2).$$

■

7.2.6 Theorem *Let $f:(0,\infty)\to[0,\infty)$ be a matrix n-concave function. Then f is also n-monotone.*

Therefore every positive matrix concave function defined on $(0,\infty)$ is also matrix monotone.

Proof. Let A and B be strictly positive $n\times n$ matrices with $A<B$ and fix arbitrarily λ in $(0,1)$. Since λB can be represented as a convex combination of two strictly positive matrices,

$$\lambda B = \lambda A + (1-\lambda)\left(\lambda(1-\lambda)^{-1}(B-A)\right),$$

and f is n-concave, we obtain

$$f(\lambda B) \ge \lambda f(A) + (1-\lambda) f\left(\lambda(1-\lambda)^{-1}(B-A)\right) \ge \lambda f(A).$$

Since f is continuous we obtain $f(B)\ge f(A)$ by passing to the limit as $\lambda\to 1$.

■

An important family of matrix monotone functions is as follows.

7.2.7 Theorem (The Löwner–Heinz inequality) *The function* $f(x) = x^p$ *is matrix monotone on* $[0, \infty)$ *if and only if* $p \in [0, 1]$.

Proof. The necessity of the condition $p \in [0, 1]$ follows from Theorem 7.2.5 because the function x^p is concave only in this range of the parameter p. The sufficiency part is done along the elegant arguments of G. K. Pedersen [466] and F. Hansen [237].

Suppose that $A, B \in \mathcal{A}(H)$ and $A \geq B > 0$. Consider the set $\mathcal{E}$ of all exponents $p \in [0, 1]$ such that $A^p \geq B^p$. Clearly, $0, 1 \in \mathcal{E}$ and $\mathcal{E}$ is a closed set because the map $p \to A^p$ is continuous. We will show that $\mathcal{E}$ is a dense subset of $[0, 1]$, which implies their equality. This is done by proving that

$$p, q \in \mathcal{E} \text{ implies } \frac{p+q}{2} \in \mathcal{E}.$$

From $A^p \geq B^p$ we infer that $A^{-p/2} B^p A^{-p/2} \leq I$ and thus

$$\left\| B^{p/2} A^{-p/2} \right\|^2 = \left\| \left(B^{p/2} A^{-p/2} \right)^T B^{p/2} A^{-p/2} \right\| = \left\| A^{-p/2} B^p A^{-p/2} \right\| \leq 1.$$

Hence $\left\| B^{p/2} A^{-p/2} \right\| \leq 1$ and similarly $\left\| B^{q/2} A^{-q/2} \right\| \leq 1$. Therefore

$$\begin{aligned} 1 &\geq \left\| \left(B^{p/2} A^{-p/2} \right) \left(B^{q/2} A^{-q/2} \right) \right\| = \left\| A^{-p/2} B^{(p+q)/2} A^{-q/2} \right\| \\ &\geq r_{sp} \left(A^{-p/2} B^{(p+q)/2} A^{-q/2} \right) \\ &= r_{sp} \left(A^{-(p+q)/4} B^{(p+q)/2} A^{-(p+q)/4} \right) r_{sp}(C) \\ &= \left\| A^{-(p+q)/4} B^{(p+q)/2} A^{-(p+q)/4} \right\|. \end{aligned}$$

Here $r_{sp}(C) = \sup \{|\lambda| : \lambda \in \sigma(C)\}$ represents the spectral radius. We have $r_{sp}(C) = \|C\|$ if C is a symmetric operator. See Theorem 3.6.4. Then $I \geq A^{-(p+q)/4} B^{(p+q)/2} A^{-(p+q)/4}$, equivalently, $A^{(p+q)/2} \geq B^{(p+q)/2}$. In conclusion, $(p+q)/2 \in \mathcal{E}$ and the proof is complete. ∎

Löwner has obtained Theorem 7.2.7 as a consequence of his integral representation of matrix monotone functions: a real-valued continuous function f on $[0, \infty)$ is matrix monotone if and only if it has a representation of the form

$$f(x) = a + bx + \int_0^\infty \frac{\lambda x}{x + \lambda} \mathrm{d}\mu(\lambda) \tag{7.4}$$

with $a \in \mathbb{R}$, $b \geq 0$ and a positive Borel measure μ on $(0, \infty)$ such that

$$\int_0^\infty \frac{\lambda}{1 + \lambda} \mathrm{d}\mu(\lambda) < \infty.$$

See F. Hiai and D. Petz [253], Theorem 4.41, p. 169, for details.

Another immediate consequence of the integral representation (7.4) is the following result:

7.2.8 Lemma *If $f : \mathbb{R}_+ \to \mathbb{R}$ is matrix monotone, then $xf(x)$ is matrix convex. In particular, the function $x\log(x)$ is matrix convex on $\mathbb{R}_+$.*

The analogue of Jensen's inequality in the framework of matrix convex functions makes necessary the consideration of noncommutative convex combinations of n-tuples of symmetric operators. This remark is due to F. Hansen and G. K. Pedersen [240], who proved the following important result:

7.2.9 Theorem (Jensen's operator inequality) *For a continuous function f defined on an interval I the following two conditions are equivalent:*

(a) *f is matrix convex;*

(b) *For each integer number $n \geq 1$ we have the inequality*

$$f\left(\sum_{k=1}^{n}(C_k)^T A_k C_k\right) \leq \sum_{k=1}^{n}(C_k)^T f(A_k)C_k \tag{7.5}$$

for every n-tuple $(A_1, ..., A_n)$ of symmetric operators on an arbitrary real Hilbert space H, with spectra contained in I, and every n-tuple $(C_1, ..., C_n)$ of bounded operators on H with $\sum_{k=1}^{n}(C_k)^T C_k = I$.

Moreover, this theorem admits the following contractive version: Let f be a continuous function defined on an interval I and suppose that $0 \in I$. Then f is matrix convex and $f(0) \leq 0$ if and only if for every natural number $n \geq 1$, the inequality (7.5) is valid for every n-tuple $(A_1, ..., A_n)$ of symmetric operators on an arbitrary real Hilbert space H, with spectra contained in I, and every n-tuple $(C_1, ..., C_n)$ of bounded operators on H with $\sum_{k=1}^{n}(C_k)^T C_k \leq I$.

A nice application of Theorem 7.2.9 is offered by the noncommutative analogue of the notion of generalized perspective function introduced first by E. G. Effros [174] (in the commutative case) and then by A. Ebadian, I. Nikoufar and M. E. Gordji [172] (in the general case). We will consider here only a simplified version of their results, which refers to the perspective function associated to a continuous function $f : [0, \infty) \to \mathbb{R}$:

$$\tilde{f} : \mathcal{A}(H)_+ \times \mathcal{A}(H)_{++} \to \mathbb{R}, \quad \tilde{f}(A, B) = B^{1/2} f\left(B^{-1/2} A B^{-1/2}\right) B^{1/2}.$$

As was noticed by E. G. Effros and F. Hansen [175], the function $\tilde{f}$ is the unique extension of the corresponding commutative perspective function associated to f that preserves the properties of homogeneity and convexity.

7.2.10 Theorem *The function f is matrix convex/concave if and only if the function $\tilde{f}$ is jointly convex/concave.*

Proof. Suppose first that f is matrix convex and consider $A_1, A_2 \in \mathcal{A}(H)_+$, $B_1, B_2 \in \mathcal{A}(H)_{++}$ and $\lambda \in [0, 1]$. Put $A = (1 - \lambda)A_1 + \lambda A_2$ and $B = C_1 =$

$(1-\lambda)B_1+\lambda B_2$. Then the operators $C_1=((1-\lambda)B_1)^{1/2}B^{-1/2}$ and $C_2=(\lambda B_2)^{1/2}B^{-1/2}$ verify $(C_1)^T C_1+(C_2)^T C_2=I$. According to Theorem 7.2.9,

$$\begin{aligned}\tilde{f}(A,B)&=B^{1/2}f\left(B^{-1/2}AB^{-1/2}\right)B^{1/2}\\&=B^{1/2}f((C_1)^T B_1^{-1/2}A_1B_1^{-1/2}C_1+(C_2)^T B_2^{-1/2}A_2B_2^{-1/2}C_2)B^{1/2}\\&\leq B^{1/2}\left[(C_1)^T f(B_1^{-1/2}A_1B_1^{-1/2})C_1+(C_2)^T f(B_2^{-1/2}A_2B_2^{-1/2})C_2\right]B^{1/2}\\&=(1-\lambda)B_1^{1/2}f(B_1^{-1/2}A_1B_1^{-1/2})B_1^{1/2}+\lambda B_2^{1/2}f(B_2^{-1/2}A_2B_2^{-1/2})B_2^{1/2}\\&=(1-\lambda)\tilde{f}(A_1,B_1)+\lambda\tilde{f}(A_2,B_2),\end{aligned}$$

and the proof of joint convexity of $\tilde{f}$ is done.

The converse implication is straightforward since $f(A)=\tilde{f}(A,I)$. ∎

The aforementioned paper of E. G. Effros includes a number of deep results that can be derived from Theorem 7.2.10. An example is offered by the case of *Kullback–Leibler divergence* of ρ from σ,

$$D_{KL}(\rho||\sigma)=\operatorname{trace}(\rho\log\rho-\rho\log\sigma),$$

where $\rho,\sigma\in\left\{A\in\operatorname{Sym}^{++}(n,\mathbb{R}):\operatorname{trace}(A)=1\right\}$.

7.2.11 Corollary (G. Lindblad [340]) *The Kullback–Leibler divergence is a jointly convex function.*

Proof. We apply Theorem 7.2.10 for the real Hilbert space $H=\mathrm{M}_n(\mathbb{R})$ (endowed with the Frobenius norm) and the continuous function $f(t)=t\log t$; according to Lemma 7.2.8, this function is operator convex on $\mathcal{A}(\mathcal{H})_+$. The operators $A,B:\mathrm{M}_n(\mathbb{R})\to\mathrm{M}_n(\mathbb{R})$ given by $A(X)=\rho X$ and $B(X)=X\sigma$ are positive and commute; for example,

$$\langle A(X),X\rangle_{HS}=\operatorname{trace}(X^T\rho X)=\operatorname{trace}(\left(\rho^{1/2}X\right)^T\left(\rho^{1/2}X\right))\geq 0$$

for all $X\in\mathrm{M}_n(\mathbb{R})$, hence A is positive. Therefore,

$$\begin{aligned}\langle\tilde{f}(A,B)(I),I\rangle_{HS}&=\langle(A\log A-A\log B)(I),I\rangle_{HS}\\&=\operatorname{trace}(\rho\log\rho-\rho\log\sigma)=D_{KL}(\rho||\sigma)\end{aligned}$$

is jointly convex. ∎

Lindblad's original argument for Corollary 7.2.11 is sketched in Exercise 10.

We end this section by mentioning an interesting connection between the matrix monotone functions and the Bernstein functions.

7.2.12 Theorem *Every strictly positive matrix monotone function defined on* $(0,\infty)$ *is a Bernstein function.*

For details (and the converse of Theorem 7.2.12) see R. L. Schilling, R. Song, and Z. Vondraček, [521], Theorem 12.17, p. 195. An alternative approach is offered by B. Simon [543].

Exercises

1. The property of matrix convexity of a continuous function $f : (a, b) \to \mathbb{R}$ is equivalent to the property of matrix convexity of all orders. Prove that f is matrix convex if, and only if, the matrix functions $A \to f(A)$ are convex on $\{A \in \mathrm{Sym}(n, \mathbb{R}) : \sigma(A) \subset (a, b)\}$, whenever $n \geq 1$.

   A similar statement holds in the case of matrix monotone functions.

   [*Hint*: If H is a real Hilbert space having a countably infinite orthonormal basis $(\mathbf{u}_n)_n$, one denotes by P_n the orthogonal projection on $\mathrm{span}\{\mathbf{u}_1, ..., \mathbf{u}_n\}$. Then for every $A \in \mathcal{A}(H)$ we have $A_n = P_n A P_n + ((a+b)/2)(I - P_n) \to A$ pointwise, whence $f(A_n) \to f(A)$ pointwise. On the other hand, by the functional calculus, $f(A_n) = f(P_n A P_n) + f((a+b)/2)(I - P_n)$.]

2. (a) Prove that the functions $f(x) = x^2$ is matrix convex on the whole real line, but it is not matrix monotone on any positive interval.

   (b) Prove that the function $f(x) = x^p$ is operator convex on $[0, \infty)$ for $1 \leq p \leq 2$ and use the formula

   $$A \log A = \lim_{p \to 1+} \frac{A^p - A}{p - 1}$$

   to conclude that the function $t \log t$ is also operator convex on the same interval.

3. (F. A. Berezin) Let P be an orthogonal projection in $\mathbb{R}^n$ and $A \in \mathrm{Sym}(n, \mathbb{R})$. Prove that

   $$\mathrm{trace}(Pf(PAP)P) \leq \mathrm{trace}(Pf(A)P),$$

   for every convex function $f : \mathbb{R} \to \mathbb{R}$.

   [*Hint:* Use Lemma 2.4.1 and Lemma 2.4.2 to reduce the proof to the case where f is the absolute value function.]

4. (Peierls' inequality) Suppose that $A \in \mathrm{Sym}(n, \mathbb{R})$ is a symmetric matrix whose spectrum is contained in the interval $[m, M]$. Infer from the spectral decomposition of symmetric matrices that for every continuous convex function $f : [m, M] \to \mathbb{R}$ and every $\mathbf{x} \in \mathbb{R}^n$ with $\|\mathbf{x}\| = 1$ we have

   $$f(\langle A\mathbf{x}, \mathbf{x}\rangle) \leq \langle f(A)\mathbf{x}, \mathbf{x}\rangle.$$

   An immediate consequence is the fact that

   $$\sum_{k=1}^{n} f(\langle A\mathbf{u}_k, \mathbf{u}_k\rangle) \leq \mathrm{trace} f(A),$$

   for every orthonormal base $(\mathbf{u}_k)_k$ of $\mathbb{R}^n$.

5. Infer from Peierls' inequality that if $f : \mathbb{R} \to \mathbb{R}$ is a convex function, then the function
$$g(A) = \operatorname{trace} f(A)$$
is convex on $\operatorname{Sym}(n, \mathbb{R})$.

6. (The Peierls–Bogoliubov inequality). Suppose that $A, B \in \operatorname{Sym}(n, \mathbb{R})$ and $f : \mathbb{R} \to \mathbb{R}$ is a convex function. Prove that
$$f\left(\frac{\operatorname{trace}\left(Ae^B\right)}{\operatorname{trace} e^B}\right) \leq \frac{\operatorname{trace}\left(f(A)e^B\right)}{\operatorname{trace} e^B}.$$
[*Hint:* Replacing B by $B+\lambda I$ (if necessary) we may assume that $\operatorname{trace}\left(e^B\right) = 1$. Then, if $e^B = \sum_k e^{\lambda_k}\langle \cdot, \mathbf{v}_k\rangle \mathbf{v}_k$ is the spectral decomposition of e^B, we have $\sum_k e^{\lambda_k} = 1$ and
$$\operatorname{trace}\left(f(A)e^B\right) = \sum_k e^{\lambda_k}\langle f(A)\mathbf{v}_k, \mathbf{v}_k\rangle.$$
Continue by using the result of Exercise 5.]

7. (Hadamard determinant inequality) Infer from Peierls' inequality the Hadamard determinant inequality, that is,
$$\det(A) \leq \prod_{k=1}^n a_{kk}$$
for every $A \in \operatorname{Sym}^{++}(n, \mathbb{R})$, $A = (a_{ij})_{i,j}$.
[*Hint:* Apply the result of Exercise 4 for $f(x) = -\log x$, the operator associated to A, and the natural basis of $\mathbb{R}^n$. Notice that $e^{\operatorname{trace}(\log A)} = \det(A)$.]

8. (An operator form of the Hermite–Hadamard inequality due to B. Mond and J. Pečarić [392]) Suppose that H is a real Hilbert space and $A \in \mathcal{A}(H)$ is a self-adjoint bounded operator whose spectrum is contained in the interval $[m, M]$. Prove that for every continuous convex function $f : [m, M] \to \mathbb{R}$ and every vector $\mathbf{x} \in H$ with $\|\mathbf{x}\| = 1$, we have
$$f(\langle A\mathbf{x}, \mathbf{x}\rangle) \leq \langle f(A)\mathbf{x}, \mathbf{x}\rangle \leq \frac{M - \langle A\mathbf{x}, \mathbf{x}\rangle}{M - m} f(m) + \frac{\langle A\mathbf{x}, \mathbf{x}\rangle - m}{M - m} f(M).$$
[*Hint:* For the left-hand side inequality, apply Theorem 2.1.5 to deduce for each $\varepsilon > 0$ the existence of an affine function $mt+n$ such that $f(t) \geq mt+n$ for all $t \in [m, M]$, and $m\langle A\mathbf{x}, \mathbf{x}\rangle + n \geq f(\langle A\mathbf{x}, \mathbf{x}\rangle) - \varepsilon$. For the right-hand side inequality, notice that $\frac{M-t}{M-m} f(m) + \frac{t-m}{M-m} f(M) - f(t) \geq 0$, so, according to the continuous functional calculus it follows that
$$\frac{M - A}{M - m} f(m) + \frac{A - m}{M - m} f(M) - f(A) \geq 0.$$

9. Infer from Theorem 7.2.10 the following special case of an inequality due to E. H. Lieb: If $0 < p < 1$, then the function $F(A, B) = \operatorname{trace}\left(A^p B^{1-p}\right)$ is jointly concave on the strictly positive $n \times n$ dimensional matrices $A,\ B$.

   [*Hint:* Adapt the argument of Corollary 7.2.11, by taking into account the fact that the function $-t^p$ is matrix convex.]

   Remark. The *Lieb concavity inequality* asserts that for every $n \times n$ dimensional matrix X and every $0 < t < 1$, the function

   $$f(A, B) = \operatorname{trace}(X^* A^t X B^{1-t}),$$

   defined on pairs of positive matrices, is jointly concave. See B. Simon [541] Theorem 8.10, p. 73, for the proof.

10. According to Exercise 9, the map $(A, B) \to \frac{1}{p-1} \operatorname{trace}\left(A^p B^{1-p} - A\right)$ is jointly convex. Prove that

    $$\lim_{p \to 1} \frac{\operatorname{trace}\left(A^p B^{1-p} - A\right)}{p-1} = \operatorname{trace}(A \log A - A \log B) = D_{KL}(A||B),$$

    and conclude that $D_{KL}(A||B)$ is jointly convex (since convexity is preserved by passing to the limit).

11. Infer from Theorem 7.2.10 that the function $(A, B) \to AB^{-1}A$ is jointly convex on $\operatorname{Sym}(n, \mathbb{R}) \times \operatorname{Sym}^{++}(n, \mathbb{R})$.

12. (Convexity inequality of O. Klein). Let $A,\ B$ be symmetric real matrices with spectrum in the domain of definition of the continuous convex function f. Prove that

    $$\operatorname{trace}\left[f(A) - f(B) - (A - B)\, f'(B)\right] \geq 0.$$

13. (A convex, but not an operator convex function) Prove that the function

    $$f : \mathrm{M}_{m \times n}(\mathbb{R}) \to \operatorname{Sym}(n, \mathbb{R}), \quad f(A) = A^T A,$$

    is convex (though it is not operator convex).

7.3 The Trace Metric of $\operatorname{Sym}^{++}(n, \mathbb{R})$

In every Hilbert space H, the middle point of a linear segment $[x, y]$ can be characterized as the unique point z such that

$$d(x, y)^2 + 4d(z, w)^2 = 2d(x, w)^2 + 2d(y, w)^2 \quad \text{for all } w \in H;$$

see the parallelogram law. Clearly, this remark also works when H is replaced by a convex subset of it.

$\mathrm{Sym}^{++}(n,\mathbb{R})$ (which is an open convex subset of the Hilbert space $\mathrm{Sym}(n,\mathbb{R})$) admits another metric structure that allows the presence of a valuable substitute of the parallelogram law. We refer here to the *trace metric*, defined on $\mathrm{Sym}^{++}(n,\mathbb{R})$ by the formula

$$\delta(A,B)=\left(\sum_{k=1}^{n}\log^2\lambda_k\right)^{1/2}, \tag{7.6}$$

where $\lambda_1,\ldots,\lambda_n$ are the eigenvalues of AB^{-1}. The fact that this metric is well defined is based on the property of invariance of the spectrum relative to similarity, that is,

$$\sigma(C)=\sigma(X^{-1}CX)$$

for every $C\in \mathrm{M}_n(\mathbb{R})$ and every X in $\mathrm{GL}(n,\mathbb{R})$ (the group of all nonsingular matrices belonging to $\mathrm{M}_n(\mathbb{R})$). Indeed, the matrix AB^{-1} is similar with

$$A^{-1/2}(AB^{-1})A^{1/2}=A^{1/2}B^{-1/2}(A^{1/2}B^{-1/2})^{\star}>0,$$

and this fact assures the strict positivity of the eigenvalues of AB^{-1}. As a consequence, the trace metric is invariant under the natural action of the group $\mathrm{GL}(n,\mathbb{R})$ on $\mathrm{Sym}^{++}(n,\mathbb{R})$ via the bijective maps

$$\Gamma_C(A)=C^TAC,\quad A\in\mathrm{Sym}^{++}(n,\mathbb{R}),$$

associated to each $C\in\mathrm{GL}(n,\mathbb{R})$. Indeed,

$$\delta(\Gamma_C(A),\Gamma_C(B))=\delta\left(A,B\right).$$

In dimension 1, $\mathrm{Sym}^{++}(n,\mathbb{R})$ coincides with the interval $(0,\infty)$ and the trace metric is given by the formula

$$\delta(a,b)=\left|\log a-\log b\right|.$$

Then, for every two points $a,b\in(0,\infty)$, there is a unique point $c\in(0,\infty)$ such that

$$\delta(a,c)=\delta(c,b)=\frac{1}{2}\delta(a,b).$$

This unique point c is nothing but the geometric mean $\sqrt{ab}$ of a and b. Remarkably, all these facts extend to all dimensions.

The *geometric mean* $A\#_{1/2}B$ of two strictly positive matrices A and B was introduced by W. Pusz and S. L. Woronowicz [491] as the unique strictly positive solution

$$X=A^{1/2}(A^{-1/2}BA^{-1/2})^{1/2}A^{1/2} \tag{7.7}$$

of the so called *Riccati equation,*

$$XA^{-1}X=B.$$

That $A\#_{1/2}B$ is such a solution is trivial. For uniqueness, suppose that X and Y are two strictly positive matrices such that $XA^{-1}X = B$ and $YA^{-1}Y = B$. Then

$$\left(A^{-1/2}XA^{-1/2}\right)^2 = A^{-1/2}\left(XA^{-1}X\right)A^{-1/2}$$
$$= A^{-1/2}\left(YA^{-1}Y\right)A^{-1/2} = \left(A^{-1/2}YA^{-1/2}\right)^2.$$

Because a strictly positive matrix has a unique strictly positive square root it follows that $A^{-1/2}XA^{-1/2} = A^{-1/2}YA^{-1/2}$. This easily implies that $X = Y$.

Some simple but useful properties are listed below:

$$A\#_{1/2}B = (AB)^{1/2} \quad \text{if } A \text{ and } B \text{ commute;}$$
$$A\#_{1/2}B = B\#_{1/2}A;$$
$$(C^TAC)\#_{1/2}(C^TBC) = C^T(A\#_{1/2}B)C \quad \text{for all } C \in \mathrm{GL}(n,\mathbb{R}).$$

See Exercises 1-3.

The *geometric-arithmetic mean inequality* for strictly positive matrices,

$$A\#_{1/2}B \leq \frac{A+B}{2},$$

can be easily derived from Lemma 3.5.6, on the simultaneous diagonalization of noncommuting matrices.

7.3.1 Theorem *Given two matrices A and B in $\mathrm{Sym}^{++}(n,\mathbb{R})$, their geometric mean $A\#_{1/2}B$ is the unique matrix X in $\mathrm{Sym}^{++}(n,\mathbb{R})$ such that*

$$\delta(A,X) = \delta(B,X) = \frac{1}{2}\,\delta(A,B).$$

The proof follows from the following result:

7.3.2 Lemma *Suppose that A and B are two strictly positive matrices. Then, the map*

$$G_{(A,B)} : \mathrm{Sym}^{++}(n,\mathbb{R}) \to \mathrm{Sym}^{++}(n,\mathbb{R}),$$

defined by the formula $G_{(A,B)}X = (A\#_{1/2}B)X^{-1}(A\#_{1/2}B)$, has the following properties:

(a) *$G_{(A,B)}$ is a bijective isometry relative to the trace metric;*
(b) *$G_{(A,B)}(A) = B$ and $G_{(A,B)}(B) = A$;*
(c) *$A\#_{1/2}B$ is the unique fixed point of $G_{(A,B)}$ and*

$$\delta(G_{(A,B)}(X),X) = 2\delta(A\#_{1/2}B,X)$$

Proof. The proof of the assertions (a) and (b) is straightforward.

As concerns the first part of the assertion (c), if X is a fixed point of $G_{(A,B)}$, then

$$(A\#_{1/2}B)X^{-1}(A\#_{1/2}B) = X,$$

which yields $\left(X^{-1/2}(A\#_{1/2}B)X^{-1/2}\right)^2 = \mathrm{I}$ and $X^{-1/2}(A\#_{1/2}B)X^{-1/2} = \mathrm{I}$, by the uniqueness of the square root. Thus $X = A\#_{1/2}B$.

The second part of (c) is equivalent to the fact that

$$\delta((A\#_{1/2}B)X^{-1}(A\#_{1/2}B), X) = 2\delta(X, A\#_{1/2}B)$$

for every $X \in \mathrm{Sym}^{++}(n,\mathbb{R})$. This follows from the definition of the trace metric, by noticing that $\sigma(C^2) = \{\lambda^2 : \lambda \in \sigma(C)\}$ for all C in $\mathrm{Sym}^{++}(n,\mathbb{R})$. ■

The Hilbert space $\mathrm{Sym}(n,\mathbb{R})$ (endowed with the Frobenius norm) and the metric space $\mathrm{Sym}^{++}(n,\mathbb{R})$ can be related via the exponential map exp, which is given by the formula

$$\exp : \mathrm{Sym}(n,\mathbb{R}) \to \mathrm{Sym}^{++}(n,\mathbb{R}), \quad \exp(A) = \sum\nolimits_{k=1}^{r} \mathrm{e}^{\lambda_k} P_k,$$

provided that A has the spectral decomposition $A = \sum_{k=1}^{r} \lambda_k P_k$. Clearly this map is a continuous bijection; its continuity follows from Weyl's perturbation Theorem 6.4.7. If we denote its inverse by log, then

$$\delta(A,B) = \left\| \log(A^{-1/2}BA^{-1/2}) \right\|_{HS}.$$

It is worth mentioning that the above definition of the exponential map agrees with that used in functional analysis,

$$\exp(A) = \sum\nolimits_{k=1}^{\infty} \frac{1}{k!} A^k.$$

7.3.3 Lemma *The exponential is a continuous map from* $\mathrm{Sym}(n,\mathbb{R})$ (*endowed with the Frobenius norm*) *onto the space* $\mathrm{Sym}^{++}(n,\mathbb{R})$ (*endowed with the trace metric*).

Proof. Indeed, if $A_m \to A$ in $\mathrm{Sym}(n,\mathbb{R})$, then $\mathrm{e}^{-A_m}\mathrm{e}^{A}$ converges to I in the Frobenius norm, which implies that $\lambda_k(\mathrm{e}^{-A_m}\mathrm{e}^{A}) \to 1$ for each $k \in \{1, ...n\}$. Taking into account the definition of the trace metric, we conclude that $\delta\left(\mathrm{e}^{A_m}, \mathrm{e}^{A}\right)$ converges to 0. ■

In order to outline deeper properties of the metric space $\mathrm{Sym}^{++}(n,\mathbb{R})$ we shall need the following technical result:

7.3.4 Lemma (The exponential metric increasing property) *We have*

$$\|\log A - \log B\|_{HS} \leq \delta(A,B) \quad \textit{for all } A, B \in \mathrm{Sym}^{++}(n,\mathbb{R}), \tag{7.8}$$

The equality occurs when A *and* B *commute with each other.*

For details see R. Bhatia [64], Theorem 6.1.4 and Proposition 6.1.5. A proof of the inequality (7.8) (not including the equality part) makes the objective of Exercise 6.

7.3.5 Corollary *The metric space* $\mathrm{Sym}^{++}(n,\mathbb{R})$ *is complete relative to the trace metric.*

Proof. Let $(A_m)_m$ be a Cauchy sequence in $\mathrm{Sym}^{++}(n,\mathbb{R})$. By the exponential metric increasing property, the operators $B_m = \log A_m$ form a Cauchy sequence in $\mathrm{Sym}(n,\mathbb{R})$, so $B_m \to B$ for some B in $\mathrm{Sym}(n,\mathbb{R})$. By Lemma 7.3.3, the sequence $(A_m)_m$ converges to $A = \mathrm{e}^B$ in the metric space $\mathrm{Sym}^{++}(n,\mathbb{R})$. ∎

Another basic feature of the geometry of $\mathrm{Sym}^{++}(n,\mathbb{R})$ is as follows.

7.3.6 Theorem (The semiparallelogram law) *Let A and B be two matrices in $\mathrm{Sym}^{++}(n,\mathbb{R})$ and let $M = A\#_{1/2}B$. Then, for every matrix $C \in \mathrm{Sym}^{++}(n,\mathbb{R})$, we have*

$$\delta^2(M,C) \le \frac{\delta^2(A,C)+\delta^2(B,C)}{2} - \frac{1}{4}\delta^2(A,B).$$

Proof. By applying the isometry $\Gamma_{M^{-1/2}}$ to all matrices involved we may assume that $M = I$. Then

$$\delta(M,C) = \|\log M - \log C\|_{HS} \quad \text{and} \quad \delta(A,B) = \|\log A - \log B\|_{HS},$$

according to the exponential metric increasing property. By the parallelogram law in $\mathrm{Sym}(n,\mathbb{R})$,

$$\begin{aligned}\delta^2(M,C) &= \|\log M - \log C\|_{HS}^2 \\ &= \frac{\|\log A - \log C\|_{HS}^2 + \|\log B - \log C\|_{HS}^2}{2} - \frac{\|\log A - \log B\|_{HS}^2}{4} \\ &= \frac{\|\log A - \log C\|_{HS}^2 + \|\log B - \log C\|_{HS}^2}{2} - \frac{1}{4}\delta^2(A,B).\end{aligned}$$

The proof ends by taking again into account the exponential metric increasing property. ∎

The discussion above can be summarized saying that $\mathrm{Sym}^{++}(n,\mathbb{R})$ endowed with the trace metric is a global NPC space in the sense of the following definition:

7.3.7 Definition *A metric space $M = (M,d)$ is called a global NPC space (that is, a space with a global nonpositive curvature) if it is complete and for each pair of points $x_0, x_1 \in M$ there exists a point $y \in M$ with the property that for all $z \in M$ we have*

$$d^2(z,y) \le \frac{1}{2}d^2(z,x_0) + \frac{1}{2}d^2(z,x_1) - \frac{1}{4}d^2(x_0,x_1). \tag{7.9}$$

The point y that appears in Definition 7.3.7 is called the *midpoint* of x_0 and x_1; it is the unique point $x_{1/2}$ in M such that

$$d(x_0,x_{1/2}) = d(x_{1/2},x_1) = \frac{1}{2}d(x_0,x_1).$$

The existence of geodesics in the space $\mathrm{Sym}^{++}(n,\mathbb{R})$ follows from the following result, whose proof is borrowed from K. T. Sturm [558].

7.3.8 Theorem *If $M=(M,d)$ is a global NPC space, then each pair of points $x_0,x_1\in M$ can be connected by a geodesic, that is, by a rectifiable and continuous map $\gamma:[0,1]\to M$ such that $\gamma(0)=x_0$, $\gamma(1)=x_1$ and the length of $\gamma|_{[s,t]}$ is $d(\gamma(s),\gamma(t))$ for all $0\le s\le t\le 1$. Moreover, this geodesic is unique and the inequality (7.9) can be extended as*

$$d^2(z,\gamma(t))\le(1-t)d^2(z,x_0)+td^2(z,x_1)-t(1-t)d^2(x_0,x_1), \tag{7.10}$$

for every $z\in M$ and every $t\in[0,1]$.

Proof. Given $x_0,x_1\in M$, we infer from Definition 7.3.7 the existence of their midpoint $x_{1/2}$. Then the points $x_{1/4}$ and $x_{3/4}$ are obtained as midpoints of x_0 and $x_{1/2}$ and $x_{1/2}$ and x_1, respectively. In this way, we obtain the points x_t for all dyadic $t\in[0,1]$ and, clearly, $d(x_s,x_t)=d(x_s,x_r)+d(x_r,x_t)$ for all dyadic $0\le s<r<t\le 1$. By completeness of M, it yields the existence of $x_t\in M$ for all $t\in[0,1]$ such that $\gamma:[0,1]\to M$, $\gamma(t)=x_t$, is a geodesic connecting x_0 and x_1. The uniqueness of the geodesic follows from the uniqueness of the midpoints.

As concerns the inequality (7.10), it suffices to prove it for dyadic $t\in[0,1]$. It obviously holds for $t=0$ and $t=1$. Assuming that it holds for all $t=k2^{-n}$ with $k=0,1,...,2^n$, we will show that (7.10) also holds for all $t=k2^{-n-1}$ with $k=0,1,...,2^{n+1}$. The case of even k is covered by our assumption, so it remains the consider the case of points $t=k2^{-n-1}$ with k an odd number. Put $\Delta=2^{-n-1}$. Then, according to the definition of global NPC spaces, for all $z\in M$ we have

$$d^2(z,x_t)\le\frac{1}{2}d^2(z,x_{t-\Delta})+\frac{1}{2}d^2(z,x_{t+\Delta})-\frac{1}{4}d^2(x_{t-\Delta},x_{t+\Delta}),$$

and, by (7.10), for multiples of 2^{-n}, we have

$$\begin{aligned}d^2(z,x_{t\pm\Delta})\le{}&(1-t\mp\Delta)d^2(z,x_0)+(t\pm\Delta)d^2(z,x_1)\\&-(t\pm\Delta)(1-t\mp\Delta)d^2(x_0,x_1),\end{aligned}$$

for all $z\in M$. Combining them, we conclude that

$$d^2(z,x_t)\le(1-t)d^2(z,x_0)+td^2(z,x_1)-t(1-t)d^2(x_0,x_1).$$

■

Among the Banach spaces, only Hilbert spaces are global NPC spaces. In their case, the geodesics are the line segments.

In the case of $\mathrm{Sym}^{++}(n,\mathbb{R})$, the geodesic connecting the points A and B is parametrized by the function

$$\gamma(t)=A\#_tB,\quad t\in[0,1],$$

where $A\#_tB=A^{1/2}(A^{-1/2}BA^{-1/2})^tA^{1/2}$.

Exercises

1. Use spectral decomposition to prove that $A\#_{1/2}B = (AB)^{1/2}$ if the strictly positive matrices A and B commute.

2. Infer from the Riccati equation that

$$\left(A\#_{1/2}B\right)^{-1} = \left(A^{-1}\right)\#_{1/2}\left(B^{-1}\right) \text{ and } \left(A\#_{1/2}B\right)^{-1} = \left(B^{-1}\right)\#_{1/2}\left(A^{-1}\right)$$

for all $A, B \in \mathrm{Sym}^{++}(n,\mathbb{R})$. This implies the commutativity of the geometric mean,

$$A\#_{1/2}B = B\#_{1/2}A.$$

3. Prove that the transformations $\Gamma_C(A) = C^T AC$ preserve the geometric mean, that is,

$$(C^T AC)\#_{1/2}(C^T BC) = C^T(A\#_{1/2}B)C$$

for all $A, B \in \mathrm{Sym}^{++}(n,\mathbb{R})$ and $C \in \mathrm{GL}(n,\mathbb{R})$.

[*Hint:* Use the fact that the geometric mean of two strictly positive matrices A and B is the unique strictly positive solution of the equation $XA^{-1}X = B$.]

4. Suppose that $A, B \in \mathrm{Sym}^{++}(n,\mathbb{R})$ and consider the following two iterative sequences defined by $A_0 = A$, $B_0 = B$ and

$$A_{n+1} = 2(A_n^{-1} + B_n^{-1})^{-1}, \quad B_{n+1} = (A_n + B_n)/2 \qquad \text{for } n \geq 0.$$

Prove that

(a) $A_n \leq A_{n+1} \leq A\#_{1/2}B \leq B_{n+1} \leq B_n \leq$ for all $n \geq 1$;

(b) Both sequences $(A_n)_n$ and $(B_n)_n$ converge to $A\#_{1/2}B$.

[*Hint:* Apply Lemma 3.5.6 to reduce ourselves to the case where $A = I$ and B is a diagonal matrix.]

5. Infer from the preceding exercise the monotonicity property of the geometric mean: if $A \leq A'$ and $B \leq B'$ in $\mathrm{Sym}^{++}(n,\mathbb{R})$, then

$$A\#_{1/2}B \leq A'\#_{1/2}B'.$$

An immediate consequence is the following special case of the Löwner–Heinz Theorem: if $A, B \in \mathrm{Sym}^{+}(n,\mathbb{R})$ and $A \leq B$, then $A^{1/2} \leq B^{1/2}$.

6. (R. Bhatia [64]) The aim of this exercise is to sketch a proof of the exponential metric increasing property based on the formula (7.11) of J.-B. Duhamel, F. J. Dyson, R. P. Feynman, and J. Schwinger, concerning the differential of the exponential.

(a) Let $A \in \mathrm{Sym}(n, \mathbb{R})$. The differential of the exponential map is defined by the formula $\mathrm{de}^A(B) = \frac{\mathrm{d}}{\mathrm{d}t}\mathrm{e}^{A+tB}\big|_{t=0}$. Prove that

$$\mathrm{de}^A(B) = \int_0^1 \mathrm{e}^{tA} B \mathrm{e}^{(1-t)A} \mathrm{d}t. \tag{7.11}$$

(b) Suppose that $X = (x_{ij})_{i,j} \in \mathrm{Sym}(n, \mathbb{R})$ and $A \in \mathrm{Sym}^{++}(n, \mathbb{R})$ is a diagonal matrix with eigenvalues $\lambda_1, ..., \lambda_n$. Prove that the entries of the matrix $\int_0^1 A^t X A^{1-t} \mathrm{d}t$ are of the form

$$\left(\int_0^1 \lambda_i^t \lambda_j^{1-t} \mathrm{d}t \right) x_i x_j = L(\lambda_i, \lambda_j)\, x_i x_j,$$

where $L(\lambda_i, \lambda_j)$ denotes the logarithmic mean of λ_i and λ_j. Infer that in this case

$$\left\| A^{1/2} X A^{1/2} \right\|_{HS} \leq \left\| \int_0^1 A^t X A^{1-t} \mathrm{d}t \right\|_{HS}.$$

(c) Use the assertion (b) to prove that for all $A, B \in \mathrm{Sym}(n, \mathbb{R})$ we have

$$\|B\|_{HS} \leq \left\| \mathrm{e}^{-A/2} \mathrm{de}^A(B) \mathrm{e}^{-A/2} \right\|_{HS}.$$

[*Hint:* (b) The logarithmic mean is greater than or equal with the geometric mean. (c) Apply (b) for $X = \mathrm{e}^{-A/2} B \mathrm{e}^{-A/2}$.]

7.4 Geodesic Convexity in Global NPC Spaces

In any global NPC space $M = (M, d)$, one can introduce the concepts of convex set and convex functions by using the geodesics.

7.4.1 Definition *A set $C \subset M$ is called convex if for every pair of points x_0 and x_1 in C, the image of every geodesic $\gamma : [0, 1] \to M$ joining $x_0 = \gamma(0)$ and $x_1 = \gamma(1)$ is contained in C.*

A function $\varphi : C \to \mathbb{R}$ is called convex if the function $\varphi \circ \gamma : [0, 1] \to \mathbb{R}$ is convex for each geodesic $\gamma : [0, 1] \to C$, $\gamma(t) = \gamma_t$, that is,

$$\varphi(\gamma_t) \leq (1 - t)\varphi(\gamma_0) + t\varphi(\gamma_1)$$

for all $t \in [0, 1]$.

The related notions of log-convex function, strictly convex function, strongly convex function, etc. can be introduced in the same manner, imitating the case of normed linear spaces.

According to Theorem 7.3.8, for z arbitrarily fixed in M, the function

$$f(x) = d^2(x, z)$$

is strongly convex on M. As a consequence, the balls are convex sets.

One can even prove that the distance function d is convex on $M \times M$ as a function of both variables. For details, see K. T. Sturm [558], Corollary 2.5. The global NPC product structure on $M \times M$ is provided by the product metric

$$\omega((x_0, x_1), (y_0, y_1)) = \left(d^2(x_0, y_0) + d^2(x_1, y_1)\right)^{1/2}.$$

Every closed convex subset of a global NPC space is a space of the same nature.

The determinant function is convex on $\mathrm{Sym}^{++}(n, \mathbb{R})$ due to the identity

$$\det A^{1/2} \left(A^{-1/2} B A^{-1/2}\right)^t A^{1/2} = (\det A)^{1-t} (\det B)^t$$

and the *AM-GM* inequality.

Every positive linear functional on $\mathrm{M}_n(\mathbb{R})$ induces a log-convex function on $\mathrm{Sym}^{++}(n, \mathbb{R})$ in the sense of Definition 7.4.1. In particular, the trace is a log-convex function. See Exercise 3, at the end of this section.

7.4.2 Remark *More examples of convex functions defined on the global NPC space* $\mathrm{Sym}^{++}(n, \mathbb{R})$ *can be obtained by taking into account the following facts:*

If $f : \mathrm{Sym}^{++}(n, \mathbb{R}) \to \mathbb{R}$ *is convex and* $g : \mathbb{R} \to \mathbb{R}$ *is increasing and convex, then* $g \circ f$ *is convex.*

If $\Psi : \mathrm{M}_n(\mathbb{R}) \to \mathrm{M}_n(\mathbb{R})$ *is a strictly positive linear map and* f *is increasing and convex, then* $f \circ \Psi|_{\mathrm{Sym}^{++}(n,\mathbb{R})}$ *is convex.*

Many results mentioned in Chapter 4 to work in the framework of linear normed spaces remain valid also in the context of NPC spaces.

For example, any continuous midpoint convex function $f : M \to \mathbb{R}$ is convex.

A function $f : M \to \mathbb{R}$ is convex if, and only if, its epigraph is a convex set. All level sets of a convex function are convex sets.

A function $f : M \to \mathbb{R}$ is lower semicontinuous if, and only if, its level sets are convex (if, and only if, its epigraph is closed).

One serious difference between usual convexity and convexity in global NPC spaces concerns the concept of convex combination. The key remark indicating the right approach of this problem is offered by the following alternative definition of the notion of midpoint.

7.4.3 Lemma *Let* x_0 *and* x_1 *be a pair of points in the global NPC space* M. *Then their midpoint* $x_{1/2}$ *is the unique minimizer of the strongly convex function*

$$V(x) = \frac{1}{2} d^2(x, x_0) + \frac{1}{2} d^2(x, x_1).$$

Proof. According to the definition of a global NPC space, we have

$$d^2(x, x_{1/2}) \leq V(x) - \frac{1}{4} d^2(x_0, x_1) = V(x) - V(x_{1/2}),$$

whence we infer that $x_{1/2}$ is a minimizer of V. Since V is strictly convex, we conclude that $x_{1/2}$ is in fact the unique minimizer of V. ■

This remark suggests to define the NPC analog $\lambda_1 x_1 \boxplus \cdots \boxplus \lambda_n x_n$ of a convex combination $\sum_{k=1}^n \lambda_k x_k$ as the unique point minimizing the strongly convex function

$$W(x) = \sum_{k=1}^n \lambda_k d^2(x, x_k).$$

It turns out that this point is nothing but the barycenter $\mathrm{bar}(\mu)$ of the discrete Borel probability measure $\mu = \sum_{k=1}^n \lambda_k \delta_{x_k}$.

The discrete case of Jensen's inequality in the context of global NPC spaces can be settled by adapting the argument of Proposition 4.2.2.

7.4.4 Proposition *If f is a convex function on the global NPC space M, then for every family of points $x_1, ..., x_n$ in M and every family of positive weights $\lambda_1, ..., \lambda_n$ with $\sum_{k=1}^n \lambda_k = 1$ we have*

$$f(\lambda_1 x_1 \boxplus \cdots \boxplus \lambda_n x_n) \leq \sum_{k=1}^n \lambda_k f(x_k).$$

The existence of barycenters in more general situations needs the following result.

7.4.5 Lemma *If M is a global NPC space, then every lower semicontinuous strongly convex function $f : M \to \mathbb{R}$ which is bounded below admits a unique minimizer $z \in M$.*

Proof. Let $(x_n)_n$ be a sequence of points such that

$$\lim_{n\to\infty} f(x_n) = \inf_{x\in M} f(x) = \alpha.$$

Assuming that f is strongly convex with parameter $C > 0$, we have

$$\alpha \leq f\left(\frac{1}{2}x_m \boxplus \frac{1}{2}x_n\right) \leq \frac{f(x_m) + f(x_n)}{2} - \frac{C}{4}d^2(x_m, x_n),$$

whence we infer that $(x_n)_n$ is a Cauchy sequence. Due to the completeness of M there exists a point x such that $x_n \to x$ as $n \to \infty$. Then $f(x) = \alpha$ because f is lower semicontinuous and its lower bound is α.

The uniqueness of the minimizer is a consequence of the strict convexity of f. ■

Under the circumstances of Lemma 7.4.5 we denote

$$z = \arg\min_{x\in M} f(x).$$

The notion of barycenter for an arbitrary Borel probability measure μ defined on a global NPC space M was introduced by E. Cartan [120], inspired by Gauss' least squares method.

We denote by $\mathcal{P}^2(M)$ the set of all Borel probability measures μ on M under which all functions $d^2(\cdot, z)$ are integrable.

7.4.6 Definition *The barycenter $\mu \in \mathcal{P}^2(M)$ is defined by the formula*

$$\mathrm{bar}(\mu) = \arg\min_{z \in M} \frac{1}{2} \int_M d^2(z,x) \mathrm{d}\mu(x).$$

The fact that the notion of barycenter is well defined follows from Lemma 7.4.5, by noticing that strong convexity of the function

$$I(x) = \int_M d^2(z,x) \mathrm{d}\mu(z).$$

The generalization of Jensen's inequality to the context of global NPC spaces was done by J. Jost [281]. We state it here in the formulation of J. Eells and B. Fuglede [173], Proposition 12.3, p. 242:

7.4.7 Theorem (Jensen's Inequality). *For any lower semicontinuous convex function $f : M \to \mathbb{R}$ and any Borel probability measure $\mu \in \mathcal{P}^2(M)$ we have the inequality*

$$f(\mathrm{bar}(\mu)) \leq \int_M f(x) \mathrm{d}\mu(x),$$

provided the right-hand side is well defined.

The proof of Eells and Fuglede is based on the following result concerning barycenters:

7.4.8 Lemma *If a Borel probability measure μ is supported by a convex closed set K, then its barycenter* $\mathrm{bar}(\mu)$ *lies in K.*

The proof of Lemma 7.4.8 is an immediate consequence of the basic separation Theorem 3.3.1.

Exercises

1. Prove that for every three matrices A, B, C in $\mathrm{Sym}^{++}(n, \mathbb{R})$ we have

$$\delta(A \#_{1/2} B, A \#_{1/2} C) \leq \frac{\delta(A,B)}{2},$$

   where δ denotes the trace metric.

2. Prove that the geometric mean $A \#_{1/2} B$ is continuous as a function of pairs of strictly positive matrices.

3. Suppose that $A, B, A', B' \in \mathrm{Sym}^{++}(n, \mathbb{R})$. Prove that the function

$$f(t) = \delta(A \#_t B, A' \#_t B')$$

   is convex on [0,1].

   [*Hint*: It suffices to show that f is midpoint convex.]

4. Prove that every positive linear functional on $\mathrm{M}_n(\mathbb{R})$ induces a log-convex function on $\mathrm{Sym}^{++}(n,\mathbb{R})$ in the sense of Definition 7.4.1.

   [*Hint*: Reduce the problem to the case of pure states, that is, to the case of functionals $\omega : A \to \langle A\mathbf{u}, \mathbf{u}\rangle$ associated to the unit vectors $\mathbf{u} \in \mathbb{R}^n$.]

5. (The variance inequality) Suppose that M is a global NPC space and $\mu \in \mathcal{P}^2(M)$. Prove that for all $z \in M$,

$$\int_M d^2(z,x)\mathrm{d}\mu(x) \geq d^2(z,\mathrm{bar}(\mu)) + \int_M d^2(\mathrm{bar}(\mu),x)\mathrm{d}\mu(x).$$

   Compare to Lagrange's barycentric identity. It is remarkable that the above inequality characterizes the global NPC spaces among the complete metric spaces. See K. T. Sturm [558], Theorem 4.9.

7.5 Comments

The theory of convex spectral functions goes back to Minkowski's determinant inequality and to von Neumann's paper from 1937, relating the concepts of unitarily invariant matrix norm and symmetric gauge function. See [413]. A norm on $\mathrm{M}_N(\mathbb{C})$ is *unitarily invariant* if $\|UAV\| = \|A\|$ for all unitary U and V. Examples are the operator norm and the Frobenius norm. More generally, if $f : \mathbb{R}^N \to \mathbb{R}$ is a *symmetric gauge function*, that is, a norm invariant under permutations on $\mathbb{R}^N$ such that $f(\mathbf{x}) = f(|\mathbf{x}|)$ for every $\mathbf{x}$, then

$$\|A\|_f = f(s_1(A)..., s_N(A))$$

is a unitarily invariant matrix norm. Von Neumann has proved that every unitarily invariant matrix norm can be written in this form. This was generalized to the infinite-dimensional setting by I. C. Gohberg and M. G. Krein. See the book of B. Simon [541] for details.

The differential calculus and the Legendre-Fenchel duality in the context of spectral functions are treated in details by A. S. Lewis [331], [332]. See also J. M. Borwein and J. Vanderwerff [90] and the references therein.

The notion of spectral function in the framework of C^*-algebras was studied by F. Hansen [238] in connection with Jensen type inequalities for convex functions of several operator variables and the construction of multivariate geometric means.

The subject of matrix (and operator) convexity is important in quantum statistical mechanics, quantum information theory, free convexity etc. Our presentation is based on the expository papers of E. A. Carlen [117] and F. Hansen [237].

More advanced material is made available by the books of R. Bhatia [61], B. Simon [541, 542] and J. Watrous [581].

The extension of trace inequalities from the case of matrices to that of operators is possible for the operators of the trace class. Recall that a bounded linear

operator A (acting on the separable Hilbert space H) is said to be of *trace class* (or *nuclear*) if

$$\|A\|_1 = \sum_n \langle |A|\, \mathbf{u}_n, \mathbf{u}_n \rangle < \infty \tag{7.12}$$

for some (and hence all) orthonormal bases $(\mathbf{u}_n)_n$ of H. In this case, the *trace* of A is given by the formula

$$\operatorname{trace} A = \sum_n \langle A\mathbf{u}_n, \mathbf{u}_n \rangle,$$

and is independent of the choice of the orthonormal basis. A result due to V. B. Lidskii asserts that

$$\operatorname{trace} A = \sum_n \lambda_n(A),$$

where $\lambda_1(A), \lambda_2(A), \lambda_3(A), \ldots$ are the eigenvalues of A, enumerated with their algebraic multiplicities. Therefore, a positive self-adjoint operator is of trace class if, and only if, $\operatorname{trace} A < \infty$.

Notice that every trace class operator is compact (but not every compact operator is of trace class).

The trace class operators $A : H \to H$ constitute a Banach space relative to the *nuclear norm* $\|\cdot\|_1$ defined by the formula (7.12). This space, denoted $N(H, H)$, is the dual of the Banach space $K(H, H)$, of all compact operators from H into itself, endowed with the operator norm. The dual of $N(H, H)$ is the space $L(H, H)$, of all continuous linear operators.

A nice survey of the trace class operators (including details of all results mentioned above) is offered by the book of B. Simon [541].

The world of matrix inequalities is full of many exotic examples, which seems to appear from nowhere. So is the case of *Golden–Thompson inequality,* which asserts that if $A, B \in \operatorname{Sym}(N, \mathbb{R})$, then

$$\operatorname{trace}\left(\mathrm{e}^{A+B}\right) \leq \operatorname{trace}\left(\mathrm{e}^{A}\mathrm{e}^{B}\right).$$

For the proof and generalizations see the aforementioned books of R. Bhatia and B. Simon. Interestingly, the det companion of this inequality is an identity,

$$\det\left(\mathrm{e}^{A+B}\right) = \det\left(\mathrm{e}^{A}\mathrm{e}^{B}\right).$$

The theory of global NPC spaces was initiated by A. D. Alexandrov and Yu. G. Reshetnyak and was strongly influenced by the work of M. Gromov. A very readable introductory paper is that by J. D. Lawson and Y. Lim [326]. Some more advanced material clarifying many geometric aspects is available in the books of R. Bhatia [64] and J. Jost [281]. A probabilistic view of the theory of global NPC spaces is offered by the excellent survey of K. T. Sturm [558].

During the last two decades much activity was done to extend the concept of mean of several variables to the framework of global NPC spaces. See the paper of J. D. Lawson and Y. Lim [327] and references therein.

An extension of the majorization inequality of Hardy–Littlewood–Pólya to the context of global NPC spaces can be found in a paper by C. P. Niculescu and I. Roventa [440].



Chapter 8

Duality and Convex Optimization

Convex optimization is one of the main applications of the theory of convexity and Legendre–Fenchel duality is a basic tool, making more flexible the approach of many concrete problems. The diet problem, the transportation problem and the optimal assignment problem are among the many problems that during the Second World War and immediately after led L. Kantorovich, T. C. Koopmans, F. L. Hitchcock, and G. B. Danzig to develop the mathematical theory of linear programming. Soon it was realized that most results extend to the framework of convex functions, which marked the birth of convex programming. Later on, W. Fenchel, R. T. Rockafellar, and J. J. Moreau laid the foundations of convex analysis.

Since even the briefest presentation of the aims and scope of convex optimization requires more than a single chapter, our discussion here will be limited to its basic aspects. The following nice books provide a considerably more systematic treatment of the subject and many more significant applications: V. Barbu and T. Precupanu [37], J. M. Borwein and A. S. Lewis [89], J. M. Borwein and J. Vanderwerff [90], F. H. Clarke [137], I. Ekeland and R. Temam [177], R. T. Rockafellar and R. J.-B. Wets [506].

8.1 Legendre–Fenchel Duality

An immediate consequence of the separation theorems presented in Section 3.3 is the fact that the closed convex sets can be described either as a union of points or as an intersection of closed half-spaces. This dual description has important consequences when applied to epigraphs.

Given a real Banach space E, the *conjugate* (or the *Legendre–Fenchel transform*) of a proper function $f\colon E \to \mathbb{R}\cup\{\infty\}$ is the function $f^*\colon E^* \to \mathbb{R}\cup\{\infty\}$ defined by the formula

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C. P. Niculescu and L.-E. Persson, *Convex Functions and Their Applications*,
CMS/CAIMS Books in Mathematics 14, https://doi.org/10.1007/978-3-031-71967-7_8

$$f^*(x^*) = \sup\{\langle \mathbf{x}, x^*\rangle - f(\mathbf{x}) : \mathbf{x} \in E\}. \tag{8.1}$$

Here $\langle \mathbf{x}, x^*\rangle = x^*(\mathbf{x})$ represents the duality map between E and E^*. It is clear that the formula (8.1) can be restated as

$$f^*(x^*) = \sup\{\langle \mathbf{x}, x^*\rangle - f(\mathbf{x}) : \mathbf{x} \in \operatorname{dom}(f)\},$$

a fact which shows that the function f^* is convex and lower semicontinuous since it is the pointwise supremum of the family of continuous affine functions $x^* \to \langle \mathbf{x}, x^*\rangle - f(\mathbf{x})$. Moreover, f^* takes values in $\mathbb{R} \cup \{\infty\}$. Notice that these properties of f^* do not assume that f is convex.

Geometrically, the Legendre–Fenchel conjugate solves the problem of finding for a given $x^* \in E^*$, the greatest affine function $y = \langle \mathbf{x}, x^*\rangle - \alpha$ majorized by f and having the gradient x^*. The solution is $\mathbf{y} = \langle \mathbf{x}, x^*\rangle - f^*(x^*)$.

The following result extends Young's inequality (1.5). See Example 8.1.6.

8.1.1 Proposition (The Fenchel–Young inequality) *If $f : E \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper convex function, then so is f^* and*

$$f(\mathbf{x}) + f^*(x^*) \geq x^*(\mathbf{x}) \quad \textit{for all } \mathbf{x} \in E, x^* \in E^*. \tag{8.2}$$

Equality holds if, and only if, $x^* \in \partial f(\mathbf{x})$ (and also if and only if $\mathbf{x} \in \partial f^*(x^*)$, see Theorem 8.1.2 below).

Proof. According to Theorem 4.3.2, f admits a continuous affine minorant h. Since this minorant is of the form $h(\mathbf{x}) = z^*(\mathbf{x}) + \alpha$ for suitable $z^* \in E^*$ and $\alpha \in \mathbb{R}$, we have $f^*(z^*) \leq \sup\{z^*(\mathbf{x}) - h(\mathbf{x}) : \mathbf{x} \in E\} \leq -\alpha$, so f^* is a proper function. The inequality (8.2) is now clear. As concerns the equality case, notice the following string of equivalent assertions:

$$\begin{aligned} x^*(\mathbf{x}) = f(\mathbf{x}) + f^*(x^*) &\Leftrightarrow x^*(\mathbf{x}) \geq f(\mathbf{x}) + f^*(x^*) \\ &\Leftrightarrow x^*(\mathbf{x}) \geq f(\mathbf{x}) + x^*(\mathbf{y}) - f(\mathbf{y}) \text{ for all } \mathbf{y} \in E \\ &\Leftrightarrow f(\mathbf{y}) \geq f(\mathbf{x}) + x^*(\mathbf{y} - \mathbf{x}) \text{ for all } \mathbf{y} \in E \\ &\Leftrightarrow x^* \in \partial f(\mathbf{x}). \end{aligned}$$

■

By definition, the *biconjugate* f^{**} of f is given by the formula

$$f^{**}(\mathbf{x}) := \sup\{x^*(\mathbf{x}) - f^*(x^*) : x^* \in E^*\}, \quad \mathbf{x} \in E.$$

Clearly,

$$f^{**}(\mathbf{x}) \leq f(\mathbf{x}).$$

In general,

$$f \leq g \text{ implies } f^* \geq g^*$$

and thus $f^{**} \leq g^{**}$. The circumstances under which $f = f^{**}$ make the objective of the following result.

8.1.2 Theorem (The Fenchel–Hörmander–Moreau theorem on biconjugation) *Suppose that $f: E \to \mathbb{R}\cup\{\infty\}$ is a proper function. Then the following assertions are equivalent:*

(a) *f is lower semicontinuous and convex;*
(b) *$f = g^*$ for some proper function g;*
(c) *$f = f^{**}$.*

Proof. It is obvious that (c) implies (b), which in turn implies (a). As concerns the implication (a) $\Rightarrow$ (c), notice that every continuous affine minorant h of f verifies $h = h^{**} \leq f^{**} \leq f$. According to Theorem 4.3.2, f is the pointwise supremum of the family of its continuous and affine minorants and thus $f = f^{**}$.

Alternatively, the proof of the implication (a) $\Rightarrow$ (c) is a simple consequence of the Fenchel–Young inequality when $\operatorname{dom} f = E$. Indeed, in this case $\partial f(\mathbf{x}) \neq \emptyset$ for every $\mathbf{x}$, so by choosing $x^* \in \partial f(\mathbf{x})$, we have

$$f(\mathbf{x}) = x^*(\mathbf{x}) - f^*(x^*) \leq \sup_{\mathbf{y}^*} [y^*(\mathbf{x}) - f^*(y^*)] = f^{**}(\mathbf{x}),$$

whence we infer that $f = f^{**}$. The general case can be fixed via an approximation argument. We will come back to this fact in Section 8.6, Exercise 6. ■

8.1.3 Corollary *If E is a real Banach space and $f: E \to \mathbb{R} \cup \{\infty\}$ is a proper convex function, then $f^{**} = \operatorname{cl} f$.*

Combining Proposition 8.1.1 and Theorem 8.1.2, one obtains the following inversion rule for subgradients: Under the assumptions of Corollary 8.1.3, we have

$$\partial f^* = (\partial f)^{-1} \text{ and } \partial f = (\partial f^*)^{-1};$$

see Section 4.3 for the inverse of a set-valued map.

Since any real Hilbert space H is self-dual, we infer from Theorem 8.1.2, that the Legendre–Fenchel conjugate $f \to f^*$ is an involution of the set $\Gamma(H)$ (of all lower semicontinuous proper convex functions $f: H \to \mathbb{R} \cup \{\infty\}$). It acts according to the formulas

$$f^*(\mathbf{v}) = \sup\{\langle \mathbf{x}, \mathbf{v}\rangle - f(\mathbf{x}) : \mathbf{x} \in H\}$$
$$f(\mathbf{x}) = f^{**}(\mathbf{x}) = \sup\{\langle \mathbf{x}, \mathbf{v}\rangle - f^*(\mathbf{v}) : \mathbf{v} \in H\}.$$

8.1.4 Example *For any nonempty convex subset C of a real Banach space, the conjugate of the indicator function ι_C is the support function h_C. According to Theorem 8.1.2, if C is also closed, then the conjugate of h_C is ι_C.*

In the particular case when C is a convex cone, we have $(\iota_C)^* = h_{C^\circ}$, so $(\iota_C)^{**} = \iota_{C^{\circ\circ}}$. Therefore $C^{\circ\circ} = \operatorname{cl} C$, a fact that illustrates the bipolar theorem.

8.1.5 Example *If E is a real Banach space, then the conjugate of the norm function $\mathrm{N}(\mathbf{x}) = \|\mathbf{x}\|$ is the indicator function of the closed unit ball of E^*, that is,*

$$\mathrm{N}^*(x^*) = \sup\{x^*(\mathbf{x}) - \|\mathbf{x}\| : \mathbf{x} \in E\} = \begin{cases} 0 & \text{if } \|x^*\| \leq 1 \\ \infty & \text{if } \|x^*\| > 1. \end{cases}$$

Indeed, if $\|x^*\| \le 1$, *then* $x^*(\mathbf{x}) - \|\mathbf{x}\| \le \|\mathbf{x}\| (\|x^*\| - 1) \le 0$ *and the supremum is attained at* $\mathbf{x} = 0$. *If* $\|x^*\| > 1$, *then there exists an element* $\mathbf{x} \in E$ *such that* $\|\mathbf{x}\| = 1$ *and* $\varepsilon = x^*(\mathbf{x}) - 1 > 0$. *Then* $x^*(n\mathbf{x}) - \|n\mathbf{x}\| = n\varepsilon$ *for every* $n \in \mathbb{N}$, *whence* $\mathrm{N}^*(x^*) = \infty$.

8.1.6 Example *Suppose that* $\varphi : \mathbb{R} \to \mathbb{R}$ *is an even function and* $\varphi \in \Gamma(\mathbb{R})$. *Then, for every real Banach space* E, *the functions*

$$F(\mathbf{x}) = \varphi(\|\mathbf{x}\|), \quad \mathbf{x} \in E,$$

and

$$G(x^*) = \varphi^*(\|x^*\|), \quad x^* \in E^*,$$

are conjugate to each other. Indeed, both functions are lower semicontinuous, convex and proper, so it suffices to show that $F^* = G$. *This is done as follows:*

$$\begin{aligned} F^*(x^*) &= \sup_{\mathbf{x}\in E} (\langle \mathbf{x}, x^*\rangle - F(\mathbf{x})) \\ &= \sup_{t>0} \sup_{\mathbf{x}\in E, \|\mathbf{x}\|=t} (\langle \mathbf{x}, x^*\rangle - \varphi(\|\mathbf{x}\|)) \\ &= \sup_{t>0} (t\|x^*\| - \varphi(t)) \\ &= \sup_{t\in\mathbb{R}} (t\|x^*\| - \varphi(t)) = \varphi^*(\|x^*\|) = G(x^*). \end{aligned}$$

If $p \in (1, \infty)$, *then, as was noticed in Exercise 5, Section 2.1, the real function* $\varphi(t) = |t|^p / p$ *is an example of even continuous convex function whose conjugate is* $\varphi^*(t) = |t|^q / q$, *where* $1/p + 1/q = 1$. *Therefore,*

$$F(\mathbf{x}) = \frac{1}{p}\|\mathbf{x}\|^p \quad \text{and} \quad G(x^*) = \frac{1}{q}\|x^*\|^q$$

are conjugate to each other in every real Banach space.

Combining this with Exercise 6 in Section 2.1, we obtain that $\frac{1}{2}\|x\|^2$ *is the only function on the Euclidean space* $\mathbb{R}^N$ *whose Legendre–Fenchel conjugate is equal to itself.*

In the case of continuously differentiable convex functions, one can easily compute f^* by using the differential of f. A simple example is as follows:

8.1.7 Example *Consider the strictly convex function* $f(\mathbf{x}) = \frac{1}{2}\langle A\mathbf{x}, \mathbf{x}\rangle$, *associated to a matrix* $A \in Sym^{++}(N, \mathbb{R})$. *For each* $\mathbf{p} \in \mathbb{R}^N$, *the function* $f_{\mathbf{p}} : \mathbf{x} \to \langle \mathbf{x}, \mathbf{p}\rangle - \frac{1}{2}\langle A\mathbf{x}, \mathbf{x}\rangle$ *is strictly concave and differentiable, so it attains its maximum at the point where* $\nabla f_{\mathbf{p}}(\mathbf{x}) = \mathbf{p} - A\mathbf{x} = 0$, *that is, at* $\mathbf{x} = A^{-1}\mathbf{p}$. *Therefore*

$$f^*(\mathbf{p}) = \frac{1}{2}\langle A^{-1}\mathbf{p}, \mathbf{p}\rangle \quad \text{for } \mathbf{p} \in \mathbb{R}^N.$$

For $A = \mathrm{I}$, *we infer that the function* $\varphi(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|^2$ *is self-conjugate (that is, equal to its Legendre–Fenchel conjugate). Moreover,* φ *is the only function*

defined on a finite-dimensional real Hilbert space H with this property. Indeed, from the Fenchel–Young inequality it follows that $\varphi(\mathbf{x}) \geq \frac{1}{2}\|\mathbf{x}\|^2$, whence, by conjugation we deduce the other inequality, $\varphi(\mathbf{x}) = \varphi^*(\mathbf{x}) \leq \frac{1}{2}\|\mathbf{x}\|^2$.

Example 8.1.7 illustrates the classical Legendre conjugation. This applies only to continuously differentiable strictly convex functions and it was W. Fenchel who extended the applicability of this transform using the formula (8.1).

8.1.8 Theorem (The classical Legendre conjugation) *Let $f : \mathbb{R}^N \to \mathbb{R}$ be a function of class C^2, whose Hessian matrix is strictly positive at every point $\mathbf{x} \in \mathbb{R}^N$. Then the conjugate of f is a function $f^* : \mathbb{R}^N \to \mathbb{R}$ of the same nature. More precisely, the following properties hold true:*

(a) *the gradient map $\mathbf{x} \to \nabla f(\mathbf{x})$ is a homeomorphism from $\mathbb{R}^N$ onto itself;*

(b) *$f^\star(\mathbf{p}) = \langle \mathbf{p}, (\nabla f)^{-1}\,\mathbf{p}\rangle - f((\nabla f)^{-1}\,\mathbf{p})$ for every $\mathbf{p} \in \mathbb{R}^N$;*

(c) *$f^\star$ is a C^1 function and $\nabla f^\star = (\nabla f)^{-1}$;*

(d) *the matrices $Hess_{\mathbf{x}} f$ and $Hess_{\mathbf{p}} f^*$ are inverses to each other when $\mathbf{p} = \nabla f(\mathbf{x})$ (or, equivalently, when $\mathbf{x} = \nabla f^\star(\mathbf{p})$).*

Therefore, the classical Legendre conjugation can be interpreted as an encoding of the function's epigraph in terms of its supporting hyperplanes.

Proof. (a) By our hypotheses, f is strongly convex and supercoercive; see 5.5.2, and Proposition 5.4.8. Hence, for every $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$ with $\mathbf{x} \neq \mathbf{y}$ and $\mathbf{y} \neq 0$, the function $g(t) = f(\mathbf{x} + t\mathbf{y})$ is strictly convex on $\mathbb{R}$, whence $g'(1) - g'(0) = \langle \nabla f(\mathbf{x} + \mathbf{y}) - \nabla f(\mathbf{y}), \mathbf{y}\rangle > 0$. This shows that ∇f is one-to-one. Let $\mathbf{p} \in \mathbb{R}^N$. Since $g(\mathbf{x}) = f(\mathbf{x}) - \langle \mathbf{x}, \mathbf{p}\rangle$ is convex, coercive and of class C^1, it attains a global minimum at a point $\mathbf{x}$ for which $\nabla g(\mathbf{x}) = \nabla f(\mathbf{x}) - \mathbf{p} = 0$. Hence the gradient map $\nabla f(\mathbf{x})$ is also onto. The subgradient inequality $f(\mathbf{0}) + \langle \nabla f(\mathbf{x}), \mathbf{x}\rangle \geq f(\mathbf{x})$ and the property of supercoercivity of f yield $\|\nabla f(\mathbf{x})\| \to \infty$ as $\|\mathbf{x}\| \to \infty$. Therefore the inverse image under ∇f of every compact set is compact too, a fact which assures the continuity of $(\nabla f)^{-1}$. For (b), notice that the reasoning above demonstrates that the point $\mathbf{x}$ at which the map $\langle \mathbf{x}, \mathbf{p}\rangle - f(\mathbf{x})$ attains its supremum verifies $\nabla f(\mathbf{x}) - \mathbf{p} = 0$, whence $f^\star(\mathbf{p}) = \langle \mathbf{p}, (\nabla f)^{-1}\,\mathbf{p}\rangle - f((\nabla f)^{-1}\,\mathbf{p})$. The assertions (c) and (d) can be deduced by differentiation, using the inverse mapping theorem. ■

The classical Legendre conjugation plays an important role in the theory of Hamilton–Jacobi equations. See Section 8.7.

An example of conjugacy involving the Hilbert space of symmetric matrices is offered by the following result.

8.1.9 Theorem (A. S. Lewis [330]) *Suppose that $f : \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ is a symmetric function and $\Lambda : \mathrm{Sym}(N, \mathbb{R}) \to \mathbb{R}$ is the eigenvalues map that associates to each symmetric matrix A the string $\lambda^\downarrow(A)$, of its eigenvalues with multiplicities, listed downward. Then*

$$(f \circ \Lambda)^* = f^* \circ \Lambda.$$

Proof. According to Corollary 6.4.11, for every matrix $Y \in \mathrm{Sym}(N,\mathbb{R})$ we have

$$\begin{aligned}(f\circ\Lambda)^*(Y) &= \sup\{\operatorname{trace} XY - f(\Lambda(X)) : X\in\mathrm{Sym}(N,\mathbb{R})\}\\ &\le \sup\{\langle\Lambda(X),\Lambda(Y)\rangle - f(\Lambda(X)) : X\in\mathrm{Sym}(N,\mathbb{R})\}\\ &\le \sup\left\{\langle\mathbf{x},\Lambda(Y)\rangle - f(\mathbf{x}) : \mathbf{x}\in\mathbb{R}^N\right\} = f^*(\Lambda(Y)).\end{aligned}$$

For the reverse inequality, notice that Y admits a factorization of the form $Y = U\Lambda(Y)U^*$, with $U\in\mathrm{O}(N)$, which implies that

$$\begin{aligned}f^*(\Lambda(Y)) &= \sup\left\{\langle\mathbf{x},\Lambda(Y)\rangle - f(\mathbf{x}) : \mathbf{x}\in\mathbb{R}^N\right\}\\ &= \sup\left\{\operatorname{trace}(\operatorname{Diag}(\mathbf{x})U^*YU) - f(\mathbf{x}) : \mathbf{x}\in\mathbb{R}^N\right\}\\ &= \sup\left\{\operatorname{trace}(U\operatorname{Diag}(\mathbf{x})U^*Y) - f(\Lambda(U\operatorname{Diag}(\mathbf{x})U^*)) : \mathbf{x}\in\mathbb{R}^N\right\}\\ &\le \sup\{\operatorname{trace}(XY) - f(\Lambda(X)) : X\in\mathrm{Sym}(N,\mathbb{R})\} = (f\circ\Lambda)^*(Y).\end{aligned}$$

■

8.1.10 Corollary *Suppose that $f:\mathbb{R}^N\to\mathbb{R}\cup\{\infty\}$ is a lower semicontinuous proper convex function. If f is also symmetric, then the function $f\circ\Lambda$ is lower semicontinuous and convex.*

Proof. Indeed, $f = f^{**}$ by Theorem 8.1.2 and since f^* is symmetric we have $f\circ\Lambda = (f^*)^*\circ\Lambda = (f^*\circ\Lambda)^*$ by Theorem 8.1.9. Therefore $f\circ\Lambda$ is a conjugate function and this fact implies its convexity and also its lower semicontinuity. ■

8.1.11 Corollary *Suppose that $f:\mathbb{R}^N\to\mathbb{R}\cup\{\infty\}$ is a symmetric convex function and $X,Y\in\mathrm{Sym}(N,\mathbb{R})$. Then $Y\in\partial(f\circ\Lambda)(X)$ if, and only if, $\Lambda(Y)\in\partial f(\Lambda(X))$ and there exists an orthogonal matrix V such that $V^*XV = \operatorname{Diag}(\Lambda(X))$ and $V^*YV = \operatorname{Diag}(\Lambda(Y))$.*

Proof. According to Proposition 8.1.1 and Theorem 8.1.9, $Y\in\partial(f\circ\Lambda)(X)$ if, and only if,

$$\operatorname{trace} XY = (f\circ\Lambda)(X) + (f\circ\Lambda)^*(Y) = f(\Lambda(X)) + f^*(\Lambda(Y)).$$

On the other hand, $f(\Lambda(X)) + f^*(\Lambda(Y)) \ge \langle\Lambda(X),\Lambda(Y)\rangle$ and $\langle\Lambda(X),\Lambda(Y)\rangle \ge \operatorname{trace} XY$, by Corollary 6.4.11. This shows that $Y\in\partial(f\circ\Lambda)(X)$ if, and only if, $\Lambda(Y)\in\partial f(\Lambda(X))$. The second part of the statement of Corollary 8.1.11 follows from Corollary 6.4.11. ■

Since the conjugate of the lower semicontinuous proper convex function

$$f:\mathbb{R}^N\to\mathbb{R}\cup\{\infty\},\quad f(\mathbf{x}) = \begin{cases} -\sum_{k=1}^N \log x_k & \text{if } \mathbf{x}\in\mathbb{R}^N_{++}\\ \infty & \text{otherwise,}\end{cases}$$

is the function

$$f^*:\mathbb{R}^N\to\mathbb{R}\cup\{\infty\},\quad f^*(\mathbf{x}) = \begin{cases} f(-\mathbf{x}) - N & \text{if } \mathbf{x}\in -\mathbb{R}^N_{++}\\ \infty & \text{otherwise,}\end{cases}$$

it follows from Theorem 8.1.9 that the conjugate of its spectral function

$$F : \mathrm{Sym}(N,\mathbb{R}) \to \mathbb{R}\cup\{\infty\}, \quad F(A) = \begin{cases} -\log\det A & \text{if } A \in \mathrm{Sym}^{++}(N,\mathbb{R}) \\ \infty & \text{otherwise,} \end{cases}$$

verifies $F^*(A) = -\log\det(-A) - N$ for all $A \in -\mathrm{Sym}^{++}(N,\mathbb{R})$.

Exercises

1. Suppose that H is a Hilbert space and $f\colon H \to \mathbb{R}\cup\{\infty\}$ is a lower semicontinuous proper convex function. Prove that:
   (a) the Legendre–Fenchel conjugate of $f(\mathbf{x}) + \alpha$ is $f^*(\mathbf{p}) - \alpha$;
   (b) if $\lambda > 0$, then the Legendre–Fenchel conjugate of $\lambda f(\mathbf{x})$ is $\lambda f^*(\mathbf{p}/\lambda)$;
   (c) if $\lambda > 0$, then the Legendre–Fenchel conjugate of $\lambda f(\mathbf{x}/\lambda)$ is $\lambda f^*(\mathbf{p})$;
   (d) the Legendre–Fenchel conjugate of $f(\mathbf{x}) + \langle \mathbf{x}, \mathbf{q}\rangle$ is $f^*(\mathbf{p} - \mathbf{q})$.

2. (a) Suppose that $h(\mathbf{x}) = \langle \mathbf{x}, \mathbf{a}\rangle + \mathbf{b}$ is an affine function on the Hilbert space H. Prove that $\mathrm{dom}h^* = \{\mathbf{a}\}$, $h^*(\mathbf{a}) = -\mathbf{b}$ and $h = h^{**}$.
   (b) Consider the linear-quadratic function $f(\mathbf{x}) = \frac{1}{2}\langle A\mathbf{x}, \mathbf{x}\rangle + \langle \mathbf{x}, \mathbf{a}\rangle + \mathbf{b}$, where $A : H \to H$ is a symmetric, positive, and invertible operator and $\mathbf{a}, \mathbf{b} \in H$. Prove that $f^*(\mathbf{x}) = \langle A^{-1}(\mathbf{v} - \mathbf{a}), \mathbf{v} - \mathbf{a}\rangle$ for all $\mathbf{v} \in H$ (and $f = f^{**}$).

3. Let $f_i : \mathbb{R}^{n_i} \to \mathbb{R}\cup\{\infty\}$ $(i = 1, ..., n)$ be lower semicontinuous proper convex functions. Prove that the conjugate of their direct sum $\oplus_{i=1}^n f_i$ is the direct sum $\oplus_{i=1}^n f_i^*$ of their conjugates.

4. Let E be a real Banach space. Prove that for every family of functions $f_\alpha : E \to \mathbb{R}\cup\{\infty\}$ $(\alpha \in A)$ we have

$$(\inf_{\alpha\in A} f_\alpha)^* = \sup_{\alpha\in A} f_\alpha^* \quad \textit{and} \quad (\sup_{\alpha\in A} f_\alpha)^* \le \inf_{\alpha\in A} f_\alpha^*.$$

5. (Support function) The notion of a support function can be attached to any nonempty convex set C in $\mathbb{R}^N$, by defining it as the conjugate of the indicator function of C, that is, $h_C = \iota_C^*$. Prove that the support function of $C = \{(x,y) \in \mathbb{R}^2 : x + y^2/2 \le 0\}$ is $h_C(x,y) = y^2/2x$ if $x > 0$, $h_C(0,0) = 0$ and $h_C(x,y) = \infty$ otherwise. Infer that h_C is a lower semicontinuous proper convex function.

6. Calculate the support function for the convex set

$$C = \{A \in \mathrm{Sym}^{++}(N,\mathbb{R}) : \mathrm{trace}(A) = 1\}.$$

7. Suppose that $g, h : \mathbb{R}^N \to \mathbb{R}\cup\{\infty\}$ are two proper functions such that h is lower semicontinuous and convex. Prove that the conjugate of the function

$$f(\mathbf{x}) = \begin{cases} g(\mathbf{x}) - h(\mathbf{x}) & \text{if } \mathbf{x} \in \mathrm{dom}g \\ \infty & \text{if } \mathbf{x} \notin \mathrm{dom}g \end{cases}$$

is given by the formula

$$f^*(\mathbf{y}) = \sup_{\mathbf{x}\in \mathrm{dom} g} \left(\langle \mathbf{x}, \mathbf{y}\rangle - g(\mathbf{x}) + h(\mathbf{x})\right) = \sup_{\mathbf{z}\in \mathrm{dom} h^*} \left(g^*(\mathbf{y}+\mathbf{z}) - h^*(\mathbf{z})\right).$$

[*Hint:* Notice that $h = h^{**}$.]

8. (The Tolland–Singer formula) Suppose that $g, h : \mathbb{R}^N \to \mathbb{R}\cup\{\infty\}$ are two lower semicontinuous proper convex functions. Infer from the preceding exercise that

$$\inf_{\mathbf{x}\in \mathrm{dom} g} \left(g(\mathbf{x}) - h(\mathbf{x})\right) = \sup_{\mathbf{z}\in \mathrm{dom} h^*} \left(g^*(\mathbf{z}) - h^*(\mathbf{z})\right).$$

9. (F. Barthe [40]) Let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be vectors in $\mathbb{R}^N$ ($m \geq N$), and $c_1, \ldots, c_m$ be positive numbers such that $\sum_{k=1}^m c_k = N$. For $\lambda = (\lambda_1, \ldots, \lambda_m)$ in $(0,\infty)^m$, consider the following two norms on $\mathbb{R}^N$:

$$M_\lambda(\mathbf{x}) = \inf\left\{ \left(\sum_{k=1}^m c_k \theta_k^2/\lambda_k \right)^{1/2} : \mathbf{x} = \sum_{k=1}^m c_k \theta_k v_k,\ \theta_k \in \mathbb{R} \right\}$$

and

$$N_\lambda(\mathbf{x}) = \left(\sum_{k=1}^m c_k \lambda_k \langle \mathbf{x}, \mathbf{v}_k\rangle^2 \right)^{1/2}$$

Prove that $\mathcal{E}_\lambda = \{\mathbf{x} \in \mathbb{R}^N : M_\lambda(\mathbf{x}) \leq 1\}$ is the polar of the ellipsoid $\mathcal{F}_\lambda = \left\{\mathbf{x} \in \mathbb{R}^N : N_\lambda(\mathbf{x}) \leq 1\right\}$ and

$$\mathrm{Vol}_N(\mathcal{E}_\lambda)\mathrm{Vol}_N(\mathcal{F}_\lambda) = \mathrm{Vol}_N(B)^2,$$

where B denotes the closed unit ball of the Euclidean space $\mathbb{R}^N$.

[*Hint*: Notice that the support function of the polar of $\mathcal{F}_\lambda$ is

$$h_{\mathcal{F}_\lambda^\circ}(\mathbf{x}) = \left(\sum_{k=1}^m c_k \lambda_k \langle \mathbf{x}, \mathbf{v}_k\rangle^2 \right)^{1/2},$$

which equals the support function of $\mathcal{E}_\lambda$,

$$h_{\mathcal{E}_\lambda}(\mathbf{x}) = \sup\left\{ \sum_{k=1}^m c_k \theta_k \langle \mathbf{x}, \mathbf{v}_k\rangle : \sum_{k=1}^m c_k \theta_k^2/\lambda_k \leq 1, \theta_k \in \mathbb{R} \right\}.]$$

10. (D. Kramkov and W. Schachermayer) Suppose that $U : (0,\infty) \to \mathbb{R}$ is an increasing, strictly concave and continuously differentiable function that satisfies the Inada conditions, that is,

$$U'(0) = \lim_{x\to 0} U'(x) = \infty \text{ and } U'(\infty) = \lim_{x\to\infty} U'(x) = 0.$$

Prove that

$$V(y) = \sup_{x>0} \{U(x) - xy\}, \quad y > 0$$

is a continuously differentiable, decreasing, and strictly convex function satisfying $V'(0) = -\infty$, $V'(\infty) = 0$, $V(0) = U(\infty)$, and $V(\infty) = U(0)$. Moreover,

$$U(x) = \inf_{y>0} (V(y) - xy), \quad x > 0.$$

8.2 The Correspondence of Properties under Duality

Conjugacy offers a convenient way to get information about the behavior of a function through its conjugate. For example, it allows us to recognize properties like coercivity and convexity, or to compute the optimal value of a lower semicontinuous proper convex function.

Throughout this section, H denotes a real Hilbert space.

8.2.1 Theorem (J. J. Moreau and R. T. Rockafellar) *A lower semicontinuous proper convex function $f\colon H \to \mathbb{R} \cup \{\infty\}$ has bounded level sets if and only if its conjugate is continuous at the origin.*

Proof. By Lemma 5.4.3, the function f has bounded level sets if and only if it verifies an inequality of the form $f(\mathbf{x}) \geq \alpha \|\mathbf{x}\| + b$ with $a, b \in \mathbb{R}$ and $\alpha > 0$. Taking conjugates, this is equivalent to f^* being bounded above on a neighborhood of 0, which in turn is equivalent to f^* being continuous at 0. See Proposition 4.1.11. ■

8.2.2 Corollary (Hörmander–Moreau–Rockafellar) *Under the hypotheses of Theorem 8.2.1, if $\mathbf{y} \in H$, then $f - \langle \cdot, \mathbf{y} \rangle$ is coercive if and only if f^* is continuous at $\mathbf{y}$.*

The duality of supercoercive convex functions makes the objective of Exercises 1–3.

8.2.3 Theorem (V. Soloviev) *Suppose that $g\colon H \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper function. If the conjugate of g is differentiable, then g is necessarily convex.*

The details can be found in the paper of J.-B. Hiriart-Urruty [254].

There is a dual correspondence between strict convexity of a function and the smoothness of its conjugate.

8.2.4 Theorem (a) *If $f : H \to \mathbb{R}$ is a strictly convex and supercoercive function, then f^* is continuously differentiable on H.*

(b) *If $g : H \to \mathbb{R}$ is a supercoercive, differentiable, and convex function, then g^* is strictly convex on H.*

Proof. (a) Since f is supercoercive, the effective domain of f^* is the whole space H. See Exercise 1 at the end of this section. Then, with the help of Exercise 7, Section 4.3, we infer that $\partial f^*(\mathbf{x})$ is a singleton for every $\mathbf{x}$. According to Corollary 5.1.10, this implies that f is differentiable. The continuity of the differential follows from Exercise 3, Section 5.2.

For (b), if g^* fails to be strictly convex, then (by Exercise 7, Section 4.3) there exist distinct points $\mathbf{x}$ and $\mathbf{y}$ and $\mathbf{z} \in \partial g^*(\mathbf{x}) \cap \partial g^*(\mathbf{y})$. This implies that $\mathbf{x}, \mathbf{y} \in \partial g(\mathbf{z})$, a contradiction with differentiability of g. ■

A stronger version of Theorem 8.2.4 will be presented in the Comments section at the end of this chapter.

Let f be a lower semicontinuous proper convex function defined on the real Hilbert space H and let $\mathbf{x}_0 \in \mathrm{dom} f$ be arbitrarily fixed. The *recession function* of f is defined by the formula

$$(\mathrm{rec} f)(\mathbf{y}) = \sup_{t>0} \frac{f(\mathbf{x}_0 + t\mathbf{y}) - f(\mathbf{x}_0)}{t} = \lim_{t\to\infty} \frac{f(\mathbf{x}_0 + t\mathbf{y}) - f(\mathbf{x}_0)}{t},$$

the second equality being motivated by the three chords inequality. See Remark 1.4.1. The formula defining the recession function is independent of $\mathbf{x}_0$. See Exercise 7 (at the end of this section), based on an equivalent definition of recession functions, used by H. H. Bauschke and P. L. Combettes in [43], Section 9.4.

The recession function is proper, lower semicontinuous, and sublinear. If f is bounded below, then $\mathrm{rec} f \geq 0$.

8.2.5 Proposition (P. J. Laurent [325]) *The recession function of f is the support function of* $\mathrm{dom}\, f^*$, *while* $\mathrm{rec}\,(f^*)$ *is the support function of* $\mathrm{dom}\, f$.

Proof. According to Theorem 8.1.2, it suffices to prove the second assertion. Let $\mathbf{u} \in \mathrm{dom} f^*$ be arbitrarily fixed. Then, for every $\mathbf{v} \in H$ and $t > 0$, we have

$$\begin{aligned} f^*(\mathbf{u} + t\mathbf{v}) &= \sup\{\langle \mathbf{y}, \mathbf{u} + t\mathbf{v}\rangle - f(\mathbf{y}) : \mathbf{y} \in \mathrm{dom} f\} \\ &\leq \sup\{\langle \mathbf{y}, \mathbf{u}\rangle - f(\mathbf{y}) : \mathbf{y} \in \mathrm{dom} f\} + t \sup_{\mathbf{y}\in\mathrm{dom} f} \langle \mathbf{y}, \mathbf{v}\rangle \\ &\leq f^*(\mathbf{u}) + t h_{\mathrm{dom} f}(\mathbf{v}), \end{aligned}$$

whence $\mathrm{rec}\,(f^*) \leq h_{\mathrm{dom} f}$. On the other hand, for $\mathbf{v} \in H$ and $\alpha \in \mathbb{R}$ with $(\mathrm{rec} f^*)(\mathbf{v}) < \alpha$ one has $f^*(\mathbf{u} + t\mathbf{v}) \leq f^*(\mathbf{u}) + t\alpha$ for all $t > 0$, hence for any $\mathbf{x} \in H$,

$$\begin{aligned} f(\mathbf{x}) &\geq \langle \mathbf{x}, \mathbf{u} + t\mathbf{v}\rangle - f^*(\mathbf{u} + t\mathbf{v}) \\ &\geq \langle \mathbf{x}, \mathbf{u}\rangle - f^*(\mathbf{u}) + t\,(\langle \mathbf{x}, \mathbf{v}\rangle - \alpha)\,. \end{aligned}$$

This yields $\langle \mathbf{x}, \mathbf{v}\rangle \leq \alpha$ for all $x \in \mathrm{dom} f$, whence $\sigma_{\mathrm{dom} f} \leq \mathrm{rec}\,(f^*)$. Therefore $\mathrm{rec}\,(f^*) = h_{\mathrm{dom} f}$. ■

We next discuss the duality between strong convexity and strong smoothness.

8.2.6 Theorem (a) *If $f : \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous α-strongly convex function, then its Legendre–Fenchel conjugate f^* is an $(1/\alpha)$-strongly smooth and convex function whose domain is $\mathbb{R}^N$.*

(b) *If $f : \mathbb{R}^N \to \mathbb{R}$ is convex and L-strongly smooth, then f^* is $1/L$-strongly convex function on any convex subset C of $\mathrm{dom} f^*$.*

A generalization of this result to the framework of ω-convex functions is given in Exercise 5.

Proof. (a) According to Proposition 5.4.8, for each $\mathbf{y} \in \mathbb{R}^N$, the maximizer of $\sup_{\mathbf{u}\in \mathrm{dom} f} (\langle \mathbf{y}, \mathbf{u}\rangle - f(\mathbf{u}))$ exists and is unique. Therefore, the domain of f^* is $\mathbb{R}^N$.

By Proposition 8.1.1, a point $\mathbf{x}$ maximizes $\langle \mathbf{y}, \mathbf{u}\rangle - f(\mathbf{u})$ if and only if $\mathbf{y} \in \partial f(\mathbf{x})$ which is equivalent to $\mathbf{x} \in \partial f(\mathbf{y})$. Since $\mathbf{x}$ is a unique maximizer, it follows that $\partial f(\mathbf{y}^*)$ contains exactly one element, so by Corollary 5.1.10, the function f^* is differentiable and

$$\nabla f^*(\mathbf{y}) = \mathbf{x} = \operatorname*{arg\,max}_{\mathbf{u}\in \mathrm{dom} f} (\langle \mathbf{y}, \mathbf{u}\rangle - f(\mathbf{u})) .$$

Since f is α-strongly convex, from Theorem 5.5.2 we infer that

$$\langle \nabla f(\mathbf{x}_1) - \nabla f(\mathbf{x}_2), \mathbf{x}_1 - \mathbf{x}_2\rangle \geq \alpha \|\mathbf{x}_1 - \mathbf{x}_2\|^2 \quad \text{for all } \mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^N.$$

Put $\mathbf{y}_1 = \nabla f(\mathbf{x}_1)$ and $\mathbf{y}_2 = \nabla f(\mathbf{x}_2)$. Since $\mathbf{y}_i \in \partial f(\mathbf{x}_i)$ $(i = 1, 2)$ is equivalent to $\mathbf{x}_i \in \partial f^*(\mathbf{y}_i)$ $(i = 1, 2)$, the last inequality implies that

$$\langle \mathbf{y}_1 - \mathbf{y}_2, \nabla f(\mathbf{y}_1) - \nabla f^*(\mathbf{y}_2)\rangle \geq \alpha \|\nabla f(\mathbf{y}_1) - \nabla f^*(\mathbf{y}_2)\|^2 \quad \text{for all } \mathbf{y}_1, \mathbf{y}_2 \in \mathbb{R}^N.$$

By the Cauchy–Bunyakovsky–Schwarz inequality,

$$\|\mathbf{y}_1 - \mathbf{y}_2\| \, \|\nabla f(\mathbf{y}_1) - \nabla f^*(\mathbf{y}_2)\| \geq \langle \mathbf{y}_1 - \mathbf{y}_2, \nabla f(\mathbf{y}_1) - \nabla f^*(\mathbf{y}_2)\rangle,$$

which allows us to conclude that

$$\|\nabla f(\mathbf{y}_1) - \nabla f^*(\mathbf{y}_2)\| \leq \frac{1}{\alpha} \|\mathbf{y}_1 - \mathbf{y}_2\| \quad \text{for all } \mathbf{y}_1, \mathbf{y}_2 \in \mathbb{R}^N.$$

(b) Denote $g(\mathbf{y}) = f(\mathbf{x} + \mathbf{y}) - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{y}\rangle$. Then,

$$\begin{aligned}
g^*(\mathbf{u}) &= \sup_{\mathbf{v}} \{\langle \mathbf{v}, \mathbf{u}\rangle - g(\mathbf{v})\} \\
&= \sup_{\mathbf{v}} \{\langle \mathbf{v}, \mathbf{u}+\nabla f(\mathbf{x})\rangle - f(\mathbf{x} + \mathbf{v}) + f(\mathbf{x})\} \\
&= \sup_{\mathbf{w}} \{\langle \mathbf{w} - \mathbf{x}, \mathbf{u}+\nabla f(\mathbf{x})\rangle - f(\mathbf{w}) + f(\mathbf{x})\} \\
&= \sup_{\mathbf{w}} \{\langle \mathbf{w}, \mathbf{u}+\nabla f(\mathbf{x})\rangle - f(\mathbf{w})\} + f(\mathbf{x}) - \langle \mathbf{x}, \mathbf{u}+\nabla f(\mathbf{x})\rangle \\
&= f^* (\mathbf{u}+\nabla f(\mathbf{x})) + f(\mathbf{x}) - \langle \mathbf{x}, \nabla f(\mathbf{x})\rangle - \langle \mathbf{x}, \mathbf{u}\rangle \\
&= f^* (\mathbf{u}+\nabla f(\mathbf{x})) - f^* (\nabla f(\mathbf{x})) - \langle \mathbf{x}, \mathbf{u}\rangle,
\end{aligned}$$

the last equality being motivated by the fact that

$$f(\mathbf{x}) + f^* (\nabla f(\mathbf{x})) = \langle \mathbf{x}, \nabla f(\mathbf{x})\rangle.$$

See Proposition 8.1.1. By the smoothness assumption, $g(\mathbf{y}) \leq \frac{L}{2}\|\mathbf{y}\|^2$, which implies that $g^*(\mathbf{u}) \geq \frac{1}{2L}\|\mathbf{u}\|^2$ because the half-squared norm is self-conjugated. See Example 8.1.6. Therefore

$$f^*(\mathbf{u}+\nabla f(\mathbf{x})) - f^*(\nabla f(\mathbf{x})) - \langle \mathbf{x}, \mathbf{u}\rangle \geq \frac{1}{2L}\|\mathbf{u}\|^2 \tag{8.3}$$

for all $\mathbf{u}, \mathbf{x} \in \mathbb{R}^N$.

Our next goal is to prove that for every point $\mathbf{u}'$ in the relative interior of the domain of f^* if $\mathbf{x} \in \partial f^*(\mathbf{u}')$, then $\mathbf{u}' = \nabla f(\mathbf{x})$. Indeed, according to Proposition 8.1.1,

$$\langle \mathbf{x}, \mathbf{u}'\rangle = f(\mathbf{x}) + f^*(\mathbf{u}') \text{ and } \langle \mathbf{x}, \nabla f(\mathbf{x})\rangle = f(\mathbf{x}) + f^*(\nabla f(\mathbf{x})),$$

whence $\langle \mathbf{x}, \mathbf{u}' - \nabla f(\mathbf{x})\rangle = f^*(\mathbf{u}') - f^*(\nabla f(\mathbf{x}))$. By taking into account the inequality (8.3), we have

$$\begin{aligned} 0 &= f^*(\mathbf{u}') - f^*(\nabla f(\mathbf{x})) - \langle \mathbf{x}, \mathbf{u}' - \nabla f(\mathbf{x})\rangle \\ &= f^*(\mathbf{u}' - \nabla f(\mathbf{x}) + \nabla f(\mathbf{x})) - f^*(\nabla f(\mathbf{x})) - \langle \mathbf{x}, \mathbf{u}' - \nabla f(\mathbf{x})\rangle \geq \frac{1}{2L}\|\nabla f(\mathbf{x}) - \mathbf{u}'\|^2, \end{aligned}$$

so that $\mathbf{u}' = \nabla f(\mathbf{x})$.

Now, let $\mathbf{v}_1, \mathbf{v}_2 \in C$ and $\lambda \in (0,1)$. Denote $\mathbf{v} = (1-\lambda)\mathbf{v}_1 + \lambda \mathbf{v}_2$ and choose $\mathbf{x} \in \partial f^*(\mathbf{v})$. Then $\mathbf{v} = \nabla f^*(\mathbf{x})$ by the previous remark, so that, by applying (8.3) two times (for $\mathbf{u}$ replaced by $\mathbf{v}_1 - \mathbf{v}$ and $\mathbf{u}$ replaced by $\mathbf{v}_2 - \mathbf{v}$), we infer that

$$\begin{aligned} f^*(\mathbf{v}_1) - f^*(\mathbf{v}) - \langle \mathbf{x}, \mathbf{v}_1 - \mathbf{v}\rangle &\geq \frac{1}{2L}\|\mathbf{v}_1 - \mathbf{v}\|^2, \\ f^*(\mathbf{v}_2) - f^*(\mathbf{v}) - \langle \mathbf{x}, \mathbf{v}_2 - \mathbf{v}\rangle &\geq \frac{1}{2L}\|\mathbf{v}_2 - \mathbf{v}\|^2, \end{aligned}$$

whence $(1-\lambda)f^*(\mathbf{v}_1) + \lambda f^*(\mathbf{v}_2) - f^*(\mathbf{v}) \geq \frac{\lambda(1-\lambda)}{2L}\|\mathbf{v}_1 - \mathbf{v}_2\|^2$ and the proof is complete. ■

If we extend the von Neumann entropy to $\mathrm{Sym}(n, \mathbb{R})$ via the formula

$$S(A) = \begin{cases} -\operatorname{trace}(A \log A) & \text{if } A \in \mathbf{S}_n \\ -\infty & \text{otherwise,} \end{cases}$$

then one can show that the functions $\log \operatorname{trace}(\mathrm{e}^A)$ and $-S(A)$ are conjugate to each other.

8.2.7 Theorem *For all* $A \in \mathrm{Sym}(n, \mathbb{R})$,

$$-S(A) = \sup\left\{\operatorname{trace}(AH) - \log \operatorname{trace}(\mathrm{e}^H) : H \in \mathrm{Sym}(n, \mathbb{R})\right\}$$

and

$$\log \operatorname{trace}(\mathrm{e}^H) = \sup\left\{\operatorname{trace}(AH) + S(\mathrm{A}) : A \in \mathrm{Sym}(n, \mathbb{R})\right\}.$$

A very short argument is available in the paper of E. A. Carlen [117], Theorem 2.13.

8.2.8 Remark *By combining the Theorems* 8.2.6 *and* 8.2.7 *with the fact that the von Neumann Entropy is* $1/2$*-strongly convex, we infer that the noncommutative analogue of the log-sum-exp function is* 2*-strongly smooth.*

Exercises

1. Suppose that f is a lower semicontinuous proper convex function defined on $\mathbb{R}^N$. Prove that $\mathrm{dom} f^* = \mathbb{R}^N$ if f is supercoercive.

   [*Hint*: According to Theorem 4.3.2, f admits an affine minorant.]

2. Prove that the conjugate of a supercoercive convex function $f : \mathbb{R}^N \to \mathbb{R}$ is also supercoercive.

   [*Hint*: According to Exercise 1, f^* is real-valued and defined on $\mathbb{R}^N$. Then, infer from Fenchel–Young inequality that $f^*(\mathbf{y})/\|\mathbf{y}\| \geq \alpha\|\mathbf{y}\| - f(\alpha\mathbf{y}/\|\mathbf{y}\|)$, whenever $\alpha > 0$ and $\mathbf{y} \in \mathbb{R}^N \setminus \{\mathbf{0}\}$.]

3. Suppose that $f : \mathbb{R}^N \to \mathbb{R}$ is a lower semicontinuous proper convex function. Prove that f (respectively f^*) is supercoercive if, and only if, f^* (respectively f) is bounded on bounded sets.

   Remark. Combining Corollary 8.2.2 with Exercise 3, one can easily prove that a function $h : \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ is supercoercive if, and only if, the function $h(\cdot) - \langle \cdot, x^* \rangle$ has bounded level sets whenever $x^* \in \mathbb{R}^N$.

4. Suppose $f : \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper convex function. Prove that f is Lipschitz with Lipschitz constant $\mathrm{Lip}(f) = K > 0$ if, and only if, $\mathrm{dom} f^*$ is included in the closed ball $\bar{B}_K(\mathbf{0})$.

5. (C. Zălinescu [597]) Let E be a reflexive Banach space, $f : E \to \mathbb{R}$ a continuous convex function and $\omega : \mathbb{R}_+ \to \mathbb{R}_+$ a function such that $\omega(t) > 0$ for $t > 0$. Then the following two assertions are equivalent:

   (a) $f(y) \geq f(x) + x^*(y - x) + \omega(\|x - y\|)$ for all $x, y \in E$ and $x^* \in \partial f(x)$;

   (b) $f^*(y^*) \leq f^*(x^*) + (y^* - x^*)(x) + \omega^*(\|x^* - y^*\|)$ for all $x \in E$ and $x^*, y^* \in E^*$ with $x^* \in \partial f(x)$.

6. Let C be a nonempty closed convex set in a Hilbert space H. The *recession cone* of C is the cone $\mathrm{rec} C$, consisting of all vectors $\mathbf{x} \in H$ such that $\mathbf{x}_0 + \lambda\mathbf{x} \in C$ for all $\mathbf{x}_0 \in C$ and $\lambda > 0$. Prove that:

   (a) $\mathrm{rec} C = \{\mathbf{x} \in H : \mathbf{x} + C \subset C\}$;

   (b) $\mathrm{rec} C = \{\mathbf{x} : \mathbf{x}_0 + \lambda\mathbf{x} \in C \text{ for all } \lambda > 0\} = \bigcap_{\lambda > 0} \lambda\,(C - \mathbf{x}_0)$, where $\mathbf{x}_0$ is arbitrarily fixed in C.

7. The recession function attached to a lower semicontinuous, convex, and proper function $f : H \to \mathbb{R} \cup \{\infty\}$ can be defined as the function $\mathrm{rec} f$ whose epigraph is the recession cone of $\mathrm{epi} f$. Prove that

$$(\mathrm{rec} f)(\mathbf{y}) = \sup_{x \in \mathrm{dom} f} [f(\mathbf{x} + t\mathbf{y}) - f(\mathbf{x})]$$

and then recover the formula

$$(\mathrm{rec}f)(\mathbf{y}) = \sup_{t>0} \frac{f(\mathbf{x}_0 + t\mathbf{y}) - f(\mathbf{x}_0)}{t} = \lim_{t\to\infty} \frac{f(\mathbf{x}_0 + t\mathbf{y}) - f(\mathbf{x}_0)}{t},$$

where $\mathbf{x}_0 \in \mathrm{dom} f$ is arbitrarily fixed.

[*Hint*: A point $(\mathbf{y}, \lambda) \in H\times\ \mathbb{R}$ belongs to $\mathrm{epi}(\mathrm{rec}f)$ if, and only if, it belongs to $\mathrm{rec}(\mathrm{epi}f)$, that is, for $(\mathbf{x}_0, f(\mathbf{x}_0)) \in \mathrm{epi}f$ arbitrarily fixed, we have $(\mathbf{x}_0, f(\mathbf{x}_0))+t(\mathbf{y}, \lambda) \in \mathrm{epi}\ f$ for all $t > 0$. Consider the function $\varphi(t) = (f(\mathbf{x}_0+t\mathbf{y})-f(\mathbf{x}_0))/t$. Since f is convex, the three chords inequality implies that the function φ is increasing and $\alpha = \sup_{t>0} \varphi(t) = \lim_{t\to\infty} \varphi(t) \leq \infty$. Thus the condition $(\mathbf{y}, \lambda) \in H \times \mathbb{R}$ is equivalent to $\lambda \in [\alpha, \infty)$ and $\mathbf{y} \in \mathrm{dom}(\mathrm{rec}f)$ if, and only if, $\alpha \in \mathbb{R}$ (in which case $(\mathrm{rec}f)(\mathbf{y}) = \alpha$).]

8. (A proof of the Bunt–Motzkin Theorem via Asplund's function) Given a nonempty closed subset S of $\mathbb{R}^n$ one can associate to it the function

$$\varphi_S(\mathbf{x}) = \frac{1}{2}\,(\|\mathbf{x}\|^2 - d_S^2(\mathbf{x})),$$

where $d_S(\mathbf{x}) = \inf\{\|\mathbf{x} - \mathbf{s}\| : \mathbf{s} \in S\}$ represents the distance from $\mathbf{x}$ to S. Prove that:

(a) $\varphi_S(\mathbf{x}) = \sup\{\langle \mathbf{x}, \mathbf{s}\rangle - \frac{1}{2}\|\mathbf{s}\|^2 : \mathbf{s} \in S\}$, which implies that φ_S is a convex function.

(b) φ_S is the conjugate of the function $f_S(\mathbf{x}) = \|\mathbf{x}\|^2/2$ if $\mathbf{x} \in S$ and $f_S(\mathbf{x}) = \infty$ otherwise.

Infer from Theorem 8.2.3 the Bunt–Motzkin Theorem 3.2.2.

9. (A second proof of the Bunt–Motzkin Theorem) Let C be a nonempty closed subset of $\mathbb{R}^N$ and let

$$d_C : \mathbb{R}^N \to \mathbb{R}, \quad d_C(\mathbf{x}) = \inf\{\|\mathbf{x} - \mathbf{y}\| : \mathbf{y} \in C\}$$

be the distance function from C.

(a) Prove that $\varphi = d_C^2$ verifies the relation

$$\varphi(\mathbf{x} + \mathbf{y}) = \varphi(\mathbf{x}) + \varphi'(\mathbf{x}; \mathbf{y}) + \varepsilon(\mathbf{y})\|\mathbf{y}\|,$$

where $\varphi'(\mathbf{x}; \mathbf{y}) = \min\{\langle 2\mathbf{y}, \mathbf{x} - \mathbf{z}\rangle : \mathbf{z} \in \mathcal{P}_C(\mathbf{x})\}$ and $\lim_{\mathbf{y}\to\mathbf{0}} \varepsilon(\mathbf{y}) = \varepsilon(\mathbf{0}) = 0$. In particular, φ is Gâteaux differentiable everywhere. Here $\mathcal{P}_C(\mathbf{x})$ denotes the set of best approximation to $\mathbf{x}$ from C.

(b) Suppose that C is also convex. Infer the formula

$$\nabla \frac{d_C^2}{2}(\mathbf{x}) = \mathbf{x} - P_C(\mathbf{x}) \quad \text{for all } \mathbf{x} \in \mathbb{R}^N.$$

(c) Prove that d_C is differentiable at a point $\mathbf{x} \in \mathbb{R}^N\backslash C$ if, and only if, $\mathcal{P}_C(\mathbf{x})$ is a singleton.

(d) Consider the function $f_C(\mathbf{x}) = \|\mathbf{x}\|^2/2$ if $\mathbf{x} \in C$, and $f_C(\mathbf{x}) = \infty$ if $\mathbf{x} \in \mathbb{R}^n\backslash C$ (where C is a nonempty closed subset of $\mathbb{R}^N$). Notice that $f_C^*(\mathbf{y}) = [\|\mathbf{y}\|^2 - d_C^2(\mathbf{y})]/2$ and infer from Theorem 8.2.3 the conclusion of Bunt–Motzkin Theorem 3.2.2.

8.3 The Convex Programming Problem

The aim of this section is to discuss the problem of minimizing a convex function over a convex set defined by a system of convex inequalities. The main result is the equivalence of this problem to the so-called saddle-point problem. Assuming the differentiability of the functions concerned, the solution of the saddle-point problem is characterized by the Karush–Kuhn–Tucker conditions, which will be made explicit in Theorem 8.3.2 below. In what follows $f, g_1, \ldots, g_m$ will denote convex functions on $\mathbb{R}^N$. The *convex programming problem* for these data is to minimize $f(x)$ over the convex set

$$X = \{\mathbf{x} \in \mathbb{R}^N : \mathbf{x} \geq \mathbf{0},\ g_1(\mathbf{x}) \leq 0, \ldots, g_m(\mathbf{x}) \leq 0\}.$$

In optimization theory, f represents a *cost*, which is minimized over the *feasible set* X. Among the constraints $g_k \leq 0$ we may include equalities $g(\mathbf{x}) = 0$ with g a convex function defined on $\mathbb{R}^N$ (since an equality of this form is equivalent to two inequalities, $g(\mathbf{x}) \leq 0$ and $-g(\mathbf{x}) \leq 0$).

A particular case is the canonical form of the *linear programming problem*. In this problem we seek to maximize a linear function

$$L(\mathbf{x}) = -\langle \mathbf{x}, \mathbf{c} \rangle = -\sum_{k=1}^{N} c_k x_k$$

subject to the constraints

$$\mathbf{x} \geq \mathbf{0} \quad \text{and} \quad A\mathbf{x} \leq \mathbf{b},$$

where A is a real $m \times N$-dimensional matrix and $\mathbf{b} \in \mathbb{R}^m$. Notice that this problem can be easily converted into a minimization problem, by replacing L by $-L$. According to Theorem 5.4.11, L attains its global maximum at an extreme point of the convex set $\{\mathbf{x} : \mathbf{x} \geq \mathbf{0},\ A\mathbf{x} \leq \mathbf{b}\}$. This point can be found by using numerical algorithms.

Linear programming has many applications in planning, scheduling, resource allocation and design in transportation, communication, banking, energy, health and agriculture etc. This motivates a very active area of research for finding more efficient algorithms.

The following example is due to E. Stiefel. Consider a matrix $A = (a_{ij})_{i,j} \in \mathrm{M}_{m \times N}(\mathbb{R})$ and a vector $\mathbf{b} \in \mathbb{R}^m$ such that the system $A\mathbf{x} = \mathbf{b}$ has no solution. Typically this occurs when we have more equations than unknowns. The error in the equation of rank i is a function of the form

$$e_i(\mathbf{x}) = \sum_{j=1}^{N} a_{ij} x_j - b_i.$$

The problem of Chebyshev approximation is to minimize the maximum absolute error

$$Z = \max\{|e_i(\mathbf{x})| : i = 1, \ldots, m\}.$$

Letting Z be a new unknown, this problem can be read as

$$\text{minimize } Z$$

subject to the inequalities

$$-Z \leq \sum_{j=1}^{N} a_{ij}x_j - b_i \leq Z \quad (i = 1, \ldots, m),$$

which can be easily converted into a canonical linear programming problem.

We pass now to the convex programming problem. As in the case of any constrained extremal problem, one can apply the method of Lagrange multipliers in order to eliminate the constraints (at the cost of increasing the number of variables). An excellent exposition of the theoretical aspects of this method is offered by the paper of B. H. Pourciau [489].

The *Lagrangian function* associated with the convex programming problem is the function

$$F(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}) + y_1 g_1(\mathbf{x}) + \cdots + y_m g_m(\mathbf{x})$$

of $N+m$ real variables $x_1, \ldots, x_N, y_1, \ldots, y_m$ (the components of $\mathbf{x}$ and respectively of $\mathbf{y}$). A *saddle point* of F is any point $(\mathbf{x}^0, \mathbf{y}^0)$ of $\mathbb{R}^N \times \mathbb{R}^m$ such that

$$\mathbf{x}^0 \geq \mathbf{0}, \quad \mathbf{y}^0 \geq \mathbf{0}$$

and

$$F(\mathbf{x}^0, \mathbf{y}) \leq F(\mathbf{x}^0, \mathbf{y}^0) \leq F(\mathbf{x}, \mathbf{y}^0)$$

for all $\mathbf{x} \geq \mathbf{0}$, $\mathbf{y} \geq \mathbf{0}$. $F(x^0, y^0)$ appears as a maximum for the affine function $F(x^0, y)$ and a minimum for the convex function $F(x, y^0)$.

The saddle points of F will provide solutions to the convex programming problem that generates F:

8.3.1 Theorem *Let $(\mathbf{x}^0, \mathbf{y}^0)$ be a saddle point of the Lagrangian function F. Then $\mathbf{x}^0$ is a solution to the convex programming problem and*

$$f(\mathbf{x}^0) = F(\mathbf{x}^0, \mathbf{y}^0).$$

Proof. The condition $F(\mathbf{x}^0, \mathbf{y}) \le F(\mathbf{x}^0, \mathbf{y}^0)$ yields

$$y_1 g_1(\mathbf{x}^0) + \cdots + y_m g_m(\mathbf{x}^0) \le y_1^0 g_1(\mathbf{x}^0) + \cdots + y_m^0 g_m(\mathbf{x}^0).$$

By keeping $y_2, \ldots, y_m$ fixed and taking the limit as $y_1 \to \infty$ we infer that $g_1(\mathbf{x}^0) \le 0$. Similarly, $g_2(\mathbf{x}^0) \le 0, \ldots, g_m(\mathbf{x}^0) \le 0$. Thus $\mathbf{x}^0$ belongs to the feasible set X. From $F(\mathbf{x}^0, \mathbf{0}) \le F(\mathbf{x}^0, \mathbf{y}^0)$ and from the definition of X we infer

$$0 \le y_1^0 g_1(\mathbf{x}^0) + \cdots + y_m^0 g_m(\mathbf{x}^0) \le 0,$$

that is, $y_1^0 g_1(\mathbf{x}^0) + \cdots + y_m^0 g_m(\mathbf{x}^0) = 0$. Then $f(\mathbf{x}^0) = F(\mathbf{x}^0, \mathbf{y}^0)$. Since $F(\mathbf{x}^0, \mathbf{y}^0) \le F(\mathbf{x}, \mathbf{y}^0)$ for all $x \ge 0$, we have

$$f(\mathbf{x}^0) \le f(\mathbf{x}) + y_1^0 g_1(\mathbf{x}) + \cdots + y_m^0 g_m(\mathbf{x}) \le f(\mathbf{x})$$

for all $\mathbf{x}$ in X, which shows that $\mathbf{x}^0$ is a solution to the convex programming problem. ■

8.3.2 Theorem (The Karush–Kuhn–Tucker conditions) *Suppose that the convex functions $f, g_1, \ldots, g_m$ are differentiable on $\mathbb{R}^N$. Then $(\mathbf{x}^0, \mathbf{y}^0)$ is a saddle point of the Lagrangian function F if, and only if,*

$$\mathbf{x}^0 \ge \mathbf{0}, \tag{8.4}$$

$$\frac{\partial F}{\partial x_k}(\mathbf{x}^0, \mathbf{y}^0) \ge 0, \quad \textit{for } k = 1, \ldots, N, \tag{8.5}$$

$$\frac{\partial F}{\partial x_k}(\mathbf{x}^0, \mathbf{y}^0) = 0 \quad \textit{whenever } x_k^0 > 0, \tag{8.6}$$

and

$$\mathbf{y}^0 \ge \mathbf{0}, \tag{8.7}$$

$$\frac{\partial F}{\partial y_j}(\mathbf{x}^0, \mathbf{y}^0) = g_j(\mathbf{x}^0) \le 0, \quad \textit{for } j = 1, \ldots, m, \tag{8.8}$$

$$\frac{\partial F}{\partial y_j}(\mathbf{x}^0, \mathbf{y}^0) = 0 \quad \textit{whenever } y_j^0 > 0. \tag{8.9}$$

Proof. If $(\mathbf{x}^0, \mathbf{y}^0)$ is a saddle point of F, then the conditions (8.4) and (8.7) are clearly fulfilled. Also,

$$F(\mathbf{x}^0 + t\mathbf{e}_k, \mathbf{y}^0) \ge F(\mathbf{x}^0, \mathbf{y}^0) \quad \text{for all } t \ge -x_k^0.$$

If $x_k^0 = 0$, then

$$\frac{\partial F}{\partial x_k}(\mathbf{x}^0, \mathbf{y}^0) = \lim_{t \to 0+} \frac{F(\mathbf{x}^0 + t\mathbf{e}_k, \mathbf{y}^0) - F(\mathbf{x}^0, \mathbf{y}^0)}{t} \ge 0.$$

If $x_k^0 > 0$, then $\frac{\partial F}{\partial x_k}(\mathbf{x}^0, \mathbf{y}^0) = 0$ by Fermat's theorem on interior extrema. In a similar way, one can prove (8.8) and (8.9). Suppose now that the conditions (8.4)–(8.9) are satisfied. As $F(\mathbf{x}, \mathbf{y}^0)$ is a differentiable convex function of $\mathbf{x}$ (being a linear combination, with positive coefficients, of such functions), it verifies the assumptions of Theorem 5.3.1. Taking into account the conditions (8.4)–(8.6), we are led to

$$\begin{aligned} F(\mathbf{x}, \mathbf{y}^0) &\geq F(\mathbf{x}^0, \mathbf{y}^0) + \langle \mathbf{x} - \mathbf{x}^0, \nabla_x F(\mathbf{x}^0, \mathbf{y}^0) \rangle \\ &= F(\mathbf{x}^0, \mathbf{y}^0) + \sum_{k=1}^{n} (x_k - x_k^0) \, \frac{\partial F}{\partial x_k}(\mathbf{x}^0, \mathbf{y}^0) \\ &= F(\mathbf{x}^0, \mathbf{y}^0) + \sum_{k=1}^{n} x_k \, \frac{\partial F}{\partial x_k}(\mathbf{x}^0, \mathbf{y}^0) \geq F(\mathbf{x}^0, \mathbf{y}^0), \end{aligned}$$

for all $\mathbf{x} \geq \mathbf{0}$. On the other hand, by (8.7)–(8.9), for $\mathbf{y} \geq \mathbf{0}$ we have

$$\begin{aligned} F(\mathbf{x}^0, \mathbf{y}) &= F(\mathbf{x}^0, \mathbf{y}^0) + \sum_{j=1}^{m} (y_j - y_j^0) \, g_j(\mathbf{x}^0) \\ &= F(\mathbf{x}^0, \mathbf{y}^0) + \sum_{j=1}^{m} y_j g_j(\mathbf{x}^0) \\ &\leq F(\mathbf{x}^0, \mathbf{y}^0). \end{aligned}$$

Consequently, $(\mathbf{x}^0, \mathbf{y}^0)$ is a saddle point of F. ■

We shall illustrate Theorem 8.3.2 by the following example:

$$\text{minimize } (x_1 - 2)^2 + (x_2 + 1)^2 \quad \text{subject to } \ 0 \leq x_1 \leq 1 \text{ and } 0 \leq x_2 \leq 2.$$

Here $f(x_1, x_2) = (x_1 - 2)^2 + (x_2 + 1)^2$, $g_1(x_1, x_2) = x_1 - 1$ and $g_2(x_1, x_2) = x_2 - 2$. The Lagrangian function attached to this problem is

$$F(x_1, x_2, y_1, y_2) = (x_1 - 2)^2 + (x_2 + 1)^2 + y_1(x_1 - 1) + y_2(x_2 - 2)$$

and the Karush–Kuhn–Tucker conditions give us the equations

$$\begin{cases} x_1(2x_1 - 4 + y_1) = 0, \\ x_2(2x_2 + 2 + y_2) = 0, \\ y_1(x_1 - 1) = 0, \\ y_2(x_2 - 2) = 0, \end{cases} \tag{8.10}$$

and the inequalities

$$\begin{cases} 2x_1 - 4 + y_1 \geq 0, \\ 2x_2 + 2 + y_2 \geq 0, \\ 0 \leq x_1 \leq 1 \text{ and } 0 \leq x_2 \leq 2, \\ y_1, y_2 \geq 0. \end{cases} \tag{8.11}$$

The system of equations (8.10) admits 9 solutions,

$$(1,0,2,0),\ (1,2,2,-6),\ (1,-1,2,0),\ (0,0,0,0),\ (2,0,0,0),$$
$$(0,-1,0,0),\ (2,-1,0,0),\ (0,0,0,-1),\ (2,0,0,-1),$$

of which only $(1,0,2,0)$ verifies the inequalities (8.11). Consequently,

$$\inf_{\substack{0\le x_1\le 1\\ 0\le x_2\le 2}} f(x_1,x_2) = f(1,0) = 2.$$

We next indicate a fairly general situation under which the convex programming problem is equivalent to the saddle-point problem. For this, we shall need the following convex analogue of Farkas' Theorem of the alternative (see Exercise 3, Section 3.3):

8.3.3 Lemma *Let $f_1,\dots,f_m$ be convex functions defined on a nonempty convex set Y in $\mathbb{R}^N$. Then either there exists $\mathbf{y}$ in Y such that*

$$f_1(\mathbf{y}) < 0,\dots,f_m(\mathbf{y}) < 0,$$

or there exist positive numbers $a_1,\dots,a_m$, not all zero, such that

$$a_1 f_1(\mathbf{y}) + \cdots + a_m f_m(\mathbf{y}) \ge 0 \quad \textit{for all } \mathbf{y}\in Y.$$

Proof. Assume that the first alternative does not work and consider the set

$$C = \Big\{(t_1,\dots,t_m)\in\mathbb{R}^m : \text{there is } \mathbf{y}\in Y \text{ with } f_k(\mathbf{y}) < t_k \text{ for all } k=1,\dots,m\Big\}.$$

Then C is an open convex set that does not contain the origin of $\mathbb{R}^m$. According to Theorem 3.3.2, the set C and the origin can be separated by a closed hyperplane, that is, there exist scalars $a_1,\dots,a_m$ not all zero, such that for all $\mathbf{y}\in Y$ and all $\varepsilon_1,\dots,\varepsilon_m > 0$,

$$a_1(f_1(\mathbf{y})+\varepsilon_1) + \cdots + a_m(f_m(\mathbf{y})+\varepsilon_m) \ge 0. \tag{8.12}$$

Keeping $\varepsilon_2,\dots,\varepsilon_m$ fixed and letting $\varepsilon_1\to\infty$, we infer that $a_1\ge 0$. Similarly, $a_2\ge 0,\dots,a_m\ge 0$. Letting $\varepsilon_1\to 0,\dots,\varepsilon_m\to 0$ in (8.12) we conclude that $a_1 f_1(\mathbf{y}) + \cdots + a_m f_m(\mathbf{y}) \ge 0$ for all $\mathbf{y}$ in Y. ■

8.3.4 Theorem (Slater's condition) *Suppose that $\mathbf{x}^0$ is a solution of the convex programming problem. If there exists $\bar{\mathbf{x}}\ge 0$ such that $g_1(\bar{\mathbf{x}}) < 0,\dots,g_m(\bar{\mathbf{x}}) < 0$, then one can find an $\mathbf{y}^0$ in $\mathbb{R}^m$ for which $(\mathbf{x}^0,\mathbf{y}^0)$ is a saddle point of the associated Lagrangian function F.*

Proof. By Lemma 8.3.3, applied to the functions $g_1,\dots,g_m, f-f(\mathbf{x}^0)$ and the set $Y=\mathbb{R}^N_+$, we can find $a_1,\dots,a_m,a_0\ge 0$, not all zero, such that

$$a_1 g_1(\mathbf{x}) + \cdots + a_m g_m(\mathbf{x}) + a_0(f(\mathbf{x}) - f(\mathbf{x}^0)) \ge 0 \tag{8.13}$$

for all $\mathbf{x} \geq \mathbf{0}$. A moment's reflection shows that $a_0 > 0$. Put $y_j^0 = a_j/a_0$ and $\mathbf{y}^0 = (y_1^0, \ldots, y_m^0)$. By (8.13) we infer that $f(\mathbf{x}^0) \leq f(\mathbf{x}) + \sum_{j=1}^m y_j^0 g_j(\mathbf{x}) = F(\mathbf{x}, \mathbf{y}^0)$ for all $\mathbf{x} \geq \mathbf{0}$. Particularly, for $\mathbf{x} = \mathbf{x}^0$, this yields

$$f(\mathbf{x}^0) \leq f(\mathbf{x}^0) + \sum_{j=1}^m y_j^0 g_j(\mathbf{x}^0) \leq f(\mathbf{x}^0)$$

that is, $\sum_{j=1}^m y_j^0 g_j(\mathbf{x}^0) = 0$, whence $F(\mathbf{x}^0, \mathbf{y}^0) = f(\mathbf{x}^0) \leq F(\mathbf{x}, \mathbf{y}^0)$ for all $\mathbf{x} \geq \mathbf{0}$. On the other hand, for $\mathbf{y} \geq 0$ we have

$$F(\mathbf{x}^0, \mathbf{y}^0) = f(\mathbf{x}^0) \geq f(\mathbf{x}^0) + \sum_{j=1}^m y_j g_j(\mathbf{x}^0) = F(\mathbf{x}^0, \mathbf{y}),$$

so that $(\mathbf{x}^0, \mathbf{y}^0)$ is a saddle point. ■

A nice geometric application of the convex programming (more precisely, of the quadratic programming), was noted by J. Franklin [202], in his beautiful introduction to mathematical methods of economics. It is about a problem of J. Sylvester, requiring the least circle which contains a given set of points in the plane. Suppose the given points are $\mathbf{a}_1, \ldots, \mathbf{a}_m$. They lie inside the circle of center $\mathbf{x}$ and radius r if

$$\|\mathbf{a}_k - \mathbf{x}\|^2 \leq r^2 \quad \text{for } k = 1, \ldots, m. \tag{8.14}$$

We want to find $\mathbf{x}$ and r so as to minimize r. Letting

$$x_0 = \frac{1}{2}(r^2 - \|x\|^2),$$

we can replace the quadratic constraints (8.14) by linear ones,

$$x_0 + \langle \mathbf{a}_k, \mathbf{x} \rangle \geq b_k \quad \text{for } k = 1, \ldots, m;$$

here $b_k = \|\mathbf{a}_k\|^2/2$. In this way, Sylvester's problem becomes a problem of quadratic programming,

$$\text{minimize } 2x_0 + x_1^2 + x_2^2,$$

subject to the m linear inequalities

$$x_0 + a_{k1}x_1 + a_{k2}x_2 \geq b_k \quad (k = 1, \ldots, m).$$

There are known many numerical algorithms for linear programming. We mention here the simplex algorithm, the interior point algorithms, and the ellipsoid method. See the book of J. Matoušek and B. Gärtner [370] for an introduction to this topic.

We continue with a short discussion about the duality in linear programming. Given the linear problem

$$\mathcal{P} : \text{minimize } f(\mathbf{x}) = \langle \mathbf{c}, \mathbf{x} \rangle \text{ subject to } A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq \mathbf{0}, \tag{8.15}$$

to which we will refer to as the *primal* problem, one associates to it the *dual* problem

$$\mathcal{P}^*\text{: maximize } g(\mathbf{y}) = -\langle \mathbf{b}, \mathbf{y}\rangle \text{ subject to } A^T\mathbf{y} \geq -\mathbf{c}, \mathbf{y} \geq \mathbf{0}. \tag{8.16}$$

It is easy to verify that the dual of the dual problem is the primal problem.

The interest in the dual problem is motivated by Theorem 8.3.5 and the fact that the dual problem is often easier to solve. For example, the dual problem has as many variables as constraints are there in the primal problem. This may reduce the dimensionality of the problem, sometimes considerably. Here is an example due to Jiangqi Dai: Given the real numbers $a_1, ..., a_{1000}, b_1, ..., b_{1000}, p, q$, consider the primal problem

$$\begin{aligned} &\min_{t_1,\ldots,t_{1000}\in\mathbb{R}} (t_1 + \cdots + t_{1000}) \\ \text{subject to} \quad &t_1a_1 + \cdots + t_{1000}a_{1000} \geq p \\ &t_1b_1 + \cdots + t_{1000}b_{1000} \geq q \\ &t_i \geq 0 \end{aligned}$$

Its dual is

$$\begin{aligned} &\max (\lambda p + \mu q) \\ \text{subject to} \quad &\lambda a_i + \mu b_i \leq 1 (\forall i) \\ &\lambda, \mu \geq 0 \end{aligned}$$

The following classical result relating the two problems can be found for example in the books of Ch. L. Byrne [112], p. 177, and A. W. Roberts and D. E. Varberg [504], pp. 148–149.

8.3.5 Theorem (The duality theorem of linear programming) *Consider the primal and dual problems $\mathcal{P}$ and $\mathcal{P}^*$ with feasible sets respectively $\mathcal{F}$ and $\mathcal{F}^*$. Then:*

(a) *If $\mathbf{x} \in \mathcal{F}$ and $\mathbf{y} \in \mathcal{F}^*$, then $f(\mathbf{x}) \leq g(\mathbf{y})$.*

(b) *If for some $\bar{\mathbf{x}} \in \mathcal{F}$ and some $\bar{\mathbf{y}} \in \mathcal{F}^*$ we have $f(\bar{\mathbf{x}}) = g(\bar{\mathbf{y}})$, then $\bar{\mathbf{x}}$ is an optimal solution to $\mathcal{P}$ and $\bar{\mathbf{y}}$ is an optimal solution to $\mathcal{P}^*$.*

(c) *Conversely, if one of the problems $\mathcal{P}$ or $\mathcal{P}^*$ has an optimal solution $\bar{\mathbf{x}}$ or $\bar{\mathbf{y}}$, then so does the other and $f(\bar{\mathbf{x}}) = g(\bar{\mathbf{y}})$.*

(d) *If $\mathcal{F} \neq \emptyset$ and f is bounded above on $\mathcal{F}$, or $\mathcal{F}^* \neq \emptyset$ and g is bounded below on $\mathcal{F}^*$, then both problems $\mathcal{P}$ and $\mathcal{P}^*$ have optimal solutions.*

An extension of Theorem 8.3.5 to the general setting of optimization problems will be presented in the next section.

Exercises

1. Minimize $x^2 + y^2 - 6x - 4y$, subject to $x \geq 0$, $y \geq 0$ and $x^2 + y^2 \leq 1$.

2. Prove the assertions (a) and (b) of Theorem 8.3.5.

3. Every introductory textbook on linear programming includes a proof of the fact that every primal problem (8.15) can be put in standard form,

$$\mathcal{P}_s : \text{minimize } \tilde{f}(\tilde{\mathbf{x}}) = \langle \tilde{\mathbf{c}}, \tilde{\mathbf{x}} \rangle \quad \text{subject to } \tilde{A}\tilde{\mathbf{x}} = \tilde{\mathbf{b}}, \tilde{\mathbf{x}} \geq \mathbf{0},$$

with $\tilde{\mathbf{b}} \geq \mathbf{0}$. See, for example, Ch. L. Byrne [112]. Use the theorems of the alternative to provide conditions under which this problem admits feasible solutions.

4. (Orthogonal projection) Consider a point $\mathbf{x}_0$ in the Euclidean space $\mathbb{R}^N$ such that $\|\mathbf{x}_0\| > 1$. Solve the minimization problem

$$\min_{\|\mathbf{x}\| \leq 1} \|\mathbf{x} - \mathbf{x}_0\|^2$$

by considering the dual problem

$$\max_{\lambda > 0} \min_{\|\mathbf{x}\| \leq 1} \left\{ \|\mathbf{x} - \mathbf{x}_0\|^2 + \lambda \left(\|\mathbf{x}\|^2 - 1 \right) \right\}.$$

5. (The convex multiplier rule of S. Karlin and H. Uzawa) Suppose that $f, g_1, \ldots, g_m$ are convex functions on $\mathbb{R}^N$ and that there is a point $\mathbf{x}_0 \in \mathbb{R}^N$ such that $g_i(\mathbf{x}_0) < 0$ for $i = 1, \ldots, m$. Then $\mathbf{a} \in \mathbb{R}^N$ is a solution of the convex programming problem for these data if, and only if, there is a vector $\mathbf{w} = (w_1, \ldots, w_m) \in \mathbb{R}^m_+$ such that

$$\mathbf{0} \in \partial f(\mathbf{a}) + w_1 \partial g_1(\mathbf{a}) + \cdots + w_m \partial g_m(\mathbf{a})$$

and

$$g_i(\mathbf{a}) \leq 0, \quad w_i g_i(\mathbf{a}) = 0 \quad \text{for } i = 1, \ldots, m.$$

8.4 Dualization of Optimization Problems

We start with an easy (but important) lemma.

8.4.1 Lemma (Dual properties in minimization) *Suppose that E is a real Hilbert space and $f : E \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper convex function. Then:*

(a) $\inf f = -f^*(\mathbf{0})$ *and* $\arg\min f = \partial f^*(\mathbf{0})$.

(b) $\arg\min f = \{\bar{\mathbf{x}}\}$ *if and only if f^* is differentiable at $\mathbf{0}$ with* $\nabla f^*(\mathbf{0}) = \bar{\mathbf{x}}$.

Proof. The assertion (a) is clear. The assertion (b) follows from Theorem 5.1.11. ■

The above result can be embedded into a very general duality scheme concerning optimality under convex perturbations.

Let E and F be two real Hilbert spaces and consider the product space $E \times F$ endowed with the Hilbertian norm associated to the scalar product

$$\langle (\mathbf{x}, \mathbf{y}), (\mathbf{u}, \mathbf{v}) \rangle = \langle \mathbf{x}, \mathbf{u} \rangle + \langle \mathbf{y}, \mathbf{v} \rangle \quad \text{for } (\mathbf{x}, \mathbf{y}), (\mathbf{u}, \mathbf{v}) \in E \times F.$$

Given a function $\Phi : E \times F \to \overline{\mathbb{R}}$, $\Phi = \Phi(\mathbf{x}, \mathbf{y})$, we associate to it the functions

$$\varphi(\mathbf{x}) = \Phi(\mathbf{x}, \mathbf{0}) \quad \text{and} \quad \psi(\mathbf{v}) = -\Phi^*(\mathbf{0}, \mathbf{v}),$$

and the following two optimization problems: the *primal* problem, which is the minimization problem

$$(\mathcal{P}) \quad \inf_{\mathbf{x}\in E} \varphi(\mathbf{x}),$$

and the *dual* problem, which is the maximization problem

$$(\mathcal{P}^*) \quad \sup_{\mathbf{v}\in F} \psi(\mathbf{v}).$$

Here

$$\Phi^*(\mathbf{u}, \mathbf{v}) = \sup \{\langle \mathbf{x}, \mathbf{u}\rangle + \langle \mathbf{y}, \mathbf{v}\rangle - \Phi(\mathbf{x}, \mathbf{y}) : (\mathbf{x}, \mathbf{y}) \in E \times F\}$$

and hence

$$\Phi^*(\mathbf{0}, \mathbf{v}) = \sup \{\langle \mathbf{y}, \mathbf{v}\rangle - \Phi(\mathbf{x}, \mathbf{y}) : (\mathbf{x}, \mathbf{y}) \in E \times F\}.$$

Since $\Phi^{***} = \Phi^*$, the duals of higher order of $(\mathcal{P})$ coincide either with $(\mathcal{P})$ or with $(\mathcal{P}^*)$ (depending on parity). When Φ is a lower semicontinuous proper convex function, then we have a complete duality between $(\mathcal{P})$ and $(\mathcal{P}^*)$, because they are dual to each other.

The function Φ appears as the source of *perturbations* for a given function f. The Fenchel–Moreau duality theorem deals with the perturbations of the form

$$\Phi(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}) - g(A\mathbf{x} - \mathbf{y}),$$

where

(FM1) $f : E \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper convex function,

(FM2) $g : F \to \mathbb{R} \cup \{\infty\}$ is an upper semicontinuous proper concave function,

and

(FM3) $A : E \to F$ is a continuous linear operator.

In this case,

$$\begin{aligned}
\Phi^*(\mathbf{u}, \mathbf{v}) &= \sup_{(\mathbf{x},\mathbf{y})\in E\times F} \{\langle \mathbf{x}, \mathbf{u}\rangle + \langle \mathbf{y}, \mathbf{v}\rangle - f(\mathbf{x}) + g(A\mathbf{x} - \mathbf{y})\} \\
&= \sup_{\mathbf{x}\in E} \sup_{\mathbf{z}\in F} \{\langle \mathbf{x}, \mathbf{u}\rangle + \langle A\mathbf{x}, \mathbf{v}\rangle - f(\mathbf{x}) + g(\mathbf{z}) - \langle \mathbf{z}, \mathbf{v}\rangle\} \\
&= \sup_{\mathbf{x}\in E} \{\langle \mathbf{x}, \mathbf{u}\rangle + \langle \mathbf{x}, A^*\mathbf{v}\rangle - f(\mathbf{x})\} - \inf_{\mathbf{z}\in F} \{\langle \mathbf{z}, \mathbf{v}\rangle - g(\mathbf{z})\} \\
&= f^*(A^*\mathbf{v} + \mathbf{u}) - g_*(\mathbf{v}),
\end{aligned}$$

where A^* denotes the adjoint operator (see R. Bhatia [63], p. 111, for details) and

$$g_*(\mathbf{v}) = \inf_{\mathbf{z}\in F} \{\langle \mathbf{z}, \mathbf{v}\rangle - g(\mathbf{z})\}$$

represents the *concave conjugate* of g. Therefore,

$$\Phi^*(\mathbf{0}, \mathbf{v}) = f^*\left(A^*\mathbf{v}\right) - g_*(\mathbf{v}).$$

Now, consider the minimization problem

$$p = \inf\{f(\mathbf{x}) - g(A\mathbf{x}) : \mathbf{x} \in X\} \tag{8.17}$$

and its dual maximization problem

$$d = \sup\left\{-f^*(A^*\mathbf{y}) + g_*(\mathbf{y}) : \mathbf{y} \in Y\right\}, \tag{8.18}$$

under the hypotheses

$$\mathbf{0} \in A(\mathrm{dom} f) - \mathrm{dom} g \quad \text{ and } \quad \mathbf{0} \in A^*(\mathrm{dom} g^*) - \mathrm{dom} f^*,$$

which assure that p and d are finite; indeed, according to the Fenchel–Young inequality we have

$$f(\mathbf{x}) - g(A\mathbf{x}) + f^*(A^*\mathbf{y}) - g_*(\mathbf{y}) \geq \langle \mathbf{x}, A^*\mathbf{y}\rangle - \langle A\mathbf{x}, \mathbf{y}\rangle \geq 0,$$

which yields

$$-\infty < d \leq p < \infty.$$

The following result provides sufficient conditions under which $p = d$.

8.4.2 Theorem (Fenchel–Moreau duality theorem) *Suppose that f, g and A verify the conditions* (FM1)-(FM3) *above.*

If there is a point $\mathbf{x}_0 \in dom f$ such that g is continuous at $A\mathbf{x}_0$, then

$$\inf_{\mathbf{x}\in E} \{f(\mathbf{x}) - g(A\mathbf{x})\} = \sup_{\mathbf{y}\in F} \{-f^*(A^*\mathbf{y}) + g_*(\mathbf{y})\}$$

and the supremum in the dual maximization problem is attained.

Proof. Consider the auxiliary function $h : F \to \overline{\mathbb{R}}$ defined by

$$h(\mathbf{y}) = \inf_{\mathbf{x}\in E} \Phi(\mathbf{x}, \mathbf{y}) = \inf_{\mathbf{x}\in E} \{f(\mathbf{x}) - g(A\mathbf{x} - \mathbf{y})\}.$$

This is a lower semicontinuous proper convex function with domain dom $h = A(\mathrm{dom} f) - \mathrm{dom} g$. Since

$$\begin{aligned} h^*(\mathbf{y}) &= \sup_{\mathbf{v}\in F} \{\langle \mathbf{v}, \mathbf{y}\rangle - h(\mathbf{v})\} = \sup_{\mathbf{v}\in F} \{\langle \mathbf{v}, \mathbf{y}\rangle - \inf_{\mathbf{x}\in E} \Phi(\mathbf{x}, \mathbf{v})\} \\ &= \sup_{(\mathbf{x},\mathbf{v})\in E\times F} \{\langle \mathbf{v}, \mathbf{y}\rangle - \Phi(\mathbf{x}, \mathbf{v})\} = \Phi^*(\mathbf{0}, \mathbf{y}), \end{aligned}$$

it follows that

$$d = \sup_{\mathbf{y}\in F} -\Phi^*(\mathbf{0},\mathbf{y}) = \sup_{\mathbf{y}\in F} \{\langle \mathbf{0},\mathbf{y}\rangle - h^*(\mathbf{y})\} = h^{**}(\mathbf{0}).$$

Since

$$p = h(\mathbf{0}),$$

the equality $p = d$ reduces to $h(\mathbf{0}) = h^{**}(\mathbf{0})$.

We next show that h is bounded above on a suitable neighborhood of the origin. Indeed, since g is concave and continuous at $A\mathbf{x}_0$, it follows that the function $\mathbf{y} \to g(A\mathbf{x}_0 - \mathbf{y})$ is bounded below on a neighborhood V of the origin. Taking into account that $h(\mathbf{y}) \leq f(\mathbf{x}_0) - g(A\mathbf{x}_0 - \mathbf{y})$, we conclude that the function h is bounded above on V (and thus is continuous there by Proposition 4.1.11).

By Corollary 8.1.3, $h^{**} = \operatorname{cl} h$, while from the continuity of h at the origin we infer that

$$\operatorname{cl} h(\mathbf{0}) = \liminf_{\mathbf{x}\to\mathbf{0}} h(\mathbf{x}) = h(\mathbf{0}).$$

Therefore $h(\mathbf{0}) = h^{**}(\mathbf{0})$ and the proof of the equality $p = d$ is done.

An element $\mathbf{y}_0 \in F$ is a solution to the dual problem if, and only if,

$$-\Phi^*(\mathbf{0},\mathbf{y}_0) \geq -\Phi^*(\mathbf{0},\mathbf{y}) \quad \text{for all } \mathbf{y} \in F,$$

equivalently, $h^*(\mathbf{y}) \geq h^*(\mathbf{y}_0)$ for all $\mathbf{y} \in F$. This means that y_0 is a point of minimum for h^*, a fact that can be restated as $\mathbf{0} \in \partial h^*(\mathbf{y}_0)$. Because h^* is lower semicontinuous and convex, we infer from Proposition 8.1.1 that this latter condition is equivalent to $\mathbf{y}_0 \in \partial h^{**}(\mathbf{0})$. Therefore the supremum in the dual maximization problem is attained. ■

For $E = F$ and A equal to the identity of E, we obtain:

8.4.3 Corollary (W. Fenchel) *Let f and $-g$ be lower semicontinuous proper convex functions on E such that $\operatorname{dom} f \cap \operatorname{dom} g$ contains a point where f and g are continuous. Then*

$$\inf\{f(\mathbf{x}) - g(\mathbf{x}) : \mathbf{x} \in E\} = \max\{-f^*(\mathbf{y}) + g_*(\mathbf{y}) : \mathbf{y} \in E\}.$$

The infimum is attained, for example, if there is a point $\bar{\mathbf{x}}$ such that $\partial f(\bar{\mathbf{x}}) \cap \partial g(\bar{\mathbf{x}}) \neq \emptyset$.

Last, but not least, it is worth noticing that the above argument for Fenchel–Moreau duality Theorem 8.4.2 and its Corollary 8.4.3 can be easily adapted to obtain their extension to the framework of Banach spaces. The details are left to the reader and make the objective of Exercise 1.

The aforementioned extension plays an important role in the theory of optimal mass transportation, nicely presented by C. Villani in his books [575] and [576]. A very brief presentation of the key points of this connection is as follows.

Consider two Borel probability measures μ and ν on $\mathbb{R}^N$ and a lower semicontinuous positive function $c : \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}$ (called *cost* function). Kantorovich's mass transportation problem consists in the minimization of the function

$$I(\pi) = \int_{\mathbb{R}^N \times \mathbb{R}^N} c(\mathbf{x}, \mathbf{y}) \mathrm{d}\pi(\mathbf{x}, \mathbf{y}),$$

over the nonempty convex set $\Pi(\mu, \nu)$ of all Borel probability measures π on $\mathbb{R}^N \times \mathbb{R}^N$, with marginals μ and ν. This means that

$$\pi(A, \mathbb{R}^N) = \mu(A) \quad \text{for all } A \in \mathcal{B}(\mathbb{R}^N)$$

and

$$\pi(\mathbb{R}^N, B) = \nu(B) \quad \text{for all } B \in \mathcal{B}(\mathbb{R}^N).$$

It is easy to show that $\pi \in \Pi(\mu, \nu)$ if, and only if, it is a positive Borel measure on $\mathbb{R}^N \times \mathbb{R}^N$ such that for all pairs $(\varphi, \psi) \in L^\infty(\mu) \times L^\infty(\nu)$ we have

$$\int_{\mathbb{R}^N \times \mathbb{R}^N} (\varphi(\mathbf{x}) + \psi(\mathbf{y}))\, \mathrm{d}\pi(\mathbf{x}, \mathbf{y}) = \int_{\mathbb{R}^N} \varphi(\mathbf{x}) \mathrm{d}\mu(\mathbf{x}) + \int_{\mathbb{R}^N} \psi(\mathbf{y}) \mathrm{d}\nu(\mathbf{y}).$$

The measures π should be viewed as *transportation plans* and the values $I(\pi)$ as *costs of transportation*. The existence of an optimal transportation plan is assured by the following result:

8.4.4 Lemma *The function $I(\pi)$ admits a minimizer.*

Proof. Since $I(\pi) \geq 0$ for all π, there is a sequence $(\pi_n)_n$ such that $(I(\pi_n))$ converges to $\min I(\pi)$. The set $\Pi(\mu, \nu)$ is a weak-star compact subset of the set $\mathrm{Prob}(\mathbb{R}^N \times \mathbb{R}^N)$, of all Borel probabilities on $\mathbb{R}^N \times \mathbb{R}^N$. Use the Banach–Alaoglu Theorem B.1.6. Therefore, we may assume (by passing to a subsequence if necessary) that $(\pi_n)_n$ is weak-star convergent to some π_∞ in $\Pi(\mu, \nu)$. The next step is to prove that $I(\pi)$ is sequentially weak-star lower semicontinuous.

Choose $\rho_R \in C_c(\mathbb{R}^N \times \mathbb{R}^N)$ such that $0 \leq \rho_R \leq 1$ and $\rho_R = 1$ on the ball of center 0 and radius $R > 0$. Then

$$\begin{aligned}
\int_{\mathbb{R}^N \times \mathbb{R}^N} \rho_R(\mathbf{x}, \mathbf{y}) c(\mathbf{x}, \mathbf{y}) \mathrm{d}\pi_\infty(\mathbf{x}, \mathbf{y}) &= \lim_{n \to \infty} \int_{\mathbb{R}^N \times \mathbb{R}^N} \rho_R(\mathbf{x}, \mathbf{y}) c(\mathbf{x}, \mathbf{y}) \mathrm{d}\pi_n(\mathbf{x}, \mathbf{y}) \\
&\leq \lim_{n \to \infty} \int_{\mathbb{R}^N \times \mathbb{R}^N} c(\mathbf{x}, \mathbf{y}) \mathrm{d}\pi_n(\mathbf{x}, \mathbf{y}) \\
&= \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^N \times \mathbb{R}^N} c(\mathbf{x}, \mathbf{y}) \mathrm{d}\pi(\mathbf{x}, \mathbf{y}).
\end{aligned}$$

Letting $R \to \infty$, we obtain that

$$\int_{\mathbb{R}^N \times \mathbb{R}^N} c(\mathbf{x}, \mathbf{y}) \mathrm{d}\pi_\infty(\mathbf{x}, \mathbf{y}) \leq \inf_{\pi \in \Pi(\mu, \nu)} \int_{\mathbb{R}^N \times \mathbb{R}^N} c(\mathbf{x}, \mathbf{y}) \mathrm{d}\pi(\mathbf{x}, \mathbf{y}),$$

whence we conclude that π_∞ is a minimizer for $I(\pi)$. ∎

As was already noticed in this section, a convex minimization problem like that concerning the function $I(\pi)$ admits a dual formulation involving the function

$$J(\varphi,\psi)=\int_{\mathbb{R}^N}\varphi(\mathbf{x})\mathrm{d}\mu(\mathbf{x})+\int_{\mathbb{R}^N}\psi(\mathbf{y})\mathrm{d}\nu(\mathbf{y}),$$

which is defined on the nonempty convex set $\mathcal{F}_c$, of all functions $(\varphi,\psi)\in L^1(\mu)\times L^1(\nu)$ such that

$$\varphi(\mathbf{x})+\psi(\mathbf{y})\leq c(\mathbf{x},\mathbf{y})$$

for μ -almost all $\mathbf{x}\in X$ and ν-almost all $\mathbf{y}\in Y$.

8.4.5 Theorem (Kantorovich duality) *We have that*

$$\inf_{\pi\in\Pi(\mu,\nu)}I(\pi)=\sup_{(\varphi,\psi)\in\mathcal{F}_c}J(\varphi,\psi).$$

C. Villani [575] uses the Fenchel–Moreau duality theorem to settle the particular case of Theorem 8.4.5 where both μ and ν have compact supports. The general case follows via an approximation argument.

In the case of quadratic costs $c(\mathbf{x},\mathbf{y})=\|\mathbf{x}-\mathbf{y}\|^2$, it is shown that $\pi\in\Pi(\mu,\nu)$ is an optimal transportation plan if, and only if, there exists a lower semicontinuous proper convex function φ such that for π-almost all $(\mathbf{x},\mathbf{y})$ we have

$$\mathbf{y}\in\partial\varphi(\mathbf{x}).$$

See [575], Theorem 2.12, p. 66 and W. Gangbo and R. J. McCann [212]. An important ingredient is Rockafellar's characterization of convexity via cyclic monotonicity. See Theorem 4.3.14.

Does there exist a map $T:\mathbb{R}^N\to\mathbb{R}^N$ such that the optimal plan π is of the form $\pi=(\mathrm{I}\times T)\,\#\mu$? The answer is provided by Brenier's theorem, which asserts that if μ is an absolutely continuous probability measure and $c(\mathbf{x},\mathbf{y})=\|\mathbf{x}-\mathbf{y}\|^2$, then there exists a convex function $\varphi:\mathbb{R}^N\to\mathbb{R}\cup\{\infty\}$ whose gradient $T=\nabla\varphi$ pushes μ forward to ν. Apart from changes on a set of μ-measure zero, T is the only map to arise in this way. The details can be found either in Y. Brenier's paper [97] or in the aforementioned books of C. Villani.

H. Brezis [101] succeeded in the presentation of a straightforward (but illuminating) case of the Monge–Kantorovich problem, sketched in Exercise 4. He based his argument on the following combinatorial result, attributed to S. N. Afriat.

8.4.6 Lemma *If all the diagonal elements of a matrix $B=(b_{ij})_{i,j}\in\mathrm{M}_m(\mathbb{R})$ vanish and $\sum_{i=1}^m b_{i\sigma(i)}\geq 0$ for all permutations $\sigma\in\Pi(m)$, then there exist real numbers $\lambda_1,...,\lambda_m$ such that*

$$\lambda_i-\lambda_j\leq b_{ij}\ \textit{for all}\ i,j\in\{1,...,m\}.$$

Exercises

1. Illustrate the Fenchel–Moreau duality Theorem 8.4.2 for the following pairs of functions defined on $\mathbb{R}$: (a) $f(x) = x^2$ and $g(x) = x - 1$. (b) $f(x) = x^2$ and $g(x) = -(x-1)^2$.

2. Suppose that $f : \mathbb{R}^N \to \mathbb{R} \cup \{-\infty\}$ is a proper function and $A : \mathbb{R}^N \to \mathbb{R}^M$ is a linear map. Prove that for every $\mathbf{b} \in \mathbb{R}^M$, the following inequality
$$\inf\{f(\mathbf{x}) : A\mathbf{x} = \mathbf{b}\} \geq \sup\left\{\langle \mathbf{b}, \mathbf{y}\rangle - f^*(A^*\mathbf{y}) : \mathbf{y} \in \mathbb{R}^M\right\}$$
holds. If, in addition, f is convex and $\mathbf{b}$ belongs to the interior of $A(\mathrm{dom} f)$, prove that the equality holds and the supremum is attained when finite.

3. Adapt the argument of Fenchel–Moreau duality Theorem 8.4.2 and of its Corollary 8.4.3, to extend them to the framework of Banach spaces.

4. (H. Brezis [101]) Consider two finite sets $X = \{P_1, ..., P_m\}$ and $Y = \{Q_1, ..., Q_m\}$ and a map $c : X \times Y \to \mathbb{R}$. In this simplified framework, Monge's problem is to find a permutation for which the following minimum is reached
$$M = \min_{\sigma \in \Pi(m)} \sum_{k=1}^{m} c(P_k, Q_{\sigma(k)}).$$
Consider also Kantorovich's problem to determine
$$K = \min_{A} \left\{ \sum_{i,j=1}^{m} a_{ij} c(P_i, Q_j) : A = (a_{ij})_{i,j} \text{ is doubly stochastic} \right\}$$
and the dual of Kantorovich's problem, the determination of
$$D = \sup_{\substack{\varphi: X \to \mathbb{R} \\ \psi: Y \to \mathbb{R}}} \left\{ \sum_{i=1}^{m} (\varphi(P_i) - \psi(Q_i)) : \varphi(x) - \psi(y) \leq c(x, y), \forall x \in X, y \in Y \right\}.$$
The aim of this exercise is to prove that
$$M = K = D,$$
a fact that illustrates Theorem 8.3.5 and its connection to Monge's problem.

(a) Prove that $K \leq M$ and $D \leq K$ (which reduces the proof to the fact that $M \leq D$).

(b) Show that it is possible to relabel the points Q_j such that
$$M = \sum_{i=1}^{m} c(P_i, Q_i) \leq \sum_{j=1}^{m} c(P_j, Q_{\sigma(j)})$$
for every permutation σ.

(c) By the definition of D it remains to find functions $\varphi : X \to \mathbb{R}$ and $\psi : Y \to \mathbb{R}$ such that

$$\sum_{i=1}^{m} (\varphi(P_i) - \psi(Q_i)) \geq M \text{ and } (\varphi(P_i) - \psi(Q_j) \leq c(P_i, Q_j) \text{ for all } i, j. \tag{8.19}$$

Consider the matrix B of coefficients $b_{ij} = c(P_i, Q_j) - c(P_i, Q_i)$ and choose $\lambda_1, ..., \lambda_m$ as in Lemma 8.4.6. Then prove that the functions defined by $\psi(Q_i) = \lambda_i$ and $\varphi(P_i) = \lambda_i + c(P_i, Q_i)$ verify the inequalities (8.19).

8.5 The Ky Fan Minimax Inequality

Brouwer's fixed point theorem asserts that every continuous map of a nonempty compact convex subset of $\mathbb{R}^N$ into itself has a fixed point. The book of J. Franklin [201] includes several elegant proofs of this result (including that by A. Garsia, based on Green's theorem).

The following consequence of Brouwer's fixed point theorem is known as the *KKM theorem* :

8.5.1 Theorem (The Knaster–Kuratowski–Mazurkiewicz theorem) *Suppose that X is a nonempty subset of $\mathbb{R}^N$ and M is a function which associates to each $\mathbf{x} \in X$ a closed nonempty subset $M(\mathbf{x})$ of X. If*

$$\operatorname{conv}(F) \subset \bigcup_{\mathbf{x} \in F} M(\mathbf{x})$$

for every finite subset $F \subset X$, then $\bigcap_{\mathbf{x} \in F} M(\mathbf{x}) \neq \emptyset$ for every finite subset $F \subset X$. Moreover, $\bigcap_{\mathbf{x} \in X} M(\mathbf{x}) \neq \emptyset$ if X is compact.

Proof. If $\bigcap_{\mathbf{x} \in F} M(\mathbf{x})$ is empty for some finite subset F, then consider the map

$$\mathbf{y} \in \operatorname{conv}(F) \to \Big[\sum_{\mathbf{x} \in F} d_{M(\mathbf{x})}(\mathbf{y})\mathbf{x}\Big] \Big/ \Big[\sum_{\mathbf{x} \in F} d_{M(\mathbf{x})}(\mathbf{y})\Big].$$

By Brouwer's fixed point theorem, this map admits a fixed point $\mathbf{z}$. Letting $G = \{\mathbf{x} \in F : \mathbf{z} \notin M(\mathbf{x})\}$, then $\mathbf{z}$ should be in $\operatorname{conv}(G)$, and this leads to a contradiction. ■

In turn, the KKM theorem yields the finite-dimensional case of the *Ky Fan minimax inequality* [188]:

8.5.2 Theorem *Suppose that C is a nonempty, compact, and convex subset of $\mathbb{R}^N$. If $f\colon C \times C \to \mathbb{R}$, $f = f(\mathbf{x}, \mathbf{y})$, is quasiconcave in the first variable and lower semicontinuous in the second variable, then there exists a point $\bar{\mathbf{y}}$ in C such that*

$$\sup_{\mathbf{x} \in C} f(\mathbf{x}, \bar{\mathbf{y}}) \leq \sup_{\mathbf{x} \in C} f(\mathbf{x}, \mathbf{x}).$$

Proof. Put $\alpha = \sup_{\mathbf{x}\in C} f(\mathbf{x},\mathbf{x})$ and consider the family of sets $M(\mathbf{x}) = \{\mathbf{y} \in C : f(\mathbf{x},\mathbf{y}) \leq \alpha\}$ for $\mathbf{x} \in C$. Clearly, $\mathbf{x} \in M(\mathbf{x})$ for every $\mathbf{x}$ and the sets $M(\mathbf{x})$ are closed because the function $f(\mathbf{x},\mathbf{y})$ is lower semicontinuous in the second variable. For every convex combination $\sum_{k=1}^{n} \lambda_k \mathbf{x}_k$ of points in C we have

$$\min_k f(\mathbf{x}_k,\mathbf{y}) \leq f\left(\sum\nolimits_{k=1}^{n} \lambda_k \mathbf{x}_k, \mathbf{y}\right),$$

whence every point in $M(\sum_{k=1}^{n} \lambda_k \mathbf{x}_k)$ (in particular, $\sum_{k=1}^{n} \lambda_k \mathbf{x}_k$) must be in one of the sets $M(\mathbf{x}_k)$. Hence the KKM theorem applies and this implies that $\bigcap_{\mathbf{x}\in C} M(\mathbf{x}) \neq \emptyset$. The conclusion of Theorem 8.5.2 is now obvious. ■

It is worth noticing that all three results stated above (Brouwer's fixed point theorem, KKM theorem and the Ky Fan minimax inequality) can be deduced from each other. See Exercise 1.

The Ky Fan minimax inequality has many important applications to mathematical economics and game theory, from which we mention here the following one:

8.5.3 Theorem (The minimax theorem of von Neumann) *Let X and Y be compact convex subsets (of suitable Euclidean spaces) and let $f : X \times Y \to \mathbb{R}$ be a function which verifies the following two conditions:*

(a) $\mathbf{x} \to f(\mathbf{x},\mathbf{y})$ *is upper semicontinuous and quasiconcave on X for each fixed $\mathbf{y} \in Y$;*

(b) $\mathbf{y} \to f(\mathbf{x},\mathbf{y})$ *is lower semicontinuous and quasiconvex on Y for each fixed $\mathbf{x} \in X$.*

Then

$$\max_{\mathbf{x}\in X} \min_{\mathbf{y}\in Y} f(\mathbf{x},\mathbf{y}) = \min_{\mathbf{y}\in Y} \max_{\mathbf{x}\in X} f(\mathbf{x},\mathbf{y}).$$

Proof. The inequality $\leq$ is straightforward. As concerns the reverse inequality, notice that the function

$$\varphi : (X \times Y) \times (X \times Y) \to \mathbb{R}, \quad \varphi\left((\mathbf{u},\mathbf{v}),(\mathbf{x},\mathbf{y})\right) = f(\mathbf{u},\mathbf{y}) - f(\mathbf{x},\mathbf{v}),$$

is quasiconcave with respect to the variable $(\mathbf{u},\mathbf{v})$ and lower semicontinuous with respect to the variable $(\mathbf{x},\mathbf{y})$. According to the Ky Fan minimax inequality, there exists a point $(\bar{\mathbf{x}},\bar{\mathbf{y}})$ such that

$$\varphi\left((\mathbf{u},\mathbf{v}),(\bar{\mathbf{x}},\bar{\mathbf{y}})\right) \leq \sup_{(\mathbf{u},\mathbf{v})\in X\times Y} \varphi\left((\mathbf{u},\mathbf{v}),(\mathbf{u},\mathbf{v})\right) = 0.$$

Therefore $f(\mathbf{u},\bar{\mathbf{y}}) \leq f(\bar{\mathbf{x}},\mathbf{v})$ for all $\mathbf{u} \in X$, $\mathbf{v} \in Y$, which yields

$$f(\mathbf{x},\bar{\mathbf{y}}) \leq f(\bar{\mathbf{x}},\bar{\mathbf{y}}) \leq f(\bar{\mathbf{x}},\mathbf{y}) \quad \text{for all } \mathbf{x} \in X,\ \mathbf{y} \in Y.$$

As a consequence,

$$\min_{\mathbf{y}\in Y} \max_{\mathbf{x}\in X} f(\mathbf{x},\mathbf{y}) \leq \max_{\mathbf{x}\in X} f(\mathbf{x},\bar{\mathbf{y}}) = f(\bar{\mathbf{x}},\bar{\mathbf{y}}) = \min_{\mathbf{y}\in Y} f(\bar{\mathbf{x}},\mathbf{y}) \leq \max_{\mathbf{x}\in X} \min_{\mathbf{y}\in Y} f(\mathbf{x},\mathbf{y}),$$

and the proof of the minimax equality is complete. ■

To understand the meaning of Theorem 8.5.3 from the point of view of the game theory, let us consider the class of two-person zero-sum games, that is, games with only two players in which one player wins what the other player loses. X is the set of strategies of Player I and Y is the set of strategies of Player II. Player I chooses $x \in X$ and Player II chooses $y \in Y$, each unaware of the choice of the other. Then their choices are made known and I wins the amount $f(x, y)$ from II. According to Theorem 8.5.3, the maximum value of the minimum expected gain for one player is equal to the minimum value of the maximum expected loss for the other; moreover each player has a mixed strategy which realizes this equality.

The next application of the Ky Fan minimax inequality refers to the existence of Nash equilibrium. A group of players are in *Nash equilibrium* if each one is making the best decision possible, taking into account the decisions of the others in the game as long as the other parties' decisions remain unchanged.

8.5.4 Theorem (Nash equilibrium) *Consider the product set* $C = C_1 \times \cdots \times C_n$, *where each set* C_k *is a nonempty, compact and convex subset of* $\mathbb{R}^N$. *Consider also continuous functions* $f_1, \ldots, f_n \colon C \to \mathbb{R}$ *such that, for each* k, *the function*

$$x_k \in C_k \to f_k(y_1, \ldots, y_{k-1}, x_k, y_k, \ldots, y_n)$$

is convex on C_k *for all* $y_i \in C_i$, $i \neq k$. *Then there exists an element* $c = (c_1, \ldots, c_n) \in C$ *such that*

$$f_k(c) \leq f_k(c_1, \ldots, x_k, \ldots, c_n) \quad \text{for all } x_k \in C_k,\ k \in \{1, \ldots, n\}.$$

Proof. Apply the Ky Fan minimax inequality to the function

$$f(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^{n} [f_k(\mathbf{y}) - f_k(y_1, \ldots, x_k, \ldots, y_n)].$$

■

A very attractive introduction to game theory is provided by the book of P. D. Straffin [556].

We end with a variant of the Ky Fan minimax inequality, that proves useful for establishing delicate inequalities. See J. Diestel, H. Jarchow, A. Tonge [161], p. 191.

8.5.5 Theorem (Ky Fan Lemma [188]) *Let* E *be a separated locally convex space, and let* C *be a compact convex subset of* E. *Let* M *be a set of functions on* C *with values in* $\mathbb{R} \cup \{\infty\}$, *having the following three properties:*

(a) *Each* $f \in M$ *is convex and lower semicontinuous.*

(b) *If* $g \in \operatorname{conv}(M)$, *then there is* $f \in M$ *with* $g(\mathbf{x}) \leq f(\mathbf{x})$ *for all* $\mathbf{x} \in C$.

(c) *There is an* $r \in \mathbb{R}$ *such that each* $f \in M$ *has a value less than or equal to* r.

Then there is a point $\bar{\mathbf{x}} \in C$ *such that* $f(\bar{\mathbf{x}}) \leq r$ *for all* $f \in M$.

Proof. For $f \in M$ and $\varepsilon > 0$ denote

$$S(f, \varepsilon) := \{\mathbf{x} \in C : f(\mathbf{x}) \leq r + \varepsilon\}.$$

By (c), each set $S(f, \varepsilon)$ is non empty; it is also closed since f is lower semicontinuous. We have to prove that the intersection of the whole family of sets $S(f, \varepsilon)$ is nonempty. Since C is a compact set, it suffices to prove that the intersection of every finite subfamily $S(f_1, \varepsilon_1), ..., S(f_n, \varepsilon_n)$ is nonempty.

For this, consider the following two auxiliary subsets of $\mathbb{R}^n$: the open and convex set $A = \prod_{k=1}^n (-\infty, r + \varepsilon_k)$ and the convex hull B of

$$\{(f_k(\mathbf{x}))_{k=1}^n : \mathbf{x} \in C\} \bigcap \mathbb{R}^n;$$

the properties (b) and (c) ensure that B is nonempty. We shall prove (by contradiction) that $A \cap B \neq \emptyset$.

Assume that $A \cap B = \emptyset$. Then, by the Hahn–Banach Separation Theorem, there are $\mu = (\mu_1, ..., \mu_n) \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ such that $\langle \mu, \mathbf{a} \rangle < \lambda \leq \langle \mu, \mathbf{b} \rangle$ for all $\mathbf{a} \in A$ and $\mathbf{b} \in B$. The strict inequality forces $\mu \neq \mathbf{0}$, and so we may, without loss of generality, normalize μ so that $\sum_{k=1}^n |\mu_k| = 1$. Necessarily, $\mu_k \geq 0$ for each k. To check this, note that, no matter how we choose $\rho \leq r$, the point $\mathbf{a} = \sum_{j \neq k} r\mathbf{e}_j + \rho \mathbf{e}_k$ belongs to A and so $\langle \mu, \mathbf{a} \rangle = r \sum_{j \neq k} \mu_j + \rho \mu_k < \lambda$. Letting $\rho \to -\infty$, we see that the case $\mu_k < 0$ contradicts the existence of the upper bound λ. Therefore, $\mu_k \geq 0$.

This allows us to use hypothesis (b) to find an $f \in M$ such that for $\mathbf{x} \in C$, either $f(\mathbf{x}) = \infty$, or $f(\mathbf{x}) < \infty$ and

$$\begin{aligned} f(\mathbf{x}) &\geq \sum\nolimits_{k=1}^n \mu_k f_k(\mathbf{x}) = \langle \mu, (f_k(\mathbf{x}))_{k=1}^n \rangle \\ &\geq \lambda > \langle \mu, (r + \varepsilon_k/2)_{k=1}^n \rangle = \sum\nolimits_{k=1}^n \mu_k (r + \varepsilon_k/2) > r. \end{aligned}$$

This contradicts hypothesis (c). Therefore $A \cap B \neq \emptyset$.

To end the proof, choose $s = (s_1, ..., s_n)$ in $A \cap B$. By the definition of B, we can write $s_k = \sum_{j=1}^n \alpha_j f_k(\mathbf{x}_j)$ $(1 \leq k \leq n)$, where the $\mathbf{x}_j$'s are in C and the α_j's are positive and sum to 1. Since C is convex, the point $\mathbf{x} = \sum_{j=1}^n \alpha_j \mathbf{x}_j$ belongs to C. We have $f_k(\mathbf{x}) \leq \sum_{j=1}^n \alpha_j f_k(\mathbf{x}_j) = s_k$ $(1 \leq k \leq n)$, due to the convexity of the functions f_k. Finally, since $s \in A$, we conclude that $\mathbf{x} \in S(f_k, \varepsilon_k)$ whenever $k \in \{1, ..., n\}$. ■

Exercises

1. Prove that the Brouwer fixed point theorem, the KKM theorem and the Ky Fan minimax inequality are equivalent in the sense that each one implies the other ones.

   [*Hint*: For example, in order to show that the Ky Fan minimax inequality implies Brouwer's fixed point theorem, consider a continuous self-map f of a compact convex set $C \subset \mathbb{R}^N$ and attach to it the function $\varphi : C \times C \to \mathbb{R}$ given by $\varphi(\mathbf{x}, \mathbf{y}) = \langle \mathbf{x} - f(\mathbf{x}), \mathbf{x} - \mathbf{y} \rangle$.]

2. (The minimax flavor of the Fenchel–Moreau duality theorem) Consider the perturbation function $\Phi(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}) - g(A\mathbf{x} - \mathbf{y})$, as in the statement of Theorem 8.4.2, and attached to it the *Hamiltonian* $\mathcal{H} : E \times F \to \mathbb{R}$, defined by

$$\mathcal{H}(\mathbf{x}, \mathbf{v}) = \sup \{\langle \mathbf{y}, \mathbf{v} \rangle - \Phi(\mathbf{x}, \mathbf{y}) : \mathbf{y} \in F\}.$$

Notice that $\mathcal{H}(\mathbf{x}, \mathbf{v})$ is lower semicontinuous and convex with respect to the second variable (being a conjugate with respect to this variable) and $\mathcal{H}(\mathbf{x}, \mathbf{v})$ is lower semicontinuous and concave with respect to the first variable. Prove that the following two assertions are equivalent for a pair $(\bar{\mathbf{x}}, \bar{\mathbf{v}}) \in E \times F$:

(a) $\mathcal{H}(\mathbf{x}, \bar{\mathbf{v}}) \leq \mathcal{H}(\bar{\mathbf{x}}, \bar{\mathbf{v}}) \leq \mathcal{H}(\bar{\mathbf{x}}, \mathbf{v})$ for all $(\mathbf{x}, \mathbf{v}) \in E \times F$ (that is, $(\bar{\mathbf{x}}, \bar{\mathbf{v}})$ is a saddle point for the function $\mathcal{H}(\mathbf{x}, \mathbf{v})$);

(b) $\bar{\mathbf{x}}$ is an optimal solution to the primal problem (8.17), $\bar{\mathbf{v}}$ is an optimal solution to the dual problem (8.18) and the extreme values p and d are equal.

3. Infer from the von Neumann minimax theorem the equality $f = f^{**}$ stated in the Fenchel–Moreau theorem on biconjugation.

8.6 The Moreau–Yosida Approximation

A convex analogue of mollification is provided by the *infimal convolution*, which for two proper convex functions f, g defined on a real Hilbert space H is defined by the formula

$$(f \square g)(\mathbf{x}) = \inf\{f(\mathbf{x} - \mathbf{y}) + g(\mathbf{y}) : \mathbf{y} \in H\};$$

the value $-\infty$ is allowed. Clearly, $f \square g = g \square f$. The fact that $f \square g$ is a convex function is left as Exercise 1.

If $(f \square g)(\mathbf{x}) > -\infty$ for all $\mathbf{x}$, then $f \square g$ is a proper convex function. For example, this happens when both functions f and g are positive or, more generally, when there exists an affine function $h \colon H \to \mathbb{R}$ such that $f \geq h$ and $g \geq h$.

Notice that the distance function from a convex set C is an example of infimal convolution:

$$d_C(\mathbf{x}) = \inf_{\mathbf{y} \in C} \|\mathbf{x} - \mathbf{y}\| = (\|\cdot\| \square \iota_C)(x).$$

8.6.1 Lemma *If f and g are two proper functions defined on H, then*

$$(f \square g)^* = f^* + g^*.$$

Proof. Indeed, for $\mathbf{z}$ arbitrarily fixed in H we have

$$\begin{aligned}(f\square g)^*(\mathbf{z}) &= \sup_{\mathbf{x}\in H}\left[\langle \mathbf{x},\mathbf{z}\rangle - (f\square g)(\mathbf{x})\right]\\ &= \sup_{\mathbf{x}\in H}\left[\langle \mathbf{x},\mathbf{z}\rangle - \inf_{\mathbf{y}\in H}\left(f(\mathbf{y})+g(\mathbf{x}-\mathbf{y})\right)\right]\\ &= \sup_{\mathbf{x}\in H}\sup_{\mathbf{y}\in H}\left[\langle \mathbf{y},\mathbf{z}\rangle - f(\mathbf{y}) + \langle \mathbf{x}-\mathbf{y},\mathbf{z}\rangle - g(\mathbf{x}-\mathbf{y})\right]\\ &= \sup_{\mathbf{y}\in H}\sup_{\mathbf{v}\in H}\left(\left[\langle \mathbf{y},\mathbf{z}\rangle - f(\mathbf{y})\right] + \left[\langle \mathbf{v},\mathbf{z}\rangle - g(\mathbf{v})\right]\right)\\ &= f^*(\mathbf{z}) + g^*(\mathbf{z}).\end{aligned}$$

■

See Exercise 3 for a dual property.

A useful way to approximate from below a lower semicontinuous proper convex function $f\colon H\to\mathbb{R}\cup\{\infty\}$ is provided by the *Moreau–Yosida approximation*. This is done via the functions

$$\begin{aligned}f_\lambda(\mathbf{x}) &= \left(f\square\frac{1}{2\lambda}\|\cdot\|^2\right)(\mathbf{x})\\ &= \inf_{\mathbf{y}\in H}\left(f(\mathbf{y})+\frac{1}{2\lambda}\|\mathbf{x}-\mathbf{y}\|^2\right)\end{aligned}$$

for $\mathbf{x}\in H$ and $\lambda>0$. The functions f_λ are well-defined and *finite* for all $\mathbf{x}\in H$ due to the existence of an affine minorant $\langle\cdot,\mathbf{z}\rangle+\alpha$ for f (assured by Theorem 4.3.1). Indeed,

$$\begin{aligned}f(\mathbf{y})+\frac{1}{2\lambda}\|\mathbf{x}-\mathbf{y}\|^2 &\geq \langle \mathbf{y},\mathbf{z}\rangle+\alpha+\frac{1}{2\lambda}\|\mathbf{x}-\mathbf{y}\|^2\\ &\geq \langle \mathbf{y}-\mathbf{x},\mathbf{z}\rangle+\langle \mathbf{x},\mathbf{z}\rangle+\alpha+\frac{1}{2\lambda}\|\mathbf{x}-\mathbf{y}\|^2\\ &\geq \langle \mathbf{x},\mathbf{z}\rangle+\alpha-\left\|\frac{\mathbf{y}-\mathbf{x}}{\sqrt{\lambda}}\right\|\left\|\mathbf{z}\sqrt{\lambda}\right\|+\frac{1}{2\lambda}\|\mathbf{x}-\mathbf{y}\|^2\\ &\geq \langle \mathbf{x},\mathbf{z}\rangle+\alpha-\frac{\lambda}{2}\|\mathbf{z}\|^2,\end{aligned}$$

for all $\mathbf{y}\in H$.

Notice that the infimum in the definition of $f_\lambda(\mathbf{x})$ is actually a minimum, that is, there is $\mathbf{x}_\lambda\in H$ such that

$$f_\lambda(\mathbf{x}) = f(\mathbf{x}_\lambda)+\frac{1}{2\lambda}\|\mathbf{x}-\mathbf{x}_\lambda\|^2 \quad \text{for } \mathbf{x}\in H. \tag{8.20}$$

To prove this, consider a minimizing sequence $\mathbf{x}_n\in H$ such that

$$f(\mathbf{x}_n)+\frac{1}{2\lambda}\|\mathbf{x}-\mathbf{x}_n\|^2 \leq f_\lambda(\mathbf{x})+\frac{1}{n} \tag{8.21}$$

for all indices $n \geq 1$. According to the parallelogram law,

$$\|\mathbf{x}_m - \mathbf{x}_n\|^2 = 2\|\mathbf{x}_m - \mathbf{x}\|^2 + 2\|\mathbf{x}_n - \mathbf{x}\|^2 - 4\left\|\mathbf{x} - \frac{\mathbf{x}_m + \mathbf{x}_n}{2}\right\|^2,$$

and combing this fact with the inequality (8.21) and the convexity of f we infer that

$$\begin{aligned}\|\mathbf{x}_m - \mathbf{x}_n\|^2 &\leq 4\lambda\left(\frac{1}{m} + \frac{1}{n} + 2f_\lambda(\mathbf{x}) - f(\mathbf{x}_m) - f(\mathbf{x}_n)\right)\\ &\quad + 8\lambda\left(f\left(\frac{\mathbf{x}_m + \mathbf{x}_n}{2}\right) - f_\lambda(\mathbf{x})\right)\\ &= 4\lambda\left(\frac{1}{m} + \frac{1}{n} + 2f\left(\frac{\mathbf{x}_m + \mathbf{x}_n}{2}\right) - f(\mathbf{x}_m) - f(\mathbf{x}_n)\right)\\ &\leq 4\lambda\left(\frac{1}{m} + \frac{1}{n}\right).\end{aligned}$$

Thus $(\mathbf{x}_n)_n$ is convergent to an element $\mathbf{x}_\lambda$ of H since H is complete. Taking into account that f is lower semicontinuous we obtain

$$f(\mathbf{x}_\lambda) + \frac{1}{2\lambda}\|\mathbf{x} - \mathbf{x}_\lambda\|^2 \leq \liminf_{n\to\infty}\left(f(\mathbf{x}_n) + \frac{1}{2\lambda}\|\mathbf{x} - \mathbf{x}_n\|^2\right) \leq f_\lambda(\mathbf{x}),$$

whence $f_\lambda(\mathbf{x}) = f(\mathbf{x}_\lambda) + \frac{1}{2\lambda}\|\mathbf{x} - \mathbf{x}_\lambda\|^2$.

8.6.2 Lemma *The solutions $\mathbf{x}_\lambda$ of the equation (8.20) are the same as the solutions of the variational inequality*

$$\frac{1}{\lambda}\langle \mathbf{x}_\lambda - \mathbf{x}, \mathbf{x}_\lambda - \mathbf{y}\rangle + f(\mathbf{x}_\lambda) - f(\mathbf{y}) \leq 0 \quad \text{for all } \mathbf{y} \in H. \tag{8.22}$$

Proof. Indeed, if $\mathbf{x}_\lambda$ is a solution of the equation (8.20) and $\mathbf{y} \in H$, then for every $\theta \in (0,1)$ we have $\mathbf{x}_\lambda + \theta(\mathbf{y} - \mathbf{x}_\lambda) = (1-\theta)\mathbf{x}_\lambda + \theta\mathbf{y}$, so that

$$\begin{aligned}f(\mathbf{x}_\lambda) + \frac{1}{2\lambda}\|\mathbf{x} - \mathbf{x}_\lambda\|^2 &\leq f(\mathbf{x}_\lambda + \theta(\mathbf{y} - \mathbf{x}_\lambda)) + \frac{1}{2\lambda}\|\mathbf{x} - \mathbf{x}_\lambda - \theta(\mathbf{y} - \mathbf{x}_\lambda)\|^2\\ &\leq (1-\theta)f(\mathbf{x}_\lambda) + \theta f(y) + \frac{1}{2\lambda}\|\mathbf{x} - \mathbf{x}_\lambda\|^2\\ &\quad - \frac{\theta}{\lambda}\langle \mathbf{x} - \mathbf{x}_\lambda, \mathbf{y} - \mathbf{x}_\lambda\rangle + \frac{\theta^2}{2\lambda}\|y - \mathbf{x}_\lambda\|^2,\end{aligned}$$

whence

$$f(\mathbf{x}_\lambda) - f(\mathbf{y}) + \frac{1}{\lambda}\langle \mathbf{x}_\lambda - \mathbf{x}, \mathbf{x}_\lambda - \mathbf{y}\rangle \leq \frac{\theta}{2\lambda}\|\mathbf{y} - \mathbf{x}_\lambda\|^2.$$

Passing to the limit as $\theta \to 0$, we infer that $\mathbf{x}_\lambda$ is a solution of the variational inequality (8.22). Conversely, if $\mathbf{x}_\lambda$ is a solution of the variational inequality

(8.22), then

$$\begin{aligned} 0 &\geq f(\mathbf{x}_\lambda) - f(\mathbf{y}) + \frac{1}{\lambda}\langle \mathbf{x}_\lambda - \mathbf{x}, \mathbf{x}_\lambda - \mathbf{y}\rangle \\ &\geq f(\mathbf{x}_\lambda) + \frac{1}{2\lambda}\|\mathbf{x} - \mathbf{x}_\lambda\|^2 - f(\mathbf{y}) - \frac{1}{2\lambda}\|\mathbf{x} - \mathbf{y}\|^2, \end{aligned}$$

for all $\mathbf{y} \in H$ and the proof is done. ■

8.6.3 Corollary *The equation* (8.20) *has a unique solution.*

Proof. The existence part was already noticed. As concerns the uniqueness, suppose that $\mathbf{x}'$ is another solution of the equation (8.20). According to Lemma 8.6.2, it follows that

$$\frac{1}{\lambda}\langle \mathbf{x}_\lambda - \mathbf{x}, \mathbf{x}_\lambda - \mathbf{x}'\rangle + f(\mathbf{x}_\lambda) - f(\mathbf{x}') \leq 0$$

and

$$\frac{1}{\lambda}\langle \mathbf{x}' - \mathbf{x}, \mathbf{x}' - \mathbf{x}_\lambda\rangle + f(\mathbf{x}') - f(\mathbf{x}_\lambda) \leq 0.$$

Summing up one obtains $\|\mathbf{x}_\lambda - \mathbf{x}'\| \leq 0$, that is, $\mathbf{x}' = \mathbf{x}_\lambda$. The proof is complete. ■

In what follows we will deal with the maps

$$J_\lambda : H \to \mathrm{dom} f, \quad J_\lambda(\mathbf{x}) = \mathbf{x}_\lambda,$$

and

$$A_\lambda : H \to H, \quad A_\lambda(\mathbf{x}) = \frac{1}{\lambda}\left(\mathbf{x} - J_\lambda(\mathbf{x})\right).$$

According to Lemma 8.6.2,

$$\frac{1}{\lambda}\langle J_\lambda(\mathbf{x}) - \mathbf{x}, J_\lambda(\mathbf{x}) - \mathbf{y}\rangle + f(J_\lambda(\mathbf{x})) - f(\mathbf{y}) \leq 0 \quad \text{for all } \mathbf{y} \in H, \tag{8.23}$$

which implies that

$$A_\lambda(\mathbf{x}) \in \partial f(J_\lambda(\mathbf{x})) \quad \text{for all } \mathbf{x} \in H \tag{8.24}$$

and

$$\frac{1}{\lambda}\langle J_\lambda(\mathbf{x}) - \mathbf{x}, J_\lambda(\mathbf{x}) - J_\lambda(\mathbf{y})\rangle + f(J_\lambda(\mathbf{x})) - f(J_\lambda(\mathbf{y})) \leq 0.$$

Adding the last inequality to its companion obtained by interchanging $\mathbf{x}$ and $\mathbf{y}$, we obtain

$$\langle J_\lambda(\mathbf{x}) - J_\lambda(\mathbf{y}), \mathbf{x} - \mathbf{y}\rangle \geq \|\mathbf{x} - \mathbf{y}\|^2. \tag{8.25}$$

Since

$$\begin{aligned} \|\mathbf{x} - \mathbf{y}\|^2 &= \|(\mathbf{x} - J_\lambda(\mathbf{x})) - (\mathbf{y} - J_\lambda(\mathbf{y})) + J_\lambda(\mathbf{x}) - J_\lambda(\mathbf{y})\|^2 \\ &= \|(\mathbf{x} - J_\lambda(\mathbf{x})) - (\mathbf{y} - J_\lambda(\mathbf{y}))\|^2 + \|J_\lambda(\mathbf{x}) - J_\lambda(\mathbf{y})\|^2 \\ &\quad + 2\langle (\mathbf{x} - J_\lambda(\mathbf{x})) - (\mathbf{y} - J_\lambda(\mathbf{y})), J_\lambda(\mathbf{x}) - J_\lambda(\mathbf{y})\rangle, \end{aligned}$$

we infer (based on inequality (8.25)) that

$$\|\mathbf{x}-\mathbf{y}\|^2 \geq \|(\mathbf{x}-J_\lambda(\mathbf{x}))-(\mathbf{y}-J_\lambda(\mathbf{y}))\|^2 + \|J_\lambda(\mathbf{x})-J_\lambda(\mathbf{y})\|^2 . \tag{8.26}$$

This yields the following result:

8.6.4 Lemma *The maps J_λ and $\mathrm{I}-J_\lambda$ are both Lipschitz, with Lipschitz constant less than* 1.

We are now in a position to present the key features of the Moreau–Yosida approximants:

8.6.5 Theorem *Suppose that $f\colon H \to \mathbb{R}\cup\{\infty\}$ is a lower semicontinuous proper convex function. Then the Moreau–Yosida approximants f_λ are Fréchet differentiable convex functions on H and*

$$\nabla f_\lambda(\mathbf{x}) = \frac{1}{\lambda}(\mathbf{x}-J_\lambda(\mathbf{x})) \in \partial f(J_\lambda(\mathbf{x})).$$

Moreover, $f_\lambda \to f$ as $\lambda \to 0$.

Proof. We first prove that

$$A_\lambda(\mathbf{x}) = \frac{1}{\lambda}\,(\mathbf{x}-J_\lambda(\mathbf{x})) \in \partial f_\lambda(\mathbf{x}) \quad \text{ for all } \mathbf{x}\in H. \tag{8.27}$$

Indeed, since $A_\lambda(\mathbf{x}) \in \partial f(J_\lambda(\mathbf{x}))$, we have

$$\begin{aligned} f_\lambda(\mathbf{x}) - f_\lambda(\mathbf{y}) &= f\left(J_\lambda(\mathbf{x})\right) - f\left(J_\lambda(\mathbf{y})\right) + \frac{\lambda}{2}\|A_\lambda(\mathbf{x})\|^2 - \frac{\lambda}{2}\|A_\lambda(\mathbf{y})\|^2 \\ &\leq \langle A_\lambda(\mathbf{x}), J_\lambda(\mathbf{x})-J_\lambda(\mathbf{y})\rangle + \frac{\lambda}{2}\|A_\lambda(\mathbf{x})\|^2 - \frac{\lambda}{2}\|A_\lambda(\mathbf{y})\|^2 \\ &\overset{J_\lambda = \mathrm{I}-\lambda A_\lambda}{=} \langle A_\lambda(\mathbf{x}), \mathbf{x}-\mathbf{y}\rangle - \lambda\langle A_\lambda(\mathbf{x}), A_\lambda(\mathbf{x})-A_\lambda(\mathbf{y})\rangle \\ &\qquad + \frac{\lambda}{2}\|A_\lambda(\mathbf{x})\|^2 - \frac{\lambda}{2}\|A_\lambda(\mathbf{y})\|^2 \\ &= \langle A_\lambda(\mathbf{x}), \mathbf{x}-\mathbf{y}\rangle - \frac{\lambda}{2}\|A_\lambda(\mathbf{x})-A_\lambda(\mathbf{y})\|^2 \leq \langle A_\lambda(\mathbf{x}), \mathbf{x}-\mathbf{y}\rangle, \end{aligned}$$

and this implies the relation (8.27). Therefore

$$\begin{aligned} f_\lambda(\mathbf{x}) - f_\lambda(\mathbf{y}) &\geq \langle A_\lambda(\mathbf{y}), \mathbf{x}-\mathbf{y}\rangle \\ &= \langle A_\lambda(\mathbf{x}), \mathbf{x}-\mathbf{y}\rangle - \langle A_\lambda(\mathbf{x})-A_\lambda(\mathbf{y}), \mathbf{x}-\mathbf{y}\rangle \\ &\geq \langle A_\lambda(\mathbf{x}), \mathbf{x}-\mathbf{y}\rangle - \|A_\lambda(\mathbf{x})-A_\lambda(\mathbf{y})\|\,\|\mathbf{x}-\mathbf{y}\| \\ &\geq \langle A_\lambda(\mathbf{x}), \mathbf{x}-\mathbf{y}\rangle - \frac{1}{\lambda}\|\mathbf{x}-\mathbf{y}\|^2, \end{aligned}$$

according to Lemma 8.6.4. Thus

$$0 \geq f_\lambda(\mathbf{x}) - f_\lambda(\mathbf{y}) - \langle A_\lambda(\mathbf{x}), \mathbf{x}-\mathbf{y}\rangle \geq -\frac{1}{\lambda}\|\mathbf{x}-\mathbf{y}\|^2,$$

equivalently,

$$\frac{|f_\lambda(\mathbf{x}) - f_\lambda(\mathbf{y}) - \langle A_\lambda(\mathbf{x}), \mathbf{x} - \mathbf{y}\rangle|}{\|\mathbf{x} - \mathbf{y}\|} \leq \frac{1}{\lambda}\|\mathbf{x} - \mathbf{y}\|,$$

which shows that f_λ is Fréchet differentiable and $\nabla f_\lambda(\mathbf{x}) = A_\lambda(\mathbf{x})$. We pass now to the problem of convergence of Moreau–Yosida approximants. Let $\mathbf{x} \in \mathrm{dom}(f)$ and $\mathbf{x}^* \in \mathrm{dom} f^*$. Then

$$f(J_\lambda(\mathbf{x})) + \frac{1}{2\lambda}\|\mathbf{x} - J_\lambda(\mathbf{x})\|^2 = f_\lambda(\mathbf{x}) \leq f(\mathbf{x})$$

and

$$-f(J_\lambda(\mathbf{x})) \leq f^*(\mathbf{x}^*) - \langle J_\lambda(\mathbf{x}), \mathbf{x}^*\rangle,$$

whence

$$\begin{aligned}\frac{1}{2\lambda}\|\mathbf{x} - J_\lambda(\mathbf{x})\|^2 &\leq f(\mathbf{x}) + f^*(\mathbf{x}^*) - \langle \mathbf{x}, \mathbf{x}^*\rangle - \langle J_\lambda(\mathbf{x}) - \mathbf{x}, \mathbf{x}^*\rangle \\ &\leq \frac{1}{4\lambda}\|\mathbf{x} - J_\lambda(\mathbf{x})\|^2 + f(\mathbf{x}) + f^*(\mathbf{x}^*) - \langle \mathbf{x}, \mathbf{x}^*\rangle + \lambda\|\mathbf{x}^*\|^2,\end{aligned}$$

since

$$-\langle J_\lambda(\mathbf{x}) - \mathbf{x}, \mathbf{x}^*\rangle \leq \|J_\lambda(\mathbf{x}) - \mathbf{x}\|\,\|\mathbf{x}^*\| \leq \frac{1}{4\lambda}\|\mathbf{x} - J_\lambda(\mathbf{x})\|^2 + \lambda\|\mathbf{x}^*\|^2.$$

Therefore

$$\|\mathbf{x} - J_\lambda(\mathbf{x})\|^2 \leq 4\lambda\left(f(\mathbf{x}) + f^*(\mathbf{x}^*) - \langle \mathbf{x}, \mathbf{x}^*\rangle + \lambda\|\mathbf{x}^*\|^2\right),$$

which implies that $\lim_{\lambda\to 0} J_\lambda(\mathbf{x}) \to \mathbf{x}$. Since f is lower semicontinuous and $f(J_\lambda(\mathbf{x})) \leq f_\lambda(\mathbf{x}) \leq f(\mathbf{x})$ we infer that

$$f(\mathbf{x}) \leq \liminf_{\lambda\to 0} f(J_\lambda(\mathbf{x})) \leq \liminf_{\lambda\to 0} f_\lambda(\mathbf{x}) \leq \limsup_{\lambda\to 0} f_\lambda(\mathbf{x}) \leq f(\mathbf{x}),$$

that is, $\lim_{\lambda\to 0} f_\lambda(\mathbf{x}) = f(\mathbf{x})$. To end the proof, we have to prove that $f_\lambda(\mathbf{x})$ tends to $f(\mathbf{x})$ also in the case where $f(\mathbf{x}) = \infty$. Suppose, by reductio ad absurdum, that there exists a sequence λ_n converging to 0 such that $f_{\lambda_n}(\mathbf{x}) \leq C < \infty$ for all n. Then $f\left(J_{\lambda_n}(\mathbf{x})\right) \leq f_{\lambda_n}(\mathbf{x}) \leq C$ for all n. Since $J_{\lambda_n}(\mathbf{x}) \to \mathbf{x}$ and f is lower semicontinuous, this would imply that $f(\mathbf{x}) \leq C$, a contradiction. The proof is complete. ■

The lower semicontinuous proper convex functions f are determined by their Moreau–Yosida approximants. Indeed, according to Lemma 8.6.1 and Example 8.1.6,

$$f_\lambda^* = f^* + \frac{\lambda}{2}\|\cdot\|^2, \tag{8.28}$$

so if two lower semicontinuous proper convex functions $f, g\colon H \to \mathbb{R} \cup \{\infty\}$ verify the condition $f_\lambda = g_\lambda$ for some $\lambda > 0$, then necessarily they must be equal.

The convergence of the functions f_λ to f is much better when f is Lipschitz. See Corollary 8.6.7.

8.6.6 Lemma *Suppose that $f : H \to \mathbb{R}$ is a Lipschitz convex function and $(f_n)_n$ is a sequence of convex functions on H such that $f_n \leq f$ for all n. If $f_n^* \to f^*$ uniformly, then $f_n \to f$ uniformly on bounded subsets of H.*

Proof. Let A be a bounded subset of H and let $\varepsilon > 0$. Since f is bounded on bounded sets, it follows that $\partial f(A)$ is bounded. See Exercise 6, Section 4.3. Since $f_n^* \to f^*$ uniformly, there exists a natural number N such that $f_n^*(\mathbf{y}) - f^*(\mathbf{y}) < \varepsilon$ for all $n \geq N$ and $\mathbf{y} \in \partial f(A)$. Then, for all $\mathbf{x} \in A$ and $\mathbf{y} \in \partial f(\mathbf{x})$, we have

$$f^*(\mathbf{y}) = \langle \mathbf{x}, \mathbf{y} \rangle - f(\mathbf{x}) \geq f_n^*(\mathbf{y}) - \varepsilon \geq \langle \mathbf{x}, \mathbf{y} \rangle - f_n(\mathbf{x}) - \varepsilon,$$

whenever $n \geq N$. Therefore $f(\mathbf{x}) - \varepsilon \leq f_n(\mathbf{x}) \leq f(\mathbf{x})$ for all $\mathbf{x} \in A$ and $n \geq N$. The proof is complete. ■

8.6.7 Corollary *If $f : H \to \mathbb{R}$ is a Lipschitz convex function, then the Moreau–Yosida approximants f_λ converge uniformly to f on bounded sets as $\lambda \to 0$.*

The infimal convolution and the Moreau–Yosida approximation are useful in the theory of Hamilton–Jacobi equations.

Exercises

1. (Huber function). Prove that the Moreau–Yosida approximants of the absolute value function, $f(x) = |x|$ for $x \in \mathbb{R}$, are given by the formulas
$$f_\lambda(x) = \begin{cases} x^2/(2\lambda) & \text{for } |x| \leq \lambda \\ |x| - \lambda/2 & \text{for } |x| \geq \lambda. \end{cases}$$

2. Prove that the infimal convolution $f \square g$ of two proper convex functions $f, g \colon H \to \mathbb{R} \cup \{\infty\}$ is a convex function.

3. Suppose that H is a Hilbert space and $f, g \colon H \to \mathbb{R} \cup \{\infty\}$ are two proper convex functions. Prove that
$$(f + g)^* = \operatorname{cl}(f^* \square g^*).$$
Here the closure operation is superfluous when $0 \in \operatorname{int}(\operatorname{dom} f - \operatorname{dom} g)$ (in particular, when $\operatorname{int}(\operatorname{dom} f) \cap \operatorname{dom} g \neq \emptyset$ or $\operatorname{dom} f \cap \operatorname{int}(\operatorname{dom} g) \neq \emptyset$). See Corollary 8.4.3.

4. Suppose that H is a Hilbert space and $f : H \to \mathbb{R} \cup \{\infty\}$ is a lower semicontinuous proper convex function. Prove that the Moreau–Yosida approximants $f_\lambda(\mathbf{x})$ tend to $-f^*(\mathbf{0}) = \inf\{f(\mathbf{y}) : \mathbf{y} \in H\}$ as $\lambda \to \infty$.

5. If f is as in Exercise 4, prove that for all $\lambda > 0$,
$$\inf_{\mathbf{x} \in H} \left(f(\mathbf{x}) + \frac{1}{2\lambda} \|\mathbf{x}\|^2 \right) + \inf_{\mathbf{y} \in H} \left(f^*(\mathbf{y}) + \frac{1}{2\lambda} \|\mathbf{y}\|^2 \right) = 0.$$

6. Use the Moreau–Yosida approximation to complete the proof of the implication (a) ⇒ (c) in Theorem 8.1.2, when $E = \mathbb{R}^N$.

   [*Hint*: Notice that $(f_\lambda)^{**} = f_\lambda$ (since $\mathrm{dom} f_\lambda = \mathbb{R}^N$) and

   $$f^{**}(\mathbf{x}) \geq \liminf_{\lambda \to 0} f_\lambda^{**}(\mathbf{x}) = \liminf_{\lambda \to 0} f_\lambda(\mathbf{x}) = f(\mathbf{x}).\ ]$$

7. (J. J. Moreau [397]) Given a lower semicontinuous proper convex function $f : H \to \mathbb{R} \cup \{\infty\}$, the *proximity* operator of f is

   $$\mathrm{prox}_f : H \to H, \quad \mathrm{prox}_f(\mathbf{x}) = \arg\min_{\mathbf{y} \in H} \left(f(\mathbf{y}) + \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|^2 \right).$$

   Prove that:

   (a) For every $\mathbf{x}, \mathbf{p} \in H$, we have $\mathbf{p} = \mathrm{prox}_f(\mathbf{x})$ if, and only if, $\mathbf{x} - \mathbf{p} \in \partial f(\mathbf{p})$;

   (b) If C is a nonempty closed convex subset of H and $f = \iota_C$, then $\mathrm{prox}_f = P_C$ (the orthogonal projection on C);

   (c) If $\alpha > 0$ and $\mathbf{x} \in H$, then $\mathbf{x} = \mathrm{prox}_{\alpha f} \mathbf{x} + \alpha \mathrm{prox}_{f^*/\alpha}(\mathbf{x}/\alpha)$.

8.7 The Hopf–Lax Formula

Suppose that $\mathcal{L} : \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}$, $\mathcal{L} = \mathcal{L}(\mathbf{x}, \mathbf{q})$, is a given smooth function, referred to as the *Lagrangian*; its variables represent usually the *generalized position* $\mathbf{x}$ and *generalized velocity* $\mathbf{q}$. Consider the variational problem of minimizing the *action* functional,

$$I(\mathbf{w}) = \int_0^t \mathcal{L}(\mathbf{w}, \dot{\mathbf{w}}) ds,$$

over the class of admissible functions,

$$\mathcal{A} = \left\{ \mathbf{w} \in C^2\left([0,t], \mathbb{R}^N\right) : \mathbf{w}(0) = \mathbf{x}_0 \text{ and } \mathbf{w}(t) = \mathbf{x} \right\}.$$

Here $\mathbf{x}_0$ and $\mathbf{x}$ are two given points in $\mathbb{R}^N$ and $\dot{\mathbf{w}}$ denotes the derivative of w with respect to the variable s (classical notation in mechanics introduced by Newton). As well known (for example, see L. C. Evans [183], pp. 117–118), the minimizer $\mathbf{u}$, if it exists, must verify the Euler–Lagrange vector equation,

$$-\frac{d}{ds} \left(\nabla_{\mathbf{q}} \mathcal{L}(\mathbf{u}(s), \dot{\mathbf{u}}(s)) \right) + \nabla_{\mathbf{x}} \mathcal{L}(\mathbf{u}(s), \dot{\mathbf{u}}(s)) = 0 \quad \text{for } s \in [0,t], \tag{8.29}$$

and the boundary conditions $u(0) = \mathbf{x}_0$ and $u(t) = \mathbf{x}$. Here

$$\nabla_{\mathbf{q}} = \left(\frac{\partial}{\partial q_1}, ..., \frac{\partial}{\partial q_N} \right)$$

and

$$\nabla_{\mathbf{x}} = \left(\frac{\partial}{\partial x_1}, ..., \frac{\partial}{\partial x_N} \right).$$

Thus (8.29) is equivalent to a system of N scalar equations of second order:

$$-\frac{d}{ds}\left(\frac{\partial \mathcal{L}}{\partial q_i}(\mathbf{u}(s), \dot{\mathbf{u}}(s))\right) + \frac{\partial \mathcal{L}}{\partial x_i}(\mathbf{u}(s), \dot{\mathbf{u}}(s)) = 0 \quad \text{for } s \in [0,t] \text{ and } i = 1, ..., N.$$

It can be converted into a system of $2N$ scalar equations of first order by *replacing* the generalized velocity with *generalized momentum*,

$$\mathbf{p}(s) = D_{\mathbf{q}}\mathcal{L}(\mathbf{u}(s), \dot{\mathbf{u}}(s)), \tag{8.30}$$

and *assuming* that the equation (8.30) can be inverted to yield a unique smooth function

$$\mathbf{q} = \mathbf{q}(\mathbf{x}, \mathbf{p}).$$

Under these circumstances, we attach to the Lagrangian $\mathcal{L}$ the *Hamiltonian* function,

$$\mathcal{H}(\mathbf{x}, \mathbf{p}) = \langle \mathbf{p}, \mathbf{q}(\mathbf{x}, \mathbf{p}) \rangle - \mathcal{L}(\mathbf{x}, \mathbf{q}(\mathbf{x}, \mathbf{p})), \tag{8.31}$$

whose meaning in mechanics is that of *total energy*.

A straightforward computation yields the following fact:

8.7.1 Lemma *The Euler–Lagrange equation* (8.29) *is equivalent to Hamilton's equations*

$$\begin{cases} \dot{\mathbf{x}}(s) = \nabla_p \mathcal{H}(\mathbf{x}(s), \mathbf{p}(s)) \\ \dot{\mathbf{p}}(s) = -\nabla_x \mathcal{H}(\mathbf{x}(s), \mathbf{p}(s)) \end{cases}$$

for $s \in [0,t]$.

Notice also that the function $s \to \mathcal{H}(\mathbf{x}(s), \mathbf{p}(s))$ is constant. Indeed,

$$\begin{aligned} \frac{d}{ds}\mathcal{H}(\mathbf{x}(s), \mathbf{p}(s)) &= \sum_{i=1}^{N}\left(\frac{\partial \mathcal{H}}{\partial p_i}\dot{p}_i + \frac{\partial \mathcal{H}}{\partial x_i}\dot{x}_i\right) \\ &= \sum_{i=1}^{N}\frac{\partial \mathcal{H}}{\partial p_i}\left(-\frac{\partial \mathcal{H}}{\partial x_i}\right) + \sum_{i=1}^{N}\frac{\partial \mathcal{H}}{\partial x_i}\left(\frac{\partial \mathcal{H}}{\partial p_i}\right) = 0. \end{aligned}$$

Comparing the formula (8.31) with the classical Legendre conjugation (see Theorem 8.1.8), we see that the true meaning of the Hamiltonian is that of Legendre–Fenchel conjugate of the Lagrangian with respect to the second variable.

Therefore, the natural framework that assures not only the existence of a unique global minimizer for the action functional but also the transformation of the Euler–Lagrange equation into Hamilton's equations can be briefly described as follows:

(H) The Lagrangian $\mathcal{L} : \mathbb{R}^N \times \mathbb{R}^N \to \mathbb{R}$, $\mathcal{L} = \mathcal{L}(\mathbf{x}, \mathbf{q})$, is a function of class C^2, strictly convex and supercoercive with respect to the second variable.

The existence of a unique global minimizer of the action functional follows from Theorem 5.4.1 and Corollary 5.4.4.

According to Theorem 8.1.8, one can introduce the Hamiltonian associated to $\mathcal{L}$ via the formula

$$\mathcal{H}(\mathbf{x},\mathbf{p}) = \sup\left\{\langle \mathbf{q},\mathbf{p}\rangle - \mathcal{L}(\mathbf{x},\mathbf{q}) : \mathbf{q}\in\mathbb{R}^N\right\}.$$

This is a continuously differentiable strictly convex function. See Theorem 8.2.4. Also, the map

$$(\mathbf{x},\mathbf{v}) \to (\mathbf{x}, \nabla_{\mathbf{v}}\mathcal{L}(\mathbf{x},\mathbf{q}))$$

is bijective and of class C^1. Hence, under the hypothesis (H), the Euler–Lagrange equation can be indeed restated in terms of Hamilton's equations.

8.7.2 Remark *The motion $\mathbf{x}=\mathbf{x}(t)$ of a mass-m particle in the potential $V(\mathbf{x})$ outlines the following conjugate pair,*

$$\mathcal{L}(\mathbf{x},\mathbf{q}) = \frac{1}{2}m\left\|\mathbf{q}\right\|^2 - V(\mathbf{x}) \text{ and } \mathcal{H}(\mathbf{x},\mathbf{p}) = \frac{1}{2m}\left\|\mathbf{p}\right\|^2 + V(\mathbf{x}).$$

In this case, the Euler–Lagrange equation coincide with Newton's law of motion

$$m\frac{\mathrm{d}^2\mathbf{x}}{\mathrm{d}t^2} = -\nabla V.$$

Following L. C. Evans [183], we next discuss the Hopf–Lax formula, which provides a solution for the Hamilton–Jacobi equation with given initial data, in the particular case of Hamiltonians that depend only on the variable $\mathbf{p}$.

8.7.3 Theorem *Assume that $\mathcal{H}:\mathbb{R}^N\to\mathbb{R}$, $\mathcal{H}=\mathcal{H}(\mathbf{p})$, is a supercoercive convex function of class C^2 with conjugate $\mathcal{L}=\mathcal{L}(\mathbf{q})$, and $u_0:\mathbb{R}^N\to\mathbb{R}$ is a Lipschitz function. Then the function $u=u(t,\mathbf{x})$ given by the Hopf–Lax formula,*

$$u(t,\mathbf{x}) = \begin{cases} \min\limits_{\mathbf{y}\in\mathbb{R}^N}\left(t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right) + u_0(\mathbf{y})\right) & \text{for } (t,\mathbf{x})\in(0,\infty)\times\mathbb{R}^N \\ u_0(\mathbf{x}) & \text{for } t=0 \text{ and } \mathbf{x}\in\mathbb{R}^N \end{cases} \tag{8.32}$$

has the following properties:

(a) *u is a Lipschitz function, differentiable almost everywhere in $(0,\infty)\times\mathbb{R}^N$;*

(b) *At every point $(t,\mathbf{x})\in(0,\infty)\times\mathbb{R}^N$ where u is differentiable it verifies the Hamilton–Jacobi equation*

$$u_t(t,\mathbf{x}) + \mathcal{H}(\nabla u(t,\mathbf{x})) = 0; \tag{8.33}$$

(c) *u verifies the initial condition*

$$\lim_{t\to 0} u(t,\mathbf{x}) = u_0(\mathbf{x}) \quad \text{for } \mathbf{x}\in\mathbb{R}^N. \tag{8.34}$$

The source of the Hopf–Lax formula is provided by the variational approach of the Hamilton–Jacobi equation (8.33), when accompanied by the initial condition (8.34):

8.7.4 Lemma (The Bellman principle) *The Hopf–Lax formula* (8.32) *represents the solution of the following minimization problem:*

$$u(t,\mathbf{x}) = \inf_{w\in\mathcal{A}(t,\mathbf{x})}\left\{\int_0^t \mathcal{L}(\dot{\mathbf{w}}(s))\mathrm{d}s + u_0(\mathbf{w}(\mathbf{0}))\right\},$$

where $\mathcal{A}(t,\mathbf{x}) = \left\{\mathbf{w} : \mathbf{w}\in C^1\left([0,t],\mathbb{R}^N\right) \text{ and } \mathbf{w}(t)=\mathbf{x}\right\}$.

Proof. Indeed, since the function $\mathbf{w}(s) = \mathbf{y} + s(\mathbf{x}-\mathbf{y})/t$ belongs to $\mathcal{A}(t,\mathbf{x}))$, we have

$$u(t,\mathbf{x}) \le \int_0^t \mathcal{L}(\dot{\mathbf{w}}(s))\mathrm{d}s + u_0(\mathbf{y}) = t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right) + u_0(\mathbf{y}),$$

whence $u(t,\mathbf{x}) \le \inf_{\mathbf{y}\in\mathbb{R}^N}\left(t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right) + u_0(\mathbf{y})\right)$. On the other hand, for every smooth function $\mathbf{w}$ such that $\mathbf{w}(0)=\mathbf{y}$ and $\mathbf{w}(t)=\mathbf{x}$, we infer from Jensen's inequality that

$$\mathcal{L}\left(\frac{1}{t}\int_0^t \dot{\mathbf{w}}(s)\mathrm{d}s\right) \le \frac{1}{t}\int_0^t \mathcal{L}(\dot{\mathbf{w}}(s))\mathrm{d}s.$$

Thus

$$t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right) + u_0(\mathbf{y}) \le \int_0^t \mathcal{L}(\dot{\mathbf{w}}(s))\mathrm{d}s + u_0(\mathbf{y}),$$

which yields $u(t,\mathbf{x}) \ge \inf_{\mathbf{y}\in\mathbb{R}^N}\left(t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right) + u_0(\mathbf{y})\right)$, and thus the equality

$$u(t,\mathbf{x}) = \inf_{\mathbf{y}\in\mathbb{R}^N}\left(t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right) + u_0(\mathbf{y})\right).$$

The fact that this infimum is attained follows from Corollary 5.4.4. ∎

We pass now to the proof of Theorem 8.7.3, that makes the objective of Lemma 8.7.5, and Lemma 8.7.7.

8.7.5 Lemma *The function* $u = u(t,\mathbf{x})$ *given by the Hopf–Lax formula verifies the initial condition* (8.34) *and is Lipschitz with respect to both variables.*

Proof. Notice first that $u(t,\mathbf{x}) \le t\mathcal{L}(\mathbf{0}) + u_0(\mathbf{x})$, that is,

$$u(t,\mathbf{x}) - u_0(\mathbf{x}) \le t\mathcal{L}(\mathbf{0}).$$

On the other hand,

$$\begin{aligned} u(t,\mathbf{x}) &= \min_{y\in\mathbb{R}^N}\left(t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right)+u_0(\mathbf{y})\right)\\ &= u_0(\mathbf{x})+\min_{y\in\mathbb{R}^N}\left(t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right)+u_0(\mathbf{y})-u_0(\mathbf{x})\right)\\ &\geq u_0(\mathbf{x})-\max_{y\in\mathbb{R}^N}\left(\|\mathbf{y}-\mathbf{x}\|\,\mathrm{Lip}(u_0)-t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right)\right)\\ &= u_0(\mathbf{x})-t\max_{\mathbf{z}\in\mathbb{R}^N}\left(\|\mathbf{z}\|\,\mathrm{Lip}(u_0)-\mathcal{L}\left(\mathbf{z}\right)\right)\\ &= u_0(\mathbf{x})-t\max_{\mathbf{v}\in\overline{B}(\mathbf{0},\mathrm{Lip}(u_0))}\left(\max_{\mathbf{z}}\left(\langle\mathbf{v},\mathbf{z}\rangle-\mathcal{L}\left(\mathbf{z}\right)\right)\right)\\ &= u_0(\mathbf{x})-t\max_{\mathbf{v}\in\overline{B}(\mathbf{0},\mathrm{Lip}(u_0))}\mathcal{H}\left(v\right), \end{aligned}$$

where $\overline{B}(\mathbf{0},\mathrm{Lip}(u_0))$ denotes the closed ball centered at the origin and having the radius $\mathrm{Lip}(u_0)$. Therefore

$$|u(t,\mathbf{x})-u_0(\mathbf{x})|\leq Ct \quad \text{ for all } (t,\mathbf{x})\in(0,\infty)\times\mathbb{R}^N,$$

where

$$C=\max\left\{|\mathcal{L}(0)|\,,\max_{\mathbf{v}\in\overline{B}(\mathbf{0},\mathrm{Lip}(u_0))}|\mathcal{H}\left(\mathbf{v}\right)|\right\}.$$

The inside maximum is motivated by the continuity of $\mathcal{H}$.

As concerns the Lipschitzianity of u, let's fix arbitrarily $t,\bar{t}>0$ and points $\mathbf{x},\bar{\mathbf{x}}$ in $\mathbb{R}^N$ and choose $\mathbf{y}$ such that

$$u(t,\mathbf{x})=t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right)+u_0(\mathbf{y}).$$

Then

$$\begin{aligned} \mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right)&=\frac{u(t,\mathbf{x})-u_0(\mathbf{x})}{t}+\frac{u_0(\mathbf{x})-u_0(\mathbf{y})}{t}\\ &\leq\mathcal{L}(\mathbf{0})+\mathrm{Lip}(u_0)\frac{\|\mathbf{x}-\mathbf{y}\|}{t}. \end{aligned}\tag{8.35}$$

Since $\mathcal{L}$ is supercoercive, there is a constant $C_1>0$ (that depends on $\mathcal{L}$ and $\mathrm{Lip}(u_0)$) such that

$$\mathcal{L}(\mathbf{q})\geq\mathcal{L}(\mathbf{0})+\mathrm{Lip}(u_0)\,\|\mathbf{q}\| \quad \text{ for all } \|\mathbf{q}\|>C_1.$$

Combining this fact with the inequality (8.35) we infer that $\|\mathbf{x}-\mathbf{y}\|/t\leq C_1$. Letting $\bar{\mathbf{y}}=\bar{\mathbf{x}}-(\bar{t}/t)\,(\mathbf{x}-\mathbf{y})$, we have

$$\frac{\mathbf{x}-\mathbf{y}}{t}=\frac{\bar{\mathbf{x}}-\bar{\mathbf{y}}}{\bar{t}} \text{ and } \bar{\mathbf{y}}-\mathbf{y}=\bar{\mathbf{x}}-\mathbf{x}+(t-\bar{t})\,\frac{\mathbf{x}-\mathbf{y}}{t}.$$

Hence

$$\begin{aligned} u(\bar{t},\bar{\mathbf{x}}) - u(t,\mathbf{x}) &\le \bar{t}\mathcal{L}\left(\frac{\bar{\mathbf{x}}-\bar{\mathbf{y}}}{\bar{t}}\right) + u_0(\bar{\mathbf{y}}) - t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right) - u_0(\mathbf{y}) \\ &= (\bar{t}-t)\,\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t}\right) + u_0(\bar{\mathbf{y}}) - u_0(\mathbf{y}) \\ &\le |\bar{t}-t| \max_{\|\mathbf{q}\|\le C_1} \mathcal{L}(\mathbf{q}) + \mathrm{Lip}(u_0)\,\|\bar{\mathbf{y}}-\mathbf{y}\| \\ &\le \left(C_1 \mathrm{Lip}(u_0) + \max_{\|\mathbf{q}\|\le C_1} \mathcal{L}(\mathbf{q})\right) |\bar{t}-t| + \mathrm{Lip}(u_0)\,\|\bar{\mathbf{y}}-\mathbf{y}\|\,. \end{aligned}$$

Interchanging the roles of $(t,\mathbf{x})$ and $(\bar{t},\bar{\mathbf{x}})$, we conclude that u is a Lipschitz function. ■

The proof of the assertion (b) of Theorem 8.7.3 needs the following technical result, describing a semigroup property:

8.7.6 Lemma *For each* $\mathbf{x} \in \mathbb{R}^N$ *and* $s \in [0,t)$ *we have*

$$u(t,\mathbf{x}) = \min_{\mathbf{y}\in\mathbb{R}^N} \left\{ (t-s)\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t-s}\right) + u(s,\mathbf{y}) \right\}.$$

Proof. (L. C. Evans [183], p. 125) Choose $\mathbf{z}$ in $\mathbb{R}^N$ such that

$$u(t,\mathbf{x}) = t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{z}}{t}\right) + u_0(\mathbf{z}).$$

For $\mathbf{y} = \frac{s}{t}\mathbf{x} + (1-\frac{s}{t})\mathbf{z}$ we have $(\mathbf{x}-\mathbf{y})/(t-s) = (\mathbf{x}-\mathbf{z})/t = (\mathbf{y}-\mathbf{z})/s$ and thus

$$\begin{aligned} (t-s)\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t-s}\right) + u(s,\mathbf{y}) &\le (t-s)\,\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{z}}{t}\right) + s\mathcal{L}\left(\frac{\mathbf{y}-\mathbf{z}}{s}\right) + u_0(\mathbf{z}) \\ &\le t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{z}}{t}\right) + u_0(\mathbf{z}) = u(t,\mathbf{x}). \end{aligned}$$

This shows that $\min_{\mathbf{y}\in\mathbb{R}^N} \left\{ (t-s)\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t-s}\right) + u(s,\mathbf{y}) \right\} \le u(t,\mathbf{x})$. For the reverse inequality, let's fix arbitrarily $\mathbf{y} \in \mathbb{R}^N$ and $s \in (0,t)$. Choose $\mathbf{w} \in \mathbb{R}^N$ such that

$$u(s,\mathbf{y}) = s\mathcal{L}\left(\frac{\mathbf{y}-\mathbf{w}}{s}\right) + u_0(\mathbf{w}).$$

Since $\mathcal{L}$ is a convex function and

$$\frac{\mathbf{x}-\mathbf{w}}{t} = (1-\frac{s}{t})\frac{\mathbf{x}-\mathbf{y}}{t-s} + \frac{s}{t}\frac{\mathbf{y}-\mathbf{w}}{s},$$

we have

$$\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{w}}{t}\right) \le (1-\frac{s}{t})\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t-s}\right) + \frac{s}{t}\mathcal{L}\left(\frac{\mathbf{y}-\mathbf{w}}{s}\right),$$

which implies that

$$u(t,\mathbf{x}) \le t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{w}}{t}\right) + u_0(\mathbf{w}) \le (t-s)\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t-s}\right) + s\mathcal{L}\left(\frac{\mathbf{y}-\mathbf{w}}{s}\right) + u_0(\mathbf{w})$$
$$= (t-s)\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{y}}{t-s}\right) + u(s,\mathbf{y}).$$

The conclusion is now obvious. ■

Combining Lemma 8.7.5 with Rademacher's theorem (see Appendix D, Theorem D.1.1) we infer that the Hopf–Lax formula yields a function u, which is differentiable almost everywhere.

8.7.7 Lemma *At every point $(t,\mathbf{x}) \in (0,\infty)\times\mathbb{R}^N$ where the function $u = u(t,\mathbf{x})$ defined by the Hopf–Lax formula is differentiable, it verifies the Hamilton–Jacobi equation*

$$u_t(t,\mathbf{x}) + \mathcal{H}(\nabla u(t,\mathbf{x})) = 0.$$

Proof. Let $(t,\mathbf{x}) \in (0,\infty)\times\mathbb{R}^N$ be a point at which the function $u = u(t,\mathbf{x})$ is differentiable. Fix $\mathbf{q}\in\mathbb{R}^N$ and $h>0$. According to Lemma 8.7.6,

$$u(t+h,\mathbf{x}+h\mathbf{q}) = \min_{\mathbf{y}\in\mathbb{R}^N}\left\{h\mathcal{L}\left(\frac{\mathbf{x}+h\mathbf{q}-\mathbf{y}}{h}\right) + u(t,\mathbf{y})\right\}$$
$$\le h\mathcal{L}(\mathbf{q}) + u(t,\mathbf{x}),$$

which implies that

$$\frac{u(t+h,\mathbf{x}+h\mathbf{q}) - u(t,\mathbf{x})}{h} \le \mathcal{L}(\mathbf{q}).$$

Passing to the limit as $h\to 0$, we obtain $\langle\nabla u(t,\mathbf{x}),\mathbf{q}\rangle + u_t(t,\mathbf{x}) \le \mathcal{L}(\mathbf{q})$ (for all $\mathbf{q}\in\mathbb{R}^N$). Therefore

$$u_t(t,\mathbf{x}) + \mathcal{H}(\nabla u(t,\mathbf{x})) = u_t(t,\mathbf{x}) + \max_{\mathbf{q}\in\mathbb{R}^N}\{\langle\nabla u(t,\mathbf{x}),\mathbf{q}\rangle - \mathcal{L}(\mathbf{q})\} \le 0.$$

In order to prove the other inequality, choose $\mathbf{z}$ such that $u(t,\mathbf{x}) = t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{z}}{t}\right) + u_0(\mathbf{z})$. Fix $h>0$ and put $s=t-h$ and $\mathbf{y} = (s/t)\mathbf{x} + (1-s/t)\,\mathbf{z}$. Then $(\mathbf{x}-\mathbf{z})/t = (\mathbf{y}-\mathbf{z})/s$, which yields

$$u(t,\mathbf{x}) - u(s,\mathbf{y}) \ge \left[t\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{z}}{t}\right) + u_0(\mathbf{z})\right] - \left[s\mathcal{L}\left(\frac{\mathbf{y}-\mathbf{z}}{s}\right) + u_0(\mathbf{z})\right]$$
$$= (t-s)\mathcal{L}\left(\frac{\mathbf{x}-\mathbf{z}}{t}\right),$$

equivalently,

$$\frac{u(t,\mathbf{x}) - u\left(t-h,(1-h/t)\mathbf{x} + (h/t)\,\mathbf{z}\right)}{h} \ge \mathcal{L}\left(\frac{\mathbf{x}-\mathbf{z}}{t}\right).$$

Passing to the limit as $h \to 0$, we obtain that

$$\langle \nabla u(t,\mathbf{x}), \frac{\mathbf{x}-\mathbf{z}}{t} \rangle + u_t(t,\mathbf{x}) \geq \mathcal{L}\left(\frac{\mathbf{x}-\mathbf{z}}{t}\right).$$

Hence

$$\begin{aligned} u_t(t,\mathbf{x}) + \mathcal{H}(\nabla u(t,\mathbf{x})) &= u_t(t,\mathbf{x}) + \max_{\mathbf{q}\in\mathbb{R}^N} \{\langle \nabla u(t,\mathbf{x}), \mathbf{q}\rangle - \mathcal{L}(\mathbf{q})\} \\ &\geq u_t(t,\mathbf{x}) + \langle \nabla u(t,\mathbf{x}), \frac{\mathbf{x}-\mathbf{z}}{t}\rangle - \mathcal{L}\left(\frac{\mathbf{x}-\mathbf{z}}{t}\right) \geq 0, \end{aligned}$$

which ends the proof. ■

The proof of Theorem 8.7.3 is now an obvious consequence of Lemma 8.7.5, and Lemma 8.7.7.

As is shown in Exercise 2, the initial-value problem for the Hamilton–Jacobi equation does not have a unique *weak solution*, that is, a solution that verifies the assertions (b) and (c) of Theorem 8.7.3. This situation imposed the consideration of stronger concepts of solution, like *viscosity solution* in the sense of M. G. Crandall and P.-L. Lions. Full details and applications can be found in the books by P. Cannarsa and C. Sinestrari [114] and L. C. Evans [183] and also in the survey by M. G. Crandall, H. Ishii and P.-L. Lions [144].

Exercises

1. Prove that the Hopf–Lax formula can be formulated via the inf-convolution operation as follows:

$$u(t,\mathbf{x}) = \left((t\mathcal{H})^* \,\square u_0\right)(\mathbf{x}).$$

2. Consider the initial-value problem for the Hamilton–Jacobi equation,

$$\begin{aligned} &u_t + |u_x|^2 = 0 \text{ for a.e. } (t,x) \in (0,\infty)\times\mathbb{R} \\ &\lim_{t\to 0} u(t,x) = 0 \text{ for } x \in \mathbb{R}. \end{aligned}$$

   Prove that this problem admits an infinity of solutions.

   [*Hint:* For $\alpha \geq 0$, consider the function $u_\alpha(t,x) = 0$ for $|x| \geq at$ and $u_\alpha(t,x) = \alpha\left|x - \alpha^2 t\right|$ for $|x| \leq at$.]

3. Call a continuous function $f : \mathbb{R}^N \to \mathbb{R}$ *semiconcave* if there is a constant C such that

$$f(\mathbf{x}+\mathbf{z}) - 2f(\mathbf{x}) + f(\mathbf{x}-\mathbf{z}) \leq C\,\|\mathbf{z}\|^2,$$

   for all $\mathbf{x}, \mathbf{z} \in \mathbb{R}^N$. Prove that a function f of class C^2 is semiconcave if, and only if, $\sup_{\mathbf{x}\in\mathbb{R}^N} \left\|d^2 f(\mathbf{x})\right\| < \infty$.

4. Under the hypothesis of Theorem 8.7.3, show that the Hopf–Lax formula yields a semiconcave solution if the initial data is semiconcave. One can prove that this is actually the *only* semiconcave weak solution. See L. C. Evans [183], Theorem 7, p. 132.

8.8 Comments

The notion of duality is one of the central concepts both in geometry and in analysis. As was noticed by S. Artstein-Avidan and V. Milman [24], in many examples of duality, the standard definitions arise, explicitly, from two very simple and natural properties: involution, and order reversion. In particular, this applies to the Legendre–Fenchel conjugation.

8.8.1 Theorem *Consider a map*

$$\mathcal{T} : \Gamma(\mathbb{R}^N) \to \Gamma(\mathbb{R}^N)$$

verifying the following two properties:

(a) $\mathcal{T}\mathcal{T}\varphi = \mathcal{T}\varphi$;

(b) $\varphi \leq \psi$ *implies* $\mathcal{T}\psi \geq T\varphi$.

Then, $\mathcal{T}$ *is essentially the classical Legendre–Fenchel conjugation, namely there exists a constant* $C_0 \in \mathbb{R}$*, a vector* $\mathbf{v}_0 \in \mathbb{R}^N$ *, and an invertible symmetric linear transformation* $B : \mathbb{R}^N \to \mathbb{R}^N$ *such that*

$$(\mathcal{T}\varphi)(\mathbf{x}) = \varphi^*(B\mathbf{x} + \mathbf{v}_0) + \langle \mathbf{x}, \mathbf{v}_0 \rangle + C_0.$$

A nice introduction to the classical theory of Legendre conjugation from the point of view of physicists (as well as its importance to classical mechanics, statistical mechanics, and thermodynamics) can be found in the paper of R. K. P. Zia et al. [601].

A connection between the Legendre–Fenchel conjugation and the Laplace transform is outlined by the tropical analysis. For details, see the paper of G. L. Litvinov et al [344].

Theorem 8.1.2, on biconjugation, is due to W. Fenchel [194] (in the case of Euclidean spaces) and to J. J. Moreau [396] (in the general case). Notice that L. Hörmander [267] proved the theorem of biconjugation in the case of support and indicator functions.

The dual correspondence between the strict convexity of a function and the smoothness of its conjugate, as was noticed in Theorem 8.2.4, admits a much more general version (at the cost of a considerably more involving argument):

8.8.2 Theorem (R. T. Rockafellar [505]) *The following properties are equivalent for a lower semicontinuous proper convex function* $f : \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$:

(a) f *is strictly convex on every convex subset of* $\operatorname{dom} \partial f$ *(in particular, on* $\operatorname{ri}(\operatorname{dom} f)$*);*

(b) f^* *is differentiable on the open convex set* $\operatorname{int}(\operatorname{dom} f^*)$ *which is nonempty, but* $\partial f^*(\mathbf{x}) = \emptyset$ *for all points* $\mathbf{x} \in (\operatorname{dom} f^*) \setminus \operatorname{int}(\operatorname{dom} f^*)$*, if any.*

Moreover, due to Theorem 8.1.2, the roles of f *and* f^* *can be interchanged.*

For details, see R. T. Rockafellar and R. J.-B. Wets [506], Theorem 11.13, pp. 483–484.

The Fenchel–Moreau duality theorem 8.4.2 was proved by W. Fenchel [194] in the case of Euclidean spaces and by J. J. Moreau [398] in the general case.

The development of the theory of optimal transportation led to a natural generalization of the concepts of convexity and duality, by ascribing them to arbitrary cost functions. Some of the historical developments of this problem and some of the basic results are described by W. Gangbo and R. J. McCann [212], L. Rüschendorf [515], S. T. Rachev and L. Rüschendorf [492], and C. Villani [575], [576].

Let X and Y be two nonempty sets and let $c : X \times Y \to \mathbb{R} \cup \{\infty\}$, $c = c(x, y)$, be a function (referred to as a *cost* function). For $c(x, y) = \langle x, y \rangle$ we recover the classical theory. A proper function $\varphi : X \to \mathbb{R} \cup \{\infty\}$ is said to be *c-convex* if it admits a representation of the form

$$\varphi(x) = \sup_{y \in Y} \{c(x, y) + \psi(y)\}$$

for some function ψ. The *c-conjugate* φ^c of φ is defined by the formula

$$\varphi^c(y) = \sup_{x \in X} \{c(x, y) - \varphi(x)\}.$$

Continuing with the *double conjugate*,

$$\varphi^{cc}(y) = \sup_{y \in X} \{c(x, y) - \varphi^c(y)\},$$

we see that φ^c and φ^{cc} are c-convex and, moreover, φ is c-convex if, and only if, $\varphi = \varphi^{cc}$. In this context, the Fenchel–Young inequality take the form

$$\varphi(x) + \varphi^c(y) \geq c(x, y).$$

The concept of *c-subdifferential* (associated to a c-convex φ) is introduced by the formula

$$\partial_c \varphi(x) = \{y : \varphi(z) \geq \varphi(x) + c(z, y) - c(x, y) \text{ for all } z \in \mathrm{dom}\varphi\}.$$

As was noticed by W. Gangbo and R. J. McCann [212], if $c = c(x)$ is locally Lipschitz on $\mathbb{R}^N$, then the $c(x - y)$-convex functions are differentiable almost everywhere; for an extension see the book of C. Villani [576]. This book also includes a proof of the following description of c-concave functions:

8.8.3 Theorem *When $c = c(\mathbf{x})$ is a convex function defined on $\mathbb{R}^N$, then the $c(\mathbf{x}-\mathbf{y})$-concave functions are precisely the viscosity solutions of the Hamilton–Jacobi equation*

$$u_t + c^*(\nabla u) = 0$$

at time $t = 1$ (or, equivalently, at a time $t \geq 1$).

Last, but not least, let us mention that the theory of optimal transportation received a great impetus in the 90s when Y. Brenier used the Monge–Kantorovich formulation of optimal transportation to derive a far-reaching generalization of the polar factorization. Let Ω be an open bounded subset of $\mathbb{R}^N$. A mapping $\mathbf{u} : \Omega \to \mathbb{R}^N$ is said to be *nondegenerate* if the inverse image $\mathbf{u}^{-1}(N)$ of every zero measure set $N \subset \mathbb{R}^N$ has also zero measure.

8.8.4 Theorem (Y. Brenier [97]) *Every nondegenerate vector field*

$$\mathbf{u} \in L^\infty(\Omega, \mathbb{R}^N)$$

can be decomposed as $\mathbf{u}(x) = \nabla\varphi \circ S(x)$ *almost everywhere in* Ω, *where* $\varphi : \mathbb{R}^N \to \mathbb{R}$ *is a convex function and* $S : \bar{\Omega} \to \bar{\Omega}$ *is a measure preserving transformation.*

A map $T\colon \mathbb{R}^N \to \mathbb{R}^N$ (defined μ -almost everywhere) pushes μ forward to ν (equivalently, transports μ onto ν) if $\nu(B) = \mu(T^{-1}(B))$ for every Borel set B in $\mathbb{R}^N$. Y. Brenier [97] found a very special map pushing forward one probability to another. His result was reconsidered by R. J. McCann [373], [374], who noticed that the absolutely continuous Borel probability measures can be transported by maps of the form $T = \nabla\varphi$, where φ is convex. These maps are usually referred to as *Brenier maps*. The differentiability properties of T (motivated by the existence of the Alexandrov Hessian of φ) make an easy handling of T possible. For example, if

$$\mu(B) = \int_B f(\mathbf{x})\,\mathrm{d}\mathbf{x} \quad \text{and} \quad \nu(B) = \int_B g(\mathbf{x})\,\mathrm{d}\mathbf{x},$$

and $T = \nabla\varphi$ is the Brenier map (pushing μ forward to ν), then

$$\int_{\mathbb{R}^N} h(\mathbf{y})g(\mathbf{y})\,\mathrm{d}\mathbf{y} = \int_{\mathbb{R}^N} h(\nabla\varphi(\mathbf{x}))f(\mathbf{x})\,\mathrm{d}\mathbf{x}$$

for all bounded Borel functions $h\colon \mathbb{R} \to \mathbb{R}_+$. Assuming the change of variable $\mathbf{y} = \nabla\varphi(\mathbf{x})$ is working, the last formula leads to the so-called *Monge–Ampère equation*,

$$f(\mathbf{x}) = \det(\nabla^2\varphi(\mathbf{x})) \cdot g(\nabla\varphi(\mathbf{x})). \tag{8.36}$$

As was noted by R. J. McCann [374], this equation is valid in general, provided that $\mathrm{Hess}_x\varphi$ is replaced by the Alexandrov Hessian of φ.

When $N = 1$, a map $T = T(t)$ that transports μ to ν is obtained by defining $T(t)$ as the smallest number such that

$$\int_{-\infty}^{t} f(x)\,\mathrm{d}x = \int_{-\infty}^{T(t)} g(x)\,\mathrm{d}x.$$

This is the key parametrization trick used in the proof of Theorem 4.7.6 (first noticed in the above form by R. Henstock and A. M. Macbeath [250]).



Chapter 9

Special Topics in Majorization Theory

The primary aim of this chapter is to discuss the connection between the Hermite–Hadamard double inequality and Choquet's theory. Noticed first by C. P. Niculescu [420], [421] (during the conference *Inequalities* 2001, in Timişoara), this connection led him to a partial extension of the majorization theory beyond the classical case of probability measures, using the so-called Steffensen–Popoviciu measures. Their main feature is to offer a large framework under which the Jensen–Steffensen inequality remains available. As a consequence, one obtains the extension of the left-hand side of Hermite–Hadamard inequality to a context involving signed Borel measures on arbitrary compact convex sets. A similar extension of the right-hand side of this inequality is known only in dimension 1, the higher dimensional case being still open.

9.1 Steffensen–Popoviciu Measures

The Jensen–Steffensen inequality (see Theorem 2.4.4) reveals an important case when Jensen's inequality works beyond the framework of positive measures. The aim of this section is to provide more insight into the relation between signed measures and Jensen's inequality.

9.1.1 Definition *Let K be a compact convex subset of a real locally convex Hausdorff space E. A Steffensen–Popoviciu measure on K is any real Borel measure μ on K such that $\mu(K) > 0$ and*

$$\int_K f(x)\,\mathrm{d}\mu(x) \geq 0,$$

for every positive, continuous, and convex function $f : K \to \mathbb{R}$.

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C. P. Niculescu and L.-E. Persson, *Convex Functions and Their Applications*,
CMS/CAIMS Books in Mathematics 14, https://doi.org/10.1007/978-3-031-71967-7_9

Clearly, every Borel probability measure is an example of Steffensen–Popoviciu measure.

Since every continuous convex function $f : K \to \mathbb{R}$ is the pointwise supremum of the family of its continuous and affine minorants (see Theorem 4.3.2), it follows that a real Borel measure μ on K is Steffensen–Popoviciu if, and only if, $\mu(K) > 0$ and

$$\int_K (x^*(x) + \alpha)^+ \, \mathrm{d}\mu(x) \geq 0 \quad \text{for all } x^* \in E^* \text{ and } \alpha \in \mathbb{R}. \tag{9.1}$$

The following result (due independently to T. Popoviciu [486], and A. M. Fink [197]) gives us a complete characterization of this class of measures in the case where K is a compact interval.

9.1.2 Theorem *Let μ be a real Borel measure on an interval $[a, b]$ such that $\mu([a, b]) > 0$. Then μ is a Steffensen–Popoviciu measure if, and only if, it verifies the following condition of endpoints positivity:*

$$\int_a^t (t - x) \, \mathrm{d}\mu(x) \geq 0 \text{ and } \int_t^b (x - t) \, \mathrm{d}\mu(x) \geq 0 \tag{9.2}$$

for every $t \in [a, b]$.

Proof. The necessity of (9.2) is clear. For the sufficiency part, notice that any positive, continuous, and convex function $f : [a, b] \to \mathbb{R}$ is the uniform limit of a sequence $(g_n)_n$ of positive, piecewise linear, and convex functions. See Lemma 2.4.1. Or, according to Lemma 2.4.2, every such function g_n can be represented as a finite combination with positive coefficients of functions of the form 1, $(x - t)^+$ and $(t - x)^+$. Thus

$$\int_a^b f(x) \, \mathrm{d}\mu(x) = \lim_{n \to \infty} \int_a^b g_n(x) \, \mathrm{d}\mu(x) \geq 0.$$

■

An alternative argument, based on the integral representation of convex functions on intervals, was noticed by A. M. Fink [197].

9.1.3 Corollary *Suppose that $x_1 \leq \cdots \leq x_n$ are real points and $p_1, \ldots, p_n$ are real weights. Then the discrete measure $\mu = \sum_{k=1}^n p_k \delta_{x_k}$ is a Steffensen–Popoviciu measure if, and only if,*

$$\sum_{k=1}^n p_k > 0, \quad \sum_{k=1}^m p_k (x_m - x_k) \geq 0 \quad \text{and} \quad \sum_{k=m}^n p_k (x_k - x_m) \geq 0 \tag{9.3}$$

for all $m \in \{1, \ldots, n\}$.

9.1.4 Corollary (J. F. Steffensen [550]) *Suppose that* $x_1 \leq \cdots \leq x_n$ *are real points and* $p_1, ..., p_n$ *are real weights. Then, the discrete measure* $\mu = \sum_{k=1}^{n} p_k \, \delta_{x_k}$ *is a Steffensen–Popoviciu measure if*

$$\sum_{k=1}^{n} p_k > 0 \quad \text{and} \quad 0 \leq \sum_{k=1}^{m} p_k \leq \sum_{k=1}^{n} p_k \quad \text{for every } m \in \{1, ..., n\}. \tag{9.4}$$

Proof. Indeed, according to Theorem 9.1.2, we have to prove that (9.4) $\Rightarrow$ (9.3). This follows from Abel's summation formula (the discrete analogue of integration by parts). ■

The following result shows that Steffensen's condition (9.4) is stronger than (9.3).

9.1.5 Theorem *Suppose that* $p : [a, b] \to \mathbb{R}$ *is an integrable function. Then*

$$\int_a^b f(x)p(x)\mathrm{d}x \geq 0$$

for all positive, absolutely continuous, and quasiconvex functions $f : [a, b] \to \mathbb{R}$ *if, and only if,*

$$\int_a^x p(t)\mathrm{d}t \geq 0 \text{ and } \int_x^b p(t)\mathrm{d}t \geq 0 \quad \text{for every } x \in [a, b]. \tag{9.5}$$

The absolutely continuous measures $\mu = p(x)\mathrm{d}x$ that verify $\mu([a, b]) > 0$ and also the condition (9.5) will be referred to as *strong Steffensen–Popoviciu measures.*

Proof. *Sufficiency.* According to Exercise 14 at the end of Section 1.1, there exists a point $c \in [a, b]$ such that f is decreasing on $[a, c]$ and increasing on $[c, b]$. Then

$$\begin{aligned}
\int_a^b f(x)p(x)\mathrm{d}x &= \int_a^c f(x)p(x)\mathrm{d}x + \int_c^b f(x)p(x)\mathrm{d}x \\
&= \int_a^c f(x)\mathrm{d}\left(\int_a^x p(t)\mathrm{d}t\right) - \int_c^b f(x)\mathrm{d}\left(\int_x^b p(t)\mathrm{d}t\right) \\
&= \left[f(x)\int_a^x p(t)\mathrm{d}t\right]\Bigg|_a^c - \int_a^c f'(x)\left(\int_a^x p(t)\mathrm{d}t\right)\mathrm{d}x \\
&\quad - \left[f(x)\int_x^b p(t)\mathrm{d}t\right]\Bigg|_c^b + \int_c^b f'(x)\left(\int_x^b p(t)\mathrm{d}t\right)\mathrm{d}x \\
&= f(c)\int_a^c p(t)\mathrm{d}t + \int_a^c (-f'(x))\left(\int_a^x p(t)\mathrm{d}t\right)\mathrm{d}x \\
&\qquad + f(c)\int_c^b p(t)\mathrm{d}t + \int_c^b f'(x)\left(\int_x^b p(t)\mathrm{d}t\right)\mathrm{d}x \geq 0,
\end{aligned}$$

as a sum of positive numbers. The integration by parts for absolutely continuous functions is motivated by Theorem 18.19, p. 287, in the monograph of E. Hewitt and K. Stromberg [252].

Necessity. Assuming $x_0 \in (a, b)$ and $\varepsilon > 0$ sufficiently small, the function

$$L_\varepsilon(x) = \begin{cases} 1 & \text{if } x \in [a, x_0 - \varepsilon] \\ -(x - x_0)/\varepsilon & \text{if } x \in [x_0 - \varepsilon, x_0] \\ 0 & \text{if } x \in [x_0, b] \end{cases}$$

is positive, decreasing and also absolutely continuous. Therefore

$$\int_a^{x_0} p(t)\mathrm{d}t = \lim_{\varepsilon \to 0} \int_a^b L_\varepsilon(x)p(x)\mathrm{d}x \geq 0.$$

In a similar way one can prove that $\int_{x_0}^b p(t)\mathrm{d}t \geq 0$. ■

Since every continuous convex function $f : [a, b] \to \mathbb{R}$ is absolutely continuous, Theorem 9.1.5 implies that every strong Steffensen–Popoviciu measure is also a Steffensen–Popoviciu measure. However, the inclusion is strict. Indeed, a straightforward computation shows that $\left(x^2 + \lambda\right) \mathrm{d}x$ is an example of a Steffensen–Popoviciu measure on the interval $[-1, 1]$ if, and only if, $\lambda > -1/3$, and it is a strong Steffensen–Popoviciu measure if, and only if, $\lambda \geq -1/4$.

Several other examples of strong Steffensen–Popoviciu measures on $[-1, 1]$ are

$$\left(x^2 - \frac{1}{6}\right) \mathrm{d}x, \quad \left(x^2 \pm \frac{x}{2}\right) \mathrm{d}x, \quad \left(x^4 - \frac{1}{25}\right) \mathrm{d}x, \quad \left(x^2 - \frac{1}{6}\right)^3 \mathrm{d}x.$$

See Exercise 1, at the end of this section, for more examples.

Using the pushing-forward technique of constructing image measures, one can indicate examples of (strong) Steffensen–Popoviciu measures supported by an arbitrarily given compact interval $[a, b]$ (with $a < b$). Notice also that these measures can be glued. For example, if $p(x)\mathrm{d}x$ and $q(x)\mathrm{d}x$ are two Steffensen–Popoviciu measures on the intervals $[a, c]$ and $[c, b]$ respectively, then

$$\left(p(x)\chi_{[a,c]} + q(x)\chi_{[c,b]}\right) \mathrm{d}x$$

is a Steffensen–Popoviciu measure on $[a, b]$.

A simple example of Steffensen–Popoviciu measures in dimension 2 is indicated below.

9.1.6 Example *If $p(x)\mathrm{d}x$ and $q(y)\mathrm{d}y$ are Steffensen–Popoviciu measures on the intervals $[a, b]$ and $[c, d]$ respectively, then $(p(x) + q(y))\,\mathrm{d}x\mathrm{d}y$ is a measure of the same type on $[a, b] \times [c, d]$. In fact, if $f : [a, b] \times [c, d] \to \mathbb{R}$ is a positive convex function, then*

$$x \to \left(\int_c^d f(x, y)\mathrm{d}y\right) \text{ and } y \to \left(\int_a^b f(x, y)\mathrm{d}x\right)$$

are also positive convex functions and thus

$$\int_a^b \int_c^d f(x,y)\,(p(x)+q(y))\,\mathrm{d}y\mathrm{d}x = \int_a^b \left(\int_c^d f(x,y)dy\right) p(x)dx + \int_c^d \left(\int_a^b f(x,y)\mathrm{d}x\right) q(y)\mathrm{d}y \geq 0$$

as a sum of positive numbers.

Using appropriate affine changes of variables, one can construct Steffensen–Popoviciu measures on every quadrilateral convex domain. Here it is worth noticing that the superposition of a convex function and an affine function is also a convex function.

The following result is a two-dimensional analogue of Theorem 9.1.5. It proves the existence of strong Steffensen–Popoviciu measures on compact two-dimensional intervals.

9.1.7 Theorem *Suppose that* $p : [a,b] \to \mathbb{R}$ *and* $q : [c,d] \to \mathbb{R}$ *are continuous functions such that at least one of the following two conditions is fulfilled:*

(i) $p \geq 0$ *and* $0 \leq \int_c^y q(s)\mathrm{d}s \leq \int_c^d q(s)\mathrm{d}s$ *for all* $y \in [c,d]$;

(ii) $q \geq 0$ *and* $0 \leq \int_a^x p(t)\mathrm{d}t \leq \int_a^b p(t)\mathrm{d}t$ *for all* $x \in [a,b]$.

Then

$$\int_a^b \int_c^d f(x,y)p(x)q(y)\mathrm{d}y\mathrm{d}x \geq 0$$

for every positive, continuously differentiable, and quasiconvex function $f : [a,b] \times [c,d] \to \mathbb{R}$.

The condition of continuous differentiability on f can be replaced by that of absolute continuity in the sense of Carathéodory. See J. Šremr [564] for details.

Proof. Assuming (i) (to make a choice), we continue by choosing a continuous path $x \to c^*(x)$ such that

$$f(x,c^*(x)) = \min\{f(x,y) : c \leq y \leq d\} \text{ for each } x \in [a,b].$$

Then $y \to f(x,y)$ is decreasing on $[c,c^*(x)]$ and increasing on $[c^*(x),d]$, which implies

$$\frac{\partial f}{\partial y}(x,y) \leq 0 \text{ on } [c,c^*(x)] \quad \text{and} \quad \frac{\partial f}{\partial y}(x,y) \geq 0 \text{ on } [c^*(x),d].$$

We have

$$
\begin{aligned}
I &= \int_a^b \int_c^d f(x,y)p(x)q(y)\mathrm{d}y\mathrm{d}x \\
&\qquad = \int_a^b \left[\int_c^{c^*(x)} f(x,y)\mathrm{d}\left(\int_c^y q(t)\mathrm{d}t \right) \right] p(x)\mathrm{d}x \\
&\qquad - \int_a^b \left[\int_{c^*(x)}^d f(x,y)\mathrm{d}\left(\int_y^d q(t)\mathrm{d}t \right) \right] p(x)\mathrm{d}x \\
&= \int_a^b \left[f(x,y) \int_c^y q(t)\mathrm{d}t \Big|_c^{c^*(x)} - \int_c^{c^*(x)} \left(\frac{\partial f}{\partial y}(x,y) \int_c^y q(t)\mathrm{d}t \right) \mathrm{d}y \right] p(x)\mathrm{d}x \\
&- \int_a^b \left[f(x,y) \int_y^d q(t)\mathrm{d}t \Big|_{c^*(x)}^d - \int_{c^*(x)}^d \left(\frac{\partial f}{\partial y}(x,y) \int_y^d q(t)\mathrm{d}t \right) \mathrm{d}y \right] p(x)\mathrm{d}x,
\end{aligned}
$$

so that

$$
\begin{aligned}
I &= \int_a^b f(x,c^*(x)) \left(\int_c^{c^*(x)} q(t)\mathrm{d}t \right) p(x)\mathrm{d}x \\
&\qquad + \int_a^b \left[\int_c^{c^*(x)} \left(\frac{-\partial f}{\partial y}(x,y) \int_c^y q(t)\mathrm{d}t \right) \mathrm{d}y \right] p(x)\mathrm{d}x \\
&\qquad + \int_a^b f(x,c^*(x)) \left(\int_{c^*(x)}^d q(t)\mathrm{d}t \right) p(x)\mathrm{d}x \\
&\qquad + \int_a^b \left[\int_{c^*(x)}^d \left(\frac{\partial f}{\partial y}(x,y) \int_y^d q(t)\mathrm{d}t \right) \mathrm{d}y \right] p(x)\mathrm{d}x \\
&= \int_c^d q(t)\mathrm{d}t \int_a^b f(x,c^*(x))p(x)\mathrm{d}x - \int_a^b \left[\int_c^{c^*(x)} \left(\frac{\partial f}{\partial y}(x,y) \int_c^y q(t)\mathrm{d}t \right) \mathrm{d}y \right] p(x)\mathrm{d}x \\
&\qquad + \int_a^b \int_{c^*(x)}^d \left[\left(\frac{\partial f}{\partial y}(x,y) \int_y^d q(t)dt \right) dy \right] p(x)dx \geq 0
\end{aligned}
$$

as a sum of positive numbers. The proof is done. ■

Theorem 9.1.7 works, for example, for

$$
p(x) = 1 \text{ and } q(y) = \left(\frac{2y-c-d}{d-c} \right)^2 - \frac{1}{4},
$$

as well as for

$$
p(x) = \left(\frac{2x-a-b}{b-a} \right)^2 - \frac{1}{4} \text{ and } q(y) = 1.
$$

Combining these two cases we infer that

$$\int_a^b \int_c^d f(x,y) \left(\left(\frac{2x-a-b}{b-a} \right)^2 + \left(\frac{2y-c-d}{d-c} \right)^2 - \frac{1}{2} \right) \mathrm{d}y\mathrm{d}x \geq 0$$

for every positive, continuously differentiable, and quasiconvex function f defined on $[a,b] \times [c,d]$.

Exercises

1. Verify that the following real measures are strong Steffensen–Popoviciu on the indicated interval:

$$\left(4x^3 - 3x\right) \mathrm{d}x \quad \text{on} \quad \left[-\frac{1}{2}\sqrt{3}, 1.5 \right]; \sin x \mathrm{d}x \quad \text{on } [0, 3\pi];$$

$$\left(1 - \frac{1-x}{\pi} - \sin \pi x \right) \mathrm{d}x \quad \text{on } [0,1];$$

$$\left[\left(\frac{2x-a-b}{b-a} \right)^2 + \lambda \right] \mathrm{d}x \quad \text{on } [a,b] \quad (\text{if } \lambda \geq -1/4);$$

$$\left[\left(\frac{2x-a-b}{b-a} \right)^2 - \lambda \frac{2x-a-b}{b-a} \right] \mathrm{d}x \quad \text{on } [a,b] \quad (\text{if } |\lambda| \leq 2/3).$$

2. Suppose that $p(x)\mathrm{d}x$ is a Steffensen–Popoviciu measure on the interval $[a,b]$ and $q(y)\mathrm{d}y$ is a positive measure on the interval $[c,d]$. Prove that $p(x)q(y)\mathrm{d}x\mathrm{d}y$ is a Steffensen–Popoviciu measure on $[a,b] \times [c,d]$. Infer that every compact N-dimensional interval $\prod_{k=1}^N [a_k, b_k]$ carries a Steffensen–Popoviciu measure which is not a positive Borel measure.

3. Suppose that $p(x)\mathrm{d}x$ is a Steffensen–Popoviciu measure on the interval $[0,R]$. Prove that

$$\frac{p(\sqrt{x^2+y^2})}{\sqrt{x^2+y^2}} \mathrm{d}x\mathrm{d}y$$

is a Steffensen–Popoviciu measure on the compact disc

$$\overline{D}_R(0) = \left\{ (x,y) : x^2 + y^2 \leq R \right\}.$$

4. (Steffensen's inequalities [549]) Let $g\colon [a,b] \to \mathbb{R}$ be an integrable function such that $\lambda = \int_a^b g(t)\,\mathrm{d}t \in (0, b-a]$. Then the following two conditions are equivalent:

   (i) $0 \leq \int_a^x g(t)\,\mathrm{d}t \leq x-a$ and $0 \leq \int_x^b g(t)\,\mathrm{d}t \leq b-x$, for all $x \in [a,b]$;

   (ii) $\int_a^{a+\lambda} f(t)\,\mathrm{d}t \leq \int_a^b f(t)g(t)\,\mathrm{d}t \leq \int_{b-\lambda}^b f(t)\,\mathrm{d}t$, for all increasing functions

$f\colon [a,b] \to \mathbb{R}$.

Remark. Condition (i) asserts that $g(x)\mathrm{d}x$ and $(1-g(x))\mathrm{d}x$ are strong Steffensen-Popoviciu measures on $[a,b]$.

5. (A stronger version of Iyengar's inequality [435]) Consider a Riemann integrable function $f\colon [a,b] \to \mathbb{R}$ such that the slopes of the lines AC and CB, joining the endpoints $A(a,f(a))$ and $B(b,f(b))$ to any other point $C(x,f(x))$ of the graph of f, vary between $-M$ and M. Then:

$$\left|\frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x - \frac{f(a)+f(b)}{2}\right| \le \frac{M}{4}(b-a) - \frac{(f(b)-f(a))^2}{4M(b-a)}.$$

[*Hint*: Reduce the problem to the case where f is piecewise linear. Then notice that the function $g(t) = (f'(t)+M)/(2M)$ verifies the condition (a) in the preceding exercise.]

6. A real Borel measure μ on an interval $[a,b]$ is said to be a *dual Steffensen–Popoviciu measure* if $\mu([a,b]) > 0$ and

$$\int_a^b f(x)\,\mathrm{d}\mu(x) \ge 0$$

for every positive, continuous and concave function $f : [a,b] \to \mathbb{R}$. Prove that $\left(\left(\frac{2x-a-b}{b-a}\right)^2 - \frac{1}{6}\right)\mathrm{d}x$ is an example of such a measure. For applications, see C. P. Niculescu [427].

7. (R. Apéry) Let f be a decreasing function on $(0,\infty)$ and g be an integrable function on $[0,\infty)$ such that $0 \le g \le A$ for a suitable strictly positive constant A. Prove that

$$\int_0^\infty f(x)g(x)\,\mathrm{d}x \le A\int_0^\lambda f(x)\,\mathrm{d}x,$$

where $\lambda = (\int_0^\infty g\,\mathrm{d}x)/A$.

9.2 The Barycenter of a Steffensen–Popoviciu Measure

Throughout this section K will denote a (nonempty) compact convex subset of a real locally convex Hausdorff space E, and $C(K)$ will denote the space of all real-valued continuous functions on K. We want to relate the geometry of K with the cone $Cvx(K)$, of all real-valued continuous convex functions defined on K.

The linear space $Cvx(K)-Cvx(K)$ generated by this cone is dense in $C(K)$. In fact, due to the formula

$$\sup\{f_1-g_1, f_2-g_2\} = \sup\{f_1+g_2, f_2+g_1\} - (g_1+g_2),$$

the set $Cvx(K) - Cvx(K)$ is a linear sublattice of $C(K)$ which contains the unit and separates the points of K (since $Cvx(K)$ contains all restrictions to K of the functionals $x^\star \in E^\star$). Thus, the Stone–Weierstrass theorem applies. See [252], Theorem 7.29, p. 94.

We shall also need the space

$$A(K) = Cvx(K) \bigcap -Cvx(K),$$

of all real-valued continuous affine functions on K. This is a rich space, as the following result shows:

9.2.1 Lemma *The space $A(K)$ contains*

$$E^*|_K + \mathbb{R} \cdot 1 = \{x^*|_K + \alpha : x^* \in E^* \text{ and } \alpha \in \mathbb{R}\}$$

as a dense subspace.

Proof. Let $f \in A(K)$ and $\varepsilon > 0$. The following two subsets of $E \times \mathbb{R}$,

$$J_1 = \{(\mathbf{x}, f(\mathbf{x})) : \mathbf{x} \in K\}$$

and

$$J_2 = \{(\mathbf{x}, f(\mathbf{x}) + \varepsilon) : \mathbf{x} \in K\},$$

are nonempty, compact, convex, and disjoint. By a geometric version of the Hahn–Banach theorem (see Theorem B.2.4), there exists a continuous linear functional L on $E \times \mathbb{R}$ and a number $\lambda \in \mathbb{R}$ such that

$$\sup L(J_1) < \lambda < \inf L(J_2).$$

Hence the equation $L(\mathbf{x}, g(\mathbf{x})) = \lambda$ defines an element $g \in E^*|_K + \mathbb{R} \cdot 1$ such that

$$f(\mathbf{x}) < g(\mathbf{x}) < f(\mathbf{x}) + \varepsilon \quad \text{for all } \mathbf{x} \in K.$$

In fact, $\lambda = L(\mathbf{x}, 0) + g(\mathbf{x})L(0, 1)$, and thus $g(\mathbf{x}) = (\lambda - L(\mathbf{x}, 0))/L(0, 1)$. This solves the approximation (within ε) of f by elements of $E^*|_K + \mathbb{R} \cdot 1$. ■

The following example shows that the inclusion $E^*|_K + \mathbb{R} \cdot 1 \subset A(K)$ may be strict. For this, consider the set,

$$S = \{(a_n)_n : |a_n| \leq 1/n^2 \text{ for every } n\},$$

viewed as a subset of ℓ^2 endowed with the weak topology. Then S is compact and convex and the function $f((a_n)_n) = \sum_n a_n$ defines an element of $A(S)$. Moreover, $f(0) = 0$. However, there is no y in ℓ^2 such that $f(x) = \langle x, y \rangle$ for all $x \in S$.

Given a function f in $C(K)$, one can attach to it a *lower envelope*,

$$\check{f}(\mathbf{x}) = \sup\{h(\mathbf{x}) : h \in A(K) \text{ and } f \geq h\},$$

and an upper envelope,

$$\widehat{f}(\mathbf{x}) = \inf\{h(\mathbf{x}) : h \in A(K) \text{ and } h \geq f\}.$$

They are related by formulas of the form

$$\check{f} = -\widehat{(-f)},$$

so it suffices to investigate the properties of one type of envelope, say the upper ones:

9.2.2 Lemma *The upper envelope $\widehat{f}$ of a function $f \in C(K)$ is concave, bounded, and upper semicontinuous. Moreover:*

(a) *$f \leq \widehat{f}$ and $f = \widehat{f}$ if f is concave;*

(b) *if $f, g \in C(K)$, then $\widehat{f+g} \leq \widehat{f} + \widehat{g}$ with equality if $g \in A(K)$; also, $\widehat{\alpha f} = \alpha \widehat{f}$ if $\alpha \geq 0$;*

(c) *the map $f \to \widehat{f}$ is nonexpansive in the sense that $|\widehat{f} - \widehat{g}| \leq \|f - g\|$.*

Proof. Most of the proof of this lemma follows directly from the definitions. We shall concentrate here on the less obvious assertion, namely the second part of (a). It may be proved by reductio ad absurdum. Assume that $f(\mathbf{x}_0) < \widehat{f}(\mathbf{x}_0)$ for some $\mathbf{x}_0 \in K$. By Theorem B.2.4, there exists a closed hyperplane which strictly separates the convex sets $K_1 = \{(\mathbf{x}_0, \widehat{f}(\mathbf{x}_0))\}$ and $K_2 = \{(\mathbf{x}, r) : f(\mathbf{x}) \geq r\}$. This means the existence of a continuous linear functional L on $E \times \mathbb{R}$ and of a scalar λ such that

$$\sup_{(\mathbf{x},r)\in K_2} L(\mathbf{x}, r) < \lambda < L(\mathbf{x}_0, \widehat{f}(\mathbf{x}_0)). \tag{9.6}$$

Then $L(\mathbf{x}_0, \widehat{f}(\mathbf{x}_0)) > L(\mathbf{x}_0, f(\mathbf{x}_0))$, which yields $L(0, 1) > 0$. The function

$$h(\mathbf{x}) = \frac{\lambda - L(\mathbf{x}, 0)}{L(0, 1)}$$

belongs to $A(K)$ and $L(\mathbf{x}, h(\mathbf{x})) = \lambda$ for all $\mathbf{x}$. By (9.6), we infer that $h > f$ and $h(\mathbf{x}_0) < \widehat{f}(\mathbf{x}_0)$, a contradiction. ∎

The connection between the points of a compact convex set K and the positive functionals on $C(K)$ is made visible through the concept of barycenter. In what follows we will show that this concept can be attached not only to probability Borel measures but also to any Steffensen–Popoviciu measure.

As above, K denotes a compact convex set in a real locally convex Hausdorff space E.

9.2.3 Lemma *For every Steffensen–Popoviciu measure μ on K there exists and is unique a point* $\mathrm{bar}(\mu)$ *in K such that*

$$x^*(\mathrm{bar}(\mu)) = \frac{1}{\mu(K)} \int_K x^*(\mathbf{x})\, \mathrm{d}\mu(\mathbf{x}) \tag{9.7}$$

for all continuous linear functionals x^* on E.

The point $\mathrm{bar}(\mu)$ *is called the barycenter of μ.*

The uniqueness of the point $\operatorname{bar}(\mu)$ is a consequence of the separability of the topology of E. See Corollary B.1.5. In the case of the Euclidean spaces the norm and the weak convergence coincide, so the formula (9.7) reduces to

$$\operatorname{bar}(\mu) = \frac{1}{\mu(K)} \int_K \mathbf{x}\, \mathrm{d}\mu(\mathbf{x}).$$

Due to Lemma 9.2.1, the equality (9.7) extends to all $f \in A(K)$.

Proof of Lemma 9.2.3. We have to prove that

$$\Big(\bigcap_{f \in E^*} H_f \Big) \cap K \neq \emptyset,$$

where $\mu(f)$ denotes the integral of f with respect to μ and H_f denotes the closed hyperplane $\{\mathbf{x} : f(\mathbf{x}) = \mu(f)/\mu(K)\}$ associated to $f \in E^*$. Since K is compact, it suffices to prove that

$$\Big(\bigcap_{k=1}^{n} H_{f_k} \Big) \cap K \neq \emptyset$$

for every finite family $f_1, \ldots, f_n$ of functionals in E^*. Equivalently, attaching to each such family $f_1, \ldots, f_n$ of functionals the operator

$$T\colon K \to \mathbb{R}^n, \quad T(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_n(\mathbf{x})),$$

we have to prove that $T(K)$ contains the point $\mathbf{p} = \frac{1}{\mu(K)}(\mu(f_1), \ldots, \mu(f_n))$. In fact, if $\mathbf{p} \notin T(K)$, then a separation argument yields a point $\mathbf{a} = (a_1, \ldots, a_n)$ in $\mathbb{R}^n$ such that

$$\langle \mathbf{p}, \mathbf{a} \rangle > \sup_{\mathbf{x} \in K} \langle T(\mathbf{x}), \mathbf{a} \rangle,$$

that is,

$$\frac{1}{\mu(K)} \sum_{k=1}^{n} a_k \mu(f_k) > \sup_{\mathbf{x} \in K} \sum_{k=1}^{n} a_k f_k(\mathbf{x}).$$

Then $g = \sum_{k=1}^{n} a_k f_k$ will provide an example of a continuous affine function on K for which $\mu(g)/\mu(K) > \sup_{\mathbf{x} \in K} g(\mathbf{x})$, a fact which contradicts the property of μ to be a Steffensen–Popoviciu measure. ■

The extension of Jensen's inequality to the context of Steffensen-Popoviciu measure is straightforward.

9.2.4 Theorem (The generalized Jensen–Steffensen inequality) *Suppose that μ is a Steffensen–Popoviciu measure on the compact convex set K. Then*

$$f(\operatorname{bar}(\mu)) \leq \frac{1}{\mu(K)} \int_K f(\mathbf{x})\, \mathrm{d}\mu(\mathbf{x})$$

for all continuous convex functions $f\colon K \to \mathbb{R}$.

Proof. Indeed, according to Lemma 9.2.2,

$$\begin{aligned} f(\mathrm{bar}(\mu)) &= \sup\{h(\mathrm{bar}(\mu)) : h \in A(K),\ h \le f\} \\ &= \sup\left\{ \frac{1}{\mu(K)} \int_K h \,\mathrm{d}\mu \;\middle|\; h \in A(K),\ h \le f \right\} \\ &\le \frac{1}{\mu(K)} \int_K f \,\mathrm{d}\mu. \end{aligned}$$

■

The classical Jensen–Steffensen inequality (see Theorem 2.4.4) represents the particular case where μ is a strong Steffensen–Popoviciu measure on a compact interval.

We end this section with a monotonicity result.

9.2.5 Proposition *Suppose that μ is a Borel probability measure on K and $f\colon K \to \mathbb{R}$ is a continuous convex function. Then the function*

$$M(t) = \int_K f(t\mathbf{x} + (1-t)\mathrm{bar}(\mu)) \,\mathrm{d}\mu(\mathbf{x})$$

is convex and increasing on $[0, 1]$.

When $E = \mathbb{R}^n$ and μ is the Lebesgue measure, the value of M at $t > 0$ equals the arithmetic mean of $f|_{K_t}$, that is,

$$M(t) = \frac{1}{\mu(K_t)} \int_{K_t} f(\mathbf{x}) \,\mathrm{d}\mu(\mathbf{x});$$

here K_t denotes the image of K through the mapping $x \to tx + (1-t)\mathrm{bar}(\mu)$. Proposition 9.2.5 tells us that the arithmetic mean of $f|_{K_t}$ decreases to $f(\mathrm{bar}(\mu))$ when K_t shrinks to $\mathrm{bar}(\mu)$. The proof is based on the following approximation argument:

9.2.6 Lemma (Barycentric approximation) *Every Borel probability measure μ on K is the pointwise limit of a net of discrete Borel probability measures μ_α, each having the same barycenter as μ.*

Lemma 9.2.6 yields an immediate derivation of Jensen's integral inequality from Jensen's discrete inequality.

Proof. We have to prove that for each $\varepsilon > 0$ and each finite family $f_1, \ldots, f_n$ of continuous real functions on K there exists a discrete Borel probability measure ν such that

$$\mathrm{bar}(\nu) = \mathrm{bar}(\mu) \quad \text{and} \quad \sup_{1 \le k \le n} |\nu(f_k) - \mu(f_k)| < \varepsilon.$$

As K is compact and convex and the functions f_k are continuous, there exists a finite covering $(D_\alpha)_\alpha$ of K by open convex sets such that the oscillation of

each of the functions f_k on each set D_α is less than ε. Let $(\varphi_\alpha)_\alpha$ be a partition of unity, subordinated to the covering $(D_\alpha)_\alpha$ and put

$$\nu = \sum_\alpha \mu(\varphi_\alpha)\delta_{\mathbf{x}(\alpha)},$$

where $\mathbf{x}(\alpha)$ is the barycenter of the measure $f \to \mu(\varphi_\alpha f)/\mu(\varphi_\alpha)$. As D_α is convex and the support of φ_α is included in D_α, we have $\mathbf{x}(\alpha) \in \overline{D}_\alpha$. On the other hand,

$$\mu(h) = \sum_\alpha \mu(h\varphi_\alpha) = \sum_\alpha \frac{\mu(h\varphi_\alpha)}{\mu(\varphi_\alpha)}\,\mu(\varphi_\alpha) = \sum_\alpha h(\mathbf{x}(\alpha))\,\mu(\varphi_\alpha) = \nu(h)$$

for all continuous affine functions $h\colon K \to \mathbb{R}$. Consequently, μ and ν have the same barycenter. Finally, for each k,

$$\begin{aligned} |\nu(f_k) - \mu(f_k)| &= \Big|\sum_\alpha \mu(\varphi_\alpha) f_k(\mathbf{x}(\alpha)) - \sum_\alpha \mu(\varphi_\alpha f_k)\Big| \\ &= \Big|\sum_\alpha \mu(\varphi_\alpha)\Big[f_k(\mathbf{x}(\alpha)) - \frac{\mu(\varphi_\alpha f_k)}{\mu(\varphi_\alpha)}\Big]\Big| \\ &\le \varepsilon \cdot \sum_\alpha \mu(\varphi_\alpha) = \varepsilon. \end{aligned}$$

■

Proof of Proposition 9.2.5. A straightforward computation shows that $M(t)$ is convex and $M(t) \le M(1)$. Then, assuming the inequality $M(0) \le M(t)$, from the convexity of $M(t)$ we infer that

$$\frac{M(t) - M(s)}{t - s} \ge \frac{M(s) - M(0)}{s} \ge 0$$

for all $0 \le s < t \le 1$ that is, $M(t)$ is increasing. To end the proof, it remains to show that $M(t) \ge M(0) = f(\mathrm{bar}(\mu))$. For this, choose a net $(\mu_\alpha)_\alpha$ of discrete Borel probability measures on K as in Lemma 9.2.6 above. By Lemma 1.1.11,

$$f(\mathrm{bar}(\mu)) \le \int_K f(t\mathbf{x} + (1-t)\mathrm{bar}(\mu))\,\mathrm{d}\mu_\alpha(\mathbf{x}) \quad \text{for all } \alpha$$

and thus the desired conclusion follows by passing to the limit over α. ■

The question whether Lemma 9.2.6 remains true in the context of Steffensen–Popoviciu measures is open.

Exercises

1. Suppose that μ is a Steffensen–Popoviciu measure on a compact convex set K. Prove that

   $$\|\mu|_{A(K)}\| = \mu(K).$$

   Then, consider the particular case where $K = [-1, 1]$ and $\mathrm{d}\mu = (x^2 + a)\mathrm{d}x$ to show that $\mu/\mu(K)$, as a functional on $C(K)$, can have a norm larger than 1.

2. Suppose that $\mathrm{d}\mu = p(x)\mathrm{d}x$ is an absolutely continuous Steffensen–Popoviciu measure on an interval $[a,b]$ and let $T : [a,b] \to [a,b]$ be a measurable transformation such that $\mu(T^{-1}(A)) = \mu(A)$ for every measurable set A; in other words, μ is T-invariant. Is the barycenter of μ necessarily a fixed point of T?

3. (Barycenter from symmetry) The measure $(x^2 - 1/6)\mathrm{d}x$ defined on $[-1,1]$ is symmetric with respect to the origin and thus this point is its barycenter. Apply a similar argument to find the barycenter of the Steffensen–Popoviciu measure $(x^2 - 1/6)(y + 1/y)\mathrm{d}x\mathrm{d}y$ defined on $[-1,1] \times [1/2,2]$.

4. Suppose that μ is a strong Steffensen–Popoviciu measure on an interval $[a,b]$. Does the generalized Jensen–Steffensen inequality work for all continuous quasiconvex functions defined on $[a,b]$?

9.3 Majorization via Choquet Order

Measure majorization in the setting of Steffensen–Popoviciu measures extends the case of discrete Borel probability measures, already discussed in Section 6.8.

9.3.1 Definition *Given two Steffensen–Popoviciu measures μ and ν on a compact convex set K, we say that μ is majorized by ν in Choquet order (abbreviated, $\mu \prec_{Ch} \nu$) if*

$$\frac{1}{\mu(K)} \int_K f(\mathbf{x})\, \mathrm{d}\mu(\mathbf{x}) \leq \frac{1}{\nu(K)} \int_K f(\mathbf{x})\, \mathrm{d}\nu(\mathbf{x})$$

for all continuous convex functions $f \colon K \to \mathbb{R}$.

The relation $\prec_{Ch}$ is indeed a partial order relation. Clearly, it is transitive and reflexive; the fact that $\mu \prec_{Ch} \nu$ and $\nu \prec_{Ch} \mu$ imply $\mu = \nu$ follows from the density of $Cvx(K) - Cvx(K)$ in $C(K)$.

If f is in $A(K)$, then both f and $-f$ are in $Cvx(K)$, so that $\mu/\mu(K)$ and $\nu/\nu(K)$ coincide on $A(K)$. As a consequence, they have the same barycenter.

As we will show in the next section, the Choquet order detects when one probability measure is more diffuse than another one in the sense that its support lies closer to the extreme boundary $\operatorname{ext} K$.

9.3.2 Lemma *Suppose that μ and ν are two Steffensen–Popoviciu measures such that $\mu(K) = \nu(K) = 1$ and*

$$\int_K f(\mathbf{x})\, \mathrm{d}\mu(\mathbf{x}) \leq \int_K f(\mathbf{x})\, \mathrm{d}\nu(\mathbf{x})$$

for all functions $f \in Cvx(K)$, $f \geq 0$. Then $\mu \prec_{Ch} \nu$.

In particular, $\delta_{\mathrm{bar}(\nu)} \prec_{Ch} \nu$ and $\delta_{\mathrm{bar}(\nu)}$ is the smallest Steffensen–Popoviciu measure of total mass 1 that is majorized by ν.

Proof. The trick is to observe that $\lambda_{\varepsilon,\mathbf{z}} = \nu - \mu + \varepsilon\delta_{\mathbf{z}}$ is a Steffensen–Popoviciu measure, whenever $\varepsilon > 0$ and $\mathbf{z} \in K$. According to the Jensen–Steffensen inequality, if $f \in Cvx(K)$, then

$$\int_K f(\mathbf{x})\,\mathrm{d}\nu(\mathbf{x}) - \int_K f(\mathbf{x})\,\mathrm{d}\mu(\mathbf{x}) + \varepsilon f(\mathbf{z}) \geq \varepsilon f(\mathrm{bar}(\lambda_{\varepsilon,\mathbf{z}})) \geq \varepsilon \min_{\mathbf{x}\in K} f(\mathbf{x}),$$

whence by letting $\varepsilon \to 0$ we infer that

$$\int_K f(\mathbf{x})\,\mathrm{d}\nu(\mathbf{x}) - \int_K f(\mathbf{x})\,\mathrm{d}\mu(\mathbf{x}) \geq 0.$$

■

The following variant of Lemma 9.3.2 in the case of Stieltjes measures is due to S. Karlin and A. Novikoff [288]:

9.3.3 Lemma *Let $F\colon [a,b] \to \mathbb{R}$ be a function with bounded variation such that $F(a) = 0$. Then*

$$\int_a^b f(x)\,\mathrm{d}F(x) \geq 0 \quad \textit{for all } f \in Cvx(K)$$

if, and only if, the following three conditions are fulfilled:

$$F(b) = 0, \quad \int_a^b F(x)\,\mathrm{d}x = 0, \quad \textit{and} \quad \int_a^x F(t)\,\mathrm{d}t \geq 0 \quad \textit{for all } x \in (a,b).$$

Proof. Via an approximation argument, we may restrict to the case where f is also piecewise linear. Then, by using twice integration by parts, we get

$$\int_a^b f(x)\,\mathrm{d}F(x) = -\int_a^b F(x)f'(x)\,\mathrm{d}x = \int_a^b \left(\int_a^x F(t)\,\mathrm{d}t\right) f''(x)\,\mathrm{d}x,$$

whence the sufficiency part is proved. For the necessity, notice that $\int_a^x F(t)\,\mathrm{d}t < 0$ for some $x \in (a,b)$ yields an interval I around x on which the integral is still negative. Choosing f such that $f'' = 0$ outside I, the above equalities lead to a contradiction. The necessity of the other two conditions follows by checking our statement for $f = 1$, -1, $x - a$, and $a - x$ (in this order). ■

9.3.4 Corollary *Let $F, G\colon [a,b] \to \mathbb{R}$ be two functions with bounded variation such that $F(a) = G(a)$. Then, in order that*

$$\int_a^b f(x)\,\mathrm{d}F(x) \leq \int_a^b f(x)\,\mathrm{d}G(x)$$

for all continuous convex functions $f\colon [a,b] \to \mathbb{R}$, it is necessary and sufficient that F and G verify the following three conditions:

$$\begin{aligned} F(b) &= G(b); \\ \int_a^x F(t)\,\mathrm{d}t \le \int_a^x G(t)\,\mathrm{d}t &\quad \textit{for all } x \in (a,b); \textit{ and} \\ \int_a^b F(t)\,\mathrm{d}t &= \int_a^b G(t)\,\mathrm{d}t. \end{aligned}$$

9.3.5 Corollary *Let $f, g \in L^1[a,b]$. Then $f\mathrm{d}x \prec_{Ch} g\mathrm{d}x$ if and only if the following conditions are fulfilled:*

$$\begin{aligned} &\int_a^b f(x)\,\mathrm{d}x = \int_a^b g(x)\,\mathrm{d}x; \quad \int_a^b xf(x)\,\mathrm{d}x = \int_a^b xg(x)\,\mathrm{d}x; \\ &\int_a^x (x-t)f(t)\,\mathrm{d}t \le \int_a^x (x-t)g(t)\,\mathrm{d}t, \quad \textit{for all } x \in [a,b]. \end{aligned}$$

A generalization of the Karlin–Novikoff Lemma 9.3.3 to the case of convex functions of higher order was obtained by T. Rajba in her paper [496].

Exercises

1. A diffusion process from the middle point across the unit interval is driven by the following stochastic process:

$$P_t = (1-t)\delta_{1/2} + \frac{1}{2}\chi_{[1/2-t,1/2+t]} \text{ for } t \in [0,1].$$

   Prove that $P_t \prec_{Ch} P_s$, whenever $0 \le s \le t \le 1$.

2. Let μ and ν be two signed Borel measures on $[a,b]$ such that $\mu([a,b]) = \nu([a,b]) = 1$. Prove that $\mu \prec_{Ch} \nu$ if, and only if,

$$\int_a^t (t-x)\,\mathrm{d}\mu(x) \le \int_a^t (t-x)\,\mathrm{d}\nu(x) \quad \text{and} \quad \int_t^b (x-t)\,\mathrm{d}\mu(x) \le \int_t^b (x-t)\,\mathrm{d}\nu(x)$$

   for all $t \in [a,b]$.

3. Let $p\colon [0,1] \to R$ be a continuous function which is increasing on $[0,1/2]$ and satisfies the condition $f(x) = f(1-x)$. Prove that

$$\int_0^1 f(x)p(x)\,\mathrm{d}x \le \left(\int_0^1 f(x)\,\mathrm{d}x\right)\left(\int_0^1 p(x)\,\mathrm{d}x\right)$$

   for all $f \in Cvx\,([0,1])$. Infer that

$$\int_0^1 x(1-x)f(x)\,\mathrm{d}x \le \frac{1}{6}\int_0^1 f(x)\,\mathrm{d}x$$

and

$$\int_0^\pi f(x)\sin x\,\mathrm{d}x \le \frac{2}{\pi}\int_0^\pi f(x)\,\mathrm{d}x,$$

provided that f is convex on appropriate intervals.

[*Hint*: It suffices to verify the conditions of Corollary 9.3.5 for $f = p$ and $g = \int_0^1 p(x)\,\mathrm{d}x$. The third condition in Corollary 9.3.5 reads as

$$\frac{x^2}{2}\int_0^1 p(t)\,\mathrm{d}t \ge \int_0^x \int_0^t p(s)\,\mathrm{d}s\mathrm{d}t \quad \text{for all } x \in [0,1].$$

For $x \in [0, 1/2]$, we have to observe that $\int_0^x p(t)\,\mathrm{d}t$ is a convex function on $[0, 1/2]$, which yields

$$\frac{1}{x}\int_0^x p(t)\,\mathrm{d}t \le 2\int_0^{1/2} p(t)\,\mathrm{d}t = \int_0^1 p(t)\,\mathrm{d}t \quad \text{for all } x \in [0,1/2]. \quad]$$

9.4 Choquet's Theorem

The aim of this section is to present a full extension of the Hermite–Hadamard inequality (1.21) to the framework of continuous convex functions defined on arbitrary compact convex spaces (when the mean values are computed via Borel probability measures). This is done by combining the generalized Jensen–Steffensen inequality (see Theorem 9.2.4) with the majorization theorems of G. Choquet, E. Bishop, and K. de Leeuw. We start with the metrizable case, following the classical approach initiated by G. Choquet [131].

9.4.1 Theorem (Choquet's theorem: the metrizable case) *Let μ be a Borel probability measure on a metrizable compact convex subset K of a locally convex Hausdorff space E. Then there exists a Borel probability measure λ on K such that the following two conditions are verified:*

(a) $\mu \prec_{Ch} \lambda$ *and λ and μ have the same barycenter;*

(b) *the set* $\operatorname{ext} K$*, of all extreme points of K, is a Borel set and λ is concentrated on* $\operatorname{ext} K$ *(that is, $\lambda(K \backslash \operatorname{ext} K) = 0$).*

9.4.2 Corollary (Choquet's representation) *Given a point $\boldsymbol{x} \in K$, there exists a Borel probability measure λ concentrated on* $\operatorname{ext} K$ *such that for all affine functions f defined on K we have*

$$f(\mathbf{x}) = \int_{\operatorname{ext} K} f(\mathbf{u})\,\mathrm{d}\lambda.$$

For the proof, apply Theorem 9.4.1 to $\mu = \delta_{\mathbf{x}}$. Corollary 9.4.2 provides an infinite-dimensional extension of Minkowski's Theorem 3.3.5.

9.4.3 Corollary (The generalized Hermite–Hadamard inequality; the metrizable case) *Under the hypotheses of Theorem 9.4.1, we have*

$$f(\mathrm{bar}(\mu)) \le \int_K f(\mathbf{x})\,\mathrm{d}\mu \le \int_{\mathrm{ext}\,K} f(\mathbf{u})\,\mathrm{d}\lambda \tag{9.8}$$

for every continuous convex function $f\colon K \to \mathbb{R}$.

Notice that the right part of (9.8) reflects the maximum principle for convex functions.

In general, the measure λ is not unique, except for the case of simplices; see the book of R. R. Phelps [474], Section 10, for details.

As was noticed in Theorem 9.2.4, the left-hand side inequality in the formula (9.8) works in the more general framework of Steffensen–Popoviciu measures. This is no longer true for the right-hand side inequality. In fact, $\mu = \delta_{-1} - \delta_0 + \delta_1$ provides an example of Steffensen-Popoviciu measure on $K = [-1, 1]$ which is not majorized by any Steffensen–Popoviciu measure concentrated on $\mathrm{ext}\,K$ (that is, by any convex combination of δ_{-1} and δ_1). However, as will be shown in Section 9.6, the Hermite–Hadamard inequality remains valid for a subclass of Steffensen–Popoviciu measures, still larger than the class of probability measures.

Proof of Theorem 9.4.1. This will be done in four steps.

Step 1. We start by proving that $\mathrm{ext}\,K$ is a countable intersection of open sets (and thus it is a Borel set). Here the assumption on metrizability is essential. Suppose that the topology of K is given by the metric d and for each integer $n \ge 1$ consider the set

$$K_n = \Big\{x : \mathbf{x} = \frac{\mathbf{y}+\mathbf{z}}{2},\ \text{with } \mathbf{y}, \mathbf{z} \in K \text{ and } d(\mathbf{y}, \mathbf{z}) \ge 1/2^n\Big\}.$$

Clearly, $\mathrm{ext}\,K = K \setminus \bigcup_n K_n$ and an easy compactness argument shows that each K_n is closed. Consequently, $\mathrm{ext}\,K = \bigcap_n \complement K_n$ is a countable intersection of open sets.

Step 2. We may choose a maximal Borel probability measure $\lambda \succ \mu$. To show that Zorn's lemma may be applied, consider a chain $C = (\lambda_\alpha)_\alpha$ in

$$\mathcal{P} = \{\lambda : \lambda \succ \mu, \lambda \text{ Borel probability measure on } K\}.$$

As $(\lambda_\alpha)_\alpha$ is contained in the weak-star compact set

$$\{\lambda : \lambda \in C(K)^*,\ \lambda \ge 0,\ \lambda(1) = 1\},$$

by a compactness argument we may find a subnet $(\lambda_\beta)_\beta$ which converges to a measure $\tilde{\lambda}$ in the weak-star topology. A moment's reflection shows that $\tilde{\lambda}$ is an upper bound for C. Consequently, we may apply Zorn's lemma to choose a maximal Borel probability measure $\lambda \succ \mu$. It remains to prove that λ does the job.

Step 3. Since K is metrizable, it follows that $C(K)$ (and thus $A(K)$) is separable. This is a consequence of Urysohn's lemma in general topology. See for example, [252], Theorem 6.80, p. 75. Every sequence $(h_n)_n$ of affine functions with $\|h_n\| = 1$, which is dense in the unit sphere of $A(K)$, separates the points of K in the sense that for every $\mathbf{x} \neq \mathbf{y}$ in K there is an h_n such that $h_n(\mathbf{x}) \neq h_n(\mathbf{y})$. Consequently, the function

$$\varphi = \sum_{n=1}^{\infty} 2^{-n} h_n^2$$

is continuous and strictly convex, from which it follows that

$$\mathcal{E} = \{\mathbf{x} : \varphi(\mathbf{x}) = \widehat{\varphi}(\mathbf{x})\} \subset \operatorname{ext} K.$$

In fact, if $\mathbf{x} = (\mathbf{y} + \mathbf{z})/2$, where $\mathbf{y}$ and $\mathbf{z}$ are distinct points of K, then the strict convexity of φ implies that

$$\varphi(\mathbf{x}) < \frac{\varphi(\mathbf{y}) + \varphi(\mathbf{z})}{2} \leq \frac{\widehat{\varphi}(\mathbf{y}) + \widehat{\varphi}(\mathbf{z})}{2} \leq \widehat{\varphi}(\mathbf{x}).$$

Step 4. As a consequence of the maximality of λ, we shall show that

$$\lambda(\varphi) = \lambda(\widehat{\varphi}). \tag{9.9}$$

Then $\widehat{\varphi} - \varphi \geq 0$ and $\lambda(\widehat{\varphi} - \varphi) = 0$, which yields $\lambda(\{\mathbf{x} : \varphi(\mathbf{x}) = \widehat{\varphi}(\mathbf{x})\}) = 0$. Hence λ is concentrated on E. The proof of (9.9) is based on Lemma 9.2.2. Consider the sublinear functional $q\colon C(K) \to \mathbb{R}$, given by $q(f) = \lambda(\hat{f})$, and the linear functional L defined on $A(K) + \mathbb{R} \cdot \varphi$ by $L(h + \alpha\varphi) = \lambda(h) + \alpha\lambda(\widehat{\varphi})$. By Lemma 9.2.2, if $\alpha \geq 0$, then $L(h + \alpha\varphi) = q(h + \alpha\varphi)$, while if $\alpha < 0$, then

$$0 = \widehat{\alpha\varphi - \alpha\varphi} \leq \widehat{\alpha\varphi} + \widehat{(-\alpha\varphi)} = \widehat{\alpha\varphi} - \alpha\widehat{\varphi},$$

which yields

$$L(h + \alpha\varphi) = \lambda(h + \alpha\widehat{\varphi}) \leq \lambda(\widehat{h + \alpha\varphi}) = q(h + \alpha\varphi).$$

By the Hahn–Banach extension theorem, there exists a linear extension ω of L to $C(K)$ such that $\omega \leq q$. If $f \leq 0$, then $\hat{f} \leq 0$, so $\omega(f) \leq q(f) = \lambda(\hat{f}) \leq 0$. Therefore $\omega \geq 0$ and the Riesz–Kakutani representation theorem allows us to identify ω with a suitable Borel probability measure on K. If f is in $Cvx(K)$, then $-f$ is concave and Lemma 9.2.2 yields

$$\omega(-f) \leq q(-f) = \lambda(\widehat{-f}) = \lambda(-f)$$

that is, $\lambda \prec \omega$. Or, λ is maximal, which forces $\omega = \lambda$. Consequently,

$$\lambda(\varphi) = \omega(\varphi) = L(\varphi) = \lambda(\overline{\varphi}),$$

which ends the proof. ■

As E. Bishop and K. de Leeuw [69] noticed, if K is non-metrizable, then $\operatorname{ext} K$ need not to be a Borel set. However, they were able to prove a Choquet-type theorem. By combining their argument (as presented in [474], p. 17) with Theorem 9.4.1 above, one can prove the following more general result:

9.4.4 Theorem (The Choquet–Bishop–de Leeuw theorem [69]) *Let μ be a Borel measure on a compact convex subset K of a locally convex Hausdorff space E. Then there exists a Borel probability measure λ on K such that the following two conditions are fulfilled:*

(a) $\lambda \succ \mu$ *and* λ *and* μ *have the same barycenter;*

(b) λ *vanishes on every Baire subset of K which is disjoint from the set of extreme points of K.*

Choquet's theory has deep applications to many areas of mathematics such as function algebras, invariant measures, and potential theory. R. R. Phelps' book [474] contains a good account of this matter.

Three examples illustrating the generalized form of the Hermite–Hadamard inequality are presented below.

9.4.5 Example *Let f be a continuous convex function defined on an N-dimensional simplex $K = [\mathbf{a}_0, \dots, \mathbf{a}_N]$ in $\mathbb{R}^N$ and let μ be a Borel measure on K. Then*

$$f(\mathrm{bar}(\mu)) \le \frac{1}{\mu(K)} \int_K f(\mathbf{x})\, \mathrm{d}\mu \le \frac{1}{\mathrm{Vol}_N(K)} \sum_{k=0}^{n} \mathrm{Vol}_n([\mathbf{a}_0, \dots, \widehat{\mathbf{a}_k}, \dots, \mathbf{a}_n] \cdot f(\mathbf{a}_k).$$

Here $[\mathbf{a}_0, \dots, \widehat{\mathbf{a}_k}, \dots, \mathbf{a}_N]$ denotes the subsimplex obtained by replacing $\mathbf{a}_k$ by $\mathrm{bar}(\mu)$; this is the subsimplex opposite to $\mathbf{a}_k$, when adding $\mathrm{bar}(\mu)$ as a new vertex.

9.4.6 Example *In the case of closed balls $K = \overline{B}_R(\mathbf{a})$ in $\mathbb{R}^N$, $\mathrm{ext}\, K$ coincides with the sphere $S_R(\mathbf{a})$. According to Corollary 9.4.3, if $f\colon \overline{B}_R(\mathbf{a}) \to \mathbb{R}$ is a continuous convex function, then*

$$f(a) \le \frac{1}{\mathrm{Vol}_N \overline{B}_R(\mathbf{a})} \iiint_{\overline{B}_R(\mathbf{a})} f(\mathbf{x})\, \mathrm{d}V \le \frac{1}{\mathrm{Area}_{N-1} S_R(\mathbf{a})} \iint_{S_R(\mathbf{a})} f(\mathbf{x})\, \mathrm{d}S. \tag{9.10}$$

Results similar to Example 9.4.6 also work in the case of subharmonic functions. See the Comments at the end of this chapter. As noticed by P. Montel [394], in the context of C^2-functions on open convex sets in $\mathbb{R}^n$, the class of subharmonic functions is strictly larger than the class of convex functions. For example, the function $2x^2 - y^2$ is subharmonic but not convex on $\mathbb{R}^2$.

Many inequalities relating weighted means represent averages over the probability simplex:

$$\Sigma(N) = \{\mathbf{u} = (u_1, \dots, u_N) : u_1, \dots, u_N \ge 0,\ u_1 + \cdots + u_N = 1\}.$$

This set is compact and convex and its extreme points are the "corners"

$$(1, 0, \dots, 0), (0, 1, \dots, 0) \dots, (0, 0, \dots, 1).$$

9.4.7 Example *If $f\colon [a,b] \to \mathbb{R}$ is a continuous convex function, then for every N-tuple $\mathbf{x} = (x_1, \dots, x_N)$ of elements of $[a,b]$ and every Borel probability measure μ on $\Sigma(N)$, we have*

$$f\Big(\sum_{k=1}^{N} w_k x_k\Big) \le \int_{\Sigma(N)} f(\langle \mathbf{x}, \mathbf{u}\rangle)\, \mathrm{d}\mu \le \sum_{k=1}^{N} w_k f(x_k). \tag{9.11}$$

Here $(w_1, \dots, w_N)$ denotes the barycenter of μ. The above inequalities work in the reverse direction if f is a concave function.

Under the hypotheses of Example 9.4.7, the weighted identric mean $I(x,\mu)$ is defined by the formula

$$I(\mathbf{x},\mu) = \exp \int_{\Sigma(N)} \log(\langle \mathbf{x}, \mathbf{u}\rangle)\, \mathrm{d}\mu(\mathbf{u}), \quad \mathbf{x} \in \mathbb{R}^N_{++},$$

while the weighted logarithmic mean $L(\mathbf{x},\mu)$ is defined by the formula

$$L(\mathbf{x},\mu) = \left(\int_{\Sigma(N)} \frac{1}{\langle \mathbf{x}, \mathbf{u}\rangle}\, \mathrm{d}\mu(\mathbf{u}) \right)^{-1}, \quad \mathbf{x} \in \mathbb{R}^N_{++}.$$

The formula (9.11) easily implies that $L(\mathbf{x},\mu) \le I(\mathbf{x},\mu)$ and that both means lie between the weighted arithmetic mean $A(\mathbf{x},\mu) = \sum_{k=1}^{N} w_k x_k$ and the weighted geometric mean $G(\mathbf{x},\mu) = \prod_{k=1}^{N} x_k^{w_k}$, that is,

$$G(\mathbf{x},\mu) \le L(\mathbf{x},\mu) \le I(\mathbf{x},\mu) \le A(\mathbf{x},\mu),$$

a fact which constitutes the weighted geometric–logarithmic–identric–arithmetic mean inequality.

An important example of a Borel probability measure on $\Sigma(N)$ is the *Dirichlet measure* of parameters $p_1, \dots, p_N > 0$,

$$\frac{\Gamma(p_1 + \cdots + p_N)}{\Gamma(p_1) \cdots \Gamma(p_N)} x_1^{p_1 - 1} \cdots x_{N-1}^{p_{N-1} - 1} (1 - x_1 - \cdots - x_{N-1})^{p_N - 1}\, \mathrm{d}x_1 \cdots \mathrm{d}x_{N-1}.$$

Its barycenter is the point $(\sum_{k=1}^{N} p_k)^{-1} \cdot (p_1, \dots, p_N)$.

Exercises

1. A higher dimensional analogue of the Hermite–Hadamard inequality [42]) Let f be a continuous concave function defined on a compact convex subset $K \subset \mathbb{R}^N$ of positive volume. Prove that

$$\frac{1}{N+1} \sup_{\mathbf{x} \in K} f(\mathbf{x}) + \frac{N}{N+1} \inf_{\mathbf{x} \in \operatorname{ext} K} f(\mathbf{x}) \le \frac{1}{\operatorname{Vol}_N(K)} \int_K f(\mathbf{x})\, \mathrm{d}\mathbf{x} \le f(\operatorname{bar}(\mu)),$$

where x_μ is the barycenter of the probability measure $\mathrm{d}\mu = \mathrm{d}x / \operatorname{Vol}_N(K)$ on K.

2. (R. R. Phelps [474]) Prove that in every normed linear space E, a sequence $(\mathbf{x}_n)_n$ is weakly convergent to $\mathbf{x}$ if, and only if, the sequence is norm bounded and $\lim_{n\to\infty} x^*(\mathbf{x}_n) = x^*(\mathbf{x})$ for every extreme point x^* of the closed unit ball in E^*. As a consequence, $f_n \to f$ weakly in $C([0,1])$ if, and only if,
$$\sup\{|f_n(t)| : t \in [0,1],\ n \in \mathbb{N}\} < \infty$$
and $f_n(t) \to f(t)$ for every $t \in [0,1]$.
[*Hint*: Let K be the closed unit ball in E^*. Then K is convex and weak-star compact (see Theorem B.1.6). Each point $\mathbf{x} \in E$ gives rise to an affine mapping $A_x \colon K \to \mathbb{R}$, $A_{\mathbf{x}}(x^*) = x^*(\mathbf{x})$. Then apply Theorem 9.4.4.]

3. (R. Haydon) Let E be a real Banach space and let K be a weak-star compact convex subset of E^* such that $\operatorname{ext} K$ is norm separable. Prove that K is the norm closed convex hull of $\operatorname{ext} K$ (and hence is itself norm separable).

4. Let K be a nonempty compact convex set (in a locally convex Hausdorff space E) and let μ be a Borel probability measure on K. If $f \in C(K)$, one denotes by $\widehat{f}$ its upper envelope. Prove that:
(a) $\widehat{f}(\mathbf{x}) = \inf\{g(\mathbf{x}) : g \in Cvx(K) \text{ and } g \geq f\}$;
(b) for every pair of functions $g_1, g_2 \in Cvx(K)$ with $g_1, g_2 \geq f$, there is a function $g \in Cvx(K)$ such that $g_1, g_2 \geq g \geq f$;
(c) $\mu(\widehat{f}) = \inf\{\mu(g) : -g \in Cvx(K) \text{ and } g \geq f\}$.

5. (G. Mokobodzki) Infer from the preceding exercise that a Borel probability measure μ on K is maximal if, and only if, $\mu(f) = \mu(\widehat{f})$ for all continuous convex functions f on K (equivalently, for all functions $f \in C(K)$).

9.5 Completely Monotone Functions on Cones

The theory of completely monotone functions can be easily extended to the context of several variables using convex analysis. In what follows V denotes a finite-dimensional real vector space and $\mathcal{C}$ an open convex cone in V with closure $\overline{\mathcal{C}}$. Its dual cone is $\mathcal{C}^* = \{\mathbf{y} \in E^* : \langle \mathbf{y}, \mathbf{x}\rangle \geq 0 \text{ for all } \mathbf{x} \in \mathcal{C}\}$. The points in C^* are the linear functionals that are nonnegative on $\overline{\mathcal{C}}$.

9.5.1 Definition *A function $f : \mathcal{C} \to \mathbb{R}_+$ is called completely monotone if f is $\mathcal{C}^\infty$ on $\mathcal{C}$ and, for all integers $k \geq 1$ and all vectors $\mathbf{v}_1, ..., \mathbf{v}_k \in \mathcal{C}$, we have*
$$(-1)^k\, \partial_{\boldsymbol{v}_1} \cdots \partial_{\boldsymbol{v}_k} f(\boldsymbol{x}) \geq 0 \quad \textit{for all } \boldsymbol{x} \in \mathcal{C}. \tag{9.12}$$
Here $\partial_{\boldsymbol{v}}$ denotes the directional derivative along the vector $\boldsymbol{v}$.

A function $f : \overline{\mathcal{C}} \to \mathbb{R}_+$ is called completely monotone if it is the continuous extension of a completely monotone function on $\mathcal{C}$.

When $\mathcal{C} = (0, \infty)^n$, the condition (9.12) means that

$$(-1)^k \frac{\partial^k f}{\partial x_{i_1} \partial x_{i_2} \cdots \partial x_{i_k}}(\mathbf{x}) \geq 0$$

for all $\mathbf{x} \in (0, \infty)^n$, all integers $k \geq 1$ and all choices of indices $i_1, \cdots, i_k$.

Alternatively, the concept of complete monotonicity can be introduced for continuous functions $f : \overline{\mathcal{C}} \to \mathbb{R}_+$ via the iterates of difference operators as follows:

$$(-1)^k \Delta_{\mathbf{v}_1} \Delta_{\mathbf{v}_2} \cdots \Delta_{\mathbf{v}_k} f(\mathbf{x}) \geq 0$$

for all vectors $\mathbf{x}, \mathbf{v}_1, ..., \mathbf{v}_k \in \mathcal{C}$ and all integers $k \geq 1$. As in the case of completely monotone functions of one real variable, these functions can be obtained as Laplace transforms of Borel measures on the dual cone.

9.5.2 Theorem (Bernstein–Hausdorff–Widder–Choquet theorem). *Let f be a nonnegative continuous function on the open convex cone $\mathcal{C}$. Then f is completely monotone if and only if it is the Laplace transform of a unique Borel measure μ supported on the dual cone $\mathcal{C}^*$, that is,*

$$f(x) = \int_{\mathcal{C}^*} e^{-\langle \boldsymbol{y}, \boldsymbol{x} \rangle} \mathrm{d}\mu(\boldsymbol{y}) \quad \text{for all } \boldsymbol{x} \in \mathcal{C}.$$

When f admits a continuous extension to $\overline{\mathcal{C}}$, the last equality works for all $x \in \overline{\mathcal{C}}$.

For the proof and further details, see Choquet [133]. Very helpful in this respect is the particular case of bounded and completely monotone functions defined on $(0, \infty)$, treated by R. R. Phelps in [474], Section 2.

9.5.3 Remark *Finding the positive Borel measure μ that makes the formula of Theorem 9.5.2 working represents a practical way for checking the complete monotonicity of f. So is the case of Riesz kernels: If $\alpha_1, \alpha_2, ..., \alpha_N > 0$, then*

$$x_1^{-\alpha_1} x_2^{-\alpha_2} \cdots x_N^{-\alpha_N} = \int_{\mathbb{R}^N_{++}} e^{-\langle \boldsymbol{y}, \boldsymbol{x} \rangle} \frac{x_1^{-\alpha_1} x_2^{-\alpha_2} \cdots x_N^{-\alpha_N}}{\Gamma(\alpha_1)\Gamma(\alpha_2) \cdots \Gamma(\alpha_N)} \mathrm{d}\boldsymbol{y}$$

for all $\boldsymbol{x} \in \mathbb{R}^N_{++}$. See [306], Proposition 2.7. A more subtle case is that of inverse powers of the determinant,

$$f(X) = (\det X)^{-\rho}, \quad X \in \mathrm{Sym}^{++}(N, \mathbb{R}),$$

for which Scott and Sokal [528] have shown that it is completely monotone if and only if $\rho \in \{0, 1/2, 1, 3/2, ..., (N-1)/2\} \cup ((N-1)/2, \infty)$. See also [306], Theorem 4.1. It is worth noticing that Siegel established in 1929 the formula

$$(\det A)^{-\rho} = \int_{\mathrm{Sym}^{++}(N,\mathbb{R})} \frac{(\det X)^{\rho} \ \exp(-\mathrm{trace} AX) \mathrm{d}X}{\pi^{n(n-1)/4} \Gamma(\rho) \Gamma(\rho - 1/2) \cdots \Gamma(\rho - (n-1)/2))}$$

for all $A \in \mathrm{Sym}^{++}(N, \mathbb{R})$ and $\rho \geq (N+1)/2$. See [535], Hilfssatz 37, p. 58.

We next extend (and improve) a result due to Sendov and Zitikis; see [529], Theorem 4.1, p. 76.

9.5.4 Theorem *Every completely monotone function $f : \mathcal{C} \to \mathbb{R}_+$ satisfies*

$$\sum\nolimits_{i=1}^{n} f(\boldsymbol{x}_i) - \sum\nolimits_{1\le i<j\le n} f(\boldsymbol{x}_i + \boldsymbol{x}_j) + \cdots + (-1)^{n-1} f\Big(\sum\nolimits_{i=1}^{n} \boldsymbol{x}_i\Big) \ge 0,$$

for every $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathcal{C}$ and $n \ge 1$. When f admits a continuous extension to $\overline{\mathcal{C}}$, then f satisfies the double inequality

$$f(0) \ge \sum\nolimits_{i=1}^{n} f(\boldsymbol{x}_i) - \sum\nolimits_{1\le i<j\le n} f(\boldsymbol{x}_i + \boldsymbol{x}_j) + \cdots + (-1)^{n-1} f\Big(\sum\nolimits_{i=1}^{n} \boldsymbol{x}_i\Big) \ge 0,$$

for every $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \overline{\mathcal{C}}$ and $n \ge 1$.

For $n = 3$, the first conclusion of Theorem 9.5.4 reads as

$$\sum\nolimits_{i=1}^{3} f(\mathbf{x}_i) - \sum\nolimits_{1\le i<j\le 3} f(\mathbf{x}_i + \mathbf{x}_j) + f\Big(\sum\nolimits_{i=1}^{3} \mathbf{x}_i\Big) \ge 0,$$

which is nothing but a Hornich–Hlawka type inequality.

The proof of Theorem 9.5.4 needs the following auxiliary result.

9.5.5 Lemma *We have*

$$P = \sum\nolimits_{i=1}^{n} e^{-\alpha_i} - \sum\nolimits_{1\le i<j\le n} e^{-(\alpha_i+\alpha_j)} + \cdots + (-1)^{n+1} e^{-\sum_{i=1}^{n}\alpha_i} \ge 0$$

and

$$Q = \sum\nolimits_{i=1}^{n} \left(1 - e^{-\alpha_i}\right) - \sum\nolimits_{1\le i<j\le n} \left(1 - e^{-(\alpha_i+\alpha_j)}\right) + \cdots + (-1)^{n+1} \left(1 - e^{-\sum_{i=1}^{n}\alpha_i}\right) \ge 0,$$

whenever $\alpha_1, \ldots, \alpha_n \in \mathbb{R}_+$ and $n \ge 2$.

Proof. Indeed,

$$P = 1 - \prod_{i=1}^{n} \left(1 - e^{-\alpha_i}\right),$$

which implies that $P \in [0, 1]$. The proof ends by noticing that

$$Q = \sum\nolimits_{i=1}^{n} (-1)^{i+1} \binom{n}{i} - P = 1 - P \ge 0.$$

■

Proof of Theorem 9.5.4. According to Theorem 9.5.2, f admits the integral representation

$$f(\mathbf{x}) = \int_{\mathcal{C}^*} e^{-\langle \mathbf{y}, \mathbf{x} \rangle} \mathrm{d}\mu(\mathbf{y}) \quad \text{for all } \mathbf{x} \in \mathcal{C},$$

where μ is a Borel measure on $\mathcal{C}^*$. Then, taking into account the first assertion of Lemma 9.5.5, we have the inequality

$$0 \leq \int_{\mathcal{C}^*} \Big[\sum\nolimits_{i=1}^{n} e^{-\langle \mathbf{x}_i, \mathbf{y}\rangle} - \sum\nolimits_{1\leq i<j\leq n} e^{-\langle \mathbf{x}_i+\mathbf{x}_j, \mathbf{y}\rangle} + \cdots + (-1)^{n-1} e^{-\langle \sum_{i=1}^{n} \mathbf{x}_i, \mathbf{y}\rangle}\Big] \, \mathrm{d}\mu(\mathbf{y})$$
$$= \sum\nolimits_{i=1}^{n} f(\mathbf{x}_i) - \sum\nolimits_{1\leq i<j\leq n} f(\mathbf{x}_i + \mathbf{x}_j) + \cdots + (-1)^{n-1} f(\sum\nolimits_{i=1}^{n} \mathbf{x}_i).$$

The case where f is defined on $\overline{\mathcal{C}}$ can be settled in the same manner, using both assertions of Lemma 9.5.5. ■

Combining Theorem 9.5.4 with the aforementioned result of A.D. Scott and A.D. Sokal (see Remark 9.5.3) one obtains the following result:

9.5.6 Corollary *If* $\rho \in \{0, 1/2, 1, 3/2, ...\} \cup [(N-1)/2, \infty)$, *then*

$$\sum\nolimits_{i=1}^{n} \det{}^{-\rho}(A_i) - \sum\nolimits_{1\leq i<j\leq n} \det{}^{-\rho}(A_i + A_j) + \cdots + (-1)^{n-1} \det{}^{-\rho}(\sum\nolimits_{i=1}^{n} A_i) \geq 0,$$

for every $A_1, \ldots, A_n \in Sym^{++}(N, \mathbb{R})$ *and* $n \geq 1$.

Given Corollary 9.5.6 one may wonder whether Theorem 9.5.4 also specializes in elementary symmetric polynomials. The situation here turns out to be more subtle, and a qualified answer follows from the discussion below. Recall that for any $m = 0, 1, ..., N$, the m-th elementary symmetric polynomial of $\mathbf{x} \in \mathbb{R}^N$ is defined by the formula

$$E_{m,N}(\mathbf{x}) = \sum\nolimits_{1\leq i_1<\cdots<i_m\leq N} x_{i_1} ... x_{i_m}.$$

Notice that det and the functions $E_{m,N}$ are hyperbolic polynomials in the sense of Gårding. Kozhasov, Michalek, and Sturmfels provided a constructive proof for the fact that any elementary symmetric polynomial $E_{m,n}$ admits a real exponent $\alpha' > 0$ such that $E_{m,n}^{-\alpha}$ is completely monotone on $\mathbb{R}^N_{++}$ for all $\alpha \geq \alpha'$. See [306], Theorem 6.4.

9.5.7 Remark *The restriction of* e^{-x} *to* $\mathbb{R}_+$ *is a completely monotone function and thus it verifies the inequality*

$$e^{-x_1} + e^{-x_2} + e^{-x_3} + e^{-(x_1+x_2+x_3)} \geq e^{-(x_1+x_2)} + e^{-(x_2+x_3)} + e^{-(x_3+x_1)}.$$

However, there are $x_1, x_2, x_3 > 0$ *such that*

$$e^{-x_1} + e^{-x_2} + e^{-x_3} + e^{-(x_1+x_2+x_3)} \not\geq e^{-(x_1+x_2)} + e^{-(x_2+x_3)} + e^{-(x_3+x_1)} + e^{0}.$$

This agrees with the fact that e^{-x} *is not* 3*-convex.*

The fact that the Bernstein functions $f : \mathbb{R}_+ \to \mathbb{R}_+$ also verify the inequalities stated in Theorem 9.5.4 follows from Theorem 9.5.2 and the Lévy–Khintchine theorem (Theorem 3.2, p. 15 of [521]), which asserts that each such function admits the integral representation

$$f(x) = a + bx + \int_0^\infty (1 - e^{-xt}) \mathrm{d}\mu(t)$$

for some constants $a, b \geq 0$ and a positive measure μ on $[0, \infty)$ such that

$$\int_0^\infty \min\{1, t\} \, \mathrm{d}\mu(t) < \infty.$$

Exercises

1. (A. D Scott and A. D. Sokal [528]) Let $\beta \in \mathbb{R}$ and $\lambda > 0$. Prove that the function
$$F_{\beta,\lambda}(u, v) = (u + v)^{-\beta} \exp\left(-\lambda \frac{uv}{u + v}\right)$$
is completely monotone on $(0, \infty)^2$ if and only if $\beta \geq 1/2$. Show that for $\beta \geq 1/2$ there exists a positive Borel measure $\mu_{\beta,\lambda}$ on $[0, \infty)^2$ such that
$$(u + v)^{-\beta} \exp\left(-\lambda \frac{uv}{u + v}\right) = \iint_{[0,\infty)^2} e^{-t_1 u - t_2 v} \mathrm{d}\mu_{\beta,\lambda}(t_1, t_2).$$

2. (A. D Scott and A. D. Sokal [528]) Let V be a finite-dimensional real vector space and let $\mathcal{C}$ be an open convex cone in V. Let f be completely monotone on $\mathcal{C} \times (0, \infty)$, and let $\beta \geq 1/2$. Prove that the function
$$g(x, y, z) = (y + z)^{-\beta} f\left(x, \frac{yz}{y + z}\right)$$
is completely monotone on $\mathcal{C} \times (0, \infty)$.

3. Prove that the set of completely monotone functions on an open convex cone $\mathcal{C}$ in $\mathbb{R}^N$ is a convex cone, closed under pointwise convergence.

4. Let $\mathcal{C}$ be an open convex cone $\mathcal{C}$ in $\mathbb{R}^N$. We write $\mathcal{M}_+(\mathcal{C})$ for the convex cone of all Borel measures that are supported in the closed dual cone $\mathcal{C}^*$. The convolution product $\mu * \nu$ of two measures $\mu, \nu \in \mathcal{M}_+(\mathcal{C})$ is defined by the formula
$$(\mu * \nu)(E) = \int_{\mathbb{R}^N} \int_{\mathbb{R}^N} \chi_E(\mathbf{y} + \mathbf{z}) d\mu(\mathbf{y}) d\nu(\mathbf{z})$$
for any Borel set $E \subset \mathbb{R}^N$.
(a) Prove that if $\mu, \nu \in \mathcal{M}_+(\mathcal{C})$, then $\mu * \nu \in \mathcal{M}_+(\mathcal{C})$ and the convolution is commutative and associative.

(b) A measure $\mu \in \mathcal{M}_+(\mathcal{C})$ has a Laplace transform if the integral

$$\mathcal{L}(\mu)(\mathbf{x}) = \int_{\mathcal{C}^*} e^{-\langle \mathbf{y}, \mathbf{x} \rangle} d\mu(\mathbf{y})$$

converges for all $\mathbf{x} \in \mathcal{C}$. Prove that if the measures $\mu, \nu \in \mathcal{M}_+(\mathcal{C})$ have Laplace transforms then so does $\mu * \nu$ and

$$\mathcal{L}(\mu * \nu) = \mathcal{L}(\mu)\mathcal{L}(\nu)$$

and conclude that the product of two completely monotone functions on $\mathcal{C}$ is completely monotone.

(c) Prove that the set of completely monotone functions is a convex cone, closed under pointwise convergence.

5. (C. P. Niculescu [430]) Let $\mathcal{C}$ be an open convex cone (in a finite-dimensional real vector space V). Infer from Theorem 9.5.2 that every completely monotone function $\Phi : \overline{\mathcal{C}} \to \mathbb{R}_+$ has positive differences of second order. Then, extend on this basis Exercise 4 in Section 6.5.

9.6 The Hermite–Hadamard Inequality for Signed Measures

In dimension 1, the Hermite–Hadamard inequality asserts that for every Borel probability measure μ on a compact interval $[a, b]$ and every continuous convex function $f : [a, b] \to \mathbb{R}$ we have

$$f(\mathrm{bar}(\mu)) \leq \int_a^b f(x)\, \mathrm{d}\mu(x) \leq \frac{b - \mathrm{bar}(\mu)}{b - a} \cdot f(a) + \frac{\mathrm{bar}(\mu) - a}{b - a} \cdot f(b). \quad (9.13)$$

See Section 2.2. The Jensen-Steffensen inequality shows that the left-hand side inequality works in the more general context of Steffensen-Popoviciu measures. Unfortunately, this does not apply to the right-hand side inequality. However, A. M. Fink [197] was able to indicate some few concrete examples of signed measures for which the double inequality (9.13) holds true. This made it possible to speak on *Hermite–Hadamard measures*, that is, on those Steffensen–Popoviciu measures of total mass 1, for which the Hermite–Hadamard inequality works in its entirety. The complete characterization of these measures combines Theorem 9.1.2 with the following result due to A. Florea and C. P. Niculescu [200].

9.6.1 Theorem *Let μ be a real Borel measure on the interval $[a, b]$ such that $\mu([a, b]) = 1$. Then the inequality*

$$\frac{b - \mathrm{bar}(\mu)}{b - a} \cdot f(a) + \frac{\mathrm{bar}(\mu) - a}{b - a} \cdot f(b) \geq \int_a^b f(x)\, \mathrm{d}\mu(x) \quad (9.14)$$

works for all continuous convex functions $f : [a, b] \to \mathbb{R}$ if, and only if,

$$\frac{b-t}{b-a}\int_a^t (x-a)\mathrm{d}\mu(x) + \frac{t-a}{b-a}\int_t^b (b-x)\mathrm{d}\mu(x) \geq 0 \tag{9.15}$$

for all $t \in [a, b]$.

Proof. By an approximation argument we may reduce ourselves to the case of convex functions of class C^2. As well known, every such function f admits an integral representation of the form

$$f(x) = \frac{b-x}{b-a}\cdot f(a) + \frac{x-a}{b-a}\cdot f(b) + \int_a^b G(x,t)f''(t)\mathrm{d}t,$$

where

$$G(x,t) = \begin{cases} -(x-a)(b-t)/(b-a) & \text{if } a \leq x \leq t \leq b \\ -(t-a)(b-x)/(b-a) & \text{if } a \leq t \leq x \leq b \end{cases}$$

represents the Green function of the operator $\frac{\mathrm{d}^2}{\mathrm{d}t^2}$, with homogeneous boundary conditions $y(a) = y(b) = 0$.

Thus, for every convex functions $f \in C^2([a, b])$, we have

$$\begin{aligned}\int_a^b f(x)\,\mathrm{d}\mu(x) &- \frac{b-\mathrm{bar}(\mu)}{b-a}\cdot f(a) + \frac{\mathrm{bar}(\mu)-a}{b-a}\cdot f(b) \\ &= \int_a^b \left[f(x) - \frac{b-x}{b-a}\cdot f(a) + \frac{x-a}{b-a}\cdot f(b)\right]\mathrm{d}\mu(x) \\ &= \int_a^b \left(\int_a^b G(x,t)f''(t)\mathrm{d}t\right)\mathrm{d}\mu(x) \\ &= \int_a^b f''(t)\left(\int_a^b G(x,t)\mathrm{d}\mu(x)\right)\mathrm{d}t = \int_a^b f''(t)y(t)\mathrm{d}t,\end{aligned}$$

where $y(t) = \int_a^b G(x,t)\mathrm{d}\mu(x)$ is a continuous function.

The differential operator $\frac{\mathrm{d}^2}{\mathrm{d}t^2}$ maps $\mathrm{Conv}([a, b]) \cap C^2([a, b])$ onto the positive cone of $C([a, b])$, so the inequality (9.14) holds true if, and only if, $y(t) \leq 0$ for all $t \in [a, b]$. This ends the proof. ■

According to Theorem 9.1.2 and Theorem 9.6.1,

$$3\left(x^2 - \frac{1}{6}\right)\mathrm{d}x, \quad \frac{3780}{499}\left(x^2 - \frac{1}{6}\right)^3\mathrm{d}x \text{ and } \frac{25}{8}\left(x^4 - \frac{1}{25}\right)\mathrm{d}x$$

are examples of Hermite–Hadamard measures on the interval $[-1, 1]$ (each of total mass 1 and barycenter 0)

Notice that $\left(x^2 - \frac{1}{4}\right)\mathrm{d}x$, which is a strong Steffensen–Popoviciu measure on $[-1, 1]$, does not fulfill the condition (9.15) and thus it is not a Hermite–Hadamard measure. However, every dual Steffensen–Popoviciu measure verifies

the condition (9.15) because it represents the integral of a positive concave function.

The extension of the Hermite–Hadamard in higher dimensions for signed measures remains an open problem. No general criterion like that given by Theorem 9.6.1 is known. A simple example proving the consistency of this problem is shown in Exercise 2.

Exercises

1. (A. M. Fink [197]) Consider the signed measure $(x^2-x)\,\mathrm{d}x$ on the interval $[-1,1]$. Its total mass is $2/3$ and its barycenter is the point -1. Prove that

$$\int_{-1}^{1} f(x)(x^2-x)\,\mathrm{d}x \le \frac{2}{3}\,f(-1)$$

for all continuous convex functions f in $C^2([-1,1])$.

Remark. One can show easily that the signed measure $(x^2-x)\,\mathrm{d}x$ is not a Steffensen–Popoviciu measure on the interval $[-1,1]$.

2. Consider the real Borel measure $\mathrm{d}\mu(x,y)=\frac{3}{2}\left(x^2-\frac{1}{6}\right)\mathrm{d}x\mathrm{d}y$ on the square $[-1,1]\times[-1,1]$. Prove that:

   (a) The total mass of this measure is 1 and its barycenter is the origin.

   (b) For every continuous convex function $f:[-1,1]\times[-1,1]\to\mathbb{R}$,

$$\begin{aligned} f(0) &\le \frac{3}{2}\int_{-1}^{1}\int_{-1}^{1} f(x,y)\left(x^2-\frac{1}{4}\right)\mathrm{d}x\mathrm{d}y \\ &\le \frac{f(-1,-1)+f(-1,1)+f(1,-1)+f(1,1)}{4}. \end{aligned}$$

9.7 Comments

The theory of Steffensen–Popoviciu measures was initiated by C. P. Niculescu [420], [421] and developed by him and his collaborators in a series of papers: [200], [432], [431], [427], [445], [381] and [447]. Except the cases mentioned explicitly in the text, all results presented in this chapter represent the contribution of this research team.

Initially, it was thought that both inequalities composing the Hermite–Hadamard inequality work in the context of Steffensen–Popoviciu measures, but soon it became clear that the extension of the right-hand side inequality is more subtle. Thus, the claim of Theorems 4 and 5 in [420] concerning the existence of maximal Borel probability measures majorizing a given Steffensen–Popoviciu measure is false. The paper by A. Florea and C. P. Niculescu [200] solved the problem in dimension 1, but still we lack a result similar to that of Theorem 9.6.1 in higher dimensions.

An interesting example of a strong Steffensen–Popoviciu measure is offered by the case of Rayleigh measures, introduced by S. Kerov [295] in a paper

dedicated to the relationship between probability distributions μ and certain bounded signed measures τ on the real line $\mathbb{R}$, satisfying the Markov–Krein identity,

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}\mu(u)}{z-u} = \exp \int_{-\infty}^{\infty} \ln \frac{1}{z-u} \mathrm{d}\tau(u) \quad \text{for } z \in \mathbb{C}, \quad \mathrm{Im} z > 0.$$

Precisely, the function $R(z)$ appearing in the right-hand side can be represented as a Cauchy–Stieltjes transform of a probability measure μ,

$$R(z) = \int_{-\infty}^{\infty} \frac{\mathrm{d}\mu(u)}{z-u},$$

whenever τ is a *Rayleigh measure*, that is, a bounded signed measure that verifies the following three properties:

$$(RM1)\ 0 \leq \tau\left(\{x : x < a\}\right) \leq 1 \quad \text{for every } a \in \mathbb{R};$$
$$(RM2)\ \tau\left((-\infty, \infty)\right) = 1;$$
$$(RM3) \int_{-\infty}^{\infty} \ln\left(1 + |x|\right) \mathrm{d}\,|\tau|\,(x) < \infty.$$

See [295], Corollary 2.4.1. An alternative approach of Rayleigh measures is offered by the notion of interlace measures. Two finite positive measures τ' and τ'' *interlace* if there exists a Rayleigh measure τ such that $\tau' = \tau^+$ and $\tau'' = \tau^-$. For example, if $\lambda_1 \leq \mu_1 \leq \lambda_2 \leq \cdots \leq \mu_{n-1} \leq \lambda_n$, then the discrete measure $\tau = \sum_{k=1}^{n} \delta_{\lambda_k} - \sum_{j=1}^{n-1} \delta_{\mu_j}$ is an example of Rayleigh measure and the corresponding Markov–Krein identity has the form

$$\sum_{k=1}^{n} \frac{p_k}{z-\lambda_k} = \frac{(z-\mu_1)\cdots(z-\mu_{n-1})}{(z-\lambda_1)\cdots(z-\lambda_n)}$$

for suitable $p_1, ..., p_n > 0$ with $\sum_{k=1}^{n} p_k = 1$.

Interesting examples of absolutely continuous Rayleigh measures can be found in the paper of D. Romik [510].

Choquet theory was borned around 1956, when G. Choquet started to announce his results in a series of papers published in Comptes Rendus de l'Académie des Sciences (Paris). His paper [131] from 1960 together with his report [132] at the International Congress of Mathematicians at Stockholm (1962) provided a clear picture about the importance and scope of the Choquet theory. R. R. Phelps [474] contributed much to the popularization of this theory with a well written, readable, and easily accessible book.

As noticed in Theorem 6.1.9, the relation of majorization $x \prec y$ (for vectors in $\mathbb{R}^N$) can be characterized by the existence of a doubly stochastic matrix P such that $x = Py$. Thinking of x and y as discrete probability measures, this fact can be rephrased as saying that y is a dilation of x. The book by R. R. Phelps [474] indicates the details of an extension (due to P. Cartier, J. M. G. Fell, and P. A. Meyer [121]) of this characterization to the general framework of Borel

probability measures on compact convex sets (in a locally convex Hausdorff space).

Following [432] we shall show that the Hermite–Hadamard inequality also works in the context of subharmonic functions.

Let Ω be a bounded open subset of $\mathbb{R}^N$ with smooth boundary. Then the Dirichlet problem

$$\begin{cases} \Delta\varphi = 1 & \text{on } \Omega \\ \varphi = 0 & \text{on } \partial\Omega \end{cases} \tag{9.16}$$

has a unique solution, which is negative on Ω, according to the maximum principle for elliptic problems. See [499]. By Green's formula, for every u in $C^2(\Omega) \cap C^1(\overline{\Omega})$ we have

$$\int_\Omega \begin{vmatrix} u & \varphi \\ \Delta u & \Delta\varphi \end{vmatrix} \mathrm{d}V = \int_{\partial\Omega} \begin{vmatrix} u & \varphi \\ \nabla u & \nabla\varphi \end{vmatrix} \cdot n \, \mathrm{d}S,$$

that is, using (9.16),

$$\begin{aligned} \int_\Omega u \, \mathrm{d}V &= \int_\Omega u\Delta\varphi \, \mathrm{d}V = \int_\Omega \varphi\Delta u \, \mathrm{d}V + \int_{\partial\Omega} u(\nabla\varphi \cdot \mathbf{n}) \, \mathrm{d}S - \int_{\partial\Omega} \varphi(\nabla u \cdot \mathbf{n}) \, \mathrm{d}S \\ &= \int_\Omega \varphi\Delta u \, \mathrm{d}V + \int_{\partial\Omega} u(\nabla\varphi \cdot \mathbf{n}) \, \mathrm{d}S \end{aligned}$$

for every $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$. Here $\mathbf{n}$ denotes the outward-pointing unit normal vector on the boundary.

We are thus led to the following result:

9.7.1 Theorem (The Hermite–Hadamard inequality for subharmonic functions) *If $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ is subharmonic (that is, $\Delta u \geq 0$ on Ω) and φ satisfies (9.16), then*

$$\int_\Omega u \, \mathrm{d}V < \int_{\partial\Omega} u(\nabla\varphi \cdot \mathbf{n}) \, \mathrm{d}S$$

except for harmonic functions (when equality occurs).

The equality case needs the remark that $\int_\Omega \varphi\Delta u \, \mathrm{d}V = 0$ yields $\varphi\Delta u = 0$ on Ω, and thus $\Delta u = 0$ on Ω; notice that $\varphi\Delta u$ is continuous and $\varphi\Delta u \leq 0$ since $\varphi < 0$ on Ω.

In the case of balls $\Omega = B_R(\mathbf{a})$ in $\mathbb{R}^3$, the solution of the problem (9.16) is $\varphi(\mathbf{x}) = (\|\mathbf{x}\|^2 - R^2)/6$ and $\nabla\varphi \cdot \mathbf{n} = \langle \mathbf{x}/3, \mathbf{x}/\|\mathbf{x}\| \rangle = \|\mathbf{x}\|/3$, so that, by combining the maximum principle for elliptic problems with the conclusion of the above theorem, we obtain the following Hermite–Hadamard type inequality for subharmonic functions:

$$u(\mathbf{a}) \leq \frac{1}{\mathrm{Vol}\overline{B}_R(\mathbf{a})} \iiint_{\overline{B}_R(\mathbf{a})} u(\mathbf{x}) \, \mathrm{d}V < \frac{1}{\mathrm{Area}S_R(\mathbf{a})} \iint_{S_R(\mathbf{a})} u(\mathbf{x}) \, \mathrm{d}S$$

for every $u \in C^2(B_R(\mathbf{a})) \cap C^1(\overline{B}_R(\mathbf{a}))$ with $\Delta u \geq 0$, which is not harmonic.

Theorem 9.7.1 still works when the Laplacian is replaced by a strictly elliptic self-adjoint linear differential operator of second order which admits a Green function. See [382].

A nice theory explaining the similarities between the convex functions and the subharmonic functions was developed by G. Choquet [132].

A nice generalization of the Bernstein–Hausdorff–Widder–Choquet Theorem 9.5.2 to the framework of harmonic analysis can be found in the book of C. Berg, J. P. R. Christensen, and P. Ressel [53], Theorem 6.4, p.131.

For new results concerning the integral inequalities associated to signed measures, see L. Horváth [265] (and references therein).

Appendix A

Generalized Convexity on Intervals

An important feature underlying the notion of convex function is the comparison of means under the action of a function. Indeed, by considering the *weighted arithmetic mean* of n variables,

$$A(x_1, \ldots, x_n; \lambda_1, \ldots, \lambda_n) = \sum_{k=1}^{n} \lambda_k x_k,$$

the convex sets are precisely the sets invariant under the action of these means, while the convex functions are the functions that verify estimates of the form

$$f\left(A(x_1, \ldots, x_n; \lambda_1, \ldots, \lambda_n)\right) \leq A(f(x_1), \ldots, f(x_n); \lambda_1, \ldots, \lambda_n). \qquad (AA)$$

This motivates to call the usual concept of convexity also (A, A)-convexity.

It is natural to ask: What is happening if we consider other means?

The interval $(0, \infty)$ is closed under the action of *weighted geometric mean* of n variables,

$$G(x_1, \ldots, x_n; \lambda_1, \ldots, \lambda_n) = \prod_{k=1}^{n} x_k^{\lambda_k},$$

a fact that allowed us to consider in Sections 1.3 and 4.1 the log-convex functions, that is, the functions $f : U \to (0, \infty)$ (defined on convex sets) that verify the inequalities

$$f\left(A(x_1, \ldots, x_n; \lambda_1, \ldots, \lambda_n)\right) \leq G(f(x_1), \ldots, f(x_n); \lambda_1, \ldots, \lambda_n); \qquad (AG)$$

this motivates the name of (A, G)-convex for the log-convex functions.

The quasiconvex functions also fit a similar description as they coincide with the (A, M_∞)-convex functions, where the *max mean* is defined by

$$M_\infty(x_1, \ldots, x_n; \lambda_1, \ldots, \lambda_n) = \max\{x_1, \ldots, x_n\}.$$

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The aim of this appendix is to give a short account on the subject of convexity according to a pair of means (acting, respectively, on the domain and the codomain) and to bring into attention new classes of convex-like functions.

A.1 Means

By a mean, we understand a procedure M to associate to each discrete random variable $X : \{1, ..., n\} \to \mathbb{R}$ having the probability distribution

$$P(\{X(k) = x_k) = \lambda_k \quad \text{for } k = 1, ..., n,$$

a number $M(x_1, ..., x_n; \lambda_1, ..., \lambda_n)$ such that

$$M(x_1, ..., x_n; \lambda_1, ..., \lambda_n) \in [\min X, \max X].$$

We make the convention to omit the weights λ_k when they are equal to each others, that is, we put

$$M(x_1, ..., x_n) = M(x_1, ..., x_n; 1/n, ..., 1/n).$$

A mean is called:

symmetric if $M(x_1, ..., x_n) = M(x_{\sigma(1)}, ..., x_{\sigma(n)})$ for every finite family $x_1, ..., x_n$ of elements of I and every permutation σ of $\{1, ..., n\}$ with $n \geq 1$.

continuous when all functions $M(x_1, ..., x_n; \lambda_1, ..., \lambda_n)$ are globally continuous;

increasing/strictly increasing when all functions $M(x_1, ..., x_n; \lambda_1, ..., \lambda_n)$ with strictly positive weights are increasing/strictly increasing in each of the variables x_k (when the others are kept fixed).

When I is one of the intervals $(0, \infty)$, $[0, \infty)$ or $(-\infty, \infty)$, it is usual to consider *homogeneous* means, that is, means for which

$$M(\alpha x_1, ..., \alpha x_n; \lambda_1, ..., \lambda_n) = \alpha M(x_1, ..., x_n; \lambda_1, ..., \lambda_n)$$

whenever $\alpha > 0$.

From a probabilistic point of view, the means represent generalizations of the concept of expectation of a random variable.

The most known class of symmetric, continuous and homogeneous means on $(0, \infty)$ is that of *power means* (also known as the *Hölder's means*):

$$M_p(x_1, ..., x_n; \lambda_1, ..., \lambda_n) = \begin{cases} \min\{x_1, ..., x_n\} & \text{if } p = -\infty \\ (\sum_{k=1}^n \lambda_k x_k^p,)^{1/p} & \text{if } p \in \mathbb{R}\backslash\{0\} \\ \prod_{k=1}^n x_k^{\lambda_k} & \text{if } p = 0 \\ \max\{x_1, ..., x_n\}, & \text{if } p = \infty. \end{cases} \tag{A.1}$$

This family brings together three of the most important means in mathematics: the *arithmetic mean* $A = M_1$, the *geometric mean* $G = M_0$ and the *harmonic*

mean $H = M_{-1}$. Notice that all Power means of real index and equal weights are strictly increasing and $M_0(s,t) = \lim_{p\to 0} M_p(s,t)$. Besides, all power means of odd index $p \geq 1$ makes sense for every family of real numbers.

Power means also admits natural extensions to all positive random variables (discrete or continuous); see Exercise 1, Section 2.2.

A somewhat more general class of means is that of quasi-arithmetic means. In the discrete case, the *quasi-arithmetic means* are associated to strictly monotonic and continuous functions $\varphi : I \to \mathbb{R}$ via the formulas

$$M_\varphi(x_1, ..., x_n; \lambda_1, ..., \lambda_n) = \varphi^{-1}\left(\sum_{k=1}^{n} \lambda_k \varphi(x_k)\right);$$

in the case of an arbitrary random variable, they can be defined as Riemann–Stieltjes integrals,

$$M_\varphi(X) = \varphi^{-1}\left(\int \varphi(x) dF(x)\right),$$

where F represents the cumulative distribution function of X.

A. Kolmogorov [304] has developed the first axiomatic theory of means, providing the ubiquity of quasi-arithmetic means. A nice account on his contribution, as well as on some recent results concerning the probabilistic applications of quasi-arithmetic means, can be found in the paper of M. de Carvalho [157].

A.1.1 Lemma (K. Knopp and B. Jessen; see [247], p. 66) *Suppose that φ and ψ are two continuous functions defined in an interval I such that φ is strictly monotonic and ψ is strictly increasing. Then: $M_\varphi(x_1, \ldots, x_n; \lambda_1, \ldots, \lambda_n) = M_\psi(x_1, \ldots, x_n; \lambda_1, \ldots, \lambda_n)$ for every family $x_1, \ldots, x_n$ of elements of I and every family $\lambda_1, \ldots, \lambda_n$ of positive numbers with $\sum_{k=1}^{n} \lambda_k = 1$ $(n \in \mathbb{N}^\star)$ if and only if $\psi \circ \varphi^{-1}$ is affine, that is, $\psi = \alpha\varphi + \beta$ for some constants α and β, with $\alpha \neq 0$.*

An immediate consequence of Lemma A.1.1 is the fact that every power mean M_p of real index is a quasi-arithmetic mean M_φ, where $\varphi(x) = \log x$, if $p = 0$, and $\varphi(x) = (x^p - 1)/p$, if $p \neq 0$.

A.1.2 Theorem (M. Nagumo, B. de Finetti and B. Jessen; see [247], p. 68) *Let φ be a continuous increasing function on $(0, \infty)$ such that the quasi-arithmetic mean M_φ is positively homogeneous. Then M_φ is a power mean.*

Proof. By Lemma A.1.1, we can replace φ by $\varphi - \varphi(1)$, so we may assume that $\varphi(1) = 0$. The same argument yields two functions α and β such that $\varphi(cx) = \alpha(c)\varphi(x) + \beta(c)$ for all $x > 0$, $c > 0$. The condition $\varphi(1) = 0$ shows that $\beta = \varphi$ so, for reasons of symmetry,

$$\varphi(cx) = \alpha(c)\varphi(x) + \varphi(c) = \alpha(x)\varphi(c) + \varphi(x).$$

Letting $c \neq 1$ fixed, we obtain that α is of the form $\alpha(x) = 1 + k\varphi(x)$ for some constant k. Then φ verifies the functional equation

$$\varphi(xy) = k\varphi(x)\varphi(y) + \varphi(x) + \varphi(y)$$

for all $x > 0$, $y > 0$. When $k = 0$ we find that $\varphi(x) = C \log x$ for some constant C, so $M_\varphi = M_0$. When $k \neq 0$ we notice that $\chi = k\varphi + 1$ verifies $\chi(xy) = \chi(x)\chi(y)$ for all $x > 0$, $y > 0$. This leads to $\varphi(x) = (x^p - 1)/k$, for some $p \neq 0$, hence $M_\varphi = M_p$. ■

Some authors (such as P. S. Bullen, D. S. Mitrinović and P. M. Vasić [106]) consider a more general concept of mean, associated to functions $M : I \times I \to I$ with the single property that

$$\min\{x, y\} \leq M(x, y) \leq \max\{x, y\} \quad \text{for all } x, y \in I.$$

One can interpret $M(x, y)$ as the mean $M(x, y; 1/2, 1/2)$ of a random variable taking only two values, x and y, with probability $1/2$. Unfortunately, the problem to indicate "natural" extensions of such means to the entire class of discrete random variables remains open. The difficulty of this problem is illustrated by the case of *logarithmic* and *identric* means, which are defined, respectively, by

$$L(x, y) = \frac{x - y}{\log x - \log y} \text{ and } I(x, y) = \frac{1}{e}\left(\frac{y^y}{x^x}\right)^{1/(y-x)}$$

for $x, y > 0$, $x \neq y$, and $L(x, x) = I(x, x) = x$ for $x > 0$. The inequalities noticed in Example 1.6.2, can be completed as follows:

$$G(x, y) < L(x, y) < M_{1/3}(x, y) < M_{2/3}(x, y) < I(x, y) < A(x, y)$$

for all $x, y > 0$, $x \neq y$. See T.-P. Lin [338] and K. Stolarsky [554] for the inner inequalities and also for the fact that the logarithmic mean and the identric mean are not power means. Since the logarithmic mean is positively homogeneous, Theorem A.1.2 allows us to conclude that this mean is not quasi-arithmetic.

A satisfactory extension of the logarithmic and the identric means for arbitrary random variables was obtained by E. Neuman [410], [412]. See also the comments by C. P. Niculescu [428].

A.2 Convexity According to a Pair of Means

According to G. Aumann [30], if M and N are means defined, respectively, on the intervals I and J, a function $f : I \to J$ is called (M, N)-*midpoint convex* if

$$f(M(x, y)) \leq N(f(x), f(y)) \text{ for all } x, y \in I;$$

it is called (M, N)-*midpoint concave* if the inequality works in the reverse way, and (M, N)-*midpoint affine* if the inequality sign is replaced by equality. The condition of midpoint affinity is essentially a functional equation and this explains why the theory of generalized convexity has much in common with the subject of functional equations.

In what follows we will be interested in a concept of convexity dealing with weighted means.

A.2.1 Definition *A function $f : I \to J$ is called* (M, N)-convex *if*

$$f(M(x, y; 1-\lambda, \lambda)) \leq N(f(x), f(y); 1-\lambda, \lambda)$$

for all $x, y \in I$ and all $\lambda \in [0, 1]$. It is called (M, N)-strictly convex when the inequality is strict whenever x and y are distinct points and $\lambda \in (0, 1)$. If $-f$ is (M, N)-convex (,respectively, strictly (M, N)-convex), then we say that f is (M, N)-concave (*respectively,* (M, N)-strictly concave).

Jensen's criterion of convexity (see Theorem 1.1.8) can be extended easily to the context of power means (and even to that of quasi-arithmetic means). Thus, in their case, every (M, N)-midpoint and continuous convex function $f : I \to J$ is also (M, N)-convex. Jensen's inequality works as well, providing the possibility to deal with rather general random variables. All these facts are consequences of the following lemma, which reduces the convexity with respect to a pair of quasi-arithmetic means to the usual convexity of a function derived via a change of variable and a change of function.

A.2.2 Lemma (Aczél correspondence principle [5]) *Suppose that φ and ψ are two continuous and strictly monotonic functions defined, respectively, on the intervals I and J. Then:*

(a) *if ψ is strictly increasing, a function $f : I \to J$ is (M_φ, M_ψ)-convex/concave if and only if $\psi \circ f \circ \varphi^{-1}$ is convex/concave on $\varphi(I)$;*

(b) *if ψ is strictly decreasing, a function $f : I \to J$ is (M_φ, M_ψ)-convex/concave if and only if $\psi \circ f \circ \varphi^{-1}$ is concave/convex on $\varphi(I)$.*

A.2.3 Corollary *Suppose that $\varphi, \psi : I \to \mathbb{R}$ are two strictly monotonic continuous functions. If ψ is strictly increasing, then*

$$M_\varphi \leq M_\psi$$

if and only if $\psi \circ \varphi^{-1}$ is convex.

Corollary A.2.3 has important consequences. For example, as was noticed by J. Lamperti [320], it yields Clarkson's inequalities as stated in Section 3.2. His basic remark is as follows:

A.2.4 Lemma *Suppose that $\Phi : [0, \infty) \to \mathbb{R}$ is a continuous increasing function with $\Phi(0) = 0$ and $\Phi(\sqrt{t})$ convex. Then*

$$\Phi(|z + w|) + \Phi(|z - w|) \geq 2\Phi(|z|) + 2\Phi(|w|), \tag{A.2}$$

for all $z, w \in C$, while if $\Phi(\sqrt{t})$ is concave, then the reverse inequality is true. Provided the convexity or concavity is strict, equality holds if and only if $zw = 0$.

Clarkson's inequalities follow for $\Phi(t) = t^p$; then the function $\Phi(\sqrt{t})$ is concave for $1 < p \leq 2$ and convex for $2 \leq p < \infty$.

Proof. When $\Phi(\sqrt{t})$ is convex, we infer from Corollary A.2.3 and the parallelogram law that

$$\Phi^{-1}\left\{\frac{\Phi(|z+w|)+\Phi(|z-w|)}{2}\right\} \geq \left\{\frac{|z+w|^2+|z-w|^2}{2}\right\}^{1/2} = (|z|^2+|w|^2)^{1/2}. \tag{A.3}$$

On the other hand, taking into account the three chords inequality (as stated in Section 1.4), we infer from the convexity of $\Phi(\sqrt{t})$ and the fact that $\Phi(0)=0$ the increasing monotonicity of $\Phi(\sqrt{t})/t$; the monotonicity is strict provided that the convexity of $\Phi(\sqrt{t})$ is strict.

Then

$$|z|^2\,\frac{\Phi(|z|)}{|z|^2}+|w|^2\,\frac{\Phi(|w|)}{|w|^2} \leq \left(|z|^2+|w|^2\right)\frac{\Phi\left((|z|^2+|w|^2)^{1/2}\right)}{|z|^2+|w|^2},$$

and the inequality is strict when $\Phi(\sqrt{t})$ is strictly convex and $zw\neq 0$. Therefore $\Phi^{-1}\left(\Phi(|z|)+\Phi(|w|)\right) \leq (|z|^2+|w|^2)^{1/2}$ and this fact together with (A.3) ends the proof in the case where $\Phi(\sqrt{t})$ is convex. The case where $\Phi(\sqrt{t})$ is concave can be treated in a similar way. ■

According to Lemma A.2.2, a function $f:(0,\alpha)\to(0,\infty)$ is (H,G)-convex (concave) on $(0,\alpha)$ if and only if $\log f(1/x)$ is convex (concave) on $(1/\alpha,\infty)$. Clearly, the "strict" variant also works.

The following example of a (H,G)-strictly log-concave function is due to D. Borwein, J. Borwein, G. Fee and R. Girgensohn [82].

A.2.5 Example *Given $\alpha>1$, the function $V_\alpha(p)=2^\alpha\frac{\Gamma(1+1/p)^\alpha}{\Gamma(1+\alpha/p)}$ verifies the inequality*

$$V_\alpha^{1-\lambda}(p)V_\alpha^{\lambda}(q) < V_\alpha\left(\frac{1}{\frac{1-\lambda}{p}+\frac{\lambda}{q}}\right),$$

for all $p,q>0$ with $p\neq q$ and all $\lambda\in(0,1)$. This example has a geometric motivation. When $\alpha>1$ is an integer and $p\geq 1$ is a real number, $V_\alpha(p)$ represents the volume of the ellipsoid $\{x\in\mathbb{R}^\alpha : \|x\|_{L^p}\leq 1\}$.

In fact, according to Lemma A.2.2, it suffices to prove that the function

$$U_\alpha(x) = -\log(V_\alpha(1/x)/2^\alpha) = \log\mathfrak{b}(1+\alpha x)-\alpha\log\mathfrak{b}(1+x)$$

is strictly convex on $(0,\infty)$ for every $\alpha>1$. Using the psi function,

$$\mathrm{Psi}(x)=\frac{d}{dx}\log\mathfrak{b}(x),$$

we have

$$U_\alpha''(x)=\alpha^2\frac{d}{dx}\mathrm{Psi}(1+\alpha x)-\alpha\frac{d}{dx}\mathrm{Psi}(1+x).$$

The condition $U''_\alpha(x) > 0$ *on* $(0,\infty)$ *is equivalent with* $(x/\alpha)U''_\alpha(x) > 0$ *on* $(0,\infty)$*, and the latter holds if the function* $x \to x\frac{d}{dx}\,\mathrm{Psi}(1+x)$ *is strictly increasing. Or, a classical formula in special function theory* [21], [224] *asserts that*

$$\frac{d}{dx}\,\mathrm{Psi}(1+x) = \int_0^\infty \frac{u\mathrm{e}^{ux}}{\mathrm{e}^u - 1}\,du,$$

whence we infer that

$$\frac{d}{dx}\left(x\frac{d}{dx}\,\mathrm{Psi}(1+x)\right) = \int_0^\infty \frac{u[(u-1)\mathrm{e}^u + 1]\mathrm{e}^{ux}}{(\mathrm{e}^u - 1)^2}\,du > 0,$$

and the statement in Example A.2.5 *follows. Notice that the volume function* $V_n(p)$ *is neither convex nor concave for* $n \geq 3$*.*

The following example is devoted to the presence of generalized convexity in the framework of hypergeometric functions.

A.2.6 Example *The Gaussian hypergeometric function (of parameters* $a, b, c > 0$*) is defined via the formula*

$$F(x) =_2 F_1(x; a, b, c) = \sum_{n=0}^{\infty} \frac{(a,n)\,(b,n)}{(c,n)n!}x^n \quad \text{for } |x| < 1,$$

where $(a,n) = a(a+1)\cdots(a+n-1)$ *if* $n \geq 1$ *and* $(a,0) = 1$*. G. D. Anderson, M. K. Vamanamurthy and M. Vuorinen* [19] *proved that if* $a + b \geq c > 2ab$ *and* $c \geq a + b - 1/2$*, then the function* $1/F(x)$ *is concave on* $(0,1)$*. This implies*

$$F\left(\frac{x+y}{2}\right) \leq \frac{1}{\frac{1}{2}\left(\frac{1}{F(x)} + \frac{1}{F(y)}\right)} \quad \text{for all } x, y \in (0,1),$$

whence it follows that the hypergeometric function $F(x)$ *is* (A,H)*-convex.*

Last but not least, Lemma A.2.2 offers a simple but powerful tool for deriving the entire theory of (M_r, M_s)-convex functions from that of usual convex functions. We will illustrate this fact in the next section, by discussing the multiplicative analogue of usual convexity.

A.3 A Case Study: Convexity According to the Geometric Mean

The following relative of usual convexity was first considered by P. Montel [394] in a beautiful paper discussing different analogues of convex functions in several variables.

A.3.1 Definition *A function $f\colon I \to J$, acting on subintervals of $(0,\infty)$, is called multiplicatively convex (equivalently, (G,G)-convex) if it verifies the condition*

$$f(x^{1-\lambda}y^{\lambda}) \le f(x)^{1-\lambda}f(y)^{\lambda}, \tag{GG}$$

whenever $x, y \in I$ and $\lambda \in [0,1]$. The related notions of multiplicatively strictly convex function, multiplicatively concave function, multiplicatively strictly concave function and multiplicatively affine function can be introduced in a similar manner.

The basic tool in translating the results known for convex functions into results valid for multiplicatively convex functions (and vice versa) is the following particular case of Lemma A.2.2:

A.3.2 Lemma *Suppose that I is a subinterval of $(0,\infty)$ and $f\colon I \to (0,\infty)$ is a multiplicatively (strictly) convex function on I. Then*

$$F = \log \circ f \circ \exp\colon \log(I) \to \mathbb{R}$$

is a (strictly) convex function. Conversely, if J is an interval of $\mathbb{R}$ and $F\colon J \to \mathbb{R}$ is a (strictly) convex function, then

$$f = \exp \circ F \circ \log\colon \exp(J) \to (0,\infty)$$

is a multiplicatively (strictly) convex function.

The corresponding "concave" variant of this lemma also works.

This approach of multiplicative convexity follows C. P. Niculescu [415], [417].

According to Lemma A.3.2, every multiplicatively convex function f has finite lateral derivatives at each interior point of its domain (and the set of all points where f is not differentiable is at most countable). As a consequence, every multiplicatively convex function is continuous at the interior points of its domain. The following result represents the multiplicative analogue of Theorem 1.1.8 (Jensen's criterion of convexity).

A.3.3 Theorem *Suppose that I is a subinterval of $(0,\infty)$ and $f\colon I \to (0,\infty)$ is a function continuous on the interior of I. Then f is multiplicatively convex if and only if*

$$f(\sqrt{xy}) \le \sqrt{f(x)f(y)} \text{ for all } x, y \in I.$$

A large class of strictly multiplicatively convex functions is indicated by the following result:

A.3.4 Proposition (G. H. Hardy, J. E. Littlewood and G. Pólya [247, Theorem 177, p. 125]) *Every polynomial $P(x)$ with positive coefficients is a multiplicatively convex function on $(0,\infty)$. More generally, every real analytic function $f(x) = \sum_{n=0}^{\infty} c_n x^n$ with positive coefficients is a multiplicatively convex function on $(0,R)$, where R denotes the radius of convergence.*

Notice that except for the case of functions Cx^n (with $C > 0$ and $n \in \mathbb{N}$), Proposition A.3.4 exhibit examples of strictly multiplicatively convex functions (which are also strictly increasing and strictly convex). So are:

exp, sinh and cosh on $(0, \infty)$;

tan, sec, csc and $\frac{1}{x} - \cot x$ on $(0, \pi/2)$;

arcsin on $(0, 1]$;

$-\log(1-x)$ and $\frac{1+x}{1-x}$ on $(0, 1)$.

See the table of series in I. S. Gradshteyn and I. M. Ryzhik [224].

By continuity, it suffices to prove only the first assertion. Suppose that $P(x) = \sum_{n=0}^{N} c_n x^n$. According to Theorem A.3.3, we have to prove that

$$x, y > 0 \text{ implies } (P(\sqrt{xy}))^2 \leq P(x)P(y),$$

or, equivalently,

$$x, y > 0 \text{ implies } (P(xy))^2 \leq P(x^2)P(y^2).$$

The later implication is an easy consequence of Cauchy–Bunyakovsky–Schwarz inequality.

The following remark collects a series of useful facts concerning the multiplicative convexity of concrete functions:

A.3.5 Remark

(a) *If a function is* log*-convex and increasing, then it is multiplicatively convex.*

(b) *If a function f is multiplicatively convex, then the function $1/f$ is multiplicatively concave (and vice versa).*

(c) *If a function f is multiplicatively convex, increasing and one-to-one, then its inverse is multiplicatively concave (and vice versa).*

(d) *If a function f is multiplicatively convex, so is $x^\alpha [f(x)]^\beta$ (for all $\alpha \in \mathbb{R}$ and all $\beta > 0$).*

(e) *If f is continuous, and one of the functions $f(x)^x$ and $f(\mathrm{e}^{1/\log x})$ is multiplicatively convex, then so is the other.*

(f) *The general form of multiplicatively affine functions is Cx^α with $C > 0$ and $\alpha \in \mathbb{R}$.*

We omit the easy proof.

The indefinite integral of a multiplicatively convex function has the same nature:

A.3.6 Proposition (P. Montel [394]) *Let $f\colon [0, a) \to [0, \infty)$ be a continuous function which is multiplicatively convex on $(0, a)$. Then*

$$F(x) = \int_0^x f(t)\, dt$$

is also continuous on $[0, a)$ and multiplicatively convex on $(0, a)$.

Proof. Due to the continuity of F, it suffices to show that

$$(F(\sqrt{xy}))^2 \le F(x)F(y) \quad \text{for all } x, y \in [0, a),$$

which is a consequence of the corresponding inequality at the level of integral sums,

$$\Big[\frac{\sqrt{xy}}{n}\sum_{k=0}^{n-1} f\Big(k\,\frac{\sqrt{xy}}{n}\Big)\Big]^2 \le \Big[\frac{x}{n}\sum_{k=0}^{n-1} f\Big(k\,\frac{x}{n}\Big)\Big]\Big[\frac{y}{n}\sum_{k=0}^{n-1} f\Big(k\,\frac{y}{n}\Big)\Big],$$

that is, of the inequality

$$\Big[\sum_{k=0}^{n-1} f\Big(k\,\frac{\sqrt{xy}}{n}\Big)\Big]^2 \le \Big[\sum_{k=0}^{n-1} f\Big(k\,\frac{x}{n}\Big)\Big]\Big[\sum_{k=0}^{n-1} f\Big(k\,\frac{y}{n}\Big)\Big].$$

To see that the later inequality holds, first notice that

$$\Big[f\Big(k\,\frac{\sqrt{xy}}{n}\Big)\Big]^2 \le \Big[f\Big(k\,\frac{x}{n}\Big)\Big]\Big[f\Big(k\,\frac{y}{n}\Big)\Big]$$

and then apply the Cauchy–Bunyakovsky–Schwarz inequality. ■

According to Proposition A.3.6, the *logarithmic integral*,

$$\mathrm{Li}(x) = \int_2^x \frac{dt}{\log t}, \quad x \ge 2,$$

is multiplicatively convex. This function is important in number theory. For example, if $\pi(x)$ counts the number of primes p such that $2 \le p \le x$, then an equivalent formulation of the Riemann hypothesis is the existence of a function $C\colon (0,\infty) \to (0,\infty)$ such that

$$|\pi(x) - \mathrm{Li}(x)| \le C(\varepsilon)x^{1/2+\varepsilon} \quad \text{for all } x \ge 2 \text{ and all } \varepsilon > 0.$$

Since the function tan is continuous on $[0, \pi/2)$ and strictly multiplicatively convex on $(0, \pi/2)$, a repeated application of Proposition A.3.6 shows that the *Lobacevski's function*

$$\mathrm{L}(x) = -\int_0^x \log\cos t\, dt$$

is strictly multiplicatively convex on $(0, \pi/2)$.

Starting with $t/(\sin t)$, (which is strictly multiplicatively convex on $(0, \pi/2]$) and then switching to $(\sin t)/t$, a similar argument leads us to the fact that the *integral sine* function,

$$\mathrm{Si}(x) = \int_0^x \frac{\sin t}{t}\, dt,$$

is strictly multiplicatively concave on $(0, \pi/2]$.

An inspection of the arguments used in the proof of Proposition A.3.6 shows that it also work in the variant of multiplicatively concave functions. As a consequence, the normal distribution

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2} \mathrm{d}t$$

is an example of multiplicatively concave functions. For more general results of this type see the paper of A. Baricz [39].

W. Jarczyk, J. Matkowski [271] have noticed the following dual version of Proposition A.3.6: If $f : (0, \infty) \to (0, \infty)$ is a strictly increasing differentiable function with f' multiplicatively convex and $\lim_{x\to 0} f(x) = 0$, then f is multiplicatively convex too.

A.3.7 Proposition *The gamma function is strictly multiplicatively convex on the interval* $[1, \infty)$.

Proof. In fact, $\log \Gamma(1+x)$ is strictly convex and increasing on $(1, \infty)$. Moreover, an increasing strictly convex function of a strictly convex function is strictly convex. Hence, $F(x) = \log \Gamma(1 + \mathrm{e}^x)$ is strictly convex on $(0, \infty)$ and thus $\Gamma(1 + x) = \exp F(\log x)$ is strictly multiplicatively convex on $[1, \infty)$. As $\Gamma(1 + x) = x\Gamma(x)$, we conclude that Γ itself is strictly multiplicatively convex on $[1, \infty)$. ■

As was noticed by T. Trif [570], the result of Proposition A.3.7 can be improved: the gamma function is strictly multiplicatively concave on $(0, \alpha]$ and strictly multiplicatively convex on $[\alpha, \infty)$, where $\alpha = 0.21609...$ is the unique positive solution of the equation $\mathrm{Psi}(x) + x\frac{d}{dx}\mathrm{Psi}(x) = 0$.

A consequence of Proposition A.3.7 is the strict multiplicative convexity of the function $\Gamma(2x + 1)/\Gamma(x + 1)$ on $[1, \infty)$. This can be seen by using the Gauss–Legendre duplication formula (see [134], Theorem 10.3.10, p. 356).

A.3.8 Remark *D. Gronau and J. Matkowski* [226] *have proved the following multiplicative analogue of the Bohr–Mollerup theorem:* If $f\colon (0, \infty) \to (0, \infty)$ verifies the functional equation

$$f(x + 1) = xf(x),$$

the normalization condition $f(1) = 1$ and f is multiplicatively convex on an interval (a, ∞), for some $a > 0$, then $f = \Gamma$.

A.3.9 Remark *The following result due to J. Matkowski* [367] *connects convexity, multiplicative convexity and* (L, L)*-convexity: every convex and multiplicatively convex function is* (L, L)*-convex. The "concave" variant of this result also works. As a consequence, the exponential function is an example of* (L, L)*-convex function on* $(0, \infty)$*, while the tangent function is an example of* (L, L)*-convex function on* $(0, \pi/2)$.

Exercises

1. Let $f\colon I \to (0,\infty)$ be a differentiable function defined on a subinterval I of $(0,\infty)$. Prove that the following assertions are equivalent:

   (a) f is multiplicatively convex;

   (b) the function $xf'(x)/f(x)$ is increasing;

   (c) f verifies the inequality

   $$\frac{f(x)}{f(y)} \geq \left(\frac{x}{y}\right)^{yf'(y)/f(y)} \quad \text{for all } x, y \in I.$$

   A similar statement works for the multiplicatively concave functions. Illustrate this fact by considering the restriction of $\sin(\cos x)$ to $(0, \pi/2)$.

2. Let $f\colon I \to (0,\infty)$ be a twice differentiable function defined on a subinterval I of $(0,\infty)$. Prove that f is multiplicatively convex if and only if it verifies the differential inequality

   $$x[f(x)f''(x) - f'^2(x)] + f(x)f'(x) \geq 0 \quad \text{for all } x > 0.$$

   Infer that the integral sine function is multiplicatively concave.

3. (H.P. Mulholland [401]) Suppose that $f : (0,\infty) \to (0,\infty)$ is a strictly increasing, continuous, convex and multiplicatively convex function such that $\lim_{x\to 0} f(x) = 0$. Prove that for all $x_1, x_2, y_1, y_2 > 0$ we have

   $$f^{-1}\left(f(x_1 + y_1) + f(x_2 + y_2)\right) \leq f^{-1}\left(f(x_1 + x_2)\right) + f^{-1}\left(f(y_1 + y_2)\right).$$

   Notice that for $f(x) = x^p$ $(p \geq 1)$ one retrieves the inequality of Minkowski.

4. (A multiplicative variant of Popoviciu's inequality) Prove that

   $$\Gamma(x)\Gamma(y)\Gamma(z)\Gamma^3(\sqrt[3]{xyz}) \geq \Gamma^2(\sqrt{xy})\Gamma^2(\sqrt{yz})\Gamma^2(\sqrt{zx})$$

   for all $x, y, z \geq 1$; equality occurs only for $x = y = z$.

5. (The multiplicative analogue of the Hermite–Hadamard inequality [428]). Assume that $f : [a,b] \to (0,\infty)$ is a multiplicatively convex function. Prove that

   $$f(I(a,b)) \leq \frac{1}{b-a}\int_a^b f(x)dx \leq \frac{1}{b-a}\int_a^b f(a)^{\frac{\log b - \log x}{\log b - \log a}} f(b)^{\frac{\log x - \log a}{\log b - \log a}} dx = \frac{L\left(af(a), bf(b)\right)}{L(a,b)}.$$

   [*Hint*: Apply the Hermite–Hadamard inequality to the function $g(t) = \log f(e^t)$.]

Appendix B

Background on Convex Sets

This appendix is devoted to Hahn–Banach theorem and its consequences to the theory of convex functions.

B.1 The Hahn-Banach Extension Theorem

Throughout, E will denote a *real* linear space.

A functional $p\colon E \to \mathbb{R}$ is *subadditive* if $p(\mathbf{x}+\mathbf{y}) \leq p(\mathbf{x}) + p(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in E$; p is *positively homogeneous* if $p(\lambda \mathbf{x}) = \lambda p(\mathbf{x})$ for each $\lambda \geq 0$ and each $\mathbf{x}$ in E; p is *sublinear* if it has both the above properties. A sublinear functional p is a seminorm if $p(\lambda \mathbf{x}) = |\lambda| p(\mathbf{x})$ for all λ. Finally, a seminorm p is a norm if

$$p(\mathbf{x}) = 0 \implies \mathbf{x} = 0.$$

If p is a sublinear functional, then $p(\mathbf{0}) = 0$ and $-p(-\mathbf{x}) \leq p(\mathbf{x})$. If p is a seminorm, then $p(\mathbf{x}) \geq 0$ for all $\mathbf{x}$ in E and $\{\mathbf{x} : p(\mathbf{x}) = 0\}$ is a linear subspace of E.

B.1.1 Theorem (The Hahn–Banach theorem) *Let p be a sublinear functional on E, let E_0 be a linear subspace of E, and let $f_0\colon E_0 \to \mathbb{R}$ be a linear functional dominated by p, that is, $f_0(\mathbf{x}) \leq p(\mathbf{x})$ for all $\mathbf{x} \in E_0$. Then f_0 has a linear extension f to E which is also dominated by p.*

Proof. We consider the set $\mathcal{P}$ of all pairs (h, H), where H is a linear subspace of E that contains E_0 and $h\colon H \to \mathbb{R}$ is a linear functional dominated by p that extends f_0. $\mathcal{P}$ is nonempty (as $(f_0, E_0) \in \mathcal{P}$). One can easily prove that $\mathcal{P}$ is inductively ordered with respect to the order relation

$$(h, H) \prec (h', H') \iff H \subset H' \text{ and } h'|_H = h,$$

so that by Zorn's lemma we infer that $\mathcal{P}$ contains a maximal element (g, G). It remains to prove that $G = E$. If $G \neq E$, then we can choose an element $\mathbf{z} \in E \backslash G$ and denote by G' the set of all elements of the form $\mathbf{x} + \lambda \mathbf{z}$, with

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C. P. Niculescu and L.-E. Persson, *Convex Functions and Their Applications*, CMS/CAIMS Books in Mathematics 14, https://doi.org/10.1007/978-3-031-71967-7

$\mathbf{x} \in G$ and $\lambda \in \mathbb{R}$. Clearly, G' is a linear space that contains G strictly and the formula

$$g'(\mathbf{x} + \lambda\mathbf{z}) = g(\mathbf{x}) + \alpha\lambda$$

defines (for every $\alpha \in \mathbb{R}$) a linear functional on G' that extends g. We shall show that α can be chosen so that g' is dominated by p (a fact that contradicts the maximality of (g, G)). In fact, g' is dominated by p if

$$g(\mathbf{x}) + \alpha\lambda \leq p(\mathbf{x} + \lambda\mathbf{z})$$

for every $\mathbf{x} \in G$ and every $\lambda \in \mathbb{R}$. If $\lambda \geq 0$, this means

$$g(\mathbf{x}) + \alpha \leq p(\mathbf{x} + \mathbf{z}) \quad \text{for every } \mathbf{x} \in G.$$

If $\lambda < 0$, we get (after simplification by $-\lambda$),

$$g(\mathbf{x}) - \alpha \leq p(\mathbf{x} - \mathbf{z}) \quad \text{for every } \mathbf{x} \in G.$$

Therefore, we have to choose α such that

$$g(\mathbf{u}) - p(\mathbf{u} - \mathbf{z}) \leq \alpha \leq p(\mathbf{v} + \mathbf{z}) - g(\mathbf{v})$$

for every $\mathbf{u}, \mathbf{v} \in G$. This choice is possible because

$$g(\mathbf{u}) + g(\mathbf{v}) = g(\mathbf{u} + \mathbf{v}) \leq p(\mathbf{u} + \mathbf{v}) \leq p(\mathbf{u} - \mathbf{z}) + p(\mathbf{v} + \mathbf{z})$$

for all $\mathbf{u}, \mathbf{v} \in G$, which yields

$$\sup_{\mathbf{u} \in G}(g(\mathbf{u}) - p(\mathbf{u} - \mathbf{z})) \leq \inf_{\mathbf{v} \in G}(p(\mathbf{v} + \mathbf{z}) - g(\mathbf{v})).$$

The proof is now complete. ■

B.1.2 Corollary *If p is a sublinear functional on a real linear space E, then for every element $\mathbf{x}_0 \in E$ there exists a linear functional $f\colon E \to \mathbb{R}$ such that $f(\mathbf{x}_0) = p(\mathbf{x}_0)$ and $f(\mathbf{x}) \leq p(\mathbf{x})$ for all $\mathbf{x}$ in E.*

Proof. Take $E_0 = \{\lambda\mathbf{x}_0 : \lambda \in \mathbb{R}\}$ and $f_0(\lambda\mathbf{x}_0) = \lambda p(\mathbf{x}_0)$ in Theorem B.1.1. ■

The continuity of a linear functional on a topological linear space E means that it is bounded in a neighborhood of the origin. We shall denote by E^* the *dual space* of E that is, the space of all continuous linear functionals on E. The bilinear map

$$\langle \cdot, \cdot \rangle : E \times E^* \to \mathbb{R}, \quad \langle \mathbf{x}, x^* \rangle = x^*(\mathbf{x})$$

is called the *duality map* between E and E^*.

In the context of normed linear spaces, the remark above allows us to define the norm of a continuous linear functional $f\colon E \to \mathbb{R}$ by the formula

$$\|f\| = \sup_{\|\mathbf{x}\| \leq 1} |f(\mathbf{x})|.$$

With respect to this norm, the dual space of a normed linear space is always complete.

It is worth noting the following variant of Theorem B.1.1:

B.1.3 Theorem (The Hahn-Banach theorem) *Let E_0 be a linear subspace of the normed linear space E, and let $f_0 \colon E_0 \to \mathbb{R}$ be a continuous linear functional. Then f_0 has a continuous linear extension f to E, with $\|f\| = \|f_0\|$.*

B.1.4 Corollary *If E is a normed linear space, then for each $\mathbf{x}_0 \in E$ with $x_0 \neq 0$ there exists a continuous linear functional $f \colon E \to \mathbb{R}$ such that $f(\mathbf{x}_0) = \|\mathbf{x}_0\|$ and $\|f\| = 1$.*

B.1.5 Corollary *If E is a normed linear space and $\mathbf{x}$ is an element of E such that $f(\mathbf{x}) = 0$ for all f in the dual space of E, then $\mathbf{x} = 0$.*

The *weak topology* on E is the locally convex topology associated to the family of seminorms

$$p_F(\mathbf{x}) = \sup\{|f(\mathbf{x})| : f \in F\},$$

where F runs over all nonempty finite subsets of E^*. A sequence $(\mathbf{x}_n)_n$ converges to $\mathbf{x}$ in the weak topology (abbreviated, $\mathbf{x}_n \overset{w}{\to} \mathbf{x}$) if and only if $f(\mathbf{x}_n) \to f(\mathbf{x})$ for every $f \in E^*$. When $E = \mathbb{R}^N$ this is the coordinate-wise convergence and agrees with the norm convergence. In general, the norm function is only *weakly lower semicontinuous*, that is,

$$\mathbf{x}_n \overset{w}{\to} \mathbf{x} \implies \|\mathbf{x}\| \leq \liminf_{n\to\infty} \|\mathbf{x}_n\|.$$

By Corollary B.1.5 it follows that E^* *separates* E in the sense that

$$\mathbf{x}, \mathbf{y} \in E \quad \text{and} \quad f(\mathbf{x}) = f(\mathbf{y}) \quad \text{for all } f \in E^* \implies \mathbf{x} = \mathbf{y}.$$

As a consequence we infer that the weak topology is separated (equivalently, Hausdorff).

For E^* we can speak of the normed topology, of the weak topology (associated to $E^{**} = (E^*)^*$) and also of the *weak-star topology*, which is associated to the family of seminorms p_F defined as above, with the difference that F runs over all nonempty finite subsets of E. The weak-star topology on E^* is separated.

A net $(f_i)_{i\in I}$ (over some directed set I) converges to f in the weak-star topology (abbreviated, $f_i \overset{w^*}{\to} f$) if and only if $f_i(\mathbf{x}) \to f(\mathbf{x})$ for all $\mathbf{x} \in E$.

B.1.6 Theorem (The Banach–Alaoglu theorem) *If E is a normed linear space, then the closed unit ball of its dual space is compact in the weak-star topology. Consequently, each net of points of this ball has a converging subnet.*

See [155, p. 47] for details.

When E is a separable normed linear space, the closed unit ball of E^* is also a metrizable space in the weak-star topology (and in this case dealing with sequences suffices as well). We come to the separability situation very often, by replacing E with a subspace generated by a suitable sequence of elements.

B.1.7 Remark *According to the Banach–Alaoglu theorem, if E is a normed linear space, then each weak-star closed subset of the closed unit ball of the dual of E is weak-star compact. This is a big source of compact convex sets in mathematics. For example, so is the set* $\mathrm{Prob}(X)$*, of all Borel probability measures on a compact Hausdorff space X. These are the regular σ-additive measures μ on the Borel subsets of X with $\mu(X) = 1$. The Riesz–Kakutani representation theorem* (see [252, p. 177]) *allows us to identify* $\mathrm{Prob}(X)$ *with the following weak-star closed subset of norm-1 functionals of $C(X)^*$:*

$$K = \{L : L \in C(X)^*,\ L(1) = 1 = \|L\|\}.$$

Notice that K consists of positive functionals*, that is,*

$$f \in C(X), \quad f \geq 0 \text{ implies } L(f) \geq 0.$$

In fact, if the range of f is included in $[0, 2r]$, then $\|f - r\| \leq r$, so that $r \geq |L(f - r)| = |L(f) - r|$, that is, $L(f) \in [0, 2r]$.

Corollary B.1.4 yields an important canonical embedding of each normed linear space E into its *second dual* $E^{**} = (E^*)^*$:

$$J_E \colon E \to E^{**}, \quad J_E(\mathbf{x})(x^*) = x^*(\mathbf{x}).$$

One can easily show that J_E is a linear isometry.

A Banach space E is said to be *reflexive* if J_E is onto (that is, if E is isometric with its second dual through J_E). Besides the finite-dimensional Banach spaces, other examples of reflexive Banach spaces are Hilbert spaces and the spaces $L^p(\mu)$ for $1 < p < \infty$. One can easily prove the following permanence properties:

(R1) Every closed subspace of a reflexive space is reflexive.

(R2) The dual of a reflexive space is also a reflexive space.

(R3) Reflexivity preserves under renorming by an equivalent norm.

Property (R3) follows from the following characterization of reflexivity:

B.1.8 Theorem (The Eberlein–Šmulyan theorem) *A Banach space E is reflexive if and only if every bounded sequence of elements of E admits a weakly converging subsequence.*

The *necessity* part is a consequence of the Banach–Alaoglu theorem (Theorem B.1.6). In fact, we may restrict ourselves to the case where E is also separable. The *sufficiency* part follows from the remark that J_E maps the closed unit ball of E into a weak-star dense (and also weak-star closed) subset of the closed unit ball of E^{**}. Full details are available in books such as those by H. W. Alt [14], M. M. Day [155] or N. Dunford and J. Schwartz [169].

B.1.9 Corollary *Every nonempty closed and convex subset of a reflexive space is proximinal.*

The proof follows easily from the Eberlein–Šmulyan theorem and the weak lower semicontinuity of the norm function.

B.2 Separation of Convex Sets

The notion of a hyperplane in a real linear space E was introduced in Section 3.3 as the translate of the kernel of a nonzero linear functional. It can be equally defined as a maximal proper affine subset. In fact, if $h\colon E \to \mathbb{R}$ is a nonzero linear functional, we may choose a $\mathbf{v} \in E$ with $h(\mathbf{v}) = 1$. Then all $\mathbf{x} \in E$ can be represented as

$$\mathbf{x} = (\mathbf{x} - h(\mathbf{x})\mathbf{v}) + h(\mathbf{x})\mathbf{v}$$

where $\mathbf{x} - h(\mathbf{x})\mathbf{v} \in \ker h$. This shows that $\ker h$ is a linear space of codimension 1, and thus all its translates are maximal proper affine subsets.

Conversely, if H is a maximal proper affine set in E and $\mathbf{x}_0 \in H$, then $-\mathbf{x}_0 + H$ is a linear subspace (necessarily of codimension 1). Hence there exists a vector $\mathbf{v} \neq \mathbf{0}$ such that E is the direct sum of $-\mathbf{x}_0 + H$ and $\mathbb{R}\mathbf{v}$, that is, all $\mathbf{x} \in E$ can be uniquely represented as

$$\mathbf{x} = (-\mathbf{x}_0 + \mathbf{y}) + \lambda\mathbf{v}$$

for suitable $\mathbf{y} \in H$ and $\lambda \in \mathbb{R}$. The formula $h(\mathbf{x}) = \lambda$ defines a linear functional h such that $h(\mathbf{v}) = 1$ and $h(\mathbf{x}) = 0$ if and only if $\mathbf{x} \in -\mathbf{x}_0 + H$. Consequently,

$$H = \{\mathbf{x} : h(\mathbf{x}) = h(\mathbf{x}_0)\}.$$

Suppose now that E is a Hausdorff linear topological space. Then the discussion above shows that the closed hyperplanes H in E coincide with the constancy sets of nonzero continuous and linear functionals. In fact, it suffices to consider the case where H is a closed subspace of codimension 1. In that case E/H is one-dimensional and thus it is algebraically and topologically isomorphic to $\mathbb{R}$. By composing such an isomorphism with the canonical projection from E onto E/H we obtain a continuous linear functional h for which $H = \ker h$.

To each hyperplane $\{\mathbf{x} : h(\mathbf{x}) = \lambda\}$ we can attach two *half-spaces*,

$$\{\mathbf{x} : h(\mathbf{x}) \leq \lambda\} \quad \text{and} \quad \{\mathbf{x} : h(\mathbf{x}) \geq \lambda\}.$$

We say that two sets A and B are *separated* by a hyperplane H if they are contained in different half-spaces. The separation is *strict* if at least one of the two sets does not intersect H.

A basic result concerning the separability by hyperplanes is as follows:

B.2.1 Theorem (Mazur's theorem) *Let K be a convex set with nonempty interior in a real linear topological Hausdorff space E and let A be an affine subset which contains no interior point of K. Then there exists a closed hyperplane H such that $H \supset A$ and $H \cap K = \emptyset$.*

In other words, there exists a continuous linear functional $h\colon E \to \mathbb{R}$ and a number $\alpha \in \mathbb{R}$ such that $h(\mathbf{x}) = \alpha$ if $\mathbf{x} \in A$ and $h(\mathbf{x}) < \alpha$ if $\mathbf{x} \in K$.

Proof. We may assume that K is a convex neighborhood of the origin since otherwise we choose an interior point $\mathbf{x}_0$ in K and replace K and A by $K - \mathbf{x}_0$ and $A - \mathbf{x}_0$, respectively. Notice that translations are isomorphisms, so they preserve the nature of K and A. Denote by E_0 the linear span of A. Then A is a hyperplane in E_0, which yields a linear functional $f_0\colon E_0 \to \mathbb{R}$ such that

$$A = \{\mathbf{x} \in E_0 : f_0(\mathbf{x}) = 1\}.$$

The Minkowski functional of K,

$$p_K(\mathbf{x}) = \inf\{\lambda > 0 : \mathbf{x} \in \lambda K\},$$

is sublinear and $\{\mathbf{x} : p_k(\mathbf{x}) < 1\}$ coincides with the interior of K. In fact, if x is an interior point of K, then $\mathbf{x} + V \subset K$ for a convex neighborhood V of the origin. Due to the continuity of the map $\lambda \to \lambda\mathbf{x}$, there must exist a $\lambda \in (0,1)$ with $\lambda\mathbf{x} \in V$. Then $\mathbf{x} + \lambda\mathbf{x} \in \mathbf{x} + V \subset K$, so that $p_K(\mathbf{x}) < 1$. Conversely, if $p_K(\mathbf{x}) < 1$, then $\mathbf{x} \in \lambda K$ for some $\lambda \in (0,1)$, which yields

$$\mathbf{x} \in \mathbf{x} + (1-\lambda)K \subset \lambda K + (1-\lambda)K = K.$$

Notice that $(1-\lambda)K$ is a neighborhood of the origin. Since A contains no interior point of K it follows that $f_0(\mathbf{x}) = 1 \le p_K(\mathbf{x})$ for all $\mathbf{x} \in A$. If $\mathbf{x} \in A$ and $\lambda > 0$, then $f_0(\lambda\mathbf{x}) \le p_K(\lambda\mathbf{x})$, while for $\lambda \le 0$ we have $f_0(\lambda\mathbf{x}) \le 0 \le p_K(\lambda\mathbf{x})$. Consequently, $f_0 \le p_K$ on E_0. By Theorem B.1.1, f_0 has a linear extension f to E such that $f \le p_K$. Put $H = \{\mathbf{x}; f(\mathbf{x}) = 1\}$. Then H is a hyperplane. Since $|f(\mathbf{x})| \le p_K(\mathbf{x}) < 1$ for x in K, it follows that f is bounded on a neighborhood of 0 and thus continuous. Therefore H is a closed hyperplane and it is clear that $H \supset A$ and $H \cap K = \emptyset$. ■

B.2.2 Corollary *If U is a nonempty open convex set and F is a linear subspace such that $F \cap U = \emptyset$, then there exists a continuous linear functional f such that $f(\mathbf{x}) = 0$ if $\mathbf{x} \in F$ and $f(\mathbf{x}) > 0$ if $\mathbf{x} \in U$.*

In order to prove a strict separation result we need the following lemma of independent interest:

B.2.3 Lemma *Suppose that K_1 and K_2 are two nonempty convex sets in a real linear topological space E with $K_1 \cap K_2 = \emptyset$. If one of them is open, then there exists a closed hyperplane separating K_1 from K_2.*

Proof. If K_1 is open, then the set

$$U = K_1 - K_2 = \bigcup_{k_2 \in K_2} (K_1 - k_2)$$

is open. Since K_1 and K_2 are convex, U is convex too. Moreover, $\mathbf{0} \notin U$ since $K_1 \cap K_2 = \emptyset$. By Corollary B.2.2 there exists a continuous linear functional f

such that $f(\mathbf{x}) > 0$ on U. Therefore $f(\mathbf{x}) > f(\mathbf{y})$ for all $\mathbf{x} \in K_1$ and all $\mathbf{y} \in K_2$. Letting

$$\alpha = \inf\{f(\mathbf{x}) : \mathbf{x} \in K_1\},$$

one can show immediately that K_1 and K_2 are separated by the closed hyperplane $H = \{\mathbf{x} : f(\mathbf{x}) = \alpha\}$. The proof is completes. ■

B.2.4 Theorem (Strong separation theorem) *Let K_1 and K_2 be two nonempty convex sets in a real locally convex Hausdorff space E such that $K_1 \cap K_2 = \emptyset$. If K_1 is compact and K_2 is closed, then there exists a closed hyperplane strictly separating K_1 from K_2.*

Particularly, if K is a closed convex set in a locally convex space E and $\mathbf{x} \in E$ is not in K, then there exists a functional $f \in E^*$ such that

$$f(\mathbf{x}) > \sup\{f(\mathbf{y}) : \mathbf{y} \in K\}.$$

Proof. By our hypothesis, there exists an open convex neighborhood W of the origin such that $(K_1 + W) \cap (K_2 + W) = \emptyset$. This follows directly by using *reductio ad absurdum*. Since the sets $K_1 + W$ and $K_2 + W$ are convex and open, from Lemma B.2.3 we infer the existence of a separating hyperplane H. A moment's reflection shows that H separates strictly K_1 from K_2. This completes the proof. ■

The *closed convex hull* of a subset A of a locally convex space E is the smallest closed convex set $\overline{\text{conv}}(A)$ containing A (that is, the intersection of all closed convex sets containing A). From Theorem B.2.4 we can infer the following result on the support of closed convex sets:

B.2.5 Corollary *If A is a nonempty subset of a real locally convex Hausdorff space E, then the closed convex hull $\overline{\text{conv}}(A)$ is the intersection of all the closed half-spaces containing A. Equivalently,*

$$\overline{\text{conv}}(A) = \bigcap_{f \in E'} \{\mathbf{x} : f(\mathbf{x}) \leq \sup_{\mathbf{y} \in A} f(\mathbf{y})\}.$$

An immediate consequence is as follows.

B.2.6 Corollary *In a real locally convex Hausdorff space E, the closed convex sets and the weakly closed convex sets are the same.*

Finally it is worth mentioning a non-topological version of the separation results above, which is important in optimization theory.

Given a set A in a linear space E, a point a of A is said to be a *core point* if for every $\mathbf{v} \in E$, $\mathbf{v} \neq \mathbf{a}$, there exists an $\varepsilon > 0$ such that $\mathbf{a} + \delta\mathbf{v} \in A$ for every δ with $|\delta| < \varepsilon$.

B.2.7 Theorem *Let K and M be two nonempty convex sets in a real linear space E. If K contains core points and M contains no core point of K, then K and M can be separated by a hyperplane.*

The details can be easily filled out by adapting the argument given in the topological case.

B.3 The Krein–Milman Theorem

Theorem 3.3.5 showed that every compact convex set in $\mathbb{R}^N$ is the convex hull of its extreme points. This result can be extended to a very general setting.

B.3.1 Theorem *Let E be a locally convex Hausdorff space and K be a nonempty compact convex subset of E. If U is a convex subset of K, open in the relative topology, such that* $\operatorname{ext} K \subset U$*, then $U = K$.*

Proof. Suppose that $U \neq K$ and consider the family $\mathcal{U}$ of all open convex sets in K which are not equal to K. By Zorn's lemma, each set $U \in \mathcal{U}$ is contained in a maximal element V of $\mathcal{U}$. For each $\mathbf{x} \in K$ and $t \in [0,1]$, let $\varphi_{\mathbf{x},t} \colon K \to K$ be the continuous map defined by $\varphi_{\mathbf{x},t}(\mathbf{y}) = t\mathbf{y} + (1-t)\mathbf{x}$. Assuming $\mathbf{x} \in V$ and $t \in [0,1)$, we shall show that $\varphi_{\mathbf{x},t}^{-1}(V)$ is an open convex set which contains V properly, hence $\varphi_{x,t}^{-1}(V) = K$. In fact, this is clear when $t = 0$. If $t \in (0,1)$, then $\varphi_{\mathbf{x},t}$ is a homeomorphism and $\varphi_{\mathbf{x},t}^{-1}(V)$ is an open convex set in K. Moreover,

$$\varphi_{\mathbf{x},t}(\overline{V}) \subset V,$$

which yields $\overline{V} \subset \varphi_{\mathbf{x},t}^{-1}(V)$, hence $\varphi_{\mathbf{x},t}^{-1}(V) = K$ by the maximality of V. Therefore $\varphi_{\mathbf{x},t}(K) \subset V$. For any open convex set W in K the intersection $V \cap W$ is also open and convex, and the maximality of V yields that either $V \cup W = V$ or $V \cup W = K$. In conclusion $K \backslash V$ is precisely a singleton $\{\mathbf{e}\}$. But such a point is necessarily an extreme point of K, which is a contradiction. ■

B.3.2 Theorem (Krein–Milman theorem) *Let K be a nonempty compact convex subset of a locally convex Hausdorff space E. Then K is the closed convex hull of* $\operatorname{ext} K$*.*

Proof. By Theorem B.2.4, the set $L = \overline{\operatorname{conv}}(\operatorname{ext} K)$ is the intersection of all open convex sets containing L. If U is an open subset of K and $U \supset L$, then $U \supset \operatorname{ext} K$. Hence $U = K$ and $L = K$. ■

The above proof of the Krein–Milman theorem yields the existence of extreme points as a consequence of the formula $K = \overline{\operatorname{conv}}(\operatorname{ext} K)$. However, this fact can be checked directly. Call a subset A of K *extremal* if it is closed, nonempty and verifies the following property:

$$\mathbf{x}, \mathbf{y} \in K \ \text{ and } \ (1-\lambda)\mathbf{x} + \lambda\mathbf{y} \in A \ \text{ for some } \lambda \in (0,1) \implies \mathbf{x}, \mathbf{y} \in A.$$

By Zorn's lemma we can choose a minimal extremal subset, say S. We show that S is a singleton (which yields an extreme point of K). In fact, if S contains more than one point, the separation Theorem B.2.4 proves the existence of a functional $f \in E^*$ which is not constant on S. But in this case the set

$$S_0 = \{\mathbf{x} \in S : f(\mathbf{x}) = \sup_{\mathbf{y} \in S} f(\mathbf{y})\}$$

will contradict the minimality of S. Now the formula $K = \overline{\text{conv}}(\text{ext}\, K)$ can easily be proved by noticing that the inclusion $\overline{\text{conv}}(\text{ext}\, K) \subset K$ cannot be strict.

B.3.3 Remark *The Krein–Milman theorem provides a unifying tool for many important results in mathematics. We mention here Bernstein's integral representation* (1.15) *of the completely monotonic functions, Bochner's integral representation of positive definite functions and Lyapunov's convexity theorem of the range of non-atomic countably additive vector measure* $\mu : \Sigma \to \mathbb{R}^N$. *Details are available in the book of B. Simon* [542].

Theorem B.3.1 also yields the following result:

B.3.4 Corollary (Bauer maximum principle) *Suppose that* K *is a nonempty compact convex set as in Theorem* B.3.1. *Then every upper semicontinuous convex function* $f\colon K \to \mathbb{R}$ *attains its supremum at an extreme point.*

Proof. Since f is upper semicontinuous, the family of sets

$$U_n = \{\mathbf{x} \in K : f(\mathbf{x}) < n\}, \quad n \in \mathbb{N},$$

provides an open covering of K, so $K = U_n$ for some n, which shows that f is bounded above. Put $M = \sup\{f(\mathbf{x}) : \mathbf{x} \in K\}$. If f does not attain its supremum at a point of $\text{ext}\, K$, then $U = \{\mathbf{x} \in K : f(\mathbf{x}) < M\}$ is an open convex set containing $\text{ext}\, K$. By Theorem B.3.1 we conclude that $U = K$, which is a contradiction. ■

It is interesting to note the following converse to Theorem B.3.1:

B.3.5 Theorem (D. P. Milman) *Suppose that* K *is a compact convex set (in a locally convex Hausdorff space* E*) and* C *is a subset of* K *such that* K *is the closed convex hull of* C. *Then the extreme points of* K *are contained in the closure of* C.

Coming back to Theorem 3.3.5, the fact that every point $\mathbf{x}$ of a compact convex set K in $\mathbb{R}^N$ is a convex combination of extreme points of K,

$$\mathbf{x} = \sum_{k=1}^{m} \lambda_k \mathbf{x}_k,$$

can be reformulated as an integral representation,

$$f(\mathbf{x}) = \sum_{k=1}^{m} \lambda_k f(\mathbf{x}_k) = \int_{\text{ext}\, K} f \, \mathrm{d}\mu \tag{B.1}$$

for all $f \in (\mathbb{R}^n)^*$. Here $\mu = \sum_{k=1}^{m} \lambda_k \delta_{\mathbf{x}_k}$ is a convex combination of Dirac measures $\delta_{\mathbf{x}_k}$ and thus μ itself is a Borel probability measure on $\text{ext}\, K$.

The integral representation (B.1) can be extended to all Borel probability measures μ on a compact convex set K (in a locally convex Hausdorff space E). We shall need some definitions.

Given a Borel probability measure μ on K, and a Borel subset $S \subset K$, we say that μ is *concentrated* on S if $\mu(K \backslash S) = 0$. For example, a Dirac measure $\delta_{\mathbf{x}}$ is concentrated on $\mathbf{x}$.

A point $\mathbf{x} \in K$ is said to be the *barycenter* of μ provided that

$$f(\mathbf{x}) = \int_K f \, \mathrm{d}\mu \quad \text{for all } f \in E^*.$$

Since the functionals separate the points of E, the point $\mathbf{x}$ is uniquely determined by μ. With this preparation, we can reformulate the Krein–Milman theorem as follows:

B.3.6 Theorem *Every point of a compact convex subset K (of a locally convex Hausdorff space E) is the barycenter of a Borel probability measure on K that is supported by the closure of the extreme points of K.*

H. Bauer pointed out that the extremal points of K are precisely the points $\mathbf{x} \in K$ for which the only Borel probability measure μ which admits $\mathbf{x}$ as a barycenter is the Dirac measure $\delta_{\mathbf{x}}$. See [474, p. 6]. This fact together with Theorem B.3.4 yields D. P. Milman's aforementioned converse of the Krein–Milman theorem. For an alternative argument see [155, pp. 103–104].

Theorem B.3.4 led G. Choquet [132] to his theory on integral representation for elements of a closed convex cone.

A Banach space E is said to have the Krein–Milman property (abbreviated, KMP) if every closed bounded convex subset of E is the closed convex hull of its extreme points. Reflexive Banach spaces (like L^p-spaces with $1 < p < \infty$) and separable conjugate Banach spaces (like ℓ^1) have KMP. However, the general structure of Banach spaces with KMP is unknown.

Appendix C

Elementary Symmetric Functions

The *elementary symmetric functions* of n variables are defined by

$$\begin{aligned} e_0(x_1, x_2, \ldots, x_n) &= 1 \\ e_1(x_1, x_2, \ldots, x_n) &= x_1 + x_2 + \cdots + x_n \\ e_2(x_1, x_2, \ldots, x_n) &= \sum_{1 \le i < j \le n} x_i x_j \\ &\vdots \\ e_n(x_1, x_2, \ldots, x_n) &= x_1 x_2 \cdots x_n. \end{aligned}$$

The different e_k being of different degrees, they are not comparable. However, they are connected by nonlinear inequalities. To state them, it is more convenient to consider their averages,

$$E_k(x_1, x_2, \ldots, x_n) = e_k(x_1, x_2, \ldots, x_n) / \binom{n}{k}$$

and to write E_k for $E_k(x_1, x_2, \ldots, x_n)$ in order to avoid excessively long formulas.

Sections C.1 and C.2 are devoted to Newton's inequalities. While the original results due to I. Newton [414] and C. Maclaurin [348] can be proved using just Rolle's theorem, the inequalities of higher order are generated by Sylvester's discriminant families (whose existence implies the semi-algebraic character of the set of all real polynomials with all roots real). Several other inequalities discovered independently by H. F. Bohnenblust, M. Marcus and L. Lopes make the objective of Section C.3. The highlights of the very recent theory of symmetric polynomial majorization are mentioned in Section C.4.

Last, but not least, let us mention that the theory of elementary symmetric polynomials is generalized by that of hyperbolic polynomials (briefly exposed in Section 6.7).

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C.1 Newton's Inequalities

The simplest set of inequalities relating the elementary symmetric functions was discovered by I. Newton [414] and C. Maclaurin [348]:

C.1.1 Theorem *Let $\mathcal{F}$ be an n-tuple of positive numbers. Then:*

$$E_k^2(\mathcal{F}) > E_{k-1}(\mathcal{F}) \cdot E_{k+1}(\mathcal{F}), \quad 1 \le k \le n-1 \tag{N}$$

and

$$E_1(\mathcal{F}) > E_2^{1/2}(\mathcal{F}) > \cdots > E_n^{1/n}(\mathcal{F}) \tag{M}$$

unless all entries of $\mathcal{F}$ coincide.

Actually Newton's inequalities (N) work for n-tuples of real, not necessarily positive, elements. An analytic proof along Maclaurin's ideas will be presented below. In Section C.2 we shall indicate an alternative argument, based on mathematical induction, which yields more Newton type inequalities, in an interpolative scheme.

The inequalities (M) can be deduced from (N) since

$$(E_0E_2)(E_1E_3)^2(E_2E_4)^3 \cdots (E_{k-1}E_{k+1})^k < E_1^2E_2^4E_3^6 \cdots E_k^{2k}$$

gives $E_{k+1}^k < E_k^{k+1}$ or, equivalently,

$$E_k^{1/k} > E_{k+1}^{1/(k+1)}.$$

Among the inequalities noticed above, the most notable is of course the *AM-GM* inequality:

$$\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right)^n \ge x_1x_2 \cdots x_n$$

for all $x_1, x_2, \ldots, x_n \ge 0$. A hundred years after C. Maclaurin, A.-L. Cauchy [124] gave his beautiful inductive argument. Notice that the *AM-GM* inequality was known to Euclid [181] in the special case where $n = 2$.

C.1.2 Remark *Newton's inequalities were intended to solve the problem of counting the number of imaginary roots of an algebraic equation. In Chapter 2 of Part 2 of* Arithmetica Universalis*, entitled* De Forma Æquationis*, Newton made (without proof) the following statement:* Given an equation with real coefficients*,*

$$a_0x^n + a_1x^{n-1} + \cdots + a_n = 0 \quad (a_0 \ne 0),$$

the number of its imaginary roots cannot be less than the number of changes of sign that occur in the sequence

$$a_0^2,\ \left(\frac{a_1}{\binom{n}{1}}\right)^2 - \frac{a_2}{\binom{n}{2}} \cdot \frac{a_0}{\binom{n}{0}},\ \ldots,\ \left(\frac{a_{n-1}}{\binom{n}{n-1}}\right)^2 - \frac{a_n}{\binom{n}{n}} \cdot \frac{a_{n-2}}{\binom{n}{n-2}},\ a_n^2.$$

Accordingly, if all the roots are real, then all the entries in the above sequence must be positive (a fact which yields Newton's inequalities).

Trying to understand which was Newton's argument, C. Maclaurin [348] gave a direct proof of the inequalities (N) and (M), but the Newton counting problem remained open until 1865, when J. Sylvester [560, 561] succeeded in proving a remarkable general result.

Quite unexpectedly, it is the real algebraic geometry (not analysis) which gives us the best understanding of Newton's inequalities. The basic fact (discovered by J. Sylvester) concerns the *semi-algebraic character* of the set of all real polynomials with all roots real:

C.1.3 Theorem (J. Sylvester) *For each natural number $n \geq 2$ there exists a set of at most $n-1$ polynomials with integer coefficients,*

$$R_{n,1}(x_1, \ldots, x_n), \ldots, R_{n,k(n)}(x_1, \ldots, x_n), \tag{R_n}$$

such that the monic real polynomials of order n,

$$P(x) = x^n + a_1 x^{n-1} + \cdots + a_n,$$

which have only real roots are precisely those for which

$$R_{n,1}(a_1, \ldots, a_n) \geq 0, \ldots, R_{n,k(n)}(a_1, \ldots, a_n) \geq 0.$$

The above result can be seen as a generalization of the well-known fact that *the roots of a quadratic polynomial $x^2 + a_1 x + a_2$ are real if and only if its discriminant*

$$D_2(1, a_1, a_2) = a_1^2 - 4a_2 \tag{D_2}$$

is positive.

Theorem C.1.3 is built on Sturm's method of counting real roots, taking into account that only the leading coefficients enter into play. It turns out that they are nothing but the principal subresultant coefficients (with convenient signs added), which are determinants extracted from the Sylvester matrix.

A set $(R_{n,k})_k^{k(n)}$ as in Theorem C.1.3 will be called a *Sylvester family* (of order n).

In Sylvester's approach, $R_{n,1}(a_1, \ldots, a_n)$ equals the *discriminant* D_n of the polynomial $P(x) = x^n + a_1 x^{n-1} + \cdots + a_n$, that is,

$$D_n = D_n(1, a_1, \ldots, a_n) = \prod_{1 \leq i < j \leq n} (x_i - x_j)^2,$$

where $x_1, \ldots, x_n$ are the roots of $P(x)$; D_n is a symmetric and homogeneous (of degree $n^2 - n$) polynomial in $\mathbb{Z}[x_1, \ldots, x_n]$. For details, see [50]. Unfortunately, at present no compact formula for D_n is known. According to [518], the number of nonzero coefficients in the expression for the discriminant increases rapidly with the degree; e.g. D_9 has 26095 terms.

For $n \in \{2, 3\}$ one can indicate Sylvester families consisting of just a single polynomial, the corresponding discriminant. An inspection of the argument given by L. Euler to solve in radicals the quartic equations allows us to write down a Sylvester family for $n = 4$. See the paper by C. P. Niculescu [416].

C.1.4 Remark *Given a Sylvester family for $n = N$, we can easily indicate such a family for each $n \in \{1, \ldots, N\}$; the trick is to replace a $P(x)$ of degree n by $x^{N-n}P(x)$, which is of degree N.*

Also, any Sylvester family $(R_{n,k})_{k=1}^{k(n)}$ (for some $n \geq 2$), allows us to decide which monic real polynomial $P(x) = x^n + a_1 x^{n-1} + \cdots + a_n$ has only positive roots. A set of (necessary and) sufficient conditions consists of the following inequalities:

$$-a_1 \geq 0, \ldots, (-1)^n a_n \geq 0$$

and

$$R_{n,1}(a_1, \ldots, a_n) \geq 0, \ldots, R_{n,k(n)}(a_1, \ldots, a_n) \geq 0.$$

In fact, under the above circumstances, $x < 0$ yields $P(x) \neq 0$.

The proof of the Newton inequalities (N) presented in the book by G. H. Hardy, J. E. Littlewood and G. Pólya [247] follows Maclaurin's argument. The basic ingredient is the following lemma, a consequence of repeated application of Rolle's theorem, which we give here under the formulation of J. Sylvester [561]:

C.1.5 Lemma *If*

$$F(x, y) = c_0 x^n + c_1 x^{n-1} y + \cdots + c_n y^n$$

is a homogeneous function of the n-th degree in x and y, which has all its roots x/y real, then the same is true for all non-identical 0 equations

$$\frac{\partial^{i+j} F}{\partial x^i \partial y^j} = 0,$$

obtained from it by partial differentiation with respect to x and y. Further, if $\mathcal{E} = 0$ is one of these equations, and it has a multiple root α, then α is also a root, of multiplicity one higher, of the equation from which $\mathcal{E}$ is derived by differentiation.

Any polynomial of the n-th degree, with real roots, can be represented as

$$E_0 x^n - \binom{n}{1} E_1 x^{n-1} + \binom{n}{2} E_1 x^{n-2} - \cdots + (-1)^n E_n$$

and we shall apply Lemma C.1.5 to the associated homogeneous polynomial

$$F(x, y) = E_0 x^n - \binom{n}{1} E_1 x^{n-1} y + \binom{n}{2} E_1 x^{n-2} y^2 - \cdots + (-1)^n E_n y^n.$$

Considering the case of the derivatives

$$\frac{\partial^{n-2}F}{\partial x^k \partial y^{n-2-k}} \quad (\text{for } k = 0, \ldots, n-2),$$

we arrive at the fact that all the quadratic polynomials

$$E_{k-1}x^2 - 2E_k xy + E_{k+1}y^2$$

for $k = 0, \ldots, n-2$ also have real roots. Consequently, the Newton inequalities express precisely this fact in the language of discriminants. That is why we shall refer to (N) as the *quadratic Newton inequalities*. An alternative argument for obtaining these inequalities is offered by R. P. Stanley [546], Theorem 2.

Stopping a step ahead, we get what S. Rosset [512] called the *cubic Newton inequalities*:

$$6E_kE_{k+1}E_{k+2}E_{k+3} + 3E_{k+1}^2E_{k+2}^2 \geq 4E_kE_{k+2}^3 + E_k^2E_{k+3}^2 + 4E_{k+1}^3E_{k+3} \quad (\mathrm{N}_3)$$

for $k = 0, \ldots, n-3$. They are motivated by the well-known fact that a cubic real polynomial

$$x^3 + a_1x^2 + a_2x + a_3$$

has only real roots if and only if its discriminant

$$\begin{aligned} D_3 &= D_3(1, a_1, a_2, a_3) \\ &= 18a_1a_2a_3 + a_1^2a_2^2 - 27a_3^2 - 4a_2^3 - 4a_1^3a_3 \end{aligned}$$

is positive. Consequently, the equation

$$E_kx^3 - 3E_{k+1}x^2y + 3E_{k+2}xy^2 - E_{k+3}y^3 = 0$$

has all its roots x/y real if and only if (N_3) holds.

S. Rosset [512] derived the inequalities (N_3) by an inductive argument and noticed that they are strictly stronger than (N). In fact, (N_3) can be rewritten as

$$4(E_{k+1}E_{k+3} - E_{k+2}^2)(E_kE_{k+2} - E_{k+1}^2) \geq (E_{k+1}E_{k+2} - E_kE_{k+3})^2$$

which yields (N).

As concerns the Newton inequalities (N_n) of order $n \geq 2$ (when applied to strings of $m \geq n$ elements), they consist of at most $n-1$ sets of relations, the first one being

$$D_n\left(1, (-1)^1\binom{n}{1}\frac{E_{k+1}}{E_k}, (-1)^2\binom{n}{2}\frac{E_{k+2}}{E_k}, \ldots, (-1)^n\binom{n}{n}\frac{E_{k+n}}{E_k}\right) \geq 0$$

for $k \in \{0, \ldots, m-n\}$.

Notice that each of these inequalities is homogeneous (for example, the last one consists of terms of weight $n^2 - n$) and the sum of all coefficients in the left-hand side is 0.

C.2 More Newton Inequalities

Our argument will yield a bit more, precisely the log concavity of the functions $E_k\colon k \to E_k(\mathcal{F})$:

C.2.1 Theorem *Suppose that $\alpha, \beta \in \mathbb{R}_+$ and $j, k \in \mathbb{N}$ are numbers such that*

$$\alpha + \beta = 1 \quad \text{and} \quad j\alpha + k\beta \in \{0, \dots, n\}.$$

Then

$$E_{j\alpha+k\beta}(\mathcal{F}) \geq E_j^{\alpha}(\mathcal{F}) \cdot E_k^{\beta}(\mathcal{F}),$$

for every n-tuple $\mathcal{F}$ of positive real numbers. Moreover, equality occurs if and only if all entries of $\mathcal{F}$ are equal.

The proof will be done by induction on the length of $\mathcal{F}$, after some preliminaries.

According to Rolle's theorem, if all roots of a polynomial $P \in \mathbb{R}[X]$ are real (respectively, real and distinct), then the same is true for its derivative P'. Given an n-tuple $\mathcal{F} = (x_1, \dots, x_n)$, we shall attach to it the polynomial

$$P_{\mathcal{F}}(x) = (x - x_1) \cdots (x - x_n) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} E_k(x_1, \dots, x_n) x^{n-k}.$$

The $(n-1)$-tuple $\mathcal{F}' = \{y_1, \dots, y_{n-1}\}$, consisting of all roots of the derivative of $P_{\mathcal{F}}(x)$ will be called the *derived n-tuple* of $\mathcal{F}$. Because

$$(x - y_1) \cdots (x - y_{n-1}) = \sum_{k=0}^{n-1} (-1)^k \binom{n-1}{k} E_k(y_1, \dots, y_{n-1}) x^{n-k}$$

and

$$\begin{aligned}(x - y_1) \cdots (x - y_{n-1}) &= \frac{1}{n} \cdot \frac{dP_{\mathcal{F}}}{dx}(x) \\ &= \sum_{k=0}^{n} (-1)^k \frac{n-k}{n} \binom{n}{k} E_k(x_1, \dots, x_n) x^{n-k-1} \\ &= \sum_{k=0}^{n-1} (-1)^k \binom{n-1}{k} E_k(x_1, \dots, x_n) x^{n-1-k}\end{aligned}$$

we are led to the following result, which enables us to reduce the number of variables when dealing with symmetric functions.

C.2.2 Lemma *$E_j(\mathcal{F}) = E_j(\mathcal{F}')$ for every $j \in \{0, \dots, |\mathcal{F}| - 1\}$.*

Another simple but useful fact is the following:

C.2.3 Lemma *Suppose that $\mathcal{F}$ is an n-tuple of real numbers and $0 \notin \mathcal{F}$. Put $\mathcal{F}^{-1} = \{1/a \mid a \in \mathcal{F}\}$. Then*

$$E_j(\mathcal{F}^{-1}) = E_{n-j}(\mathcal{F})/E_n(\mathcal{F})$$

for every $j \in \{0, \dots, n\}$.

Proof of Theorem C.2.1. For $|\mathcal{F}| = 2$ we have to prove just one inequality, namely, $x_1 x_2 \leq (x_1 + x_2)^2/4$, which is clearly valid for every $x_1, x_2 \in \mathbb{R}$; the equality occurs if and only if $x_1 = x_2$.

Suppose now that the assertion of Theorem B.2.1 holds for all k-tuples with $k \leq n-1$. Let $\mathcal{F}$ be a n-tuple of positive numbers ($n \geq 3$), let $j, k \in \mathbb{N}$, and $\alpha, \beta \in \mathbb{R}_+ \backslash \{0\}$ be numbers such that

$$\alpha + \beta = 1 \quad \text{and} \quad j\alpha + k\beta \in \{0, \dots, n\}.$$

According to Lemma C.2.2 (and our inductive hypothesis), we have

$$E_{j\alpha+k\beta}(\mathcal{F}) \geq E_j^{\alpha}(\mathcal{F}) \cdot E_k^{\beta}(\mathcal{F}),$$

except for the case where $j < k = n$ or $k < j = n$. Suppose, for example, that $j < k = n$; then necessarily $j\alpha + n\beta < n$. We have to show that

$$E_{j\alpha+n\beta}(\mathcal{F}) \geq E_j^{\alpha}(\mathcal{F}) \cdot E_n^{\beta}(\mathcal{F}).$$

If $0 \in \mathcal{F}$, then $E_n(\mathcal{F}) = 0$, and the inequality is clear; the equality occurs if and only if $E_{j\alpha+n\beta}(\mathcal{F}') = E_{j\alpha+n\beta}(\mathcal{F}) = 0$, that is (according to our inductive hypothesis), when all entries of $\mathcal{F}$ coincide.

If $0 \notin \mathcal{F}$, then by Lemma C.2.3 we have to prove that

$$E_{n-j\alpha-n\beta}(\mathcal{F}^{-1}) \geq E_{n-j}^{\alpha}(\mathcal{F}^{-1}),$$

or, equivalently (see Lemma C.2.2), $E_{n-j\alpha-n\beta}((\mathcal{F}^{-1})') \geq E_{n-j}^{\alpha}((\mathcal{F}^{-1})')$, which is true by our hypothesis. The proof is complete. ■

Notice that the argument above covers Newton's inequalities even for n-tuples of real (not necessarily positive) elements.

The general problem of comparing monomials in $E_1, \dots, E_n$ was completely solved by G. H. Hardy, J. E. Littlewood and G. Pólya in [247, Theorem 77, p. 64]:

C.2.4 Theorem *Let $\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n$ be positive numbers. Then*

$$E_1^{\alpha_1}(\mathcal{F}) \cdots E_n^{\alpha_n}(\mathcal{F}) \leq E_1^{\beta_1}(\mathcal{F}) \cdots E_n^{\beta_n}(\mathcal{F})$$

for every n-tuple $\mathcal{F}$ of positive numbers if and only if

$$\alpha_m + 2\alpha_{m+1} + \cdots + (n-m+1)\alpha_n \geq \beta_m + 2\beta_{m+1} + \cdots + (n-m+1)\beta_n$$

for $1 \leq m \leq n$, with equality when $m = 1$.

C.3 Some Results of Bohnenblust, Marcus and Lopes

The property of strict concavity of the geometric mean can be extended in the framework of elementary symmetric functions by asserting the strict concavity of all functions $e_r^{1/r}$ with $r \geq 2$. This fact, due independently to H. F. Bohnenblust and M. Marcus and L. Lopes [358], is a consequence of their subadditivity, since they are positively homogeneous. See Lemma 4.4.1.

C.3.1 Theorem *The sum of two n-tuples of strictly positive numbers $\mathcal{F} = \{x_1, \ldots, x_n\}$ and $\mathcal{G} = \{y_1, \ldots, y_n\}$ is defined by the formula $\mathcal{F} + \mathcal{G} = \{x_1 + y_1, \ldots, x_n + y_n\}$. Then*

$$e_r(\mathcal{F} + \mathcal{G})^{1/r} \geq e_r(\mathcal{F})^{1/r} + e_r(\mathcal{G})^{1/r}$$

for every $r = 1, \ldots, n$. The inequality is strict unless $r = 1$ or there exists $\lambda > 0$ such that $\mathcal{F} = \lambda\mathcal{G}$.

The proof combines a special case of Minkowski's inequality with the following lemma:

C.3.2 Lemma *Under the hypotheses of Theorem C.3.1, for $r = 1, \ldots, n$ and n-tuples of positive numbers not all zero, we have*

$$\frac{e_r(\mathcal{F} + \mathcal{G})}{e_{r-1}(\mathcal{F} + \mathcal{G})} \geq \frac{e_r(\mathcal{F})}{e_{r-1}(\mathcal{F})} + \frac{e_r(\mathcal{G})}{e_{r-1}(\mathcal{G})}.$$

The inequality is strict unless $r = 1$ or there exists a $\lambda > 0$ such that $\mathcal{F} = \lambda\mathcal{G}$.

Proof of Lemma C.3.2. For $r = 1$ the inequality is actually an equality. For $r = 2$, we have to look at the following identity:

$$\frac{e_2(\mathcal{F} + \mathcal{G})}{e_1(\mathcal{F} + \mathcal{G})} - \frac{e_2(\mathcal{F})}{e_1(\mathcal{F})} - \frac{e_2(\mathcal{G})}{e_1(\mathcal{G})} = \frac{\sum_{k=1}^n (x_k \sum_{j=1}^n y_j - y_k \sum_{j=1}^n x_j)^2}{2e_1(\mathcal{F} + \mathcal{G})\, e_1(\mathcal{F})\, e_1(\mathcal{G})}.$$

Assume now that $r > 2$. For an n-tuple $\mathcal{H} = \{z_1, \ldots, z_n\}$ we shall denote $\mathcal{H}_{\hat{k}} = \{z_1, \ldots, \widehat{z_k}, \ldots, z_n\}$, where the cap indicates omission. Then:

$$\sum_{k=1}^n x_k e_{r-1}(\mathcal{F}_{\hat{k}}) = r e_r(\mathcal{F}) \tag{C.1}$$

$$x_k e_{r-1}(\mathcal{F}_{\hat{k}}) + e_r(\mathcal{F}_{\hat{k}}) = e_r(\mathcal{F}). \tag{C.2}$$

Summing on k in (C.2) we obtain

$$n e_r(\mathcal{F}) = \sum_{k=1}^n x_k e_{r-1}(\mathcal{F}_{\hat{k}}) + \sum_{k=1}^n e_r(\mathcal{F}_{\hat{k}}),$$

and thus from (C.1) we infer that $\sum_{k=1}^{n} e_r(\mathcal{F}_{\hat{k}}) = (n-r)e_r(\mathcal{F})$. Since

$$\begin{aligned} e_r(\mathcal{F}) - e_r(\mathcal{F}_{\hat{k}}) &= x_k e_{r-1}(\mathcal{F}_{\hat{k}}) \\ &= x_k e_{r-1}(\mathcal{F}) - x_k^2 e_{r-2}(\mathcal{F}_{\hat{k}}) \end{aligned}$$

we obtain

$$re_r(\mathcal{F}) = \sum_{k=1}^{n} x_k e_{r-1}(\mathcal{F}) - \sum_{k=1}^{n} x_k^2 e_{r-2}(\mathcal{F}_{\hat{k}})$$

and thus

$$\begin{aligned} \frac{e_r(\mathcal{F})}{e_{r-1}(\mathcal{F})} &= \frac{1}{r}\Big[\sum_{k=1}^{n} x_k - \sum_{k=1}^{n} \frac{x_k^2\, e_{r-2}(\mathcal{F}_{\hat{k}})}{e_{r-1}(\mathcal{F})}\Big] \\ &= \frac{1}{r}\Big[\sum_{k=1}^{n} x_k - \sum_{k=1}^{n} \frac{x_k^2}{x_k + e_{r-1}(\mathcal{F}_{\hat{k}})/e_{r-2}(\mathcal{F}_{\hat{k}})}\Big]. \end{aligned}$$

Therefore

$$\begin{aligned} \Delta &= \frac{e_r(\mathcal{F}+\mathcal{G})}{e_{r-1}(\mathcal{F}+\mathcal{G})} - \frac{e_r(\mathcal{F})}{e_{r-1}(\mathcal{F})} - \frac{e_r(\mathcal{G})}{e_{r-1}(\mathcal{G})} \\ &= \frac{1}{r}\sum_{k=1}^{n}\Big[\frac{x_k^2}{x_k + f_{r-1}(\mathcal{F}_{\hat{k}})} + \frac{y_k^2}{y_k + f_{r-1}(\mathcal{G}_{\hat{k}})} - \frac{(x_k+y_k)^2}{x_k + y_k + f_{r-1}((\mathcal{F}+\mathcal{G})_{\hat{k}})}\Big] \end{aligned}$$

where $f_s(\mathcal{F}) = e_s(\mathcal{F})/e_{s-1}(\mathcal{F})$.

The proof ends by induction. Assume that the statement of the lemma is true for $r-1$, that is,

$$f_{r-1}((\mathcal{F}+\mathcal{G})_{\hat{k}}) > f_{r-1}(\mathcal{F}_{\hat{k}}) + f_{r-1}(\mathcal{G}_{\hat{k}}), \tag{C.3}$$

unless $\mathcal{F}_{\hat{k}}$ and $\mathcal{G}_{\hat{k}}$ are proportional (when equality holds). Then

$$\begin{aligned} \Delta &> \frac{1}{r}\sum_{k=1}^{n}\Big[\frac{x_k^2}{x_k + f_{r-1}(\mathcal{F}_{\hat{k}})} + \frac{y_k^2}{y_k + f_{r-1}(\mathcal{G}_{\hat{k}})} \\ &\qquad - \frac{(x_k+y_k)^2}{x_k + y_k + f_{r-1}(\mathcal{F}_{\hat{k}}) + f_{r-1}(\mathcal{G}_{\hat{k}})}\Big] \\ &= \frac{1}{r}\sum_{k=1}^{n} \frac{[x_k f_{r-1}(\mathcal{G}_{\hat{k}}) - y_k f_{r-1}(\mathcal{F}_{\hat{k}})]^2}{[x_k + f_{r-1}(\mathcal{F}_{\hat{k}})][y_k + f_{r-1}(\mathcal{G}_{\hat{k}})][x_k + y_k + f_{r-1}(\mathcal{F}_{\hat{k}}) + f_{r-1}(\mathcal{G}_{\hat{k}})]} \end{aligned}$$

provided that at least one of the inequalities (C.3) is strict. Thus (C.3) holds also for $r-1$ replaced by r and the proof is complete.

An alternative proof of Lemma C.3.2, based on the properties of perspective functions, was found by E. K. Godunova [222].

Proof of Theorem C.3.1. In fact, by Minkowski's inequality for $p = 0$ and Lemma C.3.2, we have

$$\begin{aligned} e_r(\mathcal{F}+\mathcal{G})^{1/r} &= \Big[\frac{e_r(\mathcal{F}+\mathcal{G})}{e_{r-1}(\mathcal{F}+\mathcal{G})}\cdot\frac{e_{r-1}(\mathcal{F}+\mathcal{G})}{e_{r-2}(\mathcal{F}+\mathcal{G})}\cdots\frac{e_1(\mathcal{F}+\mathcal{G})}{e_0(\mathcal{F}+\mathcal{G})}\Big]^{1/r} \\ &\geq \Big\{\Big[\frac{e_r(\mathcal{F})}{e_{r-1}(\mathcal{F})}+\frac{e_r(\mathcal{G})}{e_{r-1}(\mathcal{G})}\Big]\cdots\Big[\frac{e_1(\mathcal{F})}{e_0(\mathcal{F})}+\frac{e_1(\mathcal{G})}{e_0(\mathcal{G})}\Big]\Big\}^{1/r} \\ &\geq \Big(\prod_{k=1}^{r}\frac{e_k(\mathcal{F})}{e_{k-1}(\mathcal{F})}\Big)^{1/r}+\Big(\prod_{k=1}^{r}\frac{e_k(\mathcal{G})}{e_{k-1}(\mathcal{G})}\Big)^{1/r} = e_r(\mathcal{F})^{1/r}+e_r(\mathcal{G})^{1/r}. \end{aligned}$$

The problem of equality is left to the reader.

Theorem C.3.1 has important consequences to positive matrices $A \in \mathrm{M}_n(\mathbb{C})$, $A = (a_{ij})_{i,j=1}^n$. In this case all eigenvalues $\lambda_1(A), \ldots, \lambda_n(A)$ are positive and the symmetric elementary functions of them can be easily computed via the Cauchy–Binet formulae:

$$\begin{aligned} \sum_{k=1}^{n}\lambda_k(A) &= \sum_{k=1}^{n} a_{kk} \\ \sum_{i<j}\lambda_i(A)\lambda_j(A) &= \det\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22}\end{pmatrix}+\cdots+\det\begin{pmatrix} a_{n-1\,n-1} & a_{n-1\,n} \\ a_{n\,n-1} & a_{n\,n}\end{pmatrix} \\ &\vdots \\ \prod_{k=1}^{n}\lambda_k(A) &= \det(a_{ij})_{i,j=1}^n. \end{aligned}$$

In particular, we recover the result of Example 4.1.5 in Section 4.1: If A and B are positive matrices, and $\alpha \in (0,1)$, then

$$\begin{aligned} \big(\det((1-\alpha)A+\alpha B)\big)^{1/n} &\geq (1-\alpha)(\det A)^{1/n}+\alpha(\det B)^{1/n} \\ &\geq (\det A)^{(1-\alpha)/n}(\det B)^{\alpha/n}. \end{aligned}$$

Newton's inequalities (of any order) have equivalent formulations in terms of positive matrices (and their principal minors). We shall recall here the analogue of the *AM-GM* inequality: *If A is a strictly positive matrix in $\mathrm{M}_n(\mathbb{R})$, then*

$$\Big(\frac{\operatorname{trace} A}{n}\Big)^n > \det A,$$

unless A is a multiple of the unit matrix I.

C.3.3 Remark *In differential geometry, the* higher order mean curvatures *are defined as the elementary symmetric functions of the principal curvatures. In fact, if S is a hypersurface in $\mathbb{R}^n$ and p is a point of S, one considers the*

Gauss map, $g\colon p \to N(p)$, whose differential at p is diagonalized by the principal curvature directions at p,

$$dg_p(e_j) = -k_j e_j \quad \textit{for } j = 1, \ldots, n.$$

Then the j-th-order mean curvatures H_j are given by

$$\prod_{k=1}^{n-1}(1 + tk_j) = \sum_{j=0}^{n-1} \binom{n-1}{j} H_j t^j.$$

See R. Osserman [458] for details. It would be interesting to explore the applications of various inequalities of convexity to this area.

C.4 Symmetric Polynomial Majorization

Motivated by an optimization problem in solid-body mechanics, M. Bîrsan, P. Neff and J. Lankeit [70] have considered the following variant of majorization for vectors in $\mathbb{R}^n_{++}$ ($N \geq 2$). Precisely, if $x, y \in \mathbb{R}^n_{++}$, we say that x is less than or equal to y with respect to the *symmetric polynomial majorization* (that is, $x \prec_{SPM} y$) if

$$e_k(x) \leq e_k(y) \quad \text{for all } k \in \{1, ..., n-1\}$$

and

$$e_n(x) = e_n(y).$$

Accordingly, a function $f : (0, \infty) \to \mathbb{R}$ is called $\prec_{SPM}$- *convex* if

$$\sum_{k=1}^{n} f(x_k) \leq \sum_{k=1}^{n} f(y_k)$$

whenever $x \prec_{SPM} y$. The simplest examples of such functions are

$$\pm 1,\ x,\ \pm \log x,\ 1/x,$$

as well as all linear combinations $a + bx + c \log x + d/x$ with $a, c \in \mathbb{R}$ and $b, d \geq 0$. Other examples are provided by the family of functions

$$L_t(x) = \log \frac{x+t}{1+t} - \frac{\log x}{1+t^2},$$

where $t > 0$ is a parameter. Indeed,

$$\begin{aligned}\sum_{k=1}^{n} L_t(x_k) &= \log \frac{(x_1 + t) \cdots (x_n + t)}{(1+t)^n} - \frac{\log(x_1 \cdots x_n)}{1+t^2} \\ &= \log \sum_{k=1}^{n} \left(t^{n-k} e_k(x_1, \ldots, x_n)\right) - n \log(1+t) - \frac{\log e_n(x_1, \ldots, x_n)}{1+t^2},\end{aligned}$$

which makes clear the implication

$$x \prec_{SPM} y \text{ implies } \sum_{k=1}^{n} L_t(x_k) \leq \sum_{k=1}^{n} L_t(y_k).$$

As was noticed by M. Šilhavý [538], the examples listed above generate *all* $\prec_{SPM}$- convex functions. For example, so is the case of squared logarithms, which combines the $\prec_{SPM}$- convexity of the functions L_t and the following representation formula,

$$\log^2 x = 2 \int_0^\infty \left[\log \frac{x+t}{1+t} - \frac{\log x}{1+t^2} \right] \frac{dt}{t}.$$

Indeed, this formula takes place for $x = 1$ and the two sides have the same derivative for $x > 0$.

Appendix D

Second-Order Differentiability of Convex Functions

The famous result of A. D. Alexandrov [12] on the second differentiability of convex functions defined on $\mathbb{R}^N$ has received a great deal of attention during the last two decades, motivated by its connection to mass transportation problems. The proof we present here follows the argument of M. G. Crandall, H. Ishii and P.-L. Lions [144] (which in turn were inspired by a paper of F. Mignot [379]). A basic ingredient in their approach is Rademacher's theorem on the almost differentiability of Lipschitz functions defined on an Euclidean space. Its proof makes the objective of Section D.1.

There are known many other proofs of Alexandrov's theorem but all make use of the machinery of hard analysis. See, for example, G. Alberti and L. Ambrosio [9], G. Bianchi, A. Colesanti and C. Pucci [66] (which includes also some historical comments) and L. C. Evans and R. Gariepy [184] to cite just a few.

D.1 Rademacher's Theorem

The almost everywhere differentiability of a convex function defined on a convex open subset of $\mathbb{R}^N$ is the particular case of the following basic result in nonlinear analysis:

D.1.1 Theorem (Rademacher's theorem) *Every locally Lipschitz function f acting on Euclidean spaces is almost everywhere differentiable.*

Proof. Since a vector-valued function $f\colon \mathbb{R}^N \to \mathbb{R}^m$ is differentiable at a point if and only if all of its components are differentiable at that point, we may restrict ourselves to the case of real-valued functions. Also, since differentiability is a

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local property, we may as well assume that f is Lipschitz. See Kirszbraun's Theorem 3.7.4. The remainder of the proof will be done in three steps.

Step 1. Fix arbitrarily a vector $\mathbf{v} \in \mathbb{R}^N$, $\|\mathbf{v}\| = 1$. We shall show that the directional derivative $f'(\mathbf{x};\mathbf{v})$ exists for almost every $\mathbf{x} \in \mathbb{R}^N$. In fact, for each $\mathbf{x} \in \mathbb{R}^N$ consider the limits

$$\underline{D}f(\mathbf{x};\mathbf{v}) = \liminf_{t\to 0} \frac{f(\mathbf{x}+t\mathbf{v}) - f(\mathbf{x})}{t}$$

and

$$\overline{D}f(\mathbf{x};\mathbf{v}) = \limsup_{t\to 0} \frac{f(\mathbf{x}+t\mathbf{v}) - f(\mathbf{x})}{t},$$

which exist in $\overline{\mathbb{R}}$. The set

$$E_{\mathbf{v}} = \{\mathbf{x} \in \mathbb{R}^n : \underline{D}f(\mathbf{x};\mathbf{v}) < \overline{D}f(\mathbf{x};\mathbf{v})\}$$

equals the set where the directional derivative $f'(\mathbf{x};\mathbf{v})$ does not exist. As in the proof of Theorem 5.2.4 we may conclude that $E_{\mathbf{v}}$ is Lebesgue measurable. We shall show that $E_{\mathbf{v}}$ is actually Lebesgue negligible. In fact, by Lebesgue's theory on the differentiability of absolutely continuous functions (see [134], or [252]), we infer that the functions

$$g(t) = f(\mathbf{x}+t\mathbf{v}), \quad t \in \mathbb{R},$$

are differentiable almost everywhere. This implies that the Lebesgue measure of the intersection of E_v with any line L is Lebesgue negligible. Then, by Fubini's theorem, we conclude that $E_{\mathbf{v}}$ is itself Lebesgue negligible.

Step 2. According to the discussion above we know that

$$\nabla f(\mathbf{x}) = \Big(\frac{\partial f}{\partial x_1}(\mathbf{x}), \ldots, \frac{\partial f}{\partial x_n}(\mathbf{x})\Big)$$

exists almost everywhere. We shall show that

$$f'(\mathbf{x};\mathbf{v}) = \langle \mathbf{v}, \nabla f(\mathbf{x})\rangle$$

for almost every $\mathbf{x} \in \mathbb{R}^N$. In fact, for an arbitrary fixed $\varphi \in C_c^\infty(\mathbb{R}^n)$ we have

$$\int_{\mathbb{R}^n} \Big[\frac{f(\mathbf{x}+t\mathbf{v}) - f(\mathbf{x})}{t}\Big]\varphi(\mathbf{x})\,\mathrm{d}\mathbf{x} = -\int_{\mathbb{R}^n} f(\mathbf{x})\Big[\frac{\varphi(\mathbf{x}) - \varphi(\mathbf{x}-t\mathbf{v})}{t}\Big]\,\mathrm{d}\mathbf{x}.$$

Since

$$\Big|\frac{f(\mathbf{x}+t\mathbf{v}) - f(\mathbf{x})}{t}\Big| \le \mathrm{Lip}(f),$$

we can apply the dominated convergence theorem to get

$$\int_{\mathbb{R}^N} f'(\mathbf{x};\mathbf{v})\varphi(\mathbf{x})\,\mathrm{d}\mathbf{x} = -\int_{\mathbb{R}^N} f(\mathbf{x})\varphi'(\mathbf{x};\mathbf{v})\,\mathrm{d}\mathbf{x}.$$

By taking into account Fubini's theorem and the absolute continuity of f on lines we can continue as follows:

$$-\int_{\mathbb{R}^N} f(\mathbf{x})\varphi'(\mathbf{x};\mathbf{v})\,\mathrm{d}\mathbf{x} = -\sum_{k=1}^{N} v_k \int_{\mathbb{R}^N} f(\mathbf{x})\frac{\partial\varphi}{\partial x_k}(\mathbf{x})\,\mathrm{d}\mathbf{x}$$
$$= \sum_{k=1}^{N} v_k \int_{\mathbb{R}^N} \frac{\partial f}{\partial x_k}(\mathbf{x})\varphi(\mathbf{x})\,\mathrm{d}\mathbf{x} = \int_{\mathbb{R}^N} \langle \mathbf{v}, \nabla f(\mathbf{x})\rangle\varphi(\mathbf{x})\,\mathrm{d}\mathbf{x}$$

and this leads us to the formula $f'(\mathbf{x};\mathbf{v}) = \langle \mathbf{v}, \nabla f(\mathbf{x})\rangle$, as φ was arbitrarily fixed.

Step 3. Consider now a countable family $(\mathbf{u}_i)_i$ of unit vectors, which is dense in the unit sphere of $\mathbb{R}^N$. By the above reasoning we infer that the complement of each of the sets

$$A_i = \{\mathbf{x} \in \mathbb{R}^N : Df(\mathbf{x};\mathbf{u}_i) \text{ and } \nabla f(\mathbf{x}) \text{ exist and } f'(\mathbf{x};\mathbf{u}_i) = \langle \mathbf{u}_i, \nabla f(\mathbf{x})\rangle\}$$

is Lebesgue negligible, and thus the same is true for the complement of

$$A = \bigcap_{i=1}^{\infty} A_i.$$

We shall show that f is differentiable at all points of A. This will be done by considering the function

$$R(\mathbf{x},\mathbf{v},t) = \frac{f(\mathbf{x}+t\mathbf{v}) - f(\mathbf{x})}{t} - \langle \mathbf{v}, \nabla f(\mathbf{x})\rangle,$$

for $\mathbf{x} \in A$, $\mathbf{v} \in \mathbb{R}^N$, $\|\mathbf{v}\| = 1$, and $t \in \mathbb{R}\backslash\{0\}$. Since

$$|R(\mathbf{x},\mathbf{v},t) - R(\mathbf{x},\mathbf{v}',t)| \le \mathrm{Lip}(f)\cdot\|\mathbf{v}-\mathbf{v}'\| + \|\nabla f(\mathbf{x})\|\cdot\|\mathbf{v}-\mathbf{v}'\|$$
$$\le (\sqrt{N}+1)\,\mathrm{Lip}(f)\cdot\|\mathbf{v}-\mathbf{v}'\|,$$

the function $R(\mathbf{x},\mathbf{v},t)$ is Lipschitz in $\mathbf{v}$. Suppose there are given a point $\mathbf{a} \in A$ and a number $\varepsilon > 0$. Since the unit sphere of $\mathbb{R}^N$ is compact and the family $(\mathbf{u}_i)_i$ is a dense subset, we can choose a natural number n such that

$$\inf_{i\in\{0,\dots,n\}} \|\mathbf{v} - u_i\| < \frac{\varepsilon}{2(\sqrt{N}+1)\,\mathrm{Lip}(f)}$$

for all $\mathbf{v}$ with $\|\mathbf{v}\| = 1$. By the definition of A, there exists a $\delta > 0$ such that

$$|R(\mathbf{a},\mathbf{u}_i,t)| < \frac{\varepsilon}{2}$$

for all $i \in \{0,\dots,N\}$ and all $|t| < \delta$. Then

$$|R(\mathbf{a},\mathbf{v},t)| \le \inf_{i\in\{0,\dots,n\}} (|R(\mathbf{a},\mathbf{u}_i,t)| + |R(\mathbf{a},\mathbf{u}_i,t) - R(\mathbf{a},\mathbf{v},t)|)$$
$$\le \frac{\varepsilon}{2} + (\sqrt{N}+1)\,\mathrm{Lip}(f)\cdot\frac{\varepsilon}{2(\sqrt{N}+1)\,\mathrm{Lip}(f)} = \varepsilon,$$

for all $\mathbf{v}$ in the unit sphere of $\mathbb{R}^N$ and all t with $|t| < \delta$. This assures the differentiability of f at $\mathbf{a}$. ■

F. Mignot [379] proved the following generalization of Rademacher's theorem: *Every Lipschitz function from a separable Hilbert space to a Hilbert space is Gâteaux differentiable at densely many points.* However, as shown in the case of the function

$$F\colon L^2[0,1] \to \mathbb{R}, \quad F(f) = \left(\int_0^1 (f^+(t))^2 \, \mathrm{d}t \right)^{1/2},$$

the set of points of Fréchet differentiability may be empty. The hypothesis on separability is essential for the validity of Mignot's result. A counterexample is provided by the projection of $\ell^2(A)$ (for an uncountable index set A) onto the cone of positive elements.

D.2 Alexandrov's Theorem

We pass now to the problem of almost everywhere second differentiability of convex functions. In order to simplify the exposition we shall make constant use of Landau's little-o symbol, precisely,

$$f = o(g) \text{ for } \mathbf{x} \to 0 \text{ if and only if } f = hg \text{ with } \lim_{\mathbf{x}\to 0} h(\mathbf{x}) = 0.$$

D.2.1 Theorem (A. D. Alexandrov [12]) *If $f : \mathbb{R}^N \to \mathbb{R}$ is a convex function, then for almost every $\mathbf{a} \in \mathbb{R}^N$ there exists $\nabla f(\mathbf{a})$ and a symmetric matrix $\nabla^2 f(\mathbf{a})$ in $\mathrm{Sym}^+(N,\mathbb{R})$ such that*

$$\lim_{\mathbf{x}\to\mathbf{a}} \frac{f(\mathbf{x}) - f(\mathbf{a}) - \langle \nabla f(\mathbf{a}), \mathbf{x}-\mathbf{a}\rangle - \frac{1}{2}\langle \nabla^2 f(\mathbf{a})\mathbf{x}-\mathbf{a}, \mathbf{x}-\mathbf{a}\rangle}{\|\mathbf{x}-\mathbf{a}\|^2} = 0.$$

The matrix $\nabla^2 f(\mathbf{a})$ is usually referred to as the *Alexandrov Hessian* of f at $\mathbf{a}$. If $\nabla f(\mathbf{a})$ and $\nabla^2 f(\mathbf{a})$ are as in the Theorem D.2.1, then for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\|\mathbf{x}-\mathbf{a}\| < \delta \text{ implies } \sup_{\mathbf{y}\in\partial f(\mathbf{x})} \|\mathbf{y} - \nabla f(\mathbf{a}) - \nabla^2 f(\mathbf{a})(\mathbf{x}-\mathbf{a})\| \le \varepsilon \|\mathbf{x}-\mathbf{a}\|.$$

Proof. By Theorem 4.3.1, the domain of the subdifferential ∂f is the whole space $\mathbb{R}^N$, while Proposition 5.1.9 shows that $\partial f(\mathbf{x}) = \{\nabla f(\mathbf{x})\}$ for all $\mathbf{x}$ in

$$X_1 = \{\mathbf{x} \in \mathbb{R}^N : f \text{ is differentiable at } \mathbf{x}\},$$

which is a set whose complement is a negligible set. We shall show that for almost all $\mathbf{x}$ in $\mathbb{R}^N$ there exists a matrix A in $\mathrm{Sym}^+(n,\mathbb{R})$ such that

$$\mathrm{d}f(\mathbf{y}) = \mathrm{d}f(\mathbf{x}) + A(\mathbf{y}-\mathbf{x}) + o(\|\mathbf{y}-\mathbf{x}\|) \quad \text{for all } \mathbf{y} \in X_1. \tag{D.1}$$

We need the fact that $J = (I + \partial f)^{-1}$ is a nonexpansive map of $\mathbb{R}^N$ into itself. See Corollary 4.3.11. This yields a new set,

$$X_2 = \{J(\mathbf{x}) : J \text{ is differentiable at } \mathbf{x} \text{ and } \mathrm{d}J(\mathbf{x}) \text{ is nonsingular}\},$$

whose complement is also a negligible set. In fact, by Rademacher's theorem, J is differentiable almost everywhere. Since J is Lipschitz, we may apply the area formula (see L. C. Evans and R. F. Gariepy [184, Theorem 3.3.2, p. 96]) to get

$$\int_B |\det(\mathrm{d}J(\mathbf{x}))|\,\mathrm{d}\mathbf{x} = \int_{\mathbb{R}^N} \#(B \cap J^{-1}(\mathbf{y}))\,\mathrm{d}\mathbf{y} \quad \text{for all Borel sets } B \text{ in } \mathbb{R}^N,$$

where # is the counting measure. By this formula (and the fact that J is onto) we infer that the complementary set of

$$\{\mathbf{x} : J \text{ is differentiable at } \mathbf{x} \text{ and } \mathrm{d}J(\mathbf{x}) \text{ is nonsingular}\}$$

is a negligible set. On the other hand, any Lipschitz function maps negligible sets into negligible sets. See [514], Lemma 7.25. Hence X_2 is indeed a set whose complementary set is negligible. We shall show that the formula (D.1) works for all $\mathbf{x}$ in $X_3 = X_1 \cap X_2$ (which is a set with negligible complementary set). Our argument is based on the following fact concerning the solvability of nonlinear equations in $\mathbb{R}^N$: If $F\colon \overline{B}_\delta(0) \to \mathbb{R}^N$ is continuous, $0 < \varepsilon < \delta$ and $\|F(\mathbf{x}) - \mathbf{x}\| < \varepsilon$ for all $\mathbf{x} \in \mathbb{R}^N$ with $\|\mathbf{x}\| = \delta$, then $F(B_\delta(\mathbf{0})) \supset B_{\delta-\varepsilon}(\mathbf{0})$. See W. Rudin [514, Lemma 7.23] for a proof based on the Brouwer fixed point theorem. By the definition of J,

$$\mathrm{d}f(J(\mathbf{x})) = \mathbf{x} - J(\mathbf{x})$$

for all $\mathbf{x}$ with $J(\mathbf{x}) \in X_3$ ($\subset X_1$). Suppose that $J(\mathbf{x}) + \tilde{\mathbf{y}} \in X_1$, where $\tilde{\mathbf{y}}$ is small. Since J is Lipschitz and $dJ(\mathbf{x})$ is nonsingular, if $\tilde{y}$ is sufficiently small, then there exists $\tilde{\mathbf{x}}$ such that $J(\mathbf{x} + \tilde{\mathbf{x}}) = J(\mathbf{x}) + \tilde{\mathbf{y}}$. Moreover, we may choose $\tilde{\mathbf{x}}$ to verify $\|\tilde{\mathbf{x}}\| \le C\|\tilde{\mathbf{y}}\|$ for some constant $C > 0$. Use a remark above (on the solvability of nonlinear equations) and the fact that $J(\mathbf{x}+\mathbf{h}) = J(\mathbf{x}) + \mathrm{d}J(\mathbf{x})\mathbf{h} + o(\|\mathbf{h}\|)$. Since J is nonexpansive, we also have $\|\tilde{\mathbf{y}}\| \le \|\tilde{\mathbf{x}}\|$, hence $\|\tilde{\mathbf{x}}\|$ and $\|\tilde{\mathbf{y}}\|$ are comparable. Then

$$\begin{aligned}\mathrm{d}f(J(x) + \tilde{y}) &= \mathrm{d}f(J(\mathbf{x} + \tilde{\mathbf{x}})) = \mathbf{x} + \tilde{\mathbf{x}} - J(\mathbf{x} + \tilde{\mathbf{x}}) \\ &= df(J(\mathbf{x})) + (\mathrm{I} - \mathrm{d}J(\mathbf{x}))\tilde{\mathbf{x}} + o(\|\tilde{\mathbf{x}}\|).\end{aligned}$$

Due to the relation $J(\mathbf{x}) + \tilde{\mathbf{y}} = J(\mathbf{x}) + \mathrm{d}J(\mathbf{x})\tilde{\mathbf{x}} + o(\|\tilde{\mathbf{x}}\|)$ and the comparability of $\|\tilde{\mathbf{x}}\|$ and $\|\tilde{\mathbf{y}}\|$, we have

$$\tilde{\mathbf{x}} = (\mathrm{d}J(\mathbf{x}))^{-1}\tilde{\mathbf{y}} + o(\|\tilde{\mathbf{y}}\|).$$

Hence $\mathrm{d}(\mathrm{d}f)(J(\mathbf{x}))$ exists and equals $(\mathrm{d}J(\mathbf{x}))^{-1} - \mathrm{I}$. It remains to prove that

$$f(J(x) + \tilde{\mathbf{y}}) = f(J(\mathbf{x})) + \mathrm{d}f(J(\mathbf{x}))\tilde{\mathbf{y}} + \frac{1}{2}\big\langle\big((\mathrm{d}J(\mathbf{x}))^{-1} - \mathrm{I}\big)\tilde{\mathbf{y}}, \tilde{\mathbf{y}}\big\rangle + o(\|\tilde{\mathbf{y}}\|^2)$$

for $J(\mathbf{x}) \in X_3$. Letting

$$R(\tilde{\mathbf{y}}) = f(J(\mathbf{x}) + \tilde{\mathbf{y}}) - f(J(\mathbf{x})) - \mathrm{d}f(J(\mathbf{x}))\tilde{\mathbf{y}} - \frac{1}{2}\big\langle\big((\mathrm{d}J(\mathbf{x}))^{-1} - \mathrm{I}\big)\tilde{\mathbf{y}}, \tilde{\mathbf{y}}\big\rangle,$$

we get a locally Lipschitz function R such that $R(\mathbf{0}) = 0$, and for almost all small $\tilde{y}$,

$$\mathrm{d}R(\tilde{\mathbf{y}}) = o(\|\tilde{\mathbf{y}}\|).$$

By the mean value theorem we conclude that $R(\tilde{\mathbf{y}}) = o(\|\tilde{\mathbf{y}}\|^2)$ and the proof is complete. ■

The result of Theorem D.2.1 can be easily extended to conclude that every proper convex function $f\colon \mathbb{R}^N \to \mathbb{R} \cup \{\infty\}$ is twice differentiable almost everywhere on its effective domain. Alexandrov's theorem has important applications to convex geometric analysis and partial differential equations. Both Theorems D.1.1 and D.2.1 remain valid in the more general framework of semiconvex functions. A function f defined on a convex set in $\mathbb{R}^N$ is said to be *semiconvex* if $f + \lambda\|\cdot\|^2$ is a convex function for some $\lambda > 0$.

In Mignot's approach [379], Alexandrov's theorem appears as a consequence of the following differentiability property of monotone maps:

D.2.2 Theorem *(Differentiability of monotone maps). Let u be a maximal monotone map on $\mathbb{R}^n$ and let D be the set of points x such that $u(x)$ is a singleton. Then u is differentiable at almost every $a \in D$, that is, there exists an $n \times n$ matrix $\nabla u(a)$ such that*

$$\lim_{\substack{\mathbf{x}\to\mathbf{a}\\ \mathbf{y}\in u(\mathbf{x})}} \frac{\mathbf{y} - u(\mathbf{a}) - \nabla u(\mathbf{a})(\mathbf{x}-\mathbf{a})}{\|\mathbf{x}-\mathbf{a}\|} = 0.$$

In fact, if we apply this result to the subdifferential of a convex function $f\colon \mathbb{R}^n \to \mathbb{R}$, we obtain that for almost every $\mathbf{a} \in \mathbb{R}^n$ where ∂f is a singleton (that is, where f is differentiable), there exists a matrix $\nabla^2 f(\mathbf{a})$ such that

$$\lim_{\substack{\mathbf{x}\to\mathbf{a}\\ \mathbf{y}\in\partial f(\mathbf{x})}} \frac{\mathbf{y} - \nabla f(\mathbf{a}) - \nabla^2 f(\mathbf{a})(\mathbf{x}-\mathbf{a})}{\|\mathbf{x}-\mathbf{a}\|} = 0. \tag{D.2}$$

If (D.2) holds, then $A = \nabla^2 f(\mathbf{a})$ proves to be the Alexandrov Hessian of f at $\mathbf{a}$. To show this, it suffices to restrict ourselves to the case where $\mathbf{a} = 0$, $f(\mathbf{a}) = 0$ and $\nabla f(\mathbf{a}) = 0$. We shall prove that $\varphi(\mathbf{h}) = f(\mathbf{h}) - \frac{1}{2}\langle A\mathbf{h}, \mathbf{h}\rangle$ verifies $\lim_{\mathbf{h}\to 0} \varphi(\mathbf{h})/\|\mathbf{h}\|^2 = 0$. In fact, fixing an $\mathbf{h} \neq \mathbf{0}$, by the nonsmooth version of the mean value theorem (see Exercise 6, Section 5.1) we get a point y in the segment joining $\mathbf{0}$ to $\mathbf{h}$, and a $\mathbf{p} \in \partial\varphi(\mathbf{y})$ such that $\varphi(\mathbf{h}) - \varphi(\mathbf{0}) = \langle \mathbf{h}, \mathbf{p}\rangle$. Then

$$\varphi(\mathbf{h}) = \langle \mathbf{q} - A\mathbf{y}, \mathbf{h}\rangle$$

for some vector $\mathbf{q} \in \partial f(\mathbf{y})$. According to (D.2), $\lim_{\mathbf{h}\to 0} \|\mathbf{q} - A(\mathbf{y})\|/\|\mathbf{h}\| = 0$, which yields $\lim_{\mathbf{h}\to\mathbf{0}} \varphi(\mathbf{h})/\|\mathbf{h}\|^2 = 0$.

Appendix E

The Variational Approach of PDE

The aim of this appendix is to illustrate a number of problems in partial differential equations (PDE) which can be solved by seeking a global minimum of suitable convex functionals. This idea goes back to Fermat's rule for finding points of extrema.

E.1 The Minimum of Convex Functionals

An important criterion for the existence of a global minimum is the following variant of Weierstrass' extreme value theorem:

E.1.1 Theorem *Let C be a closed convex set in a reflexive Banach space V and let $J\colon C \to \mathbb{R}$ be a function which fulfills the following two conditions:*

(a) *J is weakly lower semicontinuous, that is,*

$$\mathbf{u}_n \to \mathbf{u} \text{ weakly in } V \text{ implies } J(\mathbf{u}) \le \liminf_{n\to\infty} J(\mathbf{u}_n);$$

(b) *Either C is bounded, or $\lim_{\|\mathbf{u}\|\to\infty} J(\mathbf{u}) = \infty$.*

Then J admits at least one global minimum. If, moreover, J is quasiconvex, then the points of global minimum constitutes a convex set.

Proof. Put

$$m = \inf_{\mathbf{u}\in C} J(\mathbf{u}).$$

Clearly, $m < \infty$, and there exists a sequence $(\mathbf{u}_n)_n$ of elements in C such that $J(\mathbf{u}_n) \to m$. By our hypothesis (b), the sequence $(\mathbf{u}_n)_n$ is bounded, so by Theorem B.1.8, we may assume (replacing $(\mathbf{u}_n)_n$ by a subsequence if necessary) that it is also weakly converging to an element $\mathbf{u}$ in C. Here we used the fact that C is weakly closed (which is a consequence of Corollary B.2.6). Then

$$m \le J(\mathbf{u}) \le \liminf_{n\to\infty} J(\mathbf{u}_n) = m,$$

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C. P. Niculescu and L.-E. Persson, *Convex Functions and Their Applications*, CMS/CAIMS Books in Mathematics 14, https://doi.org/10.1007/978-3-031-71967-7

and thus $\mathbf{u}$ is a global minimum. The remainder of the proof is left to the reader as an exercise. ■

Under the presence of convexity and differentiability one can prove much more powerful results.

E.1.2 Theorem *Let V be a reflexive Banach space V and let $J\colon V \to \mathbb{R}$ be a convex functional with the following properties:*

(a) *J is Gâteaux differentiable*;

(b) $\lim_{\|\mathbf{u}\|\to\infty} J(\mathbf{u}) = \infty$.

Then J admits at least one global minimum and the points of global minimum are precisely the solutions $\mathbf{u}$ of the so called Euler–Lagrange equation,

$$J'(\mathbf{u}) = 0.$$

If, moreover, J is strictly convex, then there is a unique global minimum.

Proof. First notice that J is weakly lower semicontinuous. In fact, by Theorem 5.1.9,

$$J(\mathbf{u}_n) \geq J(\mathbf{u}) + J'(\mathbf{u}; \mathbf{u}_n - \mathbf{u})$$

for all n, so if $\mathbf{u}_n \overset{w}{\to} \mathbf{u}$, then $J'(\mathbf{u}; \mathbf{u}_n - \mathbf{u}) = J'(\mathbf{u})(\mathbf{u}_n - \mathbf{u}) \to 0$ and we infer that

$$\liminf_{n\to\infty} J(\mathbf{u}_n) \geq J(\mathbf{u}).$$

Hence, according to Theorem E.1.1, J admits a global minimum. If $\mathbf{u}$ is a global minimizer, then for each $\mathbf{v} \in V$ there is a $\delta > 0$ such that

$$\frac{J(\mathbf{u} + \varepsilon\mathbf{v}) - J(\mathbf{u})}{\varepsilon} \geq 0, \quad \text{whenever } |\varepsilon| < \delta.$$

This yields $J'(\mathbf{u}; \mathbf{v}) \geq 0$. Replacing $\mathbf{v}$ by $-\mathbf{v}$, we obtain

$$-J'(\mathbf{u}; \mathbf{v}) = J'(\mathbf{u}; -\mathbf{v}) \geq 0,$$

and thus $J'(\mathbf{u}; \mathbf{v}) = 0$. Conversely, if $J'(\mathbf{u}; \mathbf{v}) = 0$ for all $\mathbf{v} \in V$, then by Theorem 5.1.9 we get

$$J(\mathbf{v}) \geq J(\mathbf{u}) + J'(\mathbf{u}, \mathbf{v} - \mathbf{u}) = J(\mathbf{u}),$$

that is, $\mathbf{u}$ is a global minimizer. ■

For example, Theorem E.1.2 applies to the convex functionals of the form

$$J(\mathbf{u}) = \frac{1}{2}\|\mathbf{u} - \mathbf{w}\|^2 + \varphi(\mathbf{u}), \quad \mathbf{u} \in V,$$

where V is an L^p-space with $p \in (1, \infty)$, or a Sobolev space, $\mathbf{w}$ is an arbitrary fixed element of V and $\varphi\colon V \to \mathbb{R}$ is a weakly lower semicontinuous convex function. These functionals play an important role in connection with the existence of solutions of partial differential equations.

The following more elaborated example is borrowed from the book of J. Céa [126]:

E.1.3 Corollary *Let Ω be a nonempty open set in $\mathbb{R}^N$ and let $p > 1$. Consider a function $g \in C^1(\mathbb{R})$ which verifies the following properties:*

(a) *$g(0) = 0$ and $g(t) \geq \alpha|t|^p$ for a suitable constant $\alpha > 0$;*

(b) *The derivative g' is increasing and $|g'(t)| \leq \beta|t|^{p-1}$ for a suitable constant $\beta > 0$.*

Then the linear space $V = L^p(\Omega) \cap L^2(\Omega)$ is reflexive when endowed with the norm

$$\|u\|_V = \|u\|_{L^p} + \|u\|_{L^2},$$

and for all $f \in L^2(\Omega)$ the functional

$$J(u) = \int_\Omega g(u(x))\,\mathrm{d}x + \frac{1}{2}\int_\Omega |u(x)|^2\,\mathrm{d}x - \int_\Omega f(x)u(x)\,\mathrm{d}x, \quad u \in V$$

is convex and Gâteaux differentiable with

$$J'(u;v) = \int_\Omega g'(u(\mathbf{x}))v(\mathbf{x})\,\mathrm{d}\mathbf{x} + \int_\Omega u(\mathbf{x})v(\mathbf{x})\,\mathrm{d}\mathbf{x} - \int_\Omega f(\mathbf{x})v(\mathbf{x})\,\mathrm{d}\mathbf{x}.$$

Moreover, J admits a unique global minimizer $\bar{u}$, which is the solution of the Euler-Lagrange equation

$$J'(u) = 0.$$

Proof. V is a closed subspace of $L^2(\Omega)$ and thus it is a reflexive space. Then notice that

$$\begin{aligned} |g(t)| &= |g(t) - g(0)| \\ &= \left|\int_0^t g'(s)\,\mathrm{d}s\right| \leq \frac{\beta}{p}\,|t|^p, \end{aligned}$$

from which it follows easily that J is well defined. Letting

$$J_1(u) = \int_\Omega g(u(x))\,\mathrm{d}x,$$

by Lagrange's mean value theorem,

$$\begin{aligned} J_1(u + tv) &= \int_\Omega g(u(\mathbf{x}) + tv(\mathbf{x}))\,\mathrm{d}\mathbf{x} \\ &= \int_\Omega g(u(\mathbf{x}))\,\mathrm{d}\mathbf{x} + t\int_\Omega g'(u(\mathbf{x}) + \tau(\mathbf{x})v(\mathbf{x}))v(\mathbf{x})\,\mathrm{d}\mathbf{x}, \end{aligned}$$

where $0 < \tau(\mathbf{x}) < t$ for all $\mathbf{x}$, provided that $t > 0$. Then

$$\frac{J_1(u + tv) - J_1(u)}{t} = \int_\Omega g'(u(\mathbf{x}) + \tau(\mathbf{x})v(\mathbf{x}))v(\mathbf{x})\,\mathrm{d}\mathbf{x},$$

and letting $t \to 0+$ we get the desired formula for $J'(u;v)$.

Again by Lagrange's mean value theorem, and the fact that g' is increasing, we have

$$\begin{aligned} J_1(v) &= J_1(u) + \int_\Omega g'(u(\mathbf{x}) + \tau(\mathbf{x})(v(\mathbf{x}) - u(\mathbf{x}))) \cdot (v(\mathbf{x}) - u(\mathbf{x}))\, \mathrm{d}\mathbf{x} \\ &\geq J_1(u) + \int_\Omega g'(u(\mathbf{x})) \cdot (v(\mathbf{x}) - u(\mathbf{x}))\, \mathrm{d}\mathbf{x} \\ &= J_1(u) + J_1'(u, v - u), \end{aligned}$$

which shows that J_1 is convex. Then the functional J is the sum of a convex function and a strictly convex function. Finally,

$$\begin{aligned} J(u) &\geq \alpha \int_\Omega |u(\mathbf{x})|^p\, \mathrm{d}\mathbf{x} + \frac{1}{2} \int_\Omega |u(\mathbf{x})|^2\, \mathrm{d}\mathbf{x} - \left| \int_\Omega f(\mathbf{x}) u(\mathbf{x})\, \mathrm{d}\mathbf{x} \right| \\ &\geq \alpha \|u\|_{L^p}^p + \frac{1}{2} \|u\|_{L^2}^2 - \|f\|_{L^2} \|u\|_{L^2}, \end{aligned}$$

from which it follows that

$$\lim_{\|u\|_V \to \infty} J(u) = \infty,$$

and the conclusion follows from Theorem E.1.2. ■

The result of Corollary E.1.3 extends (with obvious changes) to the case where V is defined as the space of all $u \in L^2(\Omega)$ such that $Au \in L^p(\Omega)$ for a given linear differential operator A. Also, we can consider finitely many functions g_k (verifying the conditions (a) and (b) of Corollary E.1.3 for different exponents $p_k > 1$) and finitely many linear differential operators A_k. In that case we shall deal with the functional

$$J(u) = \sum_{k=1}^m \int_\Omega g_k(A_k u)\, \mathrm{d}x + \frac{1}{2} \int_\Omega |u|^2\, \mathrm{d}x - \int_\Omega f u\, \mathrm{d}x,$$

defined on $V = \bigcap_{k=1}^m \left\{ u \in L^2(\Omega) : A_k u \in L^{p_k}(\Omega) \right\}$; V is a reflexive Banach space when endowed with the norm

$$\|u\|_V = \sum_{k=1}^m \|A_k u\|_{L^{p_k}} + \|u\|_{L^2}.$$

E.2 Preliminaries on Sobolev Spaces

Some basic results on Sobolev spaces are recalled here for the convenience of the reader. The details are available from many sources, for example, R. A. Adams and J. J. F. Fournier and W. P. Ziemer [602]. Let Ω be a bounded open set in $\mathbb{R}^N$ with Lipschitz boundary $\partial\Omega$, and let m and p be two strictly positive integers. The Sobolev space $W^{m,p}(\Omega)$ consists of all functions $u \in L^p(\Omega)$ which

admit weak derivatives $D^\alpha u$ in $L^p(\Omega)$, for all multi-indices α with $|\alpha| \le m$. This means the existence of functions $v_\alpha \in L^p(\Omega)$ such that

$$\int_\Omega v_\alpha \cdot \varphi \, \mathrm{d}x = (-1)^{|\alpha|} \int_\Omega u \cdot D^\alpha \varphi \, \mathrm{d}x \tag{E.1}$$

for all φ in the space $C_c^\infty(\Omega)$ and all α with $|\alpha| \le m$. Due to the denseness of $C_c^\infty(\Omega)$ in $L^p(\Omega)$, the functions v_α are uniquely defined by (E.1), and they are usually denoted as $D^\alpha u$. One can prove easily that $W^{m,p}(\Omega)$ is a separable Banach space when endowed with the norm $\|\cdot\|_{W^{m,p}}$ defined by the formula

$$\langle u, v \rangle_{W^{m,p}} = \Big(\sum_{|\alpha| \le m} \|D^\alpha u\|_{L^p}^p \Big)^{1/p},$$

where $\big\|D^0 u\big\|_{L^p} = \|u\|_{L^p}$. Notice that $C^m(\overline{\Omega})$ is a dense subspace of $W^{m,p}(\Omega)$.

It is usual to denote $W^{m,p}(\Omega)$ by $H^m(\Omega)$ for $p = 2$; this is a Hilbert space relative to the norm $\|\cdot\|_{H^m}$, associated to the scalar product

$$\langle u, v \rangle_{H^m} = \sum_{|\alpha| \le m} \int_\Omega D^\alpha u \cdot D^\alpha v \, \mathrm{d}x.$$

E.2.1 Theorem (The trace theorem) *There is a continuous linear operator*

$$\gamma = (\gamma_0, \dots, \gamma_{m-1}) \colon W^{m,p}(\Omega) \to L^p(\partial\Omega)^{m-1}$$

such that

$$\gamma_0(u) = u|_{\partial\Omega}, \quad \gamma_1(u) = \frac{\partial u}{\partial n}, \dots, \gamma_{m-1}(u) = \frac{\partial^{m-1} u}{\partial n^{m-1}}$$

for all u in $C^m(\overline{\Omega})$.

The closure of $C_c^\infty(\Omega)$ in $W^{m,p}(\Omega)$ is the Sobolev space $W_0^{m,p}(\Omega)$ (denoted $H_0^m(\Omega)$ when $p = 2$). This space coincides with the kernel of the trace operator γ, indicated in Theorem E.2.1.

On $H_0^1(\Omega)$, the norm $\|\cdot\|_{H^1}$ can be replaced by an equivalent norm,

$$\|u\|_{H_0^1} = \left(\int_\Omega \|\nabla u\|^2 \, \mathrm{d}x \right)^{1/2}.$$

In fact, there exists a constant $c > 0$ such that

$$\|u\|_{H_0^1} \le \|u\|_{H^1} \le c \|u\|_{H_0^1} \quad \text{for all } u \in H_0^1(\Omega).$$

This is a consequence of a basic inequality in partial differential equations:

E.2.2 Theorem (Poincaré's inequality) *If Ω is a bounded open subset of $\mathbb{R}^N$, then for every $p \in [1, \infty)$ there exists a constant $C > 0$ (depending only on Ω and p) such that*

$$\|u\|_{L^p} \le C \left(\int_\Omega \|\nabla u\|^p \, \mathrm{d}x \right)^{1/p}$$

for all $u \in W_0^{1,p}(\Omega)$.

Proof. Since $C_c^\infty(\Omega)$ is dense into $W_0^{1,p}(\Omega)$, it suffices to prove Poincaré's inequality for functions $u \in C_c^\infty(\Omega) \subset C_c^\infty(\mathbb{R}^N)$. The fact that Ω is bounded, yields the existence of two real numbers a and b such that

$$\Omega \subset \{\mathbf{x} = (\mathbf{x}', x_N) \in \mathbb{R}^{N-1} \times \mathbb{R} : a \le x_N \le b\}.$$

We have

$$u(x', x_N) = \int_a^{x_N} \frac{\partial u}{\partial x_N}(x', t)\, \mathrm{d}t,$$

and an application of the Rogers–Hölder inequality gives us

$$\begin{aligned} |u(\mathbf{x}', x_N)|^p &\le (x_N - a)^{p/q} \int_a^{x_N} \left| \frac{\partial u}{\partial x_N}(\mathbf{x}', t) \right|^p \mathrm{d}t \\ &\le (x_N - a)^{p/q} \int_{\mathbb{R}} \left| \frac{\partial u}{\partial x_N}(\mathbf{x}', t) \right|^p \mathrm{d}t. \end{aligned}$$

Then

$$\int_{\mathbb{R}^{N-1}} |u(\mathbf{x}', t)|^p \, \mathrm{d}\mathbf{x}' \le (x_N - a)^{p/q} \int_{\mathbb{R}^N} \left| \frac{\partial u}{\partial x_N}(\mathbf{x}) \right|^p \mathrm{d}\mathbf{x},$$

which leads to

$$\int_{\mathbb{R}^N} |u(\mathbf{x})|^p \, \mathrm{d}\mathbf{x} = \int_a^b \int_{\mathbb{R}^{N-1}} |u(\mathbf{x}', t)|^2 \, \mathrm{d}\mathbf{x}' \le \frac{(b-a)^p}{p} \int_{\mathbb{R}^N} \left| \frac{\partial u}{\partial x_N}(\mathbf{x}) \right|^p \mathrm{d}\mathbf{x}$$

and now the assertion of Theorem E.2.2 is clear. ■

By Poincaré's inequality, the inclusion $W_0^{m,p}(\Omega) \subset W^{m,p}(\Omega)$ is strict whenever Ω is bounded. Notice that $W_0^{m,p}(\mathbb{R}^N) = W^{m,p}(\mathbb{R}^N)$, due to the possibility to approximate (via mollification) the functions in $W^{m,p}(\mathbb{R}^N)$ by functions in $C_c^\infty(\mathbb{R}^N)$.

E.3 Applications to Elliptic Boundary-Value Problems

In what follows we shall illustrate the role of the variational methods in solving some problems in partial differential equations. More advanced applications may be found in books like those by B. Dacorogna [146], [147] and E. Giusti [221].

E.3.1 The Dirichlet Problem. Let Ω be a bounded open set in $\mathbb{R}^N$ and let $f \in C(\overline{\Omega})$. A function $u \in C^2(\Omega) \cap C(\overline{\Omega})$ is said to be a classical solution of the Dirichlet problem

$$\begin{cases} -\Delta u + u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{E.2}$$

provided that it satisfies the equation and the boundary condition pointwise. If u is a classical solution to this problem, then the equation $-\Delta u + u = f$ is

equivalent to

$$\int_\Omega (-\Delta u + u) \cdot v \, \mathrm{d}\mathbf{x} = \int_\Omega f \cdot v \, \mathrm{d}\mathbf{x} \quad \text{for all } v \in H_0^1(\Omega).$$

By Green's formula,

$$\int_\Omega (-\Delta u + u) \cdot v \, \mathrm{d}\mathbf{x} = -\int_{\partial\Omega} \frac{\partial u}{\partial n} \cdot v \, \mathrm{d}\mathbf{x} + \int_\Omega u \cdot v \, \mathrm{d}\mathbf{x} + \sum_{k=1}^N \int_\Omega \frac{\partial u}{\partial x_k} \cdot \frac{\partial v}{\partial x_k} \, \mathrm{d}\mathbf{x},$$

so that we arrive at the following restatement of (E.2):

$$\sum_{k=1}^N \int_\Omega \frac{\partial u}{\partial x_k} \cdot \frac{\partial v}{\partial x_k} \, \mathrm{d}\mathbf{x} + \int_\Omega u \cdot v \, \mathrm{d}\mathbf{x} = \int_\Omega f \cdot v \, \mathrm{d}\mathbf{x} \tag{E.3}$$

for all $v \in C_c^\infty(\Omega)$. It turns out that (E.3) makes sense for $u \in H_0^1(\Omega)$ and $f \in L^2(\Omega)$. We shall say that a function $u \in H_0^1(\Omega)$ is a *weak solution* for the Dirichlet problem (E.2) with $f \in L^2(\Omega)$ if it satisfies (E.3) for all $v \in H_0^1(\Omega)$. The existence and uniqueness of the weak solution for the Dirichlet problem (E.2) follows from Theorem E.1.2, applied to the functional

$$J(u) = \frac{1}{2} \|u\|_{H_0^1}^2 - \langle f, u \rangle_{L^2}, \quad u \in H_0^1(\Omega).$$

In fact, this functional is strictly convex and twice Gâteaux differentiable, with

$$J'(u; v) = \langle u, v \rangle_{H_0^1} - \langle f, v \rangle_{L^2}$$
$$J''(u; v, w) = \langle w, v \rangle_{H_0^1}.$$

According to Theorem E.1.2, the unique point of global minimum of J is the unique solution of the equation

$$J'(u; v) = 0 \quad \text{for all } v \in H_0^1(\Omega),$$

and clearly, the latter is equivalent with (E.3).

E.3.2 The Neumann Problem. Let Ω be a bounded open set in $\mathbb{R}^N$ (with Lipschitz boundary) and let $f \in C(\overline{\Omega})$. A function $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ is said to be a classical solution of the Neumann problem

$$\begin{cases} -\Delta u + u = f & \text{in } \Omega \\ \dfrac{\partial u}{\partial n} = 0 & \text{on } \partial\Omega, \end{cases} \tag{E.4}$$

provided that it satisfies the equation and the boundary condition pointwise. If u is a classical solution to this problem, then the equation $-\Delta u + u = f$ is equivalent to

$$\int_\Omega (-\Delta u + u) \cdot v \, \mathrm{d}\mathbf{x} = \int_\Omega f \cdot v \, \mathrm{d}\mathbf{x} \quad \text{for all } v \in H^1(\Omega),$$

and thus with

$$\sum_{k=1}^{N} \int_\Omega \frac{\partial u}{\partial x_k} \cdot \frac{\partial v}{\partial x_k} \, \mathrm{d}\mathbf{x} + \int_\Omega u \cdot v \, \mathrm{d}\mathbf{x} = \int_\Omega f \cdot v \, \mathrm{d}\mathbf{x} \quad \text{for all } v \in H^1(\Omega), \tag{E.5}$$

taking into account Green's formula and the boundary condition $\frac{\partial u}{\partial n} = 0$ on $\partial\Omega$. As in the case of the Dirichlet problem, we can introduce a concept of a weak solution for the Neumann problem (E.4) with $f \in L^2(\Omega)$. We shall say that a function $u \in H^1(\Omega)$ is a *weak solution* for the problem (E.4) if it satisfies (E.5) for all $v \in H^1(\Omega)$. The existence and uniqueness of the weak solution for the Neumann problem follows from Theorem E.1.2, applied to the functional

$$J(u) = \frac{1}{2} \|u\|^2_{H^1} - \langle f, u \rangle_{L^2}, \quad u \in H^1(\Omega).$$

The details are similar to those used in the case of the Dirichlet problem.

Corollary E.1.3 (and its generalization to finite families of functions g_K) allow us to prove the existence and uniqueness of the weak solutions even for some Neumann problems such as

$$\begin{cases} -\Delta u + u + u^3 = f & \text{in } \Omega \\ \dfrac{\partial u}{\partial n} = 0 & \text{on } \partial\Omega, \end{cases} \tag{E.6}$$

where $f \in L^2(\Omega)$. This corresponds to the case where

$$\begin{gathered} g_1(t) = \cdots = g_N(t) = t^2/2, \quad g_{N+1}(t) = t^4/4, \\ A_k u = \partial u / \partial x_k \text{ for } k = 1, \ldots, N, \quad A_{N+1} u = u, \\ p_1 = \cdots = p_N = 2, \; p_{N+1} = 4, \end{gathered}$$

and

$$J(u) = \frac{1}{2} \|u\|^2_{H^1} + \frac{1}{4} \|u\|^4_{L^4} - \langle f, u \rangle_{L^2}, \quad u \in V = H^1(\Omega) \cap L^4(\Omega).$$

According to Corollary E.1.3, there is a unique global minimum of J and this is obtained by solving the equation

$$J'(u; v) = 0 \quad \text{for all } v \in V,$$

that is, by solving the equation

$$\sum_{k=1}^{N} \int_\Omega \frac{\partial u}{\partial x_k} \cdot \frac{\partial v}{\partial x_k} \, \mathrm{d}\mathbf{x} + \int_\Omega u \cdot v \, \mathrm{d}\mathbf{x} + \int_\Omega u^3 \cdot v \, \mathrm{d}\mathbf{x} = \int_\Omega f \cdot v \, \mathrm{d}\mathbf{x},$$

for all $v \in V$. Notice that the latter equation represents the weak form of (E.6). The conditions under which the weak solutions provide classical solutions are discussed in textbooks like that by M. Renardy and R. C. Rogers [499].

E.4 The Galerkin Method

It is important to give here an idea how the global minimum of convex functionals can be determined via numerical algorithms. For this, consider a reflexive real Banach space V, with Schauder basis $(\mathbf{e}_k)_k$. This means that every $\mathbf{u} \in V$ admits a unique representation

$$\mathbf{u} = \sum_{k=1}^{\infty} c_k \mathbf{e}_k$$

with $c_k \in \mathbb{R}$, the convergence being in the norm topology. As a consequence, for each $n \in N$ there is a linear projection

$$P_n \colon V \to V, \quad P_n \mathbf{u} = \sum_{k=1}^{n} c_k \mathbf{e}_k.$$

Since $P_n \mathbf{u} \to \mathbf{u}$ for every $\mathbf{u}$, the Banach–Steinhaus theorem in functional analysis assures that $\sup \|P_n\| < \infty$. Consider a functional $J \colon V \to \mathbb{R}$ which is twice Gâteaux differentiable and for each $\mathbf{u} \in V$ there exist $\nabla J(\mathbf{u}) \in V^*$ and $H(\mathbf{u}) \in L(V, V^*)$ such that

$$J'(\mathbf{u}; \mathbf{v}) = \langle \nabla J(\mathbf{u}), \mathbf{v} \rangle$$
$$J''(\mathbf{u}; \mathbf{v}, \mathbf{w}) = \langle H(\mathbf{u})\mathbf{v}, \mathbf{w} \rangle$$

for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$. In addition, we assume that $H(\mathbf{u})$ satisfies estimates of the form:

$$\begin{cases} |\langle H(\mathbf{u})\mathbf{v}, \mathbf{w} \rangle| \le M \|\mathbf{v}\| \, \|\mathbf{w}\| \\ \langle H(\mathbf{u})\mathbf{v}, \mathbf{v} \rangle \ge \alpha \|\mathbf{v}\|^2 \end{cases} \tag{E.7}$$

for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$. Here M and α are positive constants. By Taylor's formula, J is strictly convex and $\lim_{\|\mathbf{u}\| \to \infty} J(\mathbf{u}) = \infty$. Then, by Theorem E.1.2, J is lower semicontinuous and admits a unique global minimum. In the Galerkin method, the global minimum $\mathbf{u}$ of J is found by a finite-dimensional approximation process. More precisely, one considers the restriction of J to $V_n = \operatorname{span}\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ and one computes the global minimum $\mathbf{u}_n$ of this restriction by solving the equation

$$\langle \nabla J(\mathbf{u}_n), \mathbf{v} \rangle = 0 \quad \text{for all } \mathbf{v} \in V_n.$$

The existence of $\mathbf{u}_n$ follows again from Theorem E.1.2. Remarkably, these minimum points approximate the global minimum $\mathbf{u}$ in the following strong way:

E.4.1 Theorem *We have*

$$\lim_{n \to \infty} \|\mathbf{u}_n - \mathbf{u}\| = 0.$$

Proof. Letting $v_n = P_n\mathbf{u}$, we know that $\mathbf{v}_n \to \mathbf{u}$. By Taylor's formula, for each n there is a $\lambda_n \in (0,1)$ such that

$$J(\mathbf{v}_n) = J(\mathbf{u}) + \langle \nabla J(\mathbf{u}), \mathbf{v}_n - \mathbf{u}\rangle + \frac{1}{2}\langle H(\mathbf{u} + \lambda_n(\mathbf{v}_n - \mathbf{u}))(\mathbf{v}_n - \mathbf{u}), \mathbf{v}_n - \mathbf{u}\rangle.$$

Combining this with the first estimate in (E.7), we get $J(\mathbf{v}_n) \to J(\mathbf{u})$. By the choice of $\mathbf{u}_n$, it yields that

$$J(\mathbf{u}) \le J(\mathbf{u}_n) \le J(\mathbf{v}_n),$$

so that $J(\mathbf{u}_n) \to J(\mathbf{u})$ too. Also, $\sup J(\mathbf{u}_n) < \infty$. Since $\lim_{\|\mathbf{u}\|\to\infty} J(\mathbf{u}) = \infty$, we deduce that the sequence $(\mathbf{u}_n)_n$ is norm bounded. According to Theorem B.1.6, it follows that $(\mathbf{u}_n)_n$ has a weak converging subsequence, say $\mathbf{u}_{k(n)} \overset{w}{\to} \mathbf{u}'$. Since J is lower semicontinuous, we have

$$J(\mathbf{u}') \le \liminf_{n\to\infty} J(\mathbf{u}_{k(n)}) \le J(\mathbf{u}),$$

from which it follows that $\mathbf{u}' = \mathbf{u}$ and $\mathbf{u}_n \overset{w}{\to} \mathbf{u}$. Again by Taylor's formula, for each n there is $\mu_n \in (0,1)$ such that

$$J(\mathbf{u}_n) = J(\mathbf{u}) + \langle \nabla J(\mathbf{u}), \mathbf{u}_n - \mathbf{u}\rangle + \frac{1}{2}\langle H(\mathbf{u} + \mu_n(\mathbf{u}_n - \mathbf{u}))(\mathbf{u}_n - \mathbf{u}), \mathbf{u}_n - \mathbf{u}\rangle.$$

This relation, when combined with the second estimate in (E.7), leads to

$$\frac{2}{\alpha}\|\mathbf{u}_n - \mathbf{u}\|^2 \le |J(\mathbf{u}_n) - J(\mathbf{u})| + |\langle \nabla J(\mathbf{u}), \mathbf{u}_n - \mathbf{u}\rangle|$$

and the conclusion of the theorem is now obvious. ■

The numerical solution of partial differential equations using discontinuous polynomial approximations (started in 1973 by W. H. Reed and T. R. Hill) led to the tremendous evolution of the *discontinuous Galerkin method*, very popular nowadays for solving problems in fluid mechanics. A nice account on this topic is made available by the book of V. Dolejčí and M. Feistauer [162].

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